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Crazy Merged 1

Sections 1–3

One merged book: the condensed books and the Paper 1–4 answers combined, section by section, with each topic appearing once. Read each chapter top to bottom. Where a topic was answered in the papers, the full paper answer sits in its place (marked "Paper answer", with the papers it was asked in), followed by a dashed box of anything extra the book had. Paper answers not tied to a chapter are at the end of each section, then the section's Rapid Revision Sheet.

Section	Title
1	Basic Principles (General Urology)
2	Oncology, DSD, Gender & Ageing
3	Infections & Inflammation

SECTION 1

Basic Principles (General Urology)

1 Personal and Professional Development of the Urologist

1 · Professionalism

- **Unprofessional behaviour in medical school** is one of the strongest predictors of later disciplinary action by medical boards.
- Surgeons in the **highest quartile of patient complaints have more surgical complications**, so unprofessional behaviour is a measurable safety signal.
- The Patient Advocacy Reporting System (PARS) uses patient complaints, and the Coworker Reporting System (CORS) uses co-worker reports. After a CORS report, most individuals are never reported again.

2 · Burnout

- **Definition (triad)**: emotional exhaustion, depersonalisation and diminished personal accomplishment, in response to chronic interpersonal stressors. It is distinct from depression.
- Prevalence is about **40%** of physicians, and urologists approach **50%**. It causes more medical error.
- Physician suicide is **1.4×** the population rate in men and **2.7×** in women.
- **20% rule**: spending less than 20% of effort on the most meaningful activity raises burnout risk about **3×**.
- Drivers are loss of autonomy, purpose and community, electronic health record time and administrative burden.

Prevention strategy	Effect
Institutional awareness seminars	Minimally effective
Practice interventions (scribes)	More effective
Restoring autonomy over schedule	Significantly reduces burnout
Mindfulness, communication training, exercise	Evidence-based reduction

> **KEY POINT**
>
> Interventions that restore autonomy, purpose or community work; awareness alone does not.

3 · Four domains of learning

Domain	Framework	Assessment
Knowledge	Bloom's taxonomy (apex: create)	Easiest; certification exams
Psychomotor	Imitation, manipulation, precision, articulation, naturalisation (novice to master)	Proctorship (often needed for robotic and laser credentialing), simulation

Domain	Framework	Assessment
Communication	TeamSTEPPS	Often untaught
Affective	Krathwohl's taxonomy (apex: characterisation by a value set)	Formative

SAQ Operating room safety : the role of surgeon.

Paper answer · P4

Type: SAQ (10 marks)
Source: 2021-Summer Paper 4, Q3a

Introduction

- About half of hospital adverse events involve surgery; half of these are preventable. The surgeon is **team leader** and ultimately responsible.
- **Swiss cheese model (Reason):** harm occurs when holes in several barriers line up, so safety needs layered defences.

WHO (World Health Organization) Surgical Safety Checklist (2008, 19 items)

Phase	Key checks
Sign in (before induction)	Identity, site, procedure, consent; site marked; allergy; airway; blood loss >500 mL
Time out (before incision)	Team introductions; confirm patient, site, procedure; surgeon states critical steps, duration, blood loss; antibiotic within 60 min; imaging displayed
Sign out (before leaving theatre)	Procedure recorded; counts correct; specimen labelled; equipment problems; recovery concerns

- Pilot in 8 hospitals: deaths fell 47%, major complications 36%.

Surgeon's responsibilities

1. **Universal Protocol** (Joint Commission, 2004): verification, **site marking** by the operating surgeon, patient awake, correct side only, time-out. Wrong-site surgery is a never event.
2. **Lead the time-out:** whole team engaged; repeat when procedure or surgeon changes.
3. **Retained surgical items** (sponges commonest): a count discrepancy raises risk about 100-fold; do not close until resolved; X-ray or RF (radiofrequency) tags.
4. **Fire safety:** the surgeon owns the **ignition source** (diathermy, light cables, laser). Let alcohol prep dry; FiO_2 (inspired oxygen fraction) ≤30% near the airway.
5. **Sharps:** double gloves, blunt needles, neutral zone (no hand-to-hand passing).
6. **Distractions:** "sterile cockpit" during critical steps (e.g. hilar clamping).
7. **Communication** (failure underlies up to 70% of events): Use **TeamSTEPPS** (Team Strategies and Tools to Enhance Performance and Patient Safety) tools: brief, debrief, **SBAR** (situation, background, assessment, recommendation), closed-loop check-back, **CUS** ("concerned, uncomfortable, safety issue"), two-challenge rule. Flatten hierarchy so juniors speak up.
8. **Self-monitoring: I'M SAFE** (illness, medication, stress, alcohol, fatigue, eating).

- **Also know:** positioning and padding; VTE (venous thromboembolism) prophylaxis; simulation.

Diagram to draw: Swiss cheese model with aligned holes.

Key points to remember:

- WHO checklist: sign in, time out, sign out.
- Universal Protocol: verify, mark, time-out.
- Count discrepancy: ~100-fold risk of a retained item.
- Fire triangle: surgeon owns ignition.

Also, from the condensed book (not in the paper answer):

- Communication errors contribute to about **70%** of medical errors and **27%** of malpractice claims.
- Situation awareness belongs to the individual, while a shared mental model belongs to the team.

KEY POINT

Certification tests the easiest domain, knowledge, but harm arises mainly in the communication, affective and psychomotor domains.

2 Evaluation of the Urologic Patient: History and Physical Examination

1 · Pain

- **Pain without tenderness is referred pain**: a non-tender painful testis may reflect an ipsilateral ureteric stone.

	Obstructive (colic)	Parenchymal
Cause	Distension of a hollow organ	Capsular stretch (pyelonephritis)
Pattern	Intermittent; restless, seeks a position	Constant; costovertebral angle tenderness; lies still

- Colic is mediated by prostaglandins and perceived in the **T11–L1** segments, which is why non-steroidal anti-inflammatory drugs (NSAIDs) work. A ureterovesical junction stone causes urgency and frequency.
- **Strangury** is stabbing pain at the end of micturition from an inflamed detrusor. An atonic bladder retains large volumes painlessly.

2 · Haematuria

SAQ Microscopic / asymptomatic haematuria and urinary biomarkers

Paper answer · P2 · P3 · merged from 2

Asked as: P3 SAQ "Asymptomatic haematuria and urinary biomarkers" (Asymptomatic Hematuria, Summer 2016 Paper III Q3d; Urinary biomarkers in diagnostic evaluation of hematuria, Summer 2021 Paper III Q3f) · P2 SAQ "Microscopic Hematuria" (Practice Set 1 Paper 2 Q3e; also 2016 Summer Paper 3 Q3d)

Definitions and epidemiology

- **MH (microscopic haematuria)**: **3 or more RBCs/HPF (red blood cells per high-power field)** on microscopy of one properly collected sample (AUA/SUFU (American Urological Association/Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction) 2020).
- **A dipstick alone is not enough**: myoglobin, semen and bacterial peroxidases give false positives.
- **GH (gross haematuria)**: visible; exclude pigmenturia and menstrual blood.
- Prevalence about 6.5%. Malignancy in about 3% of MH (over 85% of these are bladder cancer) and in 14–26% of GH (DETECT I, IDENTIFY).
- Haematuria is evaluated by **cystoscopy and upper tract imaging**; urine tests are non-invasive **adjuncts** for bladder cancer.

Causes

- **Neoplasm**: bladder (over 85% of cancers found), UTUC (upper tract urothelial carcinoma) and RCC (renal cell carcinoma) about 1 in 1000 each, prostate.
- **Benign**: BPE (benign prostatic enlargement), stones, UTI (urinary tract infection), stricture, TB (tuberculosis), schistosomiasis.
- **Medical renal** (IgA nephropathy): dysmorphic RBCs, casts, proteinuria.
- **Others**: exercise, endometriosis, sickle cell.

- Anticoagulants do not explain haematuria; evaluate as usual.

Initial evaluation

- History (smoking, occupation, cyclophosphamide, pelvic radiation, Lynch syndrome), examination, **blood pressure, creatinine**, urine culture.
- Suspected UTI: culture, treat, then **repeat urinalysis**; persistent MH means full evaluation.
- Glomerular or renal picture (dysmorphic RBCs, casts or proteinuria suggest a glomerular source): nephrology **and** urological evaluation.

AUA/SUFU 2020 risk stratification

	Low (all)	**Intermediate (any)**	**High (any)**
Age	Women under 50, men under 40	Women 50–59, men 40–59	60 or over
Smoking (pack-years)	Never or under 10	10–30	Over 30
RBCs/HPF	3–10	11–25	Over 25
Gross haematuria history	No	No	**Yes**
Cancer risk	0–0.4%	1–2.5%	About 6%
Work-up	Shared decision: repeat urinalysis in 6 months **or** cystoscopy + RUS (renal ultrasound)	Cystoscopy + RUS	Cystoscopy + **CT (computed tomography) urography**

- Additional risk factors (irritative symptoms, radiation, cyclophosphamide, occupational exposure) move a patient from low to intermediate.
- CT contraindicated: MR (magnetic resonance) urography, or retrograde pyelograms.
- Persistent MH after a negative ultrasound: CT or retrograde imaging.
- Hereditary RCC risk (VHL (von Hippel–Lindau)): upper tract imaging in any stratum.
- White-light cystoscopy. Cytology and urine markers are **not** used initially; cytology if CIS (carcinoma in situ) is suspected.
- **Gross haematuria**: cystoscopy, CT urography and cytology at any age.

Urine cytology

- **Sensitivity 31–62%, specificity 94–100%**: good for **high-grade tumours and CIS**, poor for low grade.
- **Paris System:** negative for HGUC (high-grade urothelial carcinoma), atypical, suspicious, HGUC (graded by nuclear-to-cytoplasmic ratio and nuclear features).
- Positive cytology with normal cystoscopy: random and prostatic urethral biopsies, upper tract washings.

Urinary biomarkers (sensitivity/specificity, Chou 2015 meta-analysis)

Marker	**Principle**	**Sens / spec**	**Note**
NMP22 (nuclear matrix protein 22)	Nuclear protein; lab or point-of-care	0.69/0.77	False positives: inflammation, stones

Marker	Principle	Sens / spec	Note
BTA (bladder tumour antigen) stat / TRAK	Complement factor H–related protein; dipstick or ELISA (enzyme-linked immunosorbent assay)	0.64/0.77	**False positive with blood** (factor H in serum)
UroVysion FISH (fluorescence in situ hybridisation)	Gain of chromosomes 3, 7, 17; loss of 9p21	0.63/0.87	After atypical cytology
ImmunoCyt	Antibodies to CEA (carcinoembryonic antigen) + two mucins	0.78/0.78	Highest sensitivity; operator dependent
Cxbladder	RT-PCR (reverse transcription polymerase chain reaction), 5 mRNAs	about 0.82/0.85	Needs validation

- **Also know:** Xpert Bladder (mRNA); AssureMDx (FGFR3, TERT, HRAS mutations + methylation).
- **Pattern:** more sensitive and less specific than cytology; they miss small low-grade tumours.

Role of markers in haematuria (guidelines)

- **AUA/SUFU 2020: no cytology or markers in the initial evaluation of microhaematuria**; cytology only after a negative work-up if CIS is suspected.
- **Gross haematuria:** cytology with cystoscopy and CT urography. A negative marker still leaves **≥10% cancer probability**, so cystoscopy is still needed.
- **No marker replaces cystoscopy** (AUA, EAU (European Association of Urology)); a negative test may one day help triage low-risk patients.
- **Recent advances:** urinary tumour DNA; methylation assays (Bladder EpiCheck).

Follow-up

- Negative evaluation: consider repeat urinalysis within 12 months; release if negative.
- Re-evaluate for gross haematuria, rising MH or new symptoms.

Diagram to draw: AUA/SUFU flowchart from MH to risk-based work-up (haematuria pathway), with markers shown as adjuncts.

> **Key points to remember:**
> - MH: 3 or more RBCs/HPF on microscopy of one sample; dipstick not enough.
> - Low: optional (repeat urinalysis or cystoscopy + ultrasound); intermediate: cystoscopy + ultrasound; high: cystoscopy + CT urography.
> - Any gross haematuria = high risk.
> - Cytology: specific for high grade/CIS, insensitive for low grade.
> - BTA is falsely positive with blood.
> - No markers in initial microhaematuria work-up; none replaces cystoscopy.

Also, from the condensed book (not in the paper answer):

- The commonest cause of gross haematuria over 50 years is **bladder cancer**.

Clue	Suggests
Initial haematuria	Urethra or prostate

Clue	Suggests
Total haematuria	Bladder or above
Terminal haematuria	Bladder neck
Vermiform clots / cuboid clots	Upper tract / bladder
Clears with irrigation but recurs / clears once a catheter is placed	Upper tract / prostate

3 · Lower urinary tract symptoms

- A **bladder diary** (usually 48 h) gives functional capacity (normal **300–400 mL**), nocturia and polyuria. Frequency means more than 5–6 voids a day.
- Daytime frequency without nocturia suggests a psychogenic cause. Nocturia without daytime frequency suggests nocturnal polyuria.
- Irritative symptoms may be the only sign of carcinoma in situ (CIS), so keep a low threshold for cystoscopy in smokers over 50.

*See the paper answer **IPSS score** (Section 16, Ch 149).*

4 · Urinary incontinence

Type	Mechanism	Hallmarks
Continuous	Outlet bypassed (fistula, ectopic ureter)	Constant wetness
Overflow	Volume exceeds capacity	Often nocturnal; long-standing obstruction; cured by relieving it
Stress	Sphincter weakness	Cough or sneeze; multiparity; after prostatectomy
Urge	Leakage with urgency	Overactive bladder, cystitis, occult cancer
Mixed	Stress plus urge	Treating one alone may worsen symptoms
Functional	Cannot reach the toilet	Mobility, cognition

*See the paper answer **Embryology and management of ectopic ureter.** (Section 4, Ch 52).*

*See the paper answer **Nocturnal enuresis** (Section 5, Ch 47).*

Also, from the condensed book (not in the paper answer):

- Enuresis affects about **15%** at 5 years and **1%** at 15 years. Primary enuresis beyond 6 years needs urological evaluation, and secondary enuresis should raise the possibility of abuse or bullying.

5 · Sexual and other symptoms

- **Erectile dysfunction (ED)** (National Institutes of Health 1993) is inability to attain or maintain an erection sufficient for satisfactory intercourse. **ED heralds occult coronary artery disease.**
- With low libido and ED, measure early-morning testosterone on two occasions **≥2 weeks** apart.

*See the paper answer **Premature ejaculation** (Section 9, Ch 74).*

*See the paper answer **Haematospermia** (Section 9, Ch 74).*

Also, from the condensed book (not in the paper answer):

- The AUA/Sexual Medicine Society definition of premature ejaculation is within about 2 min.
- The commonest cause of **pneumaturia** is an enterovesical fistula (diverticulitis, Crohn disease). Bloody urethral discharge suggests urethral carcinoma.

6 · Drug history

Side effect	Drugs
ED	Thiazides, propranolol
Ejaculatory dysfunction	α-blockers, methyldopa, phenothiazines
Priapism	**Trazodone**, phenothiazines, hydralazine, prazosin
Retention	Anticholinergics, calcium-channel blockers, α-agonists (pseudoephedrine), antihistamines
Reduced spermatogenesis	Alkylating agents, marijuana
Gynaecomastia	Digoxin, cimetidine, verapamil

Antithrombotic	Action	Half-life
Warfarin	Vitamin K antagonist	36–42 h
Dabigatran	Direct thrombin inhibitor	8–17 h
Rivaroxaban / apixaban	Factor Xa inhibitors	5–12 h / 12 h
Clopidogrel / ticagrelor	P2Y12 blockade, irreversible / reversible	6 h / 6–8 h

- Half-life is not duration: **clopidogrel acts for the life of the platelet**.
- **Bleeding risk** is low for cystoscopy and ureteroscopy, intermediate for transurethral resection of the prostate (TURP) and of bladder tumour (TURBT), and **high for percutaneous nephrolithotomy (PCNL)**, artificial urinary sphincter and robotic nephrectomy.

7 · Physical examination

- **Bladder**: palpable above about **150 mL** and visible at about 500 mL. **Bimanual examination under anaesthesia after TURBT** is standard of care and independently predicts pT3.
- **Prostate**: a nodule as hard as a flexed knuckle needs biopsy.
- **Torsion**: horizontal lie, high testis and short cord with pain. An absent cremasteric reflex is highly sensitive in children; a present reflex argues against torsion but does not exclude it.
- Normal testis is **4–5 × 2–3 cm**. A **hard testicular mass is cancer until proven otherwise**, whereas an epididymal mass is almost always benign.
- **Female**: a Q-tip deflection **≥30°** on Valsalva means urethral hypermobility, but it is not required to diagnose stress incontinence. Document prolapse by POP-Q.
- **Neuro-urology**: the bulbocavernosus reflex tests **S2–S4**. Saddle sensory loss with an absent reflex localises a sacral root lesion such as cauda equina.

3 Indications and Interpretation of Laboratory Testing

1 · Urinalysis: collection and gross examination

- Examine urine within **2 h** or refrigerate it at 2–8 °C. Refrigeration precipitates crystals.
- In children, suprapubic aspiration has the lowest contamination and catheterisation is often preferred. A **bag specimen is never used for culture**.
- Red urine is not always blood (beetroot, rifampicin, phenothiazines).
- Cloudy urine is most often **phosphaturia**, which clears with acetic acid; pyuria does not.

2 · Dipstick chemistry

Interferent	False negative	False positive
Ascorbic acid	Glucose, bilirubin, blood, leucocyte esterase	–
Phenazopyridine / dyes	Leucocyte esterase	Nitrite, bilirubin, ketones
Alkaline urine	–	Protein
High specific gravity	Leucocyte esterase, glucose	–
Semen, menses, exercise	–	Blood
Mesna, levodopa	–	Ketones
Contamination (commonest)	–	Leucocyte esterase and nitrite

- **Specific gravity** is 1.001–1.035 and becomes **fixed at 1.010 in renal insufficiency**.
- pH is 4.5–8.0. A pH **>7.5** with infection means a urease-splitting organism (Proteus), which leads to struvite staghorn stones. Uric acid and cystine stones form in acid urine.

*See the paper answer **Renal tubular acidosis (distal and type 2)** (Section 11, Ch 95).*

Also, from the condensed book (not in the paper answer):

- The renal threshold for glucose is a serum level of about **180 mg/dL**, and SGLT2 inhibitors cause glycosuria. Nitroprusside detects acetoacetate but **not β-hydroxybutyrate**.

3 · Haematuria

*See the paper answer **Microscopic / asymptomatic haematuria and urinary biomarkers** (Section 1, Ch 2).*

Red urine	Sediment	Supernatant	Plasma
Haematuria	RBCs	Clear	–
Haemoglobinuria	No RBCs	Red, haem positive	Red
Myoglobinuria	No RBCs	Red, haem positive	Clear
Dyes, porphyria	No RBCs	Red, haem negative	–

Plasma is red in haemoglobinuria because haemoglobin is haptoglobin-bound; small unbound myoglobin is filtered at once.

	Glomerular	Non-glomerular medical	Urological
RBC shape	Dysmorphic	Round	Round
RBC casts	Present	Absent	Absent
Proteinuria	Significant	Often significant	Minimal, even with gross bleeding

- Proteinuria of 2+–3+ (100–300 mg/dL) is almost never urological, so refer to nephrology.

SAQ Berger's Disease.

Paper answer · P1

Type: SAQ (10 marks)
Source: 2020-Summer Paper 1, Q3a

Berger's disease is IgA (immunoglobulin A) nephropathy (IgAN).

Definition

- **Commonest primary glomerulonephritis (GN) worldwide**, in children too. Dominant **mesangial IgA deposits** on immunofluorescence.
- Commonest cause of **glomerular haematuria** (about one-third). Second–third decades; male predominance.
- **IgA vasculitis** (Henoch–Schönlein purpura) is the systemic form.

Pathogenesis

- "Multi-hit" model: abnormal (galactose-deficient) IgA1 → autoantibodies → immune complexes → mesangial deposition, complement activation, proliferation. Mucosal (respiratory) infection triggers flares.

Clinical features

- **Recurrent painless macroscopic haematuria during or within days of an upper respiratory infection** ("synpharyngitic"), or after exercise. Post-streptococcal GN follows infection by weeks.
- Persistent microscopic haematuria, proteinuria, hypertension; occasional loin pain; rarely nephrotic syndrome or RPGN (rapidly progressive GN).

Evaluation (urologist's role)

- **Dysmorphic RBCs (red blood cells), RBC casts, proteinuria** → glomerular source. **Normal C3 and C4.**
- Without casts, complete the urological evaluation (imaging ± cystoscopy) to exclude a urinary tract cause.
- **Renal biopsy** confirms. Once diagnosed, repeat haematuria work-ups are not needed.

Oxford MEST-C score (biopsy)

Item	Grades
M: mesangial hypercellularity	M0, M1
E: endocapillary hypercellularity	E0, E1
S: segmental sclerosis	S0, S1
T: tubular atrophy/interstitial fibrosis	T0 ≤25%, T1 26–50%, T2 >50%
C: crescents	C0 none, C1 <25%, C2 ≥25% of glomeruli

Prognosis

- Often benign, but **20–30% reach ESKD (end-stage kidney disease)**.
- Poor: persistent proteinuria, hypertension, raised creatinine, older onset, high MEST-C.

Management

- No cure. **Supportive:** ACE (angiotensin-converting enzyme) inhibitor or ARB (angiotensin receptor blocker), BP (blood pressure) control, low salt, stop smoking.
- **Newer drugs:** SGLT2 (sodium–glucose cotransporter 2) inhibitors; sparsentan (dual endothelin–angiotensin receptor blocker); targeted-release budesonide acting on gut mucosal lymphoid tissue.
- Persistent proteinuria above about 0.75–1 g/day despite supportive care: consider a 6-month steroid course (KDIGO (Kidney Disease: Improving Global Outcomes) 2021); weigh infection risk.
- Crescentic RPGN: steroids plus cyclophosphamide.
- **Transplant:** recurs histologically often, clinically rarely.

Diagram to draw: glomerulus with mesangial IgA deposits.

> **Key points to remember:**
> - Commonest primary GN; haematuria with upper respiratory infection.
> - Normal complement; biopsy shows mesangial IgA.
> - Oxford MEST-C; 20–30% ESKD.
> - ACE inhibitor or ARB is the mainstay.

Also, from the condensed book (not in the paper answer):

- Other clues are deafness with a family history (Alport syndrome), haemoptysis (Goodpasture syndrome), and a recent throat infection with low C3 (poststreptococcal glomerulonephritis).

4 · Proteinuria

SAQ Proteinuria

Paper answer · P1

Asked as:

- Proteinuria — *Summer 2017 Paper I Q3f (SAQ)*
- Proteinuria — *Summer 2021 Paper I Q3d (SAQ)*

Type: SAQ (10 marks)
(also answered in 2017-Summer Paper 1, Q3f)

Definition

- Normal protein excretion is **<150 mg/day** (another source: **80–150 mg/day**) (albumin <30 mg/day). The filter (glomerulus) blocks (holds back) albumin and the proximal tubule reabsorbs filtered small proteins.
- Normal mix: about 30% albumin, 30% globulins, 40% tissue proteins (Tamm–Horsfall).
- **Nephrotic range:** >3.5 g/day, with hypoalbuminaemia, oedema and hyperlipidaemia.

Detection and quantification (KDIGO, Kidney Disease: Improving Global Outcomes, categories by ACR)

Category	ACR (albumin-to-creatinine ratio)	PCR (protein-to-creatinine ratio)	24-h protein
A1 normal–mild	<30 mg/g (<3 mg/mmol)	<150 mg/g (<15 mg/mmol)	<150 mg
A2 moderate	30–300 mg/g	150–500 mg/g	150–500 mg
A3 severe	>300 mg/g (>30 mg/mmol)	>500 mg/g	>500 mg
- **Spot early-morning ACR is first line**; PCR if non-albumin protein suspected; 24-h urine with creatinine check. Alternatives: PCR, 24-h urine.			
- **Dipstick** (tetrabromophenol blue) detects albumin only (mainly albumin), from 20–30 mg/dL; misses dilute urine and **light chains**. **Sulfosalicylic acid** detects (precipitates) all proteins; dipstick negative but sulfosalicylic acid positive → suspect myeloma.			

Types

Type	Mechanism	Amount	Example
Glomerular (commonest)	Leaky filter	Often >2 g; albumin ≈70% (about 70%)	Diabetes, glomerulonephritis, FSGS (focal segmental glomerulosclerosis), amyloid
Tubular	Failed reabsorption of small proteins	<2 g (seldom >2–3 g); mainly globulins	Fanconi syndrome, heavy metals, drugs
Overflow	Excess plasma protein	Variable (0.3–2 g)	**Myeloma** (Bence Jones), myoglobin, haemoglobin

By timing	Features
Transient	Fever, exercise, heart failure; commonest in children
Orthostatic	Young men, upright only, <1 g; benign (normal first-morning sample). Test with a split (supine overnight vs upright day) collection
Persistent	Mostly glomerular; needs full work-up

Evaluation

Positive dipstick → repeat; exclude UTI (urinary tract infection), fever, exercise → persistent → quantify (ACR, PCR or 24-h protein) → creatinine/eGFR (estimated glomerular filtration rate), sediment (dysmorphic red cells / RBCs, casts), ultrasound → glomerular pattern (>2 g, albumin) → nephrology ± renal biopsy; C3, C4, ANA if systemic → globulins → electrophoresis, immunofixation (myeloma)

- UTI: urinary tract infection; eGFR: estimated glomerular filtration rate; RBC: red blood cell; ANA: antinuclear antibody.

Urological relevance

- **Haematuria work-up:** proteinuria, casts or dysmorphic cells suggest (suggests) medical renal disease; urological bleeding rarely raises protein much (adds little protein).
- **Reflux nephropathy:** proteinuria marks progression to renal failure.
- **After nephrectomy or partial nephrectomy:** >50% loss of mass causes hyperfiltration, proteinuria and FSGS. AUA (American Urological Association) 2021 counts proteinuria as a reason to prefer (favour) nephron sparing and refer to nephrology.
- **Living donors:** significant proteinuria excludes donation. **Also know:** mTOR (mammalian target of rapamycin) inhibitors cause it.

Management

- Treat the cause; blood pressure control; **ACE (angiotensin-converting enzyme) inhibitor or ARB (angiotensin receptor blocker)**; SGLT2 (sodium–glucose cotransporter-2) inhibitors.

Diagram to draw: glomerulus and proximal tubule marking glomerular, tubular and overflow sites.

> **Key points to remember:**
> - Normal <150 mg/day; nephrotic >3.5 g/day.
> - ACR A2 30–300, A3 >300 mg/g.
> - Glomerular, tubular, overflow; orthostatic proteinuria is benign.
> - Dipstick misses light chains; use sulfosalicylic acid or immunofixation.

5 · Infection: leucocyte esterase, nitrite and culture

	Leucocyte esterase	Nitrite
Detects	Host response (neutrophils)	Organism (gram-negative nitrate reductase)
Performance	Sensitive, non-specific	Specific, insensitive
Classic miss	Glycosuria, proteinuria	**Enterococcus**; low counts

- If only one is positive, do microscopy and culture before antibiotics. A dipstick never replaces culture.
- Culture thresholds are **≥10^5 CFU/mL** for clean catch and **≥10^3 CFU/mL** for a catheter specimen; **any growth on suprapubic aspiration is significant**.

6 · Sediment microscopy

- Squamous cells mean contamination, while **renal tubular cells always mean renal pathology**. Eosinophils suggest acute interstitial nephritis.
- Schistosoma haematobium ova have a **terminal spike**.

Cast	Meaning
Hyaline	May be normal (exercise, heat)
RBC	Glomerulonephritis

Cast	Meaning
WBC	Pyelonephritis, interstitial nephritis
Muddy-brown granular	Acute tubular necrosis
Fatty	Nephrotic syndrome

LAQ Urinary crystals

Paper answer · P2

Type: LAQ (25 marks)
Source: Practice Set 11 Paper 2, Q2

Introduction

- Crystals are precipitated urinary solutes seen on microscopy of the sediment. They occur in both normal and diseased states.
- Their value lies in suggesting stone type, metabolic disease, infection or drug toxicity. **Cystine is the one crystal that is never normal.**

How crystals form (physicochemistry)

- Urine is normally **supersaturated** with calcium oxalate and phosphate. Crystals form only when the concentration product passes a critical level.

Saturation zone	What happens
Undersaturated (below Ksp, solubility product)	No crystals; stones may dissolve
Metastable (Ksp to Kf, formation product)	No spontaneous nucleation; growth on existing crystals; inhibitors effective
Unstable (above Kf)	Spontaneous nucleation, growth and aggregation

- **Nucleation** in vivo is mostly heterogeneous (on cell debris or other crystals). Crystals then grow, aggregate and are retained, for example on **Randall plaque**.
- **Inhibitors:** citrate, magnesium, pyrophosphate, Tamm–Horsfall protein, nephrocalcin, osteopontin. No inhibitor exists for uric acid or cystine, so these depend on pH and concentration.
- **Determinants of crystalluria:** urine volume, solute excretion, pH, temperature and inhibitors.

Examination of urine for crystals

- Fresh midstream specimen, examined within 2 hours at body or room temperature. **Refrigeration precipitates crystals** that may not redissolve.
- Dipstick pH (physiological range 4.5–8.0); centrifuged sediment; polarised light helps identify crystals.
- Phosphaturia (cloudy alkaline urine) **clears with acetic acid**; pyuria does not.
- Candida mimics calcium oxalate and red cells; budding and hyphae separate it.

Classification by urine pH and shape

Urine pH	Crystal	Shape	Clinical significance
Acidic	**Uric acid**	Rhomboid or lemon-shaped plates, rosettes; yellow-brown	Uric acid stones (pH <5.5; pKa 5.35), gout, tumour lysis
Acidic	Amorphous urates	Pink "brick-dust" granules	Usually normal (concentrated, cooled urine)
Acidic	**Cystine**	**Hexagonal** colourless plates	Always abnormal: cystinuria
Acidic or alkaline	**Calcium oxalate dihydrate**	**Envelope** (octahedral)	Common, often normal; hypercalciuria
Acidic or alkaline	Calcium oxalate monohydrate	Dumbbell or oval	Hyperoxaluria; ethylene glycol poisoning
Alkaline	**Triple phosphate (struvite)**	**Coffin-lid** prisms	Urease-producing infection (Proteus); staghorn stones
Alkaline	Amorphous phosphate	White granules	Benign phosphaturia
Alkaline	Calcium phosphate (brushite, apatite)	Needles or rosettes; amorphous	RTA (renal tubular acidosis), hyperparathyroidism
Alkaline	Ammonium biurate	"Thorny apple" spheres	Old specimens; ammonium urate stones (diarrhoea, laxative abuse, ileostomy)
Any	Cholesterol	Notched flat plates	Nephrotic syndrome, lipiduria

- **Rare metabolic crystals:** 2,8-dihydroxyadenine, brown spiculated spheres (APRT [adenine phosphoribosyltransferase] deficiency, radiolucent stones); xanthine (hereditary xanthinuria, high-dose allopurinol); tyrosine needles and leucine spheres (severe liver disease).

Drug crystals

- **Two routes:** the drug itself precipitates (direct), or it changes urine chemistry (indirect).

Drug	Crystal and features	Prevention
Indinavir (protease inhibitor)	Needles or fan-shaped plates; crystalluria in about two-thirds of new users; stones 4–13%; radiolucent, often invisible on CT	Hydration; more soluble below pH 5.5
Atazanavir	Highest stone risk of newer antiretrovirals	Hydration
Sulfonamides (sulfadiazine)	"Sheaves of wheat"	Hydration, alkalinise
Acyclovir (high-dose intravenous)	Birefringent needles; acute kidney injury	Slow infusion, hydration
Amoxicillin, ciprofloxacin (high dose)	Needles; worse in alkaline urine	Hydration
Methotrexate (high dose)	Crystal nephropathy	Alkalinise, hydration
Triamterene, ephedrine, guaifenesin, silicate antacids	Stones of drug or metabolite	Stop the drug

- **Indirect:** loop diuretics and steroids (hypercalciuria, calcium stones); acetazolamide and topiramate (alkaline urine, hypocitraturia, calcium phosphate); cytotoxics (uric acid); laxative abuse (ammonium urate).

Clinical significance

1. **Diagnosis of stone type** before stone analysis is available; crystals in a stone former help direct metabolic work-up.
2. **Recurrence risk:** persistent crystalluria in repeated early-morning samples, and large or aggregated crystals, suggest active stone disease.
3. **Infection:** struvite crystals with urine pH >7.2 signal a urease-producing organism.
4. **Acute kidney injury:** uric acid (tumour lysis), calcium oxalate monohydrate (ethylene glycol) and drug crystals cause intratubular obstruction.
5. **Monitoring therapy:** absence of cystine or uric acid crystals confirms adequate alkalinisation. - Most crystalluria in healthy people is transient and harmless.

Management principles

- Repeat on a fresh, warm specimen; confirm pH; 24-hour urine metabolic evaluation and stone analysis.
- **Fluids** for urine output above 2–2.5 L/day.
- **pH manipulation:** alkalinise (potassium citrate) for uric acid, cystine, methotrexate and sulfonamides; avoid alkali in calcium phosphate and struvite.
- Treat the cause: antibiotics and stone clearance for struvite; stop or change the drug; specific therapy (thiols for cystine, allopurinol for uric acid).

Diagram to draw: sketches of hexagon (cystine), envelope (calcium oxalate), coffin lid (struvite), rhomboid (uric acid), sheaf of wheat (sulfonamide).

> **Key points to remember:**
> - Cystine hexagons are always pathological.
> - Acidic: uric acid, cystine; alkaline: struvite, phosphates; calcium oxalate at any pH.
> - Indinavir: crystalluria in two-thirds, stones 4–13%, radiolucent.
> - Refrigerated urine precipitates crystals: examine fresh within 2 hours.

7 · Serum tests

- Check creatinine and estimated glomerular filtration rate (eGFR) whenever obstruction is suspected, because renal function sets the urgency of drainage.
- Creatinine rises only after about **50%** loss of GFR. CKD stages by eGFR (mL/min/1.73 m²) are G1 ≥90, G2 60–89, G3a 45–59, G3b 30–44, G4 15–29 and G5 <15.

8 · Tumour markers and cytology

SAQ Tumour markers in urology

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 18 SAQ Sheet, Q3d

Definition and uses

- A substance made by a tumour, measured in blood, urine or tissue.
- Uses: screening, diagnosis, staging, prognosis, response, and detecting relapse. Ideal: specific, sensitive, proportional to bulk, short half-life.

Prostate

- **PSA (prostate-specific antigen):** a kallikrein serine protease (hK3) from luminal cells; **organ-specific, not cancer-specific** (BPH (benign prostatic hyperplasia), prostatitis, instrumentation raise it). Best marker for post-treatment follow-up.
- **Refinements:** density (PSA ÷ volume; cut-offs 0.08–0.15); %free PSA (lower in cancer; for PSA 4–10 ng/mL); velocity (little value); **PHI (Prostate Health Index) = (p2PSA ÷ free PSA) × √total PSA**; 4Kscore.
- **Urine:** PCA3 (prostate cancer antigen 3) after DRE (digital rectal examination); score = PCA3 mRNA ÷ KLK3 mRNA × 1000.
- **Also know:** tissue genomic classifiers.

Testis (germ cell tumours)

Marker	Half-life	Source	Never raised in
AFP (α-fetoprotein)	5–7 days	Embryonal, yolk sac	Pure seminoma, choriocarcinoma
β-hCG (human chorionic gonadotrophin)	24–36 hours	Choriocarcinoma, embryonal; seminoma in ~15%	Yolk sac tumour
LDH (lactate dehydrogenase)	24 hours	Non-specific; reflects bulk	—

- Measure before and after orchidectomy, on chemotherapy, in follow-up. Post-orchidectomy values give the **S stage** (S1: LDH <1.5 × normal, hCG <5000 mIU/mL, AFP <1000 ng/mL) and the **IGCCCG (International Germ Cell Cancer Collaborative Group)** risk group.
- Raised AFP with "pure seminoma" histology → treat as NSGCT (non-seminomatous germ cell tumour). False positives: liver disease, marijuana, LH (luteinising hormone) cross-reaction.
- **miR-371a-3p:** sensitive new marker; not raised by teratoma.

Bladder and upper tract

- **Urine cytology:** sensitive and specific for high-grade cancer, poor for low grade; for surveillance, not screening.
- NMP22 (nuclear matrix protein 22), BTA (bladder tumour antigen), ImmunoCyt, UroVysion FISH (fluorescence in situ hybridisation): selected use only.

Others

- **Kidney and penis:** no validated serum marker.
- **Adrenal:** metanephrines, cortisol, aldosterone–renin ratio. **Neuroblastoma:** urinary VMA (vanillylmandelic acid)/HVA (homovanillic acid) raised in 90–95%.

Diagram to draw: marker decline curve after orchidectomy using half-lives.

Key points to remember:

- PSA: organ-specific, not cancer-specific.
- AFP never in pure seminoma; half-lives AFP 5–7 days, hCG 24–36 hours.
- S stage uses post-orchidectomy values.
- Cytology: good for high grade only.

SAQ Urine cytology

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 1 Paper 1, Q3b

Definition and principle

- Microscopy of shed urothelial cells (Papanicolaou, 1945).
- **Principle:** high-grade cells are poorly cohesive, shed readily and look atypical; low-grade cells look almost normal. So cytology detects **high-grade tumours and CIS (carcinoma in situ)** but misses low grade.

Specimens

- Voided urine (not the first morning sample, whose cells have degenerated), at least 30 mL; bladder washing; upper tract washings.

Performance

- Sensitivity 31–62%, specificity 94–100%; strongly grade-dependent.
- **False positives:** instrumentation, stones, infection, radiotherapy, intravesical BCG (bacille Calmette–Guérin), polyomavirus ("decoy" cells), ileal conduit.
- **False negatives:** low grade, low cellularity, degeneration.

The Paris System

- Published 2016; **2nd edition 2022**. Aim: detect **HGUC (high-grade urothelial carcinoma)**. Key criterion: **N:C (nuclear-to-cytoplasmic) ratio**, plus hyperchromasia, coarse chromatin and irregular nuclear membrane.

Category	Criteria
Non-diagnostic	Inadequate
Negative for HGUC	N:C <0.5, or atypia explained by a benign cause
AUC (atypical urothelial cells)	N:C ≥0.5 and one feature
Suspicious for HGUC	N:C ≥0.7, at least two features, few cells (<5–10)
HGUC	N:C ≥0.7, at least two features, many cells
LGUN (low-grade urothelial neoplasm)	Fibrovascular cores; rarely diagnosable
Other malignancy	Squamous, adenocarcinoma, secondary

- It reduces equivocal "atypia" reports; cancer risk rises with each category.

Clinical uses

- **Not** for screening or initial microhaematuria evaluation (AUA/SUFU [American Urological Association / Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction] 2020).

- Gross haematuria; NMIBC (non-muscle-invasive bladder cancer) surveillance (mandatory in high risk); suspected CIS; upper tract tumours.
- **Positive cytology, normal cystoscopy:** image upper tracts, selective washings ± ureteroscopy, random bladder and prostatic urethral biopsies.
- **Also know:** FISH (fluorescence in situ hybridisation; chromosomes 3, 7, 17, 9p21), NMP22 (nuclear matrix protein 22), BTA (bladder tumour antigen), CxBladder: more sensitive, less specific; none replaces cystoscopy.

Diagram to draw: normal urothelial cell beside an HGUC cell with high N:C ratio.

> **Key points to remember:**
> - High specificity, low sensitivity; good for high grade and CIS.
> - Paris System: N:C ≥0.7 for suspicious and HGUC; AUC N:C ≥0.5.
> - Positive cytology, normal cystoscopy: upper tracts, random biopsies.
> - Not for screening or initial microhaematuria.

9 · Stone, adrenal and andrology tests

*See the paper answer **Metabolic evaluation in stone disease** (Section 11, Ch 99).*

*See the paper answer **Adrenal mass evaluation, incidentaloma and adrenocortical carcinoma** (Section 14, Ch 146).*

Also, from the condensed book (not in the paper answer):

- Aldosterone **>15 ng/dL** with an aldosterone-to-renin ratio **>20** leads to adrenal vein sampling.

*See the paper answer **Semen analysis nomenclature.** (Section 9, Ch 69).*

Also, from the condensed book (not in the paper answer):

- Leucocytes **>1 million/mL** define leucocytospermia.

10 · Office tests

*See the paper answer **Uroflowmetry** (Section 7, Ch 107).*

Also, from the condensed book (not in the paper answer):

- Qmax **>15 mL/s** gives a low probability of obstruction and 10–15 mL/s is equivocal. A post-void residual **<100 mL** is generally acceptable.
- **No antibiotic prophylaxis** for simple cystoscopy or urodynamics without risk factors (age, steroids, immunodeficiency, anatomical anomaly, colonised material, long hospitalisation).

4 Basic Principles of Urologic Ultrasonography

1 · Physical principles

- Piezoelectric crystals convert alternating current into a longitudinal mechanical wave. Echo amplitude gives pixel brightness, and round-trip time gives depth. The urological range is **3.5–20 MHz**.
- Velocity equals frequency × wavelength. Speed of sound in soft tissue is **1540 m/s**, so frequency and wavelength vary inversely.

	Axial resolution	Lateral resolution
Separates	Objects one behind the other	Objects side by side
Improved by	Higher frequency	Narrow beam: focal zone at the depth of interest

> **KEY POINT**
> Absorption is proportional to frequency, so high frequency resolves finely but penetrates poorly: **7–18 MHz** for the testis and **3.5–5 MHz** for the adult kidney.

- **Reflection depends on the impedance difference at an interface, not on density.** Soft tissues cluster together (1.4–1.7 × 10^6 kg/m²·s), while air (0.0004) and bone (7.8) are extremes. Hence gel, bowel-gas obscuration and stone shadowing.

Artefacts

Artefact	Mechanism	Appearance
Acoustic shadowing	Strong reflection or attenuation	Dark band behind a stone; the stone looks crescentic, so size is inaccurate
Increased through-transmission	Less attenuation than surroundings	Bright tissue behind a simple cyst
Edging (critical angle)	Refraction along a curved surface	Shadow at the testicular pole
Reverberation	Bounce between two strong reflectors	Equally spaced echoes; the second at **twice the depth**
Comet tail	Many small reflectors (bowel gas)	Bright tail; may hide the lower renal pole
Twinkling	Rough reflective surface on colour Doppler	Colour mosaic behind a stone; detects small non-shadowing stones

2 · Modes

SAQ Doppler in urology

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 4 Paper 1, Q3b

Principle

- **Doppler shift:** sound reflected from moving red cells changes frequency in proportion to velocity and the cosine of the insonation angle: $\Delta f = 2f_0 v \cos\theta / c$. Keep the angle **below**

60°.

- **RI (resistive index) = (PSV − EDV) / PSV** (peak systolic and end-diastolic velocity); reflects downstream resistance.

Modes

Mode	Shows	Use
Colour	Velocity and direction (red towards, blue away by convention)	Flow in kidney, testis, ureteric jets
Spectral (duplex)	Waveform at one point; velocities, RI	Penile and renal arteries
Power	Amplitude only; 3–5 times more sensitive, less angle-dependent	Low flow: torsion, prostate lesions

Applications

Organ	Findings
Scrotum	**Torsion:** absent intratesticular flow (compare with the other side); epididymitis: increased flow; **varicocele:** venous reflux on Valsalva; intratesticular RI >0.6 suggests poor spermatogenesis
Penis (after intracavernosal injection)	**PSV <25 cm/s = arterial ED (erectile dysfunction)**, 25–35 equivocal, >35 normal; **EDV >5 cm/s and RI <0.75 = veno-occlusive dysfunction**; Peyronie plaque assessment
Priapism	Ischaemic: absent cavernosal flow; non-ischaemic: high flow (PSV >50 cm/s), locates the fistula
Kidney	Normal RI 0.47–0.70; **raised RI in acute obstruction** (useful in pregnancy, where change in RI is more accurate); renal artery stenosis: high velocity and parvus-tardus waveform; arteriovenous fistula after biopsy or PCNL (percutaneous nephrolithotomy)
Transplant	Arterial thrombosis: absent signal; **venous thrombosis: reversed diastolic flow**; stenosis: high velocity; rejection: loss of diastolic flow, high RI
Bladder	Ureteric jets (absent on the obstructed side)
Tumours	Vascularity of renal and testicular masses; tumour thrombus; hypervascular prostate cancer foci

- **Also know:** intraoperative micro-Doppler in microsurgical varicocelectomy; contrast-enhanced ultrasound and elastography in multiparametric ultrasound.

Limitations

- Operator- and angle-dependent; poor in obesity and deep vessels; early torsion may show collateral flow, so **Doppler cannot exclude torsion**.

Diagram to draw: spectral waveform marking PSV and EDV with the RI formula.

> **Key points to remember:**
> - RI = (PSV − EDV)/PSV; normal renal 0.47–0.70.
> - Penile PSV <25 cm/s arterial; EDV >5 cm/s veno-occlusive.
> - Absent intratesticular flow = torsion, but explore on clinical grounds.

Also, from the condensed book (not in the paper answer):

- **Elastography**: strain elastography (manual compression) is qualitative and operator-dependent. **Shear-wave elastography is quantitative** (shear waves 1–10 m/s; $E = 3\rho c^2$). Hard does not mean malignant.

3 · Contrast-enhanced ultrasound (CEUS)

LAQ Contrast agents in urology: iodinated and non-iodinated agents, complications and contrast-induced nephropathy

Paper answer · P1 · merged from 3

Asked as: P1 LAQ "Iodinated and other contrast agents (use in urology) and their complications" · P1 SAQ "Non iodinated contrast materials" · P1 SAQ "Acetylcysteine and Contrast Induced Nephropathy (CIN)"

Introduction

- Iodine (atomic number 53, K-edge 33 keV) absorbs X-rays strongly. All iodinated agents are built on a **tri-iodinated benzene ring**.
- They are filtered by the glomerulus, not secreted or reabsorbed, so they concentrate in urine. Half-life is about 2 hours with normal function; biliary excretion rises in renal failure.
- Modern practice uses **non-ionic low-osmolar** (iohexol, about 600–850 mOsm/kg) or **iso-osmolar** (iodixanol, 290) agents. High-osmolar ionic agents caused far more reactions.
- Uses: CT (computed tomography) urography, IVU (intravenous urography), pyelography, cystourethrography, angiography.
- Non-iodinated agents are used when iodinated contrast is risky (allergy, renal impairment) or with MRI (magnetic resonance imaging) and ultrasound.

Classification of iodinated agents

Class	Example	Osmolality (mOsm/kg)	Iodine : particles
High-osmolar ionic monomer	Diatrizoate	~1500–2000 (5–8 × plasma)	3 : 2
Low-osmolar non-ionic monomer	Iohexol, iopamidol	~600–850	3 : 1
Low-osmolar ionic dimer	Ioxaglate	~600	6 : 2
Iso-osmolar non-ionic dimer	Iodixanol	~290	6 : 1

- **Why it matters:** a higher ratio gives more opacity per particle. Fewer particles mean less fluid shift, pain and reactions. So non-ionic low- or iso-osmolar agents are standard intravascularly.

Types of contrast agent and their uses in urology

Agent	Principle	Urological use
Iodinated: ionic high-osmolar (diatrizoate); non-ionic low-osmolar (iohexol); iso-osmolar dimer (iodixanol, 290 mOsm/kg)	Iodine absorbs X-rays; glomerular filtration	CT urography, angiography; retrograde pyelogram, urethrogram, cystogram
Gadolinium (GBCA, gadolinium-based contrast agent)	Shortens T1 relaxation; chelated	MRI of renal mass, prostate, MR urography

Agent	Principle	Urological use
Microbubbles	Gas in shell, intravascular	CEUS (contrast-enhanced ultrasound): cysts; voiding urosonography
Carbon dioxide	Negative contrast	Angiography in CKD (chronic kidney disease)

Complications of iodinated contrast

A. Hypersensitivity (allergic-like) reactions

- Mostly non-IgE (immunoglobulin E) direct mast-cell release: idiosyncratic, not dose-related. Severe reactions: 0.22% with ionic vs 0.04% with non-ionic agents (Katayama, 1990).

Type	Timing	Features
Immediate	Within about 1 hour (most within minutes)	Urticaria, angio-oedema, bronchospasm, hypotension; anaphylaxis if severe
Non-immediate	Hours to days	Maculopapular rash; T-cell mediated, milder

- Risk factors: previous reaction (strongest), asthma, atopy, severe cardiac disease, drug allergy, female sex. **Shellfish ("iodine") allergy is not a cross-reaction.**
- **Physiological reactions:** warmth, metallic taste, nausea, vasovagal hypotension with bradycardia, arrhythmia, seizures.
- Retrograde studies are safe in allergy.

B. Contrast-associated acute kidney injury (see the separate CIN section below)

C. Others

- **Metformin:** lactic acidosis only if AKI (acute kidney injury) develops. ACR (American College of Radiology): eGFR (estimated glomerular filtration rate) ≥30 and no AKI, continue. eGFR <30 or AKI, withhold 48 hours and restart when function (creatinine) is stable.
- **Thyroid:** iodine load can trigger hyperthyroidism/thyrotoxicosis (Jod-Basedow) in Graves' disease or nodular goitre. It blocks radio-iodine scans or therapy for several weeks.
- **Extravasation:** usually minor, but large power-injected volumes can rarely cause compartment syndrome or skin necrosis. Elevate the limb, apply compresses and observe. Get a surgical review if swelling increases, blisters form, or perfusion or sensation changes.
- **Also know:** iodide "mumps"; myasthenia flare; pregnancy (crosses placenta, not teratogenic); breastfeeding can continue; phaeochromocytoma safe with IV (intravenous) non-ionic agents; dialysis patients need no urgent dialysis.

Prevention and treatment of contrast reactions

Premedication for high-risk patients / prior reaction (ACR 2023)

Regimen	Drugs
1	Prednisone 50 mg orally 13, 7 and 1 hour before + diphenhydramine 50 mg 1 hour before
2	Methylprednisolone 32 mg orally 12 and 2 hours before + diphenhydramine 50 mg 1 hour before

- First consider ultrasound, MRI or non-contrast CT. Breakthrough reactions still occur, so keep resuscitation drugs ready. Retrograde pyelography is safe even after anaphylaxis, because

little contrast is absorbed.

Treating reactions `Stop injection → airway, oxygen, call for help Urticaria → observe ± H1-antihistamine (may progress) Bronchospasm → salbutamol inhaler 2 puffs (90 µg each), up to 3 times Anaphylaxis → intramuscular (IM) adrenaline 0.3–0.5 mg (1 mg/mL) every 5–15 min (repeat), max 1 mg; IV fluids Vasovagal → legs up, IV fluids, IV atropine Observe ≥30 min; later allergy testing; record the reaction`

Contrast-induced nephropathy (CIN) and contrast-associated AKI

Definitions

- **CIN (contrast-induced nephropathy)** (older term): creatinine rise ≥25% or ≥0.5 mg/dL within 48–72 hours of iodinated contrast, with no other cause.
- Current terms (ACR):
- **CA-AKI (contrast-associated acute kidney injury):** AKI within 48 hours of contrast, whatever the cause.
- **CI-AKI (contrast-induced AKI):** the subset actually caused by contrast.
- **KDIGO (Kidney Disease: Improving Global Outcomes) AKI:** creatinine rise ≥0.3 mg/dL in 48 hours, ≥1.5 × baseline in 7 days, or urine <0.5 mL/kg/h for ≥6 hours.

Pathophysiology

- **Renal (afferent) vasoconstriction** causes **medullary hypoxia** and ischaemic tubular injury (ischaemic ATN, acute tubular necrosis), plus **direct tubular toxicity**. Usually transient; dialysis rarely needed.

Risk factors

- **Patient:** CKD (strongest), diabetes, age, dehydration, heart failure, anaemia, NSAIDs (non-steroidal anti-inflammatory drugs).
- **Procedure:** high contrast volume, repeated doses within 48–72 hours, intra-arterial route, high-osmolar agents.
- **eGFR thresholds:** real risk mainly at **eGFR <30 mL/min/1.73 m²** (IV contrast); ESUR (European Society of Urogenital Radiology) uses **<45** for intra-arterial first-pass exposure.
- ACR/NKF (National Kidney Foundation) 2020: older uncontrolled studies overstated the risk; it is rare but real.

Prevention

- Consider ultrasound or non-contrast CT instead.
- Lowest dose of a **low- or iso-osmolar** agent; weight-based dosing; a low tube voltage (80 kVp, kilovolt peak) needs less contrast. Avoid repeat doses within 48 hours; stop NSAIDs.
- **IV isotonic saline** (e.g. 1 mL/kg/h before and after) for high-risk patients (eGFR <30 not on dialysis): the only proven measure.
- **PRESERVE (2018):** sodium bicarbonate and **NAC (N-acetylcysteine)** gave no benefit over saline.
- **AMACING (2017):** no prophylaxis was non-inferior to hydration at eGFR 30–59.
- Prophylactic dialysis (removing contrast by dialysis) does not reduce risk.
- **Metformin:** withhold for 48 hours only if eGFR <30 or AKI (see above).

Acetylcysteine

Introduction

- **NAC:** acetylated form of the amino acid cysteine; a **thiol (sulfhydryl, –SH) compound**. Given orally, intravenously or by nebuliser.

Mechanisms of action

- **Glutathione precursor:** supplies cysteine, the rate-limiting substrate for **glutathione** synthesis.
- **Antioxidant:** the free –SH group scavenges reactive oxygen species.
- **Mucolytic:** –SH breaks **disulfide bonds** in mucoproteins, lowering mucus viscosity.
- **Thiol–disulfide exchange:** in theory forms a soluble mixed disulfide with cystine.

Clinical uses

- **1. Paracetamol (acetaminophen) overdose: main indication.**
- Overdose saturates normal pathways; the toxic metabolite **NAPQI** (N-acetyl-p-benzoquinone imine) builds up once glutathione is exhausted, causing hepatic necrosis.
- NAC restores glutathione. It is most effective **within 8 hours**, but still worth giving later.
- **IV 21-hour regimen:** 150 mg/kg, then 50 mg/kg over 4 h, then 100 mg/kg over 16 h. Shorter two-bag regimens are now used in some countries.
- **2. Mucolytic:** COPD (chronic obstructive pulmonary disease), cystic fibrosis.

Urological relevance

- **Contrast-induced AKI:** once widely used (early small trials, KDIGO 2012 suggestion). **PRESERVE (2018):** NAC gave no benefit over saline. It is **no longer recommended**; isotonic saline hydration is the only proven measure.
- **Cystinuria:** thiol action is theoretical; standard thiols are **tiopronin** and D-penicillamine, not NAC.
- **Cyclophosphamide cystitis: mesna** (another thiol) is the agent used.
- **Mucus after enterocystoplasty or neobladder:** sometimes used orally or in irrigation; evidence weak.
- **Male infertility:** antioxidant for oxidative stress (high ROS (reactive oxygen species), sperm DNA fragmentation); limited evidence.
- **Ischaemia–reperfusion** (renal surgery, transplant): antioxidant protection studied; not established.

Adverse effects

- **Anaphylactoid reactions** with IV loading (flushing, rash, bronchospasm): slow the infusion, give antihistamine.
- Nausea, vomiting; unpleasant sulphur smell; bronchospasm when nebulised in asthma.
- Safe in pregnancy and renal impairment for paracetamol poisoning; benefit outweighs risk.

Non-iodinated contrast agents

1. Gadolinium-based contrast agents (GBCAs)

- **Mechanism:** paramagnetic gadolinium **shortens T1 relaxation** of nearby water, giving bright enhancement. Free gadolinium is toxic, so it is **chelated** (linear or macrocyclic; macrocyclic is more stable). Usual dose 0.1 mmol/kg; renal excretion.

- **NSF (nephrogenic systemic fibrosis):** skin and organ fibrosis, joint contractures, no proven treatment. Risk: eGFR <30 mL/min/1.73 m², dialysis, AKI.

ACR group	Agents	NSF risk
I (linear)	Gadodiamide, gadopentetate, gadoversetamide	Highest
II	Gadobenate, gadoteridol, gadoterate, gadobutrol	**Little or none, even in ESRD (end-stage renal disease)**
III	Gadoxetate (liver-specific agent)	Few NSF data; recent ACR editions have moved some agents between groups, so check the current manual

- **Gadolinium deposition:** traces retained in brain (dentate nucleus) and bone, mainly with linear agents; no proven harm. Europe suspended most linear agents (2017).
- **Other:** allergic reactions rarer than with iodinated agents; not nephrotoxic at usual doses; avoid in pregnancy unless essential.
- **Uses:** renal mass, prostate mpMRI (multiparametric MRI), MR urography and angiography.

2. Ultrasound contrast (microbubbles)

- Gas core (perfluorocarbon or sulfur hexafluoride) in a lipid or albumin shell; strong harmonic echoes.
- Purely **intravascular**; gas exhaled by lungs. **No nephrotoxicity or radiation.** Rare cardiopulmonary reactions.
- **CEUS:** solid vs cystic renal mass (Bosniak), ablation monitoring, surveillance in CKD. Intravesical use: **voiding urosonography (ceVUS) for VUR (vesicoureteral reflux)**; ureteric patency after PCNL (percutaneous nephrolithotomy).

3. Other agents

- **CO_2 angiography:** negative contrast; not allergenic or nephrotoxic. Useful in CKD; **never above the diaphragm** (cerebral embolism).
- **Ferumoxytol** (iron oxide): MRI agent in renal failure. Lymphotropic iron nanoparticles help nodal staging (nodal metastases fail to darken).
- **Barium:** GI (gastrointestinal) studies only; contraindicated if perforation suspected.
- **Also know:** air as negative contrast (historical); radiotracers (^{99m}Tc, PET (positron emission tomography)) are not contrast agents; MRI safety (pacemakers, metal).

Diagrams to draw

- Tri-iodinated benzene ring, monomer versus dimer, on an osmolality scale.
- Chelated gadolinium (macrocyclic vs linear ring).
- Contrast, then vasoconstriction plus tubular toxicity, then medullary hypoxia, then AKI.
- Paracetamol to NAPQI to glutathione conjugation, with NAC replenishing glutathione.

Key points to remember:

- Iodine : particle ratio 1.5 (ionic), 3 (non-ionic monomer), 6 (iodixanol, 290 mOsm/kg).
- Non-ionic agents: severe reactions 0.04% vs 0.22% ionic.
- Anaphylaxis: IM adrenaline 0.3–0.5 mg, max 1 mg; shellfish allergy is not a risk.
- Premedication: prednisone 50 mg at 13-7-1 h or methylprednisolone 32 mg at 12-2 h, plus diphenhydramine.
- CIN = rise ≥25% or 0.5 mg/dL in 48–72 h; now CA-AKI vs CI-AKI.
- Main risk: eGFR <30 (IV); <45 intra-arterial.
- Isotonic saline only proven; PRESERVE 2018: bicarbonate and NAC useless; AMACING: no prophylaxis at eGFR 30–59.
- Metformin: withhold 48 h only if eGFR <30 or AKI.
- NAC: thiol; glutathione precursor, antioxidant, mucolytic; paracetamol antidote, best within 8 h; cystinuria uses tiopronin, not NAC.
- GBCAs shorten T1; NSF risk eGFR <30; group II agents almost no risk.
- Microbubbles: intravascular, no nephrotoxicity; ceVUS for reflux. CO_2 angiography: never above diaphragm.

Also, from the condensed book (not in the paper answer):

- Bubbles cannot cross the glomerulus and **never opacify the collecting system**.

4 · Safety and ALARA

	Mechanical	**Thermal**
Nature	Cavitation, streaming	Heating
Greatest	Near gas: lung, bowel	High frequency, focusing, long dwell, Doppler
Index	Mechanical index (MI): **≤0.7** low risk; limit time if >0.4 next to lung or bowel	Thermal index (TI) is the probability of a 1 °C rise; keep **<2** in urology

- **MI and TI are displayed estimates, not safety limits.**
- ALARA (as low as reasonably achievable) means low power, short examinations and adjusted focus. The **informed operator is the most important safety factor**.

5 · Renal ultrasound

- Normal kidney length is **10–12 cm** and width 4–5 cm, measured in the true midsagittal plane. Cortex is **>7 mm**, parenchyma **>15 mm** and intrarenal RI **0.47–0.70**.
- Cortex is slightly hypoechoic to liver; an iso- or hyperechoic cortex suggests medical renal disease.
- **Pyramids mimic calyces**: pyramids lie outside the hilar fat, and the collecting system lies inside it.
- Ultrasound-guided PCNL access is hard with a long skin-to-stone distance, a non-dilated system or a staghorn.
- After renal biopsy, complications rise with **severe hypertension, not the number of passes**. Haematuria appears within 6 h, so 6 h of observation equals admission.

*See the paper answer **Radiology for solid renal masses.** (Section 13, Ch 137).*

6 · Bladder and pelvic ultrasound

- Bladder volume = W × H × L × **0.625**.

*See the paper answer **Methods to estimate prostate size** (Section 16, Ch 149).*

Also, from the condensed book (not in the paper answer):

- **Bladder wall thickness predicts bladder outlet obstruction better than uroflowmetry**, post-void residual or prostate volume.
- **An automated bladder scanner is not an imaging study.** Fluid outside the bladder (ascites, urinoma, lymphocele) gives a wrong residual.

7 · Scrotal ultrasound

- Show both testes side by side, because **diffuse lymphoma or leukaemia is visible only by comparison**.
- Intratesticular PSV is **<10 cm/s**, and an RI **>0.6** suggests impaired spermatogenesis.

> **KEY POINT**
>
> In torsion, epididymal collaterals can appear within hours, and reactive hyperaemia can give a false-negative study. Ultrasound cannot diagnose torsion; only the surgeon can, so strong suspicion means exploration.

8 · Penile and urethral ultrasound

*See the paper answer **Doppler in urology** (Section 1, Ch 4).*

Also, from the condensed book (not in the paper answer):

- Arterial dilation **<75%** and acceleration time **>110 ms** also indicate arteriogenic ED. PSV is the most accurate measure, and arteriogenic ED flags systemic cardiovascular disease.

9 · Transrectal ultrasound (TRUS) and therapy

*See the paper answer **TRUS and prostate biopsy** (Section 17, Ch 155).*

*See the paper answer **HIFU** (Section 17, Ch 163).*

Also, from the condensed book (not in the paper answer):

- **Pregnancy**: ultrasound is first-line and safe in all trimesters, but grey-scale is poor for ureteric stones.

5 Principles of CT and Radiographic Imaging

1 · Radiation biology and safety

	Deterministic	Stochastic
Threshold	Yes	**None** (linear no-threshold model)
Dose increases	Severity	Probability (cumulative)
Mechanism	Cell loss	Mutation in an oncogene or tumour suppressor
Examples	Erythema, epilation, sterility, cataract	Cancer

- Absorbed dose is in gray (Gy). **Effective dose** (sievert, Sv) adds tissue weighting and sets exposure limits.

Study	Effective dose
Chest x-ray	**0.07 mSv**
Kidney–ureter–bladder (KUB) film / two-view abdomen	**0.7 / 1.4 mSv**
Ultra-low-dose CT abdomen–pelvis	About 1 mSv (1–2)
Low-dose CT	<3–4 mSv
Conventional CT abdomen–pelvis	**10–20 mSv**; 10 mSv carries about a 1 in 1000 cancer risk

> **KEY POINT**
>
> Stochastic risk has no threshold, so no dose is safe. Hence ALARA (as low as reasonably achievable): a risk–benefit decision before every ionising study, low-dose protocols and non-ionising alternatives.

2 · Radiographic studies

- KUB is used to follow radio-opaque stones, before shock-wave lithotripsy. CT detects more than **95%** of stones. An **upright film is most sensitive for free air**.

LAQ Describe IVU & its applications in Urology.

Paper answer · P1

Type: LAQ (25 marks)
Source: 2021-Summer Paper 1, Q2

Definition and principle

- IVU (intravenous urography; also IVP, intravenous pyelography or excretory urography) is a series of timed abdominal radiographs after intravenous iodinated contrast.
- Contrast is **freely filtered by the glomerulus and not reabsorbed**. As water is reabsorbed it is concentrated in the tubules (**nephrogram**) and then outlines the calyces, pelvis, ureters and bladder (**pyelogram**).
- So IVU shows **anatomy plus a rough, qualitative idea of function and drainage** of each kidney.
- **Current status:** largely replaced by CTU (CT [computed tomography] urography). It is still used where CT is unavailable (common in India) or when dynamic excretion and drainage

must be seen.

Contrast media

Type	Example	Feature
High-osmolar ionic	Diatrizoate	More reactions; obsolete for IV use
Low-osmolar non-ionic (standard)	Iohexol, iopamidol	Fewer reactions
Iso-osmolar non-ionic	Iodixanol	High-risk renal patients

Preparation

- Check creatinine or eGFR (estimated glomerular filtration rate), allergy, asthma, pregnancy, diabetes on metformin, myeloma.
- Laxative the night before (bowel gas hides the kidneys); short fast but **no dehydration**.
- Premedicate high-risk patients, e.g. prednisone 50 mg at 13, 7 and 1 h plus diphenhydramine 50 mg at 1 h.

Technique: film sequence

Film	Timing	Shows
Scout (control) KUB (kidney–ureter–bladder)	Before contrast	Calcification, bones, gas, technique
Contrast	Low-osmolar, about 1 mL/kg (usually 50–100 mL) IV	—
Nephrogram / nephrotomogram	About 1 min (60–90 s)	Renal size, outline, parenchyma
Pyelogram	5 min	Calyces; then apply **ureteric compression**
Compression film	10 min	Well-distended calyces and pelvis
Release (full-length) film	About 15 min	Whole ureters
Bladder and **post-void** films	20–30 min	Bladder outline, residue, lower ureters
Delayed films	Up to 24 h	Level of obstruction; **no excretion at 24 h = non-functioning kidney**
- **Also know:** films in expiration; prone film fills the ureters; oblique views; erect film for nephroptosis.		
- Compression is avoided in suspected obstruction or colic, aortic aneurysm, recent abdominal surgery and trauma.		

Contraindications

- Previous severe contrast reaction; significant renal impairment; pregnancy (relative); dehydration, myeloma.

Nephrogram patterns

Pattern	Causes
Absent	Renal artery occlusion, agenesis, avulsion
Delayed, increasingly dense	**Acute obstruction**, renal artery stenosis
Striated	Acute pyelonephritis, obstruction
Rim (cortical rim) sign	Renal infarction (capsular collaterals)
Bilateral persistent dense	Hypotension, ATN (acute tubular necrosis)

Applications in urology

Condition	IVU features
Calculi	Confirms an opacity is in the tract; **radiolucent stone = filling defect**; site, obstruction, calyceal anatomy before SWL (shock wave lithotripsy) or PCNL (percutaneous nephrolithotomy)
Acute obstruction	Delayed dense nephrogram, delayed excretion, dilatation down to the block, **standing column** of contrast, forniceal rupture
PUJ (pelviureteric junction) obstruction	Ballooned pelvis, delayed drainage, non-dilated ureter; crossing-vessel indentation
Duplex system	Upper moiety obstructed or non-functioning: lower moiety displaced, "**drooping lily**"
Ureterocele	"**Cobra head**" (spring onion) at the bladder
Horseshoe kidney	Low, medially rotated; lower calyces point medially
Retrocaval ureter	"Reverse J" or **fish-hook** at L3
Medullary sponge kidney	"**Paintbrush**" papillary striations
Papillary necrosis	**Ring sign**, "lobster claw" calyces
Genitourinary TB (tuberculosis)	**Moth-eaten** calyx, **phantom calyx**, Kerr's kink, **pipestem** or beaded ureter, contracted bladder, autonephrectomy
Urothelial tumour	Irregular filling defect, non-visualised calyx
Retroperitoneal fibrosis	Medial deviation and tapering of ureters with hydronephrosis
Pelvic lipomatosis	Pear-shaped bladder
Schistosomiasis	Ureteric striations, calcified bladder
- **Trauma: one-shot intraoperative IVP:** 2 mL/kg IV push, single film at 10 min. It confirms a functioning contralateral kidney before exploration in the unstable patient.	
- **Also know:** nephroptosis (descent >5 cm or two vertebral bodies supine to erect); urinary fistulae (ureterovaginal); renal masses (distortion); post-operative assessment; diverticula.	

Complications

- **Contrast reactions:** immediate (urticaria to anaphylaxis) or delayed (rash). Severe reaction: stop contrast, **IM (intramuscular) adrenaline 0.3–0.5 mg** every 5–15 min (maximum 1 mg), oxygen, β2-agonist. Observe 30 min.
- **CA-AKI (contrast-associated acute kidney injury):** prevent with hydration, low or iso-osmolar contrast, lowest dose.
- Radiation; metformin-related lactic acidosis in renal impairment.

Limitations and current role

- Two-dimensional; bowel gas; poor parenchymal detail; misses small masses; poor in renal failure.
- CTU is more sensitive for stones, masses and urothelial lesions, but gives a higher radiation dose than IVU. IVU remains useful in resource-limited settings.

Recent advances

- Split-bolus CTU (fewer phases); dual-energy CT (virtual unenhanced images); MR urography without radiation.

Diagram to draw: normal IVU with minor and major calyces, pelvis, ureter and the three narrowings; beside it a "drooping lily".

> **Key points to remember:**
> - Contrast is filtered, not reabsorbed: nephrogram then pyelogram.
> - Delayed, increasingly dense nephrogram = acute obstruction; no excretion at 24 h = non-functioning.
> - Classic signs: drooping lily, cobra head, fish-hook, paintbrush, ring sign, phantom calyx.
> - Severe reaction: IM adrenaline 0.3–0.5 mg; CTU has replaced IVU.

P2 *Lower urinary tract trauma in pelvic fracture — plain and CT cystography (retrograde 350 mL), flame-shaped extraperitoneal versus intraperitoneal leak*

- Passive filling by excreted contrast misses ruptures. Extraperitoneal rupture can give a "molar tooth" pattern.
- A large extraperitoneal leak prevents distension and can **hide a coexisting intraperitoneal rupture**.

3 · Fluoroscopic studies

Setting	Peak kilovoltage (kVp)	Tube current (mA)
Changes	Photon energy	Photon number
Dose	**Rises exponentially**	Rises linearly
Penetration / contrast	Increases / decreases	Unchanged / increases

- **Retrograde pyelography** is chosen when intravenous contrast is hazardous (allergy, renal dysfunction). Use about 50% contrast, because dense contrast hides small filling defects. Over-pressure causes pyelovenous and pyelolymphatic backflow and sepsis.

SAQ VCUG

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 6 Paper 1, Q3c

Definition

- **VCUG (voiding cystourethrogram; MCU, micturating cystourethrogram):** fluoroscopic study of bladder filling and voiding after catheter instillation of contrast. It shows the bladder, **posterior urethra** and UVJ (ureterovesical junction) competence.
- **Gold standard for VUR (vesicoureteral reflux)** and PUV (posterior urethral valves).

Indications

- **Children:** febrile UTI (urinary tract infection) with abnormal ultrasound or recurrent infection (EAU [European Association of Urology]: also age <12 months); bilateral hydroureteronephrosis (PUV); dilated ureter; ureterocele; neurogenic bladder.
- **Adults:** stricture (with RGU [retrograde urethrogram]), diverticula, fistula, post-operative leak, voiding dysfunction.

Technique

- **Sterile urine** first. After UTI, wait until urine is sterile and the child is improving (about a week); the old 4–6-week delay is unnecessary.
- Scout film → catheter: **5 Fr feeding tube (infants), 6–8 Fr (children), 14–16 Fr (adults)** → confirm tip position.
- **Gravity fill** (≤70 cm H_2O) to expected capacity: <2 years weight (kg) × 7; 2–14 years (age × 30) + 30 mL; >14 years 300–500 mL. (Koff formula: (age + 2) × 30 mL.)
- **Oblique views** (UVJ, prostatic urethra) → remove catheter → **void under fluoroscopy** → **post-void film**.
- **Cyclic VCUG** (2–3 fill–void cycles) detects 12–20% more VUR.

Findings

- **VUR grade (International Reflux Study, 1981):**

Grade	Appearance
I	Ureter only, not dilated
II	Pelvis and calyces, no dilatation
III	Mild–moderate dilatation, minimal blunting
IV	Moderate dilatation and tortuosity
V	Gross dilatation; papillary impressions lost

- Filling-phase reflux has a poorer resolution prognosis than voiding-only reflux.
- **PUV:** dilated, elongated posterior urethra, prominent bladder neck, trabeculation.
- Ureterocele, diverticula, residual urine; post-void pelvic retention suggests PUJ (pelviureteric junction) obstruction.

Limitations and complications

- Invasive, radiation, UTI, catheter trauma; snapshot (occult VUR); interobserver variability; autonomic dysreflexia (injury above T6).

Alternatives

- **Direct radionuclide cystography:** more sensitive, lower dose, poor anatomy.

- **CeVUS (contrast-enhanced voiding urosonography):** no radiation, accuracy close to VCUG.
- **PIC (positional instillation of contrast) cystography:** occult VUR with negative VCUG.

Diagram to draw: VUR grades I–V.

> **Key points to remember:**
> - Gold standard for VUR and PUV; sterile urine first.
> - Capacity (age × 30) + 30 mL (2–14 years); cyclic study detects 12–20% more.
> - Grades I–II non-dilating, III–V dilating.

Also, from the condensed book (not in the paper answer):

- **Retrograde urethrography (RUG)** shows the anterior urethra. Use a 45° oblique, the balloon at about 2 mL in the fossa navicularis, and 20–30 mL of contrast. The prostatic urethra does not distend normally, which is not a stricture; the posterior urethra needs a combined VCUG.
- **Fluoroscopic renal access**: bull's-eye uses a C-arm tilted 15–30° toward the surgeon, with the needle end-on over the calyx.

4 · Fluoroscopy dose reduction

- Dose is proportional to fluoroscopy time. Recording it reduces it.
- Halving the collimator diameter cuts the exposed area **fourfold**. Low-dose mode halves mA and dose. Pulsed fluoroscopy at 8, 4 and 1 pulses/s saves **73%, 87% and 97%** against continuous (30 pulses/s).
- **Inverse square law**: doubling the distance gives ¼ of the dose.
- Place the source below the patient and the detector close to the patient. Wear a **0.25–0.5 mm** lead apron, a thyroid shield and glasses; the legs receive most scatter.

> **KEY POINT**
> Surgeon dose is mostly scatter from the patient, so everything that lowers patient dose protects the team.

5 · CT principles

SAQ Newer generation CT scan

Paper answer · P1

Type: SAQ (10 marks)
Source: 2015-Summer Paper 1, Q3a

Introduction

- Computed tomography (CT) is the reference imaging for stones, masses and trauma. Attenuation is expressed in Hounsfield units (HU): water 0, air −1000.

Evolution

Generation	Feature
Helical (~1989)	Continuous table movement, volumetric data

Generation	Feature
Multidetector (1998 → 320-row)	Sub-millimetre slices, 3D reconstruction
Dual-source/dual-energy (2005)	Two energies: material analysis
Photon-counting (US approval 2021)	Energy-sorts each photon: less noise, dose and metal artefact

Dual-energy CT (DECT)

- Two energies (e.g. 80 and 140 kVp) compared pixel by pixel: **uric acid vs non-uric-acid stones**, **virtual unenhanced** images (one phase fewer) and **iodine maps** (hyperdense cyst vs enhancing mass).

Low-dose CT

Protocol	Dose
Conventional abdomen/pelvis	10–20 mSv
Low-dose	<3–4 mSv
Ultra-low-dose	1–2 mSv
- Stone sensitivity stays ~95%; use standard dose if body mass index >30. EAU (European Association of Urology): ultrasound first; AUA (American Urological Association): CT first.	

Reconstruction

- Filtered back projection is noisy; iterative reconstruction halves dose; **deep learning reconstruction** gives good images at ~1 mSv.

Contrast phases and protocols

Phase	Timing	Use
Arterial	20–30 s	Renal arteries, donor anatomy, active bleeding
Corticomedullary	30–70 s	Hypervascular (clear cell) tumours, renal vein
Nephrographic	90–120 s	Most sensitive for renal masses
Excretory	7–10 min	Collecting system, ureter, urine leak
- **CT urography:** non-contrast, nephrographic and excretory phases (~20–30 mSv). **Split-bolus** merges phases, lowering dose but may miss subtle urothelial lesions.		
- **Renal mass:** enhancement ≥20 HU = solid, 10–20 equivocal, <10 cyst; Bosniak 2019 for cysts.		
- **Adrenal:** <10 HU unenhanced = lipid-rich adenoma; otherwise 15-minute washout (absolute >60% or relative >40% = adenoma).		

Phase	Timing	Use
- **Also know:** stone HU (uric acid 200–400, calcium oxalate 600–1200; indinavir invisible); bone window for fragments; partial volume causes pseudoenhancement of small cysts; ALARA (as low as reasonably achievable) dosing, since radiation risk has no threshold.		

Diagram to draw: Timeline: axial → helical → multidetector → dual-source → photon-counting.

> **Key points to remember:**
> - Doses: conventional 10–20, low-dose <3–4, ultra-low 1–2 mSv.
> - DECT: uric acid stones, virtual unenhanced, iodine maps.
> - Enhancement ≥20 HU = solid; deep learning enables ~1 mSv CT.

Material	HU
Macroscopic fat (diagnostic of angiomyolipoma, AML)	≤ −10
Simple cyst	−9 to 20
Lipid-rich adrenal adenoma	<10
Hyperdense cyst (Bosniak II)	≥70 unenhanced

- Partial-volume **pseudoenhancement of cysts <10–15 mm** leads to overtreatment, so use thin slices and a central region of interest.

6 · Noncontrast CT (NCCT) for stones

- NCCT is the gold standard; all stones are visible except **protease-inhibitor (indinavir) stones**. Cutting tube current by up to 95% preserves accuracy, and ultra-low-dose CT gives less dose than a two-view KUB.
- **Avoid ultra-low dose** with BMI >30, very thin patients (phlebolith confusion), metal hardware or a suspected infected stone.
- CT, low-dose and ultra-low-dose CT are equally accurate (sensitivity about 95%). Ultrasound has sensitivity **84%** but specificity only **53%**; KUB sensitivity is **57%**.

Body	First imaging in suspected colic
AUA	CT in adults; standard dose if BMI >30; NCCT before PCNL or SWL
EAU	Ultrasound first, CT if inconclusive (randomised trial: less radiation, same outcomes)

7 · Contrast CT protocols

- The portal venous phase (70–90 s) shows liver and nodes for routine staging. The nephrographic phase (90–120 s) is **most sensitive for renal masses** and caval thrombus. Clear cell carcinoma enhances early and avidly; papillary enhances later.
- **Adrenal protocol** is noncontrast, portal venous and 15-min delay. Absolute washout = (PV − delayed)/(PV − unenhanced) × 100 >**60%**, and relative washout = (PV − delayed)/PV × 100 >**40%**, indicate adenoma.
- Nodal cut-offs are 10 mm for prostate cancer (poor sensitivity, so use PSMA PET-CT) and, in bladder cancer, a short axis >8 mm in the pelvis and >10 mm in the abdomen.

*See the paper answer **Radiology for solid renal masses.** (Section 13, Ch 137).*

8 · CT urography and CT cystography

- CT urography always includes an excretory phase. For poor distal ureteric opacification, **hydration is the simplest and most effective aid**, followed by a diuretic or prone position.
- Do **CT before TURBT**, because post-resection change mimics invasion. CT detects T3b/T4 but cannot separate muscle-invasive from non-invasive disease.

SAQ PET scans in urology.

Paper answer · P1

Type: SAQ (10 marks)
Source: 2018-Summer Paper 1, Q3e

Principle

- **PET (positron emission tomography):** tracer emits a positron → annihilation with an electron → **two 511-keV photons at 180°** → coincidence detection. Fused with CT (computed tomography) or MRI.
- Quantitative (**SUV, standardised uptake value**); costly. Half-lives: ^{18}F 110 min, ^{68}Ga 68 min, ^{11}C 20 min (on-site cyclotron).

Tracers

Tracer	Mechanism	Main use
^{18}F-FDG (fluorodeoxyglucose)	Glucose analogue trapped after phosphorylation (Warburg effect)	Bladder, UTUC (upper tract urothelial carcinoma), penile, seminoma
PSMA (prostate-specific membrane antigen) ligands (^{68}Ga-PSMA-11, ^{18}F-DCFPyL, ^{18}F-PSMA-1007, rhPSMA-7.3)	Bind cell-surface PSMA	Prostate staging and recurrence
^{11}C/^{18}F-choline	Cell membrane synthesis	Prostate recurrence (older)
^{18}F-fluciclovine	Amino acid transport	Prostate recurrence
^{18}F-NaF (sodium fluoride)	Bone turnover	Bone metastases

Uses by organ

Prostate cancer

- FDG is poor (low glucose metabolism). PSMA has specificity >90%.
- **Primary staging (high risk): proPSMA (2020)**: PSMA PET/CT more accurate than CT plus bone scan (92% vs 65%). Nodal sensitivity only 30–40%, so a negative scan does not exclude micrometastases.
- **BCR (biochemical recurrence):** detection rises with PSA (prostate-specific antigen); about half positive below 0.5 ng/mL. PSMA beats fluciclovine (56% vs 26%). CONDOR: management changed in 63.9%.
- **Theranostics:** PSMA-positive mCRPC (metastatic castration-resistant prostate cancer) → **^{177}Lu-PSMA-617 (VISION 2021)**.
- NCCN (National Comprehensive Cancer Network): PSMA PET preferred; outcome benefit still unproven.

Testis

- **Seminoma residual mass >3 cm:** FDG-PET **≥6 weeks after chemotherapy**. Negative scan reliable (observe); positive → consider resection.
- **Not for NSGCT (non-seminomatous germ cell tumour):** teratoma is not FDG-avid.

Bladder and UTUC

- FDG for nodal and distant staging in MIBC (muscle-invasive bladder cancer): modest sensitivity, high specificity; urinary excretion hides the primary. Use when CT is equivocal.

Kidney

- FDG not routine (clear cell RCC (renal cell carcinoma) poorly avid). **^{89}Zr-girentuximab** (anti-CAIX, carbonic anhydrase IX) identifies clear cell RCC (ZIRCON).

Penis

- FDG confirms metastases in **palpable inguinal nodes**; poor for impalpable nodes.
- **Also know:** EMPIRE-1 (fluciclovine-guided salvage radiotherapy); false positives from inflammation; neuroendocrine cancer loses PSMA.

Diagram to draw: Positron annihilation with two 511-keV photons hitting a detector ring.

> **Key points to remember:**
> - proPSMA: PSMA PET 92% vs 65% accuracy.
> - Seminoma residual >3 cm: FDG-PET at ≥6 weeks; not NSGCT.
> - FDG poor in prostate and clear cell RCC.
> - VISION: ^{177}Lu-PSMA-617 in mCRPC.

SAQ PSMA PET Ct scan

Paper answer · P4

Type: SAQ (10 marks)
Source: 2019-Summer Paper 4, Q3a

Principle

- **PSMA (prostate-specific membrane antigen):** transmembrane glycoprotein over-expressed by prostate cancer, rising with grade.
- **PET/CT (positron emission tomography/computed tomography):** positron-labelled ligand binds PSMA; uptake as SUV (standardised uptake value).
- **Tracers:** ^{68}Ga-PSMA-11 (half-life 68 min); ^{18}F-DCFPyL, ^{18}F-rhPSMA-7.3, ^{18}F-PSMA-1007 (110 min, better images; PSMA-1007 little urinary excretion).
- **Specificity >90%**; beats choline and fluciclovine.

Indications

1. Primary staging (high risk):

- **proPSMA (2020):** accuracy **92% vs 65%** vs CT + bone scan.
- Nodal sensitivity only 30–40% against PLND (pelvic lymph node dissection): a negative scan does not exclude micrometastases.
- NCCN (National Comprehensive Cancer Network): may **replace** conventional imaging; AUA (American Urological Association): add-on.

2. BCR (biochemical recurrence):

- Detection rises with PSA (prostate-specific antigen): about **half positive below 0.5 ng/mL**, nearly all above 2.
- **Calais trial:** PSMA 56% vs fluciclovine 26%. **CONDOR:** management changed in 63.9%.
- Guides salvage radiotherapy (disease outside field in about 1 in 5).

3. Intraprostatic detection: PRIMARY trial: adding PSMA PET to MRI (magnetic resonance imaging) improved negative predictive value; 5-point PRIMARY score.

4. Theranostics: selects patients for **^{177}Lu-PSMA-617**. **VISION (2021):** mCRPC (metastatic castration-resistant prostate cancer) after ARPI (androgen receptor pathway inhibitor) and taxane: OS 15.3 vs 11.3 months.

Reporting

- **PROMISE miTNM** (molecular imaging TNM): miT, miN, miM staging plus expression score (vs blood pool, liver, parotid). PSMA-RADS also used.

Pitfalls

False positive	False negative
Coeliac/stellate ganglia, fractures, Paget's disease, fibrous dysplasia	Small nodes (<5 mm)
Other tumours (RCC (renal cell carcinoma), HCC (hepatocellular carcinoma))	PSMA-negative tumours (about 5–10%)
^{18}F-PSMA-1007 benign bone uptake	Neuroendocrine transformation; masking by bladder activity

- **Also know:** no proof yet that PSMA-guided treatment improves survival (stage migration).

Diagram to draw: PSMA ligand binding the cell-surface antigen with fused PET/CT image of a nodal lesion.

> **Key points to remember:**
> - proPSMA: accuracy 92% vs 65%.
> - BCR: about half positive below PSA 0.5 ng/mL.
> - Nodal sensitivity 30–40%: negative scan does not exclude nodes.
> - VISION: ^{177}Lu-PSMA-617 improves OS in mCRPC.

9 · Iodinated contrast reactions and contrast-associated AKI

*See the paper answer **Contrast agents in urology: iodinated and non-iodinated agents, complications and contrast-induced nephropathy** (Section 1, Ch 4).*

> **KEY POINT**
> Hypersensitivity is idiosyncratic (premedication, adrenaline), whereas nephrotoxicity is dose-related (hydration, less contrast). Prophylaxis for one does nothing for the other.

6 Magnetic Resonance Imaging

1 · Physics and sequences

- Magnetic resonance imaging (MRI) gives **better soft-tissue contrast and no ionising radiation**.
- **Fluid is dark on T1 and bright on T2.**
- **Diffusion-weighted imaging (DWI)**: high cellularity restricts water movement, so tumour is bright on DWI and dark on the apparent diffusion coefficient (ADC) map.
- **Chemical shift imaging**: a voxel holding both fat and water loses signal on the out-of-phase (OP) image, which means **microscopic fat**. Fat suppression nulls **macroscopic fat**.

2 · Indications, contrast and safety

*See the paper answer **Contrast agents in urology: iodinated and non-iodinated agents, complications and contrast-induced nephropathy** (Section 1, Ch 4).*

SAQ MR urography

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 1 Paper 1, Q3a

Definition

- MRU (magnetic resonance urography): magnetic resonance imaging of kidneys, ureters and bladder giving anatomy and function **without ionising radiation**.

Techniques

	Static MRU	Excretory MRU
Sequence	Heavily T2-weighted; urine is bright	Gadolinium-enhanced T1
Contrast	None	GBCA (gadolinium-based contrast agent)
Depends on function	No	Yes; low-dose furosemide improves it
Best for	Dilated or obstructed system	Non-dilated system, excretion

- **Functional MRU (children):** hydration, bladder catheter, furosemide 1 mg/kg (maximum 20 mg), T2 images, then dynamic T1 after slow contrast injection. Patlak analysis gives differential function, GFR (glomerular filtration rate) and drainage.

Appearances

- Stones and calcification give **signal voids**; tumour, clot, gas and sloughed papillae are low-signal defects in bright urine.

Indications

- CT (computed tomography) urography contraindicated: contrast allergy, renal insufficiency, **pregnancy** (separates physiological from calculous dilatation; small stones missed).
- **Children:** complex anomalies, ectopic ureter in the incontinent girl, duplex systems, megaureter, crossing vessels in UPJ (ureteropelvic junction) obstruction, ectopic kidneys.
- Hydronephrosis: alternative to MAG3 (mercaptoacetyltriglycine) renography; equivalence not yet proven.

- Upper tract tumours and renal masses.

Advantages and limitations

Advantages	Limitations
No radiation; best anatomical detail	Cost, availability, long study (up to 1 hour)
Anatomy, function and drainage in one test	Sedation often needed under 6 years
As accurate as CTU (CT urography) for obstruction	Poor for stones; complex post-processing
Safe in iodinated contrast allergy	Pacemakers, metal implants, claustrophobia

Gadolinium safety

- **NSF (nephrogenic systemic fibrosis)** was the concern at GFR <30 mL/min/1.73 m². Group II agents (gadobenate, gadoteridol, gadoterate, gadobutrol) carry little or no risk even in end-stage disease.
- **Pregnancy:** gadolinium only if it clearly helps.
- **EAU (European Association of Urology):** CTU is preferred to MRU for upper tract urothelial cancer.

Diagram to draw: coronal static T2 MRU of a hydronephrotic kidney with a dark stone in the ureter.

> **Key points to remember:**
> - Static T2 (no contrast, function-independent) versus excretory gadolinium T1.
> - Stones are signal voids; CTU preferred for urothelial cancer.
> - Functional MRU: furosemide 1 mg/kg, Patlak analysis for split function.
> - NSF risk minimal with group II agents.

Also, from the condensed book (not in the paper answer):

- MRI is the modality of choice in **pregnancy, renal insufficiency and iodinated contrast allergy**.
- **Contraindications**: non-conditional pacemakers, ferromagnetic aneurysm clips, cochlear implants and orbital metal.

3 · Adrenal MRI

*See the paper answer **Adrenal mass evaluation, incidentaloma and adrenocortical carcinoma** (Section 14, Ch 146).*

Also, from the condensed book (not in the paper answer):

- An adenoma shows an **OP signal drop** from intracytoplasmic lipid. Lipid-poor adenomas (**10–30%**) are indeterminate and need CT washout.
- **Myelolipoma** has macroscopic fat: bright on T1, it loses signal with fat suppression and shows the India ink artefact.
- **Traps**: metastases from clear cell renal cell carcinoma (RCC), hepatocellular carcinoma or liposarcoma contain lipid and mimic adenoma. One-third of phaeochromocytomas are T2-dark, so the "light bulb" sign is unreliable.

4 · Renal mass

*See the paper answer **Radiology for solid renal masses.** (Section 13, Ch 137).*

Also, from the condensed book (not in the paper answer):

- An **absent pseudocapsule** predicts **pT3a**.
- Calcium is a signal void, so MRI shows enhancing nodules in **calcified cysts** better than CT. Tumour thrombus enhances; bland thrombus does not.

5 · MR urography and bladder

*See the paper answer **MR urography** (Section 1, Ch 6).*

SAQ VI-RADS

Paper answer · P4

Type: SAQ (10 marks)
Source: Practice Set 4 Paper 4, Q3b

Definition

- **VI-RADS (Vesical Imaging–Reporting and Data System):** a 5-point score for the likelihood of **muscle (detrusor) invasion** by bladder cancer on mpMRI (multiparametric magnetic resonance imaging). Panebianco et al., Eur Urol 2018.
- Standardises bladder MRI reporting, as PI-RADS (Prostate Imaging Reporting and Data System) does for the prostate.

Technique

- Multiplanar **T2W (T2-weighted)**, **DWI (diffusion-weighted imaging)** with ADC (apparent diffusion coefficient) map, and **DCE (dynamic contrast-enhanced)** T1.
- Done **before TURBT (transurethral resection of bladder tumour)**: resection oedema mimics invasion. Distend the bladder adequately.

Principle

- On T2W the detrusor is a **low-signal (dark) line**; urine, mucosa and submucosa are bright. The question is whether tumour **interrupts that dark line**.

Scoring

Score	Invasion	T2W	DCE	DWI
1	Highly unlikely	<1 cm, ± stalk; dark line intact	No early muscle enhancement	Muscle continuous
2	Unlikely	>1 cm with stalk and thick inner layer, or sessile; line intact	Inner layer enhances only	As 1
3	Equivocal	No category 2 features; line intact	No muscle interruption	As 1
4	Likely	**Line interrupted**	Enhancement extends into muscle	High signal into muscle
5	Highly likely	**Into perivesical fat**	Through wall into fat	High signal through wall and fat

- **Cut-off ≥3 = suspicious for MIBC (muscle-invasive bladder cancer)**; the "equivocal" category already counts as positive.

- Sensitivity and specificity roughly 80–90%; DWI is the main functional sequence.

Clinical uses

- Local staging before TURBT; earlier referral for neoadjuvant chemotherapy and cystectomy.
- Patients who cannot have iodinated contrast (renal impairment, allergy).
- **MRI-first pathway** (UK BladderPath trial) shortened time to correct treatment for MIBC.
- Response after neoadjuvant chemotherapy or bladder preservation: under study.

Limitations

- Needs an experienced radiologist; post-TURBT or BCG (bacille Calmette–Guérin) inflammation; small, bladder-neck or diverticular tumours; poor for CIS (carcinoma in situ) and nodes; does not replace histology.

Diagram to draw: bladder wall with tumour breaching the dark detrusor line (VI-RADS 4).

> **Key points to remember:**
> - Five points; tests the dark muscle line on T2W.
> - ≥3 suspicious for MIBC; 4 line broken, 5 fat.
> - Scan before TURBT.
> - Panebianco 2018; accuracy about 80–90%.

Also, from the condensed book (not in the paper answer):

- Upper tract urothelial carcinoma is homogeneous on DWI (ADC **≤451**), unlike central RCC.
- **T2 MRI is the best study for a urethral diverticulum**: it defines the neck, circumferential extent and any tumour or stone.

6 · Prostate MRI

*See the paper answer **Multiparametric MRI, PI-RADS and fusion biopsy** (Section 17, Ch 155).*

Also, from the condensed book (not in the paper answer):

- Prostate MRI has sensitivity **0.91** and specificity **0.37** for Grade Group ≥2, with negative predictive value **>90%**: it rules cancer out, not in.
- Dynamic contrast curves: type 1 persistent rise is normal, type 2 plateau is indeterminate, and **type 3 washout** is most suggestive.

Score	PI-QUAL (quality)	PRECISE (active surveillance)
1	All sequences non-diagnostic	Resolution
2	One sequence acceptable	Reduction
3	Rules in, not out	Stable
4	Rules in and out	Increase
5	All optimal	Stage progression

At PI-QUAL 1–2, give no PI-RADS score.

Trial	Design	Result	Take-home
PROMIS	MRI vs template biopsy	MRI highly sensitive	MRI before biopsy (level 1b)
PRECISION	Randomised: MRI ± targeted vs systematic biopsy	More significant, less insignificant cancer	MRI pathway standard
Göteborg-2	Screening trial	Targeted-only biopsy cut overdiagnosis	MRI-targeted screening

7 Genitourinary Tract Imaging: Basic Principles of Nuclear Medicine

1 · Principles and radionuclides

LAQ Radionuclide (isotope) renal scans: MAG3 and DMSA

Paper answer · P1 · P2 · merged from 2

Asked as: P1 LAQ "Radionuclide (isotope) scans in urology: principles, renal scans, MAG3 and applications" · P2 SAQ "Renal isotope scans and DMSA renal scan"

"Uro" is read as urology. "MAG 3" is taken to mean the Tc-99m MAG3 (mercaptoacetyltriglycine) renal scan: principle, technique, interpretation and uses.

Principles

- Nuclear medicine shows **function, not structure**. A radiotracer, given in a tiny (subpharmacological) dose, follows a physiological or molecular process; the patient is the radiation source.
- **Gamma camera** (sodium iodide crystal, photomultipliers, collimator): photon, collimator (admits photons from one direction), sodium iodide crystal (light), photomultiplier tubes, image. Planar (about 20 mm resolution) or **SPECT (single-photon emission computed tomography)**, 3D, with CT (about 10 mm).
- **PET (positron emission tomography):** the positron annihilates with an electron, giving two 511 keV photons 180° apart, detected in coincidence. This rejects noise and allows true quantitation (SUV, standardised uptake value); sharper (about 5 mm).

	Planar	SPECT	PET
Image	2D	3D, with CT (computed tomography)	3D, with CT or MRI (magnetic resonance imaging)
Resolution	About 20 mm	About 10 mm	About 5 mm
Quantitation	Semiquantitative	Semiquantitative	SUV
Main use	Renal function	Problem-solving	Cancer

- **^{99m}Tc (technetium-99m):** 6-hour half-life, single 140 keV photon: the workhorse. **^{18}F:** about 110 minutes, lowest positron energy, so the sharpest PET image. ^{68}Ga comes from a generator; ^{11}C (20 minutes) needs an on-site cyclotron.

Applications in urology (overview)

Area	Tracer	Use
Kidney	^{99m}Tc-DTPA (diethylenetriaminepentaacetic acid), -MAG3, -DMSA (dimercaptosuccinic acid)	GFR (glomerular filtration rate), split function, obstruction, scars, transplant
Bone	^{99m}Tc-MDP (methylene diphosphonate)	High-risk prostate cancer staging; low specificity

Area	Tracer	Use
Prostate	**PSMA (prostate-specific membrane antigen) PET**; choline; fluciclovine	PSMA preferred (NCCN [National Comprehensive Cancer Network]) for staging and recurrence; specificity >90%
Therapy	^{177}Lu-PSMA-617; radium-223	mCRPC (metastatic castration-resistant prostate cancer); VISION trial (2021)
FDG (fluorodeoxyglucose) PET	Glucose analogue trapped in cells	**Residual seminoma >3 cm**; penile cancer with palpable nodes; distant urothelial disease. Poor in prostate and clear cell RCC (renal cell carcinoma)
Renal mass	Sestamibi SPECT; girentuximab PET	Oncocytoma "hot" on sestamibi; clear cell RCC "hot" on girentuximab
Adrenal	^{123}I-MIBG (metaiodobenzylguanidine), ^{68}Ga-DOTATATE, FDG	Metastatic or extra-adrenal phaeochromocytoma; MIBG best in MEN-2 (multiple endocrine neoplasia type 2)
Sentinel node	^{99m}Tc-nanocolloid plus blue dye	Penile cancer, impalpable nodes
Scrotum	^{99m}Tc-pertechnetate	Torsion cold, epididymitis hot
Reflux	Direct radionuclide cystography	VUR (vesicoureteral reflux) follow-up: more sensitive, lower dose than VCUG (voiding cystourethrogram)
Infection	^{111}In-labelled leucocytes; ^{67}Ga	Occult abscess; not in transplant

Renal tracers

- Each tracer images the part of the nephron that handles it. Plasma clearance is about 20% filtration and 80% tubular secretion.

Feature	**^{99m}Tc-DTPA**	**^{99m}Tc-MAG3**	**^{99m}Tc-DMSA**
Handling	Glomerular filtration only	Protein-bound; tubular secretion, high extraction	Filtered, then retained in proximal tubules (megalin, cubilin); tubular cortical binding
Study	Dynamic	Dynamic	**Static cortical**, at 2 hours (P2: 2–4 hours)
Main use	**GFR**; captopril study; renovascular	**Split function, obstruction (first choice)**, drainage; transplant	**Pyelonephritis and scars**
Renal failure	Poor images in renal failure	Good images in renal failure	Delay imaging
Dose	5–10 mCi	5–10 mCi	5 mCi adult; 50 µCi/kg child

MAG3: principle

- **^{99m}Tc-MAG3:** protein-bound tracer cleared mainly by **proximal tubular secretion**. Extraction is nearly **60%** per pass, versus about 20% for DTPA.

- So it gives **good images even with poor GFR**, lower background and lower cost. **Agent of choice for split function and obstruction.**

Dynamic and diuretic renography (MAG3): method and interpretation

- **Preparation:** good hydration; bladder catheter for outlet obstruction or neurogenic bladder, and in children; cap a nephrostomy. Dose 185–370 MBq (5–10 mCi) intravenous bolus in adults.
- **Perfusion phase** (first minute): frames every 1–2 seconds; the kidney appears within seconds of the aorta (renal curve follows aortic upstroke).
- **Function (uptake) phase:** regions of interest with background subtraction. **Split function (DRF, differential renal function) = area under each curve at 1–3 minutes**; **normal 45–55%** (45:55 to 55:45). Normal MAG3 peak at 3–5 minutes; bladder by 10–15 minutes.
- **Drainage phase:** a rising curve needs furosemide. Upright or post-void images test gravity drainage.
- **GFR:** DTPA camera method (Gates) or plasma-sample clearance.
- **Diuretic renography ("well-tempered" diuretic renogram):** dilatation is not obstruction. Furosemide 0.5 mg/kg (adult maximum 40 mg) in adults; 1 mg/kg in children, given when the system is full; separates a patulous system from obstruction. Timing varies (F+20 classic, F+15, F0, F−15), so the report must always state when it was given.

T½ (half-clearance time) after furosemide	Meaning
<10 minutes	Not obstructed
10–20 minutes	Indeterminate
>20 minutes	Obstructed

- **Curve types (O'Reilly):** I normal; II rising (obstructed); IIIa dilated, not obstructed, washes out (falls) with diuretic; IIIb equivocal.

MAG3 uses

- PUJ (pelviureteric junction) obstruction and megaureter: **surgery if DRF <40% with poor drainage or falling DRF**; follow-up after pyeloplasty.
- Transplant: acute rejection (perfusion falls) versus delayed graft function (perfusion kept); urine leak.
- **Also know:** indirect cystography for VUR; split function before nephrectomy.

MAG3 pitfalls

- Poor function, dehydration, full bladder, wrong ROI (region of interest), mistimed diuretic, massive dilatation (capacious pelvis) and neonatal immaturity mimic obstruction. **Trend on serial scans** matters.

DMSA renal scan

Principle

- **^{99m}Tc-DMSA** is filtered and then taken up by **proximal tubular cells** (megalin and cubilin receptors). It is retained in the cortex and excreted very slowly.
- So it gives a **static map of functioning cortex**, not drainage.

Technique

- Intravenous dose: **50 µCi/kg in children, 5 mCi (185 MBq) in adults**.

- Images at **2–4 hours** (P1: 2 hours; later in renal failure): posterior and oblique views, **pinhole collimator** or **SPECT**, which is slightly more sensitive.
- Radiation about 1 mSv.

What it measures

- **DRF:** from geometric-mean counts; normal 45–55%. Better than MAG3 for ectopic, horseshoe or duplex kidneys and very poor function.
- **Cortical defects:** photopenic areas.

Indications

- **Acute pyelonephritis** (gold standard), especially in infants with febrile UTI (urinary tract infection).
- **Renal scarring** in VUR and recurrent UTI; top-down approach (DMSA first, VCUG only if abnormal).
- Function of a duplex upper moiety, ectopic or horseshoe kidney; MCDK (multicystic dysplastic kidney: no uptake).
- Deciding whether to reconstruct or remove a poorly functioning kidney.

Acute pyelonephritis versus scar

	Acute pyelonephritis	**Scar**
Timing	Within 1 week of fever (sensitivity falls by 1 month)	**≥6 months** after infection
Defect	Ill-defined, may be multiple	Sharp, with cortical thinning
Kidney size	Normal or swollen	Small or contracted (often small)
Repeat scan (at about 6 months)	Resolves (about 85%)	Persists

Goldraich scar grading

Type	**Finding**
1	No more than 2 scarred areas
2	More than 2 scars, normal tissue between
3	Generalised damage, small kidney
4	End-stage, shrunken kidney, little or no uptake

Limitations

- Cannot tell **congenital reflux dysplasia** (boys, grade IV–V VUR, no UTI) from acquired scar.
- Gross hydronephrosis gives false defects; no information on drainage; poor uptake in renal failure and proximal tubular disorders.
- **Also know:** MAG3 is best for obstruction; NICE (National Institute for Health and Care Excellence): late DMSA 4–6 months after atypical or recurrent UTI.

Other renal applications (special studies)

- **Captopril renography (DTPA):** the stenotic kidney keeps GFR by angiotensin II efferent constriction. Captopril removes this support, so uptake falls. Repeat off ACE (angiotensin-

converting enzyme) inhibitors: improvement **≥10%** off ACE inhibitors means renovascular hypertension is likely. Largely replaced by CT angiography.
- **Transplant (MAG3):** acute rejection: perfusion falls and worsens; delayed graft function: perfusion kept, improves. Shows urine leak without iodinated contrast.
- **Radionuclide (nuclear) cystography:** direct (tracer instilled into the bladder) or indirect (after MAG3). More sensitive and lower radiation dose than VCUG, but little anatomy. So VCUG first; nuclear cystography for follow-up of reflux.

Infection imaging

- **^{111}In-oxine leucocytes** (image at 24 hours): best for occult abdominal or genitourinary infection. ^{99m}Tc-HMPAO (hexamethylpropyleneamine oxime) leucocytes give better images, but urinary excretion obscures the tract. Neither in a transplant (leucocytes home to the graft).
- **Also know:** ^{67}Ga-citrate (non-specific, slow).

Oncology

- **FDG (^{18}F-fluorodeoxyglucose):** a glucose analogue phosphorylated and trapped (Warburg effect). Urinary excretion limits it to distant disease.
- **Seminoma residual mass >3 cm** after chemotherapy (EAU [European Association of Urology]). Not for NSGCT (non-seminomatous germ cell tumour), because teratoma is not FDG-avid.
- Penile cancer with palpable nodes; equivocal bladder or upper-tract metastases.
- Poor in prostate cancer and clear cell RCC.
- **Prostate: PSMA PET** is preferred (NCCN) for high-risk staging and biochemical recurrence. Specificity >90%; nodal sensitivity only 30–40%. Calais trial (2019): detection 56% (PSMA) versus 26% (fluciclovine). Bone scan (^{99m}Tc-MDP) remains conventional.
- **Renal mass: sestamibi SPECT** "hot" in oncocytoma; **girentuximab** (anti-CAIX, carbonic anhydrase IX) PET "hot" in clear cell RCC (ZIRCON trial).
- **Also know:** ^{123}I-MIBG and ^{68}Ga-DOTATATE for phaeochromocytoma; ^{99m}Tc-nanocolloid sentinel node biopsy in penile cancer; ^{99m}Tc-pertechnetate scrotal scan (torsion cold, epididymitis hot) when ultrasound is equivocal.

Therapy (theranostics)

- **^{177}Lu-PSMA-617:** beta emitter for mCRPC after ARPI (androgen receptor pathway inhibitor) and taxane: **VISION (2021)**. PSMA PET confirms the target first.
- **Radium-223:** alpha emitter for bone-only mCRPC.

Advantages and limitations

- Quantitative function, low radiation, safe in contrast allergy and pregnancy (MAG3 fetal dose <5 mGy).
- Poor anatomy; PET is costly; results depend on technique.

Diagrams to draw

- MAG3 renogram: perfusion spike, peak at 3–5 minutes, washout, 1–3 minute split window; renogram curve types I (normal), II (obstructed, rising), IIIa (dilated non-obstructed, falls after furosemide) and IIIb.
- Gamma camera chain: patient, collimator, crystal, photomultiplier, image.

- Posterior DMSA image: normal kidney and a scarred kidney with polar defects.

Key points to remember:

- ^{99m}Tc: 6 hours, 140 keV; PET: two 511 keV photons.
- DTPA filtered = GFR; MAG3 secreted = obstruction and transplant; DMSA cortical = scars.
- MAG3: tubular secretion, about 60% extraction; best for split function and drainage.
- Split function (DRF) from 1–3 minute counts; normal 45–55%; furosemide 0.5 mg/kg.
- T½ >20 minutes obstructed, <10 minutes not.
- Hydration, catheter, correct diuretic timing: the well-tempered renogram.
- DMSA binds proximal tubules: a static cortical scan at 2–4 h; gold standard for acute pyelonephritis and scars.
- Scar only if the defect persists ≥6 months; pyelonephritis resolves.
- Captopril renogram: ≥10% improvement off ACE inhibitors.
- PSMA PET preferred; FDG for residual seminoma >3 cm.
- MIBG best in MEN-2; sentinel node with ^{99m}Tc-nanocolloid.
- Sestamibi hot = oncocytoma; ^{177}Lu-PSMA-617 (VISION).

Also, from the condensed book (not in the paper answer):

- **Gallium-68 (^{68}Ga)** has a half-life of **68 min**, is generator-produced and has the **longest positron range**.
- Indium-111 (half-life **67 h**) and gallium-67 (**78 h**) suit delayed infection imaging. Gallium-67 citrate is transferrin-bound and non-specific, imaged at 48–72 h.
- Renal blood flow is about **1 L/min** (20% of cardiac output), and renal plasma flow about **600 mL/min**.

2 · Renal tracers and diuretic renography

*See the paper answer **Radionuclide studies and diuretic renography in PUJ obstruction** (Section 11, Ch 93).*

*See the paper answer **Radionuclide (isotope) renal scans: MAG3 and DMSA** (Section 1, Ch 7).*

Also, from the condensed book (not in the paper answer):

- Always check **whether and when furosemide was given**. With F−15, furosemide goes 15 min before the tracer, so diuresis is maximal when the tracer arrives.
- Captopril renography has largely been replaced by computed tomography angiography and Doppler.

ALGORITHM

1. Day 1: give oral captopril and perform a renogram; slow uptake and a low peak are abnormal.
2. If abnormal, repeat 1–2 days later off angiotensin-converting enzyme (ACE) inhibitors and calcium channel blockers.
3. Improvement of **≥10%** means a high probability of renovascular hypertension and likely benefit from revascularisation.

3 · Oncological PET, bone scan and PSMA

*See the paper answer **PET scans in urology.** (Section 1, Ch 5).*

*See the paper answer **PSMA PET Ct scan** (Section 1, Ch 5).*

Also, from the condensed book (not in the paper answer):

- **Bone scan** (^{99m}Tc-methylene diphosphonate) binds hydroxyapatite at osteoblastic activity. It is non-specific, shows a **flare** early in therapy, and a **superscan** is diffuse uptake with **absent renal activity**.
- Choline and fluciclovine are taken up by inflammation and benign prostatic hyperplasia, so they perform poorly in the prostate itself. PSMA shows normal uptake in the salivary and lacrimal glands.
- The National Comprehensive Cancer Network does not require conventional imaging before PSMA-PET.

Trial	Design	Result	Take-home
Calais	^{68}Ga-PSMA vs fluciclovine, biochemical recurrence	Detection 56% vs 26%	PSMA superior
CONDOR	^{18}F-DCFPyL, biochemical recurrence	Detection 61.5% at prostate-specific antigen (PSA) 0.8 ng/mL; management changed 64%	Detects at low PSA
EMPIRE-1	Randomised: fluciclovine-guided vs conventional salvage radiotherapy	12.5% absolute 3-year event-free gain	PET-guided planning helps
VISION	Randomised, metastatic castration-resistant, after androgen receptor inhibitor and taxane	^{177}Lu-PSMA-617 approved	Radioligand therapy

8 Evaluation and Management of Haematuria

1 · Definitions and causes

*See the paper answer **Microscopic / asymptomatic haematuria and urinary biomarkers** (Section 1, Ch 2).*

Also, from the condensed book (not in the paper answer):

- One positive urinalysis suffices because bleeding from cancer is intermittent. Specificity comes from **risk-stratifying the work-up, not from raising the threshold**.
- **Women** (male:female 4:1) present at a higher stage, because MH is put down to infection or gynaecological causes.
- The timing of gross haematuria (GH) in the stream points to the source:

Timing	Usual source
Initial	Urethra, distal to the external sphincter
Terminal	Trigone, bladder neck, prostate
Total	Bladder or upper tract
Cyclic (female)	Endometriosis

2 · Evaluation of microhaematuria (AUA/SUFU 2020)

*See the paper answer **Microscopic / asymptomatic haematuria and urinary biomarkers** (Section 1, Ch 2).*

Also, from the condensed book (not in the paper answer):

- **Renal ultrasound** detects cortical tumours but is poorly sensitive for upper tract urothelial carcinoma; this is why CT urography is reserved for high risk.
- **CT urography** phases: non-contrast for stones, nephrographic for renal masses, excretory for urothelium. The dose is **20–30 mSv**, which a split-bolus protocol reduces.
- UroVysion fluorescence in situ hybridisation targets chromosomes **3, 7, 17 and 9p21**.

3 · Gross haematuria

- The cancer yield is several-fold higher than in MH, so there is **no age threshold and no stratification**.
- GH without trauma or culture-proven infection needs **cytology, cystoscopy and CT urography**. With culture-proven infection, treat it and then repeat the urinalysis.
- In acute bleeding, assess haemodynamics, blood count, coagulation and post-void residual.
- **Clot retention**: pass a large-bore (22–24 Fr) three-way catheter, evacuate all clot manually, then run continuous bladder irrigation (CBI) until clear. Evacuation must be complete, because residual clot keeps the bladder distended so vessels cannot contract, and it fuels fibrinolysis.
- If this fails, perform cystoscopy under anaesthesia with clot evacuation, fulguration and biopsy of suspicious areas.
- Decide on stopping or reversing anticoagulation with the prescriber, weighing thrombotic risk. Never blame the anticoagulant without a full evaluation.

4 · Haemorrhagic cystitis

SAQ Haemorrhagic cystitis: diagnosis and management

Paper answer · P1 · P3 · merged from 2

Asked as: P1 SAQ "Diagnosis and management of hemorrhagic cystitis" · P3 SAQ "Management of Haemorrhagic cystitis."

Definition

- Diffuse inflammation and bleeding of the bladder mucosa, after other sources are excluded (cystoscopy, cytology, upper tract imaging).

Causes and mechanism

Cause	Mechanism
Cyclophosphamide, ifosfamide	Liver forms **acrolein**, excreted in urine, which sloughs urothelium; onset within 48 h
Pelvic radiation (about 5%; 6 months to 10 years later)	**Obliterative endarteritis**, ischaemia, fibrosis, fragile neovessels
Viral	**BK polyomavirus**; adenovirus 11 and 35; after bone marrow or renal transplant
Others	Bacterial infection; busulfan, thiotepa; penicillins; amyloidosis

Diagnosis

- History: drugs, radiotherapy, transplant, anticoagulants. Severity: microscopic, gross, clots, clot retention.
- Blood count, coagulation, renal function, urine culture, viral PCR (polymerase chain reaction) if transplanted.
- **Exclude bladder cancer first:** cytology, cystoscopy with biopsy, upper tract imaging (CT urography). Cyclophosphamide and radiation are also risk factors for bladder cancer.

Prevention

- **Mesna** binds and inactivates acrolein; hyperhydration with forced diuresis, CBI (continuous bladder irrigation).

Management (escalate by severity; stability decides the branch)

`Resuscitate, transfuse, correct coagulopathy, treat infection → large-bore 3-way catheter + CBI → cystoscopy under anaesthesia: clot evacuation, fulguration, biopsy → intravesical alum 1% → bleeding after 48 h: hyperbaric oxygen (+ nephrostomies if bleeding continues) → formalin 1%, then 4% (cystogram first) → cystectomy + conduit diversion (Unstable at any stage: internal iliac embolisation)`

Intravesical agents

Agent	Mechanism and method	Caution / key point
Alum 1% (first line)	Protein precipitation, vasoconstriction; 200–300 mL/h; no anaesthesia	Avoid in renal failure (aluminium toxicity)

Agent	Mechanism and method	Caution / key point
Aminocaproic acid	Antifibrinolytic CBI, 200 mg/L	Only in a clot-free bladder; thromboembolism
Prostaglandins (carboprost), silver nitrate, hyaluronate (second-line options)	Vasoconstriction, coagulation, GAG (glycosaminoglycan) repair	Silver nitrate: cystogram
Formalin 1–4%	Protein fixation; **80–90% success**; under anaesthesia, instilled by gravity for 10–15 min	**Cystogram to exclude reflux** (occlude ureters if present); fibrosis, contracted (small) bladder, ureteric stricture

Other measures

- **HBOT** (hyperbaric oxygen therapy): 100% oxygen at 2–3 atmospheres, 30–40 sessions; raises tissue oxygen and drives neovascularisation; 80–90% response, but many relapse. **RICH-ART** trial: improved urinary symptom scores in radiation cystitis.
- **Nephrostomies:** divert urokinase-rich urine if bleeding continues.
- **Angioembolisation:** selective anterior division of internal iliac; spare the posterior division (gluteal pain); initial control about 90%. Used if unstable.
- **Cystectomy with conduit diversion:** last resort; remove the bladder (pyocystis risk).
- **Viral:** supportive (hydration, irrigation); cidofovir (intravenous or intravesical; evidence anecdotal).
- **Also know:** tranexamic acid and laser ablation are unproven; oral pentosan polysulfate in children.

Diagram to draw: stepwise escalation ladder.

> **Key points to remember:**
> - Acrolein (prevent with mesna); radiation endarteritis; BK virus.
> - Exclude bladder cancer.
> - Alum first; formalin only after cystogram.
> - Hyperbaric oxygen for radiation cystitis (RICH-ART); embolise if unstable.

Also, from the condensed book (not in the paper answer):

- **Silver nitrate 0.5–1%** also needs a cystogram first, because it precipitates in the upper tract.
- **Aminocaproic acid** inhibits plasminogen activation (urokinase) and so blocks fibrinolysis; use it only in a clot-free bladder.
- **Formalin technique**: instil by gravity with the catheter **≤15 cm** above the pubis, up to **300 mL** or bladder capacity, and protect the urethra and skin.

5 · Prostatic haematuria

- It is a diagnosis of exclusion after a full GH work-up. **Benign prostatic hyperplasia is the commonest cause of GH in men over 60**, through increased microvessel density driven by vascular endothelial growth factor (VEGF).

- **5α-reductase inhibitors** (finasteride, dutasteride) lower VEGF and microvessel density. About **90%** improve, including anticoagulated men, with onset from 2 weeks to 9 months. Randomised trials show fewer recurrences than with watchful waiting.
- Prostate cancer bleeding usually means T4 invasion. Treat with androgen deprivation therapy with or without palliative radiotherapy.

ALGORITHM

1. Benign: start with a 5α-reductase inhibitor, then fulguration, then transurethral resection, laser vaporisation or enucleation, or simple prostatectomy, and finally prostatic artery embolisation (for poor-risk patients or bleeding after resection).
2. Malignant: start with androgen deprivation with or without radiotherapy, then fulguration or channel resection, then nephrostomy, and finally cystoprostatectomy.

6 · Urethral and upper tract bleeding

- **Urethrorrhagia** (blood at the meatus independent of voiding) means a lesion distal to the sphincter. Use retrograde urethrography and cystourethroscopy. With suspected trauma or penile fracture, **perform a urethrogram before any instrumentation**. In women, consider caruncle or diverticulum.
- **Renal arteriovenous malformation** is ~**75%** acquired (biopsy, partial nephrectomy, percutaneous nephrolithotomy, trauma). GH after a renal procedure needs prompt angiography and selective embolisation.
- **Ureteroiliac artery fistula** (radiation, vascular surgery, chronic stents) is treated by endovascular stenting.
- **Nutcracker syndrome**: the left renal vein is compressed between the aorta and the superior mesenteric artery.
- **Lateralising essential haematuria**: cystoscopy during bleeding shows the side, then ureteropyeloscopy inspects every calyx at low pressure for biopsy or fulguration.

Trial	Design	Result	Take-home
DETECT I	Prospective, UK, 3,556 referrals	Cancer in 13.8% with GH vs 3.1% with MH; 3.5% in GH under 45	Investigate GH at any age
RICH-ART	Randomised hyperbaric oxygen vs standard care, radiation cystitis	Improved urinary symptom score	Supports hyperbaric oxygen

9 Occupational Health in Urology

1 · Physical ergonomics

SAQ Operating room set-up

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 19 SAQ Sheet, Q3f (see also 2021-Summer Paper 4, Q3a for operating room safety)

Physical design

- **Zones:** protective (reception, changing) → clean (pre-op, recovery) → sterile (theatre, scrub) → disposal; one-way flow.
- **Ventilation:** positive pressure, about 20 air changes an hour, filtered air; laminar flow for implants.
- **Environment:** temperature about 20–24 °C, relative humidity about 20–60%. Patient warming: even 1–2 °C hypothermia raises SSI (surgical site infection) and blood loss (**Kurz 1996**: 3-fold wound infection).
- Power backup, piped gases, anaesthetic scavenging, radiolucent table.

Equipment layout by procedure

Setting	Arrangement
Endourology (TURP (transurethral resection of the prostate), URS (ureteroscopy), PCNL (percutaneous nephrolithotomy))	Lithotomy or prone; tower and monitor directly ahead; C-arm opposite surgeon; wires, baskets, dilators, spare scopes ready
Laparoscopic, upper abdomen	Surgeon and assistant face the abdomen; monitor opposite; scrub nurse near the foot
Laparoscopic, pelvic	30° Trendelenburg, split legs; surgeon on preferred side, assistant opposite
Robotic	Patient cart, surgeon console, vision cart; Si docks between legs, Xi from the side

- **Checklist before laparoscopic incision:** irrigation–suction, electrosurgical unit, full CO_2 cylinder plus spare, white-balanced camera, insufflator and Veress needle checked, **open tray ready**.
- Cables fixed; camera cable apart from diathermy; insufflator at eye level.

Positioning safety

- Pad bony prominences; axillary roll (lateral); avoid prolonged kidney rest and full flexion (rhabdomyolysis); no shoulder braces in steep Trendelenburg (brachial plexus).

Ergonomics and staff protection

- Monitor straight ahead, top at eye level; elbows 90–120°; table at the surgeon's pubic symphysis (laparoscopy).
- **Radiation:** ALARA (as low as reasonably achievable): time, distance, shielding; lead apron, thyroid shield, glasses; dosimeter.
- Smoke evacuation; sharps safety; vaccination.

Safety systems

- **WHO (World Health Organization) Surgical Safety Checklist** (sign in, time out, sign out); counts; fire-risk assessment (oxidiser, fuel, ignition).

Diagram to draw: theatre plan: table, tower, C-arm, anaesthetist, surgeon, scrub nurse.

> **Key points to remember:**
> - Zones with one-way flow; positive pressure, about 20 air changes an hour.
> - Monitor directly ahead at eye level; open tray ready for laparoscopy.
> - Hypothermia triples wound infection (Kurz).
> - ALARA: time, distance, shielding.

Also, from the condensed book (not in the paper answer):

- Work-related musculoskeletal disorders affect **46–90%** of urologists. They are a cumulative-load problem, so every fix either restores neutral posture or interrupts the load.
- Posture targets: in open surgery, work **5–10 cm** below the elbow with neck flexion **<25°**. In endourology, set the target between symphysis and umbilicus. At the robotic console, keep back flexion **<15°** and knees at 90–120°.
- Use steep-declination loupes (neck **<20°**) and two-piece lead that puts weight on the hips. A presbyopic surgeon should use workspace lenses, not progressives.

2 · Occupational exposures

*See the paper answer **Operating room set-up** (Section 1, Ch 9).*

*See the paper answer **Operating room safety : the role of surgeon.** (Section 1, Ch 1).*

Also, from the condensed book (not in the paper answer):

- **Fluoroscopy**: use pulsed screening at the lowest rate (1 pulse/s), collimate, and use last-image hold. Keep the source far from the patient and the image intensifier close. **Magnification increases dose.**
- **Pregnant surgeon**: wear a fetal dosimeter at the waist under the apron, use maternity or double-thickness lead, and minimise radiation cases.
- For CO_2 laser on condylomata, wear an N95 mask with the evacuator tip within **5 cm**. For intravesical chemotherapy, wear two pairs of chemotherapy gloves and use a closed-system transfer device.

3 · Physical and mental health

- **Atherosclerotic cardiovascular disease** is the most threatening modifiable cause of death in surgeons. Recommended sleep is **7–9 h**, and exercise **150 min** moderate weekly plus strength training twice.
- **Burnout** means emotional exhaustion, depersonalisation and low personal accomplishment, and affects about **50%** of urologists. Spending **≥20%** of time on meaningful work lowers it.
- A **second victim** develops guilt and post-traumatic stress after an adverse event and needs psychological first aid and peer support.

Stress zone	Features	Response
Ready (green)	Balanced, focused	Exercise, sleep, mindfulness
Reacting (yellow)	Irritable, poor concentration	Self-care, delegate
Injured (orange)	Anxiety, substance misuse, burnout	Support, less workload, time off
Ill (red)	Panic, severe depression, dependence	Mental-health professional, sick leave

Imposter syndrome is a cognitive distortion, commonest in trainees, and responds to cognitive behavioural therapy.

10 Ethics and Informed Consent

1 · Principles of medical ethics

SAQ Ethics and informed consent in urology

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 19 SAQ Sheet, Q3e

Four principles (Beauchamp and Childress, 1979)

Principle	Meaning
Autonomy	Patient's deliberated self-rule; basis of consent and confidentiality
Beneficence	Act for the patient's benefit, judged by the patient's values
Non-maleficence	Avoid harm (*primum non nocere*); requires training and honest risk data
Justice	Fair distribution of resources, benefits and risks
- Hippocratic Oath gave beneficence, non-maleficence, confidentiality; autonomy and justice are modern additions (autonomy is bounded by justice).	
- **Jonsen's four-box method:** medical indications; patient preferences; quality of life; contextual features (family, cost, law).	

Informed consent: legal evolution

- From battery to negligence. **Schloendorff (1914):** right to bodily integrity. **Salgo (1957):** term "informed consent". **Canterbury v Spence (1972): reasonable-patient** disclosure standard.
- India: **Samira Kohli (2008)**: consent for one procedure does not cover another, except in emergency.

Elements

1. Diagnosis, proposed treatment and alternatives, including no treatment.
2. Risks, benefits and uncertainty compared with alternatives.
3. **Check understanding** (most often missed).
4. Voluntary decision by a patient with capacity; documented. - Material risks: no agreed cut-off (1% proposed); disclose rarer ones by severity; lay language. - Urology examples: incontinence and erectile dysfunction after prostatectomy; retrograde ejaculation after TURP (transurethral resection of the prostate); stent symptoms; conversion to open surgery.

Exceptions

- **Emergency** in an incapable patient (presumed consent); a conscious patient may still refuse.
- **No capacity:** surrogate using substituted judgement, then best interests.

- **Minors (<18 years):** parents consent; child gives assent.

Improving consent

- Decision aids, repeat-back, simple written material.
- **Shared decision making:** two-way exchange of options and patient values; vital when several options are reasonable (e.g. localised prostate cancer).

Special areas

- **End of life:** early goals-of-care talks, advance directives, DNR (do-not-resuscitate) orders.
- **Research:** written consent, ethics committee approval, extra protection for vulnerable groups.
- **Also know:** confidentiality; conflicts of interest; futility policies.

Diagram to draw: Jonsen four-box grid.

> **Key points to remember:**
> - Four principles: autonomy, beneficence, non-maleficence, justice.
> - Canterbury v Spence: reasonable-patient standard; Samira Kohli in India.
> - Commonest gap: not checking understanding.
> - Exceptions: emergency, incapacity (surrogate), minors.

Also, from the condensed book (not in the paper answer):

- All four principles are equal and **prima facie binding**; an ethical dilemma is a conflict between them that needs prioritising. Equal treatment is not always just treatment.
- In the four boxes, **quality of life draws on three principles** (autonomy, beneficence, non-maleficence) and **contextual features map to justice**.

2 · Informed consent

*See the paper answer **Ethics and informed consent in urology** (Section 1, Ch 10).*

Also, from the condensed book (not in the paper answer):

- The law moved from **battery to negligence to a patient-centred standard**. Mohr v Williams (1905) held that consulting a doctor is not consent.
- **Disclosing uncertainty improves trust.** Decision aids reduce decisional conflict; repeat-back works but is time-consuming.

3 · Capacity, exceptions and special situations

*See the paper answer **Ethics and informed consent in urology** (Section 1, Ch 10).*

Also, from the condensed book (not in the paper answer):

- If a surrogate does not follow the patient's wishes, go to the **ethics committee** and then a court-appointed conservator.
- The **Patient Self-Determination Act (1990)** requires institutions to tell patients of their right to an advance directive.
- For research, the institutional review board may **waive consent** for minimal-risk or de-identified data.

11 Perioperative Care in Urologic Surgery

1 · Preoperative evaluation

- Test selectively: an electrocardiogram at age **>40** or with cardiac history, clotting tests only with a bleeding or anticoagulant history, and a pregnancy test in every woman of childbearing age.

American Society of Anesthesiologists (ASA)	Definition
I / II	Healthy / mild systemic disease
III	Severe disease that limits activity, not incapacitating
IV	Incapacitating, a constant threat to life
V / VI	Moribund (death within 24 h) / brain-dead donor
E	Emergency suffix (never with VI)

2 · Cardiac and other risk

- Good reserve with low-stress (endoscopic) surgery needs no further testing, whatever the history. Capacity of **≥4 METs** (metabolic equivalents, about two flights of stairs) means no invasive work-up.
- The **Revised Cardiac Risk Index** has six unweighted predictors: ischaemic heart disease, heart failure, cerebrovascular disease, high-risk surgery, preoperative insulin and creatinine **>2 mg/dL**.
- After myocardial infarction (MI), waiting 4–6 weeks before elective surgery is reasonable. Frailty predicts complications independent of age.
- A forced expiratory volume in 1 s **<0.8 L** or <30% predicted is high pulmonary risk. Child class mortality is A 10%, B 30% and C 76–82%.
- Loss of >9 kg in 3 months is severe malnutrition; preoperative parenteral nutrition helps only these patients.

*See the paper answer **Stones in pregnancy** (Section 11, Ch 96).*

Also, from the condensed book (not in the paper answer):

- The fetal malformation threshold is **10 cGy**; computed tomography gives about 1 cGy.

3 · Optimising comorbidity

- Only β-blockade has improved cardiac outcome (Mangano atenolol trial).
- Stop smoking **≥8 weeks** before surgery; stopping later may raise pulmonary risk.
- Target glucose 80–250 mg/dL. Withhold metformin on the day of surgery and sodium-glucose cotransporter-2 inhibitors 3 days before (euglycaemic ketoacidosis).
- Presume adrenal suppression with prednisone >20 mg/day for >3 weeks or cushingoid features; ≤5 mg/day needs only the usual dose. Give hydrocortisone **50–100 mg** intravenously at induction, then 25–50 mg 8-hourly for 24–48 h.

4 · Antithrombotic management

- Stop warfarin 5 days before (international normalised ratio **<1.5**). Bridge moderate- and high-risk patients with therapeutic low-molecular-weight heparin (LMWH), last dose 24 h

before. Direct oral anticoagulants need no bridging.

Thrombotic risk	Mechanical valve	Atrial fibrillation ($CHADS_2$)	Venous thromboembolism (VTE)
High	Any mitral; caged-ball or tilting-disc aortic; stroke <6 months	5–6; stroke <3 months	<3 months; severe thrombophilia
Moderate	Bileaflet aortic plus risk factor	3–4	3–12 months; active cancer
Low	Bileaflet aortic alone	0–2	Single VTE >12 months

Drug	Stop before surgery
Aspirin, clopidogrel	7–10 days
Prasugrel / ticagrelor	7 / 5 days
Apixaban, rivaroxaban	48 h (72 h if high bleeding risk)
Dabigatran	48–96 h by creatinine clearance

KEY POINT

Stopping antiplatelets early after a coronary stent carries a 25–50% risk of MI. Delay elective surgery ≥30 days after a bare-metal stent and >1 year after a drug-eluting stent.

5 · Bowel preparation, temperature and positioning

*See the paper answer **Methods of preoperative bowel preparation.** (Section 12, Ch 133).*

SAQ Patient positioning in lap urology

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 2 Paper 1, Q3a (pneumoperitoneum physiology also answered in 2019-Summer Paper 1, Q2)

Principles

- Positioning uses **gravity** to move bowel away and gives port or robot access while keeping ventilation safe.
- Injury comes from **stretch, compression or ischaemia**, so position and duration act together; nerve injury occurs in about 2.8%. Shared team responsibility (ASA [American Society of Anesthesiologists] advisory, 2000).
- Secure IV (intravenous) access, oximetry and BP (blood pressure) cuff before final positioning: tucked arms become inaccessible.

Positions

Position	Used for	Set-up	Hazards
Steep Trendelenburg, low lithotomy or split legs	RARP (robot-assisted radical prostatectomy), cystectomy	Arms tucked and padded; chest strap over egg-crate; **no shoulder braces**; tilt about 22–30°	Brachial plexus; femoral nerve (excess hip flexion); facial and laryngeal oedema; corneal abrasion; tube into right bronchus

Position	Used for	Set-up	Hazards
Modified flank (30–45°)	Transperitoneal nephrectomy, pyeloplasty	Back on gel rolls; upper arm across chest; wide tape; tilt to test	Pressure points, slippage
Full lateral (90°), table flexed	Retroperitoneoscopy, adrenalectomy	**Axillary roll** 3–4 fingerbreadths below axilla; lower leg flexed 45°, upper straight on pillows	**Rhabdomyolysis**; sciatic stretch
Prone	Posterior adrenalectomy	Pad torso, elbows, hips	Airway, face and eyes

Rhabdomyolysis

- About 0.4% (1% retroperitoneal). Risk: men, BMI (body mass index) ≥25, **kidney rest**, full flexion, long cases. Downside hip pain, brown urine, CK (creatine kinase) >5,000 units. Prevent: limit kidney rest (Campbell gives 20–30 minutes in one place, under 1 hour in another), avoid hypotension, keep cases under 5 hours.

Prevention and physiology

- Abduction ≤90°; pad ulnar groove and fibular head; recheck after every table move.
- **Head-down** raises cardiac output but splints the diaphragm; restrict fluids.
- Nerve injury on day 1: neurology review, conduction studies, physiotherapy.
- **Also know:** forearm compartment syndrome from infiltrated IV lines; Xi side-docking avoids stirrups.

Diagram to draw: full flank position with axillary roll and table break.

> **Key points to remember:**
> - Nerve injury about 2.8%; brachial plexus is commonest at risk.
> - Never shoulder braces in steep Trendelenburg.
> - Axillary roll below, not in, the axilla.
> - Rhabdomyolysis: men, BMI ≥25, kidney rest; CK >5,000.

*See the paper answer **Operating room set-up** (Section 1, Ch 9).*

Also, from the condensed book (not in the paper answer):

- Ileus beyond 72–96 h should prompt a search for obstruction or retroperitoneal haemorrhage.
- Femoral neuropathy comes from retractor blades on the psoas, not from hip position.

6 · Antibiotic prophylaxis

- Give it within **60 min** before incision, limit it to **≤24 h**, and redose after 4 h. Transrectal biopsy is a contaminated wound.

Procedure (American Urological Association, AUA)	First choice	Duration
Cystoscopy (sterile urine), shock-wave lithotripsy	Only with risk factors	Single

Procedure (American Urological Association, AUA)	First choice	Duration
Endourology, nephrectomy, prostatectomy, ileal conduit	Cefazolin	Single
Transrectal biopsy	Fluoroquinolone, or cephalosporin plus aminoglycoside	Single
Colon conduit	Cefazolin plus metronidazole	Single
Percutaneous renal surgery	First- or second-generation cephalosporin	≤24 h
Prosthesis	Aminoglycoside plus cephalosporin or vancomycin	≤24 h

7 · VTE prophylaxis

- VTE is the leading non-surgical cause of death after urological surgery, highest after cystectomy. Drug prophylaxis raises minor, not major, bleeding.
- Enoxaparin is **40 mg** daily (30 mg if creatinine clearance <30 mL/min). Unfractionated heparin (UFH) 5000 U is reversible and safe in renal failure.

AUA risk	Definition	Prophylaxis
Low	Minor surgery, <40 y	Early ambulation
Moderate	40–60 y, or minor surgery with risk factors	UFH 12-hourly or enoxaparin
High	>60 y, or 40–60 y with risk factors	UFH 8-hourly or enoxaparin
Highest	Cancer, prior VTE	Drug plus compression

A Caprini score **≥5** is highest risk. Give extended **28-day** LMWH after radical pelvic cancer surgery.

8 · Anaesthesia and transfusion

- Extubate only when the train-of-four ratio reaches **0.9**. Nitrous oxide expands closed gas spaces.
- Transfusion is often indicated at haematocrit **<21%** and rarely above 30%. Give platelets below 50,000 (100,000 for high-risk surgery).
- Transfusion-related acute lung injury is the commonest cause of transfusion death. Delirium may be the first sign of sepsis.

12 Fundamentals of Intraoperative Patient Safety

1 · Checklists, site verification and team safety

*See the paper answer **Operating room safety : the role of surgeon.** (Section 1, Ch 1).*

Also, from the condensed book (not in the paper answer):

- Wrong-site surgery occurs in about **1 in 112,000** cases, and up to 30% of root causes arise before admission.
- Mark only the correct side, never with an "X". For lower-tract endourology, mark the wristband, not the meatus. Use a professional interpreter, never family.
- Simulation-based team training changes behaviour; didactic teaching alone does not.

2 · Retained items and fire

- A retained surgical item is one found after skin closure, even in theatre. It occurs in **1 in 2800–10,000** cases; sponges are commonest, then drains, then needles.
- A correct count does not exclude a retained item, so obtain a radiograph when in doubt.
- The **Silverstein fire-risk score** gives one point each for open oxygen, an ignition source and a site above the xiphoid; **3** means high risk. Monopolar diathermy causes about 90% of fires.
- For buccal graft harvest, keep inspired oxygen ≤30%, stop nitrous oxide and warn anaesthesia before using diathermy.
- Place the diathermy return pad in full contact over muscle, away from metal implants, and prefer bipolar with pacemakers.

ALGORITHM

1. Fire: stop the procedure and pour saline or water.
2. Remove the drapes, then use an extinguisher.
3. Follow RACE: Rescue, Alert, Confine (gases off), Extinguish or evacuate.

3 · Positioning injuries (see Ch 11)

- In lithotomy, stirrups compress the common peroneal nerve at the fibular head and cause foot drop. Hip flexion with knee extension stretches the sciatic nerve.
- **Well-leg compartment syndrome** follows prolonged lithotomy (**>4 h**) with raised legs and Trendelenburg. Prevent it by limiting time or lowering the legs periodically. Suspect it with pain out of proportion and a raised creatine kinase, and treat by urgent fasciotomy.
- In the flank position, long cases and obesity risk rhabdomyolysis.

13 Incisions, Exposure and Drains

1 · Abdominal incisions

Incision	Indications	Notes
Lower midline	Prostatectomy, cystectomy, diversion, pelvic node dissection	**Commonest** urological incision
Upper or full midline	Retroperitoneal lymph node dissection (RPLND), trauma, bilateral nephrectomy	**Most versatile**; poor in obesity
Pfannenstiel	Pelvic surgery, specimen extraction	Rectus separated, not cut; **least hernia**
Gibson	Renal transplant, lower ureter	Extraperitoneal; hockey-stick extension adds hernias

- Close midline fascia with strong slowly absorbable suture, taking bites **1 cm** from the edge and 1 cm apart, with suture length to wound length **≥4:1**.
- No technique is superior for dehiscence or hernia, but avoid continuous rapidly absorbable suture.

2 · Approaches to the kidney

Incision	Indications	Trade-off
Midline transperitoneal	Trauma, inferior vena cava (IVC) thrombus, bilateral disease, horseshoe	Early vascular control; poor upper retroperitoneum
Subcostal or chevron	Radical nephrectomy, bilateral tumours, IVC thrombus	Early vascular control; liver, spleen and bowel at risk
Thoracoabdominal (9th–10th rib)	Level III thrombus; large upper-pole or adrenal mass	Best exposure; pain no worse than flank
Flank (12th supracostal or 11th transcostal)	Partial or simple nephrectomy, xanthogranulomatous pyelonephritis	Extraperitoneal; **pleura** at risk; transcostal causes flank bulge
Transverse upper abdominal	Paediatric Wilms tumour	–

KEY POINT

An anterior incision gives bilateral reach and early pedicle control at the cost of the bowel; a flank incision stays extraperitoneal at the cost of the pleura.

ALGORITHM

1. Pleural tear: pass a red rubber catheter through the tear and tie a purse-string around it.
2. Put the catheter end under water and give sustained inflation.
3. Withdraw it as bubbling stops. A large tear needs a chest tube.

- A routine chest radiograph after nephrectomy is unnecessary.

3 · Inguinal and scrotal incisions

- Radical orchidectomy uses an inguinal incision of 5–7 cm. **A scrotal incision is contraindicated in suspected testis cancer.**
- Subinguinal inguinal lymph node dissection risks flap necrosis, infection and lymphocele.

4 · Drains

- A Penrose drain is passive and suits the scrotum. Jackson-Pratt and Blake drains are active closed suction and suit most other surgery.
- Use a separate stab incision away from the inferior epigastric vessels, and secure the drain with non-absorbable suture.
- Routine drains are questioned after robotic prostatectomy, partial nephrectomy and cystectomy; drain when the findings demand it.

14 Sutures and Suturing

1 · Choosing a suture

- Braided suture knots securely but harbours bacteria; monofilament inflames less but has more memory.

Suture	Strength retained	Absorbed by
Plain / chromic gut	7–10 d / 21–28 d	70 / 90 d
Caprosyn	~10 d	56 d
Monocryl (poliglecaprone 25)	2 wk 20–40%	91–119 d
Vicryl (polyglactin 910, braided)	3 wk 40–50%	56–70 d
PDS (polydioxanone)	6 wk 35–60%	182–238 d
Maxon (polyglyconate)	6 wk 25%	180 d
Prolene, Mersilene (braided polyester)	Indefinite	Never

> **KEY POINT**
>
> Strength is lost long before absorption. Match the support needed: skin 1–2 weeks (Monocryl, gut), soft tissue 2–4 weeks (Vicryl), fascia ≥6 weeks (PDS, Maxon).

- Chromic gut is the most inflammatory absorbable suture; Caprosyn absorbs fastest.
- Silk marks the cord at orchidectomy before a possible retroperitoneal lymph node dissection.
- **Never leave non-absorbable suture or exposed metal staples in contact with urine**, because they form a stone nidus.
- V-Loc barbed suture is not for fascia; Stratafix Symmetric PDS is the only barbed suture strong enough for fascia.

2 · Needles and knots

- A taper point is commonest. A cutting needle has its edge on the inner curve; a **reverse cutting** needle has it on the outer curve and suits retention sutures.
- The smallest taper needle is RB (17 mm). A UR or GU 5/8-circle needle suits the deep pelvis.
- Prolene knots unravel, so add extra throws.

3 · Techniques

- A vertical mattress everts the edges; a figure-of-eight gives haemostasis.
- Run fascia only with slowly absorbable suture, and close port defects >8 mm.

Trial	Design	Result	Take-home
STITCH	Randomised, midline: 5 mm versus 1 cm bites	Hernia at 1 year 13% vs 21%	Small bites

15 Lower Urinary Tract Catheterisation

1 · Urethral catheters

SAQ Foley's (urethral) catheters, guidewires and DJ stents with their complications

Paper answer · P1

Asked as:

- Foley's catheter. — *Summer 2016 Paper I Q3e (SAQ)*
- Urethral catheters, guidewires and DJ stents — *Practice Set 18 SAQ Q3f (SAQ)* (also answered in part in 2016-Summer Paper 1, Q3e and 2015-Summer Paper 3, Q1)
- Complications if DJS — *Summer 2021 Paper I Q3c (SAQ) — Taken as complications of a DJ (double-J) ureteric stent.*

Type: SAQ (10 marks)

Foley's / urethral catheters

History

- **Frederic Foley**, a Boston urologist, developed the modern balloon catheter (Campbell: 1929; published in the 1930s) for **haemostasis after prostatectomy**. Reybard (1855) had made the first self-retaining one.

Design

- **Size: Fr (French) = 3 × diameter in mm.** Adults (adult) usually 14–16 Fr; clot retention 20–24 Fr three-way. Newborn 6 Fr.
- Normal calibre: (male) meatus 24 Fr (narrowest point), bladder neck 28 Fr, prostatic urethra 32 Fr, female urethra 22 Fr.
- **Balloon:** usually 10 mL; larger (30 mL) for traction. Use (inflate balloon with) **sterile water**, only after urine flows at the hub; volume is printed on the port.
- **Lumens:** one (intermittent, specimen); two-way (drainage + balloon) is routine; three-way adds irrigation for haematuria (stiff wall resists collapse). Each lumen reduces (shrinks) drainage size.
- **Tips:** straight; **coudé** (curved, for prostatic enlargement, tip up); **Councill** (open tip, over a wire).
- **Materials:** latex (cheap, allergy); silicone (long term, less urethritis and encrustation; better urethral tolerance); PVC (polyvinyl chloride); coatings (silver, antibiotic) reduce bacteriuria but not clinical infection.
- **Special:** Malecot (winged, no balloon); Lapides diagnostic (rings 1 cm apart); urodynamic.

Indications and contraindications

- **Indications:** retention, clot retention, output monitoring, long surgery, immobility, after urological surgery.
- **Absolute contraindication:** suspected urethral injury (blood at meatus, pelvic fracture). Do an RUG (retrograde urethrogram) first (**suspected urethral injury → RUG first**).
- **Relative:** AUS (artificial urinary sphincter): deactivate, then 12–14 Fr; stricture.

Technique

- Aseptic; 10–15 mL lignocaine gel. In men, advance **to the hub** and see urine before inflating. Replace the foreskin.
- **Difficult:** 18 Fr coudé → guidewire → cystoscopy, Councill over wire → suprapubic catheter.

Complications

- **CAUTI (catheter-associated urinary tract infection):** 75% of hospital UTIs; bacteriuria 5–10% per catheter-day; **biofilm** forms within minutes.
- Trauma (about 0.3% of inpatients): false passage, balloon inflated in the urethra. Also stricture, traumatic hypospadias, encrustation, paraphimosis, non-deflating balloon (puncture with a wire).

CAUTI prevention

- Review daily and remove early; closed dependent system; clean meatus with water; culture only if symptomatic. Treat only symptomatic infection and **change the catheter**.

Guidewires

Feature	Details
Uses	Access, safety wire, straightening ureter, guiding dilation and stents
Construction	Stainless-steel core with outer wrap; **nitinol** core is kink-resistant
Coating	PTFE (polytetrafluoroethylene): standard safety wire. **Hydrophilic** "glide": passes impacted stones, but too slippery to be a safety wire. Hybrid: hydrophilic tip, PTFE shaft
Size	0.018–0.038 inch (commonly 0.035/0.038); 80–260 cm, 145 cm most useful; floppy tip 1–3 cm
Tip and stiffness	Straight, angled, J; standard to super-stiff (straightens tortuous ureter)

DJ (double-J) stents: introduction

- The **double-J (double-pigtail)** stent (Finney 1978) is a self-retaining polymer tube with coils in the pelvis and bladder; adult usually 6 Fr, length by height. Consent should cover its symptoms and complications.
- **Materials:** polyurethane, silicone (hydrocoated silicone less painful); **metallic (Resonance)** for extrinsic malignant compression.
- **Indications:** obstruction (infected obstructed kidney, malignancy, pregnancy), after URS (ureteroscopy) with injury, pyeloplasty, transplant, leak.
- **Placement:** retrograde pyelogram → wire → stent over wire under fluoroscopy or vision.
- **Problems:** stent symptoms (bladder irritation and reflux) eased by α-blockers and antimuscarinics; migration; encrustation (forgotten stent); failure with extrinsic compression.

Classification of DJ stent complications

Timing	Complications
At insertion	**Misplacement**: proximal end below the obstruction; **submucosal** passage or perforation (suspect when force is needed)
Early	**Stent-related symptoms**; haematuria; infection and urosepsis; **vesicoureteric reflux** up the stent (flank pain on voiding)

Timing	Complications
Late	**Migration** (proximal or distal; short stent); **encrustation**; **stent failure** (obstruction); biofilm and colonisation; **fracture**; **forgotten stent**; rarely **ureteroarterial fistula**

Stent-related symptoms

- About **4 in 5 patients**: urgency, frequency, dysuria, haematuria, flank and groin pain; sexual and work problems.
- **Mechanism:** the **intravesical coil irritates the trigone**, and **reflux up the stent** distends the kidney on voiding.
- Measured by **USSQ** (Ureteral Stent Symptom Questionnaire).
- **Reduced by: α-blockers and antimuscarinics**; intraureteral placement; hydrocoated silicone; shortest dwell. Diameter does not matter.

Encrustation and forgotten stent

- Minerals and biofilm deposit on the stent. Risk rises with **dwell time**, infection, stone-forming urine and pregnancy. Leads to obstruction, sepsis, fracture, **ureteric avulsion on removal**, renal loss.
- **FECal grading (Forgotten, Encrusted, Calcified):** I minimal linear; II circular; III partial on one or both coils; IV complete on one coil; V both coils.
- **Management:** non-contrast CT (computed tomography); never pull by force; clear **bladder coil** (cystolithotripsy), then **ureter** (ureteroscopy), then **renal coil** (flexible ureteroscopy or PCNL [percutaneous nephrolithotomy]).

Stent failure

- Extrinsic compression (malignancy, radiation), debris, encrustation. Obstructive symptoms → image; change stent, **PCN** (percutaneous nephrostomy) or **metallic stent**.

Prevention

- Stent only when indicated; correct length; **stent register**; long-term stents changed every 3–6 months (local practice); **pregnancy: about every 4 weeks**.

Diagram to draw: cross-section of a three-way Foley; DJ stent in situ with bladder coil on the trigone and arrows showing reflux.

> **Key points to remember:**
>
> - Foley balloon catheter: designed for post-prostatectomy haemostasis.
> - Fr = 3 × diameter (mm); inflate with sterile water; coudé for prostatic enlargement, Councill over a wire.
> - Suspected urethral injury → RUG first; AUS → deactivate, 12–14 Fr.
> - CAUTI 5–10% per day; remove early; change the catheter when treating.
> - Hydrophilic wire to pass, PTFE wire to stay (safety).
> - DJ symptoms in about 80%: trigonal irritation and reflux; α-blockers and antimuscarinics help.
> - Long-dwelling stents: monitor for encrustation; encrustation rises with dwell time; FECal I–V.
> - Encrusted stent: bladder, then ureter, then kidney; keep a stent register.

Also, from the condensed book (not in the paper answer):

- Lapides showed that infection comes from stasis and overdistension, not bacteria alone, so **regular emptying, not sterility, prevents infection** in clean intermittent catheterisation.

Age	Straight (Fr)	Balloon (Fr)
Newborn	4–6	6
1–23 months	6–8	6–8
2–6 y	10	8–10
7–12 y	10–12	12–14
>12 y	14	14–16

2 · Difficult catheterisation

SAQ Outline algorithm for difficult catheter placement.

Paper answer · P1

Type: SAQ (10 marks)
Source: 2024-Winter Paper 1, Q3b

Introduction

- **Normal calibre:** meatus 24 Fr (narrowest), bladder neck 28 Fr, prostatic urethra 32 Fr.
- **Causes:** BPH (benign prostatic hyperplasia, commonest), urethral stricture, false passage, meatal stenosis, bladder neck contracture, urethral trauma, abnormal anatomy.
- **Golden rule:** escalate **blind → wire → vision → suprapubic**; never repeat forceful blind passage.

Before starting

- History (surgery, radiotherapy, AUS (artificial urinary sphincter), reconstruction), DRE (digital rectal examination).
- **Absolute contraindication:** suspected urethral injury (pelvic fracture, blood at meatus, high-riding prostate) → **RUG (retrograde urethrogram)**; pelvic fracture urethral injury → suprapubic tube.
- **Technique:** 10–15 mL lubricant (lidocaine gel), penis stretched, steady pressure; inflate only after urine flows at the hub.

Algorithm (Campbell Fig. 15.6)

```
First attempt fails ├─ Prostatic obstruction suspected │ → 18-Fr coudé, tip (marker) pointing up │ → fails → flexible cystoscopy └─ Not suspected (stricture likely) → smaller catheter → fails → hydrophilic glidewire → fails → flexible cystoscopy At cystoscopy / with wire in bladder: ├─ Stricture → wire across → 5-Fr catheter, aspirate urine to confirm │ → super-stiff wire → serial dilatation (1 Fr above catheter size) │ → Councill (open-tip) catheter over wire └─ False passage → pass under vision (scope anterior) → wire → Councill over wire Wire cannot be passed / dense stricture / urethral disruption → suprapubic catheter (ultrasound-guided percutaneous, or open)
```

- **Why 18-Fr coudé:** stiffer, larger tip area, curved tip follows the bulbar-prostatic angle; 14–16 Fr usually buckle.
- **Blood on the tip** suggests false passage: stop blind attempts.

Special situations

Situation	Approach
AUS	Deactivate; smallest catheter 12–14 Fr; remove early
Obese or female	Exposure: Trendelenburg, retract pannus, two fingers in vagina to guide
Buried penis	Infrapubic pressure; flexible cystoscopy
Neobladder, continent pouch	Never force; scope + wire; suprapubic only with imaging guidance
PUV (posterior urethral valves) neonate	6–8 Fr, or 5-Fr open-ended over glidewire; else suprapubic or vesicostomy

Also know: filiforms, balloon dilators; complications: false passage, haematuria, stricture.

Diagram to draw: coudé catheter following the bulbar-prostatic curve with tip up.

> **Key points to remember:**
> - Blind → wire → vision → suprapubic.
> - BPH: 18-Fr coudé, tip up.
> - Councill catheter over wire after dilatation.
> - Suspected urethral injury: RUG first, no blind catheter.

Also, from the condensed book (not in the paper answer):

- Membranous strictures are the most prone to false passage. Pass the cystoscope with the camera angled anteriorly.
- With a neobladder, suprapubic puncture is hazardous (intraperitoneal), so use image guidance.

3 · Suprapubic catheterisation

- It is indicated when the urethra is inaccessible (trauma, stricture) and for long-term drainage.
- Percutaneous placement is contraindicated with a violated retropubic space, bowel over the bladder, or no ultrasound.

ALGORITHM

1. Confirm a full bladder with no overlying bowel on ultrasound, and tilt the patient head-down.
2. Make a 1–2 cm incision 1–2 fingerbreadths above the symphysis.
3. Aspirate urine with a spinal needle while infiltrating local anaesthetic.
4. Insert a trocar and sheath (16 Fr catheter) or use the Seldinger technique, without over-advancing.

- Bowel injury is the key complication. An experienced person should make the **first change at 4–6 weeks**.

4 · Complications

*See the paper answer **Catheter Associated Urinary Tract Infections (CAUTI)** (Section 3, Ch 26).*

Also, from the condensed book (not in the paper answer):

- Spinal cord injury raises bladder cancer risk, but surveillance cystoscopy is not recommended.

Retained catheter	Management
Balloon will not deflate	Cut the valve; pass a stiff wire end up the channel to puncture it, or use an ultrasound-guided needle; then cystoscopy to retrieve fragments
Knotted	Cystoscopic needle puncture of the balloon
Calcified	Cystoscopic laser lithotripsy

16 Fundamentals of Upper Urinary Tract Drainage

1 · Indications

- Obstruction is the commonest indication. Others are pre-stenting before retrograde intrarenal surgery, trauma (to prevent urinoma) and instillation of chemotherapy.
- In the **Whitaker test**, a renal pelvic minus bladder pressure **>12 cm H_2O** during constant-rate infusion means obstruction.

SAQ Interventional radiology in urology and therapeutic angioembolisation

Paper answer · P1

Asked as:

- Therapeutic angioembolization. — *Practice Set 7 Paper I Q3e (SAQ)*
- Interventional urological treatment in urology — *Practice Set 8 Paper I Q3e (SAQ)*

Type: SAQ (10 marks)

Assumed meaning: interventional radiology (image-guided percutaneous and endovascular) treatment in urology.

Principles

- Guidance by ultrasound, fluoroscopy or CT (computed tomography), usually under local anaesthesia. **Seldinger technique:** needle, guidewire, then devices over the wire.
- **Beforehand:** correct coagulopathy, manage anticoagulants, urine culture, antibiotic prophylaxis.
- **Requirements:** experienced team, resuscitation in the suite; unstable non-responders need surgery.

Non-vascular procedures

Procedure	Indications and points
PCN (percutaneous nephrostomy)	Obstructed infected kidney (as effective as retrograde stent), malignant ureteric obstruction, urinary leak or fistula, access for PCNL (percutaneous nephrolithotomy). Puncture a **posterior calyx** through the avascular Brödel plane. Transfusion 0.5–4%
Antegrade stenting	When retrograde stenting fails: malignant obstruction, ileal conduit, transplant ureter
Antegrade pyelography, Whitaker test	Define strictures; pressure–flow test for obstruction
Percutaneous drainage	Renal or perinephric abscess >3 cm, urinoma, lymphocele
Image-guided biopsy	Renal mass biopsy

Vascular (endovascular) procedures: therapeutic angioembolisation

Definition

- Deliberate **endovascular occlusion** of a target vessel, usually by transfemoral selective catheterisation under fluoroscopy, to stop bleeding, devascularise a tumour or close an

abnormal vessel.

- **Superselective** embolisation of a segmental branch preserves most parenchyma.

Embolic agents

Type	Examples	Use
Temporary	Autologous clot; gelatin sponge (2–6 weeks)	Priapism, prostate, bladder bleeding
Permanent particles	PVA (polyvinyl alcohol), microspheres	Tumours, PAE (prostate artery embolisation)
Mechanical	Microcoils, vascular plugs	Pseudoaneurysm, AVF (arteriovenous fistula), varicocele
Liquid	NBCA (N-butyl cyanoacrylate) glue, absolute ethanol	Tumour ablation, recurrent priapism

Urological indications

Procedure	Indications
Renal angioembolisation	Grade IV–V renal trauma; **pseudoaneurysm or arteriovenous fistula** after PCNL or partial nephrectomy (delayed bleed); bleeding AML (angiomyolipoma, first line); palliative embolisation of RCC (renal cell carcinoma)
Internal iliac (anterior division) embolisation	Intractable bladder haemorrhage; initial control about 90%. Spare the posterior division (gluteal ischaemia)
PAE (prostate artery embolisation)	LUTS (lower urinary tract symptoms) from BPH (benign prostatic hyperplasia); EAU (European Association of Urology) for men accepting lesser outcomes; AUA (American Urological Association) 2023 may offer
Varicocele embolisation	Coils or liquid sclerosants via femoral or jugular vein
High-flow priapism	Superselective cavernosal artery embolisation; resolution ≥75%; recurrence 30–40%
PTRA (percutaneous transluminal renal angioplasty)	Fibromuscular dysplasia; transplant renal artery stenosis

- **Renal trauma:** active contrast extravasation, pseudoaneurysm, AVF; success high in grade IV, lower in grade V. Failure predictors: instability, grade V, urinary extravasation. Repeat embolisation often succeeds.
- **Iatrogenic bleeding** after PCNL (percutaneous nephrolithotomy) or partial nephrectomy: **treatment of choice**; superselective coils nearly 100% technically successful.
- **AML (angiomyolipoma): first line for acute haemorrhage**; prophylactic for >6 cm, aneurysm >5 mm, growth or pregnancy plans. In TSC (tuberous sclerosis complex), embolise then give steroids.
- **RCC:** palliation of haematuria or pain in unfit patients; preoperative use no longer routine.
- **Intractable bladder haemorrhage** (radiation cystitis, cancer): bilateral **anterior division of internal iliac**; avoid the posterior division (gluteal pain).
- **Prostate:** PAE for BPH (benign prostatic hyperplasia) and refractory prostatic haematuria.
- **High-flow priapism:** superselective cavernosal artery embolisation; resolution ≥75%; recurrence → repeat with a non-resorbable agent.

- **Also know:** varicocele (venous coils); lymphatic embolisation for chylous ascites; ADPKD (polycystic kidney) pain.

Ablation

- **Cryoablation or RFA (radiofrequency ablation)**, percutaneous, for cT1a renal masses <3 cm; biopsy first. Higher local recurrence than partial nephrectomy, salvaged by repeat ablation.

Complications

- Contrast nephropathy, access-site haematoma, coil migration.
- **Post-embolisation syndrome:** fever, flank pain, nausea, leucocytosis; up to 72 hours; self-limiting.
- **Non-target embolisation:** bowel, bladder, rectum, gluteal muscle, penile skin.
- Renal infarction or abscess; ED (erectile dysfunction) after priapism embolisation.

Diagram to draw: prone PCN through a posterior calyx along the Brödel line; renal angiogram showing segmental pseudoaneurysm before and after coil occlusion.

> **Key points to remember:**
> - PCN equals retrograde stent for decompressing an obstructed infected kidney.
> - Superselective coils: treatment of choice for pseudoaneurysm and AVF.
> - Embolisation is first line for bleeding AML and post-PCNL pseudoaneurysm or arteriovenous fistula; prophylactic >6 cm or aneurysm >5 mm.
> - Bladder bleeding: internal iliac anterior division; spare posterior division.
> - Post-embolisation syndrome is common and self-limiting.

Also, from the condensed book (not in the paper answer):

- PCN avoids general anaesthesia in the very ill, while a stent avoids an external tube and can also be placed awake under local anaesthesia.

2 · Anatomy chooses the route

Situation	Preferred approach
Horseshoe kidney	Percutaneous via a more medial upper-pole calyx; computed tomography (CT) first, as retrorenal bowel is common
Pelvic or ectopic kidney	Supine anterior, image-guided or laparoscopy-assisted
Transplant	Anterior, ultrasound-guided
Ureteropelvic junction obstruction before pyeloplasty	Antegrade, to spare the ureter
Calyceal diverticulum, ileal ureter, transureteroureterostomy, urinary diversion	Antegrade

3 · Retrograde stenting and stents

*See the paper answer **Foley's (urethral) catheters, guidewires and DJ stents with their complications** (Section 1, Ch 15).*

Also, from the condensed book (not in the paper answer):

- Adult stents are typically 4.7–8 Fr and 22–30 cm; **6 Fr, 24–26 cm** is commonest.
- Under fluoroscopy alone, place the pusher marker at the mid-symphysis in men and the lower symphysis in women. This landmark is invalid after prostatectomy or with prolapse.
- The Ureteral Stent Symptom Questionnaire (USSQ) has **38 questions in six domains**. Stent length made no difference in a large randomised trial.

4 · Percutaneous nephrostomy

P1 *Pelvicalyceal anatomy and its relevance to PCNL; Prone vs supine PCNL — posterior calyx through the papilla along Brödel's avascular line, never the pelvis; calyx choice and pleural risk; prone versus supine*

- The American Urological Association (AUA) recommends antibiotic prophylaxis for all percutaneous renal surgery.
- A 21 G needle takes only a **0.018-inch** wire, which must be upsized; an 18 G needle takes a 0.035-inch wire. Dilate short, never beyond the target calyx.
- Balloon dilators are now commonest and bleed less but work poorly in scar, where metal Alken dilators are best.

Agent	Stop before access
Aspirin (primary prevention only)	1 week; continue if there is a cardiovascular indication
Warfarin	1 week (bridge if needed)
Clopidogrel	5 days
Non-steroidal anti-inflammatory drugs	3–5 days

Tube	Size	Notes
Foley or Councill	≤24 Fr	Fill balloon partly with saline, not contrast
Cope loop	6–14 Fr	Dislodges in obesity
Nephroureteral	8–12 Fr	Rarely dislodges
Malecot	16–30 Fr	Needs a stiffener

- Smaller tubes cause less pain and leak, with no more bleeding. Clamp before removal to test distal patency, and **exchange every 3 months**.

SAQ Tubeless PCNL

Paper answer · P2 · P4 · merged from 2

Asked as: P2 SAQ "Tubeless PCNL" · P4 SAQ "Tubeless PCNL"

Definition

- PCNL (percutaneous nephrolithotomy) that ends **without a nephrostomy tube**.
- **Tubeless**: drainage by a ureteral stent (double-J, or externalised stent or string). **Totally tubeless**: no stent and no tube.
- **Wickham (1984)** did the first tubeless PCNL; **Bellman (1997)** popularised it.

Rationale

- A large nephrostomy tube causes pain, more analgesia, longer stay and higher cost. Most tracts stop bleeding without tamponade.

Selection

Suitable	Keep a nephrostomy
Complete clearance, single tract	Pyonephrosis or infected stone
Minimal bleeding	Significant bleeding
Intact collecting system	Significant perforation of (injury to) the collecting system
Supracostal access, bilateral cases (shown safe)	Residual stone needing a second look

Technique (exit strategy)

1. Confirm clearance by rigid and **flexible nephroscopy** and fluoroscopy.
2. Antegrade double-J stent, or externalised stent through the urethra, or a string.
3. Withdraw the sheath under vision; the **safety wire is removed last**.
4. Watch the tract; apply pressure (rib as backstop in thin patients). Fibrin glue in the tract did not reduce bleeding in randomised trials; some use haemostatic sealants or tract cautery.
5. Urethral catheter overnight.

Evidence

- Meta-analyses of randomised trials: shorter stay and operative time, and less pain.
- The pain benefit is over large tubes (**>16 Fr**); there is **no benefit over small 8–9 Fr tubes**; small tubes (≤10 Fr) give similar comfort without more bleeding.
- No difference in fever, infection, transfusion or stone-free rate.
- Stent discomfort: worse WiSQoL (Wisconsin Stone Quality of Life) scores than nephrostomy in two trials. Externalised stents come out before discharge without cystoscopy.
- Children: insufficient evidence.

Disadvantages and complications

- Loss of tract for a second look.
- Stent symptoms and a further procedure for removal.
- Urine leak or urinoma.
- Bleeding without tamponade: re-insert a balloon or Kaye catheter, then **angioembolisation**.

Current position

- Choose tubeless, small tube (≤10 Fr) or stent case by case; both stents and small tubes are acceptable (AUA, American Urological Association). Pairs well with mini-PCNL.

Diagram to draw: kidney with sealed tract and double-J stent.

Key points to remember:

- Tubeless = stent only; totally tubeless = neither.
- Benefit is mainly over large (>16 Fr) tubes.
- Keep a tube for bleeding, perforation, infection or second look.

5 · Complications

P2 *Complications of PCNL — immediate venous bleeding needs tamponade, bleeding at about 1 week is a pseudoaneurysm or arteriovenous fistula needing selective embolisation; pleural, colon and solid-organ injury*

- Major complications occur in about 10%; PCN transfusion is **0.5–4%**. Pleural injury needing intervention is about **5%** above the 12th rib and **25%** above the 11th.
- Colon injury occurs in about 0.1%; retrorenal colon is found in 10% prone versus 2% supine. Extraperitoneal injury is managed conservatively; peritonitis needs open repair.
- Air embolism shows as a sudden fall in end-tidal CO_2 and a mill-wheel murmur. Give 100% oxygen and position head down, right side up.

17 Principles of Urologic Endoscopy

1 · Evolution of the endoscope

LAQ Flexible endourological instruments, digital and current ureteroscopes

Paper answer · P4

Asked as:

- Briefly outline flexible endourological instrument and its role in management of urological disease — *Practice Set 1 Paper IV Q1 (LAQ)*
- Digital flexible ureteroscopes — *Summer 2019 Paper IV Q3c (SAQ)*
- Current ureter scopes. — *Summer 2022 Paper IV Q3d (SAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)

(flexible ureteroscopy in detail: see Practice Set 1 Paper 2, Q3b; digital scopes also in 2019-Summer Paper 4, Q3c)

Summer 2022 Q3d taken as "current ureteroscopes"; see also 2019-Summer Paper 4, Q3c.

Introduction

- A flexible endoscope has a **deflectable tip** and carries its image by a fibre bundle or a tip-mounted chip. It follows the natural curves of the urethra, ureter and collecting system.
- Physical basis: light travels along a curved glass fibre by **total internal reflection** (Tyndall).
- Marshall reported the first flexible URS (ureteroscopy) in 1964. It became therapeutic only when a **working channel** and **active tip deflection** were added in the early 1980s.
- **Fibreoptic flexible URS (ureteroscopy):** coherent glass-fibre bundle carries the image; fibres break → black spots; honeycomb (moiré) pattern.
- **Digital scopes** (first 2006): a sensor at the tip (**"chip-on-tip"**) converts light to an electrical signal, so the image does not degrade.

Classification of current ureteroscopes

Type	Key features
Rigid rod-lens	Larger calibre; largely historical
Semirigid	**Fibreoptic** image bundle; tip **≤7 Fr**; channel >3 Fr; often **two channels**; in-line or offset eyepiece
Flexible fibreoptic	Tip 4.5–7.5 Fr (smallest); image degrades as fibres break (black dots)
Flexible digital	"Chip-on-tip" CMOS (complementary metal-oxide semiconductor) or CCD (charge-coupled device) sensor plus LED (light-emitting diode); no camera head
Single-use flexible	Digital; LithoVue (2016) first; now several makers
Ureteral resectoscope	11.5 Fr; largely replaced by laser

Semirigid ureteroscope

- Uses **coherent fibre bundles** for the image (random fibres carry only light).
- **Two channels** are preferred: irrigation through one while an instrument is in the other, or a laser through one to break a stone held in a basket.
- **Offset eyepiece** gives a straight channel for rigid probes (pneumatic, biopsy).
- **Rule:** semirigid below the iliac vessels, flexible above.

Components

Component	Key features
Optics	**Fibreoptic:** a *coherent* image bundle carries the picture; a *random* light bundle carries illumination; cladding improves reflection. **Digital:** CCD (charge-coupled device) or CMOS (complementary metal oxide semiconductor) chip at the tip with LED (light-emitting diode) light; no camera head or light cable
Deflection	Lever → control wires through movable metal rings → wire fixed at the tip (deflection mechanism: lever pulls wires). "Intuitive" = tip moves with the lever. The reticle (notch) marks the plane of deflection
Working channel	Single channel shared by irrigation and instruments, so instruments reduce flow and deflection
Shaft	Tapered tip; passive (secondary) deflection segment behind the tip helps lower-pole access

- Flexible ureteroscope parts: optics, deflection mechanism (lever pulls wires), working channel.

Sensor technology (digital)

	CCD (charge-coupled device)	CMOS (complementary metal-oxide semiconductor)
Heat and power	More	Less (runs cooler)
Cost, speed	Higher, slower	Cheaper, faster
Status	Phased out	Current standard

- **LED (light-emitting diode)** at the tip; no camera head or light cable → better ergonomics.

Types

Instrument	Specifications	Main use
Flexible cystoscope	14–17 Fr; 37–40 cm; deflection 120–220°; channel 6.4–7.5 Fr; field about 120°	Office diagnosis and minor therapy
Flexible ureteroscope	Tip 4.5–7.5 Fr; deflection ≥270° both ways; channel **3.6 Fr**; dual-channel scope (2 × 3.3 Fr) exists	Kidney and upper ureter
Flexible nephroscope (or flexible cystoscope used through the tract)	Larger calibre; too wide for the UPJ (ureteropelvic junction)	Clearing calyces during PCNL (percutaneous nephrolithotomy)

Deflection and channel (flexible ureteroscope)

- Active deflection **≥270° both directions** (another source: **>270° both ways** in one plane); enhanced primary deflection eases lower-pole access / reaches the lower pole (ureteroinfundibular angle up to about 175°).
- Working channel **3.6 Fr**; dual-channel scopes exist.

Fibreoptic vs digital (digital versus fibreoptic)

Feature	Fibreoptic	Digital
Image	Honeycomb pattern; black dots as fibres break; degrades	Sharper, larger (magnified), stable image; no focusing or white balance
Size / Tip	Smallest tips, tapered	Larger (about 8.5 Fr), flatter; more access-sheath use
Outcome / Operative time	Reference; longer	Shorter operating time, equal (similar) stone-free rate (SFR)
Durability	Fragile; over-rotation damages fibres; repair about every 15 cases	**Not better** in later comparisons; not clearly better in trials
Processor	Platform-agnostic	Proprietary

- Surprise finding: in vitro, illumination was better with the fibreoptic cystoscope.
- Digital: sharper image, **shorter operating time, same stone-free rate**. But larger tip, so **more access sheaths and sheath complications**.
- Larger tip → more access-sheath use. Keep a fibreoptic scope for narrow UPJ (pelviureteric junction) or infundibulum.

Single-use scopes (single-use vs reusable)

- **LithoVue (2016):** first single-use digital ureteroscope; CMOS, LED, tip 7.7 Fr, shaft 9.5 Fr, 3.6 Fr channel, 270° each way. Single-use cystoscopes followed (2020).
- Rationale (reason): reusable ureteroscopes may need repair after only **3–5 cases**; reprocessing and repair add cost.
- Single-use advantages: maximal deflection each case, sterile, no reprocessing or repair; cost **lower below about 99 cases/year**.
- Best value in low-volume centres and for **large lower-pole stones**.
- Reusable: cheaper in high-volume centres; needs reprocessing (peracetic acid, glutaraldehyde, ethylene oxide).
- Cost advantage depends on case volume; environmental impact is similar to reusables. Most centres use a **hybrid fleet**.

Accessories

- **Guidewires:** 0.035/0.038 inch PTFE (polytetrafluoroethylene)-coated safety wire; hydrophilic wires to pass impacted stones (exchange, as too slippery for safety); nitinol-core and hybrid wires.
- **Access sheath** (11–16 Fr): repeated passes, lower intrarenal pressure; small risk of ureteric injury.
- **Laser fibres** 200–365 µm; **nitinol tipless baskets** ≤3 Fr; three-prong graspers; 3 Fr cup biopsy forceps; 3 Fr balloon dilating to 12 Fr; fulguration electrodes; pressure irrigation pumps.

- Flexible EHL (electrohydraulic lithotripsy) probes work; flexible pneumatic probes restrict deflection.

Role in urological disease

Area	Role
Lower tract	Haematuria work-up; bladder cancer surveillance, office biopsy and fulguration; NBI (narrow-band imaging) and blue light; stent removal or exchange; bladder stones, foreign bodies, strictures; botulinum toxin injection; diversions through a catheterisable channel; difficult ureteric access via a suprapubic tract
Stones	RIRS (retrograde intrarenal surgery): renal stones <20 mm, lower pole <10 mm; >2 cm when PCNL is unsafe
UTUC (upper tract urothelial carcinoma)	Diagnostic URS and biopsy (predicts grade in about 9 of 10); laser ablation of low-grade tumours; prefer to percutaneous route (seeding)
Obstruction	Endopyelotomy without crossing vessels; endoureterotomy for strictures <1.5 cm
Diagnosis	Positive cytology, filling defects, lateralising haematuria (haemangioma fulgurated), recurrent one-sided infection; foreign body retrieval
PCNL	Antegrade or retrograde flexible endoscopy raises stone-free rate; ECIRS (endoscopic combined intrarenal surgery)

Care and complications

- **Scope damage:** deflection wear; laser fibre scraping and perforating the channel. Pass fibres with the scope straight; **fire laser only when the fibre tip is seen touching the stone** (fire only with the fibre tip seen on the stone); leave the bladder empty to avoid buckling.
- **Sterilisation:** sterilise with ethylene oxide, glutaraldehyde or peracetic acid.
- **Patient:** ureteric injury, stricture, sepsis from high pressure. Use saline irrigation and antibiotic prophylaxis for all upper tract procedures.

Recent advances

- **Intrarenal pressure sensor** in the scope (LithoVue Elite): high pressure causes fever, SIRS (systemic inflammatory response syndrome), sepsis. Temperature sensing next.
- **AI (artificial intelligence):** stone recognition, laser setting adjustment, navigation.
- **Robotic flexible URS (ureteroscopy):** ergonomics, less radiation to surgeon.
- **Also know:** suction (access) sheaths, further miniaturisation.

Diagram to draw: cross-section of a flexible ureteroscope: chip, LED, working channel and deflection wires (scope tip cross-section showing CMOS chip, LEDs, 3.6 Fr channel and 270° deflection).

Key points to remember:

- Working channel plus active deflection made flexible scopes therapeutic.
- fURS: 3.6 Fr channel, ≥270° (>270°) deflection; cystoscope 14–17 Fr.
- Chip-on-tip CMOS digital: better image, larger tip, durability not better; shorter operation.
- Single-use cheaper below about 99 cases/year; LithoVue first single-use (2016).
- Semirigid: two channels; below the iliac vessels.
- Pressure sensors and AI are the newest developments.
- Golden rule: never fire unless the fibre tip is seen on the stone.

SAQ Simulators in urology

Paper answer · P4

Type: SAQ (10 marks)
Source: Practice Set 5 Paper 4, Q3d

Rationale

- Fewer training cases and complex technology mean skills must be learnt **outside the patient**.
- Chapter 1: simulation is used to **acquire and maintain skills and to warm up** before operating.

Types

Type	Examples	Strengths and limits
Bench / box trainer	Peg transfer, cutting, suturing; chicken oesophagus anastomosis; 3D (three-dimensional)-printed perfused hydrogel kidney	Cheap, proven skill gains; limited realism
VR (virtual reality)	LapSim, LapMentor, RobotiX Mentor, SimNow (robotic prostatectomy); endourology trainers for cystoscopy, ureteroscopy, PCNL (percutaneous nephrolithotomy), TURP (transurethral resection of the prostate)	Records metrics (precision, economy of motion), builds curricula, sets expert benchmarks; costly; poor tissue feel
Live animal	Pig	Real bleeding; ethics, cost
Cadaver	Fresh-frozen	Real human anatomy; scarce
Team (non-technical)	High-fidelity theatre crises	Communication and teamwork

Validity (needed before a simulator is trusted)

- **Face:** looks real. **Content:** teaches the right skills. **Construct:** experts score better than novices. **Concurrent:** agrees with an established test. **Predictive:** simulator score forecasts operating performance.

Evidence (Chapter 18, Chapter 12)

- VR training **beats no training or box training** for novices and improves live performance and operating time.
- Robotic simulator scores correlate with live RARP (robot-assisted radical prostatectomy) performance.

- **FRS (Fundamentals of Robotic Surgery):** a multi-institutional RCT (randomised controlled trial) showed better speed and accuracy.
- Ureteroscopy team simulation taught non-technical skills better than lectures (randomised trial).

Structured curricula and assessment

- **EAU (European Association of Urology) E-BLUS** (European Basic Laparoscopic Urological Skills): online theory, then dry-model tasks, then examination.
- **Proficiency-based progression:** move on only after reaching expert benchmarks.
- Scoring: **GEARS (Global Evaluative Assessment of Robotic Skills)** and **OSATS (Objective Structured Assessment of Technical Skills)**.

Limitations

- Cost, poor tissue feel, limited proof of better patient outcomes.

Diagram to draw: pyramid: box trainer at the base, then VR, then animal or cadaver, then supervised surgery at the top.

> **Key points to remember:**
> - Bench, VR, animal, cadaver and team simulation.
> - Validity: face, content, construct, concurrent, predictive.
> - VR beats box or no training for novices.
> - E-BLUS and FRS; proficiency-based progression; GEARS.

Also, from the condensed book (not in the paper answer):

- **Desormeaux (1853)** excised a urethral papilloma, the **first therapeutic endoscopy**. Nitze (1877) placed the light source at the tip.

2 · Optics and enhanced imaging

*See the paper answer **Flexible endourological instruments, digital and current ureteroscopes** (Section 1, Ch 17).*

*See the paper answer **CA bladder, etiology, newer investigations and treatment.** (Section 12, Ch 129).*

Also, from the condensed book (not in the paper answer):

- Narrow-band imaging (NBI) needs **no instillation**.
- Blue-light cystoscopy (Food and Drug Administration approved 2010): hexaminolevulinate is instilled **1 h** before, and blue light of 360–450 nm makes tumour fluoresce red while normal urothelium stays blue.

3 · Irrigation fluids and TUR syndrome

SAQ Irrigation fluids for TURP.

Paper answer · P2

Type: SAQ (10 marks)
Source: 2017-Summer Paper 2, Q3a

Ideal irrigant

- Transparent, **non-conducting** (monopolar), isotonic, non-toxic, non-haemolytic, cheap.
- **Why non-conducting?** In monopolar TURP (transurethral resection of the prostate), current flows loop → patient → pad; ionic saline would disperse it. **Bipolar** TURP completes the circuit in the instrument, so it uses **normal saline**.

Fluids

Fluid	Osmolality (approx.)	Points
Sterile water	0	Best vision, cheap; haemolysis, hyponatraemia; now avoided for TURP
Glycine 1.5%	~200 mOsm/kg	Commonest (UK, North America); glycine → ammonia, serine; transient blindness (retinal inhibitory neurotransmitter), encephalopathy
Sorbitol 3%	~165	Metabolised to fructose and glucose → hyperglycaemia; lactic acidosis
Mannitol 5%	~275	Near iso-osmolar; not metabolised; osmotic diuresis; volume overload in renal failure
Sorbitol 2.7% + mannitol 0.54%	~180	Continental Europe
Normal saline 0.9%	~308	Bipolar TURP, lasers (HoLEP, PVP); no TUR syndrome
- Plasma is about 290 mOsm/kg, so monopolar fluids are **hypo-osmolar**.		

TUR (transurethral resection) syndrome

- **Mechanism**: prostatic venous pressure is ~10 mmHg; irrigant at higher pressure enters open sinuses → volume overload, **dilutional hyponatraemia** and hypo-osmolality → cerebral oedema; plus glycine toxicity.
- **Features**: confusion, nausea, vomiting, visual disturbance, hypertension then hypotension, bradycardia, seizures, coma.
- **Risk factors**: gland >45 mL, resection >90 minutes (VA (Veterans Affairs) study), high irrigation pressure, open sinuses or capsular perforation.
- **Prevention**: bag height ~60 cm (each extra 10 cm more than doubles absorption), continuous-flow resectoscope, time limit, spinal anaesthesia (awake patient), bipolar TURP. Ethanol 1% tagging can monitor absorption.
- **Treatment**: stop surgery, check sodium; Na <120 mEq/L serious. Furosemide for overload; **3% saline** 100 mL boluses (max 300 mL) for neurological signs; raise sodium by 4–6 mEq/L initially, not over 12 mEq/L in 24 h (risk of central pontine myelinolysis).

Bipolar advantage

- Meta-analyses: no TUR syndrome, less transfusion and clot retention; NNT (number needed to treat) about 50 to prevent one TUR syndrome.

Diagram to draw: monopolar vs bipolar current pathways.

Key points to remember:

- Monopolar: non-ionic hypo-osmolar fluid; bipolar: saline.
- Glycine 1.5%: ~200 mOsm/kg; transient blindness.
- Risk: gland >45 mL, time >90 min; bag height 60 cm.
- 3% saline; ≤12 mEq/L rise in 24 h.

*See the paper answer **TUR syndrome** (Section 16, Ch 151).*

Also, from the condensed book (not in the paper answer):

- **Ureteroscopy always uses normal saline**, so that no hypotonic fluid is absorbed. Saline's only hazard is volume overload, without hyponatraemia.
- Serum sodium is usually **<125 mmol/L** in transurethral resection (TUR) syndrome. This chapter advises a correction of no more than 8–10 mmol/L in 24 h (the paper answer gives ≤12).

4 · Cystourethroscopy

SAQ Indications for outpatient cystourethroscopy.

Paper answer · P1

Type: SAQ (10 marks)
Source: 2024-Winter Paper 1, Q3e

Introduction

- **Flexible cystourethroscopy** under local anaesthesia is the **commonest office procedure** in urology; no lithotomy needed.
- **Prerequisites:** consent; treat any UTI (urinary tract infection) first; lidocaine gel with **≥5 minutes dwell** (ideally 10–20); no antibiotic prophylaxis unless risk factors (age, diabetes, immunosuppression, catheters, prior UTI).

Diagnostic indications

Indication	Comment
Haematuria	All **gross haematuria** (with CT (computed tomography) urography, cytology). Microhaematuria by AUA/SUFU (American Urological Association/Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction) risk: intermediate and high risk; shared decision in low risk
Bladder cancer surveillance	NMIBC (non-muscle-invasive bladder cancer): first check at 3 months; low risk then yearly; high risk 3–4-monthly for 2 years; also upper-tract follow-up of UTUC (upper tract urothelial carcinoma)
Recurrent UTI	Look for stone, diverticulum, stricture, fistula, foreign body
LUTS (lower urinary tract symptoms)	Not routine; before invasive BPH (benign prostatic hyperplasia) treatment (median lobe before urethral lift, bladder-neck contracture, stricture)
Pain syndromes	IC/BPS (interstitial cystitis/bladder pain syndrome): **Hunner lesion**; chronic pelvic pain with bladder pain or low flow
Women	Suspected urethral diverticulum, vesicovaginal fistula, mesh or suture erosion after sling, recurrent UTI after surgery

Indication	Comment
Others	Urethral stricture assessment and follow-up; abnormal urine cytology; radiation cystitis; foreign body

Therapeutic indications

- **Removal or exchange of ureteric stents** (exchange with fluoroscopy).
- **Biopsy and fulguration** of small (≤1 cm) low-grade recurrences; office laser ablation.
- **Intradetrusor botulinum toxin** injection (OAB (overactive bladder), neurogenic detrusor overactivity).
- Urethral bulking agents; stricture dilatation; small stones or foreign bodies; blue-light cystoscopy where available.

When not indicated or done in theatre

- Not needed for the index stress incontinence patient, initial OAB or uncomplicated LUTS.
- **No surveillance cystoscopy** in chronic indwelling catheters (AUA/SUFU 2021).
- Theatre if TURBT (transurethral resection of bladder tumour) needed, poor tolerance, children, or complex diversions.

Diagram to draw: haematuria risk-group flowchart.

> **Key points to remember:**
> - Haematuria and NMIBC surveillance are the two commonest indications.
> - Flexible scope, lidocaine dwell ≥5 minutes, no routine prophylaxis.
> - Therapeutic: stent removal, botulinum toxin, fulguration of small recurrences.

*See the paper answer **Flexible endourological instruments, digital and current ureteroscopes** (Section 1, Ch 17).*

Rigid instruments

- French (Charrière) gauge: **3 Fr = 1 mm** diameter. The diagnostic rigid sheath is 15–17 Fr. Choose the smallest scope that does the job.
- A rigid set has a lens, bridge, sheath and obturator. Use a 0° or 12° lens for the urethra, 25° or 30° for general work, and 70° or 120° for the dome, anterior wall and bladder neck.
- The **Albarran bridge** has a lever that deflects catheters and wires into an orifice behind a high bladder neck, a large median lobe or a reimplanted ureter.

Antibiotic prophylaxis: AUA versus EAU

Prophylaxis is driven by patient risk, not by the procedure.

	American Urological Association (AUA)	European Association of Urology (EAU)
Diagnostic cystoscopy	None unless patient risk factors are present	
TURP	Yes (<24 h)	Yes
TURBT	Yes	**Only if sepsis risk is high**
Ureteroscopy	**All patients**	According to local pathogens

	American Urological Association (AUA)	European Association of Urology (EAU)
Agents	Cefazolin or trimethoprim-sulfamethoxazole (ureteroscopy: TMP-SMX or a 1st/2nd-generation cephalosporin)	Local antibiogram
Added risk factors	Smoking, steroids, distant infection, prolonged stay, anatomical anomaly	Diabetes, weight extremes, prior urinary infection, previous instrumentation

Both share older age, malnutrition, immunodeficiency and catheters or colonised material. Treat any urinary tract infection before instrumentation.

Technique and special situations

- Prepare the genitalia with **aqueous povidone-iodine** at any age, because chlorhexidine and alcohol damage mucosa.
- **Orthotopic neobladder**: do not dilate or incise an anastomotic contracture without obstruction, as this risks incontinence. **Cutaneous pouch**: use a flexible scope through the channel for diagnosis, and treat percutaneously after imaging.

5 · Upper-tract endoscopy: indications

P2 *ESWL: mechanism, indications, factors affecting outcome — AUA/EAU stone size thresholds by site, unfavourable stone and anatomical factors*

*See the paper answer **Flexible endourological instruments, digital and current ureteroscopes** (Section 1, Ch 17).*

Also, from the condensed book (not in the paper answer):

- The EAU prefers ureteroscopy (URS) for ureteric stones **>10 mm**.
- Diagnostic URS for upper tract urothelial carcinoma does not worsen survival but **raises intravesical recurrence**, so survey the bladder.
- Endoureterotomy also fails more with an allograft, an irradiated or ureteroenteric stricture, or a poorly functioning kidney. Balloon dilation alone is about 50% durable at 1 year.

6 · Ureteroscopes

*See the paper answer **Flexible endourological instruments, digital and current ureteroscopes** (Section 1, Ch 17).*

*See the paper answer **Flexible ureteroscopy / RIRS and accessories** (Section 11, Ch 97).*

Also, from the condensed book (not in the paper answer):

- **Passive (secondary) deflection** off the upper pelvic wall lengthens the effective tip, but it **fails in gross hydronephrosis**, where there is no wall to push against.

7 · Accessories

*See the paper answer **Foley's (urethral) catheters, guidewires and DJ stents with their complications** (Section 1, Ch 15).*

LAQ LASERs in urology: holmium, thulium, KTP and newer laser technologies

Paper answer · P4 · merged from 4

Asked as: P4 LAQ "LASERs in urology: applications and newer generation laser technologies (new LASER for stone treatment)" · P4 SAQ "Holmium laser" · P4 SAQ "Thalium laser in urology" · P4 SAQ "KTP*** laser in urology"

*"Thalium" is taken to mean thulium (spelling error). "KTP***" is taken to mean the KTP (potassium titanyl phosphate) "GreenLight" laser; the asterisks are ignored.*

Principle

- LASER (light amplification by stimulated emission of radiation): a "pumped" gain medium releases identical photons. A mirror resonator amplifies them, and a partial mirror lets the beam out.
- Light is **monochromatic, coherent and collimated**, so it passes down thin fibres.
- **Chromophore** decides the effect. Water absorbs holmium and thulium light. Haemoglobin absorbs KTP light.
- Absorption sets **penetration**. Shallow penetration vaporises and cuts. Deep penetration coagulates and later sloughs.
- Coagulation occurs at about 60 °C; vaporisation at ≥100 °C.
- **Pulse duration** decides the stone mechanism. Short pulses (<10 µs) act photomechanically (shock waves). Long pulses (>250 µs: Ho:YAG [holmium:yttrium-aluminium-garnet], TFL [thulium fibre laser]) act **photothermally**.

Lasers used

Laser	Wavelength (nm)	Chromophore	Penetration	Main uses
Ho:YAG	2120 (Ch 19 quotes 2100; often quoted as 2100 nm)	Water	0.3–0.4 mm (other sources: 0.4 mm; 0.31 mm)	Stones, HoLEP, incisions, tumours
TFL	1940	Water (peak)	<0.1 mm (other sources: about 0.2 mm; 0.08 mm; chapter values 0.08–0.2 mm)	Dusting, enucleation
Tm:YAG	~2010 (other sources: 2013)	Water	0.2 mm	Vaporesection, enucleation
KTP/LBO	532	Haemoglobin	0.8 mm	PVP
Diode	940–1470	Water + haemoglobin	0.5–5 mm	Prostate vaporisation
Nd:YAG	1064	Neither strongly	Up to 10 mm	Coagulation of tumours
CO_2	10,600	Water	~0.1 mm	Skin and penile lesions

Holmium:YAG laser

Physics

- Solid-state, flashlamp-pumped, **pulsed** laser at **2120 nm** (often quoted as 2100 nm).

- Chromophore **water**; penetration about **0.4 mm**. It cuts and vaporises; heat spreads 2–3 mm and seals small vessels.
- Flexible quartz fibres; works in saline.

Mechanism on stones

- Long pulses (>250 µs) act **photothermally**, not by shock waves: long pulses heat the stone until it melts or decomposes.
- **Moses effect:** the start of each pulse makes a vapour channel; the rest of the pulse crosses it to the stone. So keep the fibre close and perpendicular.
- Breaks **all stone types**, including cystine and calcium oxalate monohydrate. It is the gold standard for stones and enucleation.

Settings

Mode	Settings	Effect
Fragmenting	0.6–1.0 J × 6–10 Hz	Pieces for basketing; more retropulsion
Dusting	0.2–0.3 J × 40–80 Hz	Fine dust; "popcorning" in a calyx
HoLEP enucleation	2.0 J × 40–50 Hz	Cutting along the capsule

- **Retropulsion** rises with pulse energy (not frequency) and falls with longer pulses or pulse modulation.
- Heat tracks power: ≤10 W in the ureter; good outflow.

Uses

- **Stones:** URS (ureteroscopy), RIRS (retrograde intrarenal surgery), PCNL (percutaneous nephrolithotomy), bladder stones.
- **BPH (benign prostatic hyperplasia):** HoLEP (holmium laser enucleation of the prostate).
- **Incisions:** strictures, bladder neck contracture, endopyelotomy, ureterocele, calyceal diverticulum neck.
- **Tumours:** UTUC (upper tract urothelial carcinoma) ablation; en bloc bladder tumour resection.
- Drawback: steep HoLEP learning curve.

Thulium lasers

Types and physics

	Tm:YAG (thulium:yttrium-aluminium-garnet)	TFL (thulium fibre laser)
Wavelength	2013 nm (P4 LAQ: ~2010 nm)	**1940 nm** (water absorption peak)
Source	Diode-pumped solid crystal	Diode-pumped thulium-doped silica fibre
Output	Continuous wave (now also pulsed)	Pulsed or continuous
Water absorption	High	About 4× holmium

- Optical penetration is the shallowest of urological lasers: about 0.2 mm (chapter values for TFL: 0.08–0.2 mm), against 0.31–0.4 mm for Ho:YAG.
- Shallow absorption gives precise vaporisation and clean incision, but **carbonisation**.

Thulium:YAG (2013 nm)

- Originally continuous wave (0.2 mm penetration, carbonisation a drawback): ThuVARP (vaporesection), ThuVAP (vaporisation), ThuVEP (vapoenucleation), **ThuLEP** (laser enucleation, 2010; mostly blunt dissection; size-independent like HoLEP).
- **Pulsed Tm:YAG:** newer pulsed solid-state version of the continuous-wave thulium laser for photothermal lithotripsy and enucleation.

TFL (1940 nm) for stones

- **Diode-pumped silica fibre** laser near the **water absorption peak** (about **4× holmium absorption**).
- **Low peak power, long pulses, very high frequency** (up to about 2000 Hz [P4 LAQ: above 2000 Hz]; holmium about 120 Hz); pulse energy as low as 0.025 J. This gives photothermal fine "dusting" and **minimal retropulsion**.
- **Thin fibres** (50–150 µm; P4 LAQ: down to about 150 µm) give better irrigation and scope deflection. Small, quiet, compact unit; no dedicated high-voltage outlet.
- Settings: start 0.6 J × 6 Hz; dusting 0.2 J × 40 Hz; avoid >40 Hz (pulses miss the stone); settings ≤40 Hz are often enough. "Paint, do not drill."
- **RCTs (randomised controlled trials):** TFL beat **low-power (30 W) holmium** (stone-free rate, time); it **equalled 120 W pulse-modulated holmium** in stone-free rate, with less retropulsion.
- Concerns: efficacy on all stone types; thermal injury at high power.

Thulium for prostate

- Techniques: ThuVARP, ThuVAP, ThuVEP, ThuLEP, ThuFLEP (thulium fibre laser enucleation).
- Settings: 70–90 W enucleation, 30 W coagulation; morcellation as for HoLEP.
- Outcomes: ThuLEP equals HoLEP; size-independent; safe on anticoagulants (AUA [American Urological Association]). Retrograde ejaculation about four-fifths.

Other thulium uses: en bloc bladder tumour resection; UTUC ablation; incision of strictures.

KTP/LBO (GreenLight) laser

Physics

- A 1064 nm Nd:YAG (neodymium:yttrium-aluminium-garnet) beam passes through a **KTP or LBO (lithium triborate) crystal**. This doubles frequency and halves wavelength to **532 nm** (green).
- Green light is absorbed by **haemoglobin**, not water. Optical penetration 0.8 mm; tissue effect 1–3 mm.
- Effect: **photoselective vaporisation** with a thin coagulation rim, so vaporisation is bloodless.

Platforms

Generation	Power
KTP	80 W
LBO HPS (High Performance System)	120 W
LBO XPS	**180 W**; 750 µm MoXy fibre (metal tip, fluid cooling)

- Side-firing fibre emits at 70°; 23–24 Fr cystoscope; saline irrigation.

PVP (photoselective vaporisation of prostate) technique

- Identify ureteric orifices; open the bladder neck first.
- Sweep 15–30° in 1–2 seconds at ≤2 mm working distance. Large bubbles mean efficient vaporisation.
- Pulsatile bleeding: coagulation mode (about 30 W). Energy >5 kJ/cc predicts complete vaporisation.

Indications

- AUA: prostates <30 cc and 30–80 cc; men at **higher bleeding risk** (anticoagulation continued). Not >80 cc (high retreatment).

GOLIATH trial

- RCT, 29 European centres, about 270 men, prostate ≤100 cc: **180 W PVP vs TURP** (transurethral resection of prostate).
- PVP **non-inferior** for IPSS (International Prostate Symptom Score), Qmax and complications at 6 months, 1 and 2 years.
- PVP: shorter catheter time and stay, fewer early reinterventions. TURP: faster. Retrograde ejaculation (about two-thirds) and 2-year reoperation equal.

Complications

- Dysuria and storage symptoms (highest of transurethral techniques); UTI (urinary tract infection) and epididymitis from coagulum; capsular perforation 0.2–1%; retrograde ejaculation.
- No TUR (transurethral resection) syndrome; transfusion very rare.
- **No tissue** for histology.

Other KTP uses: laparoscopic non-contact cutting (largely supplanted); KTP enucleation possible.

Applications by area

1. Stones

- **Ho:YAG** is the gold standard and breaks all stone types (see above). **TFL:** higher water absorption and low peak power give finer dust and less retropulsion.
- **Heat** depends on power, not laser type. Keep ≤10 W in the ureter; use good flow and an access sheath.

2. BPH

- **HoLEP:** endoscopic equivalent of open prostatectomy, **size-independent**. Enucleate at about 2 J × 40–50 Hz, then morcellate. The only endoscopic procedure superior to TURP. Less bleeding, shorter catheter time; safe on anticoagulants.
- **PVP (KTP/LBO 120–180 W):** haemoglobin absorption gives bloodless vaporisation, so anticoagulants can continue. **GOLIATH:** non-inferior to TURP. Avoid >80 cc (retreatment).
- **Thulium:** vaporisation, vaporesection, ThuLEP (equivalent to HoLEP). Diode vaporisation also used.
- Nd:YAG coagulation was abandoned (1 cm penetration, sloughing).
- AUA: HoLEP, ThuLEP or PVP for **high bleeding risk**.

3. Incisions

- Laser urethrotomy has no advantage over cold knife.

- Holmium incision of bladder neck contracture, ureteric stricture, endopyelotomy, ureterocele and calyceal diverticulum neck.

4. Urothelial tumours

- **NMIBC (non-muscle-invasive bladder cancer):** laser **en bloc resection** (Ho, Tm, KTP) gives detrusor in the specimen and no obturator jerk. Office ablation of small low-grade recurrences.
- **UTUC:** biopsy, then ablate by ureteroscopy:
- Ho:YAG: <0.5 mm, haemostatic.
- Nd:YAG: 5–6 mm coagulation, so stricture risk.
- Thulium: <0.2 mm.
- **PDT (photodynamic therapy):** a photosensitiser plus laser light makes singlet oxygen.

5. Penile and genital lesions

- PeIN (penile intraepithelial neoplasia) and small Ta/T1: **Nd:YAG preferred**, with base biopsies.
- CO_2 is too superficial for T1 (recurrence up to 50%). CO_2 or Nd:YAG for genital warts.

Newer generation laser technologies (new lasers for stone treatment)

- **Aims:** less retropulsion, finer dust, faster work.
- **1. High-power holmium (60–120 W; other sources: 100–120 W machines) with pulse modulation:** high frequency allows **dusting** (0.2–0.3 J × 40–80 Hz) and "popcorning"; long-pulse modes reduce retropulsion.
- **2. Pulse modulation (Moses technology and similar):** the first part of a pulse boils a **vapour channel**; vapour absorbs less light, so the rest reaches the stone. Moses technology splits each pulse in two: the first opens the channel, the second strikes the stone. Less retropulsion; may shorten HoLEP.
- **3. TFL (1940 nm):** see above; ThuFLEP for prostate.
- **4. Pulsed Tm:YAG (2013 nm):** see above; it also does lithotripsy.
- **Others:** thin fibres; suction sheaths; AI (artificial intelligence) settings; 180 W GreenLight KTP/LBO (532 nm) vaporisation; diode lasers; burst wave lithotripsy (investigational).

Holmium versus TFL

	Ho:YAG	TFL
Wavelength	2120 nm	1940 nm
Water absorption	High	About 4× higher
Penetration	0.31 mm	0.08 mm
Peak power	High	Low
Retropulsion	More	Minimal
Fragments	Small	Smaller
Fibre	≥200 µm	≥150 µm

Principles of settings

- **Retropulsion rises with pulse energy**, not frequency; it falls with longer pulses.
- Keep the fibre close and perpendicular to the stone.
- TFL settings ≤40 Hz often enough.

Safety

- Wavelength-specific eyewear for all, including the patient. Door signage, covered windows, trained staff and a laser safety officer.
- Check fibres; never fire inside the scope. Guard against fire and thermal (urothelial) injury.
- **Heat follows total power, not laser type:** ≤10 W in the ureter, ≤25 W in the kidney; use low power, pauses, high flow, cool irrigation and an **access sheath**; urothelial damage begins above about 43 °C.

Diagrams to draw

- Water and haemoglobin absorption curves marking 532, 1064, 1940, 2013 and 2120 nm.
- Vapour bubble (Moses channel) between fibre tip and stone.
- Side-firing fibre at 70° vaporising lateral lobe.

> **Key points to remember:**
>
> - Chromophore sets penetration: TFL shallowest, Ho 0.4 mm, Nd:YAG deepest.
> - Holmium: 2120 nm; water chromophore; pulsed; photothermal; Moses effect: fibre close and perpendicular.
> - Holmium settings: fragment 0.6–1.0 J × 6–10 Hz; dust 0.2–0.3 J × 40–80 Hz.
> - HoLEP size-independent, superior to TURP.
> - TFL 1940 nm, Tm:YAG 2013 nm; water absorption about 4× holmium.
> - TFL: low peak power, fine dust, thin fibres, minimal retropulsion; settings ≤40 Hz often enough.
> - TFL beats low-power, equals high-power pulse-modulated holmium.
> - ThuLEP is size-independent and equal to HoLEP.
> - KTP 532 nm: frequency-doubled Nd:YAG; haemoglobin chromophore.
> - GOLIATH: 180 W PVP non-inferior to TURP; best for anticoagulated men; not >80 cc.
> - Nd:YAG for PeIN; eyewear for everyone.
> - Heat depends on power: ≤10 W in the ureter, ≤25 W in the kidney; access sheath; damage above 43 °C.

LAQ Energy sources in urology and laparoscopic surgery, including bipolar energy

Paper answer · P2 · P3 · P4 · merged from 3

Asked as: P3 LAQ "Therapeutic energy sources in urology (including laparoscopic energy devices)" · P4 LAQ "Basic energy modalities in urological surgery, including bipolar energy" · P2 SAQ "Energy sources in laparoscopic urology"

Introduction and principles

- Therapeutic energy is absorbed by tissue or stone and released as **heat, mechanical stress, cavitation or a chemical reaction**. Wavelength or frequency sets penetration; absorption sets the effect.
- All haemostatic devices achieve haemostasis (seal) by **denaturing protein**, by heat (current, light) or mechanical vibration.
- Heat effects: about **60 °C** coagulates protein; **100 °C** vaporises cell water; higher temperatures carbonise.

Classification

Group	Modalities
Electrical	Monopolar and bipolar electrosurgery, vessel sealers, RFA (radiofrequency ablation), IRE (irreversible electroporation), EMDA (electromotive drug administration)
Electromagnetic	Lasers, MWA (microwave ablation), chemohyperthermia, PDT (photodynamic therapy), ionising radiation
Acoustic	Ultrasonic dissector, SWL (shock wave lithotripsy), intracorporeal ultrasonic and pneumatic lithotripsy, HIFU (high-intensity focused ultrasound), histotripsy, burst wave lithotripsy, Li-ESWT (low-intensity shock wave therapy)
Direct contact	Cryoablation, water vapour (Rezum), waterjet (Aquablation)

1. Electrosurgery: physics

- Alternating current at about **500 kHz** (radiofrequency). Mains current (50 Hz) depolarises nerve and muscle; at 500 kHz ions only oscillate and produce heat, too fast to stimulate nerve or muscle.
- **Ohm's law** ($V = I \times R$): tissue is the resistance. **Current density** decides the effect: a small tip concentrates current and heats (burns); a large pad disperses it.
- **Monopolar:** generator, active electrode, patient, dispersive pad, back to generator. Partial pad contact causes burns.
- **Bipolar:** current passes only between two electrodes (jaws) on the same instrument; less power, less spread, no dispersive pad (the patient is not in the circuit).
- **Electrocautery** (heated wire) differs: no current enters the patient.

Waveform	Character	Effect
Cut (50–80 W)	Continuous, low voltage	Cell water flash-boils; rapid boiling, clean incision, little char
Coagulation (30–50 W)	Interrupted (about 6% duty cycle), high voltage	Desiccation, deeper spread (up to 3 mm)
Fulguration (spray)	Higher voltage, arcs (sparks) across air	Non-contact surface haemostasis; surface charring of oozing beds
Blend	Cut with interruptions	Cutting plus haemostasis

Bipolar energy in detail

- Radiofrequency current (about 500 kHz). Less power and less lateral thermal spread (1–6 mm vs 5–7 mm monopolar). In endoscopy, **saline** irrigation can be used.

	Monopolar	Bipolar
Circuit	Loop to patient to pad	Active and return poles on the instrument
Irrigant	Glycine 1.5%, sorbitol–mannitol	Normal saline
TUR (transurethral resection) syndrome	Yes	Theoretically eliminated
Capacitive coupling, pad burns	Yes	No

- **Endoscopic uses:**

- **Bipolar TURP (transurethral resection of prostate):**
  - *Plasma system:* energy vaporises saline around the loop into plasma (orange glow of sodium); cuts at lower temperature while sealing vessels; platinum–iridium loop.
  - *TURis (transurethral resection in saline):* the sheath acts as return electrode; chapter notes report strictures and fistulae.
  - *True bipolar:* dual loops.
  - Outcomes: equal IPSS (International Prostate Symptom Score) and Qmax; less transfusion and clot retention; shorter catheter time and stay. NNT (number needed to treat) about 50 to prevent one TUR syndrome.
- **Bipolar TUVP (transurethral vaporisation):** PlasmaButton; AUA (American Urological Association) option for <30 and 30–80 cc.
- **Bipolar enucleation:** plasmakinetic energy can perform AEEP (anatomic endoscopic enucleation of prostate) satisfactorily; comparative data limited.
- **TURBT (transurethral resection of bladder tumour):** fewer obturator jerks and less bleeding in reviews; one RCT (randomised controlled trial) showed only less thermal artefact.
- **Laparoscopic and open uses:** bipolar forceps for pinpoint haemostasis; vessel sealers; prostatic pedicles, cystectomy.
- **Bipolar safety:** safer near pacemakers; touching tines complete the circuit without tissue. Obturator reflex reduced, not abolished.

Electrosurgery in endoscopic urology: present status

- **TURP:** still the reference BPH (benign prostatic hyperplasia) operation. Monopolar TURP needs non-conducting **1.5% glycine**, risking hyponatraemia (TUR syndrome); **bipolar** in saline matches monopolar efficacy with better safety and avoids TUR syndrome.
- **TURBT:** standard; obturator reflex risk on lateral wall. En bloc resection is increasing.
- **Obturator reflex:** lateral-wall TURBT stimulates the nerve, causing adduction and perforation. Prevent with paralysis, obturator block, partial filling, short bursts, bipolar.
- **EMDA–MMC (mitomycin C):** iontophoresis raises drug uptake in NMIBC (non-muscle-invasive bladder cancer); evidence low quality, not standard.

Laparoscopic energy devices

Device	Principle	Lateral spread	Notes
Monopolar (hook, scissors)	Tip to patient to return pad	**5–7 mm**	Versatile; stray-current risk
Bipolar	Current only between jaws	1–6 mm	No pad; no capacitive coupling
Vessel sealer (LigaSure, EnSeal, Caiman)	Bipolar current + jaw compression fuses collagen and elastin; impedance feedback stops output; built-in blade	1–3 mm	Seals vessels **≤7 mm** in one application only
Ultrasonic (Harmonic)	Blade vibrating at **25–55 kHz** (P4: about 55.5 kHz); friction and cavitation denature protein; no current through the patient	**≈1 mm**	Safe with pacemakers; jaws **>200 °C**, slow to cool, so do not touch bowel

Device	Principle	Lateral spread	Notes
Argon beam	Monopolar current through argon, non-contact	Superficial	Liver, spleen, raw surfaces; **gas embolism**

- Vessel sealers are routine in laparoscopic and robotic surgery.
- **Also know:** lasers (CO_2, KTP [potassium titanyl phosphate], holmium) now rarely used laparoscopically; laparoscopic RFA (target 60 °C) and cryoablation (below −40 °C, two freeze–thaw cycles) for small renal masses; robotic HotShears, bipolar Maryland, Vessel Sealer.

Electrosurgical injury and safety

Mechanism	Prevention
Direct (inadvertent) activation out of view	Activate only with tip in view
Direct coupling: active electrode touches another metal instrument	Whole metal tip visible
Capacitive coupling: current induced through intact insulation (metal cannula in plastic retainer)	Never mix metal and plastic ports (avoid hybrid trocars)
Insulation failure	Inspect insulation; **AEM (active electrode monitoring)**
Thermal spread to ureter, nerves or bowel	Bipolar near vital structures
Return pad burns; surgical fire (with oxygen)	Pad on muscle, away from implants; **REM (return electrode monitoring)** detects poor pad contact

- **AEM** detects insulation failure and capacitive coupling.
- Unseen bowel injuries present late (days) as perforation or peritonitis.
- Use lowest effective power (cut 50–80 W, coagulation 30–50 W), tip in view; bipolar near vital structures and in pacemaker patients.

2. Ultrasonic energy

- **Ultrasonic dissector:** see the device table; little lateral spread, but the blade stays hot.
- **Intracorporeal ultrasonic lithotripter:** vibrating probe with suction of fragments.

3. Lasers

- **Principle:** pumped gain medium in a mirrored cavity gives stimulated emission and a **monochromatic, coherent, collimated** beam.
- **Effect** depends on wavelength (chromophore: water, haemoglobin) and pulse duration.
- **Mechanisms:** photothermal (long pulses), photomechanical (short pulses, shock waves), photochemical (PDT).
- **Penetration predicts behaviour:** deep (Nd:YAG) coagulates widely with delayed sloughing; shallow (thulium) vaporises precisely with char.

Laser	nm	Penetration	Use and present status
KTP/LBO (potassium-titanyl-phosphate/lithium triborate; GreenLight)	532	0.8 mm	Prostate photovaporisation (haemoglobin); can continue anticoagulation

Laser	nm	Penetration	Use and present status
Nd:YAG (neodymium:yttrium-aluminium-garnet)	1064	10 mm	Deep coagulation (penile CIS [carcinoma in situ], warts); largely obsolete for BPH
TFL (thulium fibre laser)	1940	0.08 mm	Stones: beats low-power holmium, equals high-power; less retropulsion; enucleation (ThuLEP, thulium laser enucleation of prostate)
Tm:YAG (thulium:YAG)	2000	0.2 mm	Vaporesection, enucleation
Ho:YAG (holmium:YAG)	2120	0.31 mm	Gold standard lithotripsy; **HoLEP** (holmium laser enucleation of prostate) size-independent; strictures, UTUC (upper tract urothelial carcinoma) ablation

- Safety: eye protection for all; irrigant heating; fibre damage.

4. Acoustic energy and lithotripsy

- **SWL:** electrohydraulic, electromagnetic or piezoelectric sources. Shock waves break brittle stones by **spallation, shear and cavitation**. Slower rates (about 1 Hz) work better and injure less. **Status:** first line for renal stones ≤2 cm and favourable upper ureteric stones.
- **Intracorporeal:** pneumatic (ballistic), ultrasonic, combined, electrohydraulic, laser; pneumatic, ultrasonic and combined probes in PCNL (percutaneous nephrolithotomy); electrohydraulic now rare (ureteric injury).
- **HIFU:** focused beam (1500–10,000 W/cm^2) is absorbed and heats a rice-grain focus to necrosis; transrectal prostate. **Status:** focal or salvage prostate cancer (best for posterior tumours); investigational in guidelines.
- **Histotripsy** (cavitation liquefaction), **burst wave lithotripsy, ultrasonic propulsion:** investigational.
- **Li-ESWT:** for vasculogenic ED (erectile dysfunction) and Peyronie pain; evidence still limited.

5. Ablative energies (thermal ablation of tumours)

Modality	Mechanism	Key points and present status
Cryoablation	Argon expansion (Joule–Thomson); **below −40 °C, two freeze–thaw cycles**; visible ice ball	Small renal mass ≤3–4 cm in frail patients; focal or salvage prostate
RFA	375–900 kHz current; ionic agitation; target 60 °C; char at 150 °C	Small renal mass; oncological results similar to cryo; heat sink near vessels
MWA	915 MHz or 2.45 GHz; dielectric hysteresis	Renal mass; hotter, larger zones, less heat sink
HIFU	Focused ultrasound absorbed, causing necrosis	See acoustic energy
IRE	High-voltage microsecond DC (direct current) pulses open permanent membrane pores; non-thermal	Spares vessels and ducts; focal prostate; good continence and potency; trials ongoing

- **Cryo mechanism:** extracellular ice, then cell dehydration, then intracellular ice, then membrane rupture on thawing; ischaemia and apoptosis follow.
- **IRE mechanism:** electric pulses open pores in cell membranes that do not reseal, so cells die by apoptosis without heat; collagen scaffolds survive. It needs general anaesthesia with muscle paralysis and ECG (electrocardiogram)-synchronised pulses.

- **Guidelines:** AUA 2021: consider thermal ablation for **cT1a ≤3 cm**; biopsy before or at ablation. EAU (European Association of Urology): not routinely >3 cm (cryo >4 cm).

6. Light and chemical energy

- **PDT:** photosensitiser plus light gives singlet oxygen. **VTP (vascular-targeted photodynamic therapy, padeliporfin)** for low-risk prostate cancer: RCT (Azzouzi 2017) halved progression versus AS (active surveillance; 28% vs 58%); not widely adopted. **Photodynamic diagnosis** (blue light) aids TURBT.
- **Chemohyperthermia:** MMC with bladder wall heated to **42 °C** for NMIBC; used after BCG (bacillus Calmette-Guérin) failure or shortage; not standard. EMDA for mitomycin (see above).

7. Prostate (BPH): newer energies

- **Rezum:** steam (103 °C) releases latent heat by convection; no char, zone-confined; preserves ejaculation.
- **Aquablation:** robotic non-thermal saline waterjet; good for large glands; haemostasis needed.
- **Also know:** TUMT (transurethral microwave thermotherapy) and TUNA (transurethral needle ablation) largely abandoned.

8. Ionising radiation

- EBRT (external beam radiotherapy) and **brachytherapy** (LDR [low dose rate] or HDR [high dose rate]): standard for localised prostate cancer; trimodality bladder preservation; penile cancer (brachytherapy for small T1–T2 glans tumours).
- **Radioligand therapy** (^{177}Lu-PSMA [prostate-specific membrane antigen], VISION trial) and **radium-223** (ALSYMPCA) for mCRPC (metastatic castration-resistant prostate cancer).

Diagrams to draw

- Monopolar vs bipolar circuit (monopolar loop-to-pad beside bipolar loop with local return), with cut and coagulation waveforms.
- Laser penetration depths compared, with cryoprobe ice ball.

> **Key points to remember:**
>
> - Electrosurgery at about 500 kHz; current density decides heating.
> - Cut continuous low voltage; coagulation interrupted high voltage.
> - Spread: monopolar 5–7 mm > bipolar 1–6 > sealer (LigaSure) 1–3 > ultrasonic ≈1 mm.
> - LigaSure seals ≤7 mm; Harmonic vibrates at 25–55 kHz; ultrasonic blade hottest after use.
> - Stray current: insulation failure, direct and capacitive coupling; avoid hybrid trocars; use AEM and REM.
> - Bipolar: no pad, saline irrigation, no TUR syndrome; bipolar TURP equals monopolar efficacy with fewer complications.
> - Holmium 2120 nm: gold standard for stones and HoLEP.
> - Cryo below −40 °C, two cycles; RFA 60 °C; MWA less heat sink; thermal ablation for cT1a ≤3 cm.
> - HIFU, IRE, VTP: focal prostate therapy, still evolving.

Also, from the condensed book (not in the paper answer):

- The need for dilation falls with scope size (**<1%** for a 7.4 Fr scope). A **zero-tip balloon** dilates right up to an impacted stone.
- **Electrohydraulic lithotripsy is not first line in the ureter** (AUA), because its poorly focused shock damages the ureter and scope. A rigid pneumatic probe needs a rigid scope.
- The **three-prong grasper is the safest retrieval device**, because it releases an oversized stone. Never basket an irretrievably large stone through a single-channel flexible scope.

8 · Ureteroscopy technique

*See the paper answer **Flexible ureteroscopy / RIRS and accessories** (Section 11, Ch 97).*

ALGORITHM

1. Perform cystoscopy and place a 0.038-inch PTFE safety wire.
2. For an impacted stone, pass an angled hydrophilic wire (with or without a torque catheter), then exchange it for a PTFE wire.
3. If obstruction with infection is suspected, aspirate the pelvis. If the aspirate is turbid, place a stent and postpone.
4. Drain the bladder and proceed: semirigid scope below the iliac vessels, flexible above.

- If the orifice will not admit the scope, the safest course is to **stent and return in 1–2 weeks**. **Never dilate a segment that contains a stone**, because it may perforate.
- Formal dilation is needed in about 8–25%. Fragment an impacted stone to dust of **≤2 mm**, or basket the fragments.
- Under-rotation is the commonest novice error, and over-rotation damages fibreoptic bundles. Use a 200 µm fibre when deflection is needed.
- The **T-tilt** (15° Trendelenburg plus 15° airplane away) helps at the lower pole. Give non-steroidal anti-inflammatory drugs, with or without an alpha-blocker, after surgery.

9 · Nephroscopy and complications

- The rigid nephroscope (about 24–26 Fr) passes through a 26–30 Fr Amplatz sheath, and a flexible nephroscope reaches calyces out of line.

*See the paper answer **Flexible ureteroscopy / RIRS and accessories** (Section 11, Ch 97).*

Also, from the condensed book (not in the paper answer):

- For mucosal injury or perforation, stop and stent for 2–4 weeks. Avulsion follows dragging an oversized basketed stone, and needs open or robotic reconstruction.

18 Fundamentals of Laparoscopic and Robotic Urologic Surgery

1 · Patient selection and contraindications

Absolute contraindications	Relative contraindications (caution, specific consent)
Uncorrectable coagulopathy; intestinal obstruction (unless being treated); abdominal wall infection; massive haemoperitoneum or haemoretroperitoneum; generalised peritonitis; suspected malignant ascites	Morbid obesity; extensive previous surgery; pelvic fibrosis; organomegaly; benign ascites; pregnancy; hernia; iliac or aortic aneurysm; surgeon inexperience

KEY POINT

Every relative contraindication is a problem of safe entry or of tolerating pneumoperitoneum. The answers recur: enter elsewhere (Palmer point, retroperitoneum), enter openly (Hasson), or insufflate less.

- Severe chronic obstructive pulmonary disease (COPD) needs arterial blood gases and lung function tests. Treat arrhythmias first, because hypercarbia and acidosis destabilise the myocardium.
- **Pregnancy**: enter cephalad to the fundus (Palmer point), use **8–12 mm Hg**, and operate in the **second trimester**; it is harder after 20 weeks.
- A diaphragmatic hernia lets carbon dioxide (CO_2) reach the mediastinum. Never insufflate through an abdominal-wall hernia.
- In morbid obesity, renal and adrenal laparoscopy still beats open surgery. Leave prostatectomy and cystectomy with body mass index (BMI) >30 until later in the learning curve.

2 · Preoperative preparation and set-up

*See the paper answer **Operating room set-up** (Section 1, Ch 9).*

*See the paper answer **Patient positioning in lap urology** (Section 1, Ch 11).*

Also, from the condensed book (not in the paper answer):

- **No mechanical bowel preparation** is needed unless bowel is used, and it shows no benefit even for cystectomy.
- Enhanced recovery after surgery (ERAS) allows solids until 6 h and a carbohydrate drink 2 h before surgery. It shortens stay by about **30%** and halves complications.
- Group and screen for nephrectomy, nephroureterectomy and prostatectomy. Crossmatch for partial nephrectomy, cystectomy and caval thrombectomy early in experience.

3 · Physiology of pneumoperitoneum

LAQ Pneumoperitoneum physiology and complications of laparoscopy and retroperitoneoscopy

Paper answer · P1 · P4 · merged from 2

Asked as: P1 LAQ "Pneumoperitoneum physiology, complications of retroperitoneoscopy and laparoscopic donor nephrectomy" · P4 SAQ "General complications related to laparoscopy."

"d/t" is read as "due to" and "lap" as laparoscopy.

Introduction to pneumoperitoneum

- **Pneumoperitoneum** is insufflation of gas, usually CO_2 (carbon dioxide), into the peritoneal cavity at 10–15 mmHg to create a working space.
- Its effects have **two drivers**: **pressure** (compression of the cava, renal veins, kidneys and diaphragm) and **absorbed CO_2** (hypercarbia, acidosis, sympathetic drive). Patient position and volume status modify both.
- **CO_2** is standard: non-combustible, cheap, highly blood-soluble, so embolism is less lethal. Helium or argon in severe lung disease; N_2O (nitrous oxide) supports combustion.

Choice of insufflant

Gas	Advantages	Disadvantages
CO_2 (standard)	Non-combustible; highly blood-soluble, so embolism is less lethal; cheap	Hypercarbia, acidosis, arrhythmia; COPD (chronic obstructive pulmonary disease) patients cannot clear it
Helium, argon	Inert; no hypercarbia	Poorly soluble, so gas embolism risk; costly. Helium used in lung disease or to salvage hypercarbia
N_2O (nitrous oxide)	Less peritoneal irritation	Supports combustion; no electrosurgery
Room air, oxygen	—	Embolism, combustion; abandoned

Effects by pressure (supine, Campbell Table 18.1)

Effect	5 mmHg	10 mmHg	20 mmHg	40 mmHg
Heart rate (HR)	↑	↑	↑	↓
MAP (mean arterial pressure), SVR (systemic vascular resistance)	↑	↑	↑	↑
Cardiac output	↔/↓	↔/↑	↔/↓	↓
Venous return, cardiac output (Summer 2019 source)	—	Variable	Variable	↓ sharply
GFR (glomerular filtration rate), urine output	↔	↓	↓↓	↓↓
Arterial PCO_2	↔	↑	↑	↑
Arterial pH	↔	↔/↓ (another source: ↔)	↓	↓

- Renal function is the first casualty (from 10 mmHg). Afterload rises at every pressure. Output collapses at 40 mmHg. Acidosis builds from 20 mmHg.

Cardiovascular (haemodynamic) changes

- **Afterload rises** at every pressure: aortic and splanchnic compression plus catecholamine and **vasopressin** release raise MAP and SVR.

- **Preload depends on volume status.** In normovolaemia or hypovolaemia, pressure up to 20 mmHg compresses the IVC (inferior vena cava) and venous return falls. In hypervolaemia the cava resists and venous return may rise. At ≥40 mmHg venous return falls sharply in everyone and output collapses.
- **Arrhythmias:** hypercapnia causes tachycardia and ventricular ectopics; peritoneal stretch causes **vagal bradycardia** or asystole. A new arrhythmia may herald pneumothorax, hypoxia or gas embolism.
- **CVP (central venous pressure) reads falsely high** because intrathoracic pressure rises; do not guide fluids by it.
- Leg venous stasis: theoretical DVT (deep vein thrombosis) risk.

Respiratory changes

- **Diaphragm splinted:** FRC (functional residual capacity) and compliance fall, peak airway pressure rises; atelectasis and V/Q (ventilation–perfusion) mismatch. Dead space is unchanged. PEEP (positive end-expiratory pressure) helps in lung disease.
- **Head-down** (steep Trendelenburg) adds a further fall in vital capacity, endobronchial tube slip and facial oedema.
- **Barotrauma** (pressure >15 mmHg for long): pneumothorax, pneumomediastinum.

Acid–base changes (CO_2 absorption)

- CO_2 is absorbed through the peritoneum and, extraperitoneally, through fat and disrupted lymphatics. Absorption rises over 30–60 minutes, then plateaus.
- **Hypercarbia and respiratory acidosis** with rising end-tidal CO_2. Managed by raising minute ventilation and lowering pressure.
- **COPD** patients cannot blow it off: arterial gases if insufflation exceeds 1 hour, continued in recovery; consider helium. **Hypercarbia peaks after extubation**, as stored CO_2 is released.

Renal changes

- **Oliguria** comes from **renal vein and parenchymal compression**, not ureteric compression or hormones. Cortical flow falls; medullary flow rises up to 20 mmHg, then falls. It reverses on desufflation.
- Keep pressure ≤10 mmHg (another source: ≤10–12 mmHg); avoid fluid overload. If needed give furosemide, mannitol 12.5–25 g or dopamine 2 µg/kg/min to keep urine >100 mL/hour, rather than fluid boluses (overload, oedema). Important in live donor nephrectomy.

Other systems

- **Splanchnic:** hepatic and mesenteric flow fall (rare mesenteric thrombosis); less ileus than open surgery; no rise in reflux.
- **ICP (intracranial pressure) and IOP (intraocular pressure)** rise with hypercarbia and head-down tilt: caution in head injury, shunts and glaucoma.
- **Hormonal:** catecholamines, cortisol and **vasopressin** rise; renin–angiotensin activation adds to afterload.
- **Stress and immune response:** cortisol, catecholamines and prolactin rise as in any surgery, but hyperglycaemia, insulin resistance, catabolism and IL-6 (interleukin-6) rise less than after open surgery, which may explain faster recovery; oncological benefit unproven.

- **Also know:** hypothermia from cold dry gas; shoulder-tip pain from diaphragmatic CO_2 irritation.

Effects of position and approach

Position	Heart rate	MAP	Cardiac output	Other effects
Head-up	↑	↓	↓	Output ↓, hypotension
Head-down (Trendelenburg)	↓	↑	↑ (but lungs worse)	Venous return and output ↑; lung volumes ↓, ICP and IOP ↑
Lateral flank, kidney rest	—	—	—	Lung V/Q mismatch; rhabdomyolysis risk with kidney rest

- **Summary:** head-down raises MAP and output but worsens lungs; head-up lowers output.
- **Retroperitoneal** insufflation causes smaller swings in output and venous pressures than transperitoneal; pelvic extraperitoneal insufflation raised arterial CO_2 more.

Safe practice

- Preoperative cardiopulmonary assessment; correct arrhythmias; optimise volume.
- **Practical rule: work at 10–12 mmHg** (12 mmHg keeps stroke volume unchanged); 20 mmHg only briefly (first trocar, venous bleeding); never exceed 20 mmHg for long.
- Monitor end-tidal CO_2, saturation, airway pressure and urine output.
- **Recent advances:** low-pressure (8 mmHg) surgery with valveless insufflators (AirSeal); helium (or argon) for lung disease (poorly soluble, so embolism risk).

General complications of laparoscopy

Incidence

- Historically about 4%, mortality 0.03–0.08%. Contemporary series 13–22% because procedures are complex. **Vascular, then adjacent-organ injuries are commonest.**
- Risk factors: complex procedures, operating time >4 h, creatinine >1.5 mg/dL, learning curve.

Complications by phase

Phase	Complication	Key points
Access (Veress, first trocar)	Preperitoneal insufflation; vessel or bowel puncture; **major vascular injury (0.11–2%)**	Confirm Veress tests; open access with adhesions. Major vessel: leave trocar in, laparotomy
Insufflation	**Gas embolism**; subcutaneous emphysema; pneumothorax (1.6–4%); pneumomediastinum, pneumopericardium; hypercarbia	Embolism: see below
Secondary trocars	Epigastric bleeding; crossing swords, rollover	Transilluminate; bladeless trocars; insert under vision
Dissection	Thermal or mechanical bowel injury; vascular injury; bladder or ureter injury	Four electrosurgical mechanisms below
Positioning	Nerve injury (about 2.8%: brachial plexus, femoral, sciatic); **rhabdomyolysis** (about 1% retroperitoneal)	Padding, kidney rest <1 h, cases <5 h
Exit	Bowel entrapment, late port bleeding	Inspect each site at 5 mmHg

Phase	Complication	Key points
Postoperative	Port-site hernia (0.3–5.4%), wound infection, shoulder-tip pain, DVT	Close fascia of large ports

Physiological complications: hypercarbia and acidosis; oliguria from renal vein compression; reduced venous return; raised intracranial pressure in head-down tilt (see above).

Gas embolism

- Gas enters an opened vein (renal vein, IVC or lumbar vein). Sudden collapse; end-tidal CO_2 rises abruptly, saturation falls, then end-tidal CO_2 falls; "mill-wheel" murmur.
- Treatment: stop insufflation, **left lateral, head down**, 100% oxygen, aspirate through a central line.

Electrosurgical (thermal) bowel injury: four mechanisms

1. **Inappropriate (direct) activation** out of view.
2. **Direct coupling** with another metal instrument.
3. **Capacitive coupling** (metal cannula in a plastic retainer).
4. **Insulation failure.** - Monopolar injury presents **late (up to 18 days)** with low-grade fever and pain; resect with margin.

Missed bowel injury (Bishoff)

- Not classic peritonitis: **pain at one trocar site, leucopenia, low fever, distension, diarrhoea** at 24–48 h.
- **CT (computed tomography) with oral contrast**; free gas is normal for up to 7–9 days.

Complications of retroperitoneoscopy

Introduction

- **Retroperitoneoscopy** is laparoscopic or robotic surgery in the retroperitoneum without entering the peritoneum. It gives direct hilar access and avoids bowel, but the space must be **created**.
- **Why its complications differ:** small space with no natural cavity; space made by finger and **balloon dissection** (commercial or Gaur glove balloon) behind Gerota's fascia and in front of the psoas; fewer landmarks; dissection close to the lens; gas tracks along tissue planes; CO_2 is absorbed through fat and disrupted lymphatics, so hypercarbia appears early if the port leaks.
- Minimally invasive renal surgery: about 15% complications, mostly postoperative. Learning curve about 20 cases.

Classification (by phase of surgery)

Phase	Complications
Access and space creation	Wrong plane, balloon rupture, peritoneal tear, muscle bleeding
Insufflation	Subcutaneous emphysema, hypercarbia, pneumothorax, gas embolism
Dissection	Vascular, visceral, nerve, thermal injury
Exit and recovery	Port bleeding, hernia, rhabdomyolysis, urinoma, lymphocele

1. Access and balloon-related

- **Balloon in muscle** (wrong plane, intramuscular) gives a postoperative **flank hernia** or bulge.

- **Too anterior** gives peritoneal breach or colon injury; **too posterior** gives bleeding from quadratus lumborum or psoas.
- **Balloon rupture:** latex glove balloon breaks into many fragments (silicone leaves one) that must be retrieved.
- **Peritoneal tear:** CO_2 enters the peritoneum and collapses the small working space. Fix by venting with a Veress needle or an extra port, enlarging the tear, or converting to transperitoneal.
- Disorientation: keep the **psoas horizontal** as the landmark.

2. Gas-related

- **Subcutaneous emphysema** (commonest; commoner than transperitoneal): crepitus over flank, chest and neck; pneumoscrotum. Cause: leak around the port. Prevent with a **balloon-tip (blunt-tip) trocar**, tight incisions and 12 rather than 15 mmHg.
- **Hypercarbia**, especially with emphysema; a sealed port removes the difference from transperitoneal surgery.
- **Pneumothorax (1.6–4%, commoner retroperitoneally):** pleura is near the upper pole and 12th rib. Signs: neck and chest emphysema; tension gives hypotension, absent breath sounds and high airway pressure. Treat with needle decompression (16-G needle in the second or third intercostal space, midclavicular line), then chest drain. Pneumomediastinum and pneumopericardium (about 0.8%) also occur.
- **Gas embolism** (renal vein injury, IVC or lumbar vein): managed as above.

3. Vascular injury

- Commonest complication of minimally invasive urology; major injury <1 in 1000 (P4: 0.11–2% at access) but fatal in 8–17%. Upper-tract work carries about five times the pelvic vascular risk.
- Sites: renal artery and vein, **lumbar and gonadal veins**, adrenal vein, IVC, aorta. In the small posterior space the IVC or aorta can be mistaken for hilar vessels.
- **Management of venous bleeding:** raise pressure temporarily to 25 mmHg, suction to see the site, compress or tamponade with a sponge, haemostatic sealant, extra 5-mm port for suturing or a Satinsky clamp, hand-assist, add ports, suture or clip, or convert if uncontrolled. Arterial: clip or staple; aortic injury needs conversion. Never clip blindly.

4. Visceral injury

- **Peritoneum and colon**; duodenum (right); spleen, pancreas (left); liver; pleura; ureter; adrenal; usually by thermal spread.
- Thermal bowel injury follows the four mechanisms above; it may present up to 18 days later; diagnose by CT with oral contrast.

5. Positioning and nerve injury

- **Rhabdomyolysis** (about 1% retroperitoneal; 0.4% overall): men, BMI (body mass index) ≥25, kidney rest, full flexion, long cases. Brown urine, hip pain, CK (creatine kinase) >5,000 units. Prevent: limit kidney rest to <1 hour, cases <5 hours, avoid hypotension.
- **Nerves** (neuropraxia): brachial plexus (axillary roll), sciatic; in the field, **genitofemoral and lateral femoral cutaneous** on the psoas, ilioinguinal and iliohypogastric (thigh numbness, flank bulge).

6. Postoperative

- Bleeding or haematoma; port-site hernia or haematoma; wound infection.

- **Urinoma** (partial nephrectomy, pyeloplasty); **lymphocele** or chylous leak after hilar or node dissection.
- Delayed bowel injury; DVT.
- **Conversion** to transperitoneal or open surgery for bleeding, peritoneal tear or lost orientation.

Retroperitoneal versus transperitoneal

Retroperitoneal more	Retroperitoneal less
Emphysema, pneumothorax, hypercarbia, disorientation	Bowel injury, ileus, adhesions; peritoneum spared

Prevention

- Open (Hasson) access below the 12th rib tip; balloon wholly in the retroperitoneum; balloon-tip port; 12 mmHg; trocars under vision; respect the 12th rib and pleura; check haemostasis at low pressure.

Laparoscopic donor nephrectomy

Introduction

- First described by **Ratner (1995)**; now the preferred approach (open under 2% in the United States). **Donor safety comes first** at every stage.

Donor evaluation `Questionnaire → physical and psychosocial review, blood group, HLA (human leukocyte antigen) typing → GFR by at least two methods, one measured → metabolic, serology, cancer screening → CT angiography for vessels → cardiac review if needed`

- Exclusions: kidney disease, malignancy, active infection, cardiovascular disease, diabetes, significant obesity; bilateral stones generally exclude.

Side selection

- **Left preferred:** longer renal vein, simpler dissection. **Right** if the left has multiple arteries or complex veins, or the right kidney is smaller: **the donor keeps the better kidney.**
- **Anatomy matters:** arterial variation in 25–40%; early (prehilar) branching and accessory arteries must be seen on CT, as they may contraindicate the laparoscopic route.

Techniques

Technique	Features
Pure transperitoneal	3–4 ports; colon reflected; Pfannenstiel extraction
Hand-assisted (about two-thirds)	Hand port doubles as extraction site; tactile feedback, quick haemostasis
Retroperitoneoscopic	No bowel mobilisation; direct hilum; less room
Robotic; LESS (laparoendoscopic single-site surgery)	Wristed instruments; single incision, less pain

Operative steps (left)

1. Reflect colon; divide **gonadal, adrenal and lumbar veins**.
2. Preserve periureteric tissue and the **gonadal–ureter triangle** (ureteric blood supply).
3. Keep pressure ≤10 mmHg; hydration and mannitol or furosemide to keep urine >100 mL/hour (counters pneumoperitoneum oliguria).

4. Divide ureter, then **artery first, then vein**. Vein stapled lateral to the adrenal vein; Endo-TA (non-cutting) stapler gives extra vein length.
5. Extract quickly; cold flush. Warm ischaemia a few minutes. - **Hem-o-lok:** manufacturer contraindicates it on the donor renal artery after stump bleeds. - **Open donor nephrectomy** is kept for complex anatomy or previous surgery, more often on the right.

Advantages versus open

- Less pain, shorter stay, faster return to work, better cosmesis; **more people volunteer to donate**; equal graft function.

Disadvantages and complications

- Longer operation, learning curve, shorter vessels (right vein), slightly longer warm ischaemia.
- Complications about **16%** (mostly gastrointestinal); Clavien ≥IV 2.5%; **mortality 3 per 10,000**; conversion 0.26–3.34%, mainly for vascular injury.

Long-term donor counselling

- Function 60–80% of pre-donation; lifetime ESRD (end-stage renal disease) **90 per 10,000** versus 14 in matched non-donors (Muzaale, 2014); more gestational hypertension.
- **Also know:** kidney paired donation for ABO-incompatible or crossmatch-positive pairs; living-donor grafts survive longer than deceased-donor grafts.

Diagrams to draw

- Abdomen under pressure with arrows to diaphragm (lungs), IVC (preload), aorta (afterload) and renal vein (oliguria), plus CO_2 absorption to blood.
- Abdominal wall showing capacitive coupling through a metal cannula in a plastic anchor.
- Flank cross-section showing balloon between psoas and Gerota's fascia, with peritoneum pushed medially and port positions for retroperitoneoscopy.
- Left renal hilum with gonadal, adrenal and lumbar veins, stapled artery and vein, and the gonadal–ureter triangle.

Key points to remember:

- Two drivers: pressure and absorbed CO_2; work at 10–12 mmHg.
- Afterload rises at all pressures; output collapses at 40 mmHg.
- Oliguria from renal vein compression; reversible.
- Hypercarbia peaks after extubation; COPD needs arterial gases.
- Vascular injury is commonest; leave the trocar in place.
- Gas embolism: left lateral, head down, oxygen, aspirate.
- Missed bowel injury: trocar-site pain, leucopenia.
- Retroperitoneoscopy: space is created, not natural: wrong plane gives hernia, peritoneal tear or bleeding.
- Emphysema commonest; balloon-tip port and 12 mmHg prevent it.
- Pneumothorax 1.6–4%, commoner retroperitoneally.
- Rhabdomyolysis about 1%: limit kidney rest to <1 hour.
- Ratner 1995; left kidney preferred; hand-assisted in about two-thirds.
- Mortality 3 per 10,000; conversion 0.26–3.34%.
- Donor ESRD 90 versus 14 per 10,000 (matched non-donors).

Also, from the condensed book (not in the paper answer):

- Helium or argon can be switched to **to salvage intraoperative hypercarbia**.
- The commonest working pressure is **15 mm Hg**. A pressure of ≤10 mm Hg markedly reduces oliguria.

4 · Access

LAQ Laparoscopic access: Veress, Hasson, trocars, gases; open vs laparoscopic vs robotic prostatectomy

Paper answer · P1 · P4 · merged from 2

Asked as: P1 LAQ "Laparoscopic access (Veress, Hasson cannula, extraperitoneal space), gases, and open vs laparoscopic vs robotic radical prostatectomy" · P4 SAQ "Different types of trocars in laparoscopy."

Access in laparoscopy

A. Transperitoneal: closed (Veress)

- Veress needle: 14-gauge, spring-loaded blunt stylet. It is an entry needle, not a trocar. Check flow and tip recoil first. Two "pops": fascia, then peritoneum.
- **Sites:** umbilicus (thinnest wall; aim towards pelvis in non-obese); lower quadrants; **Palmer point** (left subcostal, midclavicular line) with suspected adhesions; avoid scars.
- **Confirmation tests:** aspiration–injection–aspiration of saline; **drop test**; advancement test (1 cm free). Then insufflate: opening pressure <8–10 mmHg, set limit 15 mmHg.
- First trocar inserted blind, or under vision with an **optical trocar** (0° lens inside a clear tip).

B. Open (Hasson)

- 2-cm infra-umbilical incision; fascia and peritoneum opened under vision; stay sutures; blunt cannula.
- Safer with adhesions or previous surgery and early in training; more gas leak.

C. Retroperitoneal

- 2-cm incision below the tip of the 12th rib; finger dissection behind Gerota's fascia, in front of psoas, then **balloon dilatation** (commercial or **Gaur** glove-on-catheter balloon), then a balloon-tip port to prevent gas leak and emphysema.

D. Extraperitoneal (pelvic): infra-umbilical, behind rectus into the Retzius space with a balloon (used for extraperitoneal prostatectomy).

E. Others: hand-assisted (6.5–7.5 cm port); single-port/LESS (laparoendoscopic single-site); direct trocar entry without pneumoperitoneum.

- **Access complications:** preperitoneal insufflation, vessel or bowel puncture, gas embolism, subcutaneous emphysema.

Hasson cannula (Hassan's canula)

Assumed to mean the Hasson cannula, used for open laparoscopic access.

Introduction

- Described by **Harrith Hasson (1971)** for open ("direct vision") laparoscopy. The first port is placed under vision instead of a blind Veress needle and trocar.

- **Parts:** outer sleeve with a cone-shaped (olive) adapter, **blunt obturator**, struts to anchor fascial stay sutures, and a gas inlet with tap.

Indications

- **Previous abdominal surgery or expected adhesions** (main indication).
- Organomegaly, pregnancy, aortic aneurysm, children, thin patients.
- Retroperitoneoscopy and extraperitoneal access (open start, then balloon dilatation).

Technique (transperitoneal)

1. 2-cm semicircular infra-umbilical incision.
2. Open fascia and then peritoneum separately, under vision, finger-sized.
3. Confirm entry by sight and finger sweep (free adhesions).
4. Place two 0 silk fascial stay sutures.
5. Insert the cannula with its **blunt obturator**.
6. Seat the cone in the defect; wrap stay sutures round the struts to seal.
7. Remove obturator; insufflate at high flow (to 12–15 mmHg).
8. At closure, tie the stay sutures to close the fascia. - **Retroperitoneal:** 2–2.5 cm incision below the 12th rib tip, finger dissection onto psoas, balloon dilatation, then port.

Advantages over Veress

- No blind puncture: fewer major vessel and bowel injuries; no preperitoneal insufflation; no gas embolism from intravascular insufflation.
- Fast insufflation; easy fascial closure (less port-site hernia).

Disadvantages

- Larger incision; **gas leak** around the port; slightly longer in obese patients.
- Early retroperitoneal use caused subcutaneous emphysema and hypercarbia, now solved by balloon-cuffed blunt ports.

Complications

- Bowel injury if adherent under the wound (repair through the same incision); wound infection; port-site hernia; vascular injury is still possible.

Evidence

- Reviews show no single entry method is clearly safest for major injury; open entry has fewer failed entries.

Instrumentation for development of the extraperitoneal space

- Extraperitoneal laparoscopy works outside the peritoneum: the **flank retroperitoneum** (kidney, adrenal, ureter) or the **pelvic prevesical (Retzius) space** (extraperitoneal prostatectomy, lymphadenectomy).
- There is no natural cavity, so a working space must be **dissected before insufflation**: finger first, then balloon.

Instrument	Description and use
Open (Hasson) access set	Scalpel, S-retractors, artery forceps
Index finger	Blunt first pocket; feels psoas behind, Gerota's fascia in front

Instrument	Description and use
Commercial balloon dissector	Transparent silicone balloon on a stiff hollow shaft; inflated with room air by bulb (maximum 800 mL, 40 pumps). Oblong for the retroperitoneum, round for the pelvis. Laparoscope inside confirms position
Gaur balloon (1992)	Size 7 glove (or glove fingers, condom) tied with 0 silk on an 8-Fr red rubber catheter; filled with saline (600–800 mL). Cheap, but no rigid shaft or view; latex can rupture into fragments
Optical (visual) obturator trocar with 0° scope	Enters under vision; yellow retroperitoneal fat confirms the plane; scope sweeps the space
Blunt-tip balloon trocar	Internal balloon plus external foam cuff gives an airtight seal; prevents gas leak, emphysema and hypercarbia

Technique: retroperitoneal

1. Full flank, table flexed; 1.5–2.5-cm incision below the 12th rib tip.
2. Split muscles; open the lumbodorsal fascia; finger dissection between psoas and Gerota's fascia.
3. Balloon inflated cephalad then caudad, in front of psoas, behind Gerota's fascia.
4. Blunt-tip balloon trocar; insufflate; secondary ports under vision.

Technique: pelvic extraperitoneal

- Infraumbilical incision; open anterior rectus sheath, separate recti; finger tunnels behind the rectus to the pubis; pierce transversalis fascia; balloon midline, then each side.

Pitfalls

- Balloon in muscle causes a flank hernia. Too anterior causes peritoneal tear or colon injury; too posterior causes psoas or quadratus bleeding.
- Peritoneal tear shrinks the space; convert if needed. Extraperitoneal CO_2 needs more ventilation.

Trocars

Definition

- A **trocar** is a **cannula (port) plus an obturator**; 3–20 mm diameter, 5–20 cm long (Chapter 18).
- Parts: obturator tip, cannula, **valve** (keeps gas in), insufflation side-arm, retention mechanism.

Classification

Basis	Types
Obturator tip	Bladed (cutting) with spring safety shield; **bladeless (blunt dilating)**; **optical (visual)**; blunt (Hasson)
Material	Reusable metal; disposable plastic
Valve	Flap or trumpet valve (needs reducer); **multiseal** (5–12 mm, no reducer); **valveless (AirSeal)**
Retention	Smooth (suture-anchored); threaded sleeve; expandable arms; **balloon (Blunt Tip)**
Special	Hasson open cannula; radially expanding; hand ports; single-site multichannel ports; robotic 8-mm metal ports

Main types explained

- **Bladed trocars:** cut through tissue; more epigastric injury and port-site hernia. Now of historical interest.
- **Bladeless (blunt dilating) trocars:** spread muscle fibres; have replaced bladed ones. About **fivefold less inferior epigastric injury**; trocar-site bleeding 3% vs 9%. Switching cut port hernia from 1.83% to 0.19%. **Fascial closure only for midline ports >10 mm** or stretched sites (all ≥5 mm in children).
- **Optical trocars:** clear tip with a 0° scope inside; each layer is seen during entry.
- **Hasson cannula:** blunt obturator and cone, fixed with fascial stay sutures after open entry; safer with adhesions; leaks more gas.
- **Radially expanding trocar:** a sleeve over the Veress needle is dilated to 12 mm; less port-site bleeding and pain (Chapter 39).
- **Balloon (Blunt Tip) trocar:** internal balloon and external foam cuff seal the wall; prevents leak and **subcutaneous emphysema** in retroperitoneoscopy.
- **AirSeal valveless port:** stable pneumoperitoneum at 8–12 mmHg despite suction.
- **Single-site ports:** TriPort; da Vinci SP (single-port) 25–27-mm guide. **Hand ports:** GelPort.

Safety points

- **Never use a metal trocar with a plastic retaining ring**: capacitive coupling burns viscera out of view.
- Insert under vision; ports 8–10 cm apart.

Gases used

Gas	Advantages	Disadvantages
CO_2 (carbon dioxide; standard)	Non-combustible, cheap, **highly blood-soluble** (embolism less lethal), quickly absorbed	Absorbed, causing hypercarbia, acidosis, arrhythmia; dangerous in COPD (chronic obstructive pulmonary disease)
Nitrous oxide	Less peritoneal pain, fewer acid–base effects	**Supports combustion**: no electrosurgery
Helium	Inert, no hypercarbia; use in lung disease	Poorly soluble, so embolism risk; costly
Argon	Inert	Embolism risk
Air, oxygen	–	Embolism, combustion: abandoned

- **Physiology:** keep pressure 10–15 mmHg (12 mmHg more physiological). Pressure reduces venous return, raises afterload, causes **oliguria** (renal vein and parenchymal compression) and splints the diaphragm.
- CO_2 absorption plateaus at 30–60 minutes; higher with extraperitoneal insufflation. Hypercarbia peaks after extubation.
- **Gas embolism:** sudden collapse, millwheel murmur, end-tidal CO_2 rise then fall. Treat: stop insufflation, **left lateral, head-down**, 100% oxygen, aspirate via central line.

Comparison of open (RRP), laparoscopic (LRP) and robotic (RALP) radical prostatectomy

- **Background:** anatomical nerve-sparing open RRP (Walsh, 1982) was the standard. First LRP (Schuessler, 1997) was too slow; French centres standardised it in 2000, but laparoscopic

suturing of the anastomosis remained a barrier. The robot (da Vinci, 2000) removed that barrier, and RALP is now dominant in the USA.

Parameter	Open RRP	LRP	RALP
Vision	Direct, loupes	2D, unstable	**3D, 10× magnified**, stable
Operating time	Shortest	Longest early	Longer than open; falls with experience
Instruments	Hands, haptic feedback	Rigid, fulcrum effect	**Wristed** (7 degrees of freedom); no haptics
Learning curve	Moderate	**Steepest** (suturing)	Shorter
Blood loss	300–1000 mL	Less (pneumoperitoneum tamponade)	Least; **transfusion 1–2%**
Hospital stay, catheter	Longer; catheter ~2 weeks	Shorter	Usually 1 day; catheter ~1 week
Pain, return to work	More	Less	Less; normal activity 3–4 weeks
Positive margins	Similar	Similar	Similar (~15%; ~9% if pT2)
Continence (1 year)	~93% (expert)	Similar	Similar; possibly earlier
Potency	Surgeon-dependent	Similar	Similar
Bladder neck stricture	Up to 16% (older series)	Lower	**<2%** (mucosa-to-mucosa)
Unique risks	Wound infection, hernia	Conversion, CO_2 effects	Steep Trendelenburg (eye, nerve), robot failure (<0.5%)
Cost	Lowest	Moderate	**Highest**

Evidence

- **Yaxley 2016 (randomised, one open vs one robotic surgeon):** equal continence, potency and early margins at 3 months; less blood loss and shorter stay with robot.
- **Coughlin 2018:** no difference in urinary or sexual function at 24 months.
- **LAP-01 (robotic vs laparoscopic):** earlier continence with robot.
- **Concept:** the **surgeon, not the platform**, decides outcome; margins depend mainly on tumour (Gleason ≥7, stage) and surgeon volume. The robot's advantages are recovery and wider reach (obese patients, prior surgery).
- **Also know:** transperitoneal vs extraperitoneal approach (no consistent difference); Retzius-sparing RALP (earlier continence, more margins); single-port (SP) robot; new robotic platforms; telesurgery.

Diagrams to draw

- Veress needle sites (umbilicus, Palmer point) and trocar layout for RALP.
- Hasson cannula with cone, stay sutures and blunt obturator in the abdominal wall.
- Flank section with balloon between psoas and Gerota's fascia.
- Trocar with obturator tip types (bladed, bladeless, optical) and valve.

Key points to remember:

- Veress tests: aspiration, drop, advancement; opening pressure <10 mmHg.
- Hasson 1971: open entry under vision; main indication previous surgery or adhesions; avoids blind Veress injuries; drawback is gas leak.
- Extraperitoneal space is created first: finger, then balloon. Gaur balloon: glove on red rubber catheter. Balloon in muscle causes hernia.
- Trocar = cannula + obturator. Bladeless: fewer epigastric injuries and hernias. Balloon trocar seals and prevents retroperitoneal emphysema.
- CO_2 standard (soluble, non-flammable); helium in COPD; N_2O supports combustion.
- Gas embolism: left lateral head-down, oxygen, aspirate.
- Yaxley 2016: equal function and margins; robot less blood loss and stay.

Also, from the condensed book (not in the paper answer):

- The Veress needle should give flow with ≤2 mm Hg at 2 L/min. **Do not lift the fascia**, because this widens the gap between fascia and peritoneum rather than the abdominal space.
- At the umbilicus, angle towards the pelvis in thin patients and go perpendicular in obese patients, as the umbilicus migrates caudally.
- A **hand port placed first** is the safest entry for the inexperienced, as the intra-abdominal hand guides the secondary trocars.

	Transperitoneal	**Retroperitoneal**
Space	Large, familiar anatomy	Small; psoas is the key landmark; direct (artery-first) hilar access
Physiology	Larger haemodynamic swings	More emphysema and pneumothorax; CO_2 absorption similar with a sealed port
Best for	Large tumours, reconstruction	Previous transperitoneal surgery; posterior tumours

Vascular injury in upper-tract surgery is about **5×** the pelvic rate.

5 · Trocars and port placement

*See the paper answer **Laparoscopic access: Veress, Hasson, trocars, gases; open vs laparoscopic vs robotic prostatectomy** (Section 1, Ch 18).*

LAQ Laparoendoscopic single-site (single-port) surgery: LESS

Paper answer · P4

Asked as:

- Laproendoscopic single site surgery: LESS — *Winter 2023 Paper IV Q1 (LAQ)*
- Single port lap. Surgeries in urology — *Practice Set 2 Paper IV Q3c (SAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)
(also answered as a long answer in 2023-Winter Paper 4, Q1)

Read as laparoendoscopic single-site surgery (LESS), i.e. single-port (single-site) laparoscopic and robotic surgery in urology. NOTES (natural orifice transluminal endoscopic surgery) is

covered in 2022-Summer Paper 4, Q1.

Definition and concept

- **LESS:** laparoscopic or robotic surgery in which all instruments and the camera enter (pass) through **one skin incision**, usually hidden in the umbilicus (umbilical) or below the waistline.
- Aim: further reduce access trauma, giving **better cosmesis, less pain and shorter stay**. Chapter 142 places it on a spectrum: multiport → hand-assisted → LESS → SP (single-port) robot → NOTES. Each step to fewer incisions buys cosmesis at the price of working space and triangulation.
- First urological single-port nephrectomies were reported in 2007; the first single-port robotic urological surgery followed in 2009.

Nomenclature (LESSCAR)

- Early names were many: SILS (single-incision laparoscopic surgery), SPA (single-port access), OPUS (one-port umbilical surgery), TUES (transumbilical endoscopic surgery), e-NOTES (embryonic NOTES).
- **LESSCAR (Laparoendoscopic Single-Site Surgery Consortium for Assessment and Research, 2008)** agreed the single umbrella term **LESS** and set standards for reporting, training and research.
- The term was agreed by LESSCAR (LESS Consortium for Assessment and Research, 2008).
- Sub-types (variants): transumbilical or extraumbilical; **pure LESS**; **"LESS + 1"** or needlescopic-assisted (an extra 2–5-mm port); **R-LESS** (robotic LESS); purpose-built **SP (single-port) robot**.

Access: ports and devices

Option	Description
Several low-profile trocars	Placed close together through one fascial incision (the future extraction incision)
Multichannel ports	TriPort 15 (two 5-mm + one 15-mm); TriPort+ (three 5-mm + one 10-mm); SILS port; GelPOINT (gel cap, trocars placed anywhere on it)
da Vinci SP entry guide	One 25–27-mm cannula carrying three articulating (wristed) 6-mm instruments + one flexible articulating camera

- Position: modified flank (transperitoneal) or full flank (retroperitoneal SP, 2.5–3-cm incision near the 12th rib tip, balloon dilatation).

The triangulation problem

- In standard laparoscopy ports are spread so instruments meet the target at an angle (the "diamond"). In LESS all shafts are **parallel and in-line** with the camera (instruments lie parallel to the camera).
- Results: **instrument clashing**, **"crossing swords"**, poor depth perception, poor retraction and hard suturing.
- Solutions:
- **Articulating instruments** (tip bends inside the abdomen) and **pre-bent or curved instruments**: they cross at the fascia, so the right hand works on the left side (**chiral manipulation**).

- **Flexible-tip or 5-mm low-profile scopes**, offset optics, extra-long instruments and instruments of **different lengths** to avoid handle clash.
- A 2-mm needlescopic instrument or extra 5-mm port (an extra needlescopic port) without hesitation.
- **Robotic platforms** restore wristed movement.

Robotic single-site surgery

Platform	Features
da Vinci Single-Site (Si/Xi)	Curved crossing cannulas; semi-rigid **non-wristed** instruments; software swaps hand control
da Vinci SP (FDA [Food and Drug Administration] approval 2018)	Purpose-built; one arm, no external collisions; instruments with elbow and wrist; 3D articulating camera; on-screen navigator shows instrument positions

- SP needs **10–25 cm between (from) cannula and (to) target** to deploy the joints; a GelPort lets instruments "float" outside the skin. Assistance through an **"SP + 1" port** or ROSI (remote-operated suction-irrigation). Limited by cost and availability.

Indications

- **Kidney:** simple and **donor nephrectomy**, radical nephrectomy, partial nephrectomy, **pyeloplasty**, nephroureterectomy, cyst decortication, cryoablation.
- **Adrenalectomy**; varicocelectomy; orchidopexy and pyeloplasty in children (paediatric).
- **Pelvis:** SP radical prostatectomy (transperitoneal, extraperitoneal); **SP transvesical simple prostatectomy** and bladder diverticulectomy.
- **Best case:** thin patient (patients), small or benign lesion (lesions), experienced surgeon.
- **Relative contraindications (avoid in):** obesity, large or hilar tumours, complex vascular anatomy, dense adhesions, unfit for pneumoperitoneum.

Outcomes versus conventional laparoscopy (Chapter 142)

- **Randomised trial (RCT, randomised controlled trial):** same operative time, blood loss, complications and stay; **less analgesia**.
- **Donor nephrectomy meta-analysis:** same stay, recovery, transfusion and conversion (otherwise equal); **lower pain scores**.
- **Renal mass:** same perioperative outcome and survival.
- **SP robotic partial nephrectomy vs multiport:** shorter stay, less opioid; same margins and complications; **longer warm ischaemia** in one review.
- An **extra port is often needed**; conversion is mostly to laparoscopy, rarely open.
- **Net effect:** gains are confined (limited) to **pain, cosmesis and stay**; oncological and functional outcomes are unchanged, but the operation is technically harder and longer.

Complications

- All laparoscopic complications (usual laparoscopic injuries: access injury, vascular and bowel injury, CO_2 effects).
- **Incisional and umbilical hernia** is higher (larger fascial defect): about 5% with da Vinci Xi single-site, lower with SP.
- Wound infection; conversion; longer operation; surgeon ergonomic strain. Prevent by careful fascial closure and case selection.

Recent advances

- **da Vinci SP** with same-day discharge pathways; **SP retroperitoneal** renal surgery.
- **Transvesical SP** prostatectomy; hidden incision endoscopic surgery (HIdES) in children.
- New single-port platforms; magnetic anchoring and guidance systems / instruments (restore triangulation).

Diagram to draw: umbilical multichannel port with articulating (crossed) instruments crossing at the fascia (chiral) versus a triangulated multiport diamond (layout).

> **Key points to remember:**
> - LESS = one incision; LESSCAR (2008) standardised the term.
> - Core problem: lost triangulation; solved by articulating or curved instruments and robots.
> - Same outcomes as laparoscopy with less pain and better cosmesis; more hernias.
> - da Vinci SP (FDA 2018): one 25–27-mm cannula, wristed instruments.

Also, from the condensed book (not in the paper answer):

- **Triangulation**: place ports in a diamond around the target, with instruments at **<55°** to the horizontal and 25–45° between them. Place ports in the midline or at least 6 cm lateral.
- Before secondary trocars, inspect the whole abdomen for entry injury. Aim each port at the target, as redirecting widens the tract and causes emphysema.
- **Hand port**: the incision in centimetres equals the surgeon's glove size. Wound infection is about 9% and midline hernia 4–7%.

6 · Robotic systems and instruments

SAQ 3D printing in urology

Paper answer · P4

Type: SAQ (10 marks)
Source: Practice Set 6 Paper 4, Q3a

Definition

- **3D (three-dimensional) printing** is additive manufacturing: a solid object is built layer by layer from a digital model.
- In urology it turns a patient's CT (computed tomography) or MRI (magnetic resonance imaging) into a physical replica of kidney, tumour, vessels or prostate.

Workflow

`Thin-slice CT/MRI (DICOM files) → segmentation of kidney, tumour, vessels, calyces → STL (standard tessellation language) mesh → printer → post-processing`

- Takes hours to days; segmentation needs expertise.

Printing technologies

Method	Principle	Use
FDM (fused deposition modelling)	Melted plastic filament extruded	Cheap, rigid models

Method	Principle	Use
SLA (stereolithography)	Laser cures liquid resin	Fine detail, transparent models
Material jetting (PolyJet)	Jetted photopolymers, multi-colour	Tumour, vessels and calyces in different colours
SLS (selective laser sintering)	Laser fuses powder	Durable parts, devices
Bioprinting	Living cells in hydrogel "bio-ink"	Tissue engineering (see Practice Set 6 Paper 4, Q2)

Applications

1. **Partial nephrectomy planning:** complex hilar or endophytic tumours. The model shows the tumour against vessels and collecting system, so the surgeon plans the approach and rehearses steps. Promising.
2. **PCNL (percutaneous nephrolithotomy) planning:** staghorn stones, anomalous kidneys; choice of calyx. Early promise; 3D reformatting did not predict stone-free rate better.
3. **Prostate:** models with MRI lesions for margin and nerve-sparing planning and focal therapy.
4. **Patient education and consent:** patients understand their anatomy and the operation better.
5. **Training and simulation:** silicone or hydrogel phantoms for PCNL puncture, partial nephrectomy, pyeloplasty, anastomosis.
6. **Patient-specific devices:** prototype stents, catheters and surgical guides.

Limitations

- Cost, printing time and segmentation errors.
- Materials do not reproduce tissue feel or bleeding.
- Evidence mostly small series; no proven outcome benefit.

Future

- Virtual and augmented reality models on the robotic console, point-of-care printing, drug-eluting printed stents.

Diagram to draw: CT → segmentation → STL → printer → colour kidney model with tumour, artery and calyces.

> **Key points to remember:**
> - Additive manufacturing from CT/MRI via segmentation and an STL file.
> - Main uses: partial nephrectomy and PCNL planning, consent, simulation.
> - Limits: cost, time, unrealistic tissue, weak outcome evidence.

SAQ Hemostasis in laparoscopic surgery

Paper answer · P3 · P4 · merged from 2

Asked as: P3 SAQ "Hemostasis in laparoscopic surgery." · P4 SAQ "Hemostasis in lap surgery"

Principles

- Devices seal by **denaturing protein**. Plan from imaging.
- Keep a **haemorrhage tray** ready: laparoscopic Satinsky, 10-mm suction, vascular suture with Lapra-Ty, haemostatic agent.

Effect of pneumoperitoneum

- CO_2 at 12–15 mmHg **tamponades venous bleeding**, hiding it.
- Raise the pressure briefly for venous bleeding: 25 mmHg (P4: 20–25 mmHg).
- Before exit, **lower the pressure and inspect** the operative site and each port to unmask hidden bleeders and port-site bleeding: about 5 mmHg (P4: 5–10 mmHg).

Energy devices

Device	Lateral thermal spread	Note
Monopolar	5–7 mm	Capacitive coupling, insulation failure, direct coupling
Bipolar	1–6 mm	Current only between the jaws
Vessel sealer (LigaSure)	1–3 mm	Seals vessels **≤7 mm**; one application only
Ultrasonic (Harmonic)	≈1 mm	No current; blade >200 °C, cools slowly
Argon beam	Superficial	Raw surfaces; gas embolism risk

Clips

- **Titanium clips:** no diathermy near them (they slip).
- **Hem-o-lok** (locking polymer): circumferential dissection, see the tip beyond the vessel, feel the snap, at least two clips on the patient side with a 2-mm cuff.
- **FDA (Food and Drug Administration) warning:** after donor renal artery stump haemorrhages, the manufacturer **contraindicated Hem-o-lok on the living-donor renal artery**. A later series found it safe with correct technique, but use a stapler there.
- **Lapra-Ty** absorbable clips act as knots (renorrhaphy).

Staplers

- Endo GIA through a 12-mm port; **grey (2.0 mm) or white (2.5 mm)** loads for renal vessels; blue or green for thicker tissue.
- **Never fire across clips** (misfire). Reinforce a bleeding staple line with clips.

Sutures and haemostatic agents

- Intracorporeal or barbed sutures.
- Adjuncts: fibrin sealant, gelatin–thrombin matrix, oxidised cellulose.

Managing bleeding

- **Venous injury** (algorithm): 1. Raise pneumoperitoneum pressure (25 mmHg; P4: 20–25 mmHg). 2. Suction to identify the bleeding point. 3. Pledget tamponade. 4. Sealant. 5. Extra 5-mm port for suturing or Satinsky. 6. Hand-assist. 7. Convert.
- Keep the patient well filled (gas embolism).
- **Arterial:** minor bleeding, tamponade; renal artery, staple; aorta, convert.
- **Port-site bleeding:** cantilever the trocar to find it; diathermy or transfascial suture. Blunt trocars reduce it.

Diagram to draw: Hem-o-lok placement with two patient-side clips and a 2-mm cuff.

Key points to remember:

- Pneumoperitoneum hides venous bleeding: inspect at low pressure before exit.
- LigaSure seals ≤7 mm; spread monopolar 5–7 mm, ultrasonic ≈1 mm.
- Hem-o-lok contraindicated on the donor renal artery.
- Never staple across clips; grey or white loads for vessels.

*See the paper answer **Energy sources in urology and laparoscopic surgery, including bipolar energy** (Section 1, Ch 17).*

Also, from the condensed book (not in the paper answer):

- EndoWrist instruments have about a **10-use** limit, and an Allen bolt opens jaws locked after failure.
- Robot camera ports are 8 mm (Xi) or 12 mm (Si). Place the assistant port caudal, not between the arms.
- The LapSac is the most tear-resistant bag for morcellation, and ring forceps puncture it least.

7 · Exiting the abdomen

ALGORITHM

1. Lower the pressure to 5 mm Hg to unmask bleeding.
2. Inspect the operative site and every port.
3. Place fascial sutures under vision and remove each port under vision, the scope last.
4. Close a hand port first, then reinsufflate.

Trocar	Adult	Child
Bladed	Close all 10–12 mm sites	Close all ≥5 mm sites
Blunt	Close midline >10 mm or stretched sites	

- With BMI >30, use a closure device. Transverse extraction incisions herniate less than vertical midline ones.

8 · Complications and management

*See the paper answer **Pneumoperitoneum physiology and complications of laparoscopy and retroperitoneoscopy** (Section 1, Ch 18).*

Also, from the condensed book (not in the paper answer):

- **Bowel** is the commonest first-trocar injury. A through-and-through injury is seen only by viewing the primary port from a secondary port, so do this routinely. Leave the trocar in, gain access elsewhere and repair, with 4–5 L antibiotic irrigation. The commonest major vessels injured are the **aorta and common iliac artery**.
- **Thermal bowel injury**: resect a monopolar burn with **6 cm** margins, excise the visible area of a bipolar burn, and repair a mechanical injury laparoscopically.
- A Veress needle in a vessel is withdrawn without manipulation and re-sited. Always repair a bladder injury, and catheterise every major case.

9 · Training

*See the paper answer **Simulators in urology** (Section 1, Ch 17).*

Also, from the condensed book (not in the paper answer):

- Robotic surgeons with **>20 cases** have better outcomes, so mentor early cases.

19 Energy Modalities in Urology

*See the paper answer **Energy sources in urology and laparoscopic surgery, including bipolar energy** (Section 1, Ch 17).*

1 · Physics essentials

Every modality depends on where the energy is absorbed.

- The five surgical effects are cutting, ablation, destruction, coagulation and fragmentation.
- Laser power equals pulse energy times frequency, so 0.5 J × 20 Hz gives 10 W. **10 bar = 1 MPa** (not 1 bar).
- High frequency penetrates shallowly and low frequency deeply, which is why a scrotal probe is linear and a renal probe curvilinear. Absorption usually rises with frequency.
- Photon energy is E = hf, so raising laser power sends more photons per second, not more energetic ones.
- Acoustic impedance is sound speed times density. A large mismatch (tissue–air, tissue–stone) reflects the wave, which explains gel coupling, gas and rib shadows and the stone acoustic shadow.
- Radiation force equals absorbed power divided by wave speed. It is negligible for light but useful for sound, and is the basis of ultrasonic stone propulsion.
- A shock is an abrupt pressure rise from nonlinear steepening, and is acoustic, not electrical. Cavitation is a vapour bubble torn open by negative pressure, whereas boiling is caused by a rise in temperature.

	Electromagnetic waves	**Elastic (acoustic) waves**
Devices	Lasers, electrosurgery, microwave, radiofrequency (RF)	Shock wave lithotripsy (SWL), ultrasound therapies, intracorporeal lithotripters
Medium and wavelength	Free space; microns (light)	Needs gas, liquid or solid; millimetres
Bioeffect	Absorption, giving heat	Heat, mechanical stress in solids, cavitation in fluids
Penetration	Short, so a fibre must reach the target	Centimetres, so transcutaneous delivery is possible

2 · Electrosurgery

*See the paper answer **Energy sources in urology and laparoscopic surgery, including bipolar energy** (Section 1, Ch 17).*

Also, from the condensed book (not in the paper answer):

- **Pad placement**: use clean, well-perfused muscle and avoid bony prominences, scars, dense hair and metal implants. A **hip prosthesis is not a contraindication**; place the pad on the opposite side.
- **Pacemakers and implantable defibrillators**: prefer bipolar. If monopolar is needed, route the current path away from the device and use short bursts. Cardiology sets pacing to asynchronous and suspends anti-tachycardia therapy, an external defibrillator must be available, and the device is rechecked after surgery.
- **Surgical fire** needs fuel (drapes, alcohol prep), an oxidiser (oxygen-rich gas) and ignition. Electrosurgery is implicated in most theatre fires.

Endoscopic resection

*See the paper answer **TURP instruments, techniques and monopolar vs bipolar** (Section 16, Ch 151).*

*See the paper answer **TUR syndrome** (Section 16, Ch 151).*

Also, from the condensed book (not in the paper answer):

- Typical monopolar resection settings are cut **180 W** and coagulation **80–120 W**.
- Ions in the irrigant disperse current away from the loop, which is why monopolar irrigant must be ion-free, inert, non-toxic and transparent.
- Damaged sheath insulation causes loss of cutting or a urethral burn, so inspect the loop before mounting.

Device-assisted intravesical therapy for non-muscle-invasive bladder cancer (NMIBC)

	Electromotive drug administration (EMDA)	Chemohyperthermia
Energy	Electric field between an intravesical electrode and abdominal pads	915-MHz microwave via catheter (Synergo) with thermocouples
Mechanism	Iontophoresis, electro-osmosis and electroporation drive mitomycin C (MMC) into the lamina propria	Bladder wall at **42 °C** raises permeability and cancer-cell heat sensitivity
Evidence	Randomised controlled trial (RCT) of EMDA-MMC with bacille Calmette–Guérin (BCG): better recurrence- and progression-free survival. Meta-analysis: recurrence only. Cochrane: low quality	Not standard

3 · Lasers

*See the paper answer **LASERs in urology: holmium, thulium, KTP and newer laser technologies** (Section 1, Ch 17).*

*See the paper answer **Lasers in BPH, including green light** (Section 16, Ch 151).*

Laser lithotripsy: optical pulse duration decides the mechanism

Optical pulse	Mechanism	Bubble pressure	Fragments
Ultrashort <500 ns (Q-switched)	Plasma	Extremely high	Large
Short 0.5–10 µs (pulsed dye; frequency-doubled double-pulse neodymium:YAG [Nd:YAG], FREDDY)	Cavitation and shock (photomechanical)	>300 bar	Large
Long >250 µs (holmium:YAG [Ho:YAG], thulium fibre laser [TFL])	Photothermal	<20 bar	Small, dust
>20 ms or continuous wave	Photothermal with carbonisation	–	Adjacent tissue injured

- **Not in the paper answer:**
- Manufacturers quote the electrical (pump) pulse duration, but the **optical pulse duration** is the one that matters. Dosimetry, not the platform, sets the mechanism.
- **Moses effect**: the first ≈ **60 µs** boil a vapour channel. The fibre should be close and perpendicular to the stone (0° gives the most ablation; parallel gives none). Ho:YAG transients are only 8–20 bar at <1 J.
- The **pulsed dye laser (504 nm)** was safe because haemoglobin acts as an energy sink, but it failed on hard stones (cystine, brushite, calcium oxalate monohydrate).

Soft-tissue lasers

- The **CO_2 laser** (10,600 nm) is used for superficial penile lesions and cannot be used intracorporeally.
- The **diode laser** (940–1490 nm) is absorbed by water and haemoglobin variably, penetrates 0.5–5 mm and is used for prostate vaporisation.
- Nd:YAG forward scatter can injure bowel beyond the bladder wall. KTP is shallow because of haemoglobin absorption, not wavelength itself.
- **GreenLight** ablates best at a **4 mm/s** sweep and **≤2 mm** working distance. Gain plateaus beyond 180 W, but 180 W shortens operating time compared with 120 W.
- **Photodynamic therapy** treats the whole urothelium after one intravesical application of photosensitiser, by generating singlet oxygen.

4 · Acoustic therapy

Heating	Cavitation
Rises with frequency (absorption)	Rises as frequency falls (longer negative phase)
Dose-dependent	**Threshold effect**; nucleates at interfaces, stones and gas, so lung and bowel are vulnerable

- Do not perform SWL within **24 h** of ultrasound microbubble contrast. Doppler twinkling artefact is caused by microbubbles on the stone.
- Acoustic waves are non-ionising, so repeated safe exposures do not accumulate.

Shock wave lithotripsy

*See the paper answer **ESWL: mechanism, indications, factors affecting outcome, complications and current status** (Section 11, Ch 98).*

LAQ Energy sources for breaking stones in urology

Paper answer · P4

Asked as:

- Describe energy sources for breaking stones in Urology — *Practice Set 1 Paper IV Q2 (LAQ)*
- Describe energy sources for braking stones in uro — *Practice Set 5 Paper IV Q1 (LAQ)*

Type: LAQ (25 marks)
(also covered in 2025-Winter Paper 4, Q2 and Q3b; also answered in Practice Set 1 Paper 4, Q2)

Read as "energy sources for breaking stones in urology".

Introduction

- Stones are **brittle**: they crack under tension and shear. Tissue is **elastic** and survives the same stress. Every lithotripter exploits this difference.
- Energy reaches the stone **extracorporeally** (SWL, shock wave lithotripsy) or **intracorporeally** through an endoscope.

Classification

Group	Sources
Extracorporeal	SWL: electrohydraulic, electromagnetic, piezoelectric; BWL (burst wave lithotripsy, investigational)
Intracorporeal mechanical	Ballistic (pneumatic), ultrasonic, combined ultrasonic–pneumatic
Intracorporeal electrical	EHL (electrohydraulic lithotripsy)
Intracorporeal light	Lasers: Ho:YAG (holmium:yttrium-aluminium-garnet), TFL (thulium fibre laser), pulsed Tm:YAG (thulium:YAG); historic pulsed dye and FREDDY

1. Shock wave lithotripsy

- **Shock wave:** peak positive pressure 30–100 MPa, then a tensile tail of about −10 MPa. It is generated at F1, focused, and delivered to the stone at F2.

Generator	Mechanism	Strength	Weakness
Electrohydraulic (original HM3, 1980)	Underwater spark at F1 of an ellipsoid reflector	Very effective	Electrode life <40,000 shocks; focus drifts
Electromagnetic (modern standard)	Coil pushes a metal membrane; lens or reflector focuses	Consistent; high energy at F2; long life	—
Piezoelectric	Many ceramic elements on a spherical dish	Long life; little pain	Small focal zone, low energy

- **Fracture mechanisms:** shear and tear forces; **spallation** at the distal face; quasistatic squeezing; **cavitation** (bubble collapse erodes the surface); dynamic squeezing (most powerful). A focal zone wider than the stone works best.

- **Technique:** 60–90 shocks/min beats 120/min, because bubble clouds shield the stone and injure the kidney; voltage ramping; bubble-free gel coupling.
- Poor results: calcium oxalate monohydrate, brushite, cystine, density >1000 HU (Hounsfield units), skin-to-stone distance >10 cm.

2. Ballistic (pneumatic) lithotripsy

- Compressed air drives a projectile against a metal probe, a **jackhammer** (about 12 Hz).
- Breaks all stones cheaply with little ureteric injury. Drawbacks: **retropulsion**; rigid probes only, as flexible probes block deflection.

3. Ultrasonic lithotripsy

- A piezoceramic transducer vibrates a hollow probe at ultrasonic frequency; contact drills the stone.
- **Suction** through the hollow probe removes fragments at once. Best for soft stones in PCNL (percutaneous nephrolithotomy); weak on hard stones; needs irrigation for cooling.

4. Combined devices

- Ultrasonic plus ballistic in one probe (for example Trilogy, ShockPulse-SE): ballistic cracks hard stones while ultrasound and suction clear fragments. Faster PCNL.

5. Electrohydraulic lithotripsy

- A spark at the electrode tip forms a **vapour bubble**; expansion and collapse send pressure waves in all directions.
- Cheap; 3–5 Fr flexible probes. Poorly focused, so it injures the ureter and the scope; weak on cystine. AUA (American Urological Association) advises against it as first-line ureteric lithotripsy; still used for bladder stones.

6. Lasers

- **Pulse duration decides mechanism.** Short pulses (<10 µs: pulsed dye, FREDDY) are photomechanical, with cavitation shocks. Long pulses (>250 µs: holmium, thulium) are **photothermal**: the stone is heated until it melts, decomposes or flashes its water to steam.

	Ho:YAG	**TFL**
Wavelength	**2120 nm** (often quoted 2100)	1940 nm (water absorption peak)
Frequency	Up to about 120 Hz	Up to about 2000 Hz
Peak power	High	Low → less retropulsion
Fibre	≥200 µm	50–150 µm (more flow and deflection)
Product	Fragments and dust	Finer dust

- **Holmium:** standard since the 1990s; breaks **every** composition; flexible quartz fibres. Fragmenting 0.6–1.0 J × 6–10 Hz; dusting 0.2–0.3 J × 40–80 Hz; "popcorning" in a calyx.
- **Moses effect:** the start of the pulse boils a vapour channel through which the rest reaches the stone; pulse modulation reduces retropulsion.
- **TFL RCTs (randomised controlled trials):** superior to low-power (30 W) holmium; equal stone-free rate to 120 W pulse-modulated holmium.
- **Retropulsion** rises with pulse energy, not frequency, and falls with longer pulses.

Safety

- Laser heat follows total power: ≤10 W in the ureter, pauses, high flow, access sheath. Eyewear for all.
- Never fire without seeing the fibre tip on the stone.
- SWL: haematoma, arrhythmia, steinstrasse. Contraindications: pregnancy, coagulopathy or anticoagulants, untreated infection, aneurysm, distal obstruction.

Choice by site

Site	Usual energy
Ureter	Laser (semirigid or flexible); pneumatic distally; SWL
Kidney	SWL; laser via flexible URS (ureteroscopy); PCNL with ultrasonic, combined or laser
Bladder	Laser, pneumatic, EHL, mechanical

Recent advances

- **BWL:** handheld ultrasound bursts (5–100 cycles) break stones without anaesthesia. **Ultrasonic propulsion** moves fragments.
- Suction sheaths; AI (artificial intelligence)-adjusted laser settings.

Diagram to draw: SWL electrohydraulic ellipsoid with F1 and F2, stone at F2.

Key points to remember:

- SWL: electromagnetic is the modern standard; slow rate (60–90/min) is better.
- Holmium 2120 nm, photothermal, breaks every composition.
- TFL 1940 nm: low peak power, finer dust, less retropulsion.
- EHL is poorly focused, so it is not first-line in the ureter.

Also, from the condensed book (not in the paper answer):

- Pulse parameters: peak negative pressure ≈ **10 MPa** lasting ≈ 5 µs, about 3 J per pulse, and about **2000 shocks** at 0.5–3 Hz.
- Each shock leaves bubbles that need ≈ **1 s** to dissolve, so faster firing builds a cavitation cloud that shields the stone and injures the kidney. Small air pockets in the coupling disproportionately reduce focal pressure.
- Piezoelectric sources focus at the centre of curvature of a spherical bowl and are the only type that produces both impulses and narrowband bursts.

Investigational acoustic stone therapies

	Burst wave lithotripsy (BWL)	Ultrasonic propulsion
Waveform	Narrowband, 5–100 cycles, 200–1000 kHz	About 1500 cycles in bursts of ≤3 s
Peak pressure and rate	5–8 MPa; ≥10 Hz	≤2.4 MPa; 50% duty cycle
Guidance and anaesthesia	Real-time ultrasound; none	Ultrasound; none
Action	Breaks stones into uniform fragments proportional to wavelength	Moves fragments by radiation force

KEY POINT

Acoustic therapy is management of the negative-pressure phase. Tension cracks brittle stones and spares ductile tissue, but the same tension nucleates bubbles that either destroy tissue deliberately (histotripsy) or shield the stone and injure the kidney (SWL fired too fast). Raise the frequency for heat and lower it for cavitation.

Soft-tissue ablation

*See the paper answer **HIFU** (Section 17, Ch 163).*

Also, from the condensed book (not in the paper answer):

- **HIFU** uses 1500–10,000 W/cm² in ~1-s pulses and is poorly visible on ultrasound, so gaps occur.
- **Cavitation histotripsy** uses >10,000 W/cm² to make a bubble cloud that liquefies tissue mechanically, and is **echogenic in real time**. **Boiling histotripsy** boils the focus within one burst (<100 ms). Both are investigational.

5 · Thermal and other ablation

*See the paper answer **Energy sources in urology and laparoscopic surgery, including bipolar energy** (Section 1, Ch 17).*

Also, from the condensed book (not in the paper answer):

- In radiofrequency ablation the tissue is heated by current, so the probe itself is not hot. Early concerns about microwave ablation (collecting-system absorption, inconsistent zones) were not borne out, and its safety and efficacy equal radiofrequency ablation.
- **Rezum** injects 103 °C steam through an 18-G needle; each **9-s** injection brings tissue to ≥70 °C and stays within the zonal compartment.

KEY POINT

Current density explains electrosurgery, pulse duration explains laser lithotripsy, penetration depth (chromophore) explains soft-tissue lasers, and negative pressure (frequency and rate) explains SWL and histotripsy. In direct-contact modalities the tissue boundary, not absorption depth, limits the lesion.

6 · Landmark trials

Trial	Design	Result	Take-home
TFL vs low-power Ho:YAG	RCT, ureteroscopy for renal stones	TFL: higher stone-free rate, shorter time, less bleeding	TFL beats 30-W holmium
TFL vs high-power Ho:YAG	RCT vs 120-W pulse-modulated Ho:YAG	Stone-free equal; TFL less retropulsion; Ho:YAG less energy	The comparator explains discordant trials
Ultrasonic propulsion of residual fragments	RCT	Fewer relapses (growth, urgent visit, surgery) at 4 years	Clearing residual fragments matters

20 Complications of Urologic Surgery

1 · Classification and context

*See the paper answer **Pneumoperitoneum physiology and complications of laparoscopy and retroperitoneoscopy** (Section 1, Ch 18).*

Also, from the condensed book (not in the paper answer):

- **Clavien–Dindo** is the recommended system and grades the **treatment required, not the event**. It omits intraoperative events and does not separate early from late.

Grade	Definition	Examples and traps
I	Deviation without drugs or intervention	Antiemetics, analgesics, electrolytes; wound opened at bedside
II	Other drugs	Blood transfusion, total parenteral nutrition
IIIa / IIIb	Surgical, endoscopic or radiological intervention	IIIa without, IIIb under general anaesthesia
IVa / IVb	Life-threatening, intensive care	IVa single organ (dialysis); IVb multiorgan. Central nervous system bleed or stroke counts; transient ischaemic attack does not
V	Death	–
Suffix "d"	Present at discharge	Flags follow-up; not a grade

2 · Positional and neuromuscular complications

*See the paper answer **Patient positioning in lap urology** (Section 1, Ch 11).*

Also, from the condensed book (not in the paper answer):

- Positional injury occurs in ≈ **1%** of laparoscopic and up to ≈ **6%** of robotic cases.
- In lithotomy the stirrup compresses the **common peroneal nerve at the fibular neck** (foot drop). Tucked arms in robotic surgery endanger the **ulnar nerve**.

Compartment syndrome and operative nerve injury

- **Lower-limb compartment syndrome** occurs in 1 in 3500 overall and up to **1 in 500** radical cystectomies. Perform fasciotomy if >**30 mmHg** and symptomatic. Pressure >50 mmHg for 3–4 h causes irreversible necrosis.
- **Genitofemoral** injury (psoas hitch, ≈ 5%) is avoided by placing sutures along the tendon. Repair an **obturator** transection with tension-free **7-0** non-absorbable epineural sutures.

3 · Urine leaks and urinary tract injury

Urine leak after partial nephrectomy

It presents at about day 13. Drain creatinine confirms it but does not locate it; computed tomography (CT) urography with delayed images shows the site.

ALGORITHM

1. Drainage and a Foley catheter.
2. Ureteric stent; drain off suction.
3. Radiological drainage of collections, then nephrostomy.
4. Nephrectomy as the last resort.

Ureteric injury

P3 *Ureteric injury: iatrogenic and upper ureteric; Uretero-vaginal fistula — danger sites, diagnosis, double-dye test, timing and repair by level*

- A partial tear is closed with 4-0/5-0 absorbable sutures over a stent for 4–8 weeks.
- Strictures are treated endoscopically if short, and surgically if **>2 cm** or after endoscopic failure.

Bladder injury and anastomotic leak

- Close a bladder injury in two layers and catheterise for 7–10 days.
- A **urethrovesical leak** after robotic prostatectomy occurs in up to ≈ 15% at days 7–10.

ALGORITHM

1. Keep the catheter patent and **never apply traction**.
2. Move the drain off suction; radiological drainage; then nephrostomy.
3. Formal repair of a major disruption **before day 7**, after which tissue will not hold sutures.

4 · Bleeding and vascular injury

P1 P3 P4 *Retroperitoneoscopy and laparoscopy complications; partial nephrectomy — leave the trocar in place, venous bleeding control; delayed haematuria means pseudoaneurysm, so angioembolise*

- Haemorrhage is the commonest intraoperative complication (≈ 2%).
- The commonest vessel injured at umbilical access is the **left common iliac vein**. Only the **internal iliac and inferior mesenteric** vessels may be ligated; all other major vessels are repaired.
- The superior mesenteric artery can be mistaken for the left renal artery in the lateral position, with mortality up to **75%**.

5 · Bowel injury

Most delayed injuries are thermal (monopolar spread >20 mm).

Injury	Management
Small tear without spillage	Primary repair
Colonic with faecal soiling	Colostomy
Thermal burn	**Always oversew, never conservative**; resect if extensive, because damage exceeds what is visible
Rectal (prostatectomy ≈ 0.5–1%; salvage ≈ 7%)	Multilayer closure, omental interposition; colostomy if irradiated or missed

Duodenal injuries perforate late even when <5 mm.

6 · Venous thromboembolism (VTE) and lymphocele

Symptomatic VTE occurs in ≈ **1%** after nephrectomy or prostatectomy and ≈ **5%** after radical cystectomy, mostly after discharge.

Caprini score: 1 point for age 41–60 or body mass index >25; 2 points for age 61–74, malignancy, or major or laparoscopic surgery >45 min; 3 points for age ≥75, personal or family VTE or thrombophilia; 5 points for stroke, arthroplasty or fracture. Risk is very low at 0, low at 1–2, moderate at 3–4 and **high at ≥5**.

Risk	Prophylaxis
Very low (transurethral, anti-incontinence)	Early ambulation
Low	Intermittent pneumatic compression (IPC)
Moderate	Low-molecular-weight heparin (LMWH), low-dose unfractionated heparin or IPC
High	Heparin plus IPC; IPC alone until haemostasis is secure if bleeding risk is high
High-risk cancer surgery	Add extended LMWH for ≈ **4 weeks** after discharge

Start pharmacological prophylaxis 2–12 h before surgery.

P1 *Lymphocele and postoperative lymphatic leak — incidence, observation, drainage, fenestration*

- Prevent it by clipping or sealing visible lymphatics and by a peritoneal flap after robotic prostatectomy.

7 · Landmark trials

Trial	Design	Result	Take-home
Preventer	Randomised, perioperative heparin after prostatectomy	No reduction in symptomatic VTE	Not mandatory in low-risk prostatectomy
Extended LMWH meta-analysis	Randomised trials, abdominopelvic cancer surgery	≥1 month of LMWH cut VTE without extra bleeding	Extend after cystectomy

21 Urologic Considerations in Pregnancy

1 · Physiological changes

System	Change	Urological relevance
Cardiovascular	Cardiac output up 30–50%	The uterus compresses the vena cava, so use a **15–20° left tilt** for every procedure
Haematological	Blood volume up 40–50%; hypercoagulable	Thromboembolism risk ~10×; white cell count up to 12,200, so it is unreliable for infection
Renal	Glomerular filtration rate up **40–65%**; creatinine ≈ **0.8 mg/dL**	A "normal" non-pregnant creatinine may mean impairment
Urine	Protein up to 300 mg/24 h; calciuria doubles	Persistent proteinuria: exclude pre-eclampsia, diabetes, renal disease and infection

Dipstick microscopic haematuria occurs in up to **20%** and is usually benign; pregnancy-specific causes are pre-eclampsia and placenta percreta.

2 · Physiological hydronephrosis

SAQ Bacteriuria in pregnancy and antibiotic policy

Paper answer · P1 · P3 · merged from 2

Asked as: P1 SAQ "Bacteriuria in pregnancy." · P3 SAQ "Baceriuria and use of antibiotic policy in pregnancy"

Read as: bacteriuria in pregnancy and the antibiotic policy for it.

Definition

- **ASB (asymptomatic bacteriuria):** $\geq 10^5$ CFU (colony-forming units)/mL of one uropathogen in a midstream sample without urinary symptoms ($\geq 10^2$ CFU/mL on a catheter sample).

Epidemiology

- Prevalence is **not raised** by pregnancy (about 2–10%; P3: about 2–7%), but ASB **persists and ascends**. Higher with low socioeconomic status, multiparity, sickle cell trait. Organisms as in non-pregnant women (mainly *E. coli*).

Why pregnancy matters (pathophysiology)

- **Progesterone** relaxes ureteric smooth muscle; the **uterus compresses the ureters at the pelvic brim** (right more). Dilatation from about 15 weeks; normal by 6 weeks postpartum.
- Result: **stasis and hydroureteronephrosis**; bacteriuria persists and ascends.
- GFR (glomerular filtration rate) rises 30–50%: creatinine and drug levels fall.
- **Untreated ASB leads to pyelonephritis in 15–45%** (historical), mostly in the third trimester (60–75% of cases). Treatment cuts this to **1–4%**.
- Pyelonephritis causes **preterm birth, low birth weight** and maternal sepsis. **Cochrane 2019:** treating ASB reduces pyelonephritis, preterm delivery and low birth weight.

Screening

- **Urine culture at 12–16 weeks** or the first visit (USPSTF [US Preventive Services Task Force], IDSA [Infectious Diseases Society of America]). Dipstick misses many cases.
- Treat all ASB found in pregnancy.

Antibiotic policy

Safe	Restricted or avoided
β-lactams (penicillins such as amoxicillin; cephalosporins such as cephalexin): throughout	**Nitrofurantoin:** stop near term (from 35 weeks): neonatal haemolysis (G6PD [glucose-6-phosphate dehydrogenase] deficiency); not for pyelonephritis
Fosfomycin: lower tract (cystitis) only	**TMP-SMX (trimethoprim-sulfamethoxazole):** avoid 1st trimester (folate, neural tube); caution near term (kernicterus)
Gentamicin: pyelonephritis	**Fluoroquinolones** (cartilage), **tetracyclines** (teeth, bone), chloramphenicol: avoid

- ACOG (American College of Obstetricians and Gynecologists) 2017: nitrofurantoin and sulfonamides first-line in 2nd–3rd trimester; 1st trimester only if no alternative.
- **ASB and cystitis:** culture-directed, **4–7 days** (cures more than a single dose); **repeat culture** after treatment.

Pyelonephritis

- **Admit**; IV (intravenous) β-lactam (ampicillin plus gentamicin, ceftriaxone or aztreonam) until afebrile, then oral (cephalexin) to complete ≥14 days (P3: 10–14 days).
- No improvement by **72 h: renal ultrasound** (stone, obstruction, abscess).

Prophylaxis

- After one pyelonephritis or two cystitis episodes: nightly nitrofurantoin or cephalexin until delivery, or close culture surveillance (Cochrane 2015: close surveillance equally effective).
- **Also know:** urinary diversion or augmentation, SCI (spinal cord injury): most treat ASB in pregnancy and consider prophylaxis.

Diagram to draw

- Right hydroureter compressed at the pelvic brim.

> **Key points to remember:**
> - Prevalence unchanged; risk of pyelonephritis raised (15–45% untreated).
> - Screen by culture at 12–16 weeks; treat all ASB for 4–7 days.
> - β-lactams safe throughout; no fluoroquinolones or tetracyclines.
> - Nitrofurantoin not near term (after 35 weeks); TMP not in 1st trimester.

Also, from the condensed book (not in the paper answer):

- It peaks in the second trimester and resolves by up to 12 weeks postpartum; its grade correlates poorly with pain.
- Symptomatic hydronephrosis without a stone is treated conservatively. In a randomised trial stenting relieved pain better but caused stent symptoms, so reserve drainage for infection, failing function or genuine obstruction.

3 · Urinary tract infection and drugs

*See the paper answer **Bacteriuria in pregnancy and antibiotic policy** (Section 1, Ch 21).*

Also, from the condensed book (not in the paper answer):

- Non-steroidal anti-inflammatory drugs are contraindicated (third-trimester ductus closure); treat colic with small frequent doses of morphine. Defer intravesical bacillus Calmette–Guérin and mitomycin C until after delivery.
- The **Pregnancy and Lactation Labeling Rule (2015)** abolished the A, B, C, D and X categories in favour of a narrative risk summary.

4 · Imaging

*See the paper answer **Stones in pregnancy** (Section 11, Ch 96).*

Also, from the condensed book (not in the paper answer):

- First-trimester exposure carries teratogenic risk; later exposure mainly raises childhood cancer risk. Needed CT or nuclear studies must not be withheld.
- Give gadolinium only if it clearly improves outcome. Iodinated contrast crosses the placenta but is not teratogenic in animals. A mercaptoacetyltriglycine (MAG3) scan gives a fetal dose of **<5 mGy**.

5 · Surgery and anaesthesia

- No anaesthetic agent is teratogenic at standard doses. **Never deny or delay necessary surgery in any trimester**, but postpone elective surgery. The **second trimester is safest**.
- Involve obstetrics, monitor the fetal heart, give steroids if the fetus is viable and preterm, and operate where caesarean is available. At laparoscopy, insufflate to **10–15 mmHg**.
- Perinatal loss is driven by the disease, not the operation.

6 · Urolithiasis

*See the paper answer **Stones in pregnancy** (Section 11, Ch 96).*

*See the paper answer **Bacteriuria in pregnancy and antibiotic policy** (Section 1, Ch 21).*

Also, from the condensed book (not in the paper answer):

- Nephrostomy carries more preterm labour than a stent. Early ureteroscopy is favoured because repeated exchanges add morbidity and still leave the stone.

7 · Malignancy

Urological cancer is rare (≈ **13 per million** pregnancies) and is diagnosed late.

- **Renal cell carcinoma** is the commonest. Operate as in non-pregnant women, timed to trimester; nephrectomy can be done at caesarean. Avoid targeted therapy and immunotherapy.
- **Angiomyolipoma** is prone to rupture: embolise if stable, operate if unstable.
- **Bladder cancer**: transurethral resection is safe at any gestation; defer intravesical therapy.

8 · Placenta accreta spectrum

Grade	Depth
Accreta	Attached to myometrium
Increta	Invades myometrium
Percreta	Through serosa into bladder, ureters or rectum

- The main risk factor is **previous caesarean**. Blood loss is 3–5 L and maternal mortality up to **7%**.
- MRI maps bladder and ureteric invasion. At cystoscopy, **do not biopsy**.
- Perform **caesarean hysterectomy at ≈ 34 weeks** with the placenta in situ and retrograde ureteric catheters. With detrusor invasion, do a partial cystectomy rather than dissect the plane. Genitourinary injury occurs in ≈ **20–29%**.

9 · Pregnancy after reconstruction

- After augmentation, pyelonephritis occurs in up to **60%**. Use **serum human chorionic gonadotrophin**, because urine tests are falsely positive after contact with bowel mucosa.

Reconstruction	Mode of delivery
Augmentation with intact bladder neck	Vaginal delivery not contraindicated
Bladder neck reconstruction or artificial sphincter	Caesarean preferred by most
Ureterosigmoidostomy	Vaginal delivery risks anal sphincter damage
Bladder exstrophy	Planned caesarean

> **KEY POINT**
>
> At caesarean after reconstruction, use a **high midline incision** with a urologist present.

- **Myelomeningocele**: latex allergy in up to 60%; folic acid **4–5 mg** daily from 3 months before conception to 12 weeks.
- **Renal transplant**: wait at least 1 year with stable function and stop mycophenolate before conception.

22 Intraoperative Consultations

1 · Principles and ethics

- The patient is unconscious and never consented to the change of plan, so the consultant protects the absent patient without blaming the colleague.
- Accepting the call creates a patient–physician relationship, so disclose the reason fully to the family.
- Four factors decide whether to act now or temporise: patient condition, procedure morbidity, long-term consequences and reversibility.

	Act now	**Temporise and defer**
Example	Low-grade ureteric injury, treated with a stent	Complex proximal high-grade injury
Morbidity / reversibility	Low / reversible	High / often irreversible
Action	Proceed and disclose afterwards	Await shared decision making

Repair complex injury at once only if deferral is the greater harm. If surgeons disagree, default to the primary surgeon unless that is harmful.

2 · Consultation time-out

ALGORITHM

1. Assess the field first, because the procedure cannot be named until it is seen.
2. Introductions, patient identity, condition, procedure, blood loss, added time and concerns.
3. Verbal agreement by all.

3 · Urethral injury

*See the paper answer **Outline algorithm for difficult catheter placement.** (Section 1, Ch 15).*

Also, from the condensed book (not in the paper answer):

- In women, repair a sling or dissection injury vaginally with an interposition flap (peritoneum, omentum or labial fat).

4 · Bladder injury

P2 *Lower urinary tract trauma in pelvic fracture — American Association for the Surgery of Trauma (AAST) bladder grades I–V*

- The bladder is the commonest organ injured in pelvic surgery. Retropubic tape injures it in **3–9%** (transobturator 0.5%), so cystoscopy is routine. **Transurethral resection of bladder tumour is the commonest source**: extraperitoneal perforation needs catheter drainage, and intraperitoneal perforation usually needs open repair.
- If injury is suspected but not seen, perform a cystotomy. Close in 1–2 absorbable layers, fill-test, and drain with a urethral catheter for **7–14 days**. No routine suprapubic tube is

needed.

> **KEY POINT**
>
> Ask three questions: is it closed, is it drained, and are the ureters intact? At the trigone, when in doubt, reimplant.

5 · Ureteric injury

*See the paper answer **Ureteric injury: iatrogenic (gynaecological) and upper ureteric injury** (Section 11, Ch 94).*

Also, from the condensed book (not in the paper answer):

- Iatrogenic injury is **75–80%** of ureteric injuries and >90% are in the distal third. Avoid distal ureteroureterostomy (poor blood supply); use reimplant, then psoas hitch, then Boari flap.
- Thermal injury extends beyond the visible zone, so debride generously rather than simply stent.

6 · Renal, bowel and vascular injury

- **PCNL** (percutaneous nephrolithotomy) is the commonest cause of renovascular injury. Venous bleeding is tamponaded by clamping the nephrostomy; arterial bleeding needs selective angioembolisation, not exploration.
- In vascular injury, apply pressure and never clamp blindly.

7 · Placenta percreta and documentation

- The **posterior bladder** is the commonest site of invasion. Never biopsy at cystoscopy (exsanguinating haematuria). Caesarean hysterectomy at **34–38 weeks** is the gold standard; ureters are involved in up to 7%.
- Write a separate note of observed facts even if no injury is found. Never write "careless" or "avoidable".

23 Assessment of Urologic and Surgical Outcomes

1 · Framework

Donabedian	Structure	Process	Outcome
Definition	Context of care	What is done	Effect on patients
Examples	Volume, nurse:bed ratio	Antibiotic and thromboprophylaxis	Mortality, readmission, quality of life
Strength	Easy and cheap	Detects problems early; less affected by case mix	Most meaningful to patients
Weakness	Largely non-modifiable	Valid only if evidence links it to outcome	Confounded, so needs risk adjustment

> **KEY POINT**
> Without case-mix adjustment, surgeons who treat high-risk patients look worse, which discourages them from treating such patients.

2 · Long-term end points and bias

SAQ Clinical trial methodology

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 3 Paper 1, Q3a

Definition

- A **clinical trial** is a prospective study that assigns people to an intervention to test its effect on health outcomes. The **RCT (randomised controlled trial)** is the gold standard because randomisation balances known and unknown confounders.

Phases

Phase	Purpose	Subjects
0	Microdose pharmacokinetics	Very few
I	Safety, dose, toxicity	20–80, often volunteers
II	Activity, efficacy signal	Patients, about 100–300
III	Comparative efficacy versus standard	Hundreds to thousands; basis of approval
IV	Post-marketing surveillance	General population

Design

- **Parallel**, **crossover** (own control; stable chronic disease), **factorial** (two questions at once), **cluster** or **adaptive**.
- Hypothesis: **superiority**, **non-inferiority** (pre-set margin) or equivalence.
- Surgical innovation follows the **IDEAL** framework (Idea, Development, Exploration, Assessment, Long-term study).

Randomisation and allocation

- Simple, **block** (equal numbers), **stratified** (by prognostic factors) or **minimisation**.
- **Allocation concealment** (central or web-based) stops selection bias.

Blinding

- Single (patient), double (plus assessor) or triple (plus analyst). Surgery: sham procedures or blinded assessors.

Endpoints

- One **primary endpoint** sets sample size; secondary endpoints are exploratory.
- **Hard** (overall survival) versus **surrogate** endpoints, valid only if they meet the **Prentice criteria**; in prostate cancer only MFS (metastasis-free survival) does.
- **PROs (patient-reported outcomes)** with validated tools; complications graded by **Clavien–Dindo** or CTCAE (Common Terminology Criteria for Adverse Events).

Sample size and analysis

- Set α (type I error, usually 0.05), power (80–90%) and expected effect.
- **ITT (intention-to-treat)** preserves randomisation; per-protocol supports non-inferiority claims.
- Kaplan–Meier curves for time-to-event data.

Ethics and reporting

- Declaration of Helsinki, ICH-GCP (International Council for Harmonisation–Good Clinical Practice), informed consent, ethics committee, DSMB (data safety monitoring board).
- **Registration** before enrolment (CTRI [Clinical Trials Registry–India]).
- **CONSORT** (Consolidated Standards of Reporting Trials): checklist plus **participant flow diagram** (25 items in the 2010 version; updated 2025).
- **Also know:** lead-time bias; Will Rogers phenomenon.

Diagram to draw: CONSORT flow diagram: enrolment → randomisation → allocation → follow-up → analysis.

> **Key points to remember:**
> - Phase III RCT with concealed allocation and blinding is the gold standard.
> - Primary endpoint drives sample size; surrogates need the Prentice criteria.
> - ITT analysis; CONSORT flow diagram for reporting.

SAQ Evidence-based medicine and critical appraisal

Paper answer · P1 · P4 · merged from 2

Asked as: P1 SAQ "Evidence-based medicine and critical appraisal of research papers" · P4 SAQ "Describe evidence based medicine."

"quaLliy" is taken to mean quality, i.e. critical appraisal.

Definition

- **EBM (evidence-based medicine):** "the conscientious, explicit and judicious use of current best evidence in making decisions about the care of individual patients" (**Sackett, 1996**).
- It integrates best research evidence, **clinical expertise** and **patient values**.

- **Critical appraisal:** systematic assessment of a paper's **validity, results and applicability**; the "Appraise" step of EBM.

Steps ("5 As")

1. **Ask** a focused question in **PICO** form (Population, Intervention, Comparison, Outcome).
2. **Acquire**: PubMed, Cochrane Library, guidelines.
3. **Appraise**: validity (randomisation, blinding), effect size (ARR [absolute risk reduction], **NNT [number needed to treat] = 1/ARR**), precision (95% CI [confidence interval]), applicability.
4. **Apply** with expertise and patient preference.
5. **Assess** (audit) the outcome.

Critical appraisal: three core questions

1. Is it valid? (internal validity)

- **PICO question**; suitable design.
- **RCTs (randomised controlled trials):** true randomisation, **allocation concealment**, comparable groups at baseline, **blinding**, complete follow-up (>80%), **ITT (intention-to-treat) analysis**.
- **Bias** checked:

Bias	Meaning
Selection	Groups differ at entry
Confounding	Third factor (comorbidity, frailty) linked to exposure and outcome; needs risk adjustment
Performance, detection, attrition	Unequal care, outcome assessment or dropouts
Lead-time, length-time; stage migration (Will Rogers)	Screening or better tests make survival look longer
Publication	Negative studies unpublished

- **Pitfalls in appraisal:** selection, confounding, lead-time, length-time, stage migration (Will Rogers) and attribution bias.

2. What are the results?

- Effect: RR (relative risk), **ARR, NNT = 1/ARR**.
- Precision: **95% CI**; p value; **sample size and power** (usually 80%; α 0.05).
- **Surrogate end points** must meet the **Prentice criteria** (prostate cancer: only metastasis-free survival does); validated outcome tools (validated PROs [patient-reported outcomes]); Clavien–Dindo grading.

3. Does it apply? (external validity)

- Similar patients and setting; benefit outweighs harm.

Levels of evidence (Oxford CEBM [Centre for Evidence-Based Medicine], as adapted by EAU [European Association of Urology])

Level	Type of evidence / design
1a	Systematic review or meta-analysis of RCTs
1b	At least one RCT

Level	Type of evidence / design
2a	Well-designed controlled study without randomisation
2b	Quasi-experimental (cohort) study
3	Non-experimental: case-control, correlation, case reports, case series
4	Expert committee reports or opinion

- **Grades:** A (consistent level 1), B (level 2–3), C (level 4 or consensus); D (some systems) = no recommendation possible.
- **GRADE** (Grading of Recommendations Assessment, Development and Evaluation): certainty high to very low; RCTs start high, observational studies low; **downgrade** for bias, inconsistency, indirectness, imprecision, publication bias; **upgrade** for large effect or dose–response. Recommendations **strong or weak** (now used by EAU).
- AUA (American Urological Association): evidence strength A/B/C; strong, moderate or conditional recommendations.

Reporting checklists and quality tools

Design	Checklist
RCT	**CONSORT** (Consolidated Standards of Reporting Trials)
Systematic review / meta-analysis	**PRISMA** (Preferred Reporting Items for Systematic reviews and Meta-Analyses)
Observational	**STROBE** (Strengthening the Reporting of Observational studies in Epidemiology)
Diagnostic accuracy	**STARD** (Standards for Reporting Diagnostic accuracy studies)
Surgical innovation	**IDEAL** framework (Idea, Development, Exploration, Assessment, Long-term study)

- **Also know:** Cochrane RoB 2 (risk-of-bias) tool; trial registration; conflicts of interest; Clavien–Dindo; validated PROs; Donabedian model.

Limitations of EBM

- Few surgical RCTs; trial patients differ from real ones; publication bias; "cookbook medicine" if patient values are ignored.

Diagram to draw

- Evidence pyramid and the three overlapping circles of EBM.

> **Key points to remember:**
> - Sackett 1996: best evidence + clinical expertise + patient values.
> - 5 As: Ask (PICO), Acquire, Appraise, Apply, Assess.
> - Level 1a = systematic review of RCTs; GRADE: strong or weak.
> - NNT = 1/ARR; beware lead-time, length-time and confounding bias.
> - Three questions: valid, results, applicable.
> - Randomisation, concealment, blinding, ITT, follow-up >80%.
> - CONSORT (RCT), PRISMA (reviews), STROBE (observational), STARD (diagnostic).

Also, from the condensed book (not in the paper answer):

- **Attribution bias** (cause of death misassigned) is the greatest limitation of disease-specific mortality, because a known cancer is often written on the certificate. **Detection bias** arises when one arm is imaged more.
- The **ICECaP** analysis (Intermediate Clinical End Points in Cancer of the Prostate) found that prostate-specific antigen (PSA) end points did not track overall survival, whereas **metastasis-free survival** did.

SAQ Recist Criteria.

Paper answer · P1

Type: SAQ (10 marks)
Source: 2024-Winter Paper 1, Q3a

Definition and evolution

- **RECIST (Response Evaluation Criteria In Solid Tumours):** standard imaging rules for tumour response and progression.
- **WHO (World Health Organization) 1981** (bidimensional) → **RECIST 2000** (unidimensional) → **RECIST 1.1 (2009)**, now standard. Used in trials for metastatic RCC (renal cell carcinoma), urothelial and prostate cancer.

Lesions at baseline (RECIST 1.1)

Category	Definition
Measurable lesion	Longest diameter **≥10 mm on CT (computed tomography)**, ≥20 mm on chest X-ray, ≥10 mm with callipers
Measurable node	**≥15 mm short axis**; 10–15 mm pathological but non-measurable; <10 mm normal
Non-measurable	Small lesions, blastic bone metastases, effusions, ascites
Target lesions	Up to **5 (maximum 2 per organ)**; largest, reproducible; **sum of diameters** recorded
Non-target	All other lesions, recorded as present or absent

Response categories

Response	Target lesions
CR (complete response)	All disappear; nodes <10 mm
PR (partial response)	**≥30% fall** in sum versus **baseline**
PD (progressive disease)	**≥20% rise** versus **nadir** and **≥5 mm absolute**, or any **new lesion**
SD (stable disease)	Neither PR nor PD

- **Overall response** combines target, non-target and new lesions; confirm response at ≥4 weeks in single-arm trials.
- **End points:** ORR (objective response rate = CR + PR), PFS (progression-free survival), duration of response.

Limitations

- Size only, ignoring necrosis (TKIs (tyrosine kinase inhibitors)); bone poorly assessed; observer variability.

Modifications

Criteria	Use
iRECIST (2017)	Immunotherapy: first progression is "unconfirmed", confirmed only on the next scan, allowing for **pseudoprogression**
Choi	Density fall counts as response
PCWG3 (Prostate Cancer Working Group 3)	Bone: 12-week scan is the new baseline (flare), then **≥2 new lesions** = progression; PSA (prostate-specific antigen) progression ≥25% and ≥2 ng/mL above nadir; soft tissue by RECIST
PERCIST	PET (positron emission tomography) metabolic response

Diagram to draw: number line of change in sum: −30% (PR), +20% (PD).

> **Key points to remember:**
> - RECIST 1.1: ≥10 mm lesion, node ≥15 mm short axis; 5 targets, 2 per organ.
> - PR ≥30% fall from baseline; PD ≥20% and ≥5 mm rise from nadir, or new lesion.
> - iRECIST for immunotherapy; PCWG3 for prostate.

3 · Complications

A complication is any deviation from the expected postoperative course. The Common Terminology Criteria for Adverse Events (CTCAE) are clinician-graded and organ-based, for oncology drugs.

CTCAE grade	Definition
1	Mild; observation only
2	Moderate; minimal intervention; limits **instrumental** activities of daily living
3	Severe; hospitalisation; limits **self-care** activities of daily living
4	Life-threatening; urgent intervention
5	Death

Clavien–Dindo grades by the therapy needed, not perceived severity, which reduces subjectivity.

Grade	Definition	Example
I	No drug, surgical or radiological treatment; antiemetics, analgesics, diuretics allowed; wound opened at bedside	Ileus after cystectomy managed with fluids
II	Other drugs, **including blood transfusion**	Transfusion after nephrectomy
IIIa	Intervention **without** general anaesthesia	Percutaneous drainage of a pelvic collection
IIIb	Intervention **under** general anaesthesia	Unplanned return to theatre
IVa	Life-threatening, intensive care; single-organ dysfunction (including stroke, not transient ischaemic attack)	Myocardial infarction after nephrolithotomy
IVb	Multiorgan dysfunction	Renal failure with respiratory distress
V	Death	—

It does not capture multiple complications, and anaesthesia type varies by institution.

4 · Predicting complication risk

Functional status, comorbidity and frailty predict complications across surgery; adjust for them in every non-randomised comparison.

ECOG	Status	Karnofsky
0	Fully active	100
1	Restricted strenuous activity; light work	80–90
2	Self-care, no work; up >50% of waking hours	60–70
3	Limited self-care; bed or chair >50%	40–50
4	Completely disabled	10–30
5	Dead	0

ECOG is the Eastern Cooperative Oncology Group scale.

- **Comorbidity** drives treatment choice, so it confounds. Administrative data showed higher mortality after transurethral resection of the prostate than open prostatectomy, but adjusting for comorbidity removed the difference.
- **Charlson Comorbidity Index** (predicts 1-year mortality): weight 1 for myocardial infarction, heart failure, vascular, pulmonary disease and diabetes; 2 for hemiplegia, renal disease and any malignancy; 3 for moderate–severe liver disease; **6** for metastatic cancer and AIDS.
- **Fried frailty phenotype**: frail means **≥3 of 5** criteria (weak grip, exhaustion, slow walk, low activity, unintentional weight loss).

5 · Patient-reported outcomes

SAQ Health-related quality of life (HRQOL) instruments in urology, including after urinary diversion

Paper answer · P1

Asked as:

- Health related quality of life instruments (HRQOL) in urology. — *Summer 2018 Paper I Q3c (SAQ)*
- Assessment of quality of life after urinary diversion — *Practice Set 3 Paper I Q3c (SAQ)*

Type: SAQ (10 marks)

Definition

- **HRQOL (HRQoL, health-related quality of life):** the parts of life affected by health: physical, emotional and social function and the patient's bother; the patient's view of how disease and treatment affect physical, emotional and social function, plus **bother**. It differs from a **symptom score**, which only counts symptoms.
- Must be **patient-reported**; doctors underestimate dysfunction (urinary, sexual and bowel dysfunction).

Types of instruments

Type	Measures	Examples	Limitation
Generic	Domains common to all diseases; allow comparison across diseases	**SF-36** (Short Form-36), SF-12, **EQ-5D** (EuroQol five-dimension; 5 items + VAS (visual analogue scale); gives utilities for QALYs (quality-adjusted life-years)), Sickness Impact Profile	Insensitive to diversion-specific issues
Cancer-specific	Cancer effects	**FACT-G** (Functional Assessment of Cancer Therapy–General), **EORTC QLQ-C30** (European Organisation for Research and Treatment of Cancer core questionnaire)	General cancer domains
Disease-specific	Domains of one disease; more responsive	See below	
Bladder-cancer-specific		**FACT-BI** (40 items), **EORTC QLQ-BLM30** (muscle-invasive module, 30 items), **BCI (Bladder Cancer Index**, 34 items: urinary, bowel, sexual domains), FACT-VCI (Vanderbilt Cystectomy Index)	Weight symptoms of one diversion, so biased in cross-diversion comparisons
Neobladder-specific		IONB-PRO (23 items)	Not usable across diversions

Urology-specific instruments

- **Prostate cancer: EPIC (Expanded Prostate Cancer Index Composite)**, EPIC-26, UCLA (University of California, Los Angeles) Prostate Cancer Index, FACT-P.
- **Bladder cancer:** FACT-BI, **Bladder Cancer Index**, EORTC QLQ-BLM30 and NMIBC-24, FACT-Vanderbilt Cystectomy Index.
- **Kidney cancer:** FKSI (FACT–Kidney Symptom Index).
- **LUTS (lower urinary tract symptoms): IPSS (International Prostate Symptom Score)** 7 items + 1 quality-of-life item; BPH Impact Index.
- **Incontinence and OAB (overactive bladder): ICIQ** (International Consultation on Incontinence Questionnaire), **OAB-q**, King's Health Questionnaire, I-QOL.
- **Sexual function: IIEF** (International Index of Erectile Function; 15 items), SHIM (5 items; no bother, so not true HRQOL), FSFI (Female Sexual Function Index).
- **Others:** NIH-CPSI (chronic prostatitis), O'Leary–Sant (interstitial cystitis), Wisconsin Stone QOL, USSQ (ureteric stent).
- **PROMIS (Patient-Reported Outcomes Measurement Information System):** free item banks, **computer-adaptive testing**, common T-score.

Psychometric properties (choosing a tool)

- A good tool is **reliable** (Cronbach α >0.7), **valid** in that population and **responsive** to change.

Property	Meaning	Threshold
Internal consistency	Items agree	Cronbach α >0.7 (>0.9 = redundant)

Property	Meaning	Threshold
Test–retest reliability	Stable over time	r >0.7
Content (face) validity	Right items present	Expert judgement
Criterion validity	Agrees with gold standard	r ≥0.7
Construct validity	Convergent and divergent	Correlations
Responsiveness	Detects real change	**MCID (minimal clinically important difference)**: AUA (American Urological Association) Symptom Index 3.1 points; otherwise ≥½ SD

- Validity is population-specific; translations need revalidation.
- Development: define domains → generate items → pilot → psychometric testing (Likert scales).

Quality of life after urinary diversion

Domains to assess

- Urinary: continence, catheterisation, stoma problems.
- **Body image**, sexual function, bowel function, social and emotional function, global HRQoL.

Evidence: ileal conduit versus neobladder

- **No diversion proven superior.** Studies are mostly cross-sectional and retrospective, without baseline data or randomisation.
- Verdict depends on instrument: generic tools mostly show **no difference** (largest group of studies); EORTC tools tend to **favour neobladder**; the urinary-focused BCI **penalises neobladder** for leakage.
- **Consistent signal:** body image favours neobladder; urinary function favours conduit.
- Large prospective cohort: global HRQoL exceeded baseline in all diversions by 6 months.
- Neobladder: daytime continence improves over 6–12 months; **night-time incontinence** (up to 50%) is the main bother.
- **Also know:** informed patients have less decision regret; most adapt to their diversion; PROMIS (Patient-Reported Outcomes Measurement Information System).

Practical approach

- Baseline and serial validated questionnaires (3, 6, 12 months), a disease-specific plus a generic tool, and realistic counselling.

Diagram to draw: Pyramid: symptoms → function → bother → overall HRQOL. Bar sketch: body image (neobladder better) versus urinary function (conduit better).

Key points to remember:

- Generic (SF-36, EQ-5D) vs disease-specific (EPIC, IPSS, ICIQ, IIEF) plus bladder-specific (FACT-Bl, BLM30, BCI) tools.
- SHIM lacks bother; I-QOL lacks symptoms.
- Reliable, valid, responsive; Cronbach α >0.7; IPSS MCID ~3 points.
- No diversion proven superior; studies flawed.
- Neobladder better body image; conduit better urinary function.

Also, from the condensed book (not in the paper answer):

- Clinicians consistently underestimate urinary, sexual and bowel dysfunction, so collect outcomes directly from patients. A symptom score counts what is wrong; an HRQOL instrument measures function plus bother.
- The International Prostate Symptom Score has 7 items scored **0–35**; the Female Sexual Function Index has **19 items in 6 domains**. Pad tests cannot separate stress from urgency incontinence.

★ Putting it together: everyday decisions

1 · Which imaging test first?

Question	First test
Renal colic	Low-dose non-contrast computed tomography (CT); ultrasound in pregnancy and children
High-risk haematuria	CT urography and cystoscopy
Renal mass	Multiphase CT or magnetic resonance imaging (MRI); enhancement >15–20 Hounsfield units is suspicious
Obstruction, split function	Mercaptoacetyltriglycine (MAG3) diuretic renogram
Prostate cancer before biopsy	Multiparametric MRI
Acute scrotum	Doppler ultrasound, never delaying exploration
Urethral stricture	Retrograde urethrogram

2 · Before theatre

ALGORITHM

1. Assess function, comorbidity and frailty.
2. Stop warfarin 5 days before (international normalised ratio <1.5), bridging only for high thrombotic risk; stop direct oral anticoagulants 48 h before without bridging; withhold metformin on the day.
3. Culture urine before instrumentation; give prophylactic antibiotics.
4. Give extended 28-day low-molecular-weight heparin after pelvic cancer surgery.
5. Complete the World Health Organization checklist.

3 · Irrigation and the infected obstructed kidney

*See the paper answer **TUR syndrome** (Section 16, Ch 151).*

Also, from the condensed book (not in the paper answer):

- Use saline for bipolar resection, laser and ureteroscopy; never use sterile water for resection, as it causes haemolysis and hyponatraemia.

Other paper questions in this section

Asked in the papers but not tied to a specific place in the book above.

LAQ Tissue engineering in urology: principles, status and recent advances

Paper answer · P1 · P3 · P4 · merged from 3

Asked as: P1 LAQ "Principles of tissue engineering and its use in urology" · P3 SAQ "Describe principles of tissue engineering and what its status in managemet of urological diseases is." · P4 LAQ "Tissue engineering in urology, including recent advances"

Definition and rationale

- **Tissue engineering** applies engineering and life sciences to build biological substitutes that restore, maintain or improve tissue function (Langer and Vacanti, 1993).
- **Regenerative medicine** is the wider term: cells, scaffolds, growth factors and gene therapy.
- **Why in urology:** bowel used for bladder and diversion causes mucus, stones, metabolic acidosis, infection and malignancy; grafts for long urethral defects are scarce. The aim is **to replace "wrong tissue" with "right tissue"**.

Principles: the triad plus bioreactor and blood supply

``` Cells + Scaffold + Signals (growth factors, mechanical cycling in a bioreactor) -> construct -> implanted -> vascularised and remodelled by the host

Biopsy -> cells expanded in culture -> seeded on scaffold (+ signals) -> conditioned in bioreactor -> implanted -> host vascularises and remodels ```

### Cell sources

| Source | Features, advantages and limitations |
|---|---|
| **Autologous differentiated cells** (urothelial and smooth muscle cells from a bladder biopsy) | Standard; no rejection; expand well in culture; unusable if the bladder is malignant or diseased |
| **Adult stem cells:** bone marrow or adipose mesenchymal, muscle-derived, **urine-derived** | Easy, non-invasive harvest; act mainly by paracrine (growth factor) effects |
| **Embryonic stem cells** | Pluripotent; ethical issues, teratoma risk |
| **iPSCs** (induced pluripotent stem cells) | Adult cells reprogrammed; patient-specific, no ethical issue; tumour risk |

- Gene-edited and gene-modified cells are being explored. Cell sheet engineering builds layers without a scaffold.

### Scaffolds (biomaterials)

- **Requirements:** biocompatible; biodegradable at the rate new tissue forms; porous for cell entry, attachment and vessel ingrowth; mechanically suited (a compliant bladder, a non-collapsing tube).

| Type | Examples | Feature |
|---|---|---|
| **Natural polymers** | Collagen, alginate, silk fibroin | Good cell recognition; weak |
```

Type	Examples	Feature
Acellular tissue matrices	Bladder acellular matrix, SIS (small intestinal submucosa), decellularised dermis	Keep native architecture and stored growth factors
Synthetic polymers	PGA (polyglycolic acid), PLA (polylactic acid), PLGA (their copolymer, poly(lactic-co-glycolic) acid)	Controlled shape and degradation; no bioactivity
Newer	Electrospun nanofibres, hydrogels, bio-ink; **"smart" scaffolds** releasing growth factors such as VEGF (vascular endothelial growth factor); silk fibroin	Mimic extracellular matrix

- **Seeding:** urothelial cells on the luminal side and smooth muscle cells on the outer side, copying native layers.
- **Unseeded (acellular)** matrices rely on host cells migrating in, so they work only for short onlay defects (about 0.5 cm for tubes in animal work). Longer or tubular gaps need **seeded** scaffolds, or they fibrose and contract.

Signals

- Growth factors (for example VEGF, for vessels) released from the scaffold, extracellular-matrix cues and mechanical stimuli direct cell growth and differentiation.

Bioreactors

- Culture chambers that perfuse nutrients and oxygen through thick constructs and apply **physiological conditioning**, such as cyclic stretch and filling to train bladder smooth muscle, so the tissue is functional before implantation.
- Types: static culture, spinner flask, rotating wall, perfusion and mechanical-stretch bioreactors. They also allow monitoring and sterile, standardised production.

Vascularisation and innervation (the main barrier)

- Oxygen diffuses only about **100–200 µm**, so thick constructs die centrally without vessels.
- Strategies: **omental wrap** (used by Atala), angiogenic factors, co-seeding endothelial cells, built-in (printed) vascular channels by bioprinting.
- Innervation and coordinated contraction cannot yet be recreated.

Clinical applications and status in urology

1. Bladder

- **Atala et al. (2006):** first seeded biodegradable bladder construct (autologous urothelial and muscle cells on collagen–PGA, wrapped in omentum) for augmentation in **7 myelomeningocele patients**; compliance improved.
- **Industry (Tengion) multi-institution phase II trial** in spina bifida: **no clinically or statistically significant improvement in compliance or capacity** (Chapter 48).
- Chapter 48 conclusion: a bladder with specialised urothelium and contractile muscle has **not** been achieved; it remains experimental.
- Chapter 111: an engineered bladder gives **storage only**; innervation for voluntary emptying cannot yet be recreated.

2. Urinary diversion: Neo-Urinary Conduit (Tengion, NCT01087687) (Chapter 135)

- Seeded biodegradable conduit to avoid bowel. Small enrolment; conduits survived about 250 days on average; **half developed stenosis or stricture.**
- Lessons: stiff, non-collapsible material; no cells formed around the connective-tissue sleeve; **no blood supply**.

3. Urethra

- Acellular bladder matrix onlay for hypospadias and short strictures: success in short onlay repairs.
- **Raya-Rivera et al. (2011, Lancet):** tubular PGA–PLGA scaffolds seeded with autologous cells in **5 boys** with posterior urethral defects; functional urethras at up to 6 years.
- Tissue-engineered oral mucosa (MukoCell) for urethroplasty, avoiding donor-site morbidity.

4. Other organs

- **Sphincter, detrusor:** autologous muscle-derived cell injection for SUI (stress urinary incontinence), investigational; for underactive detrusor, open-label studies improved voiding efficiency (Chapter 111). Biodegradable scaffold slings (Chapter 48).
- **Vagina:** engineered vaginas in MRKH (Mayer–Rokitansky–Küster–Hauser) syndrome (Raya-Rivera, 2014); off-the-shelf SIS grafts (Chapter 59).
- **Penis:** engineered corporal tissue and tunica grafts (stem cell-seeded SIS in Peyronie disease), animal or experimental stage. ED (erectile dysfunction) stem cell therapy is investigational (Chapter 72).
- **Kidney:** decellularised–recellularised scaffolds (animal); renal assist devices; kidney organoids.
- **Testis:** spermatogonial stem cell banking for prepubertal cancer patients (future, Chapter 4).

Organ	Status
Bladder	**Atala (2006):** seeded collagen–PGA construct with omental wrap in **7 myelomeningocele patients**, compliance improved. Industry phase II trial: **no significant gain in compliance or capacity**. Still experimental (Chapter 48)
Conduit	Neo-Urinary Conduit trial (NCT01087687): survived about 250 days on average; **half developed stenosis or stricture**; stiff material, no blood supply (Chapter 135)
Urethra	Acellular matrix onlay for short strictures and hypospadias; **Raya-Rivera (2011):** seeded tubular PGA–PLGA urethras in **5 boys**, functional up to 6 years; tissue-engineered oral mucosa avoids donor-site morbidity
Sphincter, detrusor	Autologous muscle-derived cells for SUI (investigational); biodegradable scaffold slings; underactive detrusor: open-label improvement in voiding efficiency (Chapter 111)
Vagina	Engineered vaginas in MRKH syndrome (Raya-Rivera, 2014); off-the-shelf SIS grafts (Chapter 59)
Penis	Corporal tissue and tunica grafts: animal or experimental; stem cells for ED investigational
Kidney	Decellularised–recellularised scaffolds (animal), kidney organoids, renal assist devices: experimental
Testis	Spermatogonial stem cell banking in prepubertal cancer patients

- **Overall status:** still experimental except short urethral onlays.

Recent advances

3D bioprinting

- Layer-by-layer deposition of **"bio-ink"** (cells in hydrogel), with or without polymer, driven by patient CT (computed tomography) or MRI (magnetic resonance imaging) data.
- Advantages: exact patient-specific shape, several cell types placed precisely, built-in channels for vessels.
- Current urological use: 3D-printed (non-living) kidney and prostate models already aid planning and teaching; bioprinted urethra and bladder constructs are still at the animal stage.

Organoids

- 3D self-organising mini-organs grown from stem cells or tumour tissue.
- **Kidney organoids** from iPSCs: model development and disease, drug toxicity testing.
- **Tumour organoids** (bladder, prostate): personalised drug testing and biology (WNT/β-catenin work, Chapter 129).

Whole-organ decellularisation

- Kidneys are stripped of cells by detergent perfusion, leaving the matrix and vascular tree, then recellularised with the patient's cells; still at the animal stage. Renal assist devices are a parallel route.

Other

- Robotic bladder auto-transplantation and robotic retrieval of a **vascularised composite bladder allograft** (Chapter 135): a transplant, not engineered, route to "right tissue" that avoids bowel but needs immunosuppression.
- Smart (growth factor-releasing) scaffolds; urine-derived stem cells; gene-edited and gene-modified cells; cell sheet engineering.

Challenges

- **Vascularisation** of thick constructs (diffusion limit), **innervation** and contractility.
- Fibrosis and contracture; scale-up, cost, culture time; regulatory approval.
- Stem cell tumour risk. Promising pilots followed by disappointing controlled trials.

Future

- Vascularised bioprinted tissue, off-the-shelf cell-free smart scaffolds, combined gene and cell therapy.

Diagram to draw

- Biopsy, then cell culture, then seeding on scaffold, then bioreactor, then implant with omental wrap.

Key points to remember:

- Triad: cells, scaffold, signals; plus bioreactor and blood supply.
- Diffusion limit about 100–200 µm: vascularity is the main barrier.
- Atala 2006 (7 patients) promising; phase II trial: no gain in compliance or capacity.
- Neo-urinary conduit: half stenosed; no blood supply.
- Raya-Rivera 2011: seeded tubular urethras in 5 boys.
- Still experimental except short urethral onlays.
- New: iPSCs, urine-derived cells, bioprinting, organoids.

LAQ Robotics in urology (including robotic vs standard laparoscopy)

Paper answer · P4

Asked as:

- Role of robotics in urology — *Summer 2015 Paper IV Q1 (LAQ)*
- Discuss Robot and its role in urology. — *Winter 2017 Paper IV Q1 (LAQ)*
- Robots in urology — *Practice Set 4 Paper IV Q2 (LAQ)*
- Robotic vs std lap surgeries — *Practice Set 3 Paper IV Q3b (SAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)
(also answered in 2017-Winter Paper 4, Q1; 2015-Summer Paper 4, Q1) (robotics also answered in 2017-Winter Paper 4, Q1 and 2015-Summer Paper 4, Q1)

Introduction

- Surgical robots are **master–slave telemanipulators**: the surgeon controls every movement; the robot acts on nothing by itself.
- Robotic surgery is **robot-assisted laparoscopy** with a **master–slave telemanipulator** (surgeon console, patient-side cart, vision cart). Access and physiology are the same; the interface differs.
- The robot did not change the operations. It removed the barrier of laparoscopic suturing, so RARP (robot-assisted radical prostatectomy) spread where laparoscopic prostatectomy had stalled. RARP now dominates in the USA.

History

- **PUMA 560 (1985)** first surgical robot; **PROBOT** first urological robot (TURP (transurethral resection of prostate)); "Mona" first human trial (1997).
- **da Vinci** FDA (Food and Drug Administration) approval and first RARP (**Binder and Kramer**) in **2000**; robotic adrenalectomy 1999; first paediatric robotic procedure **2002**; robotic RPLND (retroperitoneal lymph node dissection) 2006.

da Vinci system

1. **Surgeon console**: 3D (three-dimensional) view magnified about 10 times, motion scaling, tremor filtration.
2. **Patient-side cart**: **EndoWrist** instruments with **7 DOF (degrees of freedom)**; most have a 10-use limit; ports 8 mm, 8–10 cm apart to avoid arm collisions.
3. **Vision cart**: light source, camera electronics, energy generator.

Generation	Added feature
Standard (2000)	Three arms
S (2006) / Si (2009)	High definition, fourth arm / dual console for training
Xi (2014)	8-mm camera in any port, finer arms, side docking (docks from either side), no stirrups
SP (2018)	All instruments through one 2.5-cm cannula
da Vinci 5 (2024)	Force feedback

- Patents expired in 2019 (after patent expiry), so other platforms appeared (e.g. **Hugo, Versius** and the low-cost Indian **SSI Mantra**).
- **Also know:** ports 8–10 cm apart to avoid arm collisions; Xi docks from any side; retroperitoneal cases dock over the head.

Robotic vs standard laparoscopy: comparison

Feature	Standard laparoscopy	Robotic
Vision	Mostly 2D; assistant holds camera	**3D**, 10× magnified, surgeon controls camera
Instruments	Rigid, **4 DOF (degrees of freedom)**; fulcrum effect reverses movement	**EndoWrist, 7 DOF**, 90° articulation; intuitive movement
Precision	Tremor transmitted	**Tremor filtration, motion scaling**
Ergonomics	Standing, awkward	Seated, less fatigue
Suturing	Hard; steep learning curve	Easy; shorter learning curve
Haptic feedback	Present (limited)	Absent (force feedback only in newest system)
Port placement	Flexible	Ports 8–10 cm apart; docking time
Bedside	Surgeon at patient	Surgeon remote; emergency undocking delays
Cost	Low, reusable	High capital; instruments limited to about 10 uses
Failure	—	Robot failure about 0.4%

Advantages and disadvantages

Advantages	Disadvantages
3D vision, wristed instruments, easy suturing	High capital and instrument cost; cost-effective only at high volume
Tremor filtration, seated ergonomics	Limited haptic feedback
Less blood loss and shorter stay than open; fewer conversions than laparoscopy	Docking time, bulk, surgeon away from patient

Applications (with evidence)

1. RARP

- **Yaxley (Lancet 2016)**, RCT (randomised controlled trial): continence and potency **equal to open at 12 weeks**, with less blood loss and shorter stay; **Coughlin 2018**: equal at **24 months** on follow-up.
- Positive margins equal; the surgeon, not the platform, is the main variable. Bladder neck stricture under 2% vs up to 16% open.
- **Retzius-sparing** and **"hood"** techniques preserve continence structures and give earlier continence; ERAS (enhanced recovery after surgery) allows 23-hour or same-day discharge. Robotic surgery also widens candidacy (prior mesh, prior pelvic surgery) and allows salvage surgery.
- Versus laparoscopy: laparoscopic prostatectomy stalled on the vesicourethral anastomosis; the robot made it routine and dominant. A small RCT (randomised controlled trial; Porpiglia, 2013) favoured robotic continence and potency.

2. RAPN (robot-assisted partial nephrectomy)

- Versus laparoscopy (robotic vs laparoscopic gives): **fewer conversions**, **shorter warm ischaemia**, less (smaller) eGFR (estimated glomerular filtration rate) loss/fall and shorter stay; complications and margins similar. So more complex and hilar tumours are done minimally invasively.
- **Trifecta** = negative margin, no major complication within 30 days, warm ischaemia ≤25 minutes.
- ICG (indocyanine green) guides selective clamping.

3. RARC (robot-assisted radical cystectomy)

- **RAZOR (Lancet 2018)**: **non-inferior** 2-year PFS (progression-free survival) (72.3% vs 71.6%), with less blood loss and fewer transfusions (less transfusion).
- **iROC**: robotic cystectomy with **intracorporeal** diversion gave **more days alive and out of hospital at 90 days**, fewer VTE (venous thromboembolism) and wound complications. Continent diversion fell (11% in iROC).

4. Other procedures

Procedure	Role of the robot / why the robot helps
Pyeloplasty, ureteric reimplantation	Easy intracorporeal suturing; also in children and infants (small working space: about 1 L vs 6 L in adults); similar success to laparoscopy; easier suturing
BMG (buccal mucosa graft) ureteroplasty	Long proximal or mid-ureteric strictures
RPLND (retroperitoneal lymph node dissection)	Template, nerve-sparing; short stay
Adrenalectomy, nephroureterectomy	Equivalent to laparoscopy; adrenalectomy suits the retroperitoneal approach
Simple prostatectomy	Large glands; less bleeding than open
Sacrocolpopexy, VVF (vesicovaginal fistula) / fistula repair	Deep pelvic suturing
RAKT (robot-assisted kidney transplantation)	Fewer wound complications in obese recipients

- **Radical nephrectomy, adrenalectomy:** no outcome advantage over laparoscopy; robot only adds cost.
- Outcomes depend more on the **surgeon** than on the platform.

Choosing robot vs laparoscopy

- Robot: complex excision plus reconstruction. Laparoscopy: extirpation, low-resource settings.

Paediatric robotics

- Reconstruction (pyeloplasty, reimplantation, appendicovesicostomy) is where paediatric robotics has grown. Lower insufflation pressure; "burp" ports to enlarge the working space.

Physiology and positioning

- **CO_2 pneumoperitoneum** (12–15 mm Hg, lower in children; another source: work at 10–12 mm Hg) causes hypercarbia, acidosis, reduced venous return and oliguria.
- **Steep Trendelenburg** for pelvic surgery increases venous return and cardiac output, and raises intracranial and intraocular pressure.
- The only strict contraindication to minimally invasive surgery is inability to tolerate pneumoperitoneum.

Complications specific to robotics

Problem	Details
Steep Trendelenburg (~22–25°)	Facial and laryngeal oedema, raised IOP (intraocular pressure), corneal abrasion
Nerve injury (up to ~6%)	Brachial plexus (avoid shoulder braces); femoral, peroneal
Compartment syndrome, rhabdomyolysis	Long lithotomy or flank position; fasciotomy if >30 mmHg with symptoms
Thermal bowel injury / bowel burns	Insulation failure, capacitive coupling; presents late
Robot malfunction	About 0.4% (another source: <1%); half found before the patient enters; rarely forces conversion
Port-site hernia	Close larger port sites; transverse incisions cut it from 5.4% to 0.4% after RARP

- **Prevention**: secure inaccessible arms and lines before tilting, pad pressure points, limit tilt and lithotomy time, protect the eyes.

Training and credentialing

- Box trainer, **VR (virtual reality)** simulator (simulators), animal and cadaver labs, then proctored cases (proctoring). **FRS (Fundamentals of Robotic Surgery)** curriculum improved skills in an RCT. Five-day mini-residencies also help. Dual console and simulators for training. Proficiency comes after about 40 cases, but outcomes keep improving (outcomes improve) with volume.

Recent advances

- Single-port transvesical and extraperitoneal surgery; da Vinci SP (single port); force feedback (da Vinci 5); **AI (artificial intelligence)** video-based skill assessment; augmented reality 3D overlay; **telesurgery** (limited by latency; latency limits it).
- New platforms (Hugo RAS, Versius, Indian SSI Mantra) may cut costs.

- **Also know:** ERAS (enhanced recovery after surgery) with same-day RARP discharge; ICG (indocyanine green) fluorescence for selective clamping and ureteric perfusion.

Diagram to draw: console, vision cart and patient cart; RARP port placement; EndoWrist (7 DOF) beside a rigid instrument (4 DOF).

> **Key points to remember:**
> - da Vinci 2000; Xi 2014; SP 2018; patents expired 2019; 7 degrees of freedom.
> - Yaxley and Coughlin: RARP equal function to open at 12 weeks and 24 months; RAZOR non-inferior; iROC more days out of hospital.
> - RAPN trifecta: negative margin, no major complication, warm ischaemia ≤25 min.
> - Trendelenburg causes oedema and raised IOP; avoid shoulder braces.
> - Robot: 3D, 7 DOF, tremor filtration, but no haptics and high cost.
> - Robotic partial nephrectomy: shorter warm ischaemia, fewer conversions.
> - Robot made prostatectomy routine; no benefit for simple extirpation.
> - Surgeon matters more than platform.

LAQ Natural orifice endoscopic surgery.

Paper answer · P4

Type: LAQ (25 marks)
Source: 2022-Summer Paper 4, Q1

Taken as NOTES (natural orifice transluminal endoscopic surgery) and its relatives (hybrid NOTES, LESS [laparoendoscopic single-site surgery]).

Definition and concept

- **NOTES:** entry into the peritoneal or retroperitoneal cavity through a deliberate incision in a hollow viscus reached through a natural orifice (mouth, vagina, bladder, rectum). The aim is **"scarless" surgery**: no abdominal wall wound, so no wound pain, infection or hernia.
- It differs from **endoluminal** surgery (e.g. ureteroscopy), which stays inside the lumen.
- **Pure NOTES:** all instruments through the orifice. **Hybrid NOTES:** natural orifice plus one or more small abdominal ports. Hybrid is what is actually used in humans.
- Chapter 142 places NOTES at the end of a spectrum: transperitoneal → retroperitoneal → hand-assisted → LESS → SP (single-port) robotic → NOTES. **Each step to fewer incisions buys cosmesis and less pain at the cost of working space and triangulation.**

History (milestones)

Year	Event
2002	Transvaginal laparoscopic nephrectomy in pigs (Gettman, Cadeddu)
2004	Transgastric peritoneoscopy in pigs (Kalloo)
2006	NOSCAR (Natural Orifice Surgery Consortium for Assessment and Research) white paper listed barriers
2008–2010	Hybrid, then pure, transvaginal nephrectomy in humans

Routes of access

Route	Features	Urological use
Transvaginal (posterior colpotomy)	Commonest; easy closure under vision; large specimens pass; women only	Nephrectomy, donor nephrectomy, specimen extraction
Transgastric	Upper abdomen reachable; gastrotomy closure and leak are the main risk	Experimental (peritoneoscopy, renal surgery in animals)
Transvesical	Sterile urine; bladder dome port; small calibre; closure difficult	Peritoneoscopy (animal, Lima), transvesical access to pelvis
Transurethral	Urologist's familiar route	Experimental transurethral prostatectomy in cadavers
Transcolonic/transanal	Faecal contamination risk	Rarely urological

Hybrid NOTES and natural orifice extraction

- Standard laparoscopy does the dissection; **the vagina is the access for a camera or instrument and the extraction route** (Chapter 142). This is the only NOTES form with real clinical urological use.
- **NOSE (natural orifice specimen extraction):** kidney removed through the vagina after laparoscopic nephrectomy, avoiding the extraction incision, which is the largest and most painful wound.
- **NOTES-assisted donor nephrectomy:** laparoscopic or robotic donor nephrectomy with **transvaginal kidney extraction**. Reported benefits: less pain, faster recovery, no extraction scar; warm ischaemia acceptable.

Steps of transvaginal (hybrid) nephrectomy

1. Lithotomy then modified flank; bowel and vaginal preparation; antibiotics.
2. Umbilical port (camera or instrument) plus one or two small ports.
3. **Posterior colpotomy** under laparoscopic vision; long port or flexible scope through the vagina.
4. Colon reflection, hilar dissection, ureter and vessels stapled.
5. Kidney in a bag, delivered through the colpotomy.
6. Colpotomy closed with absorbable sutures under vision.

LESS: the close relative

- All ports through **one skin incision** (umbilicus or below the waistline) using several low-profile trocars or a multichannel port (TriPort; da Vinci SP 25–27-mm entry guide).
- Problems: **no triangulation**, instrument clashing, in-line view. Solutions: articulating or bent instruments, flexible-tip scopes, instruments of different lengths.
- Evidence (Chapter 142): randomised trial versus laparoscopy showed the same time, blood loss and complications, with less analgesia. Donor meta-analysis: lower pain scores. Oncological outcome unchanged.
- Drawback (Chapter 20): **more incisional and umbilical hernias** because of the larger single incision.
- **SP robot (FDA [Food and Drug Administration] approval 2018)** has revived single-site surgery.

vNOTES (vaginal NOTES)

- A gynaecological development: a single-port device in the vagina gives access to the peritoneum.
- **HALON RCT (randomised controlled trial):** vNOTES hysterectomy was non-inferior to laparoscopic hysterectomy, with more same-day discharge and less pain.
- Urogynaecological uses: vNOTES apical prolapse repair (uterosacral suspension), adnexal surgery.

Challenges (NOSCAR list)

- **Safe access** and **secure closure** of the viscerotomy (leak, fistula); peritoneal **infection**.
- **Spatial orientation**: retroflexed, off-axis view.
- **Platform**: flexible scopes lack stiffness, traction and triangulation.
- Control of **haemorrhage** and other complications.
- Training and cost; dyspareunia and fertility concerns after colpotomy; transvaginal route unavailable in men.
- Chapter 142: adoption is limited by **no clear therapeutic benefit**, safety concerns and very selective patients; **pure NOTES has very few urological cases.**

Recent advances

- **Flexible robotic platforms** (Chapter 39): MASTER (NOTES, haptics), ViaCath.
- **Transvesical single-port** (LESS or da Vinci SP) surgery for diverticulectomy and foreign bodies (Chapter 124).
- Magnetic anchoring and guidance systems to restore triangulation.

Also know: LESSCAR consortium standardised the term LESS; "embryonic NOTES" (umbilicus as the natural scar); hidden-incision endoscopic surgery (HIdES) in children.

Diagram to draw: sagittal female pelvis showing posterior colpotomy with a transvaginal port and kidney extraction, plus umbilical port.

> **Key points to remember:**
> - NOTES = transluminal via natural orifice; hybrid transvaginal is the only routinely used urological form.
> - Transvaginal extraction removes the most painful wound.
> - LESS: one incision, lost triangulation, more hernias.
> - Barriers: closure, infection, orientation, platform; no proven benefit yet.

SAQ Pioneers of urology: Novick, Chaussy, Hopkins, Huggins, Calne, Gaur and Asopa

Paper answer · P1 · merged from 3

Asked as: P1 SAQ "Pioneers of urology: Hopkins, Huggins, Calne, Gaur and Asopa" · P1 SAQ "Andrew Novick." · P1 SAQ "Christian Chaussey"

Names read as Prof. Harold Hopkins ("Herald"), Dr Charles Huggins, Prof. Sir Roy Calne ("Caine"), Dr D. D. Gaur ("Gaue"), Hari Asopa, Andrew C. Novick and Christian Chaussy ("Chaussey"). The question is taken to ask for each person's contributions to urology. Novick is answered mainly through nephron-sparing surgery, and Chaussy through the history and principles of SWL.

Prof. Harold Hopkins (British physicist)

- **Fibreoptics (1954, with Kapany):** coherent bundles of glass fibres transmit an image by total internal reflection (Tyndall's principle). This made flexible endoscopes possible.
- **Rod-lens system (1966):** the old scope used thin glass lenses separated by long air spaces. Hopkins replaced them with **long glass rods separated by short air gaps**. Result: more light transmission, a brighter and wider image, and a **smaller calibre**. Developed commercially with Karl Storz.
- It underpins all modern rigid endoscopes and enabled the first rigid ureteroscope (1979).

Dr Charles B. Huggins (1901–1997)

- Canadian-born American urologist, University of Chicago.
- **1941, with Clarence Hodges:** castration (or oestrogen) relieved far-advanced disease and caused **regression of metastatic prostate cancer**, with falling serum acid phosphatase, pain relief and weight gain; injected androgens made it worse. This proved that **prostate cancer is hormone (androgen) dependent**.
- Used serum **acid phosphatase** to monitor response: the first tumour marker.
- **1945:** bilateral adrenalectomy for castration-resistant disease (relapse after castration, suspecting adrenal androgens); modest, brief responses; morbidity high.
- **Nobel Prize in Physiology or Medicine, 1966** (shared with Peyton Rous) for hormonal treatment of prostate cancer: the basis of hormonal cancer therapy.
- **Legacy:** basis of all ADT (androgen deprivation therapy): orchidectomy, oestrogens, LHRH (luteinising hormone-releasing hormone) agonists and antagonists, and later abiraterone (blocking adrenal androgens).
- **Also know:** canine prostate physiology; adrenalectomy for breast cancer.

Prof. Sir Roy Calne (Cambridge transplant surgeon)

- Showed in dogs that **6-mercaptopurine and then azathioprine** prolong renal allograft survival; azathioprine with steroids became standard immunosuppression.
- Performed the **first liver transplant in the UK (1968)**.
- **First clinical use of ciclosporin** in kidney transplantation (late 1970s), the calcineurin-inhibitor era; acute rejection fell markedly.

Dr D. D. Gaur (Mumbai, India)

- **1992: balloon dissection of the retroperitoneum.** A size 7 surgeon's glove tied on an 8-Fr red rubber catheter, inflated inside the retroperitoneum, creates a working space.
- This made **retroperitoneoscopy** practical and cheap; used for ureterolithotomy, nephrectomy and other renal surgery.

Hari S. Asopa (Indian urologist)

1. Asopa dorsal inlay urethroplasty (2001)

- Single-stage graft urethroplasty for anterior urethral stricture by a **ventral sagittal urethrotomy** approach.
- **Steps:** expose the urethra ventrally without circumferential mobilisation; open the urethra ventrally through the stricture; incise the **dorsal urethral plate** in the midline down to the corpora; **inlay** a free graft (buccal mucosa or penile/preputial skin) and quilt it to the tunica albuginea; close the urethra over a catheter.

- **Rationale:** the graft lies on the well-vascularised corporal bed; no mobilisation preserves spongiosal blood supply; suits bulbar and penile strictures.
- Ranks with **Barbagli dorsal onlay** and **Kulkarni one-sided** repair among dorsal graft techniques.

2. Asopa hypospadias repair (1970s)

- One-stage repair using a **vascularised inner preputial island flap** still attached to the outer preputial skin, transferred ventrally to form the neourethra; a forerunner of the Duckett transverse preputial island flap.

Andrew C. Novick

Who he was

- A **Cleveland Clinic** urologist and long-time chairman of its urology department. He was a world authority on renal surgery, an editor of *Campbell-Walsh Urology* (9th edition), and died in 2008.

Contributions

1. **Nephron-sparing surgery (NSS) / partial nephrectomy (PN):** he established its technique, indications and long-term cancer control. His group showed tumours ≤4 cm did as well with NSS as with radical nephrectomy, supporting the T1a/T1b split.
2. **Renovascular surgery:** aortorenal, splenorenal and hepatorenal bypass; **bench (ex vivo) surgery with autotransplantation** for complex lesions.
3. **Renal transplantation** surgery.

Indications for PN

Category	Situations
Imperative	Bilateral RCC (renal cell carcinoma); solitary functioning kidney
Relative	Contralateral kidney threatened (stones, stenosis, diabetes, hypertension); hereditary RCC
Elective	cT1a (≤4 cm) with a normal other kidney; now larger selected tumours

- **Relative contraindications:** <20% of nephron mass left; pedicle encasement; venous thrombus.

Principles of NSS

1. Three-dimensional CT (computed tomography) maps tumour, vessels and collecting system.
2. **Early hilar control.** Keep warm ischaemia under about 25–30 minutes; if longer, use **ice-slush hypothermia** and mannitol.
3. Excise with a **negative margin** (width immaterial); intraoperative ultrasound for endophytic tumours.
4. **Watertight collecting-system closure**, vessel ligation, capsular renorrhaphy.
5. Drain for urine leak. - Function afterwards depends on parenchymal **quality and quantity preserved**; ischaemia matters less. At least 20–30% of one kidney is needed to avoid dialysis.

Legacy

- PN is now the AUA (American Urological Association) and EAU (European Association of Urology) standard for cT1a tumours.

Christian Chaussy (Munich): extracorporeal shock wave lithotripsy

Contribution and history

- Engineers at **Dornier** found by chance that shock waves from raindrop and micrometeorite research affected tissue.
- Chaussy, with Brendel and Schmiedt in **Munich**, developed the technique in animals and performed the **first human SWL (shock wave lithotripsy) on 7 February 1980 on the Dornier HM1**. They reported it in the *Lancet* in 1980.
- **HM3** (water bath, electrohydraulic spark, biplanar fluoroscopy): first installations Stuttgart 1983 and Indianapolis 1984. It remains the reference for efficacy.
- SWL made most open stone surgery obsolete.

Principles

- A shock wave is a short acoustic pulse: **peak positive 30–100 MPa, then a negative tail of about −10 MPa**.
- Generated at **F1**, focused onto the stone at **F2** through water or gel coupling (water and tissue have similar acoustic impedance); targeting by fluoroscopy or ultrasound.

Energy source	Feature
Electrohydraulic (HM1, HM3)	Underwater spark, ellipsoid reflector; most effective; electrode wear
Piezoelectric	Ceramic elements; long life, little pain; small focal zone, low energy
Electromagnetic (modern)	Coil and membrane with lens; consistent; dominant today

Mechanisms of fragmentation

- **Spallation:** reflected wave at the back surface creates tension.
- **Shear and tear** inside the stone.
- **Quasistatic squeezing** and **dynamic squeezing** (strongest force in initial breakage).
- **Cavitation:** collapsing bubbles erode the surface; excess bubbles shield the stone and injure tissue.

Practical points

- Better results at **60–90 shocks/min**, with power **ramping** and air-free coupling.
- Poor results: stone >1000 HU (Hounsfield units), skin-to-stone distance >10 cm, lower pole >1 cm.
- **Contraindications:** pregnancy, uncorrected coagulopathy, untreated UTI (urinary tract infection), distal obstruction, aortic aneurysm >4 cm.
- **Also know:** burst wave lithotripsy and ultrasonic propulsion are newer descendants.

Diagrams to draw

- Conventional lens scope versus Hopkins rod-lens scope (glass rods with short air gaps).
- Dorsal inlay: urethra opened ventrally, dorsal plate incised, graft on corpora (cross-section).
- PN steps: clamp, ice slush, excision, closure, renorrhaphy.
- Ellipsoid reflector with spark at F1 and stone at F2.

Key points to remember:

- Hopkins: rod-lens system 1966; fibreoptics 1954.
- Huggins and Hodges 1941: androgen dependence; castration regresses prostate cancer; Nobel 1966; 1945 adrenalectomy for castration-resistant disease.
- Calne: azathioprine, ciclosporin, first UK liver transplant 1968.
- Gaur 1992: glove-balloon retroperitoneoscopy.
- Asopa: ventral sagittal urethrotomy, dorsal plate incised, free graft inlaid (2001).
- Novick (Cleveland Clinic): nephron-sparing, renovascular and bench surgery; PN indications imperative, relative, elective (cT1a ≤4 cm).
- Warm ischaemia under about 25–30 minutes; ice slush if longer; negative margin is enough; keep ≥20–30% of one kidney.
- Chaussy, Munich: first human SWL 7 February 1980, Dornier HM1; HM3 (1983) is still the efficacy benchmark.
- Spallation, squeezing, shear, cavitation; electromagnetic sources today.

SAQ History of Laparoscopy

Paper answer · P1

Type: SAQ (10 marks)
Source: 2015-Summer Paper 1, Q3b

Introduction

- Laparoscopy is endoscopic viewing of the peritoneal cavity through a pneumoperitoneum; it grew from optics to robotic surgery.

Optics and access

Year	Pioneer	Milestone
1806	**Bozzini**	**Lichtleiter**, candle-lit light conductor
1877	Nitze	Distal electric light and lenses
1901	**Kelling**	Insufflated abdomen of a **dog**
1910	**Jacobaeus**	First human series; "laparoscopy"
1938	**Veress**	Spring-loaded needle
1966	**Hopkins**	**Rod-lens system**
1970s–80s	**Semm**	Automatic insufflator; operative laparoscopy
1970s	Hasson	Open access
1987	Mouret	Video cholecystectomy
- **Also know:** Desormeaux 1853, first therapeutic endoscopy; fibreoptics (Tyndall's principle) for flexible scopes; CO_2 became the standard insufflant.		

Urological milestones

Year	Pioneer	Milestone
1976	**Cortesi**	Impalpable testis, first urological use
1990–91	**Clayman**	**First laparoscopic nephrectomy**; nephroureterectomy 1991
1992	**Gaur (India)**	**Balloon retroperitoneal access**
1995	**Ratner**	Live-donor nephrectomy
1997	Schuessler	Laparoscopic radical prostatectomy (slow, not adopted)
2000	Guillonneau & Vallancien	Standardised prostatectomy
2000	**da Vinci**; **Binder**	Robot cleared in US; first robotic prostatectomy
2000s	**Menon**	Vattikuti Institute Prostatectomy
2007	Raman; Rane	Single-port nephrectomy

Why the robot spread

- Laparoscopic prostatectomy stalled because intracorporeal suturing was hard. The robot gave wristed instruments and a 3D magnified view.
- da Vinci generations: S 2006 (fourth arm), Si 2009 (dual console), Xi 2014 (flexible docking), SP 2018 (single port).

Indian contribution and recent advances

- Gaur's balloon made retroperitoneoscopy practical; indigenous robot SSI Mantra with telesurgery.
- New robots (Versius, Hugo, Hinotori), telesurgery, haptic feedback.

Diagram to draw: Timeline 1806 → 1901 → 1910 → 1938 → 1990 → 2000.

> **Key points to remember:**
> - Bozzini 1806; Kelling 1901 (dog); Jacobaeus 1910 (human); Veress 1938; Hopkins rod-lens.
> - Urology: Cortesi 1976; Clayman 1990; Gaur 1992; Ratner 1995.
> - da Vinci 2000; Binder first robotic prostatectomy; robot solved suturing barrier.

SAQ Flow cytometry

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 5 Paper 1, Q3d

Definition

- **Flow cytometry** measures physical and fluorescent properties of **single cells in suspension** flowing one at a time through a laser beam, thousands per second.

Principle

1. **Sample:** fresh cells, urine, bladder washings, or nuclei from paraffin blocks.
2. **Staining:** fluorochrome-labelled antibodies to CD (cluster of differentiation) antigens, or **DNA-binding dyes** (propidium iodide, DAPI [4′,6-diamidino-2-phenylindole]).
3. **Hydrodynamic focusing** lines the cells up in single file.

4. **Laser interrogation:** - **Forward scatter** = cell size. - **Side scatter** = internal complexity (granularity). - **Fluorescence** = amount of bound antibody or DNA.
5. Photomultiplier detectors → histograms and dot plots; **gating** selects populations. - **FACS (fluorescence-activated cell sorting)** physically separates selected cells.

DNA content analysis

- **DNA index** = tumour G0/G1 peak ÷ normal diploid peak. DI 1.0 = diploid; about 2.0 = tetraploid; any other = **aneuploid**.
- **S-phase fraction** estimates proliferation.
- Aneuploidy and high S-phase generally mean higher grade and worse prognosis.

Applications in urology

Area	Use
Bladder cancer	Ploidy of bladder washings: aneuploidy with high grade, CIS (carcinoma in situ), recurrence; replaced by **UroVysion FISH (fluorescence in situ hybridisation)**
Upper-tract urothelial cancer	Aneuploidy → poor long-term survival
Prostate cancer	Aneuploid tumours progress more after treatment than diploid
Neuroblastoma	**Hyperdiploid favourable**; diploid or tetraploid adverse; part of COG (Children's Oncology Group) risk
Urinalysis	Automated urine analysers count RBC (red blood cells), WBC (white blood cells) and bacteria
Transplantation	**Flow cytometric crossmatch**: more sensitive than CDC (complement-dependent cytotoxicity) crossmatch for donor-specific antibody; Luminex single-antigen beads; CD3 count to guide antithymocyte globulin
Male infertility	Sperm DNA fragmentation (SCSA [sperm chromatin structure assay]); haploid cells in testicular aspirates
Lymphoma	Immunophenotyping of testicular or renal lymphoma

Advantages and limitations

- **Advantages:** rapid, objective, quantitative, many parameters per cell.
- **Limitations:** architecture lost; near-diploid aneuploidy missed; normal cells dilute tumour peaks; ploidy now adds little independent value.

Diagram to draw: DNA histogram with diploid G0/G1 peak, S-phase and aneuploid peak.

> **Key points to remember:**
> - Forward scatter = size; side scatter = granularity; fluorescence = marker or DNA.
> - DNA index 1.0 diploid; aneuploidy = worse prognosis.
> - Flow crossmatch is more sensitive than CDC crossmatch.

SAQ Haemostatic agents in urology

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 18 SAQ Sheet, Q3e

Introduction

- Topical agents are **adjuncts** to surgical haemostasis (suture, clip, energy), used for diffuse ooze. No randomised study shows clear benefit of any single agent in partial nephrectomy.

Classification

Class	Examples	Mechanism / note
Passive (mechanical)	Gelatin sponge (Gelfoam), **oxidised regenerated cellulose (Surgicel)**, microfibrillar collagen, polysaccharide spheres	Matrix for platelets; needs intact clotting. Cellulose is acidic (antibacterial)
Active	Topical thrombin	Converts the patient's fibrinogen to fibrin
Flowable	Gelatin matrix + thrombin (**Floseal**)	Fills irregular beds, swells
Fibrin sealants	Fibrinogen + thrombin (Tisseel, Evicel)	Final clotting step; works despite coagulopathy; also seals leaks
Synthetic sealants and glues	PEG (polyethylene glycol) hydrogel (Coseal), cyanoacrylate, albumin–glutaraldehyde (BioGlue)	Seal; adhesive
Systemic	Tranexamic acid, aminocaproic acid, desmopressin	Antifibrinolytic; desmopressin for uraemic platelets

Urological uses

- **Partial nephrectomy:** renorrhaphy ± sealant or Surgicel bolster; sutureless repair in selected cases.
- **Laparoscopic venous injury:** pressure → sealant → suture.
- PCNL (percutaneous nephrolithotomy) tract sealing in tubeless cases; adrenal and spleen/liver ooze; prostatectomy and lymphadenectomy beds.
- **Fibrin glue:** ureteroscopic fulguration for upper-tract bleeding; fistula plugs.
- **Tranexamic acid:** lowers transfusion at radical cystectomy.

Intravesical agents (haemorrhagic cystitis)

Agent	Note
Alum 1% (50 g in 5 L)	First line; avoid in renal failure (aluminium toxicity)
Silver nitrate 0.5–1%	Chemical coagulation
Aminocaproic acid 200 mg/L irrigation	Thromboembolism and clot retention
Formalin 1–4%	Most effective; under anaesthesia; **cystogram first** to exclude reflux

Complications

- Swelling compresses nerves, ureter or spinal cord (gelatin, cellulose): use the least amount.
- Mimics abscess on imaging; foreign-body reaction; infection.
- Flowables entering open vessels → embolism or thrombosis.
- Bovine thrombin → antibodies and coagulopathy; human products → theoretical viral risk.
- **Also know:** energy adjuncts (bipolar, argon beam, vessel sealers); hyperbaric oxygen in radiation cystitis.

Diagram to draw: clotting cascade showing where passive agents, thrombin and fibrin sealant act.

> **Key points to remember:**
> - Agents are adjuncts, not substitutes for sutures.
> - Fibrin sealant works despite coagulopathy; passive agents need normal clotting.
> - Surgicel: acidic, may mimic abscess.
> - Alum first line; formalin last, after cystogram.

SAQ Sterilisation of urological instruments

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 19 SAQ Sheet, Q3a

Definitions

- **Cleaning:** removes debris that shields organisms; always first.
- **Disinfection:** kills most microbes but not all spores. HLD (high-level disinfection) kills all except large numbers of spores.
- **Sterilisation:** kills all life, including spores.

Spaulding classification

Category	Contact	Requirement	Urological examples
Critical	Sterile tissue, vessels	Sterilisation	Resectoscope, ureteroscope, nephroscope, laparoscopic and robotic instruments
Semi-critical	Mucosa	HLD at least (sterilise if possible)	Flexible cystoscope, TRUS (transrectal ultrasound) probe
Non-critical	Intact skin	Low-level disinfection	Abdominal ultrasound probe

Reprocessing steps

1. Bedside pre-clean; leak test for flexible scopes.
2. Manual cleaning with enzymatic detergent; brush and flush channels.
3. Rinse → disinfect or sterilise → rinse → dry channels (alcohol, air) → store hanging; record tracking.

Methods

Method	Conditions	Use and drawbacks
Steam autoclave	121 °C for 15 min (15 psi) or 134 °C for 3–4 min	Cheapest, most reliable; metal instruments, autoclavable optics
Dry heat	160 °C for 2 hours	Glass; rarely used
Ethylene oxide	Low temperature	Heat-sensitive scopes, cables; toxic, carcinogenic, long aeration
Hydrogen peroxide gas plasma	Low temperature, short cycle	No toxic residue; not for cellulose; long narrow lumens limited

Method	Conditions	Use and drawbacks
Peracetic acid (Steris)	Automated liquid, low temperature	Sporicidal; ureteroscopes
Glutaraldehyde 2%	HLD in 20 min; sterilant after about 10 hours	Cheap; irritant, sensitiser; must rinse well
Gamma irradiation	Industrial	Single-use catheters, stents, sutures

Monitoring

- **Physical:** time, temperature, pressure records.
- **Chemical indicators:** autoclave tape; Bowie–Dick test (air removal).
- **Biological indicators:** *Geobacillus stearothermophilus* spores (steam, plasma); *Bacillus atrophaeus* (ethylene oxide, dry heat).

Urology-specific points

- A laser-perforated ureteroscope channel lets fluid in during sterilisation, destroying the scope.
- Poor channel cleaning causes outbreaks (Pseudomonas, mycobacteria).
- **Prions** resist routine methods: single-use or special protocols.
- **Single-use scopes** avoid reprocessing (about $107 per case for reusable ureteroscopes).
- **Also know:** laser fibre re-sterilisation; limited-life robotic instruments.

Diagram to draw: reprocessing cycle flowchart.

> **Key points to remember:**
> - Spaulding: critical sterilise, semi-critical HLD, non-critical low-level.
> - Autoclave 121 °C 15 min or 134 °C 3–4 min.
> - Ureteroscopes: ethylene oxide, glutaraldehyde or peracetic acid.
> - Biological indicator: *G. stearothermophilus*.

SAQ Amyloidosis in urology

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 21 SAQ Sheet, Q3c

Definition

- Extracellular deposits of misfolded protein in **β-pleated sheet fibrils**. **Congo red stain gives apple-green birefringence** under polarised light.

Classification

Type	Protein	Cause
AL (amyloid light-chain; primary)	Immunoglobulin light chain	Plasma cell dyscrasia, myeloma
AA (amyloid A; secondary)	Serum amyloid A	Chronic inflammation: TB (tuberculosis), rheumatoid arthritis, osteomyelitis, pressure sores, hidradenitis
ATTR (hereditary transthyretin)	Transthyretin	Autosomal dominant neuropathy

Type	Protein	Cause
Dialysis-related	β2-microglobulin	Long-term dialysis
Localised	Usually AL	Deposits in one organ only

Urological sites

- **Kidney (systemic):** nephrotic proteinuria, CKD (chronic kidney disease), enlarged kidneys, renal vein thrombosis, proximal tubular defects.
- **Bladder (usually localised): painless gross haematuria**; yellowish plaques or nodules that **mimic carcinoma**. Secondary bladder amyloid can bleed severely; Campbell lists amyloidosis among systemic causes of haemorrhagic cystitis.
- **Ureter and renal pelvis:** stricture or filling defect, often calcified; mimics urothelial cancer; obstruction.
- **Urethra:** rare; mass, obstructive voiding, haematuria.
- **Seminal vesicles:** senile deposits in older men; can mimic cancer invasion on MRI (magnetic resonance imaging).
- **Also know:** ATTR gives neurogenic bladder (underactive detrusor, LUTS [lower urinary tract symptoms] >80%) and erectile dysfunction; adrenal insufficiency; papillary necrosis.

Evaluation

- **Biopsy is diagnostic** (avoids unnecessary nephroureterectomy or cystectomy).
- Exclude systemic disease: serum and urine protein electrophoresis, immunofixation, free light chains, echocardiogram, renal function.

Management

- **Localised bladder:** TUR (transurethral resection) or fulguration; intravesical DMSO (dimethyl sulfoxide) reported; recurrence is common, so follow-up cystoscopy is needed.
- **Ureter:** ureteroscopic resection or segmental excision; stent for obstruction.
- **Urethra:** observe if mild; dilatation or urethroplasty if obstructive.
- **Refractory bleeding:** irrigation, embolisation, cystectomy.
- **Systemic:** AL: chemotherapy (daratumumab with bortezomib regimen, ANDROMEDA trial) or stem cell transplant; AA: treat inflammation (colchicine in familial Mediterranean fever); ATTR: tafamidis, patisiran. AA amyloid can recur in a renal transplant.

Diagram to draw: Congo red stained fibrils with apple-green birefringence.

> **Key points to remember:**
> - Congo red, apple-green birefringence.
> - Localised bladder amyloid mimics cancer; biopsy then TUR.
> - Always exclude systemic AL disease.

SAQ Collin's knife.

Paper answer · P3

Type: SAQ (10 marks)
Source: 2018-Summer Paper 3, Q3b

*Taken to mean the **Collings knife**, the knife-shaped cutting electrode of the resectoscope (the reference chapters spell it both "Collins" and "Collings").*

Description

- A **hot (electrosurgical) knife electrode** fitted to the **resectoscope working element** in place of the loop.
- Thin **straight or angled (L- or hook-shaped) blade**; cuts a precise line with **cutting current** under irrigation and direct vision.
- **Monopolar** versions need **non-conducting irrigant** (glycine, sorbitol or mannitol); bipolar versions allow normal saline.
- Adult and **infant (9.5 Fr) resectoscope** sizes.
- **Principle:** it **incises rather than resects**; a deep, controlled cut releases a fibrous ring with little tissue loss.

Uses (from the chapters)

Use	Technique point
TUIP (transurethral incision of prostate)	Gland under 30 mL, no median lobe; incise at 5 and 7 o'clock from below the ureteric orifice to just proximal to the verumontanum, down to the capsule
Bladder neck contracture (after TURP (transurethral resection of prostate) or simple prostatectomy)	Incise until the scar springs open, to about 22 Fr; mitomycin C for recurrence
Ureterocele incision (older children)	Low transverse full-thickness incision near the bladder floor to reduce reflux
PUV (posterior urethral valve) ablation	Infant resectoscope only if urethral calibre is normal; incise at 5, 7 (± 12) o'clock; avoid hot-loop resection (stricture)
Bladder diverticulum neck	Incise the narrow neck so the diverticulum drains
PCNL (percutaneous nephrolithotomy) tract	Incise scarred fascia when rigid dilators fail

Alternatives

- **Cold knife (Sachse)** urethrotome: preferred for urethral stricture; hot knife unnecessary there.
- **Holmium or thulium laser**, Bugbee electrode, hook electrode, **Mohan's valvotome** (for valves).

Precautions and complications

- Stay away from the **ureteric orifices** and external sphincter; control depth (perforation, extravasation).
- TUR (transurethral resection) syndrome with monopolar irrigant if veins open.
- **Retrograde ejaculation** after bladder neck incision; stricture if the urethra is too small for the sheath.

Diagram to draw: resectoscope tip with L-shaped knife electrode, and the 5 and 7 o'clock TUIP incisions.

Key points to remember:

- Knife-shaped resectoscope electrode for incision, not resection.
- Uses: TUIP, bladder neck contracture, ureterocele, PUV, diverticulum neck.
- Monopolar needs glycine or sorbitol; bipolar uses saline.

SAQ APOGEE

Paper answer · P3

Type: SAQ (10 marks)
Source: Practice Set 7 Paper 3, Q3c

Assumed meaning: the Apogee transvaginal mesh kit (American Medical Systems) for posterior vaginal wall and vault prolapse. It is not described in the chapter notes; facts below are general.

Definition

- A **prepackaged polypropylene mesh kit** placed transvaginally to support the **posterior compartment and vaginal apex** (DeLancey levels I and II). Anterior companion: **Perigee** (transobturator).
- Mesh: **type I macroporous monofilament polypropylene** (Amid classification), the only type that integrates into tissue.

Indications (historical)

- Symptomatic posterior and vault prolapse, mostly **recurrent** or with poor native tissue, avoiding abdominal surgery.

Technique (principle)

1. Midline posterior vaginal incision; dissect the rectovaginal space to the ischial spines.
2. Two small perianal incisions lateral and posterior to the anus.
3. Curved trocars pass through the **ischiorectal fossa**, piercing the iliococcygeus near the ischial spine, and deliver the mesh arms.
4. Mesh body fixed to the apex and laid over the rectovaginal septum **tension-free**; epithelium closed without trimming.

Complications

- **Mesh exposure** (IUGA/ICS [International Urogynecological Association/International Continence Society] term; "erosion" deprecated), extrusion, **perforation** into rectum or bladder.
- **Dyspareunia**, buttock and pelvic pain, mesh contraction, defecatory dysfunction, ischiorectal haematoma or abscess, rectal injury.
- **Management:** small exposure → vaginal oestrogen, then partial excision; pain or perforation → excision of mesh and arms.

Regulatory status

- **FDA (US Food and Drug Administration) 2011:** serious complications of transvaginal POP (pelvic organ prolapse) mesh "not rare", with no clear benefit over native repair. Reclassified high-risk in 2016.
- **2019:** FDA ordered remaining manufacturers to **stop selling transvaginal POP mesh**; Apogee-type kits are off the market. Synthetic mesh improves anatomy but complicates

more (Cochrane 2016).
- This does **not** apply to mid-urethral slings or **sacrocolpopexy mesh**.

Current alternatives

- Native posterior colporrhaphy or site-specific repair; uterosacral suspension or **sacrospinous fixation**; **sacrocolpopexy** (abdominal, laparoscopic, robotic) for apical prolapse; colpocleisis in the frail.

Diagram to draw: posterior mesh with two arms through the ischiorectal fossae.

> **Key points to remember:**
> - Apogee: transvaginal posterior/apical polypropylene mesh kit.
> - Exposure, dyspareunia and pain drove withdrawal.
> - FDA 2019 stopped transvaginal POP mesh sales.

SAQ Urethral stents and metallic stents in urology

Paper answer · P4

Asked as:

- Urethral stents — *Summer 2015 Paper IV Q3d (SAQ)*
- Metallic stents in urology — *Practice Set 3 Paper IV Q3c (SAQ)*
- Urethral stents. — *Practice Set 4 Paper IV Q3f (earlier version) (SAQ)*

Type: SAQ (10 marks)

(urethral stents also answered in 2015-Summer Paper 4, Q3d; ureteric stents in 2015-Summer Paper 3, Q1)

Definition and introduction

- A urethral stent is an endoscopically placed tube that keeps the **prostatic or bulbar urethra** open, mainly in men unfit for surgery.
- Polymer stents (polyurethane, silicone) fail early under **extrinsic compression** and need frequent changes. Metal stents **resist compression** and stay longer, but suffer **encrustation, tissue ingrowth and migration**.

Classification (urethral stents)

Type	Examples	Features
Temporary	**Memokath** (nitinol), **Allium**	Not epithelialised (designed to avoid epithelial incorporation); removable; left 6–12 months
Permanent, epithelialising	**UroLume**	Urothelium grows over it; **withdrawn**
Biodegradable	Experimental	Dissolve

Classification of metallic stents (by site)

Site	Type	Example	Features
Ureter	Full-length coiled metal, double pigtail	**Resonance** (2006)	Tightly coiled alloy; changed every 12 months

Site	Type	Example	Features
Ureter	Segmental self-expanding mesh, bare or covered	Wallstent-type; covered nitinol (Allium)	Placed across malignant stricture; bare mesh allows ingrowth
Ureter	Thermo-expandable nitinol coil	Memokath 051	Removable
Prostate	Temporary thermo-expandable nitinol coil	**Memokath**	Not epithelialised; removable
Prostate, bulbar urethra	Covered self-expanding, removable	Allium	Left 6–12 months
Bulbar urethra	Permanent epithelialising mesh	**UroLume**	Withdrawn

Ureteric metal stents

- **Indications:** malignant extrinsic obstruction; retroperitoneal fibrosis (first Resonance use); benign stricture in a patient unfit for repair; ureteroenteric stricture with limited life expectancy; failed tandem polymer stents.
- **Outcomes:** full-length metal stent: up to 3 years of drainage, pooled success about 72.5%, migration about 1%, but about one-third failed in one malignant series. Ureteroenteric strictures: patency >80% at 6–22 months.
- **Problems:** encrustation, difficult exchange, ingrowth, migration.

Urethral and prostatic stents

Indications

1. BPH (benign prostatic hyperplasia) retention in men unfit for TURP (transurethral resection of prostate) / unfit for surgery.
2. Recurrent short bulbar stricture in men unfit for urethroplasty.
3. **DSD (detrusor–sphincter dyssynergia)**: formerly (once) an alternative to sphincterotomy; now **abandoned** for complications.

Technique

- Local or regional anaesthesia with endoscopic or fluoroscopic guidance. Memokath expands when warmed (with warm water) and softens when cooled (with cold water) for removal. A bulbar stent must **not cross the external sphincter**, or incontinence follows.

Complications

- **Migration** (temporary stents), **encrustation**, **hyperplastic ingrowth** with restenosis (UroLume), incontinence, pain and painful erections.

Results (outcomes)

- Prostatic stents free most unfit men from catheters short term, but many are later removed. Memokath variable. Removable bulbar stents succeed only in small series. UroLume worked early (good early results, then ingrowth) but failed long term from **hyperplastic ingrowth** and restenosis, so withdrawn, and many men still carry one.
- No stent removes **spongiofibrosis**, so none cures a stricture.

Guidelines

- **AUA (American Urological Association)**: **no (urethral) stent if urethroplasty is considered** (planned); stents make reconstruction harder.

Recent advances

- **Optilume** paclitaxel balloon (FDA (Food and Drug Administration) approval 2023 for anterior stricture): **ROBUST III**, after failed endoscopy, about 75% vs 27% stricture-free. Optilume BPH (EVEREST-1).
- **iTind** (temporary nitinol prostatic device): nitinol device left 5–7 days; AUA 2023 option for 25–75 mL glands without median lobe.
- **Also know:** biodegradable and drug-eluting stents (experimental).

Also know: a soft silicone "urethral stent" (splint) after hypospadias repair or urethroplasty is a different device; in complicated hypospadias repairs it reduces meatal stenosis and fistula.

Diagram to draw: sagittal urethra with prostatic stent and bulbar stent distal to sphincter; prostatic Memokath ending above the external sphincter.

> **Key points to remember:**
> - Temporary (Memokath, Allium) vs permanent (UroLume, withdrawn).
> - Metal resists extrinsic compression; weakness is ingrowth and encrustation.
> - Resonance: coiled, full-length, changed every 12 months.
> - Memokath: thermo-expandable nitinol, removable.
> - Migration, encrustation, ingrowth; never cross the sphincter.
> - AUA: no stent if urethroplasty planned; Optilume and iTind are newer.

SAQ Urological applications in nanotechnology.

Paper answer · P4

Type: SAQ (10 marks)
Source: 2022-Summer Paper 4, Q3a

Taken as "applications of nanotechnology in urology".

Definition

- **Nanotechnology:** design and use of materials of about **1–100 nm**. At this scale, surface area and optical, magnetic and thermal properties change.
- **Why useful:** particles pass through leaky tumour vessels (**enhanced permeability and retention**), carry drugs or tracers, and can be targeted with ligands such as PSMA (prostate-specific membrane antigen) antibodies.

Applications

Area	Application
Imaging and staging	**USPIO (ultrasmall superparamagnetic iron oxide) lymphotropic MRI (magnetic resonance imaging):** normal nodes take up particles and lose T2 signal; metastatic nodes do not (Chapter 6; investigational in penile cancer, Chapter 84). Gold-nanoparticle CT (computed tomography) contrast, quantum dots
Sentinel node	Technetium-99m **nanocolloid** for dynamic sentinel node biopsy in penile cancer (Chapter 84)

Area	Application
Diagnosis	Nanowire and nanoparticle biosensors detecting PSA (prostate-specific antigen) and urinary markers at very low levels
Drug delivery	Nanoparticle (albumin-bound) paclitaxel in urothelial and prostate cancer (Chapter 164: tolerated, no pathological complete response); **NanoPac** intratumoral paclitaxel (Chapter 163); liposomal drugs; mucoadhesive intravesical carriers; siRNA (small interfering RNA) delivery
Focal ablation	**Gold nanoshells (AuroLase):** collect in tumour vessels and are heated by near-infrared laser; about three-quarters success on 12-month biopsy (Chapter 163). Magnetic nanoparticle hyperthermia
Irreversible electroporation	Electric pulses create **nanoscale membrane pores** ("NanoKnife"); spares matrix, vessels and ducts; investigational (Chapter 143)
Devices	Silver and other nanocoatings on catheters and stents against biofilm and encrustation
Tissue engineering	Electrospun nanofibre scaffolds that mimic extracellular matrix
Stone disease	"Calcifying nanoparticles" (nanobacteria) linked to Randall plaque, not proven causal (Chapter 95); nano-CT of stones and tissue

Limitations

- Toxicity and long-term tissue accumulation; uptake by liver and spleen; manufacturing cost; regulatory hurdles. Most uses remain experimental.

Diagram to draw: nanoparticle with drug core, polymer shell and surface PSMA ligand entering a leaky tumour vessel.

> **Key points to remember:**
> - 1–100 nm; enhanced permeability and retention allows tumour targeting.
> - USPIO MRI: benign nodes lose signal; metastatic nodes do not.
> - AuroLase gold nanoshells: photothermal focal therapy.

SAQ Morcellation

Paper answer · P4

Type: SAQ (10 marks)
Source: Practice Set 1 Paper 4, Q3d

Definition

- Cutting a large specimen into small pieces so it leaves through a port, small incision or endoscope.
- **Manual:** ring forceps or Kelly clamp inside a bag. **Power (electromechanical):** rotating cylindrical blade (laparoscopy) or endoscopic tissue morcellator (prostate).

1. Endoscopic morcellation after enucleation

- Made **HoLEP (holmium laser enucleation of prostate)** practical; also ThuLEP (thulium laser enucleation).
- **Steps:** 1. Achieve good haemostasis first. 2. Pass a long nephroscope with a 5 mm channel. 3. Run two inflow lines fully open to keep the bladder distended. 4. Engage lobes with

suction; cut only in mid-bladder, away from the wall. 5. Collect fragments in a trap for histology.

- Devices: Piranha (rotating blades), VersaCut (oscillating).
- **Complications:** bladder mucosal injury (stop, release suction, disengage; mostly catheter only); intraperitoneal perforation needs laparotomy; **retained adenoma** (from malfunction or poor vision) needs repeat morcellation.

2. Laparoscopic morcellation (kidney, spleen)

- **LapSac** (nylon with polyurethane lining) is impermeable to tumour cells and bacteria.
- **Rules:** purpose-built bag; drapes; change gloves and instruments after; no deep forceps passes; keep pneumoperitoneum and watch the bag.

	Intact extraction	Morcellation
Incision	Larger	Small, better cosmesis
Pain, time, stay	Similar	Similar
Pathology	Full staging	Type and grade kept; size, capsule, vein invasion, and depth in UTUC (upper tract urothelial carcinoma) lost
Seeding	—	Rare port-site recurrence

- Splenectomy morcellation risks **splenosis** if the bag ruptures.

3. FDA 2014 warning (gynaecology)

- In 2014 the FDA (Food and Drug Administration) warned against laparoscopic power morcellation in hysterectomy or myomectomy for fibroids, because unsuspected uterine sarcoma (about 1 in 350) was spread and upstaged.
- A boxed warning followed; contained (in-bag) morcellation is advised.
- Lesson for urology: avoid uncontained power morcellation of possible malignancy.

Complications summary

- Bag rupture, seeding, lost staging, visceral or vascular injury, retained tissue.

Diagram to draw: enucleated lobes in a distended bladder with morcellator suction.

> **Key points to remember:**
> - Morcellator made HoLEP practical; distend bladder, haemostasis first.
> - Morcellation loses stage information; histology still obtained.
> - FDA 2014: uncontained power morcellation spreads occult sarcoma.

SAQ Stem cell therapy

Paper answer · P4

Type: SAQ (10 marks)
Source: Practice Set 1 Paper 4, Q3f (tissue engineering also in 2023-Winter Paper 1, Q1)

Definition

- Stem cells **self-renew** and **differentiate** into specialised cells. Therapy uses them to repair or replace damaged tissue.

Types

Type	Source	Comment
Embryonic	Blastocyst	Pluripotent; ethics, teratoma
Adult mesenchymal	Bone marrow, adipose (ADSC, adipose-derived stem cells), umbilical cord	Easy harvest; mainly paracrine
Muscle-derived (MDSC)	Skeletal muscle biopsy	Used for sphincter and detrusor
iPSCs (induced pluripotent stem cells)	Reprogrammed adult cells	Patient-specific; tumour risk
Urine-derived, spermatogonial	Urine; testis	Research

Mechanisms

- Differentiation into muscle, nerve or endothelium.
- **Paracrine action:** growth factors, anti-inflammatory, anti-fibrotic and neurotrophic signals.
- Homing to injured tissue; seeding of scaffolds.

Applications

- **SUI (stress urinary incontinence):**
- Autologous muscle-derived cells: thigh muscle biopsy → culture → transurethral or periurethral injection into the sphincter.
- Aim: regenerate the striated sphincter, not just bulk it.
- Early studies improved; placebo-controlled trials have not shown clear superiority, so it remains investigational.
- After childbirth injury, stem cell homing chemokines (CXCL12, CCL7) rise.
- **ED (erectile dysfunction):** intracavernosal bone-marrow, adipose or cord cells; animal and small uncontrolled human series only. SMSNA (Sexual Medicine Society of North America): experimental, IRB (institutional review board) protocols only.
- **Underactive detrusor:** muscle-derived cells injected into the bladder wall; two open-label studies improved voiding; research.
- **Tissue engineering:** cell-seeded scaffolds for bladder (Atala, 2006) and urethra (Raya-Rivera, 2011); adipose stem cell-seeded grafts for Peyronie disease (experimental).
- **Fertility:** spermatogonial stem cell banking in prepubertal cancer patients.
- **Kidney:** iPSC kidney organoids (research).

Challenges

- Poor cell survival and retention; tumour formation (embryonic, iPSC); standardisation, cost and regulation.
- Unregulated "stem cell clinics" offer unproven treatments outside trials, a growing ethical concern.
- **Also know:** cancer stem cells (a different concept) may drive tumour recurrence and resistance.

Diagram to draw: muscle biopsy → culture → periurethral injection into sphincter.

Key points to remember:

- Adult stem cells act mainly by paracrine signals.
- SUI muscle-derived cells and ED stem cells are investigational.
- Embryonic and iPSCs carry tumour risk.

SAQ Advances in lap surgeries

Paper answer · P4

Type: SAQ (10 marks)
Source: Practice Set 3 Paper 4, Q3a (related: LESS in 2023-Winter Paper 4, Q1; energy sources in 2015-Summer Paper 2, Q3d)

Introduction

- Since the first laparoscopic nephrectomy (Clayman, 1991), advances aim at **less access trauma, better vision, safer haemostasis, gentler physiology and faster training**.

1. Access and insufflation

- **Non-bladed dilating and optical trocars**: port-site hernia fell from 1.83% to 0.19%, wall bleeding from 0.83% to 0.16%.
- **Balloon dilators** for retroperitoneal and extraperitoneal space.
- **AirSeal** valveless system: stable pneumoperitoneum at **8–12 mmHg** despite suction, smoke evacuation, less CO_2 absorption.
- Lower pressure (12 rather than 15 mmHg): less pain and oliguria.

2. Fewer and smaller incisions

Technique	Gain	Limitation
LESS (laparoendoscopic single-site surgery)	Cosmesis, less analgesia	No triangulation, clashing
NOTES (natural orifice transluminal endoscopic surgery)	No abdominal scar (hybrid transvaginal extraction)	No proven benefit
Mini-laparoscopy (2–3 mm instruments)	Less pain, cosmesis	Instrument flexibility
Hand-assisted	Tactile feedback, fast bleeding control	Larger wound

3. Vision

- Chip-on-tip HD (high-definition) and **3D** laparoscopes; deflectable tips.
- **ICG (indocyanine green) near-infrared fluorescence**: vessel mapping for selective clamping in partial nephrectomy, ureteric identification, sentinel nodes.
- Laparoscopic ultrasound; 3D-printed planning models.

4. Haemostasis and suturing

- **Advanced bipolar sealers** (LigaSure: vessels ≤7 mm, 1–3 mm spread) and **ultrasonic shears** (about 1 mm spread, but very hot jaws).
- **Hem-o-lok** clips, **barbed sutures** (shorter anastomosis), Lapra-Ty clips, powered articulating staplers.

5. Perioperative care

- **ERAS (enhanced recovery after surgery)**: stay about 30% shorter, complications halved; no routine bowel preparation.

6. Training

- Box trainers, VR (virtual reality) simulators, EAU **E-BLUS** (European Basic Laparoscopic Urological Skills) certification, 5-day mini-residencies.

7. Robot-assisted laparoscopy

- Wristed 3D platforms (da Vinci Xi, SP 2018; Hugo, Versius, SSI Mantra) solved laparoscopic suturing (**see Q3b**).
- **Also know:** augmented reality, artificial intelligence, telesurgery.

Diagram to draw: LESS multichannel port in the umbilicus with articulating instruments.

> **Key points to remember:**
> - Blunt trocars cut port-site hernia to 0.19%.
> - AirSeal allows 8–12 mmHg working pressure.
> - ICG fluorescence guides selective clamping.
> - LESS: cosmesis at the cost of triangulation.

SAQ Artificial intelligence in urology

Paper answer · P4

Type: SAQ (10 marks)
Source: Practice Set 4 Paper 4, Q3d

Definitions

- **AI (artificial intelligence):** computer systems doing tasks that need human intelligence: recognising patterns, predicting, deciding.
- **ML (machine learning):** algorithms learn from data rather than fixed rules. **Supervised** (labelled data, e.g. tumour or not), **unsupervised** (finds clusters), reinforcement.
- **Deep learning:** multilayer **ANNs (artificial neural networks)**; **CNNs (convolutional neural networks)** are the commonest for images.
- **NLP (natural language processing)** and **LLMs (large language models)** handle text; **radiomics** extracts quantitative image features.

Development steps

`Data collection and labelling → training → internal validation → external validation → regulatory approval → use and monitoring`

Applications

Area	Examples
Imaging	CNNs detect stones on CT (computed tomography) with very high sensitivity; ANNs detect renal tumours; prostate MRI (magnetic resonance imaging) lesion detection; deep-learning reconstruction allows ultra-low-dose CT (about 1 mSv)
Pathology	Automated cancer detection and Gleason grading on digital slides; Paige Prostate was the first FDA (Food and Drug Administration)-authorised AI pathology tool (2021)

Area	Examples
Endoscopy	Real-time bladder-tumour detection at cystoscopy; future ureteroscopes setting laser energy, irrigation, pressure and temperature
Stones	Predict composition and stone-free outcome
Prognosis	Radiomics, pathomics and genomics combined for RCC (renal cell carcinoma) prognosis; AI on biopsy slides predicting benefit of ADT (androgen deprivation therapy) with radiotherapy
Robotic surgery	Video and kinematic data for skill assessment, outcome prediction, real-time guidance; future semi-autonomy
Paediatrics	Deep learning on ultrasound predicts CKD (chronic kidney disease) progression in valves; VUR (vesicoureteral reflux) infection-risk models
Others	Urodynamics interpretation; LLM chatbots for patient information

Advantages

- Fast big-data processing; consistency; decision support.

Limitations and ethics

- Needs large, high-quality labelled data; **bias** and poor generalisability (overfitting); "black box" decisions; few prospective or external validations.
- Privacy and data sharing; medico-legal liability; LLM "hallucinations"; regulation.
- AI assists, not replaces, the urologist or radiologist.

Diagram to draw: neural network: input layer (image pixels), hidden layers, output (stone or no stone).

> **Key points to remember:**
> - Deep learning is part of ML, which is part of AI; CNNs for images.
> - CNNs: very high sensitivity for CT stones.
> - Paige Prostate: first FDA-authorised AI pathology tool (2021).
> - Limits: data, bias, black box, validation, liability.

SAQ 3D laparoscopy

Paper answer · P4

Type: SAQ (10 marks)
Source: Practice Set 5 Paper 4, Q3b

Introduction

- Standard laparoscopy gives a **two-dimensional (2D)** image. Depth is judged from monocular cues (shadow, size, motion parallax, touch), which slows suturing.
- **Three-dimensional (3D) laparoscopy** restores **stereoscopic depth perception**, as in robotic surgery (Chapter 18: two-lens binocular systems give depth perception).

Principle

- Two lenses or two chip-on-tip sensors, a few millimetres apart, capture two images from slightly different angles, like the two eyes.
- The monitor shows both images; **polarised glasses** send one to each eye. The brain fuses them, and the **binocular disparity** is perceived as depth.

- **Also know:** active-shutter glasses; head-mounted displays; glasses-free (autostereoscopic) screens; 3D scopes at 0° and 30°, and with a deflectable tip; 3D 4K systems.

Advantages

- Better depth judgement and hand–eye coordination.
- **Shorter time for complex tasks**, especially intracorporeal suturing and knotting, with fewer errors.
- **Shorter learning curve**; novices gain more than experts.
- Gives robotic-type vision at much lower cost.

Disadvantages

- Visual strain, headache, dizziness or nausea in some users; glasses dim the image.
- Every team member needs glasses and a correct viewing distance and angle.
- Higher cost than 2D; 30° rotation needs care.
- No wristed instruments, tremor filtering or motion scaling: these come only with the robot.

Urological uses

- Reconstructive steps: **pyeloplasty**, vesicourethral anastomosis in laparoscopic radical prostatectomy, **partial nephrectomy** renorrhaphy, ureteric reimplantation, donor nephrectomy.

Evidence

- Randomised trials and meta-analyses in laboratory tasks and laparoscopic surgery show shorter operating and suturing times with 3D. Differences in complications and oncological outcomes are not established.

Feature	2D laparoscopy	3D laparoscopy	Robot
Depth perception	No	Yes	Yes (magnified)
Wristed instruments	No	No	Yes
Cost	Low	Moderate	High

Diagram to draw: two lenses at the scope tip, two images, polarised glasses, fused 3D image in the brain.

> **Key points to remember:**
> - Two lenses, two images, polarised glasses: binocular disparity gives depth.
> - Main gain: faster, more accurate suturing; shorter learning curve.
> - Drawbacks: eye strain, headache, cost, glasses for all.
> - Robotic vision at low cost, but without wristed instruments.

SAQ Telemedicine in urology

Paper answer · P4

Type: SAQ (10 marks)
Source: Practice Set 5 Paper 4, Q3c

Definition

- **Telemedicine** is delivery of health care at a distance using information and communication technology (video, phone, messaging, data transfer).

- **COVID-19 (coronavirus disease 2019)** distancing accelerated its adoption. Chapter 2: urologists' telehealth use rose sharply in 2019–2020.

Types

Type	Method	Urological example
Synchronous	Real-time video or phone consultation	Stone, PSA (prostate-specific antigen) and post-operative follow-up
Asynchronous (store-and-forward)	Reports, images, bladder diaries sent for later review	Scan reports, uroflow traces
Remote monitoring	Apps and home devices	Bladder diary apps, home uroflowmetry
Provider-to-provider	E-consults, telementoring, tele-tumour boards	Haematuria triage (Chapter 8)
Telesurgery	Remote robotic surgery	Experimental; limited by latency

Uses in urology

- **History-driven care:** LUTS (lower urinary tract symptoms) review, results, drug titration.
- ED (erectile dysfunction) evaluation by selected telemedicine (Chapter 72); men's health in large, sparse regions (Chapter 68).
- After mid-urethral sling, telemedicine follow-up is safe, and app-based follow-up gave higher satisfaction than office visits (Chapter 116).
- Internet-based pelvic floor training equalled a postal programme in an RCT (randomised controlled trial) (Chapter 114).

Benefits (Chapter 2)

- Access for rural and underserved patients; less travel; shorter waits; relatives join remotely; sustained satisfaction.

Limitations (Chapter 2)

- **No examination or in-office tests**: vital signs, cystoscopy, PVR (post-void residual).
- Needs connectivity and a secure, simple platform; hard for elderly or non-technical patients.
- No assistant to pre-gather results; more no-shows.
- **Medico-legal:** US licensure is state-based, and pandemic payment waivers have ended.

Indian framework

- **Telemedicine Practice Guidelines (2020)**: only a registered medical practitioner may consult; identity of patient and doctor confirmed; consent and records kept; narcotic and Schedule X drugs may not be prescribed.
- Government platform **eSanjeevani**; patient data covered by the Digital Personal Data Protection Act 2023.

Diagram to draw: patient (home or district hospital) linked by video and data to the urologist, with remote monitoring feeding in.

Key points to remember:

- Telemedicine trades examination for access.
- Best for follow-up, results and history-driven visits.
- Limits: no examination, connectivity, elderly, medico-legal.
- India: Telemedicine Practice Guidelines 2020; eSanjeevani.

SAQ Virtual and augmented reality in urology

Paper answer · P4

Type: SAQ (10 marks)
Source: Practice Set 5 Paper 4, Q3e

Definitions

- **Virtual reality (VR):** the user is immersed in a fully computer-generated environment, usually through a head-mounted display.
- **Augmented reality (AR):** digital images are laid over the real view, for example a 3D (three-dimensional) tumour model on the robotic camera image.
- **Mixed reality:** virtual holograms are anchored in the room and can be handled.

Workflow

`CT (computed tomography) / MRI (magnetic resonance imaging) -> segmentation of organ, tumour, vessels, collecting system -> 3D virtual model -> viewed in VR, 3D-printed, or overlaid (AR) -> registration to the live surgical view`

Applications

Use	Examples
Pre-operative planning	Partial nephrectomy: tumour depth, arterial branches, selective clamping. **PCNL (percutaneous nephrolithotomy)**: calyx and tract choice (Chapter 98: immersive VR and 3D printing show early promise). Prostatectomy: MRI lesion model to plan nerve-sparing
Intra-operative navigation	AR overlay in the robot console to locate an endophytic tumour, hilar vessels, or a prostate lesion near the capsule
Education	VR anatomy, 360° operation videos; simulation, see Q3d
Patient counselling	Patients understand their anatomy and planned operation; better-informed consent
Paediatrics	VR distraction reduces anxiety before surgery and procedures such as catheterisation and VCUG (voiding cystourethrogram) (Chapters 35 and 38)

Limitations

- **Registration error:** organs move and deform with breathing, pneumoperitoneum and dissection, so a rigid overlay drifts; manual alignment is still common.
- Segmentation time and cost.
- Chapter 161: promising but **resource-intensive and hard to validate**.
- Chapter 98: 3D reformatting added nothing over orthogonal CT measurements for predicting stone-free rate.
- Little outcome evidence; cybersickness with headsets.

Future

- **AI (artificial intelligence)** for automatic segmentation and real-time tracking.
- **Digital surgery:** reconstruction of the patient's own anatomy allows rehearsal before the operation (Chapter 147).

Diagram to draw: CT kidney, segmented 3D model, and overlay of the model on the robotic view of a renal tumour.

> **Key points to remember:**
>
> - VR immerses; AR overlays on reality.
> - Main uses: planning partial nephrectomy, PCNL and prostatectomy; navigation; teaching; counselling.
> - Biggest problem: registration on moving, deforming organs.
> - Evidence for outcomes still limited.

Rapid Revision Sheet · Section 1

Operating room safety : the role of surgeon. (SAQ)

- WHO checklist: sign in, time out, sign out.
- Universal Protocol: verify, mark, time-out.
- Count discrepancy: ~100-fold risk of a retained item.
- Fire triangle: surgeon owns ignition.

Microscopic / asymptomatic haematuria and urinary biomarkers (SAQ)

- MH: 3 or more RBCs/HPF on microscopy of one sample; dipstick not enough.
- Low: optional (repeat urinalysis or cystoscopy + ultrasound); intermediate: cystoscopy + ultrasound; high: cystoscopy + CT urography.
- Any gross haematuria = high risk.
- Cytology: specific for high grade/CIS, insensitive for low grade.
- BTA is falsely positive with blood.
- No markers in initial microhaematuria work-up; none replaces cystoscopy.

Berger's Disease. (SAQ)

- Commonest primary GN; haematuria with upper respiratory infection.
- Normal complement; biopsy shows mesangial IgA.
- Oxford MEST-C; 20–30% ESKD.
- ACE inhibitor or ARB is the mainstay.

Proteinuria (SAQ)

- Normal <150 mg/day; nephrotic >3.5 g/day.
- ACR A2 30–300, A3 >300 mg/g.
- Glomerular, tubular, overflow; orthostatic proteinuria is benign.
- Dipstick misses light chains; use sulfosalicylic acid or immunofixation.

Urinary crystals (LAQ)

- Cystine hexagons are always pathological.
- Acidic: uric acid, cystine; alkaline: struvite, phosphates; calcium oxalate at any pH.
- Indinavir: crystalluria in two-thirds, stones 4–13%, radiolucent.
- Refrigerated urine precipitates crystals: examine fresh within 2 hours.

Tumour markers in urology (SAQ)

- PSA: organ-specific, not cancer-specific.
- AFP never in pure seminoma; half-lives AFP 5–7 days, hCG 24–36 hours.
- S stage uses post-orchidectomy values.
- Cytology: good for high grade only.

Urine cytology (SAQ)

- High specificity, low sensitivity; good for high grade and CIS.
- Paris System: N:C ≥0.7 for suspicious and HGUC; AUC N:C ≥0.5.
- Positive cytology, normal cystoscopy: upper tracts, random biopsies.
- Not for screening or initial microhaematuria.

Doppler in urology (SAQ)

- RI = (PSV − EDV)/PSV; normal renal 0.47–0.70.
- Penile PSV <25 cm/s arterial; EDV >5 cm/s veno-occlusive.
- Absent intratesticular flow = torsion, but explore on clinical grounds.

Contrast agents in urology: iodinated and non-iodinated agents, complications and contrast-induced nephropathy (LAQ)

- Iodine : particle ratio 1.5 (ionic), 3 (non-ionic monomer), 6 (iodixanol, 290 mOsm/kg).
- Non-ionic agents: severe reactions 0.04% vs 0.22% ionic.
- Anaphylaxis: IM adrenaline 0.3–0.5 mg, max 1 mg; shellfish allergy is not a risk.
- Premedication: prednisone 50 mg at 13-7-1 h or methylprednisolone 32 mg at 12-2 h, plus diphenhydramine.
- CIN = rise ≥25% or 0.5 mg/dL in 48–72 h; now CA-AKI vs CI-AKI.
- Main CA-AKI risk: eGFR <30 (IV); <45 intra-arterial.
- Isotonic saline only proven; PRESERVE 2018: bicarbonate and NAC useless.
- Metformin: withhold 48 h only if eGFR <30 or AKI.
- NAC: thiol; glutathione precursor, antioxidant, mucolytic.
- NAC is the antidote for paracetamol overdose; best within 8 h.
- Cystinuria uses tiopronin, not NAC.
- GBCAs shorten T1; NSF risk eGFR <30; group II agents almost no risk.
- Microbubbles: intravascular, no nephrotoxicity; ceVUS for reflux.
- CO_2 angiography: never above diaphragm.

Describe IVU & its applications in Urology. (LAQ)

- Contrast is filtered, not reabsorbed: nephrogram then pyelogram.
- Delayed, increasingly dense nephrogram = acute obstruction; no excretion at 24 h = non-functioning.
- Classic signs: drooping lily, cobra head, fish-hook, paintbrush, ring sign, phantom calyx.
- Severe reaction: IM adrenaline 0.3–0.5 mg; CTU has replaced IVU.

VCUG (SAQ)

- Gold standard for VUR and PUV; sterile urine first.
- Capacity (age × 30) + 30 mL (2–14 years); cyclic study detects 12–20% more.
- Grades I–II non-dilating, III–V dilating.

Newer generation CT scan (SAQ)

- Doses: conventional 10–20, low-dose <3–4, ultra-low 1–2 mSv.
- DECT: uric acid stones, virtual unenhanced, iodine maps.
- Enhancement ≥20 HU = solid; deep learning enables ~1 mSv CT.

PET scans in urology. (SAQ)

- proPSMA: PSMA PET 92% vs 65% accuracy.
- Seminoma residual >3 cm: FDG-PET at ≥6 weeks; not NSGCT.
- FDG poor in prostate and clear cell RCC.
- VISION: ^{177}Lu-PSMA-617 in mCRPC.

PSMA PET Ct scan (SAQ)

- proPSMA: accuracy 92% vs 65%.
- BCR: about half positive below PSA 0.5 ng/mL.
- Nodal sensitivity 30–40%: negative scan does not exclude nodes.
- VISION: ^{177}Lu-PSMA-617 improves OS in mCRPC.

MR urography (SAQ)

- Static T2 (no contrast, function-independent) versus excretory gadolinium T1.
- Stones are signal voids; CTU preferred for urothelial cancer.
- Functional MRU: furosemide 1 mg/kg, Patlak analysis for split function.
- NSF risk minimal with group II agents.

VI-RADS (SAQ)

- Five points; tests the dark muscle line on T2W.
- ≥3 suspicious for MIBC; 4 line broken, 5 fat.
- Scan before TURBT.
- Panebianco 2018; accuracy about 80–90%.

Radionuclide (isotope) renal scans: MAG3 and DMSA (LAQ)

- ^{99m}Tc: 6 hours, 140 keV; PET: two 511 keV photons.
- DTPA filtered = GFR; MAG3 secreted = obstruction and transplant; DMSA cortical = scars.
- MAG3: tubular secretion, about 60% extraction; best for split function and drainage.
- Split function (DRF) from 1–3 minute counts; normal 45–55%; furosemide 0.5 mg/kg.
- T½ >20 min obstructed, <10 min not.
- Hydration, catheter, correct diuretic timing: the well-tempered renogram.
- DMSA binds proximal tubules: a static cortical scan at 2–4 h.
- DMSA is the gold standard for acute pyelonephritis and scars.
- Scar only if the defect persists ≥6 months; pyelonephritis resolves.
- Captopril renogram: ≥10% improvement off ACE inhibitors.
- PSMA PET preferred; FDG for residual seminoma >3 cm.
- MIBG best in MEN-2; sentinel node with ^{99m}Tc-nanocolloid.
- Sestamibi hot = oncocytoma; ^{177}Lu-PSMA-617 (VISION).

Haemorrhagic cystitis: diagnosis and management (SAQ)

- Acrolein (prevent with mesna); radiation endarteritis; BK virus.
- Exclude bladder cancer.
- Alum first; formalin only after cystogram.
- Hyperbaric oxygen for radiation cystitis (RICH-ART); embolise if unstable.

Operating room set-up (SAQ)

- Zones with one-way flow; positive pressure, about 20 air changes an hour.
- Monitor directly ahead at eye level; open tray ready for laparoscopy.
- Hypothermia triples wound infection (Kurz).
- ALARA: time, distance, shielding.

Ethics and informed consent in urology (SAQ)

- Four principles: autonomy, beneficence, non-maleficence, justice.

- Canterbury v Spence: reasonable-patient standard; Samira Kohli in India.
- Commonest gap: not checking understanding.
- Exceptions: emergency, incapacity (surrogate), minors.

Patient positioning in lap urology (SAQ)

- Nerve injury about 2.8%; brachial plexus is commonest at risk.
- Never shoulder braces in steep Trendelenburg.
- Axillary roll below, not in, the axilla.
- Rhabdomyolysis: men, BMI ≥25, kidney rest; CK >5,000.

Foley's (urethral) catheters, guidewires and DJ stents with their complications (SAQ)

- Foley balloon catheter: designed for post-prostatectomy haemostasis.
- Fr = 3 × diameter (mm); inflate with sterile water; coudé for prostatic enlargement, Councill over a wire.
- Suspected urethral injury → RUG first; AUS → deactivate, 12–14 Fr.
- CAUTI 5–10% per day; remove early; change the catheter when treating.
- Hydrophilic wire to pass, PTFE wire to stay (safety).
- DJ symptoms in about 80%: trigonal irritation and reflux; α-blockers and antimuscarinics help.
- Long-dwelling stents: monitor for encrustation; encrustation rises with dwell time; FECal I–V.
- Encrusted stent: bladder, then ureter, then kidney; keep a stent register.

Outline algorithm for difficult catheter placement. (SAQ)

- Blind → wire → vision → suprapubic.
- BPH: 18-Fr coudé, tip up.
- Councill catheter over wire after dilatation.
- Suspected urethral injury: RUG first, no blind catheter.

Interventional radiology in urology and therapeutic angioembolisation (SAQ)

- PCN equals retrograde stent for decompressing an obstructed infected kidney.
- Superselective coils: treatment of choice for pseudoaneurysm and AVF.
- Embolisation is first line for bleeding AML and post-PCNL pseudoaneurysm or arteriovenous fistula; prophylactic >6 cm or aneurysm >5 mm.
- Bladder bleeding: internal iliac anterior division; spare posterior division.
- Post-embolisation syndrome is common and self-limiting.

Tubeless PCNL (SAQ)

- Tubeless = stent only; totally tubeless = neither.
- Benefit is mainly over large (>16 Fr) tubes.
- Keep a tube for bleeding, perforation, infection or second look.

Flexible endourological instruments, digital and current ureteroscopes (LAQ)

- Working channel plus active deflection made flexible scopes therapeutic.
- fURS: 3.6 Fr channel, ≥270° (>270°) deflection; cystoscope 14–17 Fr.
- Chip-on-tip CMOS digital: better image, larger tip, durability not better; shorter operation.
- Single-use cheaper below about 99 cases/year; LithoVue first single-use (2016).
- Semirigid: two channels; below the iliac vessels.

- Pressure sensors and AI are the newest developments.
- Golden rule: never fire unless the fibre tip is seen on the stone.

Simulators in urology (SAQ)

- Bench, VR, animal, cadaver and team simulation.
- Validity: face, content, construct, concurrent, predictive.
- VR beats box or no training for novices.
- E-BLUS and FRS; proficiency-based progression; GEARS.

Irrigation fluids for TURP. (SAQ)

- Monopolar: non-ionic hypo-osmolar fluid; bipolar: saline.
- Glycine 1.5%: ~200 mOsm/kg; transient blindness.
- Risk: gland >45 mL, time >90 min; bag height 60 cm.
- 3% saline; ≤12 mEq/L rise in 24 h.

Indications for outpatient cystourethroscopy. (SAQ)

- Haematuria and NMIBC surveillance are the two commonest indications.
- Flexible scope, lidocaine dwell ≥5 minutes, no routine prophylaxis.
- Therapeutic: stent removal, botulinum toxin, fulguration of small recurrences.

LASERs in urology: holmium, thulium, KTP and newer laser technologies (LAQ)

- Chromophore sets penetration: TFL shallowest, Ho 0.4 mm, Nd:YAG deepest.
- Holmium: 2120 nm; water chromophore; pulsed; 0.4 mm penetration.
- Holmium lithotripsy is photothermal; Moses effect: fibre close and perpendicular.
- Moses: two-part pulse through a vapour channel.
- Holmium: fragment 0.6–1.0 J × 6–10 Hz; dust 0.2–0.3 J × 40–80 Hz.
- HoLEP size-independent, superior to TURP; PVP for anticoagulated men (GOLIATH).
- TFL 1940 nm, Tm:YAG 2013 nm; water absorption about 4× holmium.
- TFL: water peak, low peak power, fine dust, thin fibres, minimal retropulsion; settings ≤40 Hz often enough.
- TFL beats low-power, equals high-power pulse-modulated holmium.
- ThuLEP is size-independent and equal to HoLEP.
- KTP 532 nm: frequency-doubled Nd:YAG; haemoglobin chromophore.
- GOLIATH: 180 W PVP non-inferior to TURP; best for anticoagulated men; not >80 cc.
- Nd:YAG for PeIN; eyewear for everyone.
- Heat depends on power: ≤10 W in the ureter, ≤25 W in the kidney; access sheath; damage above 43 °C.

Energy sources in urology and laparoscopic surgery, including bipolar energy (LAQ)

- Electrosurgery at about 500 kHz; current density decides heating.
- Cut continuous low voltage; coagulation interrupted high voltage.
- Spread: monopolar 5–7 mm > bipolar 1–6 > sealer (LigaSure) 1–3 > ultrasonic ≈1 mm.
- LigaSure seals ≤7 mm; Harmonic vibrates at 25–55 kHz; ultrasonic blade hottest after use.
- Stray current: insulation failure, direct and capacitive coupling; avoid hybrid trocars; use AEM and REM.
- Bipolar: no pad, saline irrigation, no TUR syndrome.
- Bipolar TURP equals monopolar efficacy with fewer complications.

- Holmium 2120 nm: gold standard for stones and HoLEP.
- Cryo below −40 °C, two cycles; RFA 60 °C; MWA less heat sink.
- Thermal ablation for cT1a ≤3 cm.
- HIFU, IRE, VTP: focal prostate therapy, still evolving.

Pneumoperitoneum physiology and complications of laparoscopy and retroperitoneoscopy (LAQ)

- Two drivers: pressure and absorbed CO_2; work at 10–12 mmHg.
- Afterload rises at all pressures; output collapses at 40 mmHg.
- Oliguria from renal vein compression; reversible.
- Hypercarbia peaks after extubation; COPD needs arterial gases.
- Vascular injury is commonest; leave the trocar in place.
- Gas embolism: left lateral, head down, oxygen, aspirate.
- Missed bowel injury: trocar-site pain, leucopenia.
- Retroperitoneoscopy: space is created, not natural: wrong plane gives hernia, peritoneal tear or bleeding.
- Emphysema commonest; balloon-tip port and 12 mmHg prevent it (prevents leak).
- Pneumothorax 1.6–4%, commoner retroperitoneally.
- Rhabdomyolysis about 1%: limit kidney rest to <1 hour.
- Ratner 1995; left kidney preferred; hand-assisted in about two-thirds.
- Mortality 3 per 10,000; conversion 0.26–3.34%.
- Donor ESRD 90 versus 14 per 10,000 (matched non-donors).

Laparoscopic access: Veress, Hasson, trocars, gases; open vs laparoscopic vs robotic prostatectomy (LAQ)

- Veress tests: aspiration, drop, advancement; opening pressure <10 mmHg.
- Hasson 1971: open entry under vision.
- Main indication for Hasson: previous surgery or adhesions.
- Hasson avoids blind Veress injuries; drawback is gas leak.
- Extraperitoneal space is created first: finger, then balloon.
- Gaur balloon: glove on red rubber catheter.
- Balloon in muscle causes hernia.
- Trocar = cannula + obturator.
- Bladeless trocars: fewer epigastric injuries and hernias.
- Balloon trocar seals and prevents retroperitoneal emphysema.
- CO_2 standard (soluble, non-flammable); helium in COPD; N_2O supports combustion.
- Gas embolism: left lateral head-down, oxygen, aspirate.
- Yaxley 2016: equal function and margins; robot less blood loss and stay.

Laparoendoscopic single-site (single-port) surgery: LESS (LAQ)

- LESS = one incision; LESSCAR (2008) standardised the term.
- Core problem: lost triangulation; solved by articulating or curved instruments and robots.
- Same outcomes as laparoscopy with less pain and better cosmesis; more hernias.
- da Vinci SP (FDA 2018): one 25–27-mm cannula, wristed instruments.

3D printing in urology (SAQ)

- Additive manufacturing from CT/MRI via segmentation and an STL file.
- Main uses: partial nephrectomy and PCNL planning, consent, simulation.
- Limits: cost, time, unrealistic tissue, weak outcome evidence. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

Hemostasis in laparoscopic surgery (SAQ)

- Pneumoperitoneum hides venous bleeding: inspect at low pressure before exit.
- LigaSure seals ≤7 mm; spread monopolar 5–7 mm, ultrasonic ≈1 mm.
- Hem-o-lok contraindicated on the donor renal artery.
- Never staple across clips; grey or white loads for vessels.

Energy sources for breaking stones in urology (LAQ)

- SWL: electromagnetic is the modern standard; slow rate (60–90/min) is better.
- Holmium 2120 nm, photothermal, breaks every composition.
- TFL 1940 nm: low peak power, finer dust, less retropulsion.
- EHL is poorly focused, so it is not first-line in the ureter.

Bacteriuria in pregnancy and antibiotic policy (SAQ)

- Prevalence unchanged; risk of pyelonephritis raised (15–45% untreated).
- Screen by culture at 12–16 weeks; treat all ASB for 4–7 days.
- β-lactams safe throughout; no fluoroquinolones or tetracyclines.
- Nitrofurantoin not near term (after 35 weeks); TMP not in 1st trimester.

Clinical trial methodology (SAQ)

- Phase III RCT with concealed allocation and blinding is the gold standard.
- Primary endpoint drives sample size; surrogates need the Prentice criteria.
- ITT analysis; CONSORT flow diagram for reporting.

Evidence-based medicine and critical appraisal (SAQ)

- Sackett 1996: best evidence + clinical expertise + patient values.
- 5 As: Ask (PICO), Acquire, Appraise, Apply, Assess.
- Level 1a = systematic review of RCTs; GRADE: strong or weak.
- NNT = 1/ARR; beware lead-time, length-time and confounding bias.
- Three questions: valid, results, applicable.
- Randomisation, concealment, blinding, ITT, follow-up >80%.
- CONSORT (RCT), PRISMA (reviews), STROBE (observational), STARD (diagnostic).

Recist Criteria. (SAQ)

- RECIST 1.1: ≥10 mm lesion, node ≥15 mm short axis; 5 targets, 2 per organ.
- PR ≥30% fall from baseline; PD ≥20% and ≥5 mm rise from nadir, or new lesion.
- iRECIST for immunotherapy; PCWG3 for prostate.

Health-related quality of life (HRQOL) instruments in urology, including after urinary diversion (SAQ)

- Generic (SF-36, EQ-5D) vs disease-specific (EPIC, IPSS, ICIQ, IIEF) plus bladder-specific (FACT-BI, BLM30, BCI) tools.
- SHIM lacks bother; I-QOL lacks symptoms.

- Reliable, valid, responsive; Cronbach α >0.7; IPSS MCID ~3 points.
- No diversion proven superior; studies flawed.
- Neobladder better body image; conduit better urinary function.

Tissue engineering in urology: principles, status and recent advances (LAQ)

- Triad: cells, scaffold, signals; plus bioreactor and blood supply.
- Diffusion limit about 100–200 µm: vascularity is the main barrier.
- Atala 2006 (7 patients) promising; phase II trial: no gain in compliance or capacity.
- Neo-urinary conduit (conduit trial): half stenosed; no blood supply.
- Raya-Rivera 2011: seeded tubular urethras in 5 boys.
- Still experimental except short urethral onlays.
- New: iPSCs, urine-derived cells, bioprinting, organoids.

Robotics in urology (including robotic vs standard laparoscopy) (LAQ)

- da Vinci 2000; Xi 2014; SP 2018; patents expired 2019; 7 degrees of freedom.
- Yaxley and Coughlin: RARP equal function to open at 12 weeks and 24 months; RAZOR non-inferior; iROC more days out of hospital.
- RAPN trifecta: negative margin, no major complication, warm ischaemia ≤25 min.
- Trendelenburg causes oedema and raised IOP; avoid shoulder braces.
- Robot: 3D, 7 DOF, tremor filtration, but no haptics and high cost.
- Robotic partial nephrectomy: shorter warm ischaemia, fewer conversions.
- Robot made prostatectomy routine; no benefit for simple extirpation.
- Surgeon matters more than platform.

Natural orifice endoscopic surgery. (LAQ)

- NOTES = transluminal via natural orifice; hybrid transvaginal is the only routinely used urological form.
- Transvaginal extraction removes the most painful wound.
- LESS: one incision, lost triangulation, more hernias.
- Barriers: closure, infection, orientation, platform; no proven benefit yet.

Pioneers of urology: Novick, Chaussy, Hopkins, Huggins, Calne, Gaur and Asopa (SAQ)

- Hopkins: rod-lens system 1966; fibreoptics 1954.
- Huggins and Hodges 1941: androgen dependence; castration regresses prostate cancer; Nobel 1966.
- Huggins 1945: adrenalectomy for castration-resistant disease.
- Calne: azathioprine, ciclosporin, first UK liver transplant 1968.
- Gaur 1992: glove-balloon retroperitoneoscopy.
- Asopa: ventral sagittal urethrotomy, dorsal plate incised, free graft inlaid (2001).
- Novick (Cleveland Clinic): nephron-sparing, renovascular and bench surgery.
- PN indications: imperative, relative, elective (cT1a ≤4 cm).
- Warm ischaemia under about 25–30 minutes; ice slush if longer.
- Negative margin is enough; keep ≥20–30% of one kidney.
- Chaussy, Munich: first human SWL 7 February 1980, Dornier HM1.
- HM3 (1983) is still the efficacy benchmark.

- Spallation, squeezing, shear, cavitation; electromagnetic sources today.

History of Laparoscopy (SAQ)

- Bozzini 1806; Kelling 1901 (dog); Jacobaeus 1910 (human); Veress 1938; Hopkins rod-lens.
- Urology: Cortesi 1976; Clayman 1990; Gaur 1992; Ratner 1995.
- da Vinci 2000; Binder first robotic prostatectomy; robot solved suturing barrier.

Flow cytometry (SAQ)

- Forward scatter = size; side scatter = granularity; fluorescence = marker or DNA.
- DNA index 1.0 diploid; aneuploidy = worse prognosis.
- Flow crossmatch is more sensitive than CDC crossmatch.

Haemostatic agents in urology (SAQ)

- Agents are adjuncts, not substitutes for sutures.
- Fibrin sealant works despite coagulopathy; passive agents need normal clotting.
- Surgicel: acidic, may mimic abscess.
- Alum first line; formalin last, after cystogram.

Sterilisation of urological instruments (SAQ)

- Spaulding: critical sterilise, semi-critical HLD, non-critical low-level.
- Autoclave 121 °C 15 min or 134 °C 3–4 min.
- Ureteroscopes: ethylene oxide, glutaraldehyde or peracetic acid.
- Biological indicator: *G. stearothermophilus*.

Amyloidosis in urology (SAQ)

- Congo red, apple-green birefringence.
- Localised bladder amyloid mimics cancer; biopsy then TUR.
- Always exclude systemic AL disease. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

Collin's knife. (SAQ)

- Knife-shaped resectoscope electrode for incision, not resection.
- Uses: TUIP, bladder neck contracture, ureterocele, PUV, diverticulum neck.
- Monopolar needs glycine or sorbitol; bipolar uses saline.

APOGEE (SAQ)

- Apogee: transvaginal posterior/apical polypropylene mesh kit.
- Exposure, dyspareunia and pain drove withdrawal.
- FDA 2019 stopped transvaginal POP mesh sales.

Urethral stents and metallic stents in urology (SAQ)

- Temporary (Memokath, Allium) vs permanent (UroLume, withdrawn).
- Metal resists extrinsic compression; weakness is ingrowth and encrustation.
- Resonance: coiled, full-length, changed every 12 months.
- Memokath: thermo-expandable nitinol, removable.
- Migration, encrustation, ingrowth; never cross the sphincter.
- AUA: no stent if urethroplasty planned; Optilume and iTind are newer.

Urological applications in nanotechnology. (SAQ)

- 1–100 nm; enhanced permeability and retention allows tumour targeting.
- USPIO MRI: benign nodes lose signal; metastatic nodes do not.
- AuroLase gold nanoshells: photothermal focal therapy.

Morcellation (SAQ)

- Morcellator made HoLEP practical; distend bladder, haemostasis first.
- Morcellation loses stage information; histology still obtained.
- FDA 2014: uncontained power morcellation spreads occult sarcoma.

Stem cell therapy (SAQ)

- Adult stem cells act mainly by paracrine signals.
- SUI muscle-derived cells and ED stem cells are investigational.
- Embryonic and iPSCs carry tumour risk.

Advances in lap surgeries (SAQ)

- Blunt trocars cut port-site hernia to 0.19%.
- AirSeal allows 8–12 mmHg working pressure.
- ICG fluorescence guides selective clamping.
- LESS: cosmesis at the cost of triangulation.

Artificial intelligence in urology (SAQ)

- Deep learning is part of ML, which is part of AI; CNNs for images.
- CNNs: very high sensitivity for CT stones.
- Paige Prostate: first FDA-authorised AI pathology tool (2021).
- Limits: data, bias, black box, validation, liability.

3D laparoscopy (SAQ)

- Two lenses, two images, polarised glasses: binocular disparity gives depth.
- Main gain: faster, more accurate suturing; shorter learning curve.
- Drawbacks: eye strain, headache, cost, glasses for all.
- Robotic vision at low cost, but without wristed instruments.

Telemedicine in urology (SAQ)

- Telemedicine trades examination for access.
- Best for follow-up, results and history-driven visits.
- Limits: no examination, connectivity, elderly, medico-legal.
- India: Telemedicine Practice Guidelines 2020; eSanjeevani.

Virtual and augmented reality in urology (SAQ)

- VR immerses; AR overlays on reality.
- Main uses: planning partial nephrectomy, PCNL and prostatectomy; navigation; teaching; counselling.
- Biggest problem: registration on moving, deforming organs.
- Evidence for outcomes still limited.

SECTION 2

Oncology, DSD, Gender & Ageing

24 Basic Principles of Immunology and Immunotherapy in Urologic Oncology

1 · Innate and adaptive immunity

LAQ Immunology, immunotherapy and cancer vaccines in urological oncology (including Sipuleucel-T)

Paper answer · P2 · P4 · merged from 2

Asked as: P2 LAQ "Describe use and principles of immunology and Immunotherapy in Urological Oncology" · P4 LAQ "Rationale of using cancer vaccines in Urology. Explain their mechanism of action. Enumerate various vaccines for renal, prostate and bladder malignancies . Write a note on Sipuleucel-T."

Introduction

- Immunotherapy works by either **supplying a danger signal** that starts a Th1 response (BCG (bacille Calmette–Guérin), vaccine adjuvant) or **removing a brake** on an existing response (PD-1 (programmed death-1), CTLA-4 (cytotoxic T-lymphocyte antigen-4)).
- A **cancer vaccine** is a therapeutic agent that trains the patient's own T cells to recognise **tumour antigens**. It treats existing cancer; it does not prevent it.
- **BCG**, a tuberculosis vaccine, was among the first immune treatments for any cancer and is the original urological immunotherapy. It is non-specific; cancer vaccines aim for a **specific** anticancer response.

Principles of tumour immunology

Arm	Cells and role
Innate (rapid, pattern-based)	PAMPs (pathogen-associated molecular patterns) bind **TLRs (Toll-like receptors)**; macrophages, neutrophils; urothelium as sentinel (IFN (interferon), IL-8 (interleukin-8))
Bridge	**Dendritic cell (DC)**: takes up antigen, matures on danger signal, migrates to lymph node
Adaptive (specific, memory)	**CD8 killer** (sees class I MHC (major histocompatibility complex), ~9 amino acids); **CD4 helper** (class II); B cells
Brake	**Treg** (regulatory T cell, FoxP3+ CD4); MDSCs (myeloid-derived suppressor cells); M2 macrophages

- **Two signals**: TCR (T-cell receptor) meets MHC–peptide (signal 1) plus **CD28–B7** costimulation (signal 2). Without signal 2 the T cell becomes tolerant.
- **Th1 vs Th2**: IL-12 gives Th1 (IL-2, IFN-γ, TNF-α (tumour necrosis factor-α)), which is antitumour. IL-4 gives Th2 (antibody, chronic inflammation).
- **Checkpoints**: CTLA-4 (on Tregs) competes for B7; PD-1 on CD8 cells binds **PD-L1** (programmed death ligand-1); LAG-3 binds class II.

- **Adaptive immune resistance**: CD8 cells secrete IFN-γ, which induces PD-L1 on the tumour, which switches the same T cells off. So PD-L1 marks a suppressed ongoing response.

Immunoediting (three Es) and the rationale for immunotherapy in urology

- **Elimination** (small tumours destroyed), then **Equilibrium** (growth held), then **Escape** (clinical cancer).
- Escape is by loss of antigen or MHC, Treg expansion and checkpoint ligands. Vaccines try to restart the response.
- **Bladder:** inflammation-driven (schistosomiasis); CD8 infiltration is good; BCG success; high mutation load gives many neoantigens.
- **Kidney: CD8 paradox** (more CD8, worse outcome); rare spontaneous regression of metastases after nephrectomy; responses to IL-2 and interferon.
- **Prostate:** Treg-dominated and PD-L1 low, so checkpoint inhibitors fail; slow growth gives time to build a vaccine response; **organ-specific antigens** (PSA (prostate-specific antigen), PAP (prostatic acid phosphatase)).
- Vaccines have low toxicity and offer possible lasting memory.

BCG in NMIBC (non-muscle-invasive bladder cancer)

- Intravesical live attenuated mycobacteria.
- Mechanism (sequence): 1. Bacilli attach via **fibronectin**. 2. TLR2/4 activation → IL-8 and TNF-α. 3. Neutrophil wave → DC priming. 4. **Th1 CD4** response → memory.
- Th1 predicts success, Th2 failure. About half stay in remission for years.
- Used for intermediate/high-risk NMIBC and CIS (carcinoma in situ) with induction plus maintenance; standard for high-risk NMIBC.

Cytokines and gene/viral therapy

- **IFN-α** and **high-dose IL-2**: durable complete responses in a few mRCC (metastatic renal cell carcinoma) patients; capillary leak. Intravesical IFN-α: mixed results.
- **Nadofaragene firadenovec** (adenovirus carrying the IFN-α2b gene + Syn3): BCG-unresponsive CIS, CR (complete response) 53.4%; FDA (Food and Drug Administration) approved December 2022.
- **Cretostimogene**: oncolytic adenovirus with GM-CSF (granulocyte-macrophage colony-stimulating factor) gene.
- **N-803** (IL-15 superagonist) with BCG: FDA approved 2024 for BCG-unresponsive CIS.

Cancer vaccines: mechanism of action

- Two components: **antigen** (overexpressed in tumour) + **adjuvant** (a danger signal, often GM-CSF, which activates DCs).
- Sequence: `Intradermal antigen + adjuvant → immature DC takes up antigen; adjuvant matures it → DC migrates to lymph node → peptide on MHC I/II = signal 1; B7 costimulation = signal 2 → CD4 Th1 help (IL-2, interferon-γ) + CD8 cytotoxic T cells → T cells traffic to tumour → apoptosis of antigen-bearing cells → memory T cells`
- Without signal 2 the T cell becomes tolerant, hence the adjuvant.
- **Prime–boost:** the first dose primes, later doses boost.
- **Delayed effect:** survival may improve without tumour shrinkage or PSA fall.

- **Resistance:** PD-L1 on tumour, Tregs, MDSCs, MHC loss. This is the basis for combining vaccines with checkpoint inhibitors.

Vaccine platforms

Platform	Example
Whole-cell (autologous or allogeneic)	Reniale (RCC (renal cell carcinoma)); GVAX (prostate)
Peptide/protein	IMA901
Dendritic-cell (ex vivo loaded)	Sipuleucel-T; AGS-003 (rocapuldencel-T)
Viral vector (poxvirus)	PROSTVAC-VF; TroVax (MVA-5T4)
Heat-shock protein–peptide	Vitespen (HSPPC-96)
DNA/mRNA, personalised neoantigen	Phase I–II trials

Vaccines by organ

Renal cell carcinoma

- **IMA901:** 10 tumour-associated peptides + GM-CSF, with low-dose cyclophosphamide to deplete Tregs; HLA-A*02+ (human leucocyte antigen) metastatic RCC with sunitinib. **No OS (overall survival) benefit.**
- **AGS-003 (rocapuldencel-T):** autologous DCs loaded with tumour RNA, with sunitinib. **Terminated for lack of efficacy.**
- **TroVax** with standard first-line therapy: no OS benefit. **Vitespen** (adjuvant): no overall benefit.
- **Reniale** (adjuvant autologous tumour-cell vaccine): claimed benefit, not adopted.
- Checkpoint inhibitors, not vaccines, are today's RCC immunotherapy.

Prostate cancer

- **Sipuleucel-T:** the **only approved** vaccine (see note below).
- **PROSTVAC-VF:** vaccinia prime, fowlpox boost, encoding PSA + TRICOM (three costimulatory molecules). Phase III **closed for futility**.
- **GVAX:** allogeneic whole-cell lines secreting GM-CSF. Phase III trials **failed**.
- Also: DCVAC/PCa (phase III negative); DNA vaccines against PAP (trials).

Bladder cancer

- **BCG** (above); nadofaragene firadenovec, cretostimogene and N-803 (above).
- Peptide and neoantigen vaccines: experimental only.

Why most vaccines failed

- Tolerance to self-antigens; immunosuppressive tumour environment; dysfunctional patient DCs; bulky late disease; progression endpoints miss delayed effects.
- **Pattern:** IMA901, AGS-003, PROSTVAC and GVAX all failed; **only sipuleucel-T succeeded in phase III.**

Note on Sipuleucel-T

- **Nature:** autologous cellular immunotherapy: the patient's **CD54+ APCs (antigen-presenting cells)** activated with **PA2024**, a fusion protein of **PAP + GM-CSF**.
- **Manufacture and dosing:** 1. Leukapheresis collects PBMCs (peripheral blood mononuclear cells). 2. Cells incubated about 36 h with PA2024 at a central facility. 3. Infused

intravenously about 3 days after collection. 4. **Three doses at 2-week intervals** (one-month course).

- **Mechanism:** primes PAP-specific CD4 and CD8 T cells; later infusions boost.
- **IMPACT trial (Kantoff, NEJM 2010):** 512 men, 2:1 versus placebo, asymptomatic or minimally symptomatic mCRPC (metastatic castration-resistant prostate cancer).
- **OS 25.8 vs 21.7 months (HR (hazard ratio) 0.78)**, a gain of 4.1 months, despite placebo crossover.
- **No change in time to progression or PSA.**
- **FDA approval April 2010:** first therapeutic vaccine for a solid tumour.
- **Indication:** asymptomatic or minimally symptomatic mCRPC, good performance status, **no visceral metastases, no opioid need**; usually before chemotherapy.
- **Toxicity:** mild infusion reactions (fever, chills, fatigue); premedicate with paracetamol and antihistamine.
- **Limitations:** cost, logistics, no response marker; not available in India.

Checkpoint inhibitors

Site	Use (trial)
BCG-unresponsive CIS	Pembrolizumab if cystectomy declined/unfit (KEYNOTE-057: CR 41%)
Adjuvant MIBC (muscle-invasive bladder cancer) and UTUC (upper tract urothelial carcinoma)	Nivolumab after radical surgery (**CheckMate 274**: DFS (disease-free survival) 20.8 vs 10.8 months; included UTUC, benefit uncertain in that small subgroup). Atezolizumab (IMvigor010) negative
Metastatic UC (urothelial carcinoma)	Avelumab **maintenance** after platinum (JAVELIN Bladder 100, OS gain); pembrolizumab second line (**KEYNOTE-045**: OS 10.3 vs 7.4 months); **enfortumab vedotin (EV) + pembrolizumab**
RCC	Nivolumab + ipilimumab (CheckMate 214); IO (immuno-oncology) + TKI (tyrosine kinase inhibitor) combinations; adjuvant pembrolizumab (KEYNOTE-564, DFS HR 0.68)
Prostate	Monotherapy poor; pembrolizumab only for **MSI-H/dMMR** (microsatellite instability-high/mismatch repair-deficient) tumours

- VEGF (vascular endothelial growth factor) suppresses DCs and T-cell entry, so TKIs synergise with IO.
- **Biomarkers**: PD-L1 CPS (combined positive score) only for first-line single-agent pembrolizumab in cisplatin-ineligible UC; TMB (tumour mutational burden), MSI.

Adoptive cell therapy (brief)

- **CAR-T (chimeric antigen receptor T cells)**: T cells engineered against a surface antigen (e.g. PSMA (prostate-specific membrane antigen)). Investigational in GU (genitourinary) cancers; limited by poor trafficking, hostile microenvironment and cytokine release syndrome.

Toxicity: irAEs (immune-related adverse events)

- Colitis, hepatitis, pneumonitis, thyroiditis, hypophysitis, nephritis, myocarditis, rash.

Grade	Action
1	Continue, monitor
2	Hold; prednisolone 0.5–1 mg/kg
3–4	Stop; methylprednisolone 1–2 mg/kg; infliximab or mycophenolate if refractory

- Endocrinopathies need hormone replacement.
- BCG toxicity: cystitis, BCG sepsis.

Recent advances

- **EV-302**: enfortumab vedotin + pembrolizumab first line in metastatic UC, OS 31.5 vs 16.1 months versus platinum; now the preferred first line.
- **NIAGARA**: perioperative durvalumab with neoadjuvant cisplatin chemotherapy improved event-free and overall survival in MIBC.
- **Also know:** CheckMate 901 (nivolumab + gemcitabine–cisplatin); ADT (androgen deprivation) + IO trials; anti-CD38 and LAG-3 targets; DNA/mRNA personalised neoantigen vaccines (phase I–II).

Diagrams to draw:

- T cell–tumour synapse with TCR–MHC, CD28–B7, CTLA-4 and PD-1/PD-L1 blocked by antibodies.
- Sipuleucel-T flow: leukapheresis, PA2024 loading, reinfusion, T-cell priming.

Key points to remember:

- Three Es: elimination, equilibrium, escape; Th1 is antitumour.
- BCG is the original urological immunotherapy: TLR2/4 → neutrophils → Th1 memory.
- Vaccine = antigen + adjuvant; DC gives signal 1 (MHC–peptide) and signal 2 (costimulation).
- IMA901, AGS-003, PROSTVAC, GVAX all failed in phase III.
- Sipuleucel-T: PAP–GM-CSF; IMPACT OS 25.8 vs 21.7 months (+4.1 months); no PSA or progression effect.
- CheckMate 274 adjuvant nivolumab; EV-302 EV + pembrolizumab first line.

Also, from the condensed book (not in the paper answer):

- Th1 cytokines (IL-2, TNF-α, IFN-γ) mean **BCG success**; Th2 cytokines (IL-4, IL-5, IL-10, IL-13) mean **BCG failure**.

ALGORITHM

1. Danger signal (TNF-α, TLR ligand) reaches a DC.
2. DC stops antigen uptake; raises MHC and B7-1/B7-2.
3. DC enters afferent lymphatics (CCL21 sensed by CCR7).
4. In the node it activates CD4, CD8 and B cells.

	Class I MHC	Class II MHC
Recognised by	CD8	CD4
Distribution	Almost all cells; raised by IFN	Antigen-presenting cells (APCs)

	Class I MHC	Class II MHC
Peptide	~9 amino acids	13–25 amino acids

2 · Immune editing and escape

- Escape follows equilibrium, as antigen and MHC fall and Tregs rise.
- In adaptive immune resistance, CD8-derived IFN-γ induces tumour PD-L1, which switches off those PD-1-positive CD8 cells.

> **KEY POINT**
>
> PD-L1 is mostly induced by the immune attack, so it marks a suppressed response, not a fixed trait.

T-cell molecule	Ligand	Effect
CD28	CD80/CD86 (B7)	Activating
CTLA-4	CD80/CD86	Inhibitory; mainly on FoxP3+ Tregs
PD-1	PD-L1 (B7-H1), PD-L2	Inhibitory; mainly on CD8 tumour-infiltrating lymphocytes (TILs)
LAG-3	MHC class II	Inhibitory

3 · Inflammation and genitourinary (GU) tumours

	Bladder	Kidney	Prostate
Does inflammation cause cancer?	Strong: schistosomiasis, chronic UTI, catheters, cyclophosphamide	Weak	Probable: proliferative inflammatory atrophy beside HGPIN
CD8 infiltration	Better outcome in MIBC	**Worse outcome — renal paradox**	May be suppressive
Dominant restraint	PD-L1, rising with grade	PD-1/LAG-3-high T cells, few APCs	Tregs; PD-L1-negative tumour

4 · BCG in bladder cancer

> **ALGORITHM**
>
> 1. Instil 1–5 × 10^8 bacilli.
> 2. A fraction adheres via **fibronectin attachment protein**.
> 3. Urothelial TLR2/TLR4 release IL-8 and TNF-α.
> 4. Neutrophil wave (hours), then DC activation.
> 5. CD4 Th1 response in nodes (weeks), then memory.

- Memory explains durable remission in ~**50%** of NMIBC.
- **Nadofaragene firadenovec**, an adenovirus carrying the IFN-α2b gene with **Syn3**, is given 3-monthly for up to 12 months (CIS CR 53.4%). It was the FDA's **first gene therapy** (2022), for BCG-unresponsive NMIBC with CIS.

5 · Cancer vaccines

P4 *Rationale of using cancer vaccines in Urology — antigen + GM-CSF adjuvant; sipuleucel-T (PAP–GM-CSF, 3 doses 2-weekly, IMPACT); failed IMA901, AGS-003, PROSTVAC*

- Sipuleucel-T, the first FDA-approved solid-tumour vaccine, prolongs OS by **+4.1 months** with **no effect on time to progression**.

6 · Checkpoint blockade

Agent	Target
Nivolumab, pembrolizumab	PD-1
Avelumab, atezolizumab, durvalumab	PD-L1
Ipilimumab	CTLA-4

Bladder

- First-line pembrolizumab alone (KEYNOTE-052) requires cisplatin-ineligibility with CPS **≥10**, or platinum-ineligibility; with CPS <10 it was worse than carboplatin.
- Atezolizumab and durvalumab were withdrawn, and adjuvant atezolizumab (IMvigor010) was negative.
- **Enfortumab vedotin (EV)**, an anti-nectin-4 antibody–drug conjugate, gave a 68% response rate with pembrolizumab when cisplatin-ineligible (EV-103).
- †In EV-302, EV plus pembrolizumab roughly doubled PFS and OS versus platinum and is now first line for all.

Kidney

- CTLA-4 on Tregs and PD-1 on CD8 TILs make dual blockade synergistic.
- VEGF blocks DC maturation and T-cell entry, so TKIs complement immunotherapy.

Trial (vs sunitinib)	Combination	ORR	Median PFS, months
KEYNOTE-426	Pembrolizumab + axitinib	60% vs 40%	15.4 vs 11.1
JAVELIN Renal 101	Avelumab + axitinib (only anti-PD-L1)	52.5% vs 27.3%	13.3 vs 8.0
CLEAR	Pembrolizumab + lenvatinib (best)	71% vs 36%	23.9 vs 9.2
CheckMate 9ER	Nivolumab + cabozantinib	55.7% vs 27.1%	16.6 vs 8.3

Prostate

- In mCRPC, tumours are rarely PD-L1-positive; pembrolizumab gives a PSA response under 10%, and ipilimumab and IMbassador250 gave no OS benefit.
- The exception is **MSI-H/dMMR** tumours (1.2–12%), treated with pembrolizumab second line or later; its 2017 approval was the **first based on a biomarker, not a site**.
- †Immune-related adverse events of grade ≥2 (such as colitis or pneumonitis) need drug hold and corticosteroids; endocrinopathies need lifelong replacement.

> **KEY POINT**
>
> Successful GU immunotherapy either supplies a danger signal (BCG, adjuvant, VEGF inhibition) or removes a brake (PD-1, CTLA-4); it fails where the brake is not limiting, as in Treg-dominated prostate cancer.

7 · Landmark trials

Trial	Design	Result	Take-home
†IMPACT	Sipuleucel-T vs placebo (2:1), mCRPC	OS 25.8 vs 21.7 months	Survival without progression delay
KEYNOTE-057	Pembrolizumab, BCG-unresponsive CIS	CR 41% at 3 months	FDA 2020
KEYNOTE-045	Pembrolizumab vs chemotherapy, post-platinum	OS 10.3 vs 7.4 months	Only phase III-based second-line approval
JAVELIN Bladder 100	Avelumab maintenance after platinum	Longer OS	Standard switch maintenance
CheckMate 274	Adjuvant nivolumab vs placebo, high-risk MIBC	DFS 20.8 vs 10.8 months	Adjuvant option
CheckMate 025	Nivolumab vs everolimus, RCC after TKI	Longer OS, less toxicity	First RCC checkpoint approval
CheckMate 214	Nivolumab + ipilimumab vs sunitinib, first line	OS hazard ratio 0.63	First line intermediate/poor risk
KEYNOTE-564	Adjuvant pembrolizumab for 1 year, high-risk clear-cell RCC	Recurrence or death reduced by 32%	Adjuvant standard

25 Molecular Genetics and Cancer Biology

1 · Hallmarks and basic molecular genetics

- **Local invasion** is the one hallmark that uniquely separates malignant from benign.
- **Alternative splicing** makes ligand-independent AR variants (AR-V7), causing castration resistance.
- **Somatic** testing finds targetable mutations; **germline** testing finds hereditary syndromes for surveillance and family cascade testing.

2 · Gene alterations: tumour suppressors and oncogenes

Oncogenes need one altered allele, suppressors usually two.

SAQ Oncogenes

Paper answer · P1

Asked as:

- Oncogenes — *Winter 2017 Paper I Q3f (SAQ)*
- Oncogene — *Practice Set 6 Paper I Q3f (SAQ)* (also answered in 2017-Winter Paper 1, Q3f)

Type: SAQ (10 marks)

Definition

- **Proto-oncogene:** a normal gene that promotes cell growth and division (an "accelerator").
- **Oncogene:** its mutated or overactive form that drives (driving) uncontrolled proliferation. The change is **gain of function** and **dominant**: one altered allele is enough.
- The first human oncogene, **HRAS**, was isolated from (found in) the **T24 bladder cancer** cell line.

Mechanisms of activation

Mechanism	Effect	Example
Point (activating) mutation	Constitutively active protein (protein permanently "on")	RAS (bladder); FGFR3 (fibroblast growth factor receptor 3) in low-grade papillary bladder cancer; MET germline (germline MET) in hereditary papillary RCC (renal cell carcinoma)
Gene amplification	Increased copy number (more copies) → overexpression	MYC (prostate, bladder); HER2 (human epidermal growth factor receptor 2); AR (androgen receptor) in castration-resistant prostate cancer
Chromosomal translocation / fusion	Gene placed under a strong promoter, or fusion protein	**TMPRSS2–ERG** (androgen-driven ERG in about half of prostate cancers); BCR–ABL; TFE3 fusions (translocation RCC)
- **Also know:** promoter hypomethylation can switch oncogenes on.		

Functional classes

Class	Examples
Growth factors	PDGF (platelet-derived growth factor), sis

Class	Examples
Receptor tyrosine kinases	EGFR (epidermal growth factor receptor), HER2, FGFR3, MET, KIT (seminoma)
Signal transducers	**RAS** (binds GTP [guanosine triphosphate]; GTP-binding; mutation locks it "on"), BRAF, PI3K
Transcription factors	**MYC**, ERG, ETS family
Cell cycle / apoptosis regulators	Cyclin D1, BCL-2

Oncogene versus tumour suppressor gene

	Oncogene	Tumour suppressor
Normal role	Accelerator	Brake, DNA repair
Change	Gain of function	Loss of function
Alleles needed	One	Usually both (**Knudson two-hit**: retinoblastoma)
GU (genitourinary) examples	MYC, MET, FGFR3, ERG	RB1, TP53, VHL, PTEN

Urological significance

- **Bladder:** FGFR3 mutations in up to 80% of low-grade Ta tumours (10–20% of MIBC (muscle-invasive bladder cancer)); FGFR3 and TERT promoter mutations in >60% of low-grade papillary tumours; RAS (HRAS) mutually exclusive with FGFR3; HER2 in 9–12% of MIBC; MYC overexpressed in over half.
- **Kidney:** MET activation in papillary RCC.
- **Prostate:** TMPRSS2–ERG fusion, MYC (8q gain), AR amplification after (under) hormone therapy.
- **Testis:** KIT and KRAS mutations (seminoma).

Targeted therapy ("oncogene addiction")

- **Erdafitinib** (pan-FGFR inhibitor) for FGFR-altered (FGFR3/2-altered) advanced (metastatic) urothelial cancer; MET-active TKIs (tyrosine kinase inhibitors) in papillary RCC; AR-targeted drugs in prostate cancer; HER2-directed drugs (trastuzumab deruxtecan received a tumour-agnostic FDA approval in 2024 for HER2-positive solid tumours; disitamab vedotin is under study in urothelial cancer).

Diagram to draw: growth factor → receptor → RAS → RAF → MEK → ERK → MYC pathway.

> **Key points to remember:**
> - Oncogenes: gain of function, one allele; activated by mutation, amplification, translocation.
> - HRAS first found in T24 bladder cancer; FGFR3 in up to 80% low-grade Ta.
> - TMPRSS2–ERG in about half of prostate cancers; MET in papillary RCC.

Also, from the condensed book (not in the paper answer):

- Two-hit exceptions are **dominant negative** mutation (mutant TP53) and **haploinsufficiency** (p27/CDKN1B).

3 · Cell cycle

- The cycle takes ~24 h, and cyclin-dependent kinases (CDKs) work only with their phase-specific **cyclin**.

Phase	Complex	Function
G1	Cyclin D–CDK4/6, cyclin E–CDK2	Phosphorylate RB1; cyclin D needs sustained mitogen (Ras/Raf/MEK)
S	Cyclin A–CDK2, then cyclin A–CDK1	S-phase genes
G2/M	Cyclin B–CDK1	Phosphorylates lamins, so the nuclear envelope breaks down

Checkpoint	Trigger	Mechanism
G1/S	Mitogens, nutrients, O_2, DNA damage	p21 (CDKN1A) and 14-3-3σ inactivate G1 CDKs; TP53 triggers apoptosis if damage is unrepaired
G2/M	DNA damage (radiation)	ATM activates CHEK1/CHEK2, which sequester Cdc25; TP53 induces p21 and GADD45
Spindle assembly	Unattached chromosome	Blocks anaphase-promoting complex; failure causes non-disjunction and **aneuploidy** (near-universal in solid tumours)

- CDK inhibitors are **Cip/Kip** (p21, p27, p57), acting broadly, and **INK4** (p15, p16, p18, p19), acting on CDK4/6.

RB1 — guardian of G1/S

ALGORITHM

1. Hypophosphorylated RB1 binds E2F and HDACs, silencing S-phase genes.
2. Cyclin D–CDK4/6 and cyclin E–CDK2 hyperphosphorylate RB1.
3. E2F is released and the restriction point is passed (irreversible).

- **TP53**, activated by DNA damage or oncogene stress, causes arrest, repair, senescence or apoptosis. **MDM2** is its negative regulator; TP53 is mutated in over half of cancers.

> **KEY POINT**
>
> Cancer must break both RB1 (entry) and TP53 (integrity), so both pathways are lost in almost every tumour.

4 · DNA damage and repair

Pathway	Lesion	Key proteins	Cancer link
Nucleotide excision repair (NER)	Bulky adducts, cross-links (UV, chemicals)	XPC, XPD, ERCC1/XPF	Xeroderma pigmentosum; **ERCC2 in bladder cancer**
Base excision repair (BER)	Single-base damage, single-strand breaks	MUTYH, OGG1, PARP1/2	OGG1 variants linked to RCC
Mismatch repair (MMR)	Mispairs, indel loops at repeats	MSH2/MSH6, MLH1/PMS2	Mutation rate raised ~100-fold; Lynch
Homologous recombination (HR)	Double-strand break (DSB)	ATM, RAD51, BRCA1, BRCA2	Breast/ovarian, prostate

Pathway	Lesion	Key proteins	Cancer link
Non-homologous end joining (NHEJ)	DSB	Ku70/Ku80, DNA-PKcs, ligase IV	Translocations

- Unrepaired DSBs (marked by **γH2AX** foci) cause apoptosis, the basis of radiotherapy. **HR** copies the sister chromatid and is precise; **NHEJ** is template-free and error-prone.

> **KEY POINT**
>
> Repair defects are targets: HR loss means PARP inhibitor sensitivity, MMR loss means immunotherapy, and ERCC2 loss means cisplatin sensitivity.

5 · Apoptosis

- In apoptosis, initiator caspases 8, 9, 10 activate executioners 3, 6, 7.

	Intrinsic (mitochondrial)	Extrinsic (death receptor)
Trigger	Unrepaired DNA damage, lost survival signals	Fas ligand/TNF on Fas (CD95)/TNFR1
Sequence	Bax/Bak release cytochrome c; Apaf-1 apoptosome activates caspase-9	FADD recruits procaspase-8, giving caspase-8
Brakes	Bcl-2, Bcl-XL, Mcl-1; IAPs	FLIP
TP53	Dependent (transactivates Bax)	**Independent**, so a target when TP53 is mutated

- Bid links the two pathways. TP53 loss causes chemoradiotherapy resistance.

6 · Senescence, telomeres, autophagy

SAQ Telomerase

Paper answer · P3

Type: SAQ (10 marks)
Source: Practice Set 6 Paper 3, Q3b

Telomeres

- **Telomeres** are repetitive sequences (TTAGGG repeats) at chromosome ends. They protect ends from degradation and fusion.
- DNA polymerase cannot copy the very end of the lagging strand (end-replication problem). So about **100 base pairs are lost per division**: the "mitotic clock".
- Critical shortening triggers **replicative senescence** (Hayflick limit).

Telomerase: structure and function

- A specialised **reverse transcriptase** that adds telomeric repeats.
- Two core parts: **hTERT** (human telomerase reverse transcriptase, catalytic protein) and **hTR** (telomerase RNA, the template).
- **Normal expression:** only rapidly dividing tissues: germ cells (testis), intestinal epithelium, endometrium, activated lymphocytes. Most somatic cells lack it.

Role in cancer

```
Proliferation -> moderate shortening -> senescence/apoptosis (barrier 1) ->
checkpoint lost (p53) -> severe shortening -> chromosomal instability (barrier
```

2) -> telomerase activation -> immortal cancer cell

- **85–90% of human tumours** express high telomerase, giving unlimited replication.
- **ALT (alternative lengthening of telomeres):** telomerase-independent recombination; linked to ATRX or DAXX loss.
- **TERT promoter mutations** switch on hTERT.

Urological relevance

Organ	Finding
Bladder	TERT promoter mutation is the commonest mutation (over 60% even in PUNLMP, papillary urothelial neoplasm of low malignant potential). Urine markers: telomerase activity by **TRAP (telomeric repeat amplification protocol)**, a PCR (polymerase chain reaction)-based assay; hTERT messenger RNA; AssureMDx (FGFR3, TERT, HRAS mutations plus methylation). Inflammation gives false positives; not routine
Upper tract	hTR positive in almost all UTUC (upper tract urothelial carcinoma); rises with stage
Prostate	Short telomeres in cancer and premalignant lesions; short tumour telomeres predict recurrence; variable stromal telomeres predict cancer death

- **Also know:** senescent cells secrete SASP (senescence-associated secretory phenotype) cytokines in ageing and BPH (benign prostatic hyperplasia).

Therapy

- **Imetelstat** (telomerase inhibitor) is used in myelodysplastic syndrome.
- hTERT-based vaccines (UV1, GX301, INO-9012) are in early prostate, renal and bladder trials.

Diagram to draw: telomere-to-cancer flowchart with two barriers.

> **Key points to remember:**
> - Telomerase = hTERT (catalytic) + hTR (RNA template).
> - About 100 base pairs lost per division; 85–90% of tumours express telomerase.
> - TERT promoter mutation is the commonest mutation in bladder cancer.
> - TRAP assay detects urinary telomerase; inflammation gives false positives.

Also, from the condensed book (not in the paper answer):

- Telomerase is high in **85–90%** of tumours.
- **Senescent** cells are viable but permanently arrested; ADT induces senescence in some prostate cancers.
- **Autophagy** suppresses early tumours but sustains established ones.

7 · Epigenetics

- Epigenetic change alters expression without sequence change and is **reversible**. DNA methylation at CpG islands is maintained by DNMT1.
- Promoter hypermethylation silences suppressors as a "second hit": **GSTP1 in >90% of prostate cancer** (commonest somatic change), RASSF1A (up to 97% of bladder cancer) and p16. Global hypomethylation causes genomic instability.

- **H3K27me3 (written by EZH2)** represses genes; EZH2 is raised in mCRPC and predicts PSA recurrence.
- MicroRNA (18–25 nt) silences **many** mRNAs (**miR-371a-3p** in testis cancer) and siRNA **one**; long ncRNA HOTAIR drives castration resistance.

8 · Hypoxia, VHL and angiogenesis

ALGORITHM

1. In normoxia, HIF-α is prolyl-hydroxylated.
2. VHL (an E3 ubiquitin ligase) binds it for proteasomal degradation.
3. With hypoxia or VHL loss, HIF-α accumulates and dimerises with HIF-1β.
4. This switches on VEGF, angiopoietin and glycolytic genes.

- The **Warburg effect** (aerobic glycolysis) is best seen in clear cell RCC.

KEY POINT

VHL loss means permanent pseudohypoxia, driving angiogenesis, Warburg metabolism and immunosuppression; hence clear cell RCC responds to VEGF, HIF-2α and immune blockade.

- **Lineage plasticity** under driver targeting gives neuroendocrine prostate cancer.

9 · GU tumour genomics

Prostate

Gene	Significance
TMPRSS2–ERG	Commonest fusion (~half)
SPOP	Commonest mutated suppressor in primary tumours
TP53	Commonest alteration overall (more in advanced disease)
RB1, PTEN, TP53	Loss of ≥2 means aggressive variant, overlapping neuroendocrine
BRCA1/2	Germline plus somatic hit: give **PARP inhibitor**
ATM	Little PARP benefit
CDK12	Tandem duplications; poor response to hormones, chemo, PARP
MMR deficiency	Microsatellite instability (MSI): give immunotherapy
AR amplification	Late, under AR-targeted pressure; causes ADT resistance

Kidney

Subtype	Genomic hallmark	Therapy link
Clear cell (75–80%)	**3p loss** plus VHL, PBRM1, BAP1, SETD2	VEGF tyrosine kinase inhibitors (TKIs); belzutifan (HIF-2α)
Papillary (15–20%)	MET (7q); FH loss in subset	MET-active TKIs
Chromophobe	Mitochondrial DNA, mTOR pathway	mTOR-directed rational
Translocation	MiT/TFE fusions; young women	No targeted therapy

Subtype	Genomic hallmark	Therapy link
Sarcomatoid (any)	BAP1, CDKN2A loss	PD-L1-positive, so immunotherapy-responsive

- In clear cell RCC, VHL is mutated in 60–70%. **PBRM1** (~40%) means a better outcome and **BAP1** or **SETD2** (~10% each) a worse one.

Bladder (urothelial carcinoma, UC)

- UC has a high (APOBEC) mutation load, with TP53 most frequent.
- Of six consensus subtypes, **basal/squamous is largest (35%)**, luminal papillary 24%, neuroendocrine-like smallest 3%.
- **FGFR3** is altered in 15–20% of advanced UC and up to 37% of upper tract UC. These T-cell-poor tumours respond to **erdafitinib**, so test all advanced UC. HER2 is altered in 9–12% of MIBC.

Testis

- Testis cancer shows **12p gain** (†isochromosome 12p) and KIT/KRAS in seminoma; miR-371a-3p has >90% sensitivity and specificity.

Hereditary syndromes

SAQ Familial / hereditary urological malignancies

Paper answer · P1 · P2 · merged from 2

Asked as: P1 SAQ "Familial malignancies in urology" (Familial malignancies in Urology, Practice Set 2 Paper I Q3b; Familial malignancy in urology, Practice Set 8 Paper I Q3f) · P2 SAQ "Hereditary urological malignancies."

Principles

- **Knudson two-hit hypothesis:** every cell carries one inherited mutant tumour-suppressor allele, so one more hit causes cancer: tumours are **early, bilateral and multifocal**.
- **4–6% of RCC (renal cell carcinoma)** is familial. All familial RCC syndromes are **autosomal dominant**; only HPRC (hereditary papillary renal carcinoma) involves an activated oncogene.

Hereditary kidney cancer syndromes

Syndrome	Gene (chromosome)	Kidney tumour	Other features
VHL (von Hippel–Lindau)	VHL (3p25)	Clear cell, bilateral, multifocal (~50%)	Retinal angioma, CNS (central nervous system) haemangioblastoma, phaeochromocytoma, pancreatic cysts
HPRC	MET (7q31) oncogene	Multiple bilateral papillary (type 1, type 1-like)	None
HLRCC (hereditary leiomyomatosis and RCC)	FH (fumarate hydratase, 1q42–43)	**Aggressive**, often solitary papillary; collecting duct	Skin and uterine leiomyomas

Syndrome	Gene (chromosome)	Kidney tumour	Other features
BHD (Birt–Hogg–Dubé)	FLCN (folliculin, 17p11.2)	Chromophobe, hybrid oncocytic, oncocytoma	Fibrofolliculomas, lung cysts, pneumothorax
TSC (tuberous sclerosis complex)	TSC1 (9q34), TSC2 (16p13.3)	Multiple AMLs (angiomyolipomas); RCC 2–3% (P2: 1–3%)	Angiofibromas, cortical tubers

- **VHL type 1** (deletions, truncations): no phaeochromocytoma; **type 2** (missense): phaeochromocytoma.
- **Also know:** SDH (succinate dehydrogenase)-deficient RCC (paraganglioma); BAP1 (uveal melanoma, mesothelioma); Cowden (PTEN; breast, thyroid).

Management of familial kidney tumours

- **3 cm rule:** watch tumours <3 cm; nephron-sparing surgery at ≥3 cm (VHL, HPRC, BHD) to save function for repeat operations.
- **Exception: HLRCC**: excise promptly, with wide margins.
- TSC AML: everolimus at 3–5 cm (P1; another source: if >3 cm); embolisation or excision (partial nephrectomy) if ≥5 cm.
- **Belzutifan** (HIF-2α (hypoxia-inducible factor 2α) inhibitor), FDA (Food and Drug Administration)-approved 2021 for VHL tumours.

Other urological cancers

- **Prostate: BRCA2**, BRCA1, ATM, CHEK2, PALB2, HOXB13, mismatch repair genes. Father and brother affected: 9-fold risk. BRCA2 gives aggressive prostate cancer. Earlier screening, cautious active surveillance, **PARP (poly ADP-ribose polymerase) inhibitors**.
- **UTUC (upper tract urothelial carcinoma): Lynch syndrome** (MLH1, MSH2, MSH6, PMS2): about 14-fold risk, about 75-fold with MSH2. Suspect if age <60 or a history of Lynch cancer; screen with MMR (mismatch repair) immunohistochemistry, then germline testing.
- **Testis:** affected brother RR (relative risk) 8–12, father 2–4.
- **Phaeochromocytoma:** about one-third hereditary: MEN-2 (multiple endocrine neoplasia type 2; RET), VHL, NF1 (neurofibromatosis type 1), SDHB (highest malignancy).
- **Also know:** Wilms tumour (WT1 gene: WAGR (Wilms, aniridia, genitourinary anomalies, retardation) and Denys–Drash; Beckwith–Wiedemann).

Approach

- Young age, multifocality or family history: genetic counselling, germline testing, screening of relatives.

Diagram to draw: VHL loss leading to HIF accumulation, VEGF (vascular endothelial growth factor) production and angiogenesis.

> **Key points to remember:**
> - All familial RCC is autosomal dominant; 3 cm rule except HLRCC.
> - VHL 3p25, MET 7q31, FH 1q42, FLCN 17p11.
> - Lynch (MSH2) → UTUC; BRCA2 → aggressive prostate cancer.

Also, from the condensed book (not in the paper answer):

- VHL carries a >70% lifetime risk of clear cell RCC (belzutifan). HLRCC gives papillary type 2 RCC, Lynch gives UC and prostate cancer, and HOXB13 carriers need earlier PSA screening.

KEY POINT

The germline syndrome names the somatic driver (VHL: clear cell; MET: papillary; FH: type 2; BRCA: prostate), and the driver names the drug (HIF-2α, MET TKI, PARP, FGFR inhibitor, immunotherapy for MMR loss).

10 · Clinical NGS and biomarkers

- **Tumour-only** panels cannot separate somatic from germline variants.
- **Variant allele frequency (VAF)** depends on zygosity, copy number and purity; a heterozygous variant at 60% purity gives 30%. VAF of ~50% or ~100% suggests a **germline** variant, and single-digit VAF in cell-free DNA suggests **clonal haematopoiesis**.
- Circulating tumour DNA after treatment means minimal residual disease.
- A predictive marker needs a **biomarker–treatment interaction test** in a randomised trial; crossing lines mean predictive, parallel lines prognostic.
- Absolute benefit scales with baseline risk: a 50% relative reduction gives 15% absolute benefit at 30% risk but 1.5% at 3%.
- **HSD3B1 (1245A>C)** raises DHT synthesis, causing ADT resistance but abiraterone sensitivity.

11 · Landmark trials

Trial	Design	Result	Take-home
†PROfound	Olaparib vs enzalutamide/abiraterone, HR-altered mCRPC	Better radiographic PFS and overall survival, driven by BRCA1/2	Germline and somatic HR testing in metastatic prostate cancer
†THOR	Erdafitinib vs chemotherapy, FGFR-altered metastatic UC after immunotherapy	Longer overall survival	FGFR testing of all advanced UC

55 Management of Abnormalities of the External Genitalia in Males

1 · Normal development and penile size

- Male differentiation (weeks 9–13) needs Leydig testosterone, 5α-reductase conversion to DHT and a functional androgen receptor; any break causes undervirilisation.
- The genital tubercle forms the glans, urethral folds the shaft, and labioscrotal swellings the scrotum.
- **Stretched penile length (SPL)** runs from pubis (fat pad compressed) to glans tip. At term it is **3.5 ± 0.7 cm**; in adults 13.3 ± 1.6 cm.
- The micropenis cut-off is −2.5 SD: **1.9 cm** from term to 5 months, 1.5 cm at 30 weeks, 2.0 cm at 34 weeks, ≈ 3.7–3.9 cm at 5–11 years and 9.3 cm in adults. Compare with the cut-off, not the mean.
- Minipuberty (first 3 months) gives a second phase of penile growth. In Tanner staging, testicular change comes first (stage 2); semen analysis is done at Tanner 5.

2 · Prepuce: phimosis and paraphimosis

- **Physiological phimosis** is normal at birth: ≈ **90%** are **retractable by 3 years**, and < 1% remain phimotic at 17 years. Forceful retraction causes cicatricial phimosis.
- Treat persistent phimosis, balanitis, BXO or UTI.
- **Topical steroid**: betamethasone 0.05% twice daily for 30 days (or thrice daily for 21 days). About 1 in 6 fail.
- A retractile prepuce halves recurrent febrile UTI risk. The commonest acquired "penile lump" is **entrapped smegma**.
- **Paraphimosis** causes oedema within hours and may progress to gangrene.

ALGORITHM

1. Compress the glans with distal traction; this succeeds in most.
2. If it fails, reduce oedema: iced glove 5 min, granulated sugar 1–2 h, or punctures.
3. If still irreducible, do a dorsal slit, then circumcision.

3 · Circumcision

- The AAP (2012) found the benefits insufficient to recommend routine neonatal circumcision, but enough to justify access and payment.
- Before using a device (Gomco, Mogen, Plastibell), inspect the meatus to exclude hypospadias and megameatus. Older children need freehand surgery under GA.
- Analgesia is mandatory. **Dorsal penile nerve block is superior to EMLA**, whose prilocaine can cause methaemoglobinaemia.

Defer / avoid neonatal circumcision	Reason
Hypospadias, curvature, torsion, dorsal hood, buried or webbed penis	Prepuce needed for reconstruction
Large hydrocele / inguinal hernia	Skin pushed forward, causing secondary phimosis or a trapped penis

Defer / avoid neonatal circumcision	Reason
Coagulopathy (contraindication); hyperbilirubinaemia (some)	Haemorrhage
NICU setting	More complications

- **Benefits**: penile cancer occurs almost only in uncircumcised men. Febrile infant UTI falls, but **111** circumcisions prevent one UTI.
- There is ≈ **60%** protection against HIV acquisition in men, but no reduction in transmission to partners. HSV-2 and high-risk HPV are also lower, and no sexual dysfunction is proven.

Complications

SAQ Circumcision undone

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 13 Paper 1, Q3d

"Circumcision undone" is read as an incomplete or complicated circumcision needing revision. Foreskin restoration is mentioned briefly at the end.

Introduction

- Circumcision complications: **0.2–5%**, rising **10- to 20-fold after infancy**.
- Most late problems concern **skin quantity** (too little or too much) or the **exposed meatus**.

Inadequate ("undone") circumcision

- **Too little skin removed** or asymmetrical excision: redundant prepuce covering the glans.
- Leads to **secondary (cicatricial) phimosis**, smegma retention, balanitis, adhesions and cosmetic distress.
- A large hydrocele or hernia pushes skin forward and raises this risk.

Other complications

Complication	Management
Bleeding (commonest, about 0.1%), usually frenular	Pressure, cautery, silver nitrate or suture
Glanular adhesions; **skin bridges** (can cause chordee, torsion)	Office lysis or excision; steroids do not help
Excess skin removal, degloving	Heals by secondary intention with ointment; severe: grafts
Trapped penis (cicatrix over glans)	Betamethasone 0.05% and retraction; else relaxing incision, flaps, Z-plasty
Meatal stenosis (almost only after circumcision)	Meatotomy or meatoplasty
Subcoronal fistula; glans amputation; necrosis (cautery on clamp)	Hypospadias-type repair; reattach glans within 8 hours

Evaluation

- Original procedure (age, device); examine skin quantity, meatus, curvature; exclude hypospadias, buried or webbed penis and BXO (balanitis xerotica obliterans).

Circumcision revision

- **Indications:** redundant or asymmetrical skin, phimosis, adhesions, cosmesis.
- **Needs general anaesthesia** in children, unlike neonatal circumcision.
- **Technique (sleeve):** mark inner and outer incisions (inner 5–10 mm from the corona), excise the excess collar, haemostasis, interrupted absorbable sutures.
- Lyse adhesions and skin bridges; correct torsion or chordee; meatotomy if needed; histology if BXO suspected.

Prevention

- Separate prepuce fully and inspect the meatus before using a device (Gomco, Mogen, Plastibell); defer with hypospadias, curvature, buried or webbed penis; trained operator.

Also know: foreskin restoration

- Traction tissue expansion or skin grafting, for men wanting a foreskin; results variable.

Diagram to draw: sleeve revision showing two circumferential incisions.

> **Key points to remember:**
> - Complications 0.2–5%; bleeding commonest.
> - Too little skin: redundancy, phimosis, adhesions; too much: trapped penis, chordee.
> - Meatal stenosis almost only in circumcised boys.
> - Revision needs general anaesthesia.

Also, from the condensed book (not in the paper answer):

- An amputated glans should be re-attached within **8 h**.
- Degloving heals by re-epithelialisation, with **no immediate suturing or graft**.
- A skin bridge can cause chordee or torsion, so excise it.

> **KEY POINT**
>
> Late sequelae are problems of skin quantity (too much removed: trapped penis, chordee, torsion; too little: redundancy, adhesions) or of the newly exposed meatus (stenosis).

Meatal stenosis and BXO

P2 P1 *Management of Urethral meatal stenosis / Lichen sclerosus (BXO) — normal calibre by age, post-circumcision meatitis, upward stream, ventral meatotomy; BXO features, steroid, circumcision, buccal mucosa*

- Meatal stenosis occurs almost exclusively in **circumcised** boys.
- BXO in boys is rare under 5 years. Steroids help only mild preputial disease, so **circumcision is preferred**, with or without meatoplasty.
- About 1 in 5 with meatal BXO need further meatal surgery.

4 · Inconspicuous penis versus micropenis

KEY POINT

Stretched length is the single discriminator. An inconspicuous penis (buried, webbed, trapped) has normal SPL hidden by skin, fat or scar, so treatment is anatomical. A micropenis is small but normally formed, so it needs endocrine work-up.

Buried penis type	Mechanism	Treatment
Congenital	Inelastic dartos; skin not anchored to deep fascia	Surgery
Obesity	Suprapubic fat hides shaft	Weight loss; liposuction only in adolescents
Trapped (post-circumcision)	Cicatrix embeds penis in fat	Dilation or betamethasone with retraction; then repair under GA at ≥ 6 months

- If retracting skin lateral to the shaft exposes the glans easily, the shaft is normal.
- **Repair** excises the dysgenic dartos bands, fixes dorsal subcutaneous tissue to pubic fascia and scrotal tissue to the ventral base, and covers with unfurled prepuce, Z-plasty, flap or graft.
- **Webbed penis** (penoscrotal fusion) is repaired by transverse incision with vertical closure, or Byars' flap. Avoid neonatal circumcision.

Micropenis

- Micropenis is normally formed but ≥ 2.5 SD below the mean (< **1.9 cm** at term). Testes are often small or undescended.
- A defect before week 12 (placental hCG) gives a small penis with hypospadias; gonadotrophin deficiency after 14 weeks gives a normally formed micropenis.

Cause	Examples	Hormones
Hypogonadotrophic hypogonadism (commonest)	Kallmann, Prader–Willi, Laurence–Moon–Biedl, CHARGE, hypopituitarism, midline brain defects	Low LH and FSH
Hypergonadotrophic	Rudimentary testes, gonadal dysgenesis, Robinow	High LH/FSH; no testosterone rise after hCG
Idiopathic	Delayed gonadotrophin timing	Normal axis
Partial androgen insensitivity (rare)	—	Ambiguity commoner

ALGORITHM

1. History, examination, SPL and **karyotype**.
2. Refer to a paediatric endocrinologist; measure testosterone before and after hCG, with or without a GnRH test.
3. Screen the anterior pituitary (glucose, Na/K, cortisol, thyroid); then MRI brain.
4. Test **end-organ response: intramuscular testosterone enanthate × 3 months**. Short courses spare height; prolonged ones advance bone age. Repeat at puberty.

- Most achieve male identity, standing voiding and satisfactory sexual function.

5 · Curvature (chordee without hypospadias) and torsion

SAQ Congenital Penile curvature..

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 10 Paper 1, Q3d

Definition and principle

- Curvature of the erect penis present from birth, with an **orthotopic meatus**.
- Rigidity needs **symmetric compliance** of the tunica albuginea; curvature means one aspect is less compliant.

Embryology

- DHT (dihydrotestosterone) drives fusion of the urethral folds and maturation of spongiosum, Buck's and dartos fascia. The urethra may form normally while the mesenchyme stays deficient → dysgenetic, inelastic ventral tissue.
- Normal fetal ventral curvature resolves by 20 weeks.

Devine and Horton classification (1973)

Type	Abnormal tissue
I	Corpus spongiosum deficient (chordee without hypospadias)
II	Dysgenetic Buck's and dartos fascia; normal spongiosum
III	Dartos only
IV	Normal layers; **asymmetric corporal compliance** (true congenital curvature)
V	Congenitally short urethra (very rare; doubted)

Two clinical groups

Feature	Chordee without hypospadias (I–III)	Congenital curvature (IV)
Direction	Ventral ± torsion	Ventral, lateral (often left), rarely dorsal
Erection size	In proportion to flaccid size	**Noticeably large**
Presentation	Childhood	18–30 years, noticed at puberty
- **Versus acquired** (Peyronie disease, fracture): congenital is painless, no plaque, no shortening.		

Evaluation

- Home photographs of erection; **artificial erection** (Gittes–McLaughlin saline test or prostaglandin) in theatre.
- Grade: mild <30°, moderate 30–45°, severe ≥45°.

Management

- Surgery if curvature hinders intercourse or distresses. 1. Deglove; excise ventral dysgenetic tissue; mobilise spongiosum from glans to perineum. 2. Residual curve: **midline ventral**

septotomy under artificial erection. 3. Still curved: **dorsal plication** contralateral to the curve: Nesbit (excise ellipses, close transversely), Yachia (Heineke–Mikulicz), 16-dot or TAP (tunica albuginea plication). **12 o'clock is nerve-free**, so midline plication is safe. 4. Severe: ventral **corporotomy with graft** (dermis, tunica vaginalis).

- **Rule:** if the meatus reaches the glans tip, do not divide the urethra; the corpora usually limit straightening.
- **Results:** single operation in almost all; recurrence 2–15%, mostly after plication.

Diagram to draw: penile cross-section: nerves at 11 and 1 o'clock, plication at 12.

> **Key points to remember:**
> - Devine–Horton I–III = chordee without hypospadias; IV = corporal asymmetry.
> - Congenital curvature: young man, large erection, orthotopic meatus.
> - Dorsal 12 o'clock plication; never divide a urethra that reaches the glans tip.

Also, from the condensed book (not in the paper answer):

- Repair if sexual dysfunction is expected, with concurrent penile surgery, or if growth will worsen it.

Type	Associations	Repair (stepwise)
Ventral (commonest)	Hypospadias; dorsal hood; a prominent frenulum causes distal chordee	Deglove; excise fibrous tissue superficial to Buck's; Byars' flaps; if still curved, dorsal plication, Nesbit or corporal rotation; if the urethra is short, reconstruct it; if frenular, frenulotomy
Dorsal	Isolated, epispadias	Ventral plication; protect dorsal neurovascular bundles
Lateral	Corporal asymmetry; straight when flaccid, noticed on erection	Plicate corpus **opposite** maximal curvature

- Work superficial to deep. An **artificial erection** locates the apex and confirms correction.
- **Penile torsion** is usually **counter-clockwise (to the left)**, < 60° and skin-level. If < 90°, deglove and rotate; if ≥ 90°, also divide basal bands and place non-absorbable anchoring sutures at the corporal base.

6 · Abnormal number, masses and lymphoedema

Feature	Aphallia	Diphallia
Incidence / defect	1 in 10–30 million; genital tubercle fails; 46,XY, normal scrotum and testes	1 in 5 million; from accessory to complete duplication
Associations	Cryptorchidism, VUR, renal agenesis, horseshoe kidney, imperforate anus, cardiopulmonary; ≈ 1/3 stillbirth/neonatal death	Hypospadias, bifid scrotum, bladder duplication, renal, pubic diastasis, anal, cardiac
Work-up / decision	Karyotype, anomaly screen, MRI; then neophallus or reassignment (reassigned may develop male identity)	Renal US, VCUG, penile US/MRI; individualised

- In aphallia the urethra most often opens **postsphincteric** (anal verge), with the best survival.

- Penile cysts are commonest and are excised; **juvenile xanthogranuloma is observed for 1 year**.
- Lymphoedema may be primary (Milroy, Meige, praecox) or acquired (anogenital Crohn disease); surgery is excisional, not drainage. Haemangiomas involute, while vascular malformations persist and are resected.

7 · Urethral anomalies

P2 *Megameatus intact prepuce / Anterior urethral valve — MIP: wide meatus under complete prepuce, found at circumcision, pyramid repair; AUV: bulbar, diverticulum lip, VCUG, endoscopic ablation*

- If MIP is recognised during circumcision, stop and preserve the prepuce.
- **Urethral duplication** is usually sagittal (≈ 94%), and the **ventral channel is the functional one**. The **Effmann** classification is type I incomplete (IA distal, IB proximal, blind), type II complete (IIA two meatuses, including IIA2 "Y-type" perineal; IIB one meatus), and type III with bladder duplication.
- Duplication presents with double stream, incontinence or UTI and is evaluated by VCUG, RUG and cystoscopy. **Never use the hypoplastic dorsal channel as the main urethra**.
- **Congenital urethral fistula** is subcoronal with a normal meatus and is closed in layers.
- **Cowper syringocele** (Maizels types: simple, perforated, imperforate, ruptured) needs endoscopic unroofing if symptomatic.
- **Megalourethra** is **scaphoid** (commoner; spongiosum deficient) or **fusiform** (rarer; cavernosa also deficient, often lethal). Both go with prune-belly and are repaired by reduction urethroplasty.

8 · Priapism

*See the paper answer **Priapism: diagnosis and management** (Section 9, Ch 73).*

Type (children)	Cause	Treatment
Ischaemic	**Sickle cell** (commonest), leukaemia	Hydration, O_2, alkalinisation, analgesia, with or without transfusion; then aspiration with or without phenylephrine; then shunt
Non-ischaemic	Straddle injury causing an arteriocavernous fistula	Observe; if persistent, superselective embolisation
Stuttering	Sickle cell	Bedtime pseudoephedrine; oral β-agonist; long-term PDE5 inhibitor
Neonatal	Idiopathic, birth trauma, polycythaemia	None — resolves in 2–6 days

- Corporal blood gas decides urgency: ischaemic blood is dark, hypoxic and acidotic in a rigid, painful penis, while arterial blood is bright red in a non-rigid, painless one. Counsel about fibrosis and ED.

9 · Scrotal anomalies

- **Penoscrotal transposition** is failed inferomedial migration of the labioscrotal swellings. It goes with proximal hypospadias, and in the complete form **up to 75%** have upper-tract

anomalies, so do renal US and VCUG.

- Repair it with the hypospadias by **rotation-flap scrotoplasty**.

Anomaly	Feature	Association	Management
Bifid scrotum	Unfused folds	Proximal hypospadias	Scrotoplasty with hypospadias repair
Ectopic scrotum	Suprainguinal (commonest), infrainguinal, perineal	Ipsilateral renal agenesis/dysplasia, ectopic ureter; cryptorchidism; perineal lipoma usually isolated	Upper-tract US; scrotoplasty and orchidopexy at 6–12 months
Hypoplasia / agenesis	Under-developed / absent	Undescended testis (UDT); agenesis — all XY, bilateral UDT	—
Scrotoschisis	Rupture, extruded testis	Meconium peritonitis	Explore, orchidopexy, close

10 · Inguinal hernia and hydrocele

*All reflect incomplete closure of the **processus vaginalis**.*

Entity	Processus vaginalis	Plan
Indirect hernia	Patent; contents enter	Repair soon after diagnosis
Communicating hydrocele	Patent, fluid only	Repair if > 1–2 y, enlarging or fluctuating
Hydrocele of cord	Isolated patent segment	Rarely resolves; repair after 1 year
Scrotal (non-communicating)	Closed	Observe — most resolve by 1 y
Abdominoscrotal	Closed; through ring (≈ 1%; bilateral 30%)	Excise/drain; ring tightening unnecessary

- Hernia affects 1–5% of children and more with prematurity (≈ 30% if < 1 kg); right to left is 3:1. Bilateral risk is higher with prematurity, age under 1 year, cryptorchidism, ascites and VP shunts.
- An **intermittent bulge** means a patent processus, so operate. Incarceration needs immediate exploration, and reducible hernias elective repair within weeks; in preterm infants repair before discharge.

ALGORITHM

1. Skin-crease incision; open the external oblique, sparing the ilioinguinal nerve (not opened in young infants, Mitchell-Banks).
2. Split the cremaster and find the sac anteromedially; separate it from the vas and vessels.
3. Perform **high ligation** at the internal ring and tighten a wide ring.
4. Finally, **confirm the testis is scrotal**.

- Recurrence is 0.5–1% (3–6% after incarceration), from a missed or torn sac or a missed direct hernia. Other complications are iatrogenic cryptorchidism, atrophy and vas injury.
- Laparoscopic repair now matches open repair and closes the contralateral ring.
- Contralateral patency is ≈ 40% in infants, but a metachronous hernia develops in only ≈ **7%**, so routine contralateral exploration is not advised.

11 · Acute scrotum

Only cord torsion is time-critical.

Feature	Cord torsion	Appendage torsion	Epididymitis
Frequency	20–30%	**40–60% (commonest)**	5–15%
Peak age	Perinatal; 12–16 y	7–12 y	Infancy, puberty
Onset / vomiting	Acute severe; vomiting common	Sudden or insidious; uncommon	Insidious; rare
Cremasteric reflex	Usually absent	Present	Present
Signs	High-riding, horizontal lie	**Blue dot**, upper-pole nodule	Tender epididymis; pyuria 20–40%
Colour Doppler US (CDUS)	Absent/reduced flow; **whirlpool/snail sign**	Avascular upper-pole lesion, hyperaemia	Increased epididymal flow
Management	**Emergency exploration**	Ice, NSAIDs, rest	Supportive; antibiotics only if UTI

Intravaginal torsion

- The **bell-clapper** deformity is found in ≈ 12% at autopsy. Torsion affects ≈ 1 in 4000, is familial and is commoner if cryptorchid. Delay to presentation is the main cause of testis loss.
- A cremasteric reflex is present in up to 10% of torsions, so it does not exclude torsion.
- **TWIST score**: †swelling 2, hard testis 2, absent reflex 1, nausea/vomiting 1, high-riding 1. A score ≥ 5 is high risk, so go to theatre without imaging; ≤ 2 is low risk.
- On CDUS, absent flow is specific but only 63–90% sensitive. **Do not delay surgery for imaging when suspicion is high.**

Hours from onset	0–6	7–12	13–18	19–24	> 24	> 48
Orchidectomy risk	**≈ 5%**	20%	40%	60%	80%	90%

ALGORITHM

1. If torsion is suspected, explore the scrotum.
2. Detorse and apply warm packs. **Meanwhile fix the contralateral testis with non-absorbable suture**.
3. Reassess. If viable, then fix in a dartos pouch or with non-absorbable sutures. If non-viable, then do an orchidectomy; offer a prosthesis later.

- Manual detorsion fails in ≈ 1/3, and orchidopexy is still needed after success.
- Salvaged testes may still atrophy, with semen defects from ischaemia–reperfusion.
- **Intermittent torsion** precedes acute torsion in 30–50%, so offer **elective bilateral orchidopexy**.

Perinatal (extravaginal) torsion

- Cord and tunica twist together before scrotal fixation, giving a hard painless testis. Salvage is rare and it is bilateral in 5–22%; explore urgently if a normal postnatal exam was documented.

Appendage torsion, epididymitis, others

- Explore appendage torsion only if cord torsion is not excluded or pain is prolonged.
- Treat epididymitis with ice, NSAIDs and elevation, adding antibiotics if pyuria. A prepubertal boy with a positive culture needs renal US and VCUG.
- Idiopathic scrotal oedema shows the **fountain sign** and is treated with an antihistamine.

12 · Adolescent varicocele

P1 P3 *Varicocele – Pathophysiology / Varicocele: controversies — left-sided anatomy, nutcracker, grading, right/non-decompressing: image for mass; adolescent: observe, repair for asymmetry*

- Prevalence is ≈ **15%**; it appears after 10 years and peaks at Tanner 3.
- Measure testicular volume by **US (L × W × H × 0.71)**, serially. Peak retrograde flow > 38 cm/s correlates with asymmetry.
- Catch-up growth may be spontaneous, so **observation is the default**.

> **KEY POINT**
>
> **Indications for repair**: left testis > **20%** smaller (or bilateral hypotrophy), pain, abnormal semen (Tanner 5 / ≥ 18 y), or intratesticular varicocele with asymmetry. Total volume < 30 mL and low total motile count predict poor semen.

Technique	Recurrence	Hydrocele	Note
Open retroperitoneal (Palomo, mass ligation)	2–4%	Up to 30%	Lymphatics ligated
Laparoscopic, non-sparing	0–9%	11–32%	—
Laparoscopic artery/lymphatic-sparing	1–7%	0–4%	Lymphatic sparing cuts hydrocele
Microscopic subinguinal	0–10%	0–6%	Best balance; rare atrophy
Non-microscopic inguinal	7–33%	8–14%	—
Sclerotherapy (retrograde/antegrade)	6–35% / 4–12%	Occasional	Mainly for recurrence

- Hydroceles may appear 2 or more years after repair, and up to half need intervention.

13 · Epididymal and vasal anomalies

- Spermatoceles are postpubertal, contain sperm and are mostly observed.
- Bilateral absent vas suggests a **CFTR** mutation; unilateral absence suggests **renal anomalies**.

58 Management of Abnormalities of the Genitalia in Girls

1 · Embryology that explains the anomalies

- The **perineal body** splits the cloacal sphincter, so urogenital diaphragm and external anal sphincter are **both pudendal-innervated**.
- Müllerian ducts arise lateral to wolffian ducts, so a **müllerian anomaly travels with an ipsilateral renal anomaly**. Fused ducts form uterus, cervix and upper two-thirds of vagina; **sinovaginal bulbs** form the distal vagina.
- **Caudal regression** (weeks 4–5 insult) is linked to maternal diabetes; cranial extension gives VATER or MURCS (müllerian aplasia, renal aplasia, cervicothoracic somite dysplasia).

2 · Evaluation and classification

- The **pull-down manoeuvre** (labia drawn caudally and laterally) shows the distal vagina. Discharge or bleeding needs vaginoscopy under anaesthesia.
- Ultrasonography (US) shows anatomy and upper tracts; **magnetic resonance imaging (MRI) is the gold standard** for müllerian anatomy.

Category	Mechanism	Examples
Hypoplasia/agenesis	Fails to form	Mayer–Rokitansky–Küster–Hauser (MRKH)
Vertical fusion	Faulty müllerian–sinus canalisation	Imperforate hymen, transverse septum, distal atresia
Lateral fusion	Failed midline fusion/resorption	Didelphys, Herlyn–Werner–Wunderlich syndrome (HWWS)

> **KEY POINT**
>
> Decision hinge: **is it obstructive?** Obstruction needs prompt drainage; non-obstructive anomalies wait until intercourse or fertility demands surgery.

3 · Clitoris

- **Clitoromegaly** is commonest from **congenital adrenal hyperplasia (CAH)**, so check electrolytes, 17-hydroxyprogesterone and karyotype (Ch 59). A large hood with normal corpora is **pseudo-enlargement**; a **diminutive clitoris** suggests complete androgen insensitivity syndrome (CAIS).

4 · Vestibule: the interlabial mass

- **Urethral prolapse** affects prepubertal African-ancestry girls: a **doughnut with central meatus** needing no imaging. Sitz baths and topical steroid or oestrogen come first, then **circumferential excision**.
- A **Skene cyst** gives a **displaced meatus, eccentric stream** and usually ruptures.
- A **Gartner cyst / covered ectopic ureter** on the anteromedial vaginal wall is incised; upper-pole heminephrectomy follows if wetting or infection persists.
- A **prolapsed ureterocele** gives an infant retention and a pink-to-dusky mass; decompress, then reduce.

- An **ectopic ureter** below the sphincter causes **daytime wetting despite normal voiding**.

> **KEY POINT**
>
> Read every mass against the meatus: central (prolapse), displaced (Skene), separate/vaginal (Gartner, ectopic ureter, hymen, tumour). Any cyst of urethral or vaginal origin needs a **renal US, always**.

- A **labial swelling** is commonest an inguinal hernia; **childhood asymmetric labium majus enlargement (CALME)** is physiological and is observed.

5 · Vagina: obstruction and absence

Block level and uterus presence decide the operation.

Entity	Level / uterus	Clue	Treatment
Imperforate hymen	Hymen; normal	Whitish bulge (neonate); bluish bulge and amenorrhoea (adolescent)	**Transverse** incision; not aspiration
Transverse septum	Upper/middle third; normal	Normal hymen, blind vagina	Excision, Z-plasty, balloon; hollow mould
Distal atresia	Sinus; normal	Shallow dimple; distended vagina per rectum	Pull-through with or without flaps
MRKH	Bulbs/plate; absent	Hymenal fringe and pouch; 46,XX	Dilation first
Cervical agenesis	Cervix; uterus present	Obstruction; MRI	Often hysterectomy and neovagina
HWWS	Lateral fusion; didelphys	Menstruates; lateral bluish bulge	Wide septal incision

- **Imperforate hymen** is the commonest obstructive anomaly; aspiration causes ascending infection.
- **Transverse septum** (~1 in 70,000): delay causes endometriosis, and **MRI distinguishes a high septum from cervical agenesis**. **Z-plasty** makes the scar longitudinal; the main complication is **stenosis**, and a **hollow mould** is used when a uterus is present.
- **Distal atresia**: the haematocolpos acts as a tissue expander; drain, then pull through.

MRKH (müllerian agenesis)

- MRKH (~1 in 5000) is 46,XX with **functioning ovaries**, the second commonest primary amenorrhoea after gonadal dysgenesis.
- Renal anomaly occurs in ≈ **one-third**, skeletal in 10–20%. **Type A** is symmetrical without extra-müllerian anomaly; **Type B** is asymmetrical with renal, Klippel–Feil or MURCS anomalies and needs full renal and skeletal work-up.

6 · Vaginal replacement

- **Dilation is first-line** (Frank; Ingram); surgery is no better. **Vecchietti** uses an olive with laparoscopic traction.
- **McIndoe** places a split-thickness graft over a stent; it stenoses, so dilation is mandatory. Alternatives include **buccal mucosa** and **robotic peritoneal vaginoplasty**.

- **Sigmoid** is the default bowel; excise ~3 cm of intervening bowel to avoid overlapping suture lines and fix it against prolapse. **Ileum** stenoses more and is kept for an irradiated pelvis or absent colon.
- Bowel risks discharge and **adenocarcinoma**; skin risks stenosis and **squamous carcinoma**. Every neovagina needs **annual examination**.

> **KEY POINT**
>
> **Cervical atresia**: a perineum-to-uterus anastomosis loses the endocervical barrier and risks life-threatening ascending infection. Hysterectomy with neovagina is often best; otherwise use endometrial suppression and IVF with transmyometrial transfer.

7 · Lateral fusion anomalies

HWWS girls menstruate, so obstruction is missed.

- **Didelphys** has a septate vagina in up to 75%; excise it only for symptoms. Incise a **septate uterus** only for recurrent pregnancy loss.
- **HWWS** is didelphys with a unilateral imperforate hemivagina: pain **without amenorrhoea**, a lateral bluish bulge and **ipsilateral renal agenesis**. Treat by wide septal incision.

8 · Acquired disorders

- **Vaginal rhabdomyosarcoma** is a grapelike bleeding mass under 2 years with the best prognosis of genital primaries. Chemotherapy is primary; IRS trials cut surgery from **100% to 13%**.
- **Labial adhesions** are the commonest interlabial abnormality (peak 13–23 months); **80% resolve in 1 year**, so treat only symptoms. Use topical oestrogen or betamethasone 0.05% before separation.

SAQ Female circumcision

Paper answer · P1 · P2 · merged from 2

Asked as: P1 SAQ "Female circumcision" · P2 SAQ "Female Circumcision"

Taken as female genital mutilation/cutting (Chapter 58 calls it "female circumcision (infibulation)"); the answer follows WHO (World Health Organization) definitions.

Definition and epidemiology

- **FGM (female genital mutilation):** all procedures that partly or totally remove the external female genitalia, or otherwise injure them, for non-medical reasons (WHO).
- About **200 million** living women and girls are affected; about 3 million African girls a year. Mainly Africa, the Middle East and Asia (Indonesia, Malaysia); practised in some Indian communities ("khatna").
- Usually done in childhood, at 4–10 years, without anaesthesia, as a rite of passage; it predates Islam. Pharaonic type = infibulation.

WHO classification

Type	Description
I (clitoridectomy)	Partial or total removal of the clitoral glans and/or prepuce
II (excision)	Removal of the clitoral glans and labia minora, with or without the labia majora

Type	Description
III (infibulation)	Narrowing of the vaginal opening (introitus) by cutting and joining the labia into a seal, with or without clitoral excision
IV (other)	Pricking, piercing, incising, scraping, cauterisation

- The older Toubia scheme numbers differ: infibulation is Type III in WHO.

Complications

- **Immediate:** pain, haemorrhage, shock, infection (tetanus), **acute urinary retention**, urethral or rectal injury, death.
- **Urological:** slow stream, dysuria, **recurrent UTI (urinary tract infection)**, meatal obstruction under the scar, urethral stenosis, incontinence, vaginal stone from urine stasis, **vesicovaginal fistula** after obstructed labour.
- **Gynaecological and sexual:** retained menses, dysmenorrhoea, inclusion cysts, keloid, dyspareunia, apareunia, infertility.
- **Obstetric:** obstructed labour, perineal tears, postpartum haemorrhage, neonatal death.
- **Psychological:** anxiety, depression, PTSD (post-traumatic stress disorder).

Evaluation

- Sensitive, culturally respectful history; examination to classify the type; drawings or photographs for consent.
- Flow rate and residual urine; cystoscopy if fistula is suspected.

Management

- **Deinfibulation (type III):** anterior **midline division of the scar** under local or regional anaesthesia to expose the urethral meatus and introitus; raw edges oversewn (subcuticular closure of each edge). High success.
- **Timing:** ideally before pregnancy, or antenatally, or in labour. Relieves retention, UTI and dyspareunia.
- **Never reinfibulate** (illegal in the UK).
- Excise cysts, treat stenosis, repair fistula; counselling.
- Clitoral reconstruction: less predictable, limited evidence and follow-up.

Legal and ethical aspects

- WHO and the UN (United Nations) condemn FGM as a human rights violation; WHO opposes its performance by health workers.
- Illegal in many countries (e.g. UK FGM Act 2003, with mandatory reporting in girls under 18). India has no specific law.

Diagram to draw: vulva showing types I–III and the deinfibulation incision line.

> **Key points to remember:**
> - WHO types I–IV: I clitoridectomy, II excision, III infibulation, IV others.
> - Urological: retention, recurrent UTI, stenosis, stones, fistula.
> - Midline deinfibulation for type III; never reinfibulate.

Also, from the condensed book (not in the paper answer):

- **Condylomata** suggest abuse, though perinatal transmission occurs; over 50% resolve.
- An **inguinal hernia** may present CAIS (~1–2.5%); the **round ligament is absent in genetic males**. If CAIS is found, preserve the testes.

59 Differences of Sexual Development

1 · Definition and nomenclature

- In a **difference of sex development (DSD)** chromosomal, gonadal or anatomical sex is atypical. The **2006 Chicago consensus** replaced "intersex" and "hermaphrodite" with karyotype-based terms.

Old term	2006 term
Male pseudohermaphrodite	46,XY DSD
Female pseudohermaphrodite	46,XX DSD
True hermaphrodite	Ovotesticular DSD
XX male / XX sex reversal	46,XX testicular DSD
XY sex reversal	46,XY complete gonadal dysgenesis (not androgen insensitivity syndrome, AIS) **MCQ**

- Offer endocrine and genetic testing for **proximal hypospadias even with descended testes**.

2 · Normal sex development

- **SRY** upregulates **SOX9**, making Sertoli cells. Genes acting elsewhere explain syndromic DSD: **WT1** (Wilms tumour, nephropathy), **SF1** and **DAX1** (adrenal failure), SOX9 (campomelic dysplasia).
- **Sertoli anti-Müllerian hormone (AMH)** from 7–8 weeks makes Müllerian ducts regress by ≈ 10 weeks. **Leydig testosterone** from ≈ 9 weeks builds the Wolffian ducts.
- **5α-reductase 2 makes dihydrotestosterone (DHT)**, the active androgen for prostate, urogenital sinus and external genitalia.

> **KEY POINT**
>
> The testis uses two separate hormones: AMH removes Müllerian ducts, testosterone/DHT builds Wolffian and external structures. A uterus in 46,XY means Sertoli (AMH) failure, that is dysgenesis, not a synthesis or receptor defect.

3 · Classification (2006 / Chicago)

Category	Conditions
Sex chromosome DSD	47,XXY Klinefelter; 45,X Turner; 45,X/46,XY mixed gonadal dysgenesis (MGD); 46,XX/46,XY chimerism
46,XX DSD — gonadal	Gonadal dysgenesis; ovotesticular DSD; testicular DSD (SRY+, SOX9 duplication)
46,XX — androgen excess	Fetal: congenital adrenal hyperplasia (CAH: 21-hydroxylase, 11β-hydroxylase, 3β-HSD, P450 oxidoreductase); fetoplacental: aromatase deficiency; maternal: luteoma, drugs
46,XX — other	Cloacal anomalies; Mayer–Rokitansky–Küster–Hauser (MRKH); vaginal atresia

Category	Conditions
46,XY DSD — gonadal	Complete/partial gonadal dysgenesis (SRY, SOX9, SF1, WT1, DHH, MAP3K1); ovotesticular; testis regression
46,XY — androgen synthesis	LH receptor; Smith–Lemli–Opitz; StAR; 3β-HSD; 17α-hydroxylase; 17β-HSD3; 5α-reductase 2
46,XY — androgen action	Complete and partial AIS (CAIS, PAIS)
46,XY — other	Persistent Müllerian duct syndrome (PMDS); vanishing testis; isolated hypospadias; hypogonadotropic hypogonadism; cryptorchidism

- 3β-HSD and P450 oxidoreductase deficiency virilise 46,XX yet undervirilise 46,XY.

4 · 46,XX DSD

CAH is commonest and can kill a newborn through salt loss.

CAH — 21-hydroxylase deficiency

- CAH is the commonest 46,XX DSD and **commonest cause of neonatal atypical genitalia**. 21-hydroxylase deficiency (21-OHD) is **95%** of CAH; it is autosomal recessive, ≈ 1 in 15,000.
- 46,XX infants have normal ovaries and Müllerian structures, often with a **common urogenital sinus**; some look like a boy with **bilateral impalpable gonads**. Virilisation is graded **Prader I–V**.
- **17-hydroxyprogesterone (17-OHP) is raised**; measure it after **36 h**.
- A **salt-wasting crisis** (low sodium, high potassium, acidosis, shock) usually comes no earlier than **day 5–7**.
- Treat with glucocorticoid, mineralocorticoid and salt (hydrocortisone ≈ 10–15 mg/m²/day, fludrocortisone, NaCl in infancy; double-to-triple the dose in illness or surgery).
- Affected boys look normal, so **newborn screening** matters most.
- Males may get **testicular adrenal rest tumours**, which are benign and regress with more glucocorticoid but obstruct tubules, so bank sperm.
- Forms are **classic salt-wasting**, **simple virilising** and **nonclassic** (20–50% enzyme activity; hirsutism, oligomenorrhoea; may need no treatment).

Enzyme	Share	BP / K^+	46,XY	Signature
21-hydroxylase	95%	Salt wasting; low BP, high K^+	Normal	High 17-OHP
11β-hydroxylase	≈ 5%	Hypertension, low K^+	Normal	—
3β-HSD	<1%	Salt wasting	Undervirilised	High dehydroepiandrosterone (DHEA)
17α-hydroxylase	Rare	Hypertension, low K^+	Female/atypical; delayed puberty	Low androgens

Other 46,XX DSD

SAQ 46,XX male

Paper answer · P1 · P2 · merged from 2

Asked as: P1 SAQ "46-XX Male" · P2 SAQ "46 XX Male"

Assumed meaning: 46,XX testicular DSD (difference of sex development), "XX male" or de la Chapelle syndrome.

Definition and classification

- **Testes and male genitalia in a person with a 46,XX karyotype.** Old term "XX sex reversal"; renamed **46,XX testicular DSD** (Chicago consensus 2006).
- First described by **de la Chapelle (1964)**. About **1 in 20,000** males.
- In the DSD classification it is a 46,XX DSD of gonadal development, together with ovotesticular DSD; **CAH (congenital adrenal hyperplasia)** is the androgen-excess type.

Genetics

Type	Mechanism	Phenotype
SRY-positive (about 80–90%)	**SRY** (sex-determining region Y) translocated from Yp to the X chromosome during paternal meiosis	Usually normal male genitalia
SRY-negative	Testis genes switched on without SRY: **SOX9** duplication or upregulation, SOX3, loss of RSPO1 or WNT4	More hypospadias, undescended testes, ovotestis

- SRY activates SOX9, so Sertoli cells form and secrete **AMH (anti-Müllerian hormone)**: no uterus. Leydig cell testosterone virilises.
- The **AZF (azoospermia factor) region of Yq is absent**, so spermatogenesis always fails.

Clinical features

- Normal male identity; often found at **infertility work-up** for azoospermia, or prenatally (cell-free DNA 46,XX but male ultrasound).
- Small firm testes, **azoospermia**, hypergonadotropic hypogonadism, gynaecomastia in some; progressive testicular regression.
- Height normal or **shorter than average** (no Y growth genes), unlike Klinefelter.

Feature	46,XX male	Klinefelter (47,XXY)
Frequency	1 in 20,000	1 in 500–1000
Height	Normal or short	Often tall
Genital anomalies	More common	Rare
Sperm retrieval	**Never (no AZF)**	Possible by micro-TESE (microdissection testicular sperm extraction)

Evaluation

- Semen analysis (azoospermia); FSH (follicle-stimulating hormone) and LH (luteinising hormone) high, testosterone low or low-normal.
- **Karyotype** 46,XX; **FISH (fluorescence in situ hybridisation) for SRY** confirms; DSD gene panel if SRY-negative; scrotal ultrasound.
- In an infant with atypical genitalia, exclude CAH (17-hydroxyprogesterone) and ovotesticular DSD.

Management

- **Testosterone replacement** when hypogonadal (libido, bone density, muscle).

- **Infertility:** testicular sperm extraction is futile; counsel for **donor sperm or adoption**.
- Repair hypospadias or undescended testis; reduction mammoplasty for persistent gynaecomastia.
- Genetic counselling, psychological support and bone density monitoring.

Diagram to draw: X and Y chromosomes showing Xp–Yp exchange carrying SRY onto X.

> **Key points to remember:**
> - 46,XX testicular DSD: SRY on X in about 80–90%; 1 in 20,000.
> - Azoospermia always (AZF absent): no sperm retrieval.
> - Shorter than Klinefelter; FISH for SRY confirms.

Also, from the condensed book (not in the paper answer):

- Aromatase deficiency virilises mother and infant; maternal androgen exposure ends at birth, so no postnatal therapy is needed.

5 · 46,XY DSD

A uterus separates dysgenesis from synthesis or receptor defects.

Condition	Key features
Complete gonadal dysgenesis (Swyer)	Streaks, female genitalia, uterus; absent puberty; tumour 30–45%, so prophylactic gonadectomy; oestrogen; donor-ovum pregnancy
Partial gonadal dysgenesis	Dysgenetic testes, atypical genitalia, variable Müllerian structures. WT1: Denys–Drash (congenital nephrotic syndrome, Wilms 90%) or Frasier (late focal segmental glomerulosclerosis, gonadoblastoma 60%, no Wilms)
CAIS	X-linked; female genitalia, short blind vagina, no uterus; breasts by aromatisation; ≈ 50% have inguinal hernia; 1–2% of girls with hernia are CAIS
PAIS	Atypical genitalia; identity mostly male; pubertal breast growth
5α-reductase 2	Autosomal recessive; atypical/female at birth, blind pouch; **virilises at puberty**, fertility reported; testosterone:DHT **>20:1**; "guevedoces"
17β-HSD3	High androstenedione:testosterone ratio; virilises at puberty
LH receptor defect	Leydig hypoplasia; no hCG response
PMDS	AMH or AMH-receptor mutation (autosomal recessive); male genitalia with uterus and tubes; found at hernia repair or orchidopexy; 30–50% of transverse testicular ectopia
Vanishing testes	Loss after first trimester: male genitalia, impalpable testes, no Müllerian structures; undetectable testosterone/AMH, high gonadotropins at mini-puberty; testosterone from adolescence

- PAIS raised male may respond to high-dose testosterone; raised female, use a GnRH agonist or gonadectomy, then oestrogen. In CAIS, vaginal dilation usually suffices.
- **PMDS (Clarnette)**: type 1 is bilateral intra-abdominal (60–70%); type 2, testis in sac or scrotum with contralateral hernia (20–30%); type 3, transverse ectopia (10%).
- PMDS needs orchidopexy. The vasa hug the uterus, so Müllerian malignancy risk (3–8%) is weighed against careful removal sparing the vasa.

6 · Sex chromosome DSD and ovotesticular DSD

SAQ Mixed gonadal dysgenesis.

Paper answer · P3

Type: SAQ (10 marks)
Source: Practice Set 5 Paper 3, Q3e (DSD management in general in 2019-Summer Paper 3, Q3f)

Definition

- MGD (mixed gonadal dysgenesis): a **dysgenetic testis on one side and a streak gonad on the other**, with persistent Müllerian structures and variable virilisation.
- A sex chromosome DSD (difference of sex development), usually **45,X/46,XY mosaicism** (about 1 in 5,000 births). **Second commonest cause of neonatal atypical genitalia after CAH (congenital adrenal hyperplasia).**

Pathogenesis

- The Y-bearing line forms a testis; the 45,X line leaves a streak.
- AMH (anti-Müllerian hormone) and testosterone act **locally**, so anatomy is asymmetric: Wolffian duct on the **testis side**; tube and hemiuterus on the **streak side**.
- 45,X/46,XY spectrum: female (Turner + Y), atypical (MGD) or male with normal testes (about 85%).

Clinical features

- **Asymmetric genitalia:** phallus with chordee, perineal hypospadias, urogenital sinus; one testis in a hemiscrotum, the other side impalpable. Hernia may contain a uterus or tube.
- **Turner features:** short stature, coarctation, horseshoe kidney.

Evaluation (MDT, multidisciplinary team)

- **Karyotype** (many cells, for mosaicism); **17-OHP (17-hydroxyprogesterone)** to exclude CAH; LH (luteinising hormone), FSH (follicle-stimulating hormone), testosterone, AMH.
- **Pelvic ultrasound** (uterus, asymmetric gonads); genitography, endoscopy; **laparoscopy with gonadal biopsy**. No gonadectomy during diagnosis.
- Echocardiography, renal ultrasound (Turner screening).

	MGD	Ovotesticular DSD
Karyotype	45,X/46,XY	Mostly 46,XX
Gonads	Dysgenetic testis + streak	Ovary + testis, or ovotestis
Tumour risk	15–35%	About 3%

Management

- **Sex of rearing:** individualised; more androgen exposure predicts male identity; dysphoria is commoner when raised female. Avoid irreversible infant surgery.
- **Gonads:** germ cell neoplasia 15–35% (gonadoblastoma, then dysgerminoma or seminoma) in streak and testis alike. **Remove streaks** (biopsy first if tissue uncertain).
- Raised male: keep a scrotal testis under surveillance; bring down an intra-abdominal one if possible.
- Raised female: remove or relocate the testis.

- **Surgery:** male: staged hypospadias repair, orchidopexy; female: feminising surgery deferred.
- **Hormones:** pubertal sex steroids; growth hormone. Infertility usual.

Diagram to draw: asymmetric anatomy: testis with vas; streak with tube and hemiuterus.

> **Key points to remember:**
> - 45,X/46,XY; testis + streak + Müllerian remnants; asymmetric.
> - Second commonest cause of atypical genitalia after CAH.
> - Tumour risk 15–35%: remove streaks.
> - Screen for Turner features.

Also, from the condensed book (not in the paper answer):

- **Turner with Y** material: tumour 12–20%; the 2017 guideline advises gonadectomy at diagnosis, on low-level evidence.
- **Turner**: coarctation, bicuspid valve, horseshoe kidney (renal anomaly ⅓–½) and ovarian insufficiency ≈ 95%, so give lifelong cardiac and renal screening, growth hormone and oestrogen.
- **Ovotesticular DSD** has ovarian and testicular tissue; most are 46,XX, with a sinus and uterus. A **vas always accompanies a testis**, a tube an ovary.
- The ovary is usually normal and the testis dysgenetic; tumour risk is ≈ 3%. Biopsy both poles of an ovotestis and defer irreversible steps.

7 · Evaluation of the newborn with atypical genitalia

LAQ DSD: Evaluation and Management of the Newborn with Ambiguous Genitalia

Paper answer · P3

Asked as:

- Management of DSD — *Summer 2019 Paper III Q3f (SAQ)*
- Evaluation and treatment of newborn with ambiguous genitalia — *Practice Set 4 Paper III Q2 (LAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)

(DSD management also answered as an SAQ in 2019-Summer Paper 3, Q3f)

DSD read as: disorders/differences of sex development. Detrusor–sphincter dyssynergia is covered in 2016-Summer Paper 1, Q3a.

Introduction and definition

- Atypical ("ambiguous") genitalia point to a **DSD (difference/disorder of sex development)**: congenital conditions with atypical chromosomal, gonadal or anatomical sex (Chicago consensus 2006).
- It is a **medical emergency only for salt-wasting CAH (congenital adrenal hyperplasia)**; sex designation is not a "social emergency". Evaluate within 24–48 hours (neonatal evaluation, first 24–48 h).

Classification (by karyotype)

Group	Main causes (examples)
46,XX DSD	**CAH** (commonest; 21-hydroxylase 95%, 11β-hydroxylase about 5%, 3β-HSD (hydroxysteroid dehydrogenase) <1%); maternal androgens or aromatase deficiency; ovotesticular; testicular (SRY (sex-determining region Y) positive, SRY+)
46,XY DSD	Partial/complete gonadal dysgenesis; androgen synthesis defects (17β-HSD3, 5α-reductase); androgen insensitivity (CAIS/PAIS, complete/partial; PAIS = partial androgen insensitivity); LH (luteinising hormone) receptor defect
Sex chromosome DSD	45,X/46,XY MGD (mixed gonadal dysgenesis); 45,X Turner; 47,XXY Klinefelter; chimerism

History

- Consanguinity, **unexplained infant deaths** (CAH), infertility or amenorrhoea in relatives; maternal virilisation, progestogens or androgens; prenatal cfDNA (cell-free DNA) karyotype versus ultrasound discordance.

Examination (gender-neutral terms)

- Stretched phallic length, glans width, meatus, number of perineal openings, labioscrotal fusion and rugosity, pigmentation, anogenital distance. **Prader grade (stage) I–V** for virilisation.
- Dehydration, hypotension, hyperpigmentation (CAH); dysmorphism.
- **Gonads are the key sign.** Ovaries never descend, so **a palpable gonad means testicular tissue**. Bilateral impalpable gonads with atypical genitalia = CAH until proven otherwise (impalpable gonads suggest CAH). Asymmetry suggests MGD or ovotesticular DSD.
- Impalpable testis with hypospadias, or proximal hypospadias, also needs DSD work-up (Kaefer: DSD in about 30%, 50% if the testis is impalpable).

Investigations (tests)

- **Karyotype** at once (preliminary 2–3 days); **FISH (fluorescence in situ hybridisation) for X/Y/SRY** within hours.
- **CAH screen: 17-OHP (17-hydroxyprogesterone) at about 36 hours** (after 36 h; earlier values are falsely high); electrolytes and glucose daily. **Emergency:** exclude **salt-wasting CAH** (low sodium, high potassium, shock); **salt-wasting appears around day 5–7** (usually after day 5–7), so normal early electrolytes do not exclude CAH.
- **At about 1 week:**

Test	Meaning
AMH (anti-Müllerian hormone)	Present = functioning Sertoli cells
Testosterone, LH, FSH (follicle-stimulating hormone)	Testicular function (mini-puberty)
Androstenedione, DHEA-S (dehydroepiandrosterone sulfate)	Adrenal enzyme block
Testosterone:DHT (dihydrotestosterone) >20:1	5α-reductase deficiency
hCG (human chorionic gonadotropin) stimulation	No rise = no Leydig tissue or LH receptor defect

- **Pelvic ultrasound:** uterus (well seen neonatally), gonads, kidneys, cerebriform adrenals (CAH); then genitography, endoscopy, laparoscopy ± biopsy, genetic panels.
- **Genitography / endoscopy:** urogenital sinus, confluence level relative to bladder neck, cervix.
- If unclear: MRI (magnetic resonance imaging), **laparoscopy ± deep longitudinal gonadal biopsy**, gene panels (genetic panels), microarray, exome. **No gonadectomy during diagnosis (while diagnosing).**

Diagnostic algorithm

```
Palpable gonads: 0 -> CAH likely; 1 -> MGD, PAIS, ovotesticular; 2 -> 46,XY DSD (PAIS, 5α-reductase, synthesis defect) 46,XX + uterus + high 17-OHP -> CAH 46,XY, no uterus -> androgen synthesis/action defect 46,XY or 45,X/46,XY + uterus -> gonadal dysgenesis / MGD
```

- A precise diagnosis is reached in nearly all 46,XX but under 50% of 46,XY DSD.

Treatment

1. Emergency and medical management (CAH)

- Salt-wasting crisis: IV (intravenous) saline, dextrose, stress-dose hydrocortisone.
- Maintenance (Endocrine Society 2018): hydrocortisone 10–15 mg/m²/day in divided doses, fludrocortisone, salt supplements in infancy; stress dosing during illness and surgery.
- **Puberty:** oestrogen or testosterone induction if gonads fail.

2. Team and family

- **MDT (multidisciplinary team):** urologist, endocrinologist, geneticist, psychologist, neonatologist, ethicist; gender-neutral language. Congratulate parents; say "your baby", never "it"; open disclosure, peer support.

3. Sex of rearing (gender assignment)

- Weighs diagnosis, prenatal androgen exposure, likely gender identity, fertility, sexual function, family wishes and culture.
- Usually female: 46,XX CAH, CAIS (complete androgen insensitivity), complete gonadal dysgenesis. Trend male: PAIS, 5α-reductase and 17β-HSD3 deficiency.
- Micropenis (stretched length under 2 cm) alone is no reason for female rearing.

4. Surgery: timing and consent

- Irreversible surgery is ideally deferred until the child can participate (consent); parents decide with the team after balanced counselling.
- **Feminising** surgery (clitoroplasty, vaginoplasty, labioplasty) is irreversible, so deferral until the child can consent is favoured unless obstructed; severe CAH is debated.
- Clitoroplasty (ventral, nerve-sparing, glans preserved; Prader III–V); vaginoplasty by confluence (flap for low, partial/total urogenital mobilisation, pull-through for high); labioplasty. The vagina's position relative to the bladder neck, not common-channel length, decides the vaginoplasty. Stress steroids in CAH.
- **Masculinising:** proximal hypospadias repair, orchidopexy, often done in infancy (complication rates high).

5. Gonads (gonadal tumour risk)

Risk	Condition	Plan / management
High	Dysgenesis with Y, intra-abdominal (15–35%); PAIS non-scrotal (50%)	Gonadectomy
Intermediate	Turner with Y (12%)	Gonadectomy
Low	CAIS (2%); ovotesticular (3%)	Keep CAIS gonads until after puberty; ovotesticular: remove discordant tissue

Follow-up and recent advances

- Pubertal induction, psychology, fertility preservation (experimental gonadal tissue cryopreservation), transition to adult care.
- Prenatal cfDNA diagnosis; rapid exome/genome sequencing. Prenatal dexamethasone for CAH is experimental.

Diagram to draw: Prader stages I–V and the urogenital sinus with confluence level; karyotype-based DSD algorithm.

> **Key points to remember:**
> - Palpable gonad = testicular tissue; impalpable gonads = CAH until proven otherwise.
> - 17-OHP after 36 h; salt crisis around day 5–7.
> - Karyotype + uterus + gonads triad guides diagnosis; no gonadectomy while diagnosing.
> - Defer irreversible surgery; CAIS gonads kept until after puberty (till puberty).

Also, from the condensed book (not in the paper answer):

- Late presenters include CAIS, Swyer, MRKH, Turner with Y, Klinefelter and 5α-reductase deficiency; after puberty, high LH and testosterone with female genitalia means CAIS.

> **KEY POINT**
>
> **Emergency**: atypical genitalia with bilateral impalpable gonads is salt-wasting CAH until proven otherwise. Check electrolytes and 17-OHP, and watch for crisis from the end of week 1. Designation and surgery are not emergencies.

- A multidisciplinary team (MDT) reviews within **24–48 h**. A urologist without a team transfers early.
- Count palpable gonads (ovaries do not descend): none suggests CAH; one, MGD, ovotesticular DSD, PAIS or partial dysgenesis; two, 46,XY DSD.
- Hypospadias with undescended testis (Kaefer 1999): DSD ≈ 30%; 50% if testis impalpable; 65% if meatus proximal.

ALGORITHM

1. Send karyotype (≥45 cells, for mosaicism; FISH for X/Y in hours), electrolytes, and 17-OHP at ≈ 36 h.
2. Pelvic ultrasonography for uterus and gonads.
3. At ≈ 1 week: LH, FSH, testosterone, oestrogen, androstenedione, DHT, AMH, with or without hCG stimulation.
4. Genetic panel, microarray or exome.
5. If still unclear, then MRI, genitography, endoscopy, laparoscopy with or without biopsy (**no gonadectomy during diagnosis**).

- Normal AMH means functioning Sertoli (or granulosa) cells; high LH and FSH mean gonadal failure.
- A precise diagnosis is nearly always made in 46,XX DSD but in **<50%** of 46,XY.

8 · Gender assignment and MDT

- The **Money (1975)** model of early designation, early surgery and secrecy failed; shared decisions replaced it.
- Female trend: 46,XX CAH, CAIS, Swyer, LH receptor defect. Male trend: PAIS, 5α-reductase, 17β-HSD3, partial dysgenesis.
- Dysphoria is highest in 5α-reductase deficiency raised female (≈ half) and lowest in 46,XX CAH raised female.
- Micropenis (<2.5 SD; <2 cm at 0–5 months) needs an endocrine work-up; female reassignment is no longer recommended.

9 · Surgery

- **Feminising** surgery (Ch 60): clitoral surgery is irreversible and infant benefit is unproven. Ideally defer it until the individual can take part.
- **Masculinising** surgery (proximal hypospadias, orchidopexy) has high complications but is revisable, so infant repair is often chosen.
- In MRKH, dilation comes first. Uterine remnants are removed or suppressed, **never joined to a neovagina** (no cervix).

10 · Gonadal tumour risk

Risk rises with Y material (gonadoblastoma locus, TSPY), dysgenesis and intra-abdominal position, and falls with a scrotal gonad.

Risk	Condition	Germ cell tumour	Management
High	Dysgenesis with Y, intra-abdominal (Swyer, MGD)	15–35% (Swyer 30–45%)	Gonadectomy at diagnosis
	Frasier	60%	Gonadectomy
	Denys–Drash	40%	Gonadectomy
	PAIS, non-scrotal	50%	Gonadectomy
Intermediate	Turner + Y	12%	Gonadectomy (debated)
	17β-HSD3	28%	Monitor or gonadectomy

Risk	Condition	Germ cell tumour	Management
	PAIS / dysgenesis, scrotal	Unknown	Biopsy at puberty (≥30 tubules, OCT3/4)
Low	CAIS	2%	Keep until after breast development
	Ovotesticular DSD	3%	Remove discordant testicular tissue
?None	5α-reductase; Leydig hypoplasia	Unknown	Unresolved

11 · Ethics

- Respect autonomy, be transparent, and avoid irreversible procedures before consent unless necessary. Psychology is integral.

60 Surgical Management of Differences of Sexual Differentiation and Cloacal and Anorectal Malformations

1 · Principles

*See the paper answer **DSD: Evaluation and Management of the Newborn with Ambiguous Genitalia** (Section 2, Ch 59).*

Also, from the condensed book (not in the paper answer):

- Endocrine Society 2018: families with experienced surgeons decide. Never use a flap for a high confluence. Introitoplasty is needed after puberty in 25–100%.

2 · Urogenital sinus and cloaca — classification

In a **urogenital sinus** the vagina still communicates with the urinary tract (usually the mid-distal urethra), and both exit as one **common channel**.

Entity	Distinguishing feature
Genital atypia states	Commonest; mostly congenital adrenal hyperplasia (CAH)
Pure urogenital sinus	Normal external genitalia; behaves like a cloaca
Persistent cloaca	Rectum also joins, so there is **one perineal opening**
Female exstrophy	Probably a form of urogenital sinus

- CAH causes half of all newborn atypical genitalia and **90%** of 46,XX cases. Salt wasting occurs in **67–75%**; get a karyotype within **48 h**.

> **KEY POINT**
>
> **Governing principle**: **true urethral length** (bladder neck to confluence) decides the operation far more than common-channel length, in sinus and cloaca. Confluence level is a continuum.

- **Ganesan concept** MCQ: in CAH the proximal urethra is normal (bladder neck to confluence **1.5–2 cm** in infants). Virilisation lengthens the **vagina-to-meatus** distance, so a "high" CAH confluence is an illusion.

3 · Clinical evaluation

- The **first threat is metabolic**: monitor electrolytes for salt wasting before reconstruction. **Hypertension** suggests 11β-hydroxylase deficiency.
- A **suprapubic mass** (bladder, **hydrometrocolpos**, or both) is often the only early sign of a pure sinus.
- The **Huffman index** (glans width × length) is normally **<3.5 mm²**; **>10 mm²** is a concern.
- **Hydrocolpos in cloaca** forms because urine refluxes into poorly draining vagina(s). It is commoner with a common channel **>3 cm**, duplication and posterior cloaca.

Associated (cloaca)	Findings	Frequency
Renal	Solitary kidney, dysplasia, pelviureteric junction obstruction, ectopia	Majority

Associated (cloaca)	Findings	Frequency
Uterovaginal	Duplication/septation	**53–67%**
Spinal	Sacral anomalies; tethered cord ≈ one-third (mostly no skin stigmata)	Cord anomaly on magnetic resonance imaging (MRI) ≤ 43%
VACTERL; gut	Tracheo-oesophageal fistula; malrotation, duodenal atresia	Correct malrotation at colostomy

> **KEY POINT**
>
> Two early dangers: **adrenal crisis** (CAH) and **obstructive hydrocolpos** (pure sinus, cloaca). The number of perineal orifices tells you which world you are in.

4 · Imaging and endoscopy

CAH needs little imaging; a cloaca needs a full 3-D map.

Modality	CAH sinus	Pure sinus / cloaca
Ultrasonography (US)	Mandatory; **cerebriform adrenals**	Mandatory for kidneys; poor for Müllerian anatomy
Genitography	Least useful; often inaccurate	**Mandatory**, as a computed tomography (CT) or MRI cloacogram
MRI	Limited	Pelvis, spine, sphincter muscles
Endoscopy	At reconstruction	Early, separate: decompress and define

- On a **genitogram**, a **cervical impression** at the vaginal dome means female internal organs.
- Do a **gonadal biopsy** only if it would change the gender of rearing, using a **deep longitudinal incision** to reach testicular tissue within an ovotestis.
- **Prenatal cloaca** shows fetal ascites, bilobed pelvic cysts and hydronephrosis. **Always do echocardiography.**
- In duplicated vaginas the rectal fistula most often opens **within the vaginal septum**. At **early endoscopy**, **empty the vagina first**.

Common channel (Peña)	**<3 cm**	**>3 cm**
Anomalies; hydrocolpos	Fewer	More (nearly all have urological defects)
Abdominal approach	Rarely	More likely
Bladder	Better spontaneous voiding	Clean intermittent catheterisation (CIC) far more often

5 · DSD reconstruction — timing and consent

No trial compares early with late surgery.

- **For early surgery** gives pliable tissue; advocates seek a moratorium until consent.
- **Clitoroplasty** is only for severe hypertrophy (**Prader III–IV**) and is the most controversial component.

High-confluence vaginoplasty	Arguments
Early single-stage	Redundant clitoral/sinus tissue is usable; maternal oestrogen thickens tissue; a planned one-stage repair does better
After puberty	Infant repairs stenose; avoids dilation; allows consent; delay for the small vagina (**<3 cm**)
Hybrid	Clitorolabioplasty in infancy, vaginoplasty at puberty

> **KEY POINT**
>
> Without comparative data the decision is ethical, not evidential, and rests with parents and team. Endocrine Society CAH guideline (2018): families with experienced surgeons decide; the 2006 consensus limits surgery to Prader III–IV.

Body	Position
Lawson Wilkins and European paediatric endocrine societies' CAH group (LWPES/ESPE, 2002)	Surgery at 2–6 months; not between 12 months and adolescence
International intersex surgical society (ISHID) poll	Most prefer single-stage before 2 years
American Medical Association ethics council	Rejected a moratorium after 2 years' review
Advocacy groups; some US state bills	Moratorium until consent; bills not enacted

6 · Operative principles

Operate in this order: **clitoroplasty, then vaginoplasty, then labioplasty**.

Clitoral surgery	Technique	Verdict
Amputation; Lattimer recession	Excision; recession	Abandoned
Schmid (1961) MCQ	First corporeal excision preserving the neurovascular bundle	Basis of later techniques
Kogan (1983)	Subtunical excision, lateral Buck fascia incisions	—
Baskin (1999)	Defined neuroanatomy	**Ventral** incision spares the dorsal bundle
Pippi Salle (2007)	Corpora disassembled and buried in the labia	Reversible if male identity emerges

- The **Fortunoff (1964)** posterior perineal flap, now **omega-shaped**, underlies almost every repair. **Hendren and Crawford (1969)** showed that the **confluence level determines the type of vaginoplasty**.

Vaginoplasty	Indication	Principle	Pitfall
Cut-back	Labial fusion only	Incise sinus	Misses normal-calibre vagina
Flap	Low confluence	Posterior sinus opened, flap inset; confluence not moved	**Never for high confluence**: vaginal voiding, infection, incontinence

Vaginoplasty	Indication	Principle	Pitfall
Pull-through	Very high	Vagina separated; sinus becomes urethra	If the vagina falls short, add flaps
Replacement	Absent vagina	See Ch 58	—

KEY POINT

Ask: **must the confluence be moved?** A flap brings skin to a low vagina; a pull-through takes a high vagina off the urethra. A flap on a high confluence is the classic error.

7 · Low confluence repair

ALGORITHM

1. Ensure metabolic stability; give **stress-dose steroids in CAH** and antibiotics.
2. Do endoscopy; place a **Fogarty catheter in the vagina** and a Foley in the bladder.
3. Prepare the whole lower body for supine–prone rotation.

ALGORITHM

1. Place a glans traction suture.
2. Incise around the glans, keeping the inner prepuce for the hood (**second only to the glans in sensitivity**).
3. Raise the omega flap and make Y-incisions on the labia majora.
4. Deglove to the bifurcation, keeping the ventral plate intact.

- **Ventral subtunical excision**: incise Buck fascia on the **ventral-most** corpora and remove the erectile tissue, leaving the tunics and dorsal bundle untouched.

ALGORITHM

1. Flap vaginoplasty: raise the omega flap with its fat.
2. Free the posterior sinus off the rectum (**hardest step**).
3. Open the posterior sinus in the midline and **continue to normal-calibre vagina**.
4. Inset the flap.

Labioplasty: split clitoral skin forms the **labia minora**; the labia majora are moved down by **YV-plasty**.

KEY POINT

Two fixed points: never touch the dorsal neurovascular bundle; always open back to normal-calibre vagina.

8 · High confluence: pull-through

- Only ≈ **5%** of CAH cases are high; they are commoner in pure sinus.
- The **critical step** is separating the anterior vaginal wall from urethra and bladder neck, which risks sphincter and nerves.
- **ASTRA** (anterior sagittal transanorectal approach) is done prone, dividing the anterior rectal wall: excellent exposure, no colostomy, continence unaffected.

ALGORITHM

1. Position prone; incise in the midline between sinus and rectum.
2. Divide the sinus up to normal vagina.
3. Free the anterior wall off the urethra **under direct vision**.
4. Close the sinus as the urethra in 2–3 layers; mobilise the vagina.
5. Turn supine for flaps and labioplasty.

Err on the side of the vagina; if it is short, add skin flaps.

9 · Total and partial urogenital mobilisation

- **Total urogenital mobilisation (TUM, Peña 1997)** mobilises the whole sinus en bloc to the perineum without separation.
- **Never discard the sinus**: split **ventrally** it becomes the vestibule; split **dorsally** it becomes the anterior vaginal wall.

	TUM	**Partial (PUM, Rink)**
Extent	Retropubic and suprapubic	Stops at the **pubourethral ligament**
Use	Higher confluence; cloaca	Most sinus repairs
Risk	Sphincter/nerve injury, stress incontinence	Lower; convert to TUM if needed

10 · Outcomes of sinus/DSD surgery

- After infant vaginoplasty, **secondary introitoplasty** is needed after puberty in **25–100%**, because the orifice does not grow.
- **Azziz (1986)** MCQ reported poor coitus rates, but these reflect old **cut-backs**.
- **Confluence level, number of procedures and surgeon expertise** matter more than age; no data favour postpubertal surgery.
- **Elmasri meta-analysis (2018)**: female identity ≈ 89%; vaginal stenosis **27%**; only half had comfortable intercourse.

Two-procedure dilemma	**Infancy**	**After puberty**
Approach 1	Simple clitorolabioplasty	Extensive vaginoplasty
Approach 2	Extensive single-stage clitorolabiovaginoplasty	Simple introitoplasty

Both usually mean two operations, so the key factor is a **surgeon or team with a special interest**.

11 · Cloaca — initial management

- **Hendren's rule**: never do a separate rectal pull-through; repair rectum, vagina and urethra in one stage.

ALGORITHM

1. Decompress the bowel.
2. Decompress the genitourinary tract.
3. Correct nephron-destructive anomalies.
4. Then do the definitive repair.

Colostomy	Advantage	Drawback
Low loop	—	Bowel hard to obtain for vaginoplasty; interferes with pull-through
Right transverse, divided (Hendren)	Left colic supply intact; colon available for vaginal replacement	Absorbs urine, causing acidosis
Descending, divided (Levitt, Peña)	Less resorbing mucosa; enough distal colon; less prolapse	Leave a long sigmoid

- Urine absorbed by colon causes **hyperchloraemic acidosis**. **Preserve the appendix** for a later Malone antegrade continence enema (MACE).
- **Urinary decompression**: **CIC is first line**, then a cloacal cut-back; vaginostomy or vesicostomy only if CIC fails.
- **Kidneys**: structural anomalies occur in **60–83%**; chronic kidney disease (CKD) in **75%** and end-stage renal disease in **17%**. Urodynamics are abnormal in **90%**.

12 · Cloaca — definitive repair

Repair at **6–12 months**, after urinary evaluation and exclusion of hydrocolpos.

Common channel	Urethra	Procedure (Wood algorithm)
<3 cm	**>1.5 cm**	**TUM**
>3 cm	or <1.5 cm	Urogenital separation
Any	Any	Never transabdominal TUM (ischaemia)

KEY POINT

Why 1.5 cm? A urethra >1.5 cm keeps the bladder neck above the urogenital diaphragm; mobilising a shorter one drags the bladder neck below the continence mechanism and causes leakage.

ALGORITHM

1. **PSARVUP (Peña; posterior sagittal anorectovaginourethroplasty)**: place catheters in each channel; position **prone jackknife**.
2. Mark the anal site with a muscle stimulator; incise in the midline.
3. Open the rectal fistula and mobilise the rectum.
4. Do TUM or vaginal separation.
5. Close the urethra in 2–3 layers with a **fat flap**; rebuild the perineal body; centre the rectum in the sphincter.

- It is **better to enter the vagina than the rectum or urethra**; separate it from the urethra by **sharp dissection, not cautery**.
- Aftercare: Foley for 2 weeks, rectal dilation from 2–3 weeks and colostomy closure at ≈ 3 months. Almost one-third empty poorly, so start CIC if in doubt.

13 · Cloaca — results and follow-up

- Outcome depends mostly on the **spine** and **common-channel length**.
- **Urinary**: about half void spontaneously. Peña: CIC in **28%** (channel <3 cm) vs **78%** (>3 cm). Review yearly for life.

- **Renal** function is preserved by aggressive bladder management. Faecal continence needs a bowel programme, then **MACE**.
- Vaginal stenosis or obstructed menstruation occurs in **36–41%**. Pregnancy is possible, with **caesarean preferred**.

ALGORITHM

1. Do US **6–9 months after thelarche**.
2. Repeat every 6–9 months until menses are normal.

65 Adolescent and Transitional Urology

1 · Definitions

- **Adolescent urology** bridges care for complex congenital conditions; the adult urologist lacks **whole-patient knowledge** (prior operations, altered anatomy, families).
- **Transition** (Got Transition) is the planned move from child/family-centred to adult/patient-centred care: **planning, transfer and integration**. Transfer is only one step.

> **KEY POINT**
>
> Renal function and HbA1c are expected to worsen, so **stability and retention in follow-up** are success.

2 · Who needs transition

Category	Examples	Implication
1. Complex congenital	Exstrophy, posterior urethral valves, complex hypospadias, spina bifida (SB)	Lifelong care
2. Treated, now quiescent	Vesicoureteric reflux	Early-adult review; may discharge with open return
3. Cured, with sequelae	Pelvic rhabdomyosarcoma	Late-effects surveillance
4. New adult concern	Fertility, pregnancy	Adult expertise

Failed transition (loss to follow-up, reliance on the emergency department) raises morbidity and mortality because **preventive surveillance is lost**.

3 · The process

"Three Ps"	Content	Success measure
Preparation	Independence from 12–14 years; Transition Readiness Assessment Questionnaire (TRAQ)	All families involved
Plan	Follow-up schedule; self-care (catheterisation, irrigation, medications)	Shared with patient and future team
Provider of the future	May or may not be a new team	Named provider

- The **paediatric provider must initiate**, and never during a crisis; **pregnancy** is the exception and needs expedited transfer with co-management.
- Parents fear losing control, so give the patient **private time** (sexual function) and then ask what may be shared.
- Begin at about **12 years** (AAP) and complete by the late teens or early 20s; age **26** is the latest (US parental insurance ends). If independence is unachievable, plan custodianship.

Got Transition core element	Age (years)
Policy	12–14
Tracking, readiness, planning	14–18

Got Transition core element	Age (years)
Transfer of care	18–21
Transition complete (integration confirmed)	18–23

> **KEY POINT**
>
> Transfer is not the endpoint: transition ends only when integration into adult care is confirmed.

4 · Routes and models of care

Patients arrive by transitioned referral (ideal), referral without transition, a new adolescent problem, or an **emergency after discharge**, which is the hardest and often the commonest.

Model	Strength	Drawback
1a Lifelong paediatric urologist	Continuity; suits limited life expectancy	Patient outlives the provider
1b Paediatric urologist in adult department	Continuity and adult expertise	Cost, privileges, geography
2 Transfer to an adolescent/adult specialist	Specialist coordinates the MDT	Cross-institution communication
3 Direct transfer, no plan	The default	Invisible until an emergency

Lifelong and multidisciplinary need are **independent**: exstrophy may need only a urologist, whereas SB needs the largest MDT. **Nursing** is the backbone.

5 · Development and consent

- Avoid genital surgery at **3–6 years**; the 1975 AAP panel favoured genital surgery at **<6 months**.
- Augmentation waits until the child can catheterise and wants dryness.
- **Emerging adults (18–25 years)** consent for themselves unless they lack capacity.
- Patients with **SB** usually have normal intelligence but weak **working memory and executive function**.
- **Medical PTSD** causes avoidance and procedural panic; CBT may help.

6 · Setting up a service

- The core MDT includes paediatric and adult urologists, specialist nurses, a care coordinator, nephrologist and psychologist.
- Review consent **with the family, then with the patient alone**; standing orders (for example, urine cultures) avoid ED visits.

Barrier	Solution
Patient/parent: attachment, fear	Address the child directly early; readiness questionnaires
Paediatric provider: family trust	Ask "Am I the best person?"
Adult provider: **unfamiliarity with congenital conditions (biggest)**; finance	Partnerships, networks, training

120 Aging and Geriatric Urology

1 · Concepts and biology

- **Chronological age is a poor descriptor**; physiological and functional age (function, frailty, geriatric syndromes) matter. The term "geriatrics" dates from Nascher (1909) MCQ .
- **Cellular senescence** drives ageing, either by telomere shortening (**Hayflick limit**, 1961) MCQ or by stress; its secretory phenotype (SASP) causes chronic inflammation.

2 · Clinical evaluation

- **Homeostenosis** is narrowed physiological reserve, so recovery after a stressor is poorer.

Tier	Content	Significance
Basic activities of daily living (ADLs)	Bathing, dressing, grooming, toileting, feeding, ambulation	**Bathing lost first**, feeding last; loss predicts dependency and postoperative outcome
Instrumental ADLs (IADLs)	Cooking, cleaning, transport, finances, telephone, medicines	Independent living

- A simple screen is **whether the patient can bathe independently**.
- Age is seldom an independent **surgical risk**; predictors are **longer operating time, poor ASA score and urgent surgery**. Optimise rather than "clear" the patient.

ALGORITHM

1. Screen for frailty before elective major surgery (SPAQI).
2. If positive, do a diagnostic frailty assessment.
3. Then do a comprehensive geriatric assessment (CGA).

- The **ACS NSQIP/AGS checklist** has 13 items, only 2 of which are organ systems; the geriatrician-led **CGA** adds physical performance, albumin, creatinine, BNP and FEV1.

3 · Geriatric syndromes

- Geriatric syndromes feed one another, so treating one helps others. **Frailty** is decreased reserve causing vulnerability, distinct from comorbidity and disability.

	Accumulated deficits index	Fried phenotype
Method	Count failing systems	(1) weight loss >4.5 kg or 5%/year; (2) weak grip; (3) slow gait; (4) low activity; (5) exhaustion
Cut-off	Continuous	**≥3 frail; 1–2 pre-frail**
Validity	Does not predict postoperative outcome	Predicts death, dependence, falls, postoperative complications

Polypharmacy — Beers criteria

Harm often comes via the **prescribing cascade**, in which a side effect is treated with another drug.

Drug	Concern and action
Antimuscarinics (in dementia/delirium), tricyclics, first-generation antihistamines	Anticholinergic burden: delirium, falls, retention. Avoid combinations; no tricyclics in BPH

Drug	Concern and action
NSAIDs	GI bleeding; renal and heart failure at CrCl <30 mL/min
Systemic oestrogen; testosterone	Cancer, cardiovascular harm. **Vaginal oestrogen safe**; testosterone only for confirmed symptomatic hypogonadism
Ciprofloxacin, macrolides (**not azithromycin**), co-trimoxazole	Warfarin potentiation: monitor INR; dose-reduce at CrCl <30
Benzodiazepines	Cognition, gait, delirium. Avoid with opioids or ≥2 other CNS drugs
Nitrofurantoin	Listed at CrCl <30 or chronic use; usable in moderate impairment
Desmopressin; tramadol	Hyponatraemia: treat nocturia's cause; check sodium

KEY POINT

Four recurring mechanisms: anticholinergic burden, renal threshold (CrCl <30), warfarin potentiation, sodium disturbance.

Genitourinary syndrome of menopause (GSM)

- GSM causes dryness, dyspareunia and UTI **plus urgency and urgency incontinence**. Sex-steroid loss drives OAB in both sexes, including men on ADT or 5-ARIs.
- Treat with **vaginal oestrogen**. Prior breast cancer is not an absolute bar.
- A **urethral caruncle** causes painless spotting, so exclude gynaecological cancer.

4 · Incontinence management

- Incontinence is common but not normal; behavioural therapy comes first (see Voiding Dysfunction book, Ch 110). †Transient causes (DIAPPERS): delirium, infection, atrophic vaginitis, pharmaceuticals, psychological, excess output, restricted mobility, stool impaction.

*See the paper answer **Overactive bladder: pathophysiology, pharmacotherapy and non-pharmacological management** (Section 7, Ch 103).*

Also, from the condensed book (not in the paper answer):

- Beers lists **oxybutynin and fesoterodine** as "avoid" in dementia, although the paper suggests fesoterodine for the elderly; trospium has the least central penetration.
- Fewer than half persist with **antimuscarinics** at 6 months; check the post-void residual after starting.

Nursing homes

- Catheter colonisation is **98%**, so treating asymptomatic bacteriuria (ASB) causes harm.

Loeb — catheter	Loeb — no catheter	NHSN
≥1 of: fever, new costovertebral tenderness, rigors, **acute mental status change**	Acute dysuria; or fever plus new urgency, frequency, suprapubic pain, haematuria or UI	Culture ≥10^5 CFU, no other source, and clinical signs

> **KEY POINT**
>
> A positive culture alone never qualifies; catheterised residents cannot report local symptoms, so systemic signs are used.

5 · UTI and asymptomatic bacteriuria

- **A causal link between confusion and UTI is not supported** (see Infections book, Ch 26).
- ASB affects up to 22% of older women and about 10% of older men and needs no antibiotics. USPSTF gives grade D against screening; IDSA advises treatment **only in pregnancy and before urological instrumentation**.

6 · Cancer, end of life

- **Prostate**: USPSTF grade D, so no PSA screening over 70. With life expectancy **<10 years**, plan non-curative care.
- **Bladder**: restaging TURBT for T1 is often omitted in the elderly yet improves survival. BCG is less effective (immunosenescence). Cystectomy at ≥75 gives comparable results with careful care.
- **Palliative care is not withdrawal**; it means aggressive symptom control after a goals-of-care discussion.
- The Age-Friendly "4 Ms" are Mobility, Medication, Mentation and what Matters; urologists must report **elder mistreatment**.

> **KEY POINT**
>
> Undertreatment by age (skipped restaging, withheld cystectomy) is as real a risk as overtreatment. Fitness and life expectancy decide.

7 · Landmark trials

Trial	Design	Result	Take-home
ASPREE	RCT: aspirin vs placebo, healthy elderly	No disability-free survival gain; higher mortality	No primary-prevention aspirin
Testosterone Trials	RCTs, hypogonadal older men	No cognitive benefit; modest sexual-function gain	Not for cognition

Effective prevention includes vitamin D with calcium, antihypertensives, statins and exercise.

126 Management and Considerations in Transgender and Nonbinary Individuals

1 · Definitions and terminology

- **Transgender and gender-diverse (TGD)** people have a gender identity that differs from their sex assigned at birth. Assigned sex, gender identity, gender expression and sexual orientation are **independent axes**.

Term	Meaning
AFAB / AMAB	Assigned female/male at birth
Transgender man / woman	Male identity, AFAB / female identity, AMAB
Transmasculine / transfeminine	Direction of gender, not a binary endpoint
Nonbinary	Neither male nor female
Gender dysphoria	Distress from identity–assigned sex discrepancy
Gender incongruence	Diagnostic term: marked, persistent incompatibility

Not all TGD people have dysphoria; some need only social transition, others gender-affirming hormone therapy (GAHT) and/or surgery.

2 · Diagnosis and assessment

	DSM-5 (2013)	ICD-11 (2019)
Term	Gender dysphoria	**Gender incongruence** (replaced "gender identity disorder")
Criteria	Marked incongruence for ≥**6 months** , significant distress or impairment	Marked, persistent incongruence; desire for treatment of experienced gender
Chapter	Mental disorders	**Moved to Conditions Related to Sexual Health** MCQ

ICD-11 keeps a code for access but drops the mental-disorder label; **severe distress is not a prerequisite**.

WPATH Standards of Care 8 (SOC-8, 2022)

- Treat only marked, sustained incongruence, after excluding other causes and optimising coexisting conditions.
- Assess **capacity to consent**, and give **fertility counselling before any treatment**, not just surgery.
- **A single opinion** from a competent professional suffices for medical and surgical treatment.
- Before gonadectomy, **consider ≥6 months of hormones** unless not indicated.

> **KEY POINT**
>
> SOC-8 replaces gatekeeping with informed, capacity-based consent; capacity and fertility counselling stay firm because consequences are irreversible.

3 · Gender-affirming hormone therapy (GAHT)

GAHT aims for **physiological** levels of the affirmed gender in patients with persistent dysphoria, capacity, majority age and controlled comorbidities.

Direction	Agent	Dose
Transfeminine —	Oral oestradiol	2–6 mg/day
oestrogen	Transdermal oestradiol	0.025–0.2 mg/day
	Oestradiol valerate/cypionate IM	5–30 mg 2-weekly or 2–10 mg weekly
Transfeminine —	Spironolactone	100–300 mg/day
antiandrogen	Cyproterone acetate	25–50 mg/day
	GnRH agonist	3.75 mg monthly / 11.25 mg 3-monthly
Transmasculine —	Enanthate/cypionate	100–200 mg 2-weekly
testosterone	Undecanoate	1000 mg 12-weekly (rare in the US: oil embolism)
	Gel 1.6% / patch	50–100 mg/day / 2.5–7.5 mg/day

> **KEY POINT**
>
> **Why the asymmetry?** Oestrogen alone does not suppress testosterone to the female range, so transfeminine therapy needs an antiandrogen too; testosterone suppresses the ovarian axis on its own.

Transfeminine

- **Ethinyl oestradiol is not recommended** (about 20-fold thrombosis). Use 17β-oestradiol, target **100–200 pg/mL**, and **use transdermal if VTE risk** (age >45 or previous VTE).
- **Spironolactone** is commonest in the US and **cyproterone** in Europe; GnRH agonists and bicalutamide are alternatives.
- **Irreversible effects are breast development and reduced fertility.** Risks include VTE, CVD, hypertriglyceridaemia, liver dysfunction and osteoporosis.
- Monitor levels 3-monthly in year 1, then 1–2 times a year, aiming for testosterone **<50 ng/dL**; check potassium and creatinine on spironolactone.
- **After gonadectomy, reduce oestrogen by 50–75%** (titrate to LH, FSH and bone density) and **stop the antiandrogen**.

Transmasculine

- Contraindications are pregnancy, unstable coronary disease and untreated polycythaemia; oestrogen-sensitive cancer needs oncology input. **Contraception is essential** because testosterone is teratogenic.
- Target testosterone **300–1000 ng/dL**; persistent menses respond to a progestin or endometrial ablation.
- **Irreversible effects are clitoral growth, facial hair, voice change and male-pattern baldness.**
- The main risk is **erythrocytosis**, so monitor testosterone and **haematocrit** 3-monthly in year 1, then 1–2 times a year.
- **After gonadectomy, the testosterone dose is not reduced**. **Nonbinary** regimens are individualised (for example, **nandrolone** for muscle with little hair).

4 · Urological considerations

Lower urinary tract symptoms have five contributors: **natal anatomy, hormones, surgery, pelvic floor dysfunction and psychosocial factors**.

- **Testosterone-induced vaginal atrophy** (dryness, bleeding, urgency) is treated with topical vaginal oestrogen.
- **After vaginoplasty**, about a third report altered voiding (**nocturia is commonest**); meatal stenosis, stricture and fistula also occur (see Ch 127).
- **After phalloplasty**, post-void dribble comes from pooling at the native urethra–pars fixa junction; stricture and fistula are far commoner (see Ch 128).
- Treat **pelvic floor** dysfunction before surgery; this eases neovaginal dilation.
- **Tucking** causes stasis, prostatitis-like symptoms, epididymo-orchitis and UTI. Exclude atrophy, tucking and pelvic floor causes before blaming the operation.

5 · Fertility preservation

Counsel **before hormones and/or surgery**, as WPATH, ACOG, ASRM and the Endocrine Society advise.

	Transfeminine	Transmasculine
Standard	**Sperm cryopreservation**; hold GAHT for ≥**3 months** before collection	**Oocyte or embryo cryopreservation** after ovarian stimulation
Prepubertal, or cannot pause hormones	Testicular tissue cryopreservation (experimental)	Ovarian tissue; no hormone pause needed
Obstacles	Ejaculation distress: vibratory stimulation, electroejaculation or surgical retrieval	Bleeding returns; transvaginal scans cause distress

Off testosterone, menses return in 3–6 months, and most conceive within 6 months.

6 · Cancer screening — screen the organ, not the gender

- **Prostate (transfeminine)**: rare but may be aggressive. Offer shared decision-making on PSA at 55–69. **On GAHT, the upper limit is 1 ng/mL, or a rise of ≥0.3 ng/mL from nadir**, †because androgen suppression lowers PSA. After vaginoplasty, feel the prostate through the **anterior neovaginal wall**.
- **Testis**: self-examination only.
- **Cervix (transmasculine)**: cytology 3-yearly at 21–29 years; at 30–65, cytology 3-yearly or hrHPV testing 5-yearly; stop after 65. Use a small speculum and **vaginal oestradiol for 1–2 weeks before**.

7 · Affirming practice

- Keep a **detailed organ inventory** and use inclusive intake forms (name, pronouns).
- **Trauma-informed care** (SAMHSA) uses the "Four Rs": **Realise, Recognise, Respond, Resist retraumatisation**.

127 Male-to-Female Gender-Affirming Surgery: Vaginoplasty

1 · Aim and preparation

Many failures start with a patient who cannot dilate.

- **Mental health** input judges **surgical readiness**. Dysphoria-driven depression should **improve** after surgery, so it is no reason to defer; the key risk is **failure to dilate**.

Older criteria	WPATH Standards of Care 8 (SOC8)
Two assessments	**One** (inter-assessor agreement is high)
Mental health professional only	Any licensed clinician ≥ master's degree competent in transgender health
Hormones before gonadectomy	**6 months** suffices
Comorbidity "well managed"	Risks and benefits discussed
12 months' "real-life experience"	**Removed**: dangerous for those facing violence

Capacity is procedure-specific; if doubtful, a psychologist plans support.

> **KEY POINT**
>
> SOC8 replaces thresholds to clear with information to receive; capacity and fertility counselling stay mandatory.

- Stay within **1 hour's drive** of hospital for the first **3 weeks**.
- **BMI** mainly determines wound healing; no cut-off is proven, but many require **≤35 kg/m²**, and a higher BMI may suit vulvoplasty.
- **Permanent preoperative hair removal** of the scrotal–perineal graft is needed, because canal hair causes infection and hairballs.
- **Continue oestrogen perioperatively**; a 2022 meta-analysis found no added VTE risk.

Orchiectomy

- It may be the sole operation or a first stage, and afterwards spironolactone is stopped. Offer **fertility counselling and sperm banking to all**.
- Use a **vertical midline scrotal incision**; an inguinal incision may compromise the penile flap.

> **KEY POINT**
>
> Plan every orchiectomy as though vaginoplasty will follow: use a midline scrotal incision and leave cord fat for labial bulk.

2 · Penile inversion vaginoplasty (PIV)

SAQ Penile inversion vaginoplasty

Paper answer · P2 · P4 · merged from 2

Asked as: P2 SAQ "Penile inversion vaginoplasty technique." · P4 SAQ "Penile Inversion Vaginoplasty Technique"

Introduction

- The commonest gender-affirming genital operation for transgender women: a functional, aesthetic, sensate vulva and vagina.
- **Principle:** homologous penile and scrotal tissues transferred on their own blood supply.

Preparation

- **WPATH (World Professional Association for Transgender Health) SOC8 (Standards of Care, version 8):** one assessment, **6 months of hormones** before gonadectomy.
- **Permanent hair removal** of the scrotal graft area (prevents intravaginal hair).
- BMI (body mass index) usually ≤35 kg/m²; stop smoking; sperm banking (fertility preservation); continue oestrogen.
- **Choose depth:** full-depth vs **vulvoplasty (zero-depth)**: a vulva without a canal, chosen for high surgical risk, prior prostatectomy, or no wish for penetrative intercourse.

Tissue transfer (homologous tissues)

Male structure	Becomes
Glans on dorsal neurovascular bundle	**Clitoris**
Preputial mucosal collar	Clitoral hood, inner labia minora
Dorsal bulbar urethral plate (spatulated urethra)	Vestibule
Penile shaft skin (inverted)	**Neovaginal lining**
Scrotal skin (full-thickness graft)	Extra (proximal) canal lining
Lateral scrotal skin + cord fat	Labia majora

Operative steps

`Lithotomy; tranexamic acid 1 g; catheter → harvest, defat and tubularise scrotal skin graft → bilateral orchidectomy (leave cord fat for labial bulk) → deglove penis between dartos and Buck's fascia (keep dartos with skin = blood supply) → separate dorsal neurovascular bundle + glans; excise ALL corporal erectile tissue → clitoroplasty; fix at level of adductor longus tendons → shorten and spatulate urethra; excise bulbospongiosus → canal dissection between prostate and rectum: central tendon, behind prostate, preserve Denonvilliers fascia, repeated rectal checks → invert penile skin tube + graft into canal; labiaplasty → pack canal; negative-pressure dressing`

- **Why excise all erectile tissue:** residual tissue engorges and obstructs (narrows) the canal.
- Depth depends on canal dissection, not penile length.
- **Robotic peritoneal flaps** add depth with less rectal injury; **intestinal vaginoplasty** if skin is inadequate or for revision.

Postoperative care

- Pack and catheter out on about day 5.
- **Lifelong dilation**: several times daily at first, e.g. 4 times daily for 6 weeks, tapering to daily after 1 year; pelvic floor physiotherapy; continue oestrogen.

Complications and outcomes

- **Wound dehiscence** (commonest, 12–33%); granulation tissue.

- **Rectovaginal fistula** (0.8% primary, 6.3% revision): the worst complication. Urethrovaginal fistula (0.8–3.9%).
- Urethral stricture, meatal stenosis or malposition (spraying).
- **Vaginal stenosis** (poor dilation); prolapse 4–10%.
- Most achieve orgasm (median about 6 months).

Diagram to draw: sagittal view of the lined neovaginal canal between prostate and rectum.

> **Key points to remember:**
> - Glans → clitoris; penile skin → canal; scrotal skin → graft.
> - Excise all erectile tissue; keep Denonvilliers fascia intact.
> - Lifelong dilation prevents stenosis; rectovaginal fistula is the worst complication.

Also, from the condensed book (not in the paper answer):

- Pad the calves in lithotomy (**common peroneal nerve**) and give **tranexamic acid 1 g IV at the start**, as bleeding is front-loaded.

ALGORITHM

1. Foley and drain; mark adductor longus tendons (clitoral level) and dorsal midline (prevents torsion).
2. Harvest scrotal graft and small posterior rhomboid flap (relieves introital tension); defat, tubularise.
3. Orchiectomy, leaving cord fat; penile disassembly; corporal excision and clitoroplasty.
4. Urethroplasty, then canal dissection; line canal, labiaplasty, pack.

- **Disassembly**: the neurovascular bundle is best **left on the dorsal tunica**, and **all erectile tissue is excised**.
- **Urethroplasty**: excise bulbospongiosus **only after canal dissection**, as it covers a rectal injury, and spatulate downward for a seated stream.
- **Canal**: use repeated digital rectal examination, **preserve Denonvilliers fascia** and close it if breached. **Packing** ensures graft take.
- Keep the labia majora incision **≥1 cm** from the groin crease to avoid dehiscence.

Vulvoplasty (zero-depth)

Vulvoplasty suits those not wanting receptive intercourse and avoids **lifelong dilation**, but is not freely reversible. **Prior radical prostatectomy favours vulvoplasty**.

3 · Abdominoperineal and intestinal options

- **Peritoneal (Davydov-type)**: robotic dissection cut rectal injury from **~7%** to about 2.3%. The **rectum may mimic seminal vesicles**, and peritoneal defects must be **closed** to prevent hernia.
- Flaps are either **two-flap** (bladder and sigmoid peritoneum) or **pull-through** (abdominal wall peritoneum).
- **Intestinal** vaginoplasty suits severe penoscrotal hypoplasia or revision stenosis; IBD, adhesions, irradiation and short gut contraindicate it. **Dilation is still required.**

KEY POINT

Lining ladder: penile and scrotal skin first, then peritoneum, then bowel. Each adds depth but imports donor pathology; none removes the duty to dilate.

4 · Complications

Complication	Cause / key fact	Management
Wound dehiscence (commonest minor)	Introitus, dilator tension	Local care; granulation: curette or cautery plus steroid
Rectovaginal fistula (most devastating)	**0.8%** primary, 6.3% revision	Intra-op: watertight closure and bulbospongiosus flap, with or without diversion. Established: stop dilating and revise later, **or** delay closure
Urethrovaginal fistula (0.8–3.9%)	Missed injury, distal stenosis	Distal: often left. Bladder-neck (incontinence): transvaginal repair
Urethral stricture (1–6%)	Spraying, UTI, retention	Exclude pelvic floor dysfunction; dilate if short, else meatal spatulation
Urethral bulge / upward meatus	Under-debulked spongiosum	Debulk, fix meatus downward
Vaginal stenosis	Poor dilation, scar, graft loss, pelvic floor dysfunction	Pelvic floor physiotherapy **first**; then scar excision and graft, or peritoneal/bowel augmentation
Prolapse (4–10%)	Poor graft take or fixation	Early repack; peritoneal fixation acts as a colpopexy

KEY POINT

Three failure modes: violated plane (fistulas), tissue not excised (bulge), graft not taken or maintained (dehiscence, stenosis, prolapse). Only the last depends on the patient, so dilation is part of the operation.

5 · Outcomes and follow-up

- **Median time to orgasm is about 180 days**. Depth is 12–16 cm, and self-lubrication is not the norm.
- Stay is about **5 days**; on **day 5 the pack and catheter come out** and dilation is taught.
- Follow-up is weekly for 3 weeks, then at 6 weeks, 3, 6 and 12 months, then annually. †The prostate is retained; survey it via the anterior neovaginal wall.

Period	Dilation (15–20 min each)
0–6 weeks	4 × daily
6 weeks–3 months	3 × daily
3–12 months	2 × daily
>1 year	Daily; less if stable

KEY POINT

The body treats the canal as a defect to close, so stenosis is a lifelong risk. Any narrowing means more dilation, contacting the surgeon and physiotherapy.

128 Special Urologic Considerations in Transmasculine Individuals

SAQ Enlist the steps in female to male gender reassignment surgery.

Paper answer · P4

Type: SAQ (10 marks)
Source: 2024-Winter Paper 4, Q3b

Preconditions

- Gender dysphoria diagnosed per **WPATH (World Professional Association for Transgender Health)** by an MDT (multidisciplinary team).
- **Testosterone:** chapter cites **at least 12 months** before genital surgery (clitoral growth plateaus at about 2 years). WPATH SOC8 (Standards of Care 8, 2022) relaxed this to about 6 months.
- **Urological screening:** uroflowmetry, PVR (post-void residual), renal ultrasound; treat LUTD (lower urinary tract dysfunction) first.

Stage 1: removal of female structures

1. **Subcutaneous mastectomy** (usually first).
2. **Hysterectomy + bilateral salpingo-oophorectomy**, preferably laparoscopic; remove whole cervix ± upper third of vagina.

Stage 2: genitoperineal transformation

3. **Vaginectomy** (needed if the urethra is lengthened): - Hydrodissection; circumferential introital incision. - Rectovaginal plane bluntly; urethrovaginal plane sharply, **no diathermy near the sphincter**. - Paravaginal haemostasis; levators closed in midline.
4. **Urethral lengthening, fixed (perineal) part:** vestibular mucosa and labia minora tubularised over 18 Fr; covered with bulbospongiosus.
5. **Phallus construction**, either: - **Metoidioplasty:** release clitoral chordee and urethral plate; lengthen urethra to the glans (preputial or labial flap ± buccal graft). Gives a 3–8 cm phallus with erection and sensation. - **Phalloplasty: RFFF (radial forearm free flap)** or ALT (anterolateral thigh) flap with **tube-in-tube urethra**; microvascular anastomosis; **clitoral nerve coapted** to a flap nerve.
6. **Urethral anastomosis** of phallic to fixed part; neomeatus at the tip.
7. **Scrotoplasty:** labia majora flaps rotated 180° in front (Ghent type). Monsplasty and perineal closure.
8. **SPC (suprapubic catheter)** + urethral catheter; urethral catheter out at 14–21 days.

Stage 3: prostheses

9. **Testicular prostheses** (17 mL) at 6 months or more.
10. **Erectile prosthesis** after 1 year, once protective sensation returns; single cylinder fixed to pubic bone (dedicated devices, e.g. ZSI 475 FTM [female-to-male]).

Complications

- **Fistula** (phallic base), **stricture** (fixed–phallic anastomosis), hair growth, dribbling, prosthesis infection or erosion, vaginectomy bleeding or rectal/bladder injury.

Diagram to draw: three parts of the neourethra (native, fixed, phallic).

Key points to remember:

- Mastectomy → hysterectomy/oophorectomy → vaginectomy + urethroplasty + phallus + scrotum → prostheses.
- No urethral lengthening without vaginectomy (fistula risk).
- Commonest stricture: fixed–phallic anastomosis.

Also, from the condensed book (not in the paper answer):

- Haemorrhage is the commonest vaginectomy complication, and a two-stage repair is standard for long strictures.

1 · Scope

- Goals are a male appearance, **voiding standing**, sensation and penetration (see Ch 126).
- **Early puberty blockade has a surgical price**: the vagina is harder to remove and small labia majora give less scrotal tissue.

2 · Preoperative care and consent

- **Urological screening is mandatory** (uroflow, PVR, renal ultrasound), and voiding dysfunction, stones and hydronephrosis are treated first.
- SOC7 requires ≥**12 months** of testosterone before genital surgery (†SOC8: 6 months) for **clitoral hypertrophy**, which plateaus at about 2 years.

Agent	Helps	Hinders
Testosterone	Clitoral hypertrophy aids metoidioplasty	Vaginal atrophy: harder vaginectomy
GnRH agonist	Prevents female characteristics	Small labia majora: less scrotoplasty tissue

Consent covers fistula, stricture, vaginectomy bleeding and organ injury, voiding dysfunction, UTI and implant problems.

3 · Removal of female structures

- At hysterectomy the ureter lies beneath the uterine artery. †If the cervix is retained, screening continues.
- **Vaginectomy** risks haemorrhage, then bladder, then rectal injury. **Without vaginectomy there is no urethral lengthening** because of fistula risk.

ALGORITHM

1. Hydrodissect; circumferential introital incision.
2. Blunt rectovaginal plane; urethrovaginal plane **sharply, with no diathermy** (protects sphincter), then bladder plane.
3. If bladder injured: close, with or without cystoscopy.
4. Free levators, control paravaginal plexus; approximate levators.
5. Bulbospongiosus over urethra; layered closure.

Timing	Verdict
All-in-one	More major complications with phalloplasty; none extra with metoidioplasty
Early vaginectomy, phalloplasty later	Keeps blood loss away from free-flap perfusion
Two-stage (preferred)	Mastectomy, hysterectomy–oophorectomy; then vaginectomy and genital reconstruction

4 · Genital reconstruction

- The fixed (perineal) and phallic neourethra both lack spongiosum. One **dorsal clitoral nerve** is coapted for sensation.
- **Bulbospongiosus over the fixed part** reduces fistulas. Scrotoplasty flaps run on the **external pudendal artery**.

	Metoidioplasty	Phalloplasty
Phallus	Own clitoris, **3–8 cm**	Flap
Erection	Spontaneous	Needs prosthesis
Sensation	Largely kept	Depends on nerve coaptation
Penetration	Usually not	Principal goal
Donor scar	None	Visible

5 · Postoperative care and catheterisation

- The neourethral catheter comes out at **14–21 days**; the suprapubic catheter is clamped until voiding is normal.
- Catheterisation is hard because the curve into the native urethra is abrupt.

ALGORITHM

1. One gentle Coudé attempt, then one stylet-guided attempt.
2. If any difficulty, stop: flexible cystoscopy, guidewire, open-tip catheter.
3. In an emergency, place a **suprapubic catheter**.

6 · Urethral complications

- **Fistula** is commonest at the phallic base, and small ones close with diversion. If one persists >**3 months**, excise and close in layers with fat interposition.
- **Stricture** is commonest at the **fixed–phallic anastomosis**. A plateau uroflow prompts urethrography; treat after **≥6 months**.

Option	Use
Dilation	Low-grade meatal stenosis
Internal urethrotomy	Low success; never repeat
Meatotomy / Heineke–Mikulicz	Short meatal / short non-obliterative
Excision and primary anastomosis	Short, near-obliterative anastomotic
Free graft	Needs flap support or dorsal inlay
Two-stage (standard)	Long/complex; retubularise ≥6 months later

Option	Use
Flap / perineal urethrostomy	Recurrent, or no further reconstruction wanted

- **Hair** is prevented with a hairless flap or laser.
- **Post-void dribbling** is near-universal; if severe, exclude sacculation or a vaginal-remnant diverticulum.

> **KEY POINT**
>
> The neourethra is skin in fat without spongiosum: fistulas and strictures cluster at poorly perfused flap ends, heal slowly and resist endoscopic cure.

7 · Prostheses

- Insert an erectile prosthesis ≥**1 year** after phalloplasty, once there is **protective sensation** and a stable urethra.
- Infection or erosion needs urgent explantation, with reimplantation ≥6 months later.

★ Putting it together

A · Newborn with atypical genitalia

ALGORITHM

1. **Emergency is adrenal:** CAH is commonest (half of all; 90% of 46,XX). Salt wasting (low Na, high K, shock) appears at day 5–7 (see Ch 59).
2. **Tests:** karyotype (SRY), 17-hydroxyprogesterone after 36 h, electrolytes, pelvic ultrasound.
3. **Team:** †defer sex assignment until MDT evaluation.
4. **Gonads:** dysgenetic with Y (30–45% tumour risk): prophylactic gonadectomy.

B · Urogenital sinus and cloaca

KEY POINT

True urethral length outweighs common-channel length. Low confluence needs a flap; high needs pull-through or urogenital mobilisation. Clitoroplasty only for Prader III–IV, sparing the dorsal bundle. Cloaca: colostomy, decompress hydrocolpos, imaging with endoscopy, then posterior sagittal repair; seek VACTERL and tethered cord (see Ch 60).

C · Absent vagina

Cause	Key feature	Management
Mayer–Rokitansky–Küster–Hauser	46,XX, no uterus; †image kidneys	Dilation first; else McIndoe, bowel, buccal or peritoneal
Complete androgen insensitivity	46,XY, testes, no uterus	Dilation usually suffices
Septum / imperforate hymen	Uterus present, haematocolpos	Hymen: incision, not aspiration; septum: excision

D · Immunotherapy and hereditary cancer

ALGORITHM

1. **Checkpoints:** PD-L1 silences PD-1-positive T cells; CTLA-4 acts at priming. †Immune adverse events grade ≥2: hold drug, give steroids; endocrinopathies need lifelong replacement (see Ch 24).
2. **Hereditary:** VHL (clear cell; belzutifan), HLRCC (aggressive papillary), Birt–Hogg–Dubé (hybrid oncocytic), Lynch (upper-tract urothelial).
3. BRCA prostate cancer: PARP inhibitor. FGFR-altered urothelial cancer: erdafitinib (see Ch 25).

E · Older patient before surgery

KEY POINT

Function, not age, predicts outcome. Screen frailty, then do a comprehensive geriatric assessment (cognition, delirium, nutrition, falls, polypharmacy). Avoid anticholinergics (prefer a β3-agonist). For new incontinence, seek transient causes first (†DIAPPERS) (see Ch 120).

F · Transgender and nonbinary patients

ALGORITHM

1. Screen the organs present: retained prostate (†PSA lower on oestrogen) and retained cervix (see Ch 126).
2. Vaginoplasty: stenosis and rectovaginal fistula (most devastating) (see Ch 127).
3. Phalloplasty: fistula and stricture, worse without vaginectomy (see Ch 128).

Other paper questions in this section

Asked in the papers but not tied to a specific place in the book above.

LAQ Discuss in brief, present concepts in various GU cancers

Paper answer · P4

Type: LAQ (25 marks)
Source: Practice Set 4 Paper 4, Q1

Read as: the current key concepts in diagnosis and management of each GU (genitourinary) cancer: kidney, upper tract, bladder, prostate, testis and penis.

Common themes

- **De-escalation:** surveillance and organ sparing for low-risk disease.
- **Precision:** molecular classification, germline testing and better imaging (mpMRI (multiparametric MRI, magnetic resonance imaging), PSMA (prostate-specific membrane antigen) PET (positron emission tomography)).
- **Perioperative systemic therapy:** immune checkpoint inhibitors (ICIs), ADCs (antibody–drug conjugates) and targeted drugs.
- **Robotic surgery** as the default approach.

1. Kidney: RCC (renal cell carcinoma)

- **WHO (World Health Organization) 2022:** new molecularly defined types (FH (fumarate hydratase)-deficient, SDH (succinate dehydrogenase)-deficient, TFE3-rearranged RCC).
- **Mechanism:** VHL (von Hippel–Lindau) loss lets HIF (hypoxia-inducible factor) accumulate and drive VEGF (vascular endothelial growth factor). This explains why VEGF TKIs (tyrosine kinase inhibitors) and the HIF-2α inhibitor **belzutifan** work.
- **Small renal mass:** active surveillance in the elderly or unfit; renal mass biopsy; **partial nephrectomy** for cT1 to avoid CKD (chronic kidney disease); thermal ablation for small tumours.
- **Adjuvant pembrolizumab** after nephrectomy for high-risk RCC (**KEYNOTE-564**: better disease-free and overall survival).
- **Metastatic:** IMDC (International Metastatic RCC Database Consortium) risk groups. First line is an **ICI combination**: nivolumab plus ipilimumab, or an ICI plus TKI (pembrolizumab–axitinib, nivolumab–cabozantinib, pembrolizumab–lenvatinib). **CARMENA (2018):** sunitinib alone was non-inferior to cytoreductive nephrectomy plus sunitinib in intermediate/poor risk, so cytoreduction is now selective. SBRT (stereotactic body radiotherapy) for oligometastases.

2. Upper tract urothelial carcinoma (UTUC)

- **Risk stratification (EAU (European Association of Urology)):** low risk (unifocal, <2 cm, low-grade, non-invasive on CT (computed tomography) urography) goes to **kidney-sparing** ureteroscopic ablation or segmental ureterectomy; UGN-101 (mitomycin gel) is an option. High risk goes to **RNU (radical nephroureterectomy) with bladder cuff** and template lymph node dissection.
- **Single postoperative intravesical mitomycin** after RNU cuts bladder recurrence (**ODMIT-C**: 28% to 16% at 1 year).

- **POUT:** adjuvant platinum chemotherapy improved disease-free survival in pT2–T4 or N+ disease.
- **Also know:** screen for **Lynch syndrome**; aristolochic acid (Balkan) nephropathy.

3. Bladder

- **NMIBC (non-muscle-invasive bladder cancer):** EAU risk groups; en bloc TURBT (transurethral resection of bladder tumour); blue-light or narrow-band cystoscopy; re-TURBT for T1; BCG (bacillus Calmette–Guérin) induction plus maintenance. **BCG-unresponsive** disease: radical cystectomy is standard; bladder-sparing options are pembrolizumab, nadofaragene firadenovec (gene therapy) and N-803 with BCG.
- **MIBC (muscle-invasive):** VI-RADS (Vesical Imaging Reporting and Data System) on MRI; **cisplatin-based neoadjuvant chemotherapy** then cystectomy; **trimodality** bladder preservation in selected patients; adjuvant **nivolumab** (**CheckMate 274**: disease-free survival 20.8 vs 10.8 months); perioperative durvalumab with chemotherapy (**NIAGARA, 2024**) improved event-free and overall survival.
- **Metastatic: enfortumab vedotin plus pembrolizumab** (**EV-302**) replaced platinum first line; avelumab maintenance; erdafitinib for FGFR3 (fibroblast growth factor receptor 3) alterations.

4. Prostate

- **Diagnosis: MRI before biopsy** (**PRECISION**: more significant and fewer insignificant cancers); **transperineal biopsy** (less sepsis); **PSMA PET** staging (**proPSMA**: more accurate than CT plus bone scan); ISUP (International Society of Urological Pathology) grade groups; germline testing (BRCA2).
- **Localised: active surveillance** for low risk and selected favourable intermediate risk (**ProtecT, 15 years**: cancer mortality low and similar with surveillance, surgery or radiotherapy); robotic prostatectomy; **hypofractionated** radiotherapy; focal therapy in selected men.
- **Advanced:** mHSPC (metastatic hormone-sensitive) gets ADT (androgen deprivation therapy) plus an **ARPI** (androgen receptor pathway inhibitor); **triplet** with docetaxel (**PEACE-1, ARASENS**); prostate radiotherapy for low volume. EMBARK: enzalutamide for high-risk biochemical recurrence. mCRPC (metastatic castration-resistant): **PARP (poly-ADP ribose polymerase) inhibitors** for HRR (homologous recombination repair) mutations; **^{177}Lu-PSMA-617** (**VISION**: overall survival HR (hazard ratio) 0.62).

5. Testis

- **Stage I: surveillance** preferred for most seminomas and NSGCTs (non-seminomatous germ cell tumours); alternatives are one cycle of carboplatin (seminoma) or one BEP (bleomycin, etoposide, cisplatin) for high-risk NSGCT.
- **miR-371a-3p:** new serum marker for seminoma and NSGCT.
- **Low-volume stage II seminoma: primary RPLND (retroperitoneal lymph node dissection)** (SEMS, PRIMETEST) spares chemotherapy toxicity; robotic, nerve-sparing templates preserve ejaculation.
- **Also know:** IGCCCG (International Germ Cell Cancer Collaborative Group) 2021 update; survivorship care.

6. Penis

- **WHO 2022: HPV (human papillomavirus)-associated** vs **HPV-independent** squamous cell carcinoma; p16 staining; HPV vaccination for prevention.
- **Organ sparing:** topical treatment or laser for PeIN (penile intraepithelial neoplasia), glans resurfacing, wide excision, glansectomy; a negative margin of a few millimetres is enough.
- **Nodes:** for cN0 pT1b or higher, **DSNB (dynamic sentinel node biopsy)** or modified inguinal dissection. Bulky nodes: neoadjuvant TIP (paclitaxel, ifosfamide, cisplatin), then consolidative dissection.

Diagram to draw: six-row table: cancer, key diagnostic advance, localised treatment, advanced treatment.

> **Key points to remember:**
>
> - RCC: adjuvant pembrolizumab (KEYNOTE-564); first-line ICI combinations; CARMENA.
> - UTUC: single post-RNU instillation; POUT adjuvant chemotherapy.
> - Bladder: EV-302 enfortumab–pembrolizumab; CheckMate 274; NIAGARA.
> - Prostate: MRI first, PSMA PET, ARPI triplets, Lu-PSMA (VISION); testis: miR-371, RPLND for stage II seminoma.

SAQ Prepucioplasty

Paper answer · P3 · P4 · merged from 2

Asked as: P3 SAQ "Prepucioplasty" · P4 SAQ "Prepucioplasty."

Read as: preputioplasty, a foreskin-preserving operation for phimosis.

Definition

- An operation that widens the tight preputial ring but keeps the foreskin. It is an alternative to circumcision.

Background

- **Physiological phimosis** at birth is due to glans–preputial adhesions. Smegma and erections separate them. About 90% retract by 3 years; fewer than 1% are phimotic at 17 years.
- **Secondary phimosis** follows scarring (forced retraction, balanoposthitis) or BXO (balanitis xerotica obliterans).

Indications

- Persistent symptomatic phimosis: recurrent balanoposthitis, UTI (urinary tract infection), paraphimosis, painful erection.
- Failed **topical steroid** (betamethasone 0.05% twice daily for about 4 weeks with gentle retraction). About 1 in 6 fail.
- Wish to keep the foreskin (cultural reasons, cosmesis).

Contraindications

- **BXO** with scarring: circumcision is preferred. Triradiate incision with triamcinolone has been reported in selected boys.
- Physiological non-retractility needs only reassurance.
- Treat active infection first.

Techniques

Technique	Method
Dorsal slit with transverse closure	Longitudinal incision through the ring, closed transversely (Heineke–Mikulicz principle)
Triple incision	Three longitudinal cuts over the ring, each closed transversely
Y–V plasty	V-flap of outer skin advanced into a Y-cut

- **Principle:** cutting the ring lengthwise and closing it crosswise widens its circumference.
- General anaesthesia or penile block; incise only the band; confirm full retraction; absorbable sutures.

Postoperative care

- **Daily retraction from the first week**, continued for weeks, prevents re-stenosis. A topical steroid may help.

Outcomes and complications

- Most achieve a retractile foreskin. It is quicker and less painful than circumcision.
- Complications: re-phimosis (needs circumcision), bleeding, infection, poor cosmesis, paraphimosis.
- **Also know:** preputial reconstruction during hypospadias repair is also called preputioplasty; dehiscence is its risk.

Diagram to draw: dorsal incision through the ring, closed transversely.

> **Key points to remember:**
> - Steroid first; about 1 in 6 fail.
> - Cut lengthwise, close crosswise; daily retraction after.
> - BXO: circumcision preferred.

SAQ Secondary malignancies in urology

Paper answer · P1

Asked as:

- Secondary malignancies in Urology. — *Practice Set 3 Paper I Q3f (SAQ)*
- Secondary malignancies in Uro — *Practice Set 9 Paper I Q3c (SAQ)* (also answered in Practice Set 3 Paper 1, Q3f)

Type: SAQ (10 marks)

Interpreted (read) as second cancers caused by urological or oncological treatment (radiotherapy, chemotherapy, urinary diversion, immunosuppression).

Definition

- A **second (treatment-related) malignancy** is a new, histologically distinct cancer caused by treatment of a first disease. It appears after a latency of years, often **in the radiation field**.

Causes and risks

Cause	Second cancer	Key facts
Pelvic radiotherapy	**Bladder cancer**, rectal cancer, sarcoma (in-field)	Bladder RR (relative risk) 2–4 after cervical EBRT (external beam radiotherapy), 1.5–2 after prostate EBRT, smaller after brachytherapy, little with IMRT (intensity-modulated radiotherapy). Latency 15–30 years
Cyclophosphamide	**Bladder cancer**, UTUC (upper tract urothelial carcinoma)	Only chemotherapy proven to cause bladder cancer; metabolites phosphoramide mustard and acrolein. About 4.5-fold, dose-dependent (>20 g cumulative)
Urinary diversion / augmentation	Adenocarcinoma at the uro-intestinal junction	Ureterosigmoidostomy 6–29% after 10–20 years (registry 2.6%); cystoplasty 1.6%; ileal conduit 0.02%. Risk tracks colonic mucosa exposed to urine and faeces
Transplant immunosuppression	Skin cancer, PTLD (post-transplant lymphoproliferative disorder), RCC (renal cell carcinoma), bladder cancer	RCC mostly in native kidneys; bladder cancer about 3-fold and aggressive
Testicular GCT (germ cell tumour) treatment	Solid tumours, leukaemia	60–100% excess after cisplatin or radiotherapy, 200% (highest) after both; etoposide-related leukaemia
Paediatric tumours	In-field solid tumours, leukaemia	Wilms tumour 1% at 10 years

Mechanisms

- Radiation and alkylating (drugs) or topoisomerase-inhibitor chemotherapy: mutagenic DNA damage in normal tissue.
- Immunosuppression: loss of (lost) immune surveillance and oncogenic viruses (EBV [Epstein–Barr virus], HPV [human papillomavirus]).
- Diversion: urine–faeces contact at the anastomosis.

Prevention and surveillance

- IMRT, smaller fields; **mesna** with cyclophosphamide (reduces cystitis).
- **Haematuria after pelvic radiotherapy or cyclophosphamide: always cystoscopy.**
- Endoscopy from 5 years after ureterosigmoidostomy; augmentation surveillance from about 10 years; excise a defunctioned ureterosigmoid anastomosis.
- Testis cancer: fewer treatment modalities; low-dose CT (computed tomography) in surveillance.
- **Also know:** pioglitazone, aristolochic acid and phenacetin as drug-related urothelial carcinogens.

Diagram to draw: ureterosigmoid anastomosis showing (with) adenocarcinoma at the junction.

Key points to remember:

- Pelvic EBRT raises bladder cancer 1.5–4-fold; IMRT little.
- Cyclophosphamide: only chemotherapy proven to cause bladder cancer.
- Ureterosigmoidostomy cancer 6–29% after 10–20 years.
- Transplant recipients: bladder cancer about 3-fold.

SAQ Genetic counselling

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 7 Paper 1, Q3d

Assumed meaning: genetic counselling in urological practice.

Definition

- A communication process that helps patients and families understand the **medical, psychological and family implications** of a genetic condition, so they can make informed choices. It is **non-directive, voluntary and confidential**.
- **Germline testing** (blood) finds inherited variants with family implications; **somatic testing** (tumour) guides treatment.

Steps

1. **Three-generation pedigree**.
2. **Risk assessment:** inheritance and recurrence risk (autosomal dominant 50%; autosomal recessive 25%).
3. **Pre-test counselling:** which test; possible results (pathogenic, **VUS [variant of uncertain significance]**, negative); insurance and psychological harms; consent.
4. **Post-test counselling:** interpret (do not act on a VUS); **cascade testing of relatives**; surveillance; targeted therapy.
5. **Reproductive options:** prenatal diagnosis, PGT (preimplantation genetic testing), donor gametes; psychosocial support.

Urological applications

Condition	Genes	Key point
Prostate cancer	BRCA2 (> BRCA1), ATM, CHEK2, PALB2, mismatch repair (Lynch), HOXB13	NCCN (National Comprehensive Cancer Network): test metastatic, high-risk, strong family history; BRCA2: aggressive disease, PARP (poly-ADP-ribose polymerase) inhibitors, earlier screening
Hereditary RCC (renal cell carcinoma)	VHL, MET, FH, FLCN, TSC1/2, SDH, BAP1	Counsel RCC at ≤46 years, bilateral or multifocal, family history
UTUC (upper tract urothelial carcinoma)	Lynch (MSH2 highest)	Screen if <60 years or Lynch-spectrum cancer history
ADPKD (autosomal dominant polycystic kidney disease)	PKD1 (chromosome 16), PKD2 (chromosome 4)	Testing selects living related donors

Condition	Genes	Key point
Cystinuria	SLC3A1, SLC7A9 (recessive)	Screen siblings
Male infertility	Karyotype, AZF (azoospermia factor) deletion, CFTR (cystic fibrosis transmembrane conductance regulator)	AZF deletions pass to sons by ICSI (intracytoplasmic sperm injection); absent vasa: test partner for CFTR
DSD (differences of sex development)	e.g. CAH (congenital adrenal hyperplasia, recessive), AIS (androgen insensitivity, X-linked)	Recurrence and gonadal tumour risk

Ethics and law

- Confidentiality; right not to know.
- **India:** PCPNDT Act (Pre-Conception and Pre-Natal Diagnostic Techniques Act, 1994) bans disclosure of fetal sex.

Diagram to draw: three-generation pedigree with standard symbols.

> **Key points to remember:**
> - Non-directive; pre-test and post-test counselling; cascade testing.
> - Prostate: BRCA2 most important; test all metastatic men.
> - RCC ≤46 years, ADPKD, cystinuria, AZF/CFTR, DSD are classic urological indications.

SAQ Noonan syndrome

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 22 SAQ Sheet, Q3a

Definition and genetics

- Noonan syndrome is an **autosomal dominant RASopathy** (disorder of the RAS–MAPK [mitogen-activated protein kinase] signalling pathway) with variable expression; many cases are de novo. Incidence about 1 in 1,000–2,500.
- **Karyotype is normal** (46,XY or 46,XX). The phenotype resembles Turner syndrome, hence the old name "male Turner".
- **Mechanism:** gain-of-function variants increase RAS–MAPK signalling, which disturbs growth and organ development.

Gene	Approximate share	Note
PTPN11 (encodes SHP2 phosphatase)	About 50%	Pulmonary stenosis common
SOS1	About 10%	Milder growth and learning effects
RAF1	About 5–10%	Hypertrophic cardiomyopathy
RIT1, KRAS, others	Smaller	KRAS: more learning difficulty

- Related RASopathies: Costello, cardiofaciocutaneous, NF-1 (neurofibromatosis type 1).

Clinical features

- Short stature; webbed neck; hypertelorism, ptosis, low-set posteriorly rotated ears; pectus deformity.

- **Heart:** pulmonary valve stenosis (commonest), HCM (hypertrophic cardiomyopathy), septal defects.
- **Bleeding tendency:** factor XI deficiency, platelet dysfunction.
- Lymphatic dysplasia: genital (penoscrotal) lymphoedema; mild learning difficulty.

Urological aspects

- **Cryptorchidism:** the main urological feature, in most affected boys (often bilateral).
- **Testicular dysfunction:** primary Sertoli cell dysfunction (low inhibin B, high FSH [follicle-stimulating hormone]) with poor spermatogenesis, even after orchidopexy. Leydig function is usually adequate; puberty is often delayed. Female fertility is usually normal.
- **Renal anomalies (about 10%):** mostly mild renal pelvic dilatation; also duplex system, solitary kidney, hypoplasia.
- Small excess of childhood cancers (JMML [juvenile myelomonocytic leukaemia], embryonal rhabdomyosarcoma).

Evaluation

- Clinical diagnosis (van der Burgt criteria), confirmed by gene panel; karyotype excludes Turner in girls.
- Before surgery: **echocardiography, coagulation screen and factor XI level**, anaesthetic review.
- Renal and bladder ultrasound at diagnosis.

Management

- **Orchidopexy by 18 months** (EAU [European Association of Urology] and AUA [American Urological Association]); laparoscopic for non-palpable testes.
- Follow-up: pubertal hormones, semen analysis, fertility counselling.
- **Also know:** growth hormone for short stature; multidisciplinary cardiac and haematology care.

Diagram to draw: RAS–MAPK cascade with SHP2 (PTPN11) acting upstream of RAS.

> **Key points to remember:**
> - Autosomal dominant, normal karyotype; PTPN11 in about half.
> - Cryptorchidism plus Sertoli dysfunction: male subfertility.
> - Check heart and clotting (factor XI) before orchidopexy.

SAQ Gene therapy in urology

Paper answer · P4

Type: SAQ (10 marks)
Source: Practice Set 4 Paper 4, Q3f (intravesical gene therapy also answered in 2022-Summer Paper 2, Q3a)

Definition

- Delivery of nucleic acid (DNA or RNA) into cells to treat disease: add, replace or silence a gene.

Vectors

Vector	Features
Adenovirus	Efficient, non-integrating, transient expression; replication-incompetent or oncolytic
Retrovirus, lentivirus	Integrate: long expression, risk of insertional mutagenesis
AAV (adeno-associated virus)	Low immunogenicity, long expression
Non-viral	Naked plasmid, liposomes, nanoparticles: safer, less efficient
Gene silencing	siRNA (small interfering RNA), antisense; CRISPR (clustered regularly interspaced short palindromic repeats) editing experimental

Strategies in urological cancer

1. **Immunogene (cytokine) therapy: nadofaragene firadenovec**: replication-incompetent adenovirus type 5 carrying the **interferon-α2b gene** with **Syn3** surfactant, which crosses the bladder's protective barrier. The urothelium keeps secreting interferon, overcoming the 1–2-hour dwell of protein interferon. 75 mL every 3 months; CIS (carcinoma in situ) complete response **53% at 3 months, 24% at 12 months**. **FDA (Food and Drug Administration) approved December 2022** for BCG (bacille Calmette–Guérin)-unresponsive NMIBC (non-muscle-invasive bladder cancer).
2. **Oncolytic virus: cretostimogene (CG0070)**: replication-competent adenovirus; E2F promoter limits replication to RB (retinoblastoma) pathway-defective tumour cells; **GM-CSF (granulocyte-macrophage colony-stimulating factor)** transgene adds immunity.
3. **Suicide gene:** HSV-tk (herpes simplex virus thymidine kinase) turns ganciclovir or valacyclovir into a toxin; prostate cancer with radiotherapy (trials).
4. **Toxin gene under a tumour promoter:** BC-819 (diphtheria toxin A, H19 promoter).
5. **Gene-modified cells:** CAR-T (chimeric antigen receptor T cells) against PSMA (prostate-specific membrane antigen): early trials, toxic. Viral-vector vaccine PROSTVAC failed.

Benign conditions

- **OAB (overactive bladder) and ED (erectile dysfunction):** URO-902 (hMaxi-K) plasmid for the BK potassium channel relaxes smooth muscle; phase 1 only.
- **Primary hyperoxaluria type 1:** lumasiran (RNA interference against glycolate oxidase) lowers oxalate.
- Underactive detrusor: experimental.

Challenges

- Delivery barriers, vector immunity, short expression, cost, biosafety.

Diagram to draw: adenovirus → urothelial cell → interferon secretion → tumour apoptosis.

> **Key points to remember:**
> - Nadofaragene: non-replicating adenovirus + interferon-α2b + Syn3; FDA 2022.
> - CIS CR 53% at 3 months, 24% at 12 months.
> - Cretostimogene: oncolytic adenovirus + GM-CSF.
> - Lumasiran: RNA interference for primary hyperoxaluria type 1.

SAQ Molecular targeted therapy in urological cancers

Paper answer · P4

Type: SAQ (10 marks)
Source: Practice Set 6 Paper 4, Q3c

Concept

- **Targeted therapy** blocks a molecule the cancer depends on (receptor, kinase, transcription factor, DNA-repair enzyme, surface antigen), not all dividing cells.
- Classes: small-molecule TKIs (tyrosine kinase inhibitors, "-nib"), monoclonal antibodies ("-mab"), mTOR (mammalian target of rapamycin) inhibitors, HIF-2α (hypoxia-inducible factor-2α) inhibitor, PARP (poly-ADP-ribose polymerase) inhibitors, ADCs (antibody–drug conjugates), radioligands.

RCC (renal cell carcinoma)

- **Basis:** VHL (von Hippel–Lindau) loss → HIF accumulates even in normoxia → constant VEGF (vascular endothelial growth factor), PDGF (platelet-derived growth factor) and other targets.

Target	Drugs	Key evidence
VEGF receptors	Sunitinib, pazopanib, axitinib, cabozantinib (also MET, AXL), lenvatinib	COMPARZ: pazopanib noninferior to sunitinib; METEOR: cabozantinib beat everolimus
VEGF ligand	Bevacizumab	Modest PFS (progression-free survival) gain
mTOR	Temsirolimus (IV, poor risk), everolimus (oral)	Temsirolimus improved OS (overall survival) over interferon
HIF-2α	**Belzutifan**	VHL disease tumours; LITESPARK-005 beat everolimus after immunotherapy and TKI

- First line now: TKI **plus** checkpoint inhibitor. Everolimus also treats tuberous sclerosis angiomyolipoma.

Urothelial carcinoma

- **FGFR3 (fibroblast growth factor receptor 3)** alterations, typical of luminal tumours: **erdafitinib** (oral) after prior therapy.
- **ADCs: enfortumab vedotin** (Nectin-4) improved OS after platinum and immunotherapy; sacituzumab govitecan (Trop-2); HER2 (human epidermal growth factor receptor 2)-directed ADCs emerging.
- **EV-302 (2024):** enfortumab vedotin plus pembrolizumab, first line, roughly doubled OS versus platinum chemotherapy and is now standard (chapter notes predate it).

Prostate cancer (mCRPC, metastatic castration-resistant)

- **PARP inhibitors** exploit **synthetic lethality** in HRR (homologous recombination repair)-deficient tumours, especially BRCA2. **PROfound:** olaparib beat a second hormonal agent (imaging PFS 7.4 vs 3.6 months in BRCA/ATM).
- **PSMA (prostate-specific membrane antigen) radioligand: 177Lu-PSMA-617; VISION**: OS 15.3 vs 11.3 months.

Toxicity

- VEGF TKIs: hypertension, hand–foot syndrome, hypothyroidism. mTOR: mucositis, hyperglycaemia. Erdafitinib: hyperphosphataemia. Enfortumab: neuropathy, rash. PARP inhibitors: anaemia.

Limitations

- Resistance develops; cures rare; biomarker testing needed.

Diagram to draw: VHL–HIF–VEGF pathway with drug sites (TKI at receptor, bevacizumab at VEGF, mTOR and HIF-2α inhibitors).

> **Key points to remember:**
>
> - RCC: VEGF TKIs, mTOR inhibitors, belzutifan (HIF-2α).
> - Bladder: erdafitinib (FGFR3), enfortumab vedotin (Nectin-4).
> - Prostate: PARP inhibitors (BRCA2), 177Lu-PSMA (VISION).

Rapid Revision Sheet · Section 2

Immunology, immunotherapy and cancer vaccines in urological oncology (including Sipuleucel-T) (LAQ)

- Three Es: elimination, equilibrium, escape; Th1 is antitumour.
- BCG is the original urological immunotherapy: TLR2/4 → neutrophils → Th1 memory.
- Vaccine = antigen + adjuvant; DC gives signal 1 (MHC–peptide) and signal 2 (costimulation).
- IMA901, AGS-003, PROSTVAC, GVAX all failed in phase III.
- Sipuleucel-T: PAP–GM-CSF; IMPACT OS 25.8 vs 21.7 months (+4.1 months); no PSA or progression effect.
- CheckMate 274 adjuvant nivolumab; EV-302 EV + pembrolizumab first line.

Oncogenes (SAQ)

- Oncogenes: gain of function, one allele; activated by mutation, amplification, translocation.
- HRAS first found in T24 bladder cancer; FGFR3 in up to 80% low-grade Ta.
- TMPRSS2–ERG in about half of prostate cancers; MET in papillary RCC.

Telomerase (SAQ)

- Telomerase = hTERT (catalytic) + hTR (RNA template).
- About 100 base pairs lost per division; 85–90% of tumours express telomerase.
- TERT promoter mutation is the commonest mutation in bladder cancer.
- TRAP assay detects urinary telomerase; inflammation gives false positives. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

Familial / hereditary urological malignancies (SAQ)

- All familial RCC is autosomal dominant; 3 cm rule except HLRCC.
- VHL 3p25, MET 7q31, FH 1q42, FLCN 17p11.
- Lynch (MSH2) → UTUC; BRCA2 → aggressive prostate cancer.

Circumcision undone (SAQ)

- Complications 0.2–5%; bleeding commonest.
- Too little skin: redundancy, phimosis, adhesions; too much: trapped penis, chordee.
- Meatal stenosis almost only in circumcised boys.
- Revision needs general anaesthesia.

Congenital Penile curvature.. (SAQ)

- Devine–Horton I–III = chordee without hypospadias; IV = corporal asymmetry.
- Congenital curvature: young man, large erection, orthotopic meatus.
- Dorsal 12 o'clock plication; never divide a urethra that reaches the glans tip.

Female circumcision (SAQ)

- WHO types I–IV: I clitoridectomy, II excision, III infibulation, IV others.
- Urological: retention, recurrent UTI, stenosis, stones, fistula.
- Midline deinfibulation for type III; never reinfibulate.

46,XX male (SAQ)

- 46,XX testicular DSD: SRY on X in about 80–90%; 1 in 20,000.

- Azoospermia always (AZF absent): no sperm retrieval.
- Shorter than Klinefelter; FISH for SRY confirms.

Mixed gonadal dysgenesis. (SAQ)

- 45,X/46,XY; testis + streak + Müllerian remnants; asymmetric.
- Second commonest cause of atypical genitalia after CAH.
- Tumour risk 15–35%: remove streaks.
- Screen for Turner features.

DSD: Evaluation and Management of the Newborn with Ambiguous Genitalia (LAQ)

- Palpable gonad = testicular tissue; impalpable gonads = CAH until proven otherwise.
- 17-OHP after 36 h; salt crisis around day 5–7.
- Karyotype + uterus + gonads triad guides diagnosis; no gonadectomy while diagnosing.
- Defer irreversible surgery; CAIS gonads kept until after puberty (till puberty).

Penile inversion vaginoplasty (SAQ)

- Glans → clitoris; penile skin → canal; scrotal skin → graft.
- Excise all erectile tissue; keep Denonvilliers fascia intact.
- Lifelong dilation prevents stenosis; rectovaginal fistula is the worst complication.

Enlist the steps in female to male gender reassignment surgery. (SAQ)

- Mastectomy → hysterectomy/oophorectomy → vaginectomy + urethroplasty + phallus + scrotum → prostheses.
- No urethral lengthening without vaginectomy (fistula risk).
- Commonest stricture: fixed–phallic anastomosis.

Discuss in brief, present concepts in various GU cancers (LAQ)

- RCC: adjuvant pembrolizumab (KEYNOTE-564); first-line ICI combinations; CARMENA.
- UTUC: single post-RNU instillation; POUT adjuvant chemotherapy.
- Bladder: EV-302 enfortumab–pembrolizumab; CheckMate 274; NIAGARA.
- Prostate: MRI first, PSMA PET, ARPI triplets, Lu-PSMA (VISION); testis: miR-371, RPLND for stage II seminoma.

Prepucioplasty (SAQ)

- Steroid first; about 1 in 6 fail.
- Cut lengthwise, close crosswise; daily retraction after.
- BXO: circumcision preferred.

Secondary malignancies in urology (SAQ)

- Pelvic EBRT raises bladder cancer 1.5–4-fold; IMRT little.
- Cyclophosphamide: only chemotherapy proven to cause bladder cancer.
- Ureterosigmoidostomy cancer 6–29% after 10–20 years.
- Transplant recipients: bladder cancer about 3-fold.

Genetic counselling (SAQ)

- Non-directive; pre-test and post-test counselling; cascade testing.
- Prostate: BRCA2 most important; test all metastatic men.
- RCC ≤46 years, ADPKD, cystinuria, AZF/CFTR, DSD are classic urological indications.

Noonan syndrome (SAQ)

- Autosomal dominant, normal karyotype; PTPN11 in about half.
- Cryptorchidism plus Sertoli dysfunction: male subfertility.
- Check heart and clotting (factor XI) before orchidopexy. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

Gene therapy in urology (SAQ)

- Nadofaragene: non-replicating adenovirus + interferon-α2b + Syn3; FDA 2022.
- CIS CR 53% at 3 months, 24% at 12 months.
- Cretostimogene: oncolytic adenovirus + GM-CSF.
- Lumasiran: RNA interference for primary hyperoxaluria type 1.

Molecular targeted therapy in urological cancers (SAQ)

- RCC: VEGF TKIs, mTOR inhibitors, belzutifan (HIF-2α).
- Bladder: erdafitinib (FGFR3), enfortumab vedotin (Nectin-4).
- Prostate: PARP inhibitors (BRCA2), 177Lu-PSMA (VISION). *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

SECTION 3

Infections & Inflammation

26 Infections of the Urinary Tract

1 · Definitions and classification

The label decides treatment length and whether to investigate.

- **Urinary tract infection (UTI)** is the urothelial inflammatory response to bacterial invasion, usually with bacteriuria and pyuria. Urine is **not sterile**.
- **Bacteriuria** and **pyuria** are not the same as UTI; bacteriuria without pyuria is colonisation. **Sterile pyuria**: look for tuberculosis (TB) see Ch 31, stone, urothelial injury and interstitial nephritis.
- **Cystitis** causes dysuria, frequency and urgency; mimics include urethritis, vaginitis, interstitial cystitis see Ch 28 and bladder carcinoma.
- **Acute pyelonephritis** is a clinical diagnosis: chills, fever and flank pain with bacteriuria and pyuria. Flank symptoms may be absent in spinal cord injury (SCI).
- **Chronic pyelonephritis** shows a focal coarse cortical scar over a **distorted calyx**.

	Uncomplicated UTI	**Complicated UTI**
Host	Healthy, non-pregnant woman	Compromised host and/or virulent or resistant organism
Tract	Structurally and functionally normal	Structural or functional abnormality, catheter
Organism	Eradicated by short oral course	Often multidrug-resistant (MDR), so culture is mandatory

- **Complicating factors** are an anatomical or functional abnormality (obstruction, neurogenic bladder [NGB], bladder outlet obstruction [BOO]), **male sex**, pregnancy, diabetes mellitus (DM), immunosuppression, catheter, hospital acquisition and symptoms >7 days. MCQ Male sex alone makes a UTI complicated.

Term	**Definition**	**Discriminator**
Isolated	First, or one remote UTI	—
Unresolved	Therapy failed; same organism	Never cleared
Reinfection	New inoculation (~ **70%** of recurrences, often different organism)	Cleared, then new event
Bacterial persistence (relapse)	Same organism from a focus (stone, prostate)	Focus remains; recurs quickly

- **Prophylaxis** prevents reinfection of a sterilised tract; **suppression** prevents growth from an ineradicable focus.
- **Catheter-associated UTI (CAUTI)** is the commonest cause of secondary bacteraemia and ~**80%** of nosocomial UTIs.
- Uncomplicated recurrent UTI does not cause scarring, hypertension or renal failure.

2 · Pathogenesis

Route	Frequency	Features
Ascending	Commonest	Bowel flora colonise the introitus and climb; reflux not required
Haematogenous	Uncommon	S. aureus, Candida; enhanced by obstruction
Lymphatic	Rare	Severe bowel infection, retroperitoneal abscess

Organisms

- **E. coli** causes ~**85%** of community and ~50% of hospital UTIs, but less than half of complicated UTI and urosepsis.
- **S. saprophyticus** causes ~10% of cystitis in young women; complicated UTIs involve Klebsiella, Pseudomonas and E. faecalis.
- **ST131 (O25b:H4)** is an MDR E. coli clone carrying extended-spectrum β-lactamase (ESBL), with fluoroquinolone resistance as its hallmark. MCQ Enterococcus is a Gram-positive coccus, not an Enterobacteriaceae.

Virulence — fimbriae

	Type 1 pili (mannose-sensitive)	P pili (mannose-resistant)
Gene / adhesin	fim; **FimH** tip adhesin	pap operon on a pathogenicity island
Receptor	Mannosylated **uroplakins Ia/Ib**	Glycolipid (P blood-group) receptors in kidney
Tropism	Bladder, causing cystitis	Kidney, causing **pyelonephritis** (~**90%** of pyelonephritis strains)

- Adherence is the essential first step. In **phase variation** bacteria switch between piliated and non-piliated states.

Intracellular cycle (why cystitis recurs)

ALGORITHM

1. FimH binds uroplakin and the bacterium invades the umbrella cell (15 min–1 h).
2. It multiplies into an **intracellular bacterial community (IBC)**, which behaves like a biofilm (4–16 h).
3. Bacteria filament, escape and reinvade neighbouring cells (16–24 h).
4. Infected cells exfoliate, exposing deeper layers.
5. **Quiescent intracellular reservoirs (QIRs)** form (from ~2 weeks).

- QIRs are silent between attacks; urothelial disruption reactivates them, giving same-strain recurrence without a urological focus.

Host defences and susceptibility

- Lactobacilli are the introital barrier, disturbed by antimicrobials, spermicides and oestrogen loss.
- Bladder emptying is the key defence; **obstruction** makes infection life-threatening. In adults, vesicoureteral reflux (VUR) alone is harmless without high pressure or stasis.

- **DM** brings more abscess, emphysematous infection and papillary necrosis.

Renal papillary necrosis

- The main causes are **DM, obstruction with pyelonephritis, analgesic abuse and sickle cell disease**. A sloughed papilla obstructs the ureter; treat with antibiotics and early relief of obstruction.

3 · Evaluation and diagnosis

A fistula presents with pneumaturia or faecaluria, and an abscess with indolent fever and a flank mass.

	Cystitis	Vaginitis	Urethritis
Onset	Acute	Subacute	Subacute
Frequency/urgency	Prominent	Absent; no haematuria or suprapubic pain	Less marked; no fever
Clues / causes	E. coli	Discharge, odour, new partner; HSV, gonorrhoea, Chlamydia, Trichomonas, yeast, bacterial vaginosis	Discharge; initial-stream pyuria in men; N. gonorrhoeae, Chlamydia, HSV see Ch 29

- Dysuria with frequency and no vaginal discharge gives a ~**90%** probability of UTI; dysuria is the best single symptom.
- **Rule**: painless gross haematuria, or microhaematuria persisting after the infection clears, needs full upper and lower tract evaluation.
- Midstream urine is the default; a catheter sample is more accurate. With an indwelling catheter, replace it and sample the new one.

Urinalysis and culture

- **Leucocyte esterase** indicates pyuria. **Nitrite** is made by Enterobacteriaceae and is negative with Gram-positives and Pseudomonas; it is the best rule-in test. Dipsticks are better at **ruling out**.
- On microscopy, pyuria is ≥**5–10 WBC/high-power field (HPF)**, and squamous cells mean contamination. A positive culture without pyuria should be repeated on a catheter sample.

Threshold	Use
$\geq 10^5$ colony-forming units (CFU)/mL	Kass (originally pyelonephritis); ASB definition
≥**10^2 CFU/mL** of a known pathogen	Symptomatic women with cystitis; catheter specimens

- **Stone culture** predicts post-procedure sepsis better than bladder urine.

Imaging

- Imaging asks only whether there is obstruction, a collection to drain or a focus of persistence; most UTIs need none.
- **Indications in pyelonephritis**: possible obstruction, risk of papillary necrosis, prior reconstruction, **no response after 72 h**, DM, polycystic kidneys on dialysis, NGB and unusual organisms. Also image most men, compromised hosts and rapid recurrences.

Question	Best test
Stone	Non-contrast computed tomography (CT)

Question	Best test
Abscess, acute focal bacterial nephritis (AFBN), drainage planning	Contrast CT
Hydronephrosis, pyonephrosis, post-void residual (PVR)	Ultrasound (US)
Extrarenal extension	Magnetic resonance imaging (MRI)
Urethral diverticulum	Pelvic MRI with contrast
VUR; non-refluxing stump	Voiding cystourethrogram (VCUG)
Split function, scarring	Dimercaptosuccinic acid (DMSA) scan

Correctable causes of bacterial persistence (Box 26.6) include infection stones, chronic bacterial prostatitis, an infected atrophic kidney, ectopic ureter, foreign body, urethral diverticulum, an infected ureteric stump and fistula. Antibiotics cannot sterilise these, so **cure comes only by removal or drainage.**

4 · Antimicrobial principles

- In cystitis the **urinary** drug level matters; in parenchymal infection or bacteraemia, **serum and tissue** levels matter. Renal insufficiency and obstruction lower urine levels.
- **Collateral damage** means resistance, MDR colonisation, C. difficile and candidiasis, caused mainly by broad cephalosporins and quinolones.

SAQ Antimicrobial resistance

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 18 SAQ Sheet, Q3b (also answered in part in 2015-Summer Paper 1, Q3f)

Definition

- AMR (antimicrobial resistance): a microbe grows despite drug levels achievable in the patient. Reported as S (sensitive), I (intermediate), R (resistant) with the MIC (minimum inhibitory concentration).
- One of the most serious global health threats.

Types

Type	Mechanism	Relevance
Intrinsic	Species lacks the target	Predictable (enterococci and cephalosporins)
Acquired chromosomal	Mutants selected when urine level is close to MIC	Underdosing, poor compliance
Plasmid (extrachromosomal)	Horizontal transfer in bowel flora	**Commonest**; multidrug resistance after antibiotics

Mechanisms

- **Enzymes:** β-lactamases; aminoglycoside-modifying enzymes.
- **Altered target:** penicillin-binding protein; DNA gyrase mutation (quinolones).
- **Reduced entry** (porin loss) and **efflux pumps**.
- **Bypass:** enterococci scavenge folate (escape TMP-SMX (trimethoprim–sulfamethoxazole)).

Key resistant organisms

- **ESBL (extended-spectrum β-lactamase)** (plasmid, 1983): resists 3rd-generation cephalosporins; often co-resistant to fluoroquinolones and aminoglycosides. *E. coli* ST131 carries it.
- **AmpC; KPC (*Klebsiella pneumoniae* carbapenemase, 2001); NDM (New Delhi metallo-β-lactamase)**: carbapenem resistance.
- MRSA (methicillin-resistant *S. aureus*), VRE (vancomycin-resistant enterococci).

Drivers in urology

- Overuse of fluoroquinolones and cephalosporins; treating ASB (asymptomatic bacteriuria); long prophylaxis; catheters, stents, stones (biofilm).
- Transrectal biopsy: fluoroquinolone-resistant rectal *E. coli* causes sepsis.

Prevention: stewardship

1. Culture before treatment; follow the **local antibiogram**.
2. TMP-SMX only if local resistance ≤20%; oral fluoroquinolone for pyelonephritis only if ≤10%.
3. Prefer narrow, urine-confined drugs for cystitis (nitrofurantoin, fosfomycin, pivmecillinam: low resistance).
4. Shortest course; de-escalate to culture.
5. Do not treat ASB, except in pregnancy and before mucosa-breaching procedures.
6. Single-dose surgical prophylaxis; transperineal biopsy or rectal-swab-targeted prophylaxis.
7. Infection control and hand hygiene. - **Also know:** FDA (Food and Drug Administration) 2016 fluoroquinolone warning; *C. difficile*; ESBL: carbapenem, or fosfomycin/nitrofurantoin for cystitis.

Recent advances

- New β-lactam/β-lactamase inhibitor combinations (ceftazidime–avibactam, meropenem–vaborbactam); cefiderocol; gepotidacin.

Diagram to draw: Gram-negative cell showing β-lactamase, porin loss, efflux pump, altered target.

> **Key points to remember:**
> - Plasmid-mediated resistance is commonest.
> - ESBL: plasmid, co-resistant; KPC/NDM: carbapenem resistance.
> - TMP-SMX if resistance ≤20%; fluoroquinolone for pyelonephritis if ≤10%.
> - Stewardship: culture, narrow, short, no ASB treatment.

SAQ ESBL producing enterobacterial UTI

Paper answer · P1

Asked as:

- ESBL producing enterobacterial UTI — *Summer 2019 Paper I Q3f (SAQ)*
- ESBL producing enterobacterial UTI. — *Practice Set 5 Paper I Q3c (SAQ)* (also answered in 2019-Summer Paper 1, Q3f)

Type: SAQ (10 marks)

Definition

- **ESBLs (extended-spectrum β-lactamases): plasmid-encoded** enzymes that hydrolyse penicillins, **third-generation cephalosporins** and aztreonam. They are **inhibited by clavulanate** and spare **carbapenems** and cephamycins.
- Mainly *E. coli* and *Klebsiella* (Enterobacterales); first reported 1983.

Mechanism and spread

- Older types **TEM** and **SHV**; now **CTX-M** dominates worldwide. Spread through the community is linked to *E. coli* **ST131**, whose hallmark is fluoroquinolone resistance (fluoroquinolone resistance is its hallmark).
- Plasmids carry co-resistance genes: **fluoroquinolones, aminoglycosides, TMP-SMX (trimethoprim–sulfamethoxazole)**. So oral options are few.
- India has one of the highest prevalence rates.

Risk factors

- Prior antibiotics (especially cephalosporins, quinolones); recent hospital stay; **indwelling catheter**, urological instrumentation; recurrent UTI (urinary tract infection); diabetes; nursing-home residence; travel to high-prevalence areas; transrectal prostate biopsy.

Detection

- Screening: reduced susceptibility to ceftriaxone, cefotaxime or ceftazidime.
- **Phenotypic confirmation:** clavulanate synergy (**combination disc** or double-disc synergy test); automated systems.
- Genotypic PCR (polymerase chain reaction) for CTX-M.

Treatment

Infection	Options
Cystitis	**Nitrofurantoin** 5 days, **fosfomycin** 3 g single dose (E. coli), **pivmecillinam**; TMP-SMX or quinolone only if susceptible; single-dose aminoglycoside
Pyelonephritis, complicated UTI	Quinolone or TMP-SMX if susceptible; else (otherwise) **aminoglycoside**
Urosepsis, bacteraemia	**Carbapenem** (meropenem, ertapenem), then oral step-down by culture

- **MERINO trial (2018):** in ceftriaxone-resistant *E. coli*/*Klebsiella* bacteraemia, **piperacillin–tazobactam** had higher 30-day mortality than meropenem (12.3% vs 3.7%). Avoid pip-taz for serious ESBL infection.
- **Ceftolozane–tazobactam, ceftazidime–avibactam:** active but **reserved** for carbapenem-resistant organisms (stewardship).
- **Source control:** remove or change catheters; drain obstruction.
- Treat **asymptomatic bacteriuria** only before urological procedures or in pregnancy.
- **Febrile prostatitis after transrectal biopsy:** carbapenem is favoured.

Prevention

- **Antibiotic stewardship:** avoid broad-spectrum cephalosporins and quinolones; narrowest spectrum; shortest effective course; culture before treatment.
- Hand hygiene, contact precautions, catheter care.

- Pre-biopsy rectal swab culture for targeted prophylaxis; transperineal biopsy.

Also know: AmpC (not clavulanate-inhibited) and carbapenemases **KPC (*Klebsiella pneumoniae* carbapenemase)**, NDM (New Delhi metallo-β-lactamase) are wider resistance.

Diagram to draw: treatment flowchart from culture result to oral or IV (intravenous) choice.

Key points to remember:

- Plasmid CTX-M; resists 3rd-gen (third-generation) cephalosporins; inhibited by clavulanate (clavulanate inhibits).
- Oral: nitrofurantoin, fosfomycin, pivmecillinam.
- Serious infection: carbapenem; MERINO: pip-taz (piperacillin–tazobactam) inferior.
- Stewardship prevents spread.

Also, from the condensed book (not in the paper answer):

- Prior quinolone use is the strongest determinant of quinolone resistance, and plasmid resistance spreads among bowel flora.
- **Stewardship** means narrow spectrum, the shortest course, culture before treating recurrent or complicated UTI, and local antibiograms.

Agents

- Fosfomycin acts on the cell wall at a different step from β-lactams, so cross-resistance is rare. Enterococci scavenge folate, so TMP-SMX fails even when "sensitive".
- **Anti-pseudomonal** agents are antipseudomonal penicillins, **ceftazidime** (not ceftriaxone), aztreonam, aminoglycosides and fluoroquinolones.
- **Enterococcal** cover comes from aminopenicillins, piperacillin, nitrofurantoin and fosfomycin, and **never from cephalosporins**.
- Ertapenem misses Pseudomonas and Enterococcus.

Agent	Key points
Nitrofurantoin	Acts in urine only, so it is useless for pyelonephritis, prostatitis and bacteraemia. Toxicity includes pulmonary fibrosis and neuropathy. Beers 2015: avoid if creatinine clearance (CrCl) < **30 mL/min**; no long-term suppression in the elderly.
TMP-SMX	Good tissue levels; misses Enterococcus and Pseudomonas; potentiates warfarin.
Fosfomycin	3 g single dose; active against ESBL producers and vancomycin-resistant enterococci, so it is an oral option for MDR cystitis.
Fluoroquinolones	Broad, including Pseudomonas. Adjust at CrCl <30. FDA warnings include tendon rupture and aortic dissection; **2016: reserve for when there is no alternative in uncomplicated UTI**. Avoid in children and pregnancy.
Cephalosporins	Poor against enterococci; use in cystitis only if first-line agents cannot be used.
Carbapenems	Stable to ESBL and AmpC; penicillin cross-sensitivity <1%.
Aminoglycosides	Once daily 5–7 mg/kg; a nomogram level 6–14 h after the dose sets the interval at 24, 36 or 48 h. Monitor oto- and nephrotoxicity.
Aztreonam	Gram-negative spectrum without nephrotoxicity; useful in penicillin allergy and renal impairment.
Pivmecillinam	~2% E. coli resistance; Nordic first-line; not in USA.

KEY POINT

Narrow, urine-confined, flora-sparing agents (nitrofurantoin, fosfomycin, pivmecillinam) lead in cystitis; broad tissue-penetrating agents are kept for invasive infection. Resistance is made in the bowel, so breadth, duration and repetition matter more than MIC.

C. difficile infection (CDI)

- The usual culprits are cephalosporins, clindamycin, carbapenems, quinolones and TMP-SMX. Diagnosis needs ≥3 unformed stools in 24 h and toxigenic C. difficile or its toxin.

Episode (IDSA/SHEA 2021)	Treatment
Initial	**Fidaxomicin** 200 mg twice daily × 10 d; alternative oral vancomycin 125 mg four times daily × 10 d; metronidazole only if non-severe and others unavailable
First recurrence	Fidaxomicin or tapered vancomycin, with or without **bezlotoxumab** (anti-toxin B antibody)
≥2 recurrences	As above, or **faecal microbiota transplantation**
Fulminant	Oral/rectal vancomycin 500 mg four times daily and IV metronidazole

5 · Asymptomatic bacteriuria

Treating ASB usually harms; know the few groups who need it.

Group (Infectious Diseases Society of America [IDSA])	Specimen	Threshold
Women	**Two** consecutive voided, same strain	≥10^5 CFU/mL
Men	One voided	≥10^5 CFU/mL
Either sex	One catheterised	≥**10^2** CFU/mL

- Prevalence is 1–5% in premenopausal women and **100%** with a long-term catheter. Untreated ASB is harmless; treating it brings more symptomatic UTI and MDR.

Do NOT screen or treat (IDSA 2019)	Screen and treat
Premenopausal non-pregnant women; DM; elderly (community or institution); SCI; indwelling catheters; ASB with pyuria; children; renal transplant >1 month; other solid-organ transplant; **non-urological surgery (including arthroplasty and cardiac)**. No consensus: neutropenia.	**Pregnancy**; **urological procedures with anticipated mucosal bleeding** (transurethral resection [TUR], ureteroscopy, percutaneous nephrolithotomy [PCNL])

KEY POINT

ASB is treated only where bacteria are about to reach blood or tissue: pregnancy (hydroureter leading to pyelonephritis) and mucosa-breaching urological surgery. Elsewhere the resident strain is harmless or protective.

6 · Cystitis

Acute uncomplicated cystitis in women

- Diagnose on acute dysuria with a change in voiding. Fever, flank pain or no oral route should be treated as pyelonephritis.

Tier (IDSA/ESCMID)	Regimen	Caveat
First-line	**Nitrofurantoin** 100 mg twice daily × **5 d**	Not if pyelonephritis suspected
First-line	**TMP-SMX** 160/800 mg twice daily × **3 d**	Avoid if local resistance > **20%** or used in past 3 months
First-line	**Fosfomycin** 3 g single dose	Slightly lower efficacy
First-line	**Pivmecillinam** 400 mg twice daily × 3–7 d	Not in North America
Second-line	Fluoroquinolone × 3 d	Ecology and toxicity; last resort (FDA 2016)
Second-line	β-lactam (amoxicillin–clavulanate, cefpodoxime, cefadroxil) × 3–7 d	Lower efficacy; never amoxicillin alone

A **3-day** course beats a single dose and equals 7 days (7 if complicating factors). No post-treatment culture if asymptomatic.

Cystitis in men

- Male cystitis is complicated and about half have an abnormality, so evaluate and consider prostatitis see Ch 27.
- In afebrile men, **7 days is as good as 14 days** (Drekonja). The EAU suggests TMP-SMX 160/800 twice daily × 7 d, or a quinolone guided by local susceptibility.

Complicated UTI

Category (Canadian Urological Association [CUA])	Examples
Anatomical	Cystocele, fistula, diverticulum
Iatrogenic	Nosocomial, surgery, catheter
Obstruction	BOO, pelviureteric junction (PUJ) obstruction, ureteric stricture
Voiding dysfunction	Neurological disease, VUR, incontinence, high PVR
Other	Pregnancy, stones, DM, immunosuppression

Always culture and correct the abnormality. Treat **10–14 days** by culture, IV to oral once afebrile.

Emphysematous cystitis

- Bladder-wall gas, typically in an elderly diabetic woman; fever occurs in only 30–50%. **CT confirms** it.
- ~**90%** settle with broad Gram-negative cover, catheter drainage and glucose control.

Unresolved bacteriuria

Cause (descending frequency)	On-treatment culture	Action
1. Resistant from outset (recent antibiotics)	Resistant	Change drug; no further work-up
2. Emergent resistance (underdosing)	Same species, now resistant	
3. Unsuspected second pathogen	New species present from start	
4. Rapid reinfection, new resistant species	New species; suspect enterovesical fistula	
5–7. Azotaemia; papillary necrosis; giant staghorn	Same species, still **sensitive**	Renal function and imaging
8. Non-compliance, Munchausen	—	—

7 · Recurrent UTI

Most recurrences are reinfections, so the work is prevention.

- Recurrent UTI means **2 in 6 months or ≥3 in 12 months**, each culture-documented. Reinfection is a different organism, or the same one >2 weeks after treatment.
- Risk factors include intercourse, spermicide, menopause, incontinence, cystocele and raised PVR.
- Evaluate with history, pelvic examination, **culture of each episode** and PVR. **Imaging and cystoscopy are not routine** (AUA, EAU, CUA).
- **Indications for imaging with or without cystoscopy (CUA)**: prior urinary surgery or trauma; stones; gross or persisting microhaematuria; obstructive symptoms or high PVR; urea-splitters; prior pelvic malignancy; persistence despite sensitivity-based therapy; DM or immunocompromise; pneumaturia, faecaluria or anaerobes; and repeated pyelonephritis. MCQ Recurrence frequency alone is **not** an indication.
- **Struvite** stones form when urease (Proteus mirabilis) produces magnesium ammonium phosphate, with bacteria inside the stone. The AUA advises complete removal in fit patients; unfit patients get acetohydroxamic acid and suppression.

Prevention (women, AUA 2019)

ALGORITHM

1. Confirm recurrent UTI with symptoms and culture.
2. If the same organism returns at short intervals, then evaluate for persistence (upper tract imaging and cystoscopy) and remove the nidus.
3. If it is reinfection, then start behavioural measures: ≥1.5 L water a day, frequent and postcoital voiding, stop spermicide.
4. Add cranberry ≥36 mg proanthocyanidin (PAC).
5. If UTIs follow intercourse, then give a **postcoital antibiotic**. If postmenopausal, then give **vaginal oestrogen**. If neither, then consider **methenamine hippurate** (normal renal function).
6. Use continuous antibiotic prophylaxis last.

Measure	Mechanism	Evidence / use
Water ≥1.5 L/day	Washout	RCT in premenopausal women: 1.7 vs 3.2 episodes/yr
Vaginal oestrogen	Restores lactobacilli, lowers pH	Strongest non-antibiotic evidence; ~90% reduction (Raz–Stamm). **Oral oestrogen is ineffective.**
Cranberry ≥36 mg PAC (twice daily)	Type A PAC blocks **P-fimbrial** adhesion	Cochrane 2023 favourable; safe with warfarin
Methenamine hippurate 1 g twice daily	**Formaldehyde in acid urine (pH <6)**	Non-inferior to daily prophylaxis; fails in NGB. Contraindicated if CrCl <50.
D-mannose	Inhibits **FimH**	Cochrane 2022 inconclusive
Immunostimulants (OM-89, Urovac, Uromune MV140)	Bacterial lysates	Some benefit; small biased trials
Probiotics, oral oestrogen, vitamin C	—	Not effective

Antibiotic strategy	Evidence	Position
Continuous low dose 6–12 months (nitrofurantoin, TMP, TMP-SMX, cephalexin)	Works only while taken; MDR	AUA "may prescribe"; EAU: last resort
Postcoital single dose	Effective; preferred for coital UTI	Acceptable
Self-start	Old data; overuse	Prefer analgesic, sample and observe, then culture-directed treatment

KEY POINT

Recurrent UTI in women is reinfection in a receptive host, not a hidden focus. Investigate only for features of persistence or complication; prevention targets flora, pH, adhesion and washout.

8 · Kidney infections

Acute pyelonephritis

- **White cell casts** prove a renal origin. Take blood cultures if toxic, admitted or pregnant; the usual cause is P-fimbriated E. coli.

Setting (IDSA)	Regimen	Days
Outpatient, no vomiting	Ciprofloxacin 500 mg twice daily if community quinolone resistance ≤**10%** ; levofloxacin 750 mg daily; TMP-SMX double-strength twice daily only if known sensitive. With or without one IV dose of ceftriaxone 1 g or an aminoglycoside.	Cipro 7; levo 5; TMP-SMX 14
Inpatient, severe/possible sepsis	Ampicillin and gentamicin; levofloxacin; ceftriaxone 1 g daily; carbapenem (ESBL risk). Gram-positive cocci: ampicillin–sulbactam with or without an aminoglycoside.	10–14 total
Pregnancy	Ampicillin and gentamicin, aztreonam or ceftriaxone, then oral cephalexin. **No quinolone.**	10–14

- **Obstruction** needs urgent decompression by **stent or percutaneous nephrostomy**.

ALGORITHM

1. If there is no sepsis or vomiting, then send a culture, with or without US, and treat as an outpatient with oral drugs for 7–10 d.
2. If there is sepsis, then take blood and urine cultures, image, and admit for IV treatment.
3. If improved by **72 h**, then step down to oral treatment.
4. If not improved, then repeat cultures and get a **CT**, and drain any obstruction or abscess.

KEY POINT

Urine sterilises within hours but fever may lag for days. Fever beyond 72 h means obstruction or an undrained collection until CT proves otherwise. Antibiotics treat the infection; drainage treats the complication.

Acute focal bacterial nephritis (AFBN)

- AFBN is a lobar infiltrate between pyelonephritis and abscess. On contrast CT it is wedge-shaped with no wall or liquefaction, whereas an abscess is round with a liquid centre and a **ring sign**. Give IV antibiotics for ≥7 days, then 7 days oral.

Emphysematous pyelonephritis (EPN)

SAQ Emphysematous pyelonephritis: prognostic factors and conservative management

Paper answer · P1 · P3 · merged from 2

Asked as: P1 SAQ "EPN. Prognostic factors & conserv. mgt." · P3 SAQ "Prognostic indicators in emphysematous pyelonephritis."

Assumed meaning: EPN = emphysematous pyelonephritis.

Definition and pathogenesis

- EPN is an acute **necrotising** infection of the renal parenchyma and perirenal tissue by **gas-forming organisms**. It is a urological emergency.
- Organisms: *E. coli* (commonest), *Klebsiella*, *Proteus*.
- Most patients are **diabetic**; obstruction in about a quarter.
- **Mechanism (proposed):** high tissue glucose fermented to CO_2 in poorly perfused tissue.
- Reported mortality 19–43%.
- **CT (computed tomography)** is the imaging of choice.

Imaging-based prognostic classifications

1. Huang–Tseng CT classification (2000)

Class	Gas location
1	Collecting system only (emphysematous pyelitis)
2	Renal parenchyma only, no extrarenal extension
3A	Extension into the **perinephric** space
3B	Extension into the **pararenal** space
4	**Bilateral** EPN or EPN in a **solitary kidney**

2. Wan classification (CT pattern)

Type	CT	Mortality
I	Streaky or mottled gas, **no fluid**; rapid parenchymal destruction	50–60%
II	Renal or perirenal **fluid**, bubbly or loculated gas, or gas in collecting system	<20%

- **Why:** fluid means a walled-off (contained) collection that can be drained. Streaky gas without fluid means necrosis faster than the host can contain it.

Clinical and laboratory prognostic factors

- **Huang–Tseng risk factors: thrombocytopenia**, **AKI (acute kidney injury)/raised creatinine**, **altered sensorium**, **shock**.
- **Other adverse factors:** hypoalbuminaemia, bacteraemia, need for haemodialysis, polymicrobial infection.

Conservative management (now standard)

1. Resuscitation: fluids, inotropes, ICU (intensive care unit) if shocked; **tight glucose control**; correct electrolytes.

2. Antibiotics: broad-spectrum IV (intravenous), adjusted to culture; usually several weeks in total.

3. Drainage

- **Relieve obstruction:** DJ (double-J) stent or PCN (percutaneous nephrostomy).
- **PCD (percutaneous catheter drainage)** of gas and fluid collections under CT or ultrasound; multiple drains if needed; repeat CT to check.

How prognosis guides management (algorithm)

```
All: resuscitation + glucose control + IV antibiotics + relieve obstruction (stent/PCN) Class 1–2 -> antibiotics ± PCD (P3: antibiotics + PCD) Class 3–4, <2 risk factors -> antibiotics + PCD (usually succeeds) Class 3–4, >=2 risk factors, or PCD failure -> nephrectomy (emergency or delayed)
```

- Routine emergency nephrectomy carried 40–50% mortality, so drainage-first care replaced it.
- Delayed nephrectomy for a kidney that is non-functioning on renal scan.
- **Emphysematous pyelitis** (class 1, gas in the collecting system only) is often non-diabetic and often responds to antibiotics alone.

Diagram to draw: CT sketches of Huang–Tseng classes 1, 2, 3A/3B and 4.

> **Key points to remember:**
>
> - Huang–Tseng: 1 pelvis, 2 parenchyma, 3A perinephric, 3B pararenal, 4 bilateral or solitary.
> - Four risk factors: thrombocytopenia, AKI, altered sensorium, shock.
> - Wan type I (no fluid) 50–60% mortality; type II <20%.
> - Drainage first; nephrectomy for extensive disease with ≥2 risk factors or drainage failure.

Also, from the condensed book (not in the paper answer):

- EPN is mostly seen in **diabetic women**; non-diabetics are usually obstructed, with less extensive disease.

CT pattern	Prognosis
Streaky/mottled gas, **no fluid**	50–60% mortality
Fluid with bubbly/loculated gas, or gas in collecting system	<20% mortality

Huang–Tseng class †	Extent of gas	Management †
1	Collecting system only (emphysematous pyelitis)	Antibiotics and percutaneous catheter drainage (PCD) or stent
2	Parenchyma, no extrarenal extension	Antibiotics and PCD
3A	Perinephric space	PCD if <2 risk factors; nephrectomy if ≥2 or PCD fails
3B	Pararenal space	
4	Bilateral or solitary kidney	PCD (nephron-sparing)

† The risk factors are thrombocytopenia, acute kidney injury (AKI), altered consciousness and shock.

> **KEY POINT**
>
> Fluid is a good sign and streaky gas a bad one: a contained collection can be drained; diffuse gas without fluid means parenchyma is necrosing faster than the host can wall it off.

Renal and perinephric abscess

SAQ Renal and perinephric abscess

Paper answer · P1

Asked as:

- Perinephric abscess — *Summer 2017 Paper I Q3c (SAQ)*
- Treatment of renal abscess — *Winter 2017 Paper I Q3b (SAQ)*

Type: SAQ (10 marks)

Definition and types

- **Renal (cortical/corticomedullary) abscess:** pus confined to the renal parenchyma.
- **Cortical (renal carbuncle):** haematogenous *Staphylococcus aureus* from a skin source; formerly about 80% of cases.
- **Corticomedullary (now commonest):** ascending Gram-negative infection (*E. coli*, *Klebsiella*, *Proteus*) in a damaged or obstructed kidney (stones, reflux, diabetes).
- **Perinephric abscess:** pus **outside the renal capsule but within Gerota fascia**. Rupture through Gerota fascia gives a **paranephric** abscess (paranephric beyond it).

Aetiology (perinephric abscess)

- Rupture of a renal cortical abscess or pyonephrosis (commonest); extravasation of infected urine from obstruction (stone).
- **Haematogenous** spread in about a third (skin source, *S. aureus*).

- Infected perirenal haematoma or urinoma (trauma, surgery).
- Spread from bowel (Crohn disease, perforation), pancreas, spine.
- Risk factors: **diabetes (about a third)**, stones, obstruction, polycystic kidneys on dialysis.
- Organisms: *E. coli*, *Proteus*, *S. aureus*; anaerobes with bowel sources.

Clinical features

- Fever, flank pain, leucocytosis.
- Perinephric abscess is **insidious:** symptoms >5 days in most; more than a third afebrile.
- Flank pain, tenderness, flank mass (about half); psoas sign (flexed, externally rotated hip).
- **Suspect it when "pyelonephritis" persists:** pyelonephritis defervesces within 4–5 days; perinephric abscess does not.

	Pyelonephritis	**Perinephric abscess**
Symptoms before admission	≤5 days	>5 days
Fever on antibiotics	Never >4 days	≥5 days (median 7)

Investigations / diagnosis (needed to plan treatment)

- Leucocytosis, raised creatinine, pyuria; urine culture identifies the organism in only about a third; urine may be sterile (haematogenous or non-communicating abscess). Blood cultures positive in a quarter to a third.
- **CT (computed tomography): investigation of choice.** Renal abscess: round low-attenuation lesion with enhancing "ring sign". It shows the collection, its spread (psoas, flank), the cause (stone, obstruction) and plans drainage.
- Ultrasound: quick; guides drainage. Plain X-ray: loss of psoas shadow, scoliosis.
- AFBN (acute focal bacterial nephritis) is wedge-shaped without a wall and is treated medically.

Treatment

1. Antibiotics immediately (after blood and urine cultures)

Presumed source	**Empirical therapy**
Haematogenous (Gram-positive)	Penicillinase-resistant penicillin; vancomycin if allergic or MRSA (methicillin-resistant *S. aureus*)
Ascending (Gram-negative)	IV (intravenous) third-generation cephalosporin, antipseudomonal penicillin or aminoglycoside
Perinephric abscess	Aminoglycoside plus antistaphylococcal penicillin; cephalosporin or vancomycin if allergic. Adjust to culture
- Switch to oral once afebrile; continue until clinical and imaging resolution.	

2. Size-based drainage (renal abscess)

Size	**Management**
≤3 cm	Antibiotics alone (stable patient)
3–5 cm	Antibiotics alone or drainage; no consistent pathway
>5 cm	**Percutaneous drainage** (CT/ultrasound-guided) first line; may need several drains

Size	Management
Any size: immunocompromised or not responding	Percutaneous drainage

- **Perinephric abscess (if present) — lower threshold:** small (<3 cm) in an immunocompetent patient may resolve with antibiotics; drain if >3 cm. Larger or non-responding: **CT- or ultrasound-guided percutaneous drainage** (first line).
- Aspiration also distinguishes abscess from a cystic tumour; culture the pus.
- Serial ultrasound or CT until resolution. Worsening suggests perinephric spread, wrong diagnosis or resistant organism.

3. Treat the cause

- Relieve obstruction (stent or nephrostomy); treat stones once infection settles; control diabetes.

4. Surgery

- Open drainage for loculated or failed (percutaneous drainage) cases; **nephrectomy** for a non-functioning, destroyed or pyonephrotic kidney, at the same time or staged.

Outcome

- Perinephric abscess: mortality fell from 40–50% to about 12% with CT; delay in diagnosis causes most deaths.

Diagram to draw: coronal kidney showing renal, perinephric and paranephric spaces / abscess sites.

> **Key points to remember:**
> - Now mostly ascending Gram-negative abscess in a damaged kidney.
> - Perinephric: within Gerota fascia; diabetes in a third.
> - Fever beyond 4–5 days of antibiotics: get a CT.
> - CT is the investigation of choice; follow until resolution.
> - ≤3 cm antibiotics; 3–5 cm either; >5 cm percutaneous drainage.
> - Percutaneous drainage first line; nephrectomy if non-functioning.

Also, from the condensed book (not in the paper answer):

- A renal abscess ≤**3 cm** is treated with antibiotics; one >**5 cm** needs percutaneous drainage, as does any size in an immunocompromised or non-responding patient.
- A perinephric abscess has a lower drainage threshold (>3 cm); only a third are diagnosed at presentation, and delay causes nearly all deaths.

Infected hydronephrosis and pyonephrosis

- **Pyonephrosis** is infected hydronephrosis with suppurative destruction. Patients are very ill, **bacteriuria may be absent** if the ureter is blocked, and US shows a fluid–debris level.
- Give antibiotics with **immediate drainage** (stent or nephrostomy); nephrectomy if non-functioning.

SAQ Xanthogranulomatous pyelonephritis.

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 4 Paper 1, Q3a

Definition

- XGP (xanthogranulomatous pyelonephritis) is a rare, severe, **chronic** infection causing diffuse destruction of an enlarged, non-functioning kidney, with **lipid-laden foamy macrophages** (xanthoma cells).
- Up to a fifth of nephrectomies for pyelonephritis. Peak age 50–70 years; women; diabetes is a risk factor.

Pathogenesis

- **Triad: stone, obstruction, infection.** Stones in up to 80% (half staghorn).
- Low-virulence organisms (**Proteus** commonest, *E. coli*) in an obstructed, ischaemic kidney with an inadequate acute inflammatory response.

Pathology

- Diffuse (>80%) or segmental (focal).
- Gross: yellow nodules, pus-filled dilated calyces, a **contracted pelvis** encased by fibrosis, frequent **perinephric extension**.
- Microscopy: sheets of foamy histiocytes with lymphocytes, plasma cells and giant cells.
- **Malek staging:** I confined to kidney; II into perinephric fat; III into paranephric tissue or retroperitoneum.

Clinical features

- Flank pain, fever (about two-thirds), flank mass (about 60%), weight loss; persistent bacteriuria.
- Labs: anaemia, raised inflammatory markers, pyuria; liver dysfunction in up to half. One-third have **sterile urine** (prior antibiotics).

Imaging

- **CT (computed tomography), most useful:** enlarged kidney, little or no function, central calculus in a contracted pelvis; low-density cavities with rim enhancement (**"bear paw" sign**); extrarenal spread.
- Ultrasound: enlarged kidney with hypoechoic areas. DMSA (dimercaptosuccinic acid) scan quantifies lost function.
- **Differentials:** RCC (renal cell carcinoma), pyonephrosis, malakoplakia, lymphoma, tuberculosis.

Management

- Antibiotics to stabilise; drain if septic.
- **Nephrectomy** (with perinephric fat) for diffuse disease; partial nephrectomy for focal disease.
- Surgery is difficult: dense adhesions to psoas, great vessels, bowel and diaphragm. Remove the whole inflammatory mass, because simple drainage leads to **nephrocutaneous**

fistula.

- Laparoscopic nephrectomy is feasible; conversion rates are high.
- Frozen section can mimic clear cell carcinoma, and XGP can **coexist with RCC** or urothelial carcinoma. If cancer is not excluded, do nephrectomy.

Diagram to draw: CT sketch of an enlarged kidney with central staghorn and "bear paw" cavities.

> **Key points to remember:**
> - Stone, obstruction, Proteus infection → foamy macrophages.
> - CT: non-functioning enlarged kidney, contracted pelvis, "bear paw" sign.
> - Mimics RCC; treat by complete nephrectomy, not drainage.

SAQ Malakoplakia

Paper answer · P2

Type: SAQ (10 marks)
Source: Practice Set 6 Paper 2, Q3e

Definition

- "Soft plaque": a rare chronic granulomatous inflammation caused by **defective macrophage killing of bacteria**, mostly *E. coli*. Described by Michaelis and Gutmann (1902) and von Hansemann (1903).
- Commonest in the **bladder**; also ureter, kidney, prostate, testis, GI (gastrointestinal) tract, skin, lung.

Pathogenesis

- Macrophages ingest bacteria but cannot digest them (lysosomal defect, low intracellular cGMP [cyclic guanosine monophosphate]).
- Bacterial remnants become mineralised with calcium phosphate, forming laminated **Michaelis–Gutmann bodies**.
- Hosts are often immunocompromised, diabetic, chronically ill or **renal transplant recipients**.

Pathology

- Gross: soft yellow-brown plaques or nodules; renal tumour-like masses.
- Micro: sheets of large foamy histiocytes (**von Hansemann cells**) with **Michaelis–Gutmann bodies** (pathognomonic; "owl's eye" targetoid; stain with von Kossa, PAS [periodic acid–Schiff] and Prussian blue).
- MG bodies may be absent early; **α1-antitrypsin** immunostaining helps.

Clinical features

- Mostly >50 years; urinary tract **female:male 4:1**.
- Bladder: irritative LUTS (lower urinary tract symptoms), haematuria, recurrent *E. coli* UTI (urinary tract infection); plaques or masses mimicking cancer; distal ureteric stricture.
- Kidney: masses, fever, flank pain; renal vein or IVC thrombosis. Testis: epididymo-orchitis. Prostate: mimics carcinoma.

Investigations

- Urine culture (*E. coli*). Cystoscopy and **biopsy is diagnostic**.
- Imaging: enlarged kidneys with filling defects, **no stones, calcification or hydronephrosis** (unlike XGP).

	XGP (xanthogranulomatous pyelonephritis)	**Malakoplakia**
Cell	Lipid-laden foamy macrophages	Von Hansemann cells with MG bodies
Stone/obstruction	Usually (staghorn)	Absent
Laterality	Unilateral	Uni- or multifocal, bilateral
Treatment	Nephrectomy	Antibiotics; nephrectomy if unilateral, symptomatic

Management

- **Long-term antibiotics active inside macrophages:** fluoroquinolones (concentrate in macrophages), trimethoprim–sulfamethoxazole, rifampicin, doxycycline.
- **Bethanechol + ascorbic acid** raise intracellular cGMP and correct the degradative defect.
- Reduce immunosuppression where possible.
- Surgery if progression despite medical therapy: TUR (transurethral resection) of bladder lesions; **nephrectomy** for symptomatic unilateral renal disease.

Prognosis

- Bilateral or transplant-kidney disease: death often within 6 months; overall mortality can exceed 50%. Unilateral: long survival after nephrectomy.

Diagram to draw: histiocyte with Michaelis–Gutmann bodies.

> **Key points to remember:**
> - Defective macrophage digestion of *E. coli*.
> - Michaelis–Gutmann bodies are pathognomonic.
> - Fluoroquinolones + bethanechol/vitamin C; nephrectomy for unilateral renal disease.

Also, from the condensed book (not in the paper answer):

- In xanthogranulomatous pyelonephritis (XGP), urine is sterile in about a third but tissue culture is positive.

	XGP	**Malakoplakia**
Cell	Foamy lipid-laden macrophages	Von Hansemann cells, Michaelis–Gutmann bodies
Stones/obstruction	Usually	Absent
Organism	Proteus, E. coli	E. coli
Laterality	Unilateral	Uni-/multifocal, may be bilateral
Treatment	Nephrectomy	Intracellular antibiotics with or without nephrectomy

KEY POINT

Across renal infections the axis is diffuse inflammation versus contained collection. Pyelonephritis and AFBN need antibiotics; pyonephrosis and large or obstructed abscesses need drainage; when the kidney itself is the infected tissue (XGP, diffuse EPN), the answer is nephrectomy.

9 · Fournier gangrene and periurethral abscess

Skin hides the true spread; only early radical debridement saves lives.

- **Fournier gangrene** is necrotising fasciitis of the genitalia, perineum or perianal region. A source is found in ~**95%**, and the male:female ratio is ~10:1.
- Sources are **colorectal**, **urogenital** (stricture, instrumentation, catheter) and **dermal**. Risk factors are **DM**, obesity, alcoholism and immunosuppression.
- Synergistic organisms cause obliterative endarteritis, spreading along Buck's, dartos, Colles' and Scarpa's fascia.
- Pain out of proportion comes first, then crepitus and gangrene. **The skin underestimates fascial extent**, and CT must not delay surgery.

Score	Components	Use
Laboratory Risk Indicator for Necrotising Fasciitis (LRINEC)	CRP, WBC, haemoglobin, sodium, creatinine, glucose	≥**6** suspicious; ≥**8** strongly predictive
Fournier's Gangrene Severity Index (FGSI) †	Nine vital signs and blood values, each scored 0–4	>9 predicts mortality

ALGORITHM

1. On suspicion, resuscitate and start broad IV antibiotics (piperacillin–tazobactam; or a carbapenem or vancomycin with clindamycin or metronidazole).
2. Perform **immediate radical debridement** back to bleeding fascia and leave the wound open.
3. Take the patient back for a **re-look at 24–48 h**.
4. If the source is urethral, then add suprapubic diversion; if rectal, then a colostomy.
5. Consider hyperbaric oxygen as an adjunct, then reconstruct.

- **Orchidectomy is almost never needed**, because the testicular blood supply is independent of the scrotal skin. Mortality is ~20–40%.
- A **periurethral abscess** needs drainage, suprapubic diversion and antibiotics.

10 · Urosepsis †

Term (Sepsis-3)	Definition
Sepsis	Organ dysfunction from a dysregulated response to infection: rise in Sequential Organ Failure Assessment (SOFA) score ≥**2**
Quick SOFA (qSOFA)	≥2 of: respiratory rate ≥22/min, altered mentation, systolic BP ≤100 mmHg
Septic shock	Vasopressors for mean arterial pressure (MAP) ≥65 mmHg **and** lactate >2 mmol/L despite fluids

ALGORITHM

1. Hour-1 bundle: measure lactate and take blood and urine cultures.
2. Give a broad IV antibiotic within 1 h.
3. If hypotensive or lactate ≥4, then give 30 mL/kg crystalloid.
4. If still hypotensive, then start noradrenaline for MAP ≥65.
5. Achieve **source control**: decompress with a stent or nephrostomy; no definitive stone surgery now.
6. De-escalate antibiotics by culture.

† Not in the source notes. Manipulating infected stones releases endotoxin, so drain first and operate later.

11 · Special populations

Pregnancy

*See the paper answer **Bacteriuria in pregnancy and antibiotic policy** (Section 1, Ch 21).*

Also, from the condensed book (not in the paper answer):

- GFR rises by 30–50%, so creatinine falls to about **0.5 mg/dL**, and a "normal" 0.8–0.9 may mean impairment.
- Severe or MDR pyelonephritis needs a carbapenem.

Elderly

- **Do not screen for or treat ASB**; bacteriuria with confusion is not UTI. The **absence** of pyuria **excludes** bacteriuria.
- The **revised McGeer criteria** require **localising** signs and ≥10^5 CFU/mL of ≤2 species in non-catheterised residents; catheterised residents may qualify on **systemic** signs.
- Treat for 3–6 days if uncomplicated and 10–14 if febrile.

Catheter-associated UTI

SAQ Catheter Associated Urinary Tract Infections (CAUTI)

Paper answer · P1

Type: SAQ (10 marks)
Source: 2017-Winter Paper 1, Q3c

Definition

- **CAUTI (catheter-associated urinary tract infection):** symptomatic UTI (urinary tract infection) with a catheter in place >2 days (or removed within 48 hours). Distinguish from **asymptomatic bacteriuria**, which needs no treatment.

Epidemiology and risk factors

- Commonest hospital-acquired infection (about 75–80% of hospital UTIs); commonest cause of secondary bacteraemia.
- Bacteriuria develops at about **5–10% per catheter-day**; 100% with long-term catheters. Intermittent catheterisation: 1–3% per catheterisation.

- **Risk factors:** duration (most important), female sex, older age, diabetes, breaks in the closed system, no systemic antibiotics.

Pathogenesis

- Entry at insertion, then **extraluminal** (along the outer surface; commoner in women) or **intraluminal** (from the drainage bag).
- **Biofilm:** bacteria in a matrix on the catheter surface, formed within minutes; it shields them from antibiotics. Urease-producers (*Proteus*) cause encrustation.
- Organisms: *E. coli*, *Pseudomonas*, *Proteus*, *Enterococcus*; long-term catheters polymicrobial.

Diagnosis

- Fever, rigors, flank or suprapubic pain, or delirium with no other source. Pyuria does not distinguish infection.
- Culture from a **freshly replaced catheter**, not the bag. **IDSA (Infectious Diseases Society of America): ≥10^3 CFU/mL** (colony-forming units) with symptoms; surveillance (CDC (Centers for Disease Control and Prevention)) definition uses ≥10^5.

Prevention (bundle)

- **Avoid unnecessary catheters;** review daily, remove early. Aseptic insertion.
- **Closed, dependent drainage**; bag below bladder; no routine irrigation.
- Meatal cleaning with water; no prophylactic antibiotics (delay bacteriuria only 3–4 days, select resistance).
- Alternatives: **clean intermittent catheterisation**, condom or suprapubic catheter.
- Silver-alloy catheters reduce bacteriuria in short-term use; no proven fall in symptomatic CAUTI.
- Do not treat asymptomatic bacteriuria (except pregnancy, before urological surgery).

Treatment

- **Replace or remove the catheter** before taking the culture and starting antibiotics, if in place >2 weeks (biofilm source).
- Empirical, then culture-directed antibiotics: **7 days** if prompt response; **10–14 days** if delayed.

Diagram to draw: catheter in bladder showing extraluminal and intraluminal routes and biofilm.

> **Key points to remember:**
> - Bacteriuria 5–10% per catheter-day; duration is the key risk.
> - Diagnosis: symptoms + ≥10^3 CFU/mL from a fresh catheter.
> - Remove catheter early; closed drainage; treat only symptomatic CAUTI, 7 or 10–14 days.

Also, from the condensed book (not in the paper answer):

- Catheters in place >30 days carry 4–5 species, and silicone is no better than latex.

Spinal cord injury

- All UTIs in SCI are complicated, and UTI is the commonest cause of fever. Symptoms are atypical: fever, spasticity, leakage, autonomic dysreflexia † and cloudy urine.

- **Clean intermittent catheterisation (CIC)** has the fewest long-term complications and an indwelling urethral catheter the most.

Drainage (NIDRR)	Significant bacteriuria
Indwelling urethral or suprapubic catheter	Any detectable count
Intermittent catheterisation	≥10^2 CFU/mL
Catheter-free men (clean voided)	≥10^4 CFU/mL

- Do not treat ASB. Treat afebrile UTI for 5–7 days; febrile UTI needs admission and **14 days** parenteral-first treatment. Change the catheter.
- Prophylaxis does not work in NGB; for recurrent UTI, get imaging and urodynamics (UDS) and lower storage pressure.
- A catheter for >**10 years** raises the risk of bladder cancer, so these patients need surveillance.

Diabetes, renal transplant

- In DM, do not treat ASB, but keep a low threshold to image pyelonephritis.
- After transplant, do not screen for ASB beyond 1–2 months. Treat uncomplicated UTI for 5–10 days and pyelonephritis for 14–21 days. Stents raise UTI by about half, so remove them within **4 weeks**.

12 · Fungal UTI

SAQ Fungal UTI

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 18 SAQ Sheet, Q3a

Introduction and organisms

- Fungal UTI (urinary tract infection) is rising with antibiotics, catheters and immunosuppression. **Candida** is commonest: *C. albicans*, then *C. glabrata* (often fluconazole-resistant); *C. krusei* is intrinsically resistant.
- Others in AIDS (acquired immunodeficiency syndrome) or transplant patients: Aspergillus, Cryptococcus, Mucorales, Histoplasma.

Risk factors and routes

- Antibiotics, indwelling catheters and stents, diabetes, steroids, neutropenia, prematurity, parenteral nutrition, obstruction.
- **Ascending** (catheter) causes most candiduria. **Haematogenous** spread from candidaemia: in disseminated candidiasis the kidney is the organ most often involved.

Clinical spectrum

- Asymptomatic candiduria (most) → cystitis → pyelonephritis → **fungus ball (bezoar)** causing obstruction → renal abscess or candidaemia.

Diagnosis

- Microscopy: budding yeasts and pseudohyphae (may mimic red cells). Repeat culture: contamination is common.

- Children: treat when repeated cultures exceed **10,000–15,000 CFU (colony-forming units)/mL**.
- Persistent funguria: ultrasound or CT (computed tomography) for bezoar; blood cultures if high risk.

Management

Situation	Treatment
Asymptomatic	Remove or change catheter, stop antibiotics; no antifungal
Treat asymptomatic only if	Neutropenia, very-low-birth-weight infant, before a urological procedure
Cystitis	Fluconazole 200 mg (3 mg/kg) daily for 2 weeks
Pyelonephritis	Fluconazole 200–400 mg (3–6 mg/kg) daily for 2 weeks
Fluconazole-resistant (*C. glabrata*, *C. krusei*)	Amphotericin B deoxycholate 0.3–0.6 mg/kg/day for 1–7 days ± flucytosine
Resistant cystitis	Amphotericin B bladder irrigation 50 mg/L
Fungus ball	PCN (percutaneous nephrostomy) drainage + systemic antifungal + local amphotericin irrigation; endoscopic or open removal if persistent

- Fluconazole concentrates in urine; **echinocandins and lipid amphotericin do not.**
- **Before procedures:** IDSA (Infectious Diseases Society of America) 2016 treats all candiduria (fluconazole 400 mg daily for several days before and after). AUA (American Urological Association) Best Practice Statement 2019 gives single-dose antifungal prophylaxis only for urothelium-breaching surgery. The two disagree.
- **Also know:** alkalinisation for bezoar; fluconazole caution under 6 months.

Diagram to draw: budding yeast with pseudohyphae; fungus ball in renal pelvis with nephrostomy.

> **Key points to remember:**
> - Candida commonest; *C. glabrata* often fluconazole-resistant.
> - Asymptomatic candiduria: remove catheter, no antifungal (except neutropenia, neonate, procedure).
> - Fluconazole 2 weeks; echinocandins fail in urine.
> - Fungus ball: drain, antifungal, remove.

Also, from the condensed book (not in the paper answer):

- AUA BPS 2019 advises antifungal prophylaxis for high-pressure endoscopy (TURBT, TURP, therapeutic ureteroscopy) and open entry into the tract.

13 · Perioperative antimicrobial prophylaxis

One well-timed dose suffices; more only adds resistance.

Principle (AUA BPS 2019)	Rule
Indication	Any procedure breaching a tissue barrier, weighing host defence and consequences
Timing	Within **60 min** of incision; vancomycin and quinolones within 120 min

Principle (AUA BPS 2019)	Rule
Duration	**Single dose**; ≤**24 h** only for PCNL, prostheses, special cases
Agent	Local flora per antibiogram
Known infection	Sterilise urine before elective surgery (this is treatment); infected urine causes bacteraemia in ~half

- Host risk factors include age, smoking, steroids, immunodeficiency, hardware and infected stones.

Wound class	Examples	Antimicrobial prophylaxis (AP)
I clean	Scrotal/inguinal, retroperitoneal lymph node dissection (RPLND), nephrectomy without entering tract	Not needed (except groin/perineal incision, prosthesis)
II clean-contaminated	Any endoscopy; opening the urinary tract	Single dose
III contaminated	Infected stone PCNL, **transrectal biopsy**, bowel segments	Strongly recommended; anaerobic cover
IV dirty	Abscess, open trauma	Treatment, tailor to cultures

Procedure	AP?	Agent	Duration
Diagnostic cystoscopy / UDS (no risk factors)	No (EAU strongly against)	—	—
Cystoscopy with biopsy/fulguration	Host factors	TMP-SMX, amoxicillin–clavulanate, cephalosporin	Single
TURP / TURBT, laser prostate	All (TURBT: EAU only if high risk)	Cefazolin, TMP-SMX	Single (TURP short course with catheter acceptable)
Transrectal prostate biopsy	All	Fluoroquinolone or cephalosporin, with or without aminoglycoside; targeted by rectal swab	Single (≥1 day)
Ureteroscopy	All (diagnostic: uncertain)	TMP-SMX, 1st/2nd-gen cephalosporin	Single
PCNL	All	1st/2nd-gen cephalosporin; aminoglycoside and metronidazole	≤24 h
Shock-wave lithotripsy (SWL)	Only with risk factors (EAU: no)	TMP-SMX, cephalosporin	Single
Open/robotic entering tract	All	Cefazolin	Single
Colon segment	All	Cefazolin and metronidazole / cefoxitin / ertapenem, plus oral neomycin with erythromycin or metronidazole	Single

Procedure	AP?	Agent	Duration
Artificial urinary sphincter (AUS), penile prosthesis, sacral neuromodulation	All	Aminoglycoside and cephalosporin or vancomycin	≤24 h
Vaginal surgery incl. slings	All	Cefazolin/cefoxitin	Single
Penile (circumcision), clean scrotal	Likely none (scrotal: high-risk only)	Cefazolin	Single

- **Transrectal ultrasound (TRUS) biopsy** causes sepsis in 0.3–3.1%, mainly from quinolone-resistant E. coli. Use augmented or **targeted** prophylaxis; the AUA advises alternatives if local resistance is >20%.
- **UDS (SUFU 2017)**: give AP for neurogenic lower urinary tract dysfunction, raised PVR, ASB, immunosuppression, age >70, catheter or recent joint implant.
- **Catheter or stent removal**: give a single dose only if risk factors are present.
- **Endocarditis** (AHA): no AP solely to prevent it; with a high-risk valve and colonised urine, sterilise the urine first.
- **Orthopaedic hardware**: only a joint replaced <2 years ago or a high-risk host needs AP for high-bacteraemia procedures.

> **KEY POINT**
>
> Prophylaxis is justified by wound contamination × host vulnerability. Where neither is raised (diagnostic cystoscopy, sterile-urine SWL, index UDS), ecological cost outweighs benefit; otherwise give one well-timed, flora-matched dose. Anything longer is treatment.

14 · Landmark trials

Trial	Design	Result	Take-home
Raz & Stamm 1993	RCT, intravaginal oestriol vs placebo, postmenopausal recurrent UTI	0.5 vs 5.9 UTIs/patient-year; lactobacilli returned, pH fell	Vaginal oestrogen first-line after menopause
Hooton 2018	RCT, +1.5 L water/day vs none, premenopausal	Cystitis 1.7 vs 3.2/year	Hydration halves recurrences
Cai (ASB in recurrent UTI)	RCT, treat vs observe ASB in young women	Treatment increased recurrence and resistant E. coli	Do not treat ASB; E. faecalis may protect
ALTAR (methenamine)	Methenamine vs daily antibiotic prophylaxis, 12 months	Non-inferior	Non-antibiotic alternative
Drekonja 2021	RCT, afebrile male UTI, 7 vs 14 days ciprofloxacin/TMP-SMX	7 days non-inferior	7 days for afebrile men

27 Inflammatory and Pain Conditions of the Lower Genitourinary Tract: Prostatitis, Orchitis, Epididymitis and Scrotal Pain

1 · Prostatitis: definitions and classification

Only categories I and II are infections; most men have category III.

- The Drach (1978) scheme, built on Meares–Stamey cultures, was replaced by the MCQ National Institutes of Health (**NIH, 1999**) classification.

NIH category	Entity	Defining feature
I	Acute bacterial prostatitis (ABP)	Acute prostatic urinary tract infection (UTI), often febrile
II	Chronic bacterial prostatitis (CBP)	Recurrent UTI, **same organism**; well between episodes
IIIA	Chronic prostatitis/chronic pelvic pain syndrome (CP/CPPS), inflammatory	Leucocytes in expressed prostatic secretion (EPS), post-massage urine (VB3) or semen
IIIB	CP/CPPS, non-inflammatory	No leucocytes
IV	Asymptomatic inflammatory	Incidental on biopsy or semen analysis

- I and II are **5–10%**; III is **90–95%**. **IIIA vs IIIB has no clinical significance**, and the prostate may not be the pain source.
- **Category III (NIH CPCRN)** needs pelvic pain for **≥3 of the previous 6 months**, excluding uropathogens, urogenital cancer, prior radiation, stricture and neurological bladder disease.
- Histological inflammation is near-universal (**>70%** at autopsy), so it cannot define disease; **Nickel grading** records its location, extent and grade.
- **Granulomatous prostatitis** classically follows **intravesical bacillus Calmette–Guérin (BCG)**, giving a **transient benign prostate-specific antigen (PSA) rise**.

2 · Category I: acute bacterial prostatitis

- Spread is mainly **ascending urethral**. Risks include catheters, instrumentation or biopsy, obstruction and insertive anal intercourse.
- **E. coli causes 65–80%**; Pseudomonas is commoner after instrumentation. After biopsy, organisms are often fluoroquinolone (FQ)-resistant or extended-spectrum β-lactamase (ESBL) producers.

> **KEY POINT**
>
> **Why young healthy men get ABP:** the organism, not the host — virulent uropathogenic E. coli that forms **biofilm** more often than cystitis strains; biofilm blocks antibiotics, so prostatic infection needs longer, well-penetrating therapy.

- Symptoms are frequency, dysuria, perineal pain, fever and rigors, with possible retention or sepsis. In febrile male UTI, **>90%** show raised PSA or prostate volume.

Evaluation

- First ask whether it was **spontaneous or after manipulation or biopsy**, as this decides empirical therapy.
- The prostate is tender and boggy; examine **gently and never massage**. Send midstream and **blood cultures**, and measure residual urine by **ultrasound (US)**.
- **No PSA** acutely; if raised, follow it to normal (up to **6 months**). Use transrectal ultrasound (TRUS) or CT only for suspected abscess, and **never biopsy**.

Treatment

- **Outpatient care only if** the man is not systemically ill, tolerates oral drugs and is not in retention.
- Systemic illness (**EAU**): amoxicillin or a 2nd-generation cephalosporin with an aminoglycoside, or a 3rd-generation cephalosporin. Ciprofloxacin only if local resistance **<10%** and fully oral outpatient care, or β-lactam anaphylaxis.
- **Febrile prostatitis after biopsy** (AUA White Paper): **no FQ or trimethoprim–sulfamethoxazole (TMP-SMX)**; use a **carbapenem**, amikacin or 2nd/3rd-generation cephalosporin.
- **Duration**: traditionally 4 weeks, but **2 weeks** of ciprofloxacin is equivalent; TB prostatitis needs ≥**6 months**. NSAIDs and α-blockers are useful adjuncts.
- **Retention**: brief urethral catheterisation is acceptable, but use a **suprapubic catheter for long-term** drainage.

Prostatic abscess

- Suspect it with high fever, immunosuppression or **no response after 48 h**. It is hypoechoic on TRUS, and **CT** shows spread.

ALGORITHM

1. If the abscess is <1–2 cm, then treat with antibiotics.
2. If it progresses and lies peripherally, then perform US-guided percutaneous drainage (recurrence 15–33%).
3. If it is large or drainage fails, then perform **transurethral unroofing**, using intraoperative US to locate it.
4. If it extends outside the prostate, then drain it openly (rare).

3 · Category II: chronic bacterial prostatitis

Recurrent same-organism UTI in men needs a long penetrating course.

- The prostate harbours the same organism and the man is **well between episodes**, whereas CP/CPPS pain is continuous.
- Organisms are Gram-negatives and E. faecalis; biofilm explains the poor response. The role of **Chlamydia** is debated.

Localisation test	Specimens	Interpretation
Meares–Stamey 4-glass (previous)	VB1 urethra; VB2 bladder (midstream); EPS; VB3 post-massage	Prostatic if EPS/VB3 counts exceed VB1/VB2
2-glass pre/post-massage (current)	Midstream pre-massage and first 10 mL post-massage	Predicts 4-glass in **>95%**

- There is no agreed cut-off, though some use a **10-fold rise**; **do not culture semen**.
- A residual volume **>180 mL** raises infection risk. Men **<45** with a first UTI have normal imaging, but check for **stricture**.

> **KEY POINT**
>
> **Why fluoroquinolones?** Prostatic entry is passive (lipid solubility, ionisation, protein binding, size). FQs are lipid-soluble **zwitterions** that concentrate in alkaline CBP fluid and cover Gram-negatives, Gram-positives and atypicals. Tetracyclines, macrolides and trimethoprim also penetrate.

Situation (EAU)	Regimen
First line	FQ (ciprofloxacin = levofloxacin) **4–6 weeks**
FQ-resistant, TMP-SMX-sensitive	TMP-SMX **3 months** (cure 30–50% vs 60–80% with FQ)
C. trachomatis	Azithromycin 500 mg daily 3 weeks, or doxycycline 100 mg twice daily 10 days; azithromycin is better than ciprofloxacin
T. vaginalis	Metronidazole 500 mg three times daily 14 days

- A macrolide can be added because it **penetrates biofilm**. Refractory cases need suppression or TURP.

4 · CP/CPPS — aetiology

An insult (infection, stress, trauma) in a susceptible man starts **neurogenic inflammation**, which other factors maintain, so single treatments fail.

Domain	Key evidence
Infection	Bacterial counts are equal in patients and controls; in mice, pain persists after clearance
Inflammation	Biopsy inflammation does not correlate with symptoms; symptoms track **cytokines**
Neurological	**Central sensitisation**
Pelvic floor	Tenderness and **failure to relax**; tender in **81%**
Psychosocial	**Catastrophising and helplessness predict** pain
Endocrine	Blunted adrenocorticotropic hormone (ACTH) response

5 · CP/CPPS — clinical features and evaluation

SAQ Chronic (non-bacterial) prostatitis: diagnosis, classification and management

Paper answer · P1 · P4 · merged from 2

Asked as: P1 SAQ "Chronic Prostatitis : diagnosis, classification and management." · P4 SAQ "Chronic non bacterial prostatitis."

Definition and classification: NIH (National Institutes of Health), 1999

Category	Description
I	Acute bacterial prostatitis

Category	Description
II	**Chronic bacterial prostatitis (CBP)**: recurrent UTI (urinary tract infection) with the **same organism**; well between episodes
IIIA	**CP/CPPS (chronic prostatitis/chronic pelvic pain syndrome)**, **inflammatory**: leucocytes in EPS (expressed prostatic secretion), VB3 (post-massage urine) or semen
IIIB	CP/CPPS, **non-inflammatory**
IV	Asymptomatic inflammatory prostatitis (incidental)

- Chronic non-bacterial prostatitis is now called **CP/CPPS**, NIH category III: pelvic pain for **≥3 of the last 6 months** without uropathogenic infection. It accounts for 90–95% of prostatitis.
- The IIIA vs IIIB split has no treatment value.

Aetiology

- Interplay of past infection, neurogenic inflammation, **pelvic floor muscle dysfunction**, autoimmunity, psychological stress and **central sensitisation**.
- The prostate may not be the pain source.

Clinical features

- Perineal (commonest), testicular, suprapubic or penile pain; ejaculatory pain; LUTS (lower urinary tract symptoms); sexual dysfunction.

Evaluation (diagnosis of exclusion)

- **NIH-CPSI (Chronic Prostatitis Symptom Index):** pain, urinary and quality-of-life domains (score 0–43).
- **DRE (digital rectal examination):** prostate tender in fewer than half; palpate the pelvic floor for **trigger points** (tender in about half).
- **Localisation tests:**
- **Meares–Stamey 4-glass:** VB1 (urethra), VB2 (bladder), EPS, VB3 (prostate).
- **2-glass** (pre- and post-massage urine): matches the 4-glass test in >95%.
- Urinalysis and culture, PVR (post-void residual), flow rate.
- Cytology if irritative symptoms or haematuria.
- Selected patients: cystoscopy (if low flow or bladder pain), urodynamics.
- **UPOINT phenotyping: U**rinary, **P**sychosocial, **O**rgan-specific, **I**nfection, **N**eurological/systemic, **T**enderness of muscles.

Management of category II (chronic bacterial prostatitis)

- Fluoroquinolone (lipid-soluble zwitterion, enters prostate) for **4–6 weeks** (EAU (European Association of Urology)).
- Chlamydia: azithromycin. Resistant organisms: TMP-SMX (trimethoprim-sulfamethoxazole) for 3 months. Refractory: suppression or TURP (transurethral resection of prostate).

Management of category III (CP/CPPS): multimodal, UPOINT-directed

Monotherapy trials fail.

Domain	Therapy
Urinary	**α-blockers** in recent-onset (<1 year), α-blocker-naive men; antimuscarinics

Domain	Therapy
Psychosocial	CBT (cognitive behavioural therapy), stress reduction
Organ (prostate)	**Quercetin**, **pollen extract** (small RCTs, randomised controlled trials); 5-ARI (5α-reductase inhibitor) if BPH (benign prostatic hyperplasia)
Infection	Quinolone or tetracycline for ≥6 weeks only if treatment-naive and <1 year (EAU), or positive VB3; no repeated courses
Neurological	Pregabalin, amitriptyline, duloxetine
Tenderness	**Pelvic floor physiotherapy, myofascial trigger-point release**, biofeedback

- **Others: acupuncture** (beat sham in a large RCT: 61.5% vs 38.3% response), aerobic exercise, PTNS (percutaneous tibial nerve stimulation), shockwave therapy; NSAIDs (non-steroidal anti-inflammatory drugs) only in combination.
- **Avoid (not recommended):** TURP or radical prostatectomy for pain.

Diagram to draw: UPOINT snowflake with a therapy for each domain.

> **Key points to remember:**
> - NIH I–IV; III = pain ≥3 of 6 months, no infection; IIIA inflammatory, IIIB not.
> - CBP: same organism, fluoroquinolone 4–6 weeks.
> - CP/CPPS: NIH-CPSI to score, UPOINT to phenotype and treat.
> - Multimodal therapy; α-blocker plus antibiotic only early.

Also, from the condensed book (not in the paper answer):

- The **NIH Chronic Prostatitis Symptom Index (NIH-CPSI)** † scores 0–43 (pain 0–21, urinary 0–10, quality of life 0–12), and a **≥6 points** fall is meaningful.
- **PSA is not indicated**, and a raised value must never be blamed on CP/CPPS.
- In men with bladder pain, about 42% meet interstitial cystitis criteria see Ch 28, and Hunner lesions respond to fulguration.

UPOINT phenotyping

Monotherapy fails because each man has a different mix of domains. UPOINT maps them for simultaneous treatment, and the domain count tracks severity.

Domain	Identify by	Therapy
Urinary	Lower urinary tract symptoms (LUTS), PVR	α-blocker; †antimuscarinic
Psychosocial	Depression, catastrophising	Psychology, stress reduction
Organ-specific	Prostate tenderness; bladder pain	†Quercetin/pollen; bladder therapy; 5α-reductase inhibitor (5-ARI) in BPH
Infection	Positive VB3 with negative VB2	Antibiotics
Neurologic/systemic	Pain beyond pelvis, irritable bowel syndrome (IBS), fibromyalgia	Tricyclics, pregabalin, acupuncture
Tenderness	Pelvic floor trigger points	Physiotherapy, myofascial release

6 · CP/CPPS — treatment

> **KEY POINT**
>
> **Governing fact:** no monotherapy has a positive definitive trial, because each tested one agent in an unphenotyped population. Start conservative, then give **simultaneous, phenotype-directed multimodal therapy** with a multidisciplinary team.

Modality	Evidence / role
Antibiotics	Quinolone trials negative. EAU: quinolone or tetracycline ≥**6 weeks** only if **treatment-naive and <1 year**. **No repeated courses without a positive culture**.
α-blockers	Trials negative; use in newly diagnosed men with voiding symptoms. Best combination: **antibiotic and α-blocker**.
Anti-inflammatories	Celecoxib helps pain only; steroids negative.
5-ARI	Dutasteride (REDUCE): ≥6-point NIH-CPSI fall in 49% vs 37%; for older men with BPH.
Neuropathic	Pregabalin missed its endpoint; amitriptyline needs ≥50 mg.
Conservative	Diet, **aerobic exercise** and cognitive therapy.
Acupuncture	Strongest conservative evidence: sham-controlled RCT, 61.5% vs 38.3% responders at 32 weeks.
Pelvic floor physiotherapy	Myofascial trigger-point release beats global massage (57% vs 21%); refer if the pelvic floor is tender.
Neuromodulation	Tibial or sacral nerve stimulation if frequency/urgency.
Surgery	No TURP or prostatectomy. **Bladder neck incision** only for proven obstruction.

7 · Acute orchitis

- **Bacterial** orchitis (age >35) is usually Gram-negative. **Mumps** is the commonest viral cause, following parotitis, with **atrophy in about half** but rare infertility. Autoimmune and vasculitic forms are treated with steroids.
- Ask about **pain duration**, because torsion is the key differential; get urine culture, STI tests and **scrotal US**. **Follow-up is mandatory** to avoid missing torsion or tumour.

8 · Acute epididymitis

Age and sexual history predict the organism.

- Epididymal pain and swelling for **<6 weeks**; infection is **ascending**.
- **In sexually active men <35, C. trachomatis and N. gonorrhoeae dominate**; **in men >35 and children, enteric organisms dominate**. Because >40% of men under 35 grow enterics, **screen all sexually active men for STI**.
- Send culture, first-void urine and STI NAAT; **Doppler US** excludes torsion.

Likely pathogen	Regimen
Chlamydia/gonorrhoea	Ceftriaxone **500 mg IM** (intramuscular) single dose (1 g if ≥150 kg) and doxycycline 100 mg twice daily **10 days**
STI and enteric (insertive anal sex)	Ceftriaxone 500 mg IM and levofloxacin 500 mg once daily 10 days

Likely pathogen	Regimen
Enteric only	Levofloxacin 500 mg once daily 10 days

- Give rest, elevation and NSAIDs, and treat partners. With **no improvement at 72 h**, repeat US for abscess or infarction.

9 · Chronic orchitis and epididymitis

- **Granulomatous orchitis** has infective causes (TB, brucellosis, syphilis) and an **idiopathic** form usually found after orchidectomy for suspected tumour. **Malakoplakia** shows **Michaelis–Gutmann bodies**.
- **Chronic epididymitis** (≥**3 months**) has a **Nickel classification**: inflammatory (infective, TB, **amiodarone**, idiopathic), obstructive (congenital, post-infective, post-vasectomy) and epididymalgia (normal epididymis).
- **TB epididymitis** is the commonest genital site, presenting as painless swelling with a **sinus**. Give four drugs for ≥**6 months**, and operate **only on anti-TB drugs**. **A mass not shrinking after 3 weeks needs inguinal exploration**.

10 · Chronic scrotal content pain (CSCP)

A cord block predicts who benefits from denervation surgery.

CSCP is intrascrotal pain for ≥**3 months** that interferes with life; there are no evidence-based guidelines.

- Post-procedural causes are **inguinal hernia repair** (commonest) and vasectomy (~**1 in 1000**, the consent figure). Others include varicocele, intermittent torsion and referred pain from a ureteric stone.
- **Wallerian degeneration** affects the **"trifecta nerve complex"**: cremaster, perivasal sheath and posterior lipomatous tissue. Pelvic floor dysfunction is common.
- **First decide** whether the pain is isolated or part of CP/CPPS, and get a **duplex US**.
- A **spermatic cord block** uses **20 mL 0.25% bupivacaine** at the **pubic tubercle**. **>50% relief** predicts success of microsurgical denervation of the spermatic cord (MDSC).
- Medical options are NSAIDs and **neuropathic agents**, without chronic opioids. Botulinum toxin equalled placebo in an RCT.

ALGORITHM

1. If there is an identifiable lesion, then do directed surgery (hydrocelectomy, spermatocelectomy, orchidopexy, **microsurgical varicocelectomy**).
2. If the pain followed vasectomy, then offer **vasectomy reversal** (82–93%; wait ≥3 months).
3. If there is no lesion and the block is positive, then perform **MDSC**.
4. If the epididymis is focally affected, then perform epididymectomy.
5. Last resort: **inguinal orchidectomy**.

- **MDSC** is subinguinal and microscopic: divide the cord nerves, **preserve arteries and lymphatics** and ligate the internal spermatic veins. It cures 50–70%; consent for **testicular atrophy**. **Targeted trifecta MDSC** is equally effective and quicker.
- **Epididymectomy** succeeds in 50–75%. **Orchidectomy** is least reliable, and **inguinal beats scrotal** (75% vs 55%).

11 · Landmark trials

Trial	Design	Result	Take-home
NIH CPCRN ciprofloxacin/tamsulosin	RCT, 6 weeks, NIH-CPSI ≥15	No benefit vs placebo	Monotherapy fails in unselected CP/CPPS
REDUCE (CP/CPPS analysis)	Dutasteride vs placebo, 4 years	≥6-point fall 49% vs 37%	5-ARI for older men with BPH
MAPP myofascial physiotherapy	Feasibility vs global massage	57% vs 21% responders	Physiotherapy for tender pelvic floor
Acupuncture vs sham	Large RCT	61.5% vs 38.3% at 32 weeks	Evidence-based option
Botulinum toxin cord injection	Double-blind RCT	No better than local anaesthetic	Open-label series overestimate benefit

28 Interstitial Cystitis/Bladder Pain Syndrome and Related Disorders

1 · Definition and nomenclature

LAQ Interstitial cystitis / bladder pain syndrome

Paper answer · P2 · P4 · merged from 2

Asked as: P4 LAQ "Interstitial cystitis / painful bladder syndrome: recent trends in evaluation, diagnosis and management" (Discuss the recent trends in the evaluation. Diagnosis and management of interstitial cystitis, Winter 2024 Paper IV Q1 LAQ; Painful bladder syndrome management, Summer 2021 Paper IV Q3d SAQ) · P2 SAQ "Interstitial cystitis (Bladder pain syndrome)."

Read as: recent trends in the evaluation, diagnosis and management of IC/BPS (interstitial cystitis/bladder pain syndrome).

Definitions: how they have changed

- **NIDDK (National Institute of Diabetes and Digestive and Kidney Diseases, 1987):** research criteria; missed about 60% of patients, so the definition was broadened.
- **AUA (American Urological Association), current:** an unpleasant sensation (pain, pressure, discomfort) perceived to be related to the bladder, with LUTS (lower urinary tract symptoms) for **more than 6 weeks**, without infection or another identifiable cause.
- **ESSIC (European Society for the Study of Interstitial Cystitis, 2008):** chronic (**>6 months**) bladder-related pain with at least one other urinary symptom (persistent urge or frequency), after confusable diseases are excluded.
- **ICD-11 (International Classification of Diseases, 11th revision):** "chronic primary bladder pain syndrome".
- **Trend: splitting the syndrome.** "IC" is reserved for **Hunner lesion (HL) disease**, a true inflammatory disease; BPS without HL is a symptom complex. Both sit within UCPPS (urological chronic pelvic pain syndrome).

Phenotypes

- **Hunner lesion disease ("IC"):** about 20%, mainly over 50; true inflammation (dense lymphoplasmacytic infiltrate, mast cells).
- **Non-Hunner BPS (bladder pain syndrome):** bladder often histologically normal, or shows glomerulations on cystoscopy.
- Also pelvic-floor-dominant and widespread (centralised) pain phenotypes.

Pathophysiology (current view)

- Multifactorial: leaky GAG (glycosaminoglycan) layer; mast-cell activation; neurogenic inflammation; **central sensitisation**; stress-axis dysfunction; small-fibre neuropathy. No infection proven.
- **Associated conditions:** 75–80% have pain beyond the pelvis; IBS (irritable bowel syndrome, >30%), fibromyalgia, vulvodynia, endometriosis; tender pelvic floor in about 80%.

Recent trends in evaluation (AUA 2022), before treatment

- **Diagnosis of exclusion plus localisation:** tests aim to find the pain generator.
- **History:** pain worse on filling, relieved by voiding; "painful urgency" without urge incontinence, unlike OAB (overactive bladder). **Ask about ketamine.**
- **Examination: six-site pelvic-floor palpation** (anterior and posterior levator ani, obturator internus, each side); vulval swab test; DRE (digital rectal examination) in men.
- **Basic tests:** PVR (post-void residual), urinalysis, culture; cytology if risk factors; frequency–volume (voiding) chart.
- **Scores:** O'Leary–Sant ICSI/ICPI (Interstitial Cystitis Symptom/Problem Index), GUPI (Genitourinary Pain Index), pain VAS (visual analogue scale); **score pain and urinary symptoms separately.**
- **Body map:** widespread pain = 3 or more of 12 regions besides the pelvis.
- **Office cystoscopy if over 50 or HL suspected** (AUA 2022 change): HL is a stellate red area with radiating vessels, found only by looking.
- **Local anaesthetic challenge** (lidocaine with or without bupivacaine) localises pain to the bladder and selects intravesical therapy.
- **Dropped:** potassium sensitivity test (neither sensitive nor specific; obsolete); routine hydrodistension (glomerulations are non-specific); routine urodynamics (optional only).
- **Biopsy** only to exclude CIS (carcinoma in situ).

ESSIC classification

Cystoscopy with hydrodistension plus biopsy, e.g. "3C":

Cystoscopy	Biopsy
X not done; 1 normal; 2 glomerulations; 3 Hunner lesion	X not done; A normal; B inconclusive; C positive (inflammation, mastocytosis, fibrosis)

Differential diagnosis (exclude)

- UTI (urinary tract infection) including tuberculosis, CIS, stones, radiation or cyclophosphamide cystitis, **ketamine cystitis**, OAB, diverticulum, endometriosis.

Phenotype-based diagnosis and treatment (key trend)

Phenotype	Identified by	Treatment
Hunner lesion	Cystoscopy	Fulguration and/or triamcinolone; ciclosporin A if refractory
Anaesthetic-responsive bladder pain	Local anaesthetic challenge	Intravesical therapy
Pelvic-floor tenderness	Pelvic-floor examination	Myofascial physiotherapy
Widespread (centralised) pain	Body map, associated pain conditions	Systemic: amitriptyline, gabapentinoids, CBT (cognitive behavioural therapy), pain team

- **MAPP (Multi-Disciplinary Approach to the Study of Chronic Pelvic Pain) reanalysis, 2023:** hydroxyzine/PPS (pentosan polysulfate) helped localised pain; amitriptyline helped widespread pain.

Management principles (AUA 2022)

- **Old AUA (2011/2014): six tiers**, from self-care to surgery.

- **AUA 2022 dropped the tiers:** grouped options, phenotype-guided, shared decisions; multimodal care allowed early; major surgery stays last.
- Stop treatments that fail; monitor with symptom scores.

Treatment options

Group	Options
Behavioural	Education; avoid trigger foods (coffee, citrus, spicy food, alcohol); stress management; flare plans
Physical	**Myofascial pelvic-floor physiotherapy** if tender (Standard); **avoid Kegel exercises**
Oral	**Amitriptyline** 10 mg at night, rising to about 50 mg; hydroxyzine; cimetidine; **PPS 100 mg three times daily** (**pigmentary maculopathy**: baseline and periodic retinal checks)
Intravesical	**DMSO (dimethyl sulfoxide) 50%, 50 mL, 15–20 min dwell**; heparin; alkalinised lidocaine (flares); also hyaluronic acid, chondroitin
Procedures	**HL: fulguration and/or triamcinolone injection** (75–90% improve); short (1–3 min) low-pressure (60–80 cmH_2O) hydrodistension; **onabotulinumtoxinA 100 U into the trigone** (must accept self-catheterisation; risk of retention); SNM (sacral neuromodulation)
Refractory HL	**Oral ciclosporin A** (watch kidney function and blood pressure)
Last resort (<10%)	Cystoplasty (augmentation) or diversion with or without cystectomy; pain may persist

- **Do not offer:** long-term antibiotics, intravesical BCG (bacille Calmette–Guérin), long-term oral (systemic) steroids.
- Pain management: gabapentinoids; opioids only via pain specialists.
- Most hydrodistension responses last under 2 months.
- Treat associated pain conditions (IBS, fibromyalgia) with other specialists.

Recent advances

- Urinary biomarkers and functional MRI (magnetic resonance imaging) for phenotyping (research); phenotype-stratified trials.
- **Also know:** tanezumab (anti-NGF (nerve growth factor)) not adopted; liposomal intravesical botulinum toxin no better than placebo; pudendal neuromodulation.

Diagram to draw: cystoscopic stellate Hunner lesion with radiating vessels, and the four-phenotype treatment chart.

> **Key points to remember:**
> - AUA definition: bladder pain with LUTS >6 weeks, no other cause; ESSIC >6 months.
> - AUA 2022: no tiers; phenotype-based; cystoscopy if over 50 or HL suspected; anaesthetic challenge, not potassium test (obsolete).
> - HL: fulguration or triamcinolone; ciclosporin if refractory.
> - Amitriptyline, PPS, DMSO; PPS causes maculopathy; antibiotics and BCG not offered.

Also, from the condensed book (not in the paper answer):

- **IC/BPS** ranges from inflammatory disease with **Hunner lesions (HLs)** to pain with a normal bladder and many non-urological complaints.
- The **AUA** defines it as bladder-related pain, pressure or discomfort with lower urinary tract symptoms (LUTS) for **>6 weeks**, with no infection or other cause. † **ESSIC (2008)** requires **>6 months** and ≥1 urinary symptom; **type 3C** is classic HL disease.
- The **NIH/NIDDK (1987)** research criteria missed **~60%** when used clinically. The current **trend** is to call HL disease "IC" and non-HL disease "BPS".

2 · Epidemiology (headline)

- The RICE study found that **3.3–7.9 million US women** and 2.1–4.6 million men have IC/BPS-type symptoms.
- Female:male is **5:1** in clinics but **2–3:1** in populations, so men are under-diagnosed. **Ask men labelled CP/CPPS about bladder pain** see Ch 27.
- **HL prevalence** is **~20%**, rising with age.
- The course is chronic but **not progressive**, with flares, commonly after sexual activity.

> **KEY POINT**
>
> Uncontrolled series "work" but randomised controlled trials (RCTs) fail because of regression to the mean, placebo and extra attention. Expect improvement in the first 3–4 weeks of any treatment.

3 · Aetiology and pathophysiology

The cause is multifactorial, and each theory points to a treatment.

Theory	Mechanism / evidence	Treatment link
Glycosaminoglycan (GAG) layer / "leaky" urothelium	Urinary solutes such as potassium reach suburothelial nerves	Pentosan polysulfate (PPS), heparin, hyaluronic acid
Mast cells	Histamine release; marked in HL disease, but counts have **no diagnostic role**	Hydroxyzine, cimetidine, amitriptyline
Neurogenic	Neurogenic inflammation, pelvic cross-sensitisation and central sensitisation	Tricyclics, gabapentinoids, neuromodulation, CBT
Autoimmune	Excess of lupus, Sjögren syndrome and inflammatory bowel disease (IBD); lymphoplasmacytic infiltrate in HL	Ciclosporin A (CyA) for HL disease
Infection	No consistent organism; 18-week antibiotic RCT showed no benefit	No antibiotics without proven infection

> **KEY POINT**
>
> Central sensitisation explains why many BPS patients have no physical findings and do not respond to local bladder therapy — the mechanism is systemic, not vesical.

4 · Pathology: Hunner lesion vs non-Hunner phenotype

- There is **no pathognomonic histology**. Biopsy only HLs or suspicious lesions, to exclude carcinoma in situ (CIS), tuberculous and ketamine cystitis.

Feature	HL disease ("IC")	Non-HL ("BPS")
Cystoscopy	Stellate erythema, radiating vessels, central scar; bleeds on filling ("waterfall"); rare at trigone	Normal, or glomerulations after distension
Histology	Ulceration, granulation, dense **lymphoplasmacytic** infiltrate, increased mast cells	Near-normal; sparse inflammation
Pain source	Bladder — end-organ disease	Often regional/widespread
Treatment	Fulguration / triamcinolone; CyA; dimethyl sulfoxide (DMSO) works better; cystectomy succeeds	Phenotype-directed; surgery often fails

5 · Associated conditions

- **75–80%** have pain beyond the pelvis. Among overlapping pain conditions, **irritable bowel syndrome (IBS)** is commonest (>30%), followed by **fibromyalgia**, chronic fatigue and migraine.
- **High-tone pelvic floor dysfunction (HTPFD)** occurs in **~80%**. **Autoimmune** disease shows the largest relative excess; in lupus, think of **lupus cystitis**.

> **KEY POINT**
>
> Commonest is not the same as most over-represented: IBS is commonest; autoimmune disease shows the greatest relative excess. Widespread pain predicts failure of bladder-directed therapy.

6 · Clinical features

Pain linked to filling separates IC/BPS from overactive bladder.

- Suprapubic pain is **worse with filling and relieved by voiding**, driving day and night frequency.

Feature	Overactive bladder (OAB)	IC/BPS	HTPFD
Urge	Sudden; fear of leakage	Persistent, mounting, pain-driven	Persistent; pelvic pressure
Pain	Absent	Present	Present
Incontinence	Urgency incontinence	Not a feature	—
Voiding	—	Frequency, low volumes	Obstructive

7 · Diagnosis — exclusion plus localisation

Basic evaluation (AUA 2022)

- Ask about the pain–filling link, triggers, pelvic irradiation, **cyclophosphamide and ketamine**.
- Examine for vaginal and urethral pathology, do a **Q-tip test** and DRE, and **palpate the pelvic floor**.

- Test post-void residual (PVR), urinalysis and culture, with cytology if at risk. Score pain and urinary symptoms separately, for example with the O'Leary–Sant indices (trial endpoint ≥30% fall).

Confusable disorders (ESSIC-modified)

Group	Disorder	Excluded by
Malignancy	Bladder cancer/CIS; prostate, gynae, colon	Cystoscopy with or without biopsy; prostate-specific antigen (PSA); pelvic exam
Infection	Bacterial, atypical, **TB**, herpes; prostatitis	Culture; **sterile pyuria needs TB culture**
Iatrogenic/drug	Radiation, cyclophosphamide, tiaprofenic acid, **ketamine**	History; ketamine causes upper-tract change
Outflow	Bladder neck obstruction, benign prostatic obstruction, retention	PVR, uroflow, (video)urodynamics
Structural/other	Stones, diverticulum; endometriosis, vulvodynia, atrophy; IBS, IBD	Imaging, cystoscopy, exam, laparoscopy, colonoscopy
Neuromuscular	Pudendal entrapment; pelvic-floor myalgia	Nerve block; pelvic-floor exam

- **Ketamine cystitis** causes HL-like inflammation, fibrosis and **hydronephrosis (~30%)**. Suspect it in the **young (<30) with visible inflammation**, and treat by **cessation**.
- **HTPFD** gives pelvic pressure, obstructive voiding and dyspareunia, with tender **myofascial trigger points**. Missed HTPFD causes poor response, so refer to a pelvic-floor physiotherapist.
- **Urodynamics** are not standard. Compliance is **normal**, and detrusor overactivity **does not exclude IC/BPS**.

Cystoscopy and bladder tests

- **AUA 2022** advises office cystoscopy when **HL is suspected, especially over 50**. Flexible cystoscopy shows most HLs **without hydrodistension**; biopsy **any trigonal lesion**.
- Relief after an **anaesthetic challenge** shows bladder-centred pain and selects intravesical therapy.
- The **potassium sensitivity test** (**0.4 M KCl** vs water) is **not recommended**, because it is painful and non-specific.
- **Hydrodistension** under anaesthesia fills at **80 cm H_2O for 1–3 min** to show **glomerulations**, which are **non-specific**. Biopsy only **after** distension and only suspicious lesions.

8 · Phenotype-directed management

*See the paper answer **Interstitial cystitis / bladder pain syndrome** (Section 3, Ch 28).*

> **KEY POINT**
>
> MAPP reanalysis (2023) of two negative NIH RCTs: hydroxyzine/PPS helped **localised pelvic pain**; amitriptyline helped **widespread pain**. A trial negative in all-comers can hide a positive result in the right phenotype.

9 · Treatment — AUA 2022 principles

The aim is coping and quality of life, not cure.

- There are **no tiers**, only grouped options: patients may move to any category **except major surgery** without failing "lower" ones. Stop ineffective treatment and seek other pain generators.
- The **Standard** is manual (myofascial) physiotherapy for pelvic-floor tenderness. Fulguration and/or triamcinolone for HLs is a **Recommendation**. Most drugs are **Options**. Chronic opioids are not routinely recommended.

ALGORITHM

1. If symptoms last ≥6 weeks, take a history, examine (pelvic floor, Q-tip, DRE), and check PVR, urinalysis and culture, with cytology if indicated.
2. Exclude confusable disease.
3. If HL is suspected (especially >50), then do cystoscopy. If HL is found, then fulgurate or inject triamcinolone; if refractory, give CyA.
4. If there is no HL, then match the phenotype: pelvic floor, physiotherapy; lidocaine responder, intravesical therapy; widespread pain, systemic therapy and CBT.
5. Combine behavioural, oral and intravesical treatment as needed, then BTX-A or sacral neuromodulation (SNM).
6. Only if bladder-centred and all else fails (rare), consider diversion with or without cystectomy.

10 · Treatment options (AUA 2022)

*See the paper answer **Interstitial cystitis / bladder pain syndrome** (Section 3, Ch 28).*

Also, from the condensed book (not in the paper answer):

- SNM is favoured over BTX-A for HTPFD or obstructive voiding.

10.1 Behavioural and pelvic-floor therapy

- Common **diet triggers** are caffeine, alcohol, fizzy drinks, spicy food and citrus; an elimination diet identifies the patient's own triggers.
- Education and behavioural modification alone improved 45% in the NIH amitriptyline RCT.
- In the myofascial physiotherapy RCT, 10 sessions beat global massage, **59% vs 26%**.

10.2 Oral therapy

- **Amitriptyline**: start **10 mg at night, adding 10 mg weekly to ~50 mg**. In the NIH RCT it gave 55% vs 45% (not significant), but **77% vs 53%** if ≥50 mg was tolerated. **Contraindications** are long QT, conduction block, recent myocardial infarction and heart failure.
- **Hydroxyzine** (H1 blocker) 25–50 mg at night had a negative RCT; **cimetidine** 400 mg twice daily has one positive RCT.
- **PPS 100 mg three times daily** is the only FDA-approved oral IC drug; only **3–6%** reaches urine and it acts in **3–6 months**. Its phase IV RCT was negative.
- **PPS maculopathy** is cumulative-dose related. **FDA (2020)**: retinal examination for all **within 6 months**, then periodically.

- **Ciclosporin A** is for HL refractory to fulguration or triamcinolone, at **1–3 mg/kg/day in two doses**. Its main toxicities are **nephrotoxicity and hypertension**, so monitor blood pressure, renal function and levels.
- **Antibiotics**: the AUA advises **none long-term**, and only one empirical course if never treated.

10.3 Intravesical therapy

Intravesical therapy is for **bladder-centred** pain; a poor lidocaine response predicts failure of every instillation. It is an AUA Option.

Agent	Regimen	Key points
DMSO 50%	50 mL, **15–20 min dwell**, every 1–2 weeks × 4–6	FDA-approved. Better in HL. Causes garlic breath and an initial flare; increases absorption of co-instilled lidocaine.
Heparin	10,000–40,000 U, 30–60 min, self-instilled	GAG mimic; benefit takes months.
Alkalinised lidocaine	200 mg with 8.4% bicarbonate daily × 5	Immediate effect, so useful for flares; tachyphylaxis.
Hyaluronic acid, chondroitin, PPS	Weekly then monthly	Weak/negative trials

- Cocktails are no better than DMSO alone.

10.4 Hunner lesion treatment

- **AUA Recommendation**: fulguration and/or submucosal triamcinolone, with biopsy if never biopsied. **75–90%** improve, but repeat treatment is usually needed within 1–2 years.

ALGORITHM

1. Under general anaesthesia, map the HLs before distension causes bleeding.
2. Distend mildly and biopsy suspicious or trigonal lesions.
3. Fulgurate at **low volume** (Bugbee electrode), working from the **outer margin inwards**.
4. Inject triamcinolone 40 mg/mL (or 1:1 with bupivacaine/lidocaine) in 0.5-mL aliquots, 2–3 mm submucosally.
5. Fulgurate bleeders, empty the bladder and instil lidocaine–bupivacaine.

KEY POINT

Fulgurate from the margin inward: cautery erythema and oedema spread peripherally, so the true edge is lost if you start centrally.

10.5 Hydrodistension

- AUA Option: **short (1–3 min), low-pressure (60–80 cm H_2O)** distension, with benefit fading by 6 months.
- An anaesthetic capacity **<400 mL** marks a bladder-centred phenotype, often HL.

10.6 OnabotulinumtoxinA and neuromodulation

- **BTX-A** is an Option after failed conservative, oral and intravesical therapy: **100 U** into the bladder including the **trigone**, lasting about 5–9 months. The patient **must accept clean intermittent catheterisation (CIC)**.

- **SNM (S3)** is an Option and works best with **HTPFD or levator tenderness**, but IC/BPS has high explantation rates.

	BTX-A	SNM
Favoured when	Bladder-centred pain; can do CIC	Obstructive voiding, incomplete emptying, HTPFD, CIC impossible
Main risk	Retention needing CIC	Adverse events, explantation

10.7 Major surgery — last resort

- Surgery is only for **rare (<10%), selected, bladder-centred** patients failing all else, and pain may persist after cystectomy. If pain persists after a **differential spinal block** to T10, surgery will fail.

Predicts success	Predicts failure
HLs (refractory)	No HLs
Small fibrotic bladder; anaesthetic capacity <200–400 mL	High anaesthetic capacity
Profound relief with intravesical anaesthetic	Equivocal/no response
Painful filling, pain on scope touch	Urethral/neuropathic pain; HTPFD; widespread pain

- Cystectomy with a conduit leaves about three-quarters pain-free. Prefer the **simplest diversion**, as a retained bladder risks pain and pyocystis.

11 · What NOT to do (AUA)

Do not offer	Why
Long-term oral antibiotics	RCT negative; adverse events 80% vs 40%
Intravesical bacillus Calmette–Guérin (BCG)	Ineffective; DMSO superior; † BCG sepsis
High-pressure, long hydrodistension†	No benefit; rupture, sepsis †
Long-term oral glucocorticoids	No benefit; toxicity
† Intravesical resiniferatoxin	Ineffective in RCT †
Kegel strengthening	Worsens HTPFD pain
Major surgery first / non-bladder-centred pain	Irreversible; pain persists

> **KEY POINT**
>
> Nothing is worse than irreversible surgery followed by continued pain — every predictor of surgical success marks bladder-generated pain.

12 · Landmark trials

Trial	Design	Result	Take-home
Myofascial PT RCT (FitzGerald 2012)	10 sessions myofascial vs global massage	59% vs 26%	Guideline Standard
NIH amitriptyline RCT (Foster 2010)	EBMP with or without amitriptyline	55% vs 45% (not significant); 77% vs 53% if ≥50 mg	Titrate to ≥50 mg; behavioural therapy works

Trial	Design	Result	Take-home
PPS phase IV RCT	PPS vs placebo	No difference (≈40% each)	Efficacy doubtful; remains Option
MAPP phenotype reanalysis (2023)	Two negative NIH RCTs by pain distribution	Hydroxyzine/PPS localised; amitriptyline widespread	Treat by phenotype

29 Sexually Transmitted Diseases

1 · Screening, prevention and partner notification

One sexually transmitted disease (STD) often hides another.

- The nucleic acid amplification test (**NAAT**) is the standard screen; men who have sex with men (MSM) need urethral, rectal and pharyngeal tests. MCQ Test the pharynx for gonorrhoea, not chlamydia.
- **Partner notification** covers the previous **60 days** for urethritis, chlamydia and Mycoplasma genitalium; all partners for trichomoniasis; and the previous month for scabies or lice. Abstain for 7 days and until partners are treated. Test everyone at risk for HIV and syphilis.

2 · Urethritis

- **Gonococcal** urethritis makes up 5–20%. In non-gonococcal urethritis (**NGU**), C. trachomatis is commoner than M. genitalium, which is the leading cause of **persistent or recurrent** urethritis. Up to half have no pathogen.
- **Diagnosis** needs any one: discharge; Gram stain with **≥2 white blood cells (WBC) per oil-immersion field**; positive leucocyte esterase on first-void urine; or ≥10 WBC/hpf in first-void urine.
- Intracellular gram-negative diplococci in a symptomatic man confirm gonorrhoea; a negative Gram stain does not exclude it.
- **First-void urine NAAT** is preferred; **culture** is for treatment failure and children.

Pathogen	First line	Alternatives / notes	Follow-up
Gonorrhoea (CDC 2021)	**Ceftriaxone 500 mg IM once** (1 g if ≥150 kg)	Gentamicin 240 mg IM and azithromycin 2 g (β-lactam allergy); cefixime 800 mg. Europe: ceftriaxone 1 g and azithromycin 2 g. Add chlamydia cover unless excluded	Test of cure only for **pharyngeal** infection; retest at 3 months
Chlamydia / NGU	**Doxycycline 100 mg bd 7 days**	Azithromycin 1 g once	No test of cure; retest at 3 months
M. genitalium	Doxycycline 7 days, then **azithromycin** 1 g, then 500 mg daily × 3, if macrolide-sensitive; or **moxifloxacin** 400 mg × 7 days if resistant or untested	Single-dose azithromycin is inadequate	Abstain until cured
T. vaginalis (male)	Metronidazole 2 g once	Tinidazole 2 g once	Treat all partners

- Higher-dose **ceftriaxone monotherapy** reflects resistance; cephalosporins are safe unless a recent IgE-mediated penicillin reaction occurred. Treatment failure needs culture and susceptibility.
- **M. genitalium** has no cell wall, so Gram stain misses it; NAAT with a macrolide-resistance assay picks the second drug.
- There is no chlamydia test of cure, because nucleic acid persists ~3 weeks.

ALGORITHM

1. Persistent NGU: check adherence and look for an untreated partner.
2. Send NAAT for M. genitalium, with or without T. vaginalis.
3. If negative, then think of chronic pelvic pain syndrome.

KEY POINT

Men with gonorrhoea usually have symptoms; women are usually silent and present with pelvic inflammatory disease or infertility. Treating the man protects his partner.

3 · Epididymitis by age

Acute means <6 weeks; chronic (≥6 weeks) is mostly tuberculosis (see Ch 27). Always exclude **torsion**. Apply urethritis criteria and NAAT; add urine culture if enterics are likely.

Group	Organisms	Empiric regimen (10 days)
Young, sexually active	C. trachomatis, N. gonorrhoeae	Ceftriaxone 500 mg IM and doxycycline 100 mg bd
Insertive anal sex	Above, plus enterics	Ceftriaxone and levofloxacin 500 mg daily
>35 years, obstruction or instrumentation	E. coli	Levofloxacin alone

4 · Genital ulcer disease

Herpes is commonest, then syphilis. Appearance misleads, so test **every ulcer** with syphilis serology, HSV NAAT, an H. ducreyi test if endemic, and HIV. Biopsy non-healing ulcers, because carcinoma mimics them.

Disease (organism)	Lesion	Nodes	Test	Treatment
Primary syphilis (T. pallidum)	**Painless**, indurated, clean, single	Painless, rubbery, bilateral	Serology; darkfield	Benzathine penicillin G 2.4 MU IM once
Herpes (HSV-2 more than HSV-1)	**Painful** multiple vesicles that become shallow ulcers	Tender, bilateral	NAAT of unroofed base	Aciclovir and related drugs
Chancroid (H. ducreyi)	**Painful, ragged, undermined, purulent**	Tender **suppurative buboes**	Special culture; exclusion	Azithromycin 1 g once, or ceftriaxone 250 mg IM once, or ciprofloxacin 500 mg bd 3 days, or erythromycin 7 days
Lymphogranuloma venereum, LGV (C. trachomatis L1–L3)	Small, painless, transient	Painful, matted, fistulae; groove sign	Chlamydia NAAT	Doxycycline bd **21 days**
Donovanosis (Klebsiella granulomatis)	**Painless, beefy red**, bleeds easily	**Absent** (pseudobuboes)	**Donovan bodies** on crush preparation	Azithromycin 1 g weekly or 500 mg daily, ≥3 weeks **and until healed**

Genital herpes

- HSV lies latent in the **sacral ganglia** and reactivates with trauma, surgery or immunosuppression. Most transmission is from asymptomatic shedding. HSV-2 recurs more, so type it.
- Complications are aseptic meningitis and **urinary retention**; HSV-2 raises HIV risk 2–3-fold.

Drug	First episode (7–10 days)	Suppressive	Episodic
Aciclovir	400 mg tds	400 mg bd	800 mg bd 5 days
Valaciclovir	1 g bd	500 mg–1 g daily	500 mg bd 3 days
Famciclovir	250 mg tds	250 mg bd	1 g bd 1 day

- Treat every first episode. **Episodic** therapy must start within 1 day of lesions or in the prodrome.

Chancroid, donovanosis, LGV

- **Chancroid** needs a painful ulcer, a typical picture, and syphilis and HSV excluded. Aspirate or drain fluctuant buboes.
- **Donovanosis**: if antibiotics fail, biopsy, because **penile squamous cell carcinoma (SCC) mimics it**.

SAQ Lymphogranuloma venereum

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 2 Paper 1, Q3c

Definition

- **LGV (lymphogranuloma venereum)** is an invasive, lymphotropic STD (sexually transmitted disease) caused by ***Chlamydia trachomatis*** **serovars L1, L2 and L3**.
- Of 15 serovars, A–C cause trachoma and D–K anogenital infection. The L serovars invade lymphatics, causing lymphadenitis and scarring.

Epidemiology

- Endemic in the tropics, including India; increasing as **proctitis in MSM (men who have sex with men)**. Not more frequent or virulent in HIV (human immunodeficiency virus).

Clinical stages

Stage	Timing	Features
Primary	3–30 days	Small painless papule or ulcer, often unnoticed. Rectal inoculation → **proctitis** (bleeding, pain, tenesmus, discharge), now the commonest presentation
Secondary (inguinal syndrome)	2–6 weeks later	Fever, malaise; **painful, matted, unilateral inguinal or femoral buboes** that may form fistulae. **Groove sign**: nodes above and below the inguinal ligament
Tertiary	Untreated	Fibrosis and lymphatic block: **genital elephantiasis** (esthiomene in women), perirectal abscess, fistulae, **rectal stricture**

- Inguinal disease is commoner in men: the cervix and vagina drain to pelvic nodes.

Differential diagnosis of genital ulcer

- Syphilis (painless ulcer and nodes); herpes (painful vesicles); chancroid (painful ulcer, suppurative nodes); donovanosis (beefy ulcer, no nodes); **LGV: transient painless lesion, then painful matted nodes**.

Investigations

- **NAAT (nucleic acid amplification test)** of ulcer, rectal or node aspirate; LGV-specific genotyping confirms where available. Rectal chlamydia NAAT in all proctocolitis.
- Serology not routine. Screen for HIV, syphilis, gonorrhoea; offer PrEP (pre-exposure prophylaxis).

Treatment

- **Doxycycline 100 mg twice daily for 21 days** (first line).
- Alternatives: erythromycin 500 mg four times daily for 21 days, or azithromycin 1 g weekly for 3 weeks (pregnancy; consider test of cure by NAAT).
- Fluctuant buboes: aspirate through healthy skin to prevent sinus formation.
- **Late disease:** stricture dilatation or resection, fistula repair, reduction of elephantiasis.

Diagram to draw: groin with buboes above and below the inguinal ligament (groove sign).

> **Key points to remember:**
> - Serovars L1–L3; invasive and lymphotropic.
> - Proctitis is now the commonest presentation.
> - Groove sign; esthiomene; rectal stricture.
> - Doxycycline 100 mg twice daily for 21 days.

5 · Syphilis staging and serology

Stage	Features	Treatment (penicillin-allergic)
Primary (10–90 days)	Chancre and nodes; heals in 3–8 weeks	Benzathine penicillin **2.4 MU IM once** (doxycycline 14 days)
Secondary (3–5 months)	Rash on **palms and soles**, **condylomata lata**, alopecia, hepatitis, nephritis	As primary
Early latent (<1 year)	Positive serology only	As primary
Late latent / unknown	Positive serology only	Benzathine **weekly × 3** (doxycycline 28 days)
Tertiary	Gumma, aortitis, tabes dorsalis	Weekly × 3 if cerebrospinal fluid (CSF) normal
Neurosyphilis (any stage)	Central nervous system, eye, ear	**Aqueous penicillin G 18–24 MU/day IV 10–14 days**

- T. pallidum cannot be cultured, so **two serological tests** are used. The **nontreponemal test (NTT)** (VDRL, RPR) can be falsely positive; its titre falls with treatment, so it **monitors treatment**. The **treponemal test (TT)** (FTA-ABS, TP-PA, EIA) usually stays positive for life.
- A significant change is **fourfold (two dilutions)**. Use the same NTT each time.

ALGORITHM

1. **Reverse algorithm**
2. Start with a TT. If positive, then do an NTT.
3. If the NTT is negative, then do a second, different TT.
4. If that is positive and there is no past treatment, then treat as late latent syphilis.

- The **Jarisch–Herxheimer** reaction (fever and myalgia within 24 h) is treponemal lysis, not allergy; give NSAIDs.
- In primary or secondary syphilis, test for HIV and offer pre-exposure prophylaxis (PrEP) if negative.
- **Follow-up** serology is at 6 and 12 months (and 24 for latent disease). A fourfold rise means failure or reinfection, so retreat weekly × 3 and consider CSF.

6 · HPV and anogenital warts

- Human papillomavirus (HPV) **6 and 11** cause ~90% of warts. HPV **16 and 18** cause >70% of cervical cancer, and also penile and anal cancer.
- **Bowenoid papulosis** (HPV-16) is a high-grade squamous intraepithelial lesion. **Giant condyloma** (Buschke–Löwenstein) is a locally invasive verrucous carcinoma.
- Diagnosis is clinical. **Biopsy if** the diagnosis is uncertain, there is no response or worsening, the patient is immunocompromised, or the lesion is pigmented, indurated, fixed or ulcerated.

Treatment	Key points
Imiquimod 5% (patient)	3 nights/week for up to 16 weeks; wash off after 6–10 h. Weakens condoms
Podofilox 0.5% (patient)	bd for 3 days, then 4 days off. **Contraindicated in pregnancy**
Sinecatechins 15% (patient)	tds, not washed off. Weakens condoms. Not in HIV, herpes or pregnancy
Cryotherapy, trichloroacetic acid (TCA) 80–90% (provider)	Cryotherapy every 1–2 weeks; TCA weekly
Excision, curettage, CO_2 laser	For large warts; use smoke extraction

Change treatment if there is no response by 3 months; observation is acceptable.

- The **vaccine** is **nonavalent** (6, 11, 16, 18, 31, 33, 45, 52, 58). Give it at **11–12 years** (from 9), with catch-up to **26**, in both sexes. Starting <15 needs **2 doses**; ≥15 or immunocompromised needs **3 doses**.

7 · Molluscum and ectoparasites

- **Molluscum** contagiosum forms **umbilicated** papules that regress in 6–12 months; use cryotherapy, curettage or imiquimod. Extensive lesions suggest HIV.
- **Pubic lice**: use a **permethrin 1%** rinse for 10 min, or malathion 0.5% or ivermectin 250 µg/kg if resistant.
- **Scabies** causes nocturnal itch and burrows. Apply **permethrin 5%** to all skin, or give ivermectin 200 µg/kg repeated at 2 weeks. **Crusted** scabies (immunocompromised) needs topical treatment and repeated ivermectin, never lindane.

- Treat contacts from the past month and wash bedding.

8 · HIV for urologists

LAQ Describe clinical implication of AIDS on the Genito-Urinary Tract?

Paper answer · P1

Type: LAQ (25 marks)
Source: 2020-Summer Paper 1, Q1

Introduction

- AIDS (acquired immunodeficiency syndrome) is the late stage of HIV (human immunodeficiency virus) infection. HIV is a retrovirus that destroys CD4 T lymphocytes. It spreads by blood, semen, vaginal fluid and breast milk.

Stage	Features
1. Acute	2–4 weeks after infection; flu-like illness; high viral load; antibody tests may be negative
2. Clinical latency	Years; usually no symptoms
3. AIDS	**CD4 <200 cells/mm³** or an AIDS-defining condition

- **Concept:** genitourinary (GU) disease follows the falling CD4 count. Organisms become opportunistic, infections become chronic and the GU tract becomes a reservoir. With HAART (highly active antiretroviral therapy), the burden shifts to drug toxicity and diseases of ageing.

Diagnosis (urologist's role)

- Test anyone with, or at risk of, an STD (sexually transmitted disease), and every TB (tuberculosis) patient.
- Antigen–antibody immunoassay → antibody differentiation assay; if discordant, HIV RNA (acute infection).

Manifestations by site

Site	Manifestations
Genital skin and STDs	Ulcerative STDs (herpes, syphilis, chancroid) increase HIV acquisition and transmission; HSV-2 (herpes simplex virus 2) in 50–90%. Warts (HPV (human papillomavirus) 6, 11) in about 20%, may be giant; molluscum 5–18%. Perineal cellulitis (cover MRSA (methicillin-resistant *Staphylococcus aureus*)); Fournier gangrene commoner
UTI (urinary tract infection)	About fivefold commoner. Bacteriuria about 30% at CD4 <200, 11% at 200–500, none above 500. *E. coli* and other gram-negatives; *Pseudomonas* up to 30%. *Salmonella* UTI needs lifelong prophylaxis. Culture before treatment
Opportunistic UTI	No response, or CD4 <500: look for fungi (*Candida*, *Cryptococcus*, *Histoplasma*), TB, CMV (cytomegalovirus), adenovirus
Kidney	Opportunistic pyelonephritis (*Pneumocystis*, *Candida*, mycobacteria). AKI (acute kidney injury) from sepsis, dehydration, nephrotoxins (tenofovir, indinavir, amphotericin). Obstruction by stones, lymphoma, fungal balls

Site	Manifestations
GU TB	HIV raises active TB 15–20-fold. Disease is disseminated and bilateral, with **less fibrosis, fewer strictures and fewer contracted bladders**, because fibrosis needs an intact immune response. Mortality is higher. Same 6-month regimen; rifampicin lowers protease inhibitor levels
Prostate	Bacterial, TB and fungal prostatitis; prostatic abscess and urosepsis; relapse up to 70%. Cryptococcal prostatitis is a reservoir for relapsing meningitis, so give long-term antifungals
Testis and epididymis	Recurrent, chronic epididymo-orchitis (including TB). Testis harbours opportunists. Testicular atrophy and hypogonadism
Voiding	LUTS (lower urinary tract symptoms) worsen with disease stage. Retention may herald seroconversion. Neurogenic causes: HIV encephalopathy, toxoplasmosis, **CMV polyradiculopathy** (bladder and bowel dysfunction with sciatic pain: lumbar puncture, treat early), neuropathy
Sexual function	ED (erectile dysfunction) in about half of men under 50; protease inhibitors worsen it. **Protease inhibitors raise sildenafil levels 3–10-fold: start low.** Low libido, ejaculatory disorders

HIVAN (HIV-associated nephropathy)

- Nephrotic-range proteinuria and rising creatinine, often without oedema or hypertension. Typically CD4 <200 and viral load >1000 copies/mL; about 12 times commoner in Black patients.
- Biopsy: **collapsing FSGS (focal segmental glomerulosclerosis)** with tubular microcysts.
- Treatment: HAART plus ACE (angiotensin-converting enzyme) inhibitor. Without it, dialysis follows within about 10 months.

Stones

- **Indinavir** stones in up to 22% of treated patients. Indinavir precipitates in alkaline urine. Stones are radiolucent, and pure stones may be invisible even on CT (computed tomography).
- Stop the drug and hydrate: succeeds in up to 70%. Newer protease inhibitors rarely cause stones.

Malignancy

AIDS-defining	Non-AIDS-defining
Kaposi sarcoma (HHV-8 (human herpesvirus 8)), non-Hodgkin lymphoma, cervical cancer	Penile, anal, testicular, prostate cancers; Hodgkin lymphoma
Falls with HAART	Now about 70% of cancers in treated HIV

- **Kaposi sarcoma:** genital in about 3% of AIDS; glans lesions can block the meatus. Treat with ART (antiretroviral therapy) first. Local excision, laser or intralesional vinblastine; radiotherapy 15–30 Gy; systemic chemotherapy if disseminated.
- **Penile cancer:** HIV raises HPV infection; about fourfold risk of death.

Implications for urological practice

- **Universal precautions** for every patient: double gloves, no hand-to-hand passing of sharps.
- Occupational risk: percutaneous about 0.3%, mucous membrane about 0.09%.

- **PEP (post-exposure prophylaxis)** within 72 hours, for 28 days: tenofovir + emtricitabine plus raltegravir or dolutegravir. HIV test at 6 weeks, 3 and 6 months.
- CD4 <200: TMP-SMX (trimethoprim-sulfamethoxazole) *Pneumocystis* prophylaxis also lowers UTI.
- **Prevention:** male circumcision cuts heterosexual acquisition by 50–60% (three African RCTs (randomised controlled trials)); consistent condom use cuts risk about 80%; PrEP (pre-exposure prophylaxis).

Also know: IRIS (immune reconstitution inflammatory syndrome) on starting ART; serodiscordant couples: suppressed viral load with sperm washing and assisted reproduction; transplantation is possible with controlled HIV.

Diagram to draw: male GU tract outline labelled with the lesion at each site.

> **Key points to remember:**
> - AIDS = CD4 <200; UTI risk rises as CD4 falls.
> - HIV-TB: disseminated, less fibrosis, fewer strictures.
> - Indinavir stones: radiolucent, may be CT-invisible; stop drug and hydrate.
> - PEP within 72 hours for 28 days; circumcision cuts acquisition 50–60%.

Also, from the condensed book (not in the paper answer):

- Start post-exposure prophylaxis (PEP) within **72 h**, and only for mucosal, broken-skin or percutaneous contact with blood or genital fluid. Needlestick risk is **0.3%**.
- Circumcision cuts heterosexual acquisition by **50–60%** but does not protect partners. **After sexual assault** (men), give ceftriaxone 500 mg once, doxycycline 7 days, hepatitis B and HPV vaccines, and PEP within 72 h.
- PrEP is daily emtricitabine with tenofovir disoproxil fumarate, or tenofovir alafenamide in men only.

30 Cutaneous Diseases of the External Genitalia

1 · Approach to genital dermatoses

Genital rashes look alike, so biopsy early to avoid missing cancer.

- Genital skin has few reaction patterns, so mimics abound and the **biopsy threshold is low**. Ask about over-the-counter creams, and do a **full skin survey**, because extragenital lesions often give the diagnosis.
- Identify the **primary lesion**. A papule is solid and **≤0.5 cm**; larger solid lesions are nodules or plaques, and fluid-filled ones are vesicles (≤0.5 cm) or bullae. An **erosion** loses epidermis only and does not scar; an **ulcer** loses dermis too and scars.
- Bedside tests: KOH prep for hyphae or yeast, mineral oil prep for scabies, Tzanck smear for herpes, and Wood's lamp for erythrasma (coral-red) and vitiligo.

Biopsy: a 2–3 mm punch under local anaesthetic is often diagnostic; large or meatal lesions go to theatre.

2 · Principles of therapy

- Use **topical treatment first**; urologists overprescribe systemic antibiotics.
- Give **systemic antifungals** only for extensive dermatophytosis, Majocchi granuloma, severe immunocompromise or topical failure, and avoid systemic **ketoconazole** (hepatotoxicity, adrenal insufficiency).
- Give systemic prednisone for ≤3 weeks as a single morning dose.
- **Topical steroids** cause atrophy in thin genital skin, and the **prepuce acts as an occlusive dressing**. The rule is low potency and a short course; lichen sclerosus (LS) is the exception.
- Narrow-band UVB or the 308-nm excimer treats psoriasis and vitiligo. **PUVA causes genital squamous cell carcinoma (SCC)**, so it is avoided.

3 · Eczematous dermatoses

Exclude Candida, herpes, impetigo and drug reaction first.

Atopic dermatitis (AD)

- AD is relapsing, flexural and intensely pruritic; **90%** present before 5 years. Use emollients, a short topical steroid, calcineurin inhibitors and a sedating antihistamine; severe disease needs **dupilumab**.

Contact dermatitis

Feature	Irritant (ICD)	Allergic (ACD)
Mechanism	Direct cytotoxicity; dose-driven	Type IV hypersensitivity
Share	~**80%**	Minority; genital ACD is rare
Agents	Soaps, solvents, acids/alkalis	**Nickel** (commonest), neomycin, fragrances, latex, topical steroids
Picture	Burning, limited to the exposed zone	Demarcated, pruritic, weeping

Feature	Irritant (ICD)	Allergic (ACD)
Confirm	Exposure history	**Patch testing**
Treat	Avoidance, barrier	Remove allergen. If severe, a **3-week** prednisone taper (a short course rebounds)

Seborrhoeic dermatitis (SD)

- SD gives greasy-scaled plaques in sebaceous areas. It affects up to **83%** of patients with AIDS, so **extensive SD should prompt an HIV test**. A topical **antifungal** (ketoconazole) is the backbone.
- Lichen simplex chronicus (lichenification, in the LS differential) needs the itch–scratch cycle broken with a short potent steroid.

4 · Papulosquamous disorders

Tinea, erythrasma, secondary syphilis and neoplasia (Bowen disease, extramammary Paget disease (EPD), mycosis fungoides) mimic psoriasis, so "psoriasis" failing treatment needs KOH, syphilis serology and biopsy.

Psoriasis

- Psoriasis affects ~2% and forms sharp plaques with silvery scale; check nails, scalp, **umbilicus** and natal cleft. **Inverse psoriasis** loses its scale in folds, so use KOH to exclude tinea or Candida.

Step	Genital psoriasis treatment
First-line	Low-potency steroid (hydrocortisone butyrate 0.1%) continuously for **≤2 weeks**; triamcinolone 0.1% for thick plaque
Steroid-sparing	Calcipotriene, calcineurin inhibitors
Localised	308-nm excimer laser
Contraindicated	**PUVA** (dose-dependent genital SCC)
Extensive	Methotrexate, cyclosporine, apremilast, anti-TNF, ustekinumab, **ixekizumab (anti-IL-17), which has the best genital data**

Reactive arthritis (formerly Reiter)

- Reactive arthritis follows urethritis or enteric infection in **HLA-B27**-positive people. **Balanitis circinata** looks like psoriasis, so diagnose from **context**; it is self-limiting and needs only a low-potency steroid.

Lichen planus (LP)

- LP forms **polygonal, violaceous, flat-topped papules**; glans papules are white and reticular. Papular LP usually resolves within a year.
- **Erosive mucosal LP persists, scars and carries an SCC risk**, so review regularly and biopsy ulcers. Use an **ultrapotent** steroid (clobetasol 0.05%).

5 · Lichen sclerosus (LS) / BXO

SAQ Lichen sclerosus (BXO, balanitis xerotica obliterans)

Paper answer · P1

Asked as:

- Lichen sclerosis. — *Summer 2020 Paper I Q3c (SAQ)*
- BXO — *Practice Set 10 Paper I Q3b (SAQ)*

Type: SAQ (10 marks)
(also answered in 2020-Summer Paper 1, Q3c)

Taken as lichen sclerosus (LS), formerly balanitis xerotica obliterans (BXO). BXO (balanitis xerotica obliterans) is taken as genital lichen sclerosus (LS) in the male.

Definition

- Chronic, progressive, **scarring inflammatory disease of genital skin**; possibly premalignant. BXO is its late stage in men.

Epidemiology

- 6–10 times commoner in females (prepubertal and menopausal).
- Men: peak 30–50 years in one chapter, older men in another; also children (boys with phimosis). Commonly found at circumcision for phimosis after infancy. 15–20% have extragenital disease.

Aetiology

- Cause unknown. Autoimmune links (thyroid disease, vitiligo, alopecia); familial clustering.
- **Chronic occlusion by urine** under the foreskin (dribbling, obesity) definitely increases risk.
- Infection and trauma are proposed but unproven.

Clinical features

- White, atrophic, sclerotic plaques with loss of architecture; purpura, fissures; itch, pain, painful erection, poor stream, spraying.
- Sequence: prepuce → **phimosis** → glans → **meatal stenosis** (sentinel site) → fossa navicularis and penile urethral stricture.
- **Spares the proximal (bulbar and posterior) urethra.**
- Buried penis from loss of shaft skin.
- **Mechanism of stricture:** voiding against meatal stenosis forces urine into the glands of Littre → inflammation → spongiofibrosis.

Evaluation

- Clinical diagnosis. **Biopsy** if doubtful or suspicious: hyperkeratosis, epidermal atrophy, basal vacuolation, dermal oedema with homogenised collagen, band-like lymphocytic infiltrate (lymphocytes).
- Uroflowmetry; **RUG (retrograde urethrogram) before meatal surgery**, because a proximal stricture often coexists (may coexist).
- **Differential:** lichen planus, Zoon balanitis, penile intraepithelial neoplasia.

Management

Situation	Treatment
Skin disease	**Clobetasol propionate 0.05%** daily for 1–3 months (another source: for about 3 months), then taper to twice-weekly maintenance; calcineurin inhibitors second line; retinoids for refractory disease
Phimosis	**Circumcision**, usually curative

Situation	Treatment
Meatal stenosis	Dilatation with clobetasol; extended meatotomy; BMG (buccal mucosal graft) meatoplasty
Anterior stricture	**Oral mucosal graft, never genital skin** (LS recurs in skin); staged or single stage
Panurethral or unfit	**Perineal urethrostomy** (easy because the proximal urethra is spared); self-dilatation

- Recurrence 40–50% after medical treatment (recurrence is common); restricture higher than in other aetiologies (restricture rates exceed other aetiologies); treatment follows the patient's goals.

Malignancy

- Penile SCC (squamous cell carcinoma) in **2.3–9%** on long follow-up; HPV (human papillomavirus)-independent type.
- **Lifelong follow-up**; biopsy any non-healing ulcer or nodule.

Diagram to draw: penis with LS at (showing) prepuce, glans, meatus and fossa involved, proximal urethra spared.

> **Key points to remember:**
> - Genital skin disease; meatus is the sentinel site.
> - Spares proximal urethra; perineal urethrostomy easy.
> - Clobetasol first; oral mucosa, never skin, for repair.
> - SCC in 2.3–9%: lifelong follow-up.

Also, from the condensed book (not in the paper answer):

- LS is 6–10× commoner in females, and affected men are usually >60; **15–20%** have extragenital disease, and in men the **perianal skin is spared**.
- Circumcision is usually curative, but LS may recur in the scar.

ALGORITHM

1. Biopsy.
2. Give a **potent steroid (clobetasol 0.05%) for 3 months**, then taper.
3. Continue twice-weekly maintenance (or mometasone 0.1% or a calcineurin inhibitor).
4. Lifelong surveillance of non-healing fissures, ulcers, nodules.

- Recurrence is **40–50%**, so maintain proactively. Meatal disease needs meatotomy or meatoplasty.

Disease	Steroid potency	Duration
AD / psoriasis / reactive arthritis	Low	Short (psoriasis ≤2 weeks)
Seborrhoeic dermatitis	Low	Initial severe phase only
Lichen planus	Ultrapotent	Until controlled
Lichen sclerosus	Potent	3 months, then maintenance

KEY POINT

Steroid strength tracks the scarring and malignant stakes of the disease. LS is the single exception to the short-course rule.

6 · Vesiculobullous and drug eruptions

Feature	Erythema multiforme (EM) minor	Stevens–Johnson syndrome (SJS)	Fixed drug eruption (FDE)
Trigger	**HSV**	**Drugs**: sulfonamides, NSAIDs, β-lactams, anticonvulsants	Sulfonamides, NSAIDs, tetracyclines, carbamazepine
Lesion	Symmetric **targets**; palms, soles, mouth	Blisters and necrosis; genitals involved in ~**60%**	Solitary violaceous plaque, often on the glans
Course	Self-limited, recurs	Mortality up to ~**30%**	**Recurs at the same site within 24 h** of re-exposure
Sequelae	None	Meatal stenosis, strictures	Brown pigmentation
Treatment	**Suppressive aciclovir prevents it**	See algorithm	Stop the drug

ALGORITHM

1. **SJS**
2. Stop the offending drug.
3. Give burn-type supportive care in an intensive care or burn unit (lowers mortality).
4. Limited data support cyclosporine 3–5 mg/kg/day, etanercept, plasmapheresis or IVIG; systemic steroids are controversial.

KEY POINT

One drug class can cause a trivial FDE or a lethal SJS; extent and systemic illness separate them. Genital FDE mimics herpes; "same spot each time" is the clue.

Autoimmune blistering disease needs perilesional biopsy with direct immunofluorescence.

7 · Noninfectious genital ulcers

A genital ulcer is infectious until proven otherwise (see Ch 29).

Behçet disease (BD)

- BD is an **HLA-B51**-linked vasculitis, **more severe in males**, with oral and genital ulcers, uveitis, epididymitis and **pulmonary artery aneurysm**. Oral ulcer is most sensitive; **genital ulcer is most specific**. No confirmatory test exists. Treat with colchicine, steroids or **anti-TNF**.

International Criteria for BD (2014) — ≥4 points	Points
Ocular lesions; oral aphthosis; genital aphthosis	2 each
Skin lesions; central nervous system; vascular; positive pathergy (optional)	1 each

Pyoderma gangrenosum (PG)

- PG accompanies inflammatory bowel disease and myeloproliferative disease as a painful ulcer with an **undermined violaceous border**. Biopsy the edge, treat with steroids, and avoid debridement.

8 · Infections

Balanitis and balanoposthitis

ALGORITHM

1. Remove irritants and improve hygiene.
2. Give a topical antibiotic or antifungal, then a short low-potency steroid.
3. If this fails, then **biopsy** (neoplasia, Zoon, psoriasis, HPV, HSV).
4. If recurrent with phimosis, then **circumcision**.

Type	Clue	Management
Candidal	Satellite lesions; diabetes	Topical imidazole; check glucose
Irritant/allergic	Products, soaps	Avoidance
Circinate	Urethritis, HLA-B27	Low-potency steroid
Zoon (plasma cell)	Shiny red plaque, elderly uncircumcised	Biopsy, then circumcision
LS / BXO	White sclerotic glans, phimosis	Potent steroid; circumcision
Pseudoepitheliomatous keratotic micaceous balanitis (PEKMB)	Mica-like scaly plaque	Biopsy; excision

Soft-tissue infection

- Folliculitis needs hygiene. A furuncle needs incision and drainage with MRSA cover. Erysipelas has a sharp border and cellulitis an indistinct one; give antibiotics and **mark the border**.
- Fournier gangrene is a polymicrobial fascial infection with pain out of proportion, grey skin and crepitus; it needs emergency debridement (see Ch 26).

> **KEY POINT**
>
> Fournier gangrene is fascial, so skin shows a fraction of the necrosis; testes are spared. Imaging must never delay surgery. Debride to bleeding tissue, with a second look at 24–48 h.

- **Ecthyma gangrenosum** (P. aeruginosa bacteraemia in the immunocompromised) gives black eschar and needs immediate IV antipseudomonals.
- A **genital bite** needs amoxicillin–clavulanate 875/125 mg bd, irrigation and debridement.

Hidradenitis suppurativa (HS)

- HS is a **follicle-centred** disease of apocrine-bearing skin that causes abscesses, **sinus tracts** and scars, with fistulae and **SCC in chronic scars** as complications.

Hurley stage	Features	Mainstay
I	Abscess(es), no tracts or scarring	Topical clindamycin, deroofing
II	Recurrent abscesses, tracts, scarring; separated areas	Oral tetracycline, or clindamycin and rifampicin; deroofing; adalimumab if refractory
III	Diffuse interconnected tracts	Adalimumab and **wide excision**, graft or flap

- Advise stopping smoking and weight loss; **adalimumab** is given 40 mg weekly. **Repeated incision and drainage is discouraged** because the tract recurs.

Red groin rash

Feature	Candidal intertrigo	Tinea cruris	Erythrasma	Inverse psoriasis
Scrotum	**Involved**	**Spared**	—	May be involved
Look	**Satellite lesions**	Scaly advancing border, central clearing	Red-brown patch	Non-scaling plaque
Test	KOH: yeast	KOH: hyphae	Wood's lamp: **coral-red**	KOH negative
Treat	Topical imidazole **≥2 weeks**; test for diabetes	Topical azole or allylamine; terbinafine 250 mg/day if deep; no steroids	Clindamycin 1% or miconazole; oral erythromycin if severe	Low-potency steroid

> **KEY POINT**
>
> Azoles cover yeast and dermatophyte, so use them when unsure. Nystatin misses tinea; allylamines are weaker on Candida. A negative KOH from an untreated lesion means rethink the diagnosis.

Infestations

- Pubic lice and scabies both get permethrin 5%, **repeated at 1 week**; for scabies treat the whole body and the household. Ivermectin 200 µg/kg ×2 is the alternative, and lindane is disfavoured. Screen lice patients for other STDs.

9 · Premalignant and malignant lesions

SAQ Premalignant conditions in urology (including premalignant lesions of penis)

Paper answer · P1

Asked as:

- Premalignant conditions in urology — *Summer 2021 Paper I Q3a (SAQ)*
- Premalignant lesion of penis — *Practice Set 6 Paper I Q3a (SAQ)*

Type: SAQ (10 marks)

(premalignant conditions also answered in 2021-Summer Paper 1, Q3a)

Definition

- Altered tissue with a raised risk of invasive cancer, often **intraepithelial neoplasia with an intact basement membrane**. Treating it prevents cancer.
- Penis: a lesion with raised risk of invasive SCC (squamous cell carcinoma). The key entity is **PeIN (penile intraepithelial neoplasia)**: dysplastic squamous epithelium with an **intact basement membrane**.

Penis

Classification

1. PeIN (HPV-associated or HPV-independent)

- **HPV (human papillomavirus)-associated** (commonest; basaloid and warty) or differentiated (HPV-independent).
- **Mechanism (HPV):** viral E6 inactivates p53, E7 inactivates Rb (retinoblastoma protein); **p16** is the marker.
- **HPV-independent (differentiated):** arises in lichen sclerosus.
- **WHO (World Health Organization) 2022:** all PeIN is high grade; grades I–III abandoned.

Clinical entity / lesion	Site, appearance / key points	Risk of invasion
Erythroplasia of Queyrat	Glans or prepuce; red velvety (glans) plaque	**Progresses in 10–33%**
Bowen disease	Shaft; scaly red (shaft) plaque	About 5% progress
Bowenoid papulosis	Shaft papules, young men; HPV 16	Does not progress; may regress

2. Other lesions

- **Lichen sclerosus (BXO [balanitis xerotica obliterans]):** SCC (squamous cell carcinoma) in **2.3–9%** (HPV-independent type).
- **Buschke–Löwenstein tumour** (giant condyloma; verrucous carcinoma): HPV 6/11; locally invasive, locally destructive, rarely metastasises; may harbour SCC.
- **Also know:** cutaneous horn; PEKMB (pseudoepitheliomatous keratotic and micaceous balanitis, controversial); condyloma acuminatum; (chronic) phimosis; genital lichen planus; Kaposi sarcoma (HHV-8 [human herpesvirus 8], check HIV [human immunodeficiency virus]).

Evaluation

- Biopsy first: **multiple biopsies of adequate depth** to exclude invasion; p16 staining. Examine inguinal nodes; partner screening for cervical neoplasia.

Treatment (organ-preserving)

Situation	Treatment
Prepuce (preputial lesions)	Circumcision or excision with clear margin
Glans	**Topical 5-FU (5-fluorouracil) or 5% imiquimod**; **laser** (Nd:YAG [neodymium:yttrium-aluminium-garnet] preferred; CO_2 too superficial); **glans resurfacing** with skin graft
Shaft	Excision; Mohs surgery

Situation	Treatment
Lichen sclerosus	Clobetasol 0.05% for 3 months then maintenance; circumcision; lifelong follow-up
Buschke–Löwenstein	Wide excision; avoid radiotherapy (anaplastic change)

- Long-term follow-up (recurrence common after organ-preserving treatment). **Prevention:** neonatal circumcision, HPV vaccination, avoiding smoking.

Bladder

- **CIS** (carcinoma in situ): flat high-grade lesion, umbrella cells lost; untreated, **40–83%** invade muscle. Treat with BCG (bacillus Calmette–Guérin).
- **Urothelial dysplasia:** 14–19% progress; 60% to CIS with prior cancer.
- **Cystitis glandularis** (intestinal type): low risk of adenocarcinoma; resect and survey.
- **Leukoplakia** (keratinising squamous metaplasia): traditionally premalignant; Campbell now says **not** (no TP53 changes). Schistosomiasis and chronic catheters predispose to SCC.

Prostate

- **HGPIN** (high-grade prostatic intraepithelial neoplasia): precursor; isolated HGPIN no longer needs repeat biopsy. IDC-P (intraductal carcinoma of the prostate) usually means invasive cancer nearby.

Testis

- **GCNIS** (germ cell neoplasia in situ): precursor of all postpubertal GCTs (germ cell tumours) except spermatocytic tumour; **50% invasive within 5 years**. Treat by orchidectomy or **radiotherapy ≥20 Gy**.
- **Gonadoblastoma** in dysgenetic gonads with Y material; cryptorchidism.

Kidney and others

- **Also know:** papillary adenoma; acquired cystic kidney disease on dialysis; urethral diverticulum; bowel in augmentation.

Diagram to draw: normal urothelium vs dysplasia vs CIS (umbrella cells lost); penile epithelium with full-thickness atypia above an intact basement membrane.

> **Key points to remember:**
> - PeIN: intact basement membrane; WHO 2022 all high grade.
> - Queyrat 10–33% vs Bowen (about) 5% progression; bowenoid papulosis does not progress.
> - Lichen sclerosus: SCC 2.3–9%.
> - Untreated bladder CIS: 40–83% invade.
> - GCNIS: 50% invasive in 5 years; radiotherapy ≥20 Gy.

Also, from the condensed book (not in the paper answer):

- Penile intraepithelial neoplasia is HPV-related (type 16) or non-HPV (LS, chronic LP); biopsy **several areas**.
- Verrucous carcinoma needs local excision or Mohs surgery, not primary radiotherapy (anaplastic transformation).

Lesion	Treatment
SCC in situ (SCCis) — accessible (scrotum)	Excision with a **5-mm margin** (favoured)
SCCis — glans	Mohs surgery, laser, cryotherapy; topical 5-fluorouracil (5-FU) or imiquimod (higher recurrence)
Bowenoid papulosis	Observe in a reliable young man; or 5-FU, imiquimod, cryotherapy, laser
PEKMB	Biopsy (exclude verrucous carcinoma); excision or ablation

Tumour	Key features	Treatment
Basal cell carcinoma	Rare; pearly papule	Excision or Mohs
Kaposi sarcoma	HHV-8; classic, AIDS and transplant types; a glans lesion may block the meatus	Restore immunity; excision, laser or vinblastine if solitary; radiotherapy 15–30 Gy if regional; chemotherapy if disseminated
Melanoma	<1% of penile cancers	Wide excision or partial penectomy; immunotherapy if advanced
Extramammary Paget	Red plaque at apocrine sites; CEA and EMA positive; underlying carcinoma in **10–30%** (urethra, prostate, bladder, rectum)	**Mandatory search for internal malignancy**; excision or Mohs
Mycosis fungoides	Patches progress to plaques, then tumours; Sézary syndrome if in blood	Ultrapotent steroid, phototherapy

10 · Benign male genital lesions

Lesion	Site / look	Management
Pearly penile papules	Coronal rows of angiofibromas in 14–48% of young men, mistaken for condyloma	Reassure
Fordyce spots	Pin-sized shaft papules (ectopic sebaceous glands)	Reassure
Angiokeratoma of Fordyce	Red-purple scrotal papules; may bleed	Observe; laser if bleeding
Zoon balanitis	Shiny red glans plaque; elderly uncircumcised men	**Biopsy**, then **circumcision (curative)**
Median raphe cyst	Ventral, near the glans	Excision

> **KEY POINT**
>
> Zoon balanitis and erythroplasia of Queyrat are both red glans plaques in uncircumcised men: one benign and cured by circumcision, the other SCCis. A persistent red glans plaque always needs biopsy.

11 · Pigmentary disorders

- Vitiligo is accentuated by Wood's lamp; unlike LS, it has normal texture without atrophy or scarring.

31 Tuberculosis and Parasitic Infections of the Genitourinary Tract

1 · Urogenital tuberculosis — organism and pathogenesis

TB seeds the urinary tract late and heals by scarring.

LAQ Genitourinary tuberculosis: pathogenesis, diagnosis, management and thimble bladder

Paper answer · P2 · P3 · merged from 2

Asked as: P3 LAQ "Renal and genitourinary TB: etiopathogenesis, presentation, (laboratory and newer) diagnosis and management" (Describe etiopathogenisis, clinical presentation, diagnosis and management of Renal TB?, Summer 2022 Paper III Q2 LAQ; Laboratory Diagnosis of Genitourinary Tuberculosis, Winter 2023 Paper III Q3f SAQ; Describe newer diagnostic modalities for Gu Kochs. Briefly outline role of all present diagnostic tests and give diagnostic algorithm, Practice Set 6 Paper III Q3d SAQ) · P2 LAQ "Describe pathophysiology of genitourinary tuberculosis and management of thimble bladder in 30-year male."

Assumed meaning: "Gu Kochs" = genitourinary tuberculosis.

Introduction

- Caused by the MTBC (*Mycobacterium tuberculosis* complex); *M. bovis* (and BCG (bacille Calmette–Guérin)) is innately resistant to pyrazinamide.
- UG (urogenital) TB (tuberculosis) is 15–20% of pulmonary TB in developing countries; the second commonest extrapulmonary site after lymph nodes. **Kidney is involved in 80%.**
- Untreated, it destroys the kidney and contracts the bladder.

Etiopathogenesis

- **Primary infection:** inhaled bacilli spread to hilar nodes, then by lymph and blood to the well-perfused kidney and epididymis, before immunity develops (up to 12 weeks).
- **Granuloma:** macrophages, epithelioid cells and Langhans giant cells in a lymphocyte cuff, maintained by IFN-γ (interferon-γ). In 90–95% it contains the infection and it stays latent; viable bacilli stay dormant.
- **Reactivation** (lifetime 5–15%), even decades later: old age, diabetes, renal failure, HIV (human immunodeficiency virus), malnutrition, steroids or other immunosuppression. **Caseation** then releases bacilli into urine.

Route	Lesion
Haematogenous (principal)	Parenchymal disease: kidney, epididymis, prostate
Descending urinary	Luminal disease: ureteric stricture, cystitis, urethra
Ascending (intravesical BCG, about 1%)	Cystitis, prostatitis, epididymo-orchitis
Contiguous	From psoas, bowel

- **Renal sequence:** 1. Bacilli lodge in **periglomerular capillaries**. 2. Cortical granulomas coalesce. 3. **Caseation**; cavities open into calyces. 4. Papillary necrosis, abscess, calcification. 5. Bacilluria seeds ureter and bladder.
- **Autonephrectomy:** caseo-cavernous type, or fibrotic calcified "putty" kidney.

Site	Frequency	Pathology
Kidney	80%	Periglomerular granulomas, then caseating cavities, papillary necrosis, calcification; **autonephrectomy** ("putty kidney")
Ureter	—	Mural granulomas; stricture, commonest at the **VUJ (vesicoureteric junction)**; "beaded corkscrew" or "pipestem" ureter
Bladder	—	Cystitis starts around the ureteric orifices ("golf-hole" orifice); ulcers; after about a year of fibrosis it contracts (<100 mL); **thimble bladder <20 mL**
Epididymis	Commonest genital site	Starts in globus minor; beaded vas, posterior scrotal sinus
Prostate, seminal vesicles	22–49%	Calcification, abscess; infertility

- **Key concept:** healing occurs by fibrosis, and fibrosis obstructs. Infundibular and ureteric strictures and bladder contraction can **worsen during treatment**.
- **Also know:** granulomatous interstitial nephritis; miliary renal TB; amyloidosis; squamous metaplasia of the pelvis (squamous cell carcinoma risk); ESRD (end-stage renal disease) in about 7%.

Clinical presentation

- Renal TB is silent until it reaches the bladder; asymptomatic in about 8%.
- **Storage LUTS (lower urinary tract symptoms)** and dysuria (commonest); haematuria and flank pain (about one-third); colic <10% (sloughed papillae, clot, debris).
- Constitutional symptoms <20%. Recurrent "UTI (urinary tract infection)" not responding to antibiotics.
- Signs: renal angle tenderness, flank sinus, beaded vas, epididymal nodule; hypertension; renal failure if bilateral.

Diagnosis: principles and specimen

- GUTB (genitourinary tuberculosis) is paucibacillary with intermittent shedding, so **multiple samples** are needed.
- **Hierarchy of proof:** positive culture or DNA, then caseating granulomas on histology, then clinical picture with response to treatment.
- **Specimen:** early-morning first-void urine (EMU), **3–5 samples on consecutive days**, processed at once (acidic urine kills bacilli). Avoid prior fluoroquinolones.
- Also tissue, pus, FNAC (fine-needle aspiration cytology) or semen.

Laboratory tests

Test	Method	Comment / role
Urine routine	Microscopy	**Sterile (acid) pyuria** with or without microscopic haematuria in >90%
ZN (Ziehl–Neelsen) smear	Carbol fuchsin, acid-alcohol, methylene blue: red bacilli. Auramine fluorescence is more sensitive	Quick; sensitivity under 50%; saprophytic mycobacteria give false positives

Test	Method	Comment / role
Urine AFB (acid-fast bacilli) culture: gold standard	Decontaminate and concentrate. **LJ (Löwenstein–Jensen)** solid medium: 4–6 weeks. **MGIT (mycobacteria growth indicator tube)** liquid: about 10 days; use a liquid system as well	80–90% sensitivity with ideal technique; diagnosis; gives DST (drug-susceptibility testing); **the only test that monitors treatment**
NAAT (nucleic acid amplification test): GeneXpert MTB/RIF	Automated cartridge PCR (polymerase chain reaction); about 2 hours (result in 1–2 days)	Rapid diagnosis; detects MTB and **rifampicin resistance** (MDR (multidrug-resistant) TB surrogate). **Xpert Ultra** is more sensitive. Stays positive after cure, so cannot monitor treatment
TrueNat	Indian chip-based NAAT	NTEP (National TB Elimination Programme)
LPA (line probe assay)	DNA strip hybridisation	Rifampicin, isoniazid or second-line drug resistance
LF-LAM (lateral-flow lipoarabinomannan)	Urine antigen strip, under 30 minutes; point-of-care	HIV with symptoms, CD4 <200 or advanced disease (WHO (World Health Organization) 2019)
Histopathology	Biopsy (bladder, epididymis)	Caseating granulomas; NAAT on tissue raises yield
TST (tuberculin skin test)	5 TU (tuberculin units) PPD intradermal; read induration at 48–72 h	Proves infection only, not active disease; BCG false positive
IGRA (interferon-γ release assay)	ESAT-6/CFP-10 blood test (QuantiFERON, T-SPOT)	Unaffected by BCG; cannot separate latent and active; shows infection, not active disease

Newer diagnostic modalities

- **Liquid culture (MGIT):** about 10 days versus 4–6 weeks on LJ.
- **NAAT:** GeneXpert MTB/RIF (Xpert Ultra more sensitive), about 2 hours, detects rifampicin resistance; TrueNat; **LPA** for isoniazid resistance.
- **Urine LF-LAM:** strip, under 30 minutes; WHO 2019.
- **IGRA:** unaffected by BCG.
- **Imaging:** CT (computed tomography) urography; **MR (magnetic resonance) urography with DWI (diffusion-weighted imaging)** grades fibrosis.

Supportive tests

- ESR, blood count, renal function, HIV, blood sugar.
- Sputum and **chest radiograph in all** (mandatory; lung TB coexists in up to one-fifth).

Imaging and endoscopy

Imaging	Findings / role
KUB (kidney–ureter–bladder) film	Calcification (24–44%); lobar "putty" kidney

Imaging	Findings / role
IVU (intravenous urography) / CT urography	**Moth-eaten calyx**, **phantom calyx** (infundibular stenosis), cavities, hydrocalycosis, **Kerr's kink**, pipestem or beaded corkscrew ureter, ureteric strictures, calcification, small bladder; IVU replaced by CT
Ultrasound	Children, pregnancy, scrotum; follows hydronephrosis during treatment
MRU (magnetic resonance urography)	Children, pregnancy
Cystoscopy and biopsy	**"Golf-hole" ureteric orifice**, ulcers; biopsy if culture negative (yield 19–52%)
Chest radiograph	Mandatory (lung TB in up to one-fifth)

Diagnostic algorithm

```
Suspicion (sterile pyuria, storage symptoms, epididymal nodule) -> 3-5 EMU: ZN + culture (LJ + MGIT) + Xpert -> chest X-ray; CT urography Positive -> treat; DST guides regimen Negative -> cystoscopy/biopsy, aspiration + NAAT on tissue Still negative -> empirical trial if clinically compatible
```

Medical management (first step in all)

- **Multidrug therapy** because bacilli exist in different metabolic states and single drugs select resistance.
- Baseline liver and renal function, HIV, hepatitis, diabetes; DOT (directly observed therapy).
- **2HRZE/4HR:** isoniazid, rifampicin, pyrazinamide, ethambutol for 2 months, then isoniazid and rifampicin for 4 months, with **pyridoxine** against isoniazid neuropathy. India's NTEP uses 2HRZE/4HRE.
- Complicated disease: some treat 12–24 months, or ≥6 months after surgery (P2: some extend to 9–12 months). The new 4-month rifapentine–moxifloxacin regimen is not used for UG TB.
- Toxicity: hepatitis (all except ethambutol); optic neuritis (ethambutol).
- **Monitor with ultrasound** for worsening obstruction; stent or PCN (percutaneous nephrostomy) early. Steroids for strictures: anecdotal only.
- MDR-TB: second-line drugs (bedaquiline, pretomanid, linezolid).

Surgical management

- About half need surgery, **after at least 4–6 weeks of ATT (antituberculous therapy)**, when inflammation and bacillary load have settled.

Problem	Procedure
Obstruction	Early **DJ (double-J) stent** (succeeds in 41%) or **PCN**
Localised polar lesion not responding	Partial nephrectomy
Non-functioning kidney + persistent disease or hypertension	**Nephrectomy** (laparoscopic preferred; ligate artery and vein separately)
PUJ (pelviureteric junction) stricture	Pyeloplasty; ureterocalicostomy
Lower ureteric stricture	Reimplantation, psoas hitch, Boari flap; ileal ureter if long
Contracted bladder	Augmentation; substitution if <20 mL

- A stricture not improving after 6 weeks of drugs needs open repair.
- Contested: nephrectomy of a non-functioning kidney without active disease.

Thimble bladder in a 30-year-old man: assessment

Assess	How	Why
Disease activity	Repeat urine AFB culture; bladder biopsy	Do not reconstruct with active TB
Capacity and compliance	Cystogram, cystoscopy under anaesthesia, urodynamics	Size and type of reconstruction
Upper tracts	Ultrasound, CT urography, MCU (micturating cystourethrogram)	VUJ stricture or reflux needs reimplantation
Renal function	Creatinine, GFR (glomerular filtration rate), split function	Bowel reservoir needs creatinine <2 mg/dL, GFR >35–40 mL/min
Urethra, bladder neck, sphincter	Urethrogram, endoscopy	Orthotopic reservoir needs a patent, continent outlet
Patient	Dexterity, motivation, fertility wishes	May need CIC (clean intermittent catheterisation)

Thimble bladder: surgical options

Bladder state	Operation
Half the bladder diseased, capacity <100 mL	**Augmentation ileocystoplasty** ("clam" or onlay of detubularised ileum)
Severely contracted	Augmentation with **ileocaecum** or sigmoid
Thimble bladder (<20 mL)	**Substitution cystoplasty:** supratrigonal excision of the diseased bladder, keeping trigone and bladder neck, and an orthotopic detubularised ileal or ileocaecal reservoir

- **Principles:** detubularise and fold the bowel into a spherical, low-pressure reservoir (Laplace's law; interrupts peristalsis). Anastomose widely to the trigone.
- **Ureters:** if the VUJ is strictured or refluxing, excise to healthy ureter and **reimplant into the reservoir**: an antireflux tunnel under a colonic taenia, or an isoperistaltic afferent ileal limb. A Boari flap or psoas hitch is not possible in a contracted bladder.
- **Avoid orthotopic reconstruction** with poor renal function, a strictured or incompetent urethra, inability to perform CIC, active TB, or short bowel or inflammatory bowel disease. Then use a continent catheterisable pouch (Mitrofanoff) or an **ileal conduit**.
- **Also know:** laparoscopic and robotic augmentation; transurethral incision of bladder neck contracture.

Complications of bladder reconstruction

- **Mucus** and catheter blockage; **stones**; bacteriuria and UTI; **hyperchloraemic metabolic acidosis**; vitamin B12 deficiency if ileum is lost; **perforation** (needs regular emptying); incomplete emptying needing CIC; nocturnal incontinence; late tumour in the reservoir.

Follow-up

- Relapse 6.3–22% (mean about 5 years), so follow up with urine AFB culture and imaging for up to 10 years.
- After reconstruction: also creatinine, bicarbonate, B12 and ultrasound.

Diagrams to draw:

- Kidney showing caseous cavity, phantom calyx and VUJ stricture.
- Stepwise diagnostic flowchart (algorithm above) from sterile pyuria to culture and DST.
- Supratrigonal excision of a thimble bladder with an ileocaecal substitute and ureteric reimplantation.

Key points to remember:

- Kidney 80%; periglomerular seeding; bladder disease descends from the kidney; heals by fibrosis, so obstruction worsens on treatment.
- Sterile pyuria; 3–5 early-morning urine AFB cultures; culture is the gold standard and the only test that monitors treatment.
- GeneXpert (Xpert) detects rifampicin resistance in 2 hours; cannot monitor treatment.
- LF-LAM for HIV with CD4 under 200.
- TST and IGRA show infection, not active disease.
- 2HRZE/4HR with pyridoxine; surgery after 4–6 weeks of ATT.
- Thimble bladder <20 mL: augment if capacity <100 mL; substitute if <20 mL; needs creatinine <2 mg/dL and a good urethra.
- Relapse up to 22%; follow up 10 years.

Also, from the condensed book (not in the paper answer):

- UG TB is **15–20%** of pulmonary TB in developing countries. Infection stays latent in 90–95%; lifetime reactivation is **5–15%**, so UG TB often appears without active lung disease.
- **M. bovis (and bacille Calmette–Guérin, BCG) is innately pyrazinamide-resistant**.
- Isolated urogenital (UG) TB is generally **not infectious**, except during aerosol-generating procedures.
- Ascending spread: intravesical BCG causes UG TB in **~1%**.

KEY POINT

UG TB is blood-borne and lands in the two best-perfused UG organs, kidney and epididymis. Blood-borne seeding gives parenchymal disease; descending urinary spread gives luminal disease (distal ureteric stricture, cystitis). Healing by fibrosis turns an infection into an obstruction.

2 · Clinical features and pathology by site

- It is indolent, and patients are repeatedly treated for urinary tract infection (UTI). Constitutional symptoms occur in **<20%**. **Lower urinary tract symptoms dominate**; haematuria or flank pain occurs in ~⅓. **The hallmark is sterile pyuria** (>90%).

Site	Frequency / route	Signature lesion
Kidney	**80%**, haematogenous	Caseating cavities, then autonephrectomy (up to ⅓); "putty" kidney; end-stage renal failure ~7%
Ureter	Descending	Stricture at the vesicoureteric junction (VUJ); beaded corkscrew ureter. **Stricture obstruction is an important cause of renal failure**
Bladder	Descending	Starts at orifices; trigone spared. After ~1 year <100 mL; "thimble" <20 mL

Site	Frequency / route	Signature lesion
Epididymis	10–55%; second commonest; bilateral in ~⅓	Starts in the **globus minor (tail)**; posterior scrotal sinus in up to half
Vas	Contiguous	Thickened, beaded
Testis	From epididymis	Hard tumour-like mass
Prostate	22–49%	Calcified hard gland. "Prostatitis" responds partly to quinolones, then relapses
Seminal vesicles	Via vas or ejaculatory ducts	Low-volume ejaculate, azoospermia, haemospermia; infertility may be the first symptom
Urethra	1.9–4.5%	With prostatic disease; urethroscrotal fistula
Penis	Exceedingly rare	Ulcer; papulonecrotic tuberculid

3 · Diagnosis

*See the paper answer **Genitourinary tuberculosis: pathogenesis, diagnosis, management and thimble bladder** (Section 3, Ch 31).*

Also, from the condensed book (not in the paper answer):

- KUB calcification progresses from punctate to globular to ring, and IVU may show cloverleaf calyces.

ALGORITHM

1. **Hierarchy of proof**
2. If culture or MTBC DNA is positive, then the diagnosis is made.
3. If negative, then caseating granulomas on tissue in the right context confirm it.
4. If still negative, then a compatible picture, exposure and response to empirical treatment suffice.
5. Always look for coexisting pulmonary TB (up to one-fifth): chest radiograph in every patient.

- Culture sensitivity is 10–90% because shedding is intermittent. The Ziehl–Neelsen (ZN) smear detects **<50%**. Acidic urine inhibits mycobacteria, so culture promptly.
- **GeneXpert** detects rifampicin resistance, a surrogate for multidrug-resistant (MDR) TB. The WHO uses it in the initial work-up of all TB, but a negative result does not exclude TB, and it **cannot monitor treatment** because dead bacilli shed DNA.
- Fine-needle aspiration cytology of an epididymal nodule is diagnostic in ~⅔. At cystoscopy ulcers mimic cancer, and a **"golf-hole" orifice** means imaging the upper tract.
- Tuberculin skin test (**TST**) cut-offs are **≥5 mm** for contacts and the immunosuppressed, **≥10 mm** for immigrants, high-risk workers and drug injectors, and **≥15 mm** with no risk factors. The interferon-γ release assay (**IGRA**) has no BCG false positives and is preferred for people from endemic countries. Both show exposure only.

Modality	Signs / role
Plain film (KUB)	Calcification in 24–44%; **lobar "putty" kidney** is pathognomonic of end-stage disease; ureteric calcification is **intraluminal** (schistosomiasis: mural)

Modality	Signs / role
Intravenous urography (IVU)	**Moth-eaten calyx**; **phantom calyx**; **Kerr's kink**; **pipe-stem ureter**; contracted bladder. Findings are mostly **obstructive**
CT urography	Standard; granuloma may mimic renal cell carcinoma
Ultrasound (US)	**Serial monitoring of hydronephrosis during treatment**
MRI	Diffusion-weighted imaging separates pyonephrosis from hydronephrosis

KEY POINT

Imaging signs are fibrosis made visible: destruction (moth-eaten calyx, cavities) heals by scarring (phantom calyx, Kerr's kink, pipe-stem ureter, golf-hole orifice, thimble bladder). So obstruction is the commonest finding and must be watched during treatment.

4 · Medical treatment

- Combine drugs to prevent resistance. Check baseline liver function, renal function and HIV status, and use directly observed therapy (DOT).

ALGORITHM

1. **Standard 6-month short course (2HRZE / 4HR)**
2. Intensive, 2 months: daily INH (H), rifampicin (R), pyrazinamide (Z) and ethambutol (E); stop E once fully susceptible.
3. Continuation, 4 months: HR daily (or thrice weekly by DOT).
4. Give pyridoxine throughout.

Drug	Daily adult dose	Signature toxicity	Monitor / note
INH	5 mg/kg (300 mg)	Neuropathy, hepatitis	Pyridoxine 25–50 mg
Rifampicin	10 mg/kg (600 mg)	Hepatitis, **drug interactions**	Antiretroviral and immunosuppressant levels; rifabutin with antiretrovirals
Pyrazinamide	Weight-banded 1000–2000 mg	Hyperuricaemia, hepatitis	M. bovis is resistant
Ethambutol	Weight-banded 800–1600 mg	Optic neuritis (red–green vision)	Check vision; the **only non-hepatotoxic** first-line drug

- Some give 12–24 months for complicated disease, as UG TB relapses more.
- **Monitor** liver function, vision, urine culture and **serial US**, because healing fibrosis can worsen strictures and bladder contraction.
- **Steroids** are anecdotal in UG TB; consider them if a stricture worsens on treatment.

5 · Surgical treatment

*About half need surgery, now mostly **reconstructive**.*

- The goal is organ preservation. Indications are obstruction or abscess, a non-functioning infected kidney, resistant hypertension and reconstruction.
- **Timing: 4–6 weeks after starting chemotherapy**, except in emergencies.

ALGORITHM

1. If uraemia or sepsis, then decompress urgently.
2. Try a retrograde double-J stent first (preferred; succeeds in < half, 41%).
3. If that fails, then try an antegrade stent.
4. If that fails, then percutaneous nephrostomy (PCN); loculated systems may need several.
5. Then repair definitively; if the kidney is unsalvageable, then nephrectomy.

- Removing a PCN without fixing the cause leaves a tuberculous cutaneous fistula.
- **Nephrectomy is established** for a non-functioning kidney with recalcitrant TB or resistant hypertension; **routine removal** of an asymptomatic non-functioning kidney is **contested**. Partial nephrectomy suits a polar lesion unresponsive to drugs.
- At nephrectomy, expect dense fibrosis and **ligate artery and vein separately** (late arteriovenous fistula).
- **Strictures** are commonest in the lower ureter, and endoscopic results are poor. Stent early for strictures arising on treatment and **image every stricture**; if worsening, give steroids, and **if there is no improvement after 6 weeks, operate**.

Site / gap	Procedure	Key point
Pelviureteric junction (PUJ), short	Dismembered pyeloplasty; flap if longer	Harder than congenital PUJ obstruction
PUJ unreconstructable	Ureterocalicostomy (lower pole)	Cover with capsule or omentum
Upper/mid ureter	Ureteroureterostomy; Davis intubated ureterotomy	—
Lower ureter	Excise, then ureteroneocystostomy	Bladder mobilisation gains 2–3 cm
Moderate gap	Psoas hitch	Up to **5 cm**; needs good capacity
Long gap	Boari flap, with or without hitch	**10–15 cm**; poor in a contracted bladder
Multiple / recurrent	**Ileal ureter**	Native ureter unusable

KEY POINT

Trap: in TB the bladder may itself be contracted, ruling out psoas hitch and Boari flap. Assess capacity first; ileal ureter, with or without augmentation, is then the option.

*See the paper answer **Genitourinary tuberculosis: pathogenesis, diagnosis, management and thimble bladder** (Section 3, Ch 31).*

Also, from the condensed book (not in the paper answer):

- Contracted bladder: operate for intolerable symptoms or capacity **<100 mL**, once chemotherapy controls disease; a thimble bladder needs orthotopic substitution, and the spared trigone allows supratrigonal substitution.
- **Genital** surgery is only for failed medical therapy: **epididymectomy** if the testis is spared, otherwise orchidectomy. Obstructive azoospermia is often irreversible, so offer sperm retrieval.

6 · BCG-related infection, relapse, special situations

- **BCG UG TB** (~1%) causes granulomatous prostatitis, epididymo-orchitis, renal abscess or sepsis. For organ infection give INH and rifampicin for 3–6 months, since **pyrazinamide is useless**; for sepsis give triple therapy and steroids, and stop BCG permanently.
- **Relapse** occurs in **6.3–22%**, at a mean of ~5 years, so follow up for **10 years**.

Situation	Change	Duration
MDR-TB (INH and R resistant)	Second-line; BPaL (bedaquiline, pretomanid, linezolid), not yet approved for extrapulmonary TB	> 18 months; 6–9 months with BPaL
Extensively drug-resistant (XDR) TB	MDR plus fluoroquinolone and bedaquiline/linezolid resistance; 5–6 drugs	≥ 2 years
Pregnancy	HRE and pyridoxine; Z omitted in the USA (WHO includes it); breastfeeding allowed	9 months (without Z)
HIV on effective antiretroviral therapy (ART)	Standard daily regimen; rifabutin with protease inhibitors	6 months (9 if not on ART)
Renal transplant	Rifamycin with drug-level monitoring, or a rifamycin-free regimen	≥ 18 months if rifamycin-free

- **HIV** raises TB risk 15–20-fold, with more dissemination but **less fibrosis and fewer strictures**.

7 · Schistosomiasis (bilharzia)

Trapped eggs cause obstruction and squamous bladder cancer.

Species	Distribution	Target / venous home	Snail
S. haematobium	Africa, Arabian Peninsula	Vesical/pelvic plexus — UG tract	**Bulinus**
S. mansoni	Africa, Middle East, S America	Mesenteric — liver, gut	Biomphalaria
S. japonicum	China, SE Asia	Mesenteric — liver, gut	Oncomelania

ALGORITHM

1. **Life cycle**
2. Cercariae penetrate intact skin in fresh water and pass via the lungs to the liver, where they pair.
3. Paired adults move to the vesical plexus and live 3–5 years.
4. Eggs (**terminal spine**) cross the bladder wall into urine and hatch as miracidia.
5. Miracidia infect Bulinus, which releases cercariae.

- **Retained eggs cause the disease**; adults do not multiply in humans. Children aged 5–15 carry the heaviest burden.
- The **egg granuloma** forms polypoid masses (active), then fibroses and calcifies into **"sandy patches"**, leading to obstruction, metaplasia and cancer. Obstruction is **distal**; hydroureter may be **tonic** (reversible once relieved) or **atonic** (fibrotic).
- **Salmonella** shelters in the worm, causing chronic bacteriuria; treat both.
- **Bladder cancer** is rated **Group 1** by IARC. Patients are younger (40–50), and **SCC is 60–90%**; mass drug treatment in Egypt shifted SCC to urothelial cancer.

- Acute infection gives swimmer's itch and **Katayama fever**; chronic infection (from 8–12 weeks) gives **painless terminal haematuria**. Men may get haemospermia and infertility.

> **KEY POINT**
>
> Trap: a negative urine egg count does not exclude severe disease; the most damaged kidneys belong to egg-negative, often asymptomatic, inactive-stage patients.

Test	Meaning	Limitation
Urine egg microscopy — **gold standard**	Active infection	Collect **09:00–15:00** (noon peak); negative early and in light or inactive disease
Rectal snip (squash prep)	Active or past infection; egg viability	Avoids bladder biopsy risks
Serology	Exposure	Stays positive, so it **never judges cure**
Circulating antigen	Live worms	Urine CCA is poor for S. haematobium
PCR	Most sensitive	Impractical in the field

- Imaging shows wall thickening, polyps and hydronephrosis on US, and a **calcified bladder ("fetal head in the pelvis")** with mural ureteric calcification on plain film.
- **Praziquantel (PZQ)** is the only WHO-listed drug: **40 mg/kg once**, or 20 mg/kg twice 6–8 h apart, curing 60–90%. PZQ is weak on immature worms, so repeat after several weeks.
- Active polypoid obstruction resolves with PZQ alone; late fibrotic stenosis needs surgery.

Complication	Surgery
Distal / intramural stenosis	Antireflux reimplantation; balloon dilatation (restenosis common); meatoplasty for meatal stenosis
Long / multifocal stenosis	Boari flap; ileal ureter; ileal conduit; long-term nephrostomy
Contracted bladder	Ileocystoplasty or diversion, always with PZQ
Chronic deep ulcer	Partial cystectomy (fulguration fails)

8 · Filariasis

In India, filariasis commonly causes hydrocele and chyluria.

- W. bancrofti causes >90% and is the **only cause of genital filariasis**; Brugia affects the leg. Microfilariae are **nocturnally periodic**, and doxycycline kills the Wolbachia endosymbiont.
- **Acute adenolymphangitis** is a **retrograde** lymphangitis. **Funiculoepididymitis** is a recurrent painful cord swelling that spares the testis and is often mistaken for tumour.
- **Hydrocele** is the commonest chronic lesion; elephantiasis comes late.
- Diagnosis uses a midnight smear, the **circulating antigen** card, and the US "**filarial dance sign**".

LAQ Discuss the Aetiopathogenesis, Clinical feature and management of Chyluria.

Paper answer · P3

Type: LAQ (25 marks)
Source: 2024-Winter Paper 3, Q2 (also answered in 2018-Summer Paper 1, Q1, with lymphatic

anatomy)

Definition

- **Chyluria** is the passage of **chyle** (intestinal lymph rich in chylomicrons and protein) in the urine, making it milky.
- It needs an abnormal **lymphourinary fistula**, usually at a renal calyceal fornix (pyelolymphatic fistula).

Relevant lymphatic anatomy

- Intestinal lacteals → intestinal trunk → **cisterna chyli** (L1–L2) → **thoracic duct** → left subclavian–jugular junction.
- Lumbar (para-aortic) trunks from the kidneys join the cisterna. Lymph flows cephalad and right to left.
- Valves normally stop backflow from the cisterna into renal hilar lymphatics.

Aetiology

Parasitic (commonest; India, China)	**Non-parasitic**
Wuchereria bancrofti (lymphatic filariasis; Culex mosquito in urban India)	Congenital: lymphangiectasia, renal lymphangioma
Rarely other parasites	Trauma or surgery (partial nephrectomy, RPLND [retroperitoneal lymph node dissection])
	Retroperitoneal tumour, fibrosis, tuberculosis, pregnancy
- Genital and urinary filariasis is almost always *W. bancrofti*; Brugia species spare the genitalia.	

Pathogenesis

- **Life cycle:** a mosquito deposits infective larvae, which mature in central lymphatics over 6–9 months. Adults live 5–7 years in retroperitoneal and genital lymphatics. Females release microfilariae with **nocturnal periodicity**.
- **Obstructive theory:** adult worms and the host reaction cause **lymphangitis and fibrosis** of lymphatics between the intestinal trunks and thoracic duct.
- Lymphatics below the block **dilate**, their **valves fail**, and chyle **refluxes** into renal hilar and peripelvic lymphatics.
- Rising pressure ruptures the thin-walled lymphatics into a calyceal fornix, creating a **pyelolymphatic fistula**. Rarely the fistula is in the ureter or bladder.
- Chyluria appears **earlier than elephantiasis**, is usually **intermittent** and may remit spontaneously.

Clinical features

- **Milky urine**, worse after fatty meals and on standing; may clear after fasting.
- **Chylous clots:** colic, retention.
- **Haematochyluria** (blood-stained chylous urine).
- **Systemic effects** of fat and protein loss: weight loss, **hypoproteinaemia**, oedema, anaemia, fatigue, lowered immunity.
- Endemic residence, filarial fever, hydrocele or lymphoedema.

- **Severity:** mild (intermittent, no systemic effect); moderate (persistent, clots); severe (haematochyluria, clot retention, weight loss, hypoalbuminaemia).

Evaluation

1. Confirm chyle

- **Ether (chloroform) test:** milky urine shaken with ether clears, because the fat dissolves.
- **Urine triglycerides** positive; Sudan III staining shows fat globules; urine protein raised.
- Differentiate from **phosphaturia** (clears with acetic acid) and **pyuria** (leucocytes, no clearing).

2. Filarial tests

- Night (midnight) blood smear: often negative by this stage.
- **Circulating filarial antigen** (card test), eosinophilia, PCR (polymerase chain reaction); ultrasound "filarial dance sign" in scrotal lymphatics.

3. Lateralise and localise the fistula

- **Cystoscopy after a fatty meal:** milky efflux from one ureteric orifice; bilateral in some.
- **Retrograde pyelography:** pyelolymphatic backflow.
- **Lymphoscintigraphy:** non-invasive; tracer seen in the kidney and bladder. MR lymphangiography or CT; pedal lymphangiography is the historical gold standard.

4. Baseline: albumin, haemoglobin, renal function.

Management

`Mild -> diet + antifilarial drugs Moderate/persist -> add endoscopic sclerotherapy (repeatable) Severe or failed -> renal hilar lymphatic disconnection` **1. Conservative**

- **Low-fat, high-protein diet with MCT (medium-chain triglyceride) oil**, plenty of fluids. MCTs are absorbed directly into portal blood, not lacteals, so less chyle forms.
- **DEC (diethylcarbamazine)** 2 mg/kg three times daily, traditionally for 12 days (a 1-day course is as effective). It kills microfilariae but only modestly kills adults.
- **Doxycycline** (anti-*Wolbachia*, 200 mg daily for 4–8 weeks); albendazole; ivermectin.

2. Endoscopic sclerotherapy (renal pelvic instillation)

- Ureteric catheter on the affected side, head-down position. Instil **silver nitrate (0.5–1%)** or **povidone-iodine**, repeated over several sessions.
- **Mechanism:** chemical lymphangitis and fibrosis seal the fistula.
- Good initial response; recurrence common.
- **Complications:** flank pain, haematuria, papillary necrosis, ureteric stricture, anuria; rarely renal failure or death (silver nitrate).

3. Surgery: renal hilar lymphatic disconnection (chylolymphatic disconnection)

- **Indications:** severe disease, failed sclerotherapy, haematochyluria with clot retention, marked weight loss.
- **Steps:** 1. Full renal mobilisation (nephrolysis). 2. **Skeletonise the renal artery and vein**, ligating, clipping or sealing every hilar lymphatic. 3. Strip the upper ureter; remove perinephric fat. 4. Nephropexy; optional **omental wrap** around the hilum. 5. On-table cystoscopy confirms clear efflux.
- Reflux occurs only through **hilar lymphatics**, so upper-pole dissection can be limited.

- **Approach:** open, laparoscopic (transperitoneal or retroperitoneal) or robotic. **Retroperitoneoscopic** ligation beats open (shorter operation and stay, less blood loss, fewer complications).
- **Also know:** lymphovenous anastomosis; autotransplantation (rare); nephrectomy obsolete.

4. Prevention

- Mass drug administration (albendazole with DEC or ivermectin), bednets; clip lymphatics during renal and nodal surgery.

Diagram to draw: retroperitoneal lymphatics with obstructed thoracic duct, dilated hilar lymphatics and pyelolymphatic fistula at a calyceal fornix.

> **Key points to remember:**
> - Chyluria = pyelolymphatic fistula, usually from *W. bancrofti* obstruction.
> - Ether test and urine triglycerides confirm; cystoscopy after a fatty meal lateralises.
> - Diet with MCT and DEC → silver nitrate sclerotherapy → laparoscopic hilar lymphatic disconnection.
> - Chyluria precedes elephantiasis and is often intermittent.

Also, from the condensed book (not in the paper answer):

- Most chyluria remits on conservative treatment; instil sclerosant on the side identified at cystoscopy.

Drug	Action	Regimen / caution
Diethylcarbamazine (DEC)	Microfilaricidal, modest on adults	Drug of choice: 6 mg/kg/day; a single-day course equals a 12-day course. **Contraindicated if onchocerciasis or Loa loa is possible** (blindness, encephalopathy)
Albendazole	Micro- and macrofilaricidal	400 mg bd for 21 days; single dose in mass treatment
Ivermectin	Microfilaricidal	150–400 µg/kg once; caution with Loa
Doxycycline	Anti-Wolbachia; kills adults	200 mg daily for 4–8 weeks; improves lymphoedema and hydrocele; not in pregnancy or children

- Mass treatment uses albendazole with DEC in India, or with ivermectin where Loa or onchocerciasis is co-endemic.
- **Hydrocele** recurs after aspiration, so **excise the sac**. In funiculoepididymitis excise nodules and preserve the testis. Elephantiasis needs reduction scrotoplasty.
- **Chyluria** occurs when dilated retroperitoneal lymphatics rupture into the collecting system, giving milky urine with fat and protein loss. Exclude TB, tumour and trauma.

9 · Hydatid disease and other parasites

Never puncture a hydatid cyst blindly.

SAQ Hydatid disease of kidney

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 19 SAQ Sheet, Q3d

Introduction

- Cystic echinococcosis (CE) by *Echinococcus granulosus*, a cestode. Dog: definitive host; sheep and humans: intermediate.
- Ingested eggs → liver (commonest), lung, then **kidney (third site, about 2–3%)** via the systemic circulation. Usually single, unilateral, cortical.

Pathology

- Wall: **pericyst** (host fibrous layer), **laminated membrane** (ectocyst), **germinal layer** (endocyst) producing brood capsules, scolices and daughter cysts ("hydatid sand").
- **Closed** versus **open** cyst (ruptured into the collecting system → **hydatiduria**).

Clinical features

- Grows 1–2 cm a year; long asymptomatic; flank mass, dull pain.
- **Hydatiduria** ("grape-skin" membranes in urine; pathognomonic) with colic, in a minority.
- Infection; rupture (anaphylaxis). Renal function usually preserved.

Investigations

- **Serology** (ELISA (enzyme-linked immunosorbent assay)) is adjunctive only (sensitivity 60–90%); eosinophilia; Casoni test obsolete.
- **Ultrasound and CT (computed tomography):** thick-walled cyst, wall calcification, daughter cysts, detached membranes ("water-lily sign"). Stage guides treatment.

WHO (World Health Organization) stage	Appearance	Activity
CE1	Unilocular, double-line wall	Active
CE2	Multivesicular, daughter cysts	Active
CE3a / CE3b	Detached membranes / daughter cysts in solid matrix	Transitional
CE4 / CE5	Solid "ball of wool" / calcified wall	Inactive

- Retrograde pyelogram shows communication; image liver and lungs. **Never aspirate blindly** (anaphylaxis, seeding).

Management

- **Albendazole 400 mg twice daily**, 1–6 months: alone for small or inoperable cysts; around surgery to reduce seeding (± praziquantel).
- **Surgery is the mainstay** (open, laparoscopic or robotic), nephron-sparing when possible: 1. Pack the field with scolicide-soaked swabs (hypertonic saline, povidone-iodine). 2. Inject scolicide only if **not** communicating (chemical injury to the collecting system). 3. **Pericystectomy, partial nephrectomy or deroofing**; close communications; omentoplasty or drain. 4. **Nephrectomy** for a non-functioning kidney.
- **PAIR (puncture, aspiration, injection, re-aspiration):** chapter says liver only, not kidney; some series report percutaneous renal treatment.
- Calcified inactive cysts (CE4, CE5) may be observed.

Diagram to draw: layers of a hydatid cyst with daughter cysts and a communication into a calyx.

Key points to remember:

- Kidney: third site (about 2–3%).
- Hydatiduria (grape-skin membranes) is pathognomonic of a communicating cyst.
- Albendazole 400 mg twice daily around surgery; no scolicide if communicating.

Also, from the condensed book (not in the paper answer):

- Pack the field with hypertonic (20%) saline swabs, and have the anaesthetist ready for anaphylaxis.
- Remove the cyst intact (pericystectomy) or do a partial nephrectomy; nephrectomy is only for a destroyed kidney.
- Enterobiasis, amoebiasis, trichomoniasis (see Ch 29) and fungal UTI (see Ch 26) are rarer genitourinary infections.

★ Putting it together: infection decisions

A · UTI: drug matches site

Situation	Choice	Principle
Uncomplicated cystitis	Nitrofurantoin 5 days, fosfomycin 3 g once, TMP–SMX 3 days (resistance <20%), pivmecillinam	Narrow, urine-confined
Pyelonephritis, complicated UTI, men	Quinolone or cephalosporin; carbapenem if resistant	Nitrofurantoin and fosfomycin do not reach tissue
ESBL producers	Fosfomycin or nitrofurantoin for cystitis; carbapenem if invasive	Co-resistance to quinolones
Asymptomatic bacteriuria	Treat only in pregnancy and before mucosa-breaching surgery	Otherwise harmful

B · Beyond antibiotics

ALGORITHM

1. **Image**: no response by 72 h, sepsis, obstruction, gas, febrile men.
2. **Drain** any obstructed infected kidney; treat the stone later.
3. Renal abscess: if ≤3 cm, then antibiotics; 3–5 cm, variable; if >5 cm, then percutaneous drainage.
4. **Emphysematous pyelonephritis**: gas within kidney: antibiotics and drainage; if beyond it, then nephrectomy (see Ch 26).
5. **Fournier: urgent repeated radical debridement**.

C · Recurrent UTI in women

ALGORITHM

1. Confirm each episode by culture.
2. Behaviour; vaginal oestrogen if postmenopausal.
3. Methenamine, cranberry, D-mannose.
4. Self-start or post-coital antibiotic; finally continuous prophylaxis.

D · Pelvic pain syndromes compared

	CP/CPPS	IC/BPS
Core	Male pelvic pain >3 months, no infection (NIH III)	Filling-related bladder pain, no other cause
Phenotyping	UPOINT; NIH-CPSI	Hunner vs non-Hunner
Treatment	By domain: α-blocker, pelvic-floor physiotherapy, neuromodulators, psychology	Behavioural, oral (amitriptyline, PPS), intravesical, Hunner fulguration, BTX-A; diversion last

E · GU tuberculosis in four lines

ALGORITHM

1. **Suspect:** sterile pyuria, contracted bladder, beaded vas.
2. **Confirm:** three early-morning urines for AFB, culture, GeneXpert.
3. **Treat:** 2HRZE, then 4HR; watch for ureteric stricture.
4. **Operate** after 4–6 weeks of drugs (see Ch 31).

Other paper questions in this section

Asked in the papers but not tied to a specific place in the book above.

LAQ Genitourinary infections, common organisms, bacteraemia and urosepsis: pathophysiology and management

Paper answer · P1

Asked as:

- Discuss general principles of genitourinary infections, characteristics of common organisms and management of bacteremia and septic shock. — *Summer 2016 Paper I Q2 (LAQ)*
- Describe Pathophysiology of urosepsis and its management. — *Summer 2017 Paper I Q2 (LAQ)*
- Describe pzthophysiology of urosepsis and its management. — *Practice Set 11 Paper I Q1 (LAQ)* (also answered in 2017-Summer Paper 1, Q2) — *"Pzthophysiology" is read as pathophysiology.*

Type: LAQ (25 marks)

Definitions

- **UTI (urinary tract infection):** urothelial inflammation from bacterial invasion, usually with bacteriuria and pyuria. **Sterile pyuria** needs evaluation (tuberculosis, stone, tumour).
- **Uncomplicated:** healthy non-pregnant woman, normal tract. **Complicated:** men, pregnancy, obstruction, stones, catheter, neurogenic bladder, diabetes, immunosuppression.
- **Recurrent:** ≥2 in 6 months or ≥3 in 12 months. Unresolved = never cleared; reinfection = new organism; **bacterial persistence** = same organism from a focus (stone, prostate).
- **Prophylaxis** assumes a sterile tract; **suppression** controls a focus that cannot be eradicated.
- **Urosepsis:** sepsis whose source is the urogenital tract (the urinary tract or male genital organs). It is usually Gram-negative and usually follows obstruction or instrumentation.

Pathogenesis of UTI

- **Routes:** ascending from bowel flora (commonest); haematogenous (*S. aureus*, *Candida*); lymphatic (rare).
- **Virulence (uropathogenic *E. coli*): type 1 pili** (mannose-sensitive) bind bladder uroplakins and cause cystitis. **P pili** (mannose-resistant) bind renal glycolipids and are found in about 90% of pyelonephritis strains. Haemolysin damages tissue. Bacteria invade umbrella cells and form intracellular communities; quiescent reservoirs cause recurrence.
- **Principle:** a strong host is infected only by virulent strains; a compromised host (obstruction, catheter) by almost any organism.
- **Host defences:** voiding washout, lactobacilli, urine osmolality and pH, uromodulin (Tamm–Horsfall protein), urothelial exfoliation.
- **Risk factors: obstruction** (stasis impairs defences and turns UTI life-threatening), VUR (vesicoureteral reflux; with infection it scars the growing kidney), stones, catheters, diabetes, pregnancy, spermicides.
- **Diagnosis:** dipstick (nitrite, leucocyte esterase) rules out better than it rules in. Culture is the gold standard: ≥10^5 CFU (colony-forming units)/mL (Kass); ≥10^2 in a symptomatic

woman or a catheter specimen. Image (ultrasound, CT (computed tomography)) in complicated infection or fever beyond 72 hours.

Characteristics of common organisms

Organism	Key features
E. coli	~85% of community and ~50% of hospital UTI; commonest cause of urosepsis; ST131 clone is ESBL (extended-spectrum β-lactamase) and quinolone resistant
Proteus	Urease splits urea into ammonia: alkaline urine (pH >7), **struvite stones**, persistence in stones
Klebsiella	Capsule; urease; ESBL and KPC (*K. pneumoniae* carbapenemase) producers
Pseudomonas	Nosocomial, catheter biofilm; needs ceftazidime, piperacillin, aminoglycoside, fluoroquinolone or meropenem
Enterococcus	**Always resistant to cephalosporins**; amoxicillin, nitrofurantoin; VRE (vancomycin-resistant)
S. saprophyticus	~10% of cystitis in young sexually active women
S. aureus	Haematogenous renal abscess; in urine, look for bacteraemia
Anaerobes, *Candida*	Abscesses, fistulae; *Candida* with catheters and diabetes (*Candida* in catheterised diabetics)

- **Urosepsis organisms:** *E. coli* (commonest), *Proteus*, *Klebsiella*, *Pseudomonas*, *Enterococcus*; ESBL (extended-spectrum β-lactamase) producers rising; *Candida* in catheterised diabetics.
- **Also know:** emerging *Aerococcus urinae*; *Ureaplasma* and *Mycoplasma* with negative routine cultures; *Chlamydia* and tuberculosis do not grow on routine culture (sterile pyuria).
- **Antibiotic principles (see 2015-Summer Paper 1, Q3f):** urine levels matter in cystitis, tissue levels in pyelonephritis and bacteraemia. Use a narrow spectrum and short course; adjust for renal function; **drain obstruction**.

Bacteraemia, sepsis, septic shock and urosepsis

Sepsis-3 (2016)

Term (Sepsis-3, 2016)	Definition
Sepsis	Life-threatening organ dysfunction from a dysregulated host response to infection: acute rise in **SOFA (Sequential Organ Failure Assessment) score ≥2**
qSOFA (quick SOFA)	Respiratory rate ≥22/min, altered mentation, systolic blood pressure ≤100 mmHg; ≥2 = high risk (of death) (prompt only)
Septic shock	Vasopressor needed for MAP (mean arterial pressure) ≥65 mmHg **and** lactate >2 mmol/L despite (adequate) fluids; mortality >40%
- SIRS (systemic inflammatory response syndrome) criteria were dropped from the definition as non-specific.	

Causes and risk factors (urological triggers)

- **Urological triggers:** infected obstructed kidney, PCNL (percutaneous nephrolithotomy), high-pressure ureteroscopy, prostate biopsy, TURP (transurethral resection of the prostate), infected stones, emphysematous pyelonephritis, abscess.
- **Obstruction with infection:** stone, PUJ (pelviureteric junction) obstruction, tumour, pregnancy, giving infected hydronephrosis or pyonephrosis. Pyonephrosis, emphysematous pyelonephritis, renal, perinephric and prostatic abscess.
- **Parenchymal infection:** emphysematous pyelonephritis, renal or perinephric abscess, prostatic abscess, Fournier gangrene.
- **Instrumentation:** PCNL (percutaneous nephrolithotomy), ureteroscopy with high irrigation pressure, TURP (transurethral resection of prostate), prostate biopsy (sepsis 0.3–3.1%), catheters.
- **Host:** diabetes, immunosuppression, old age, neurogenic bladder.

Pathophysiology

- **Pathophysiology (summary):** endotoxin binds TLR4 (Toll-like receptor 4). Cytokines (TNF-α (tumour necrosis factor-α), interleukins IL-1 and IL-6) and nitric oxide cause vasodilatation, capillary leak, myocardial depression and DIC (disseminated intravascular coagulation), then organ failure.

1. Entry of bacteria (and endotoxin) into blood

- Obstruction causes stasis, impairs washout and raises pelvic pressure. Bacteria and endotoxin then enter the blood (circulation) by **pyelovenous and pyelolymphatic backflow**.
- High irrigation pressure at endoscopy does the same, and so does fragmenting infected stones, which release endotoxin (fragmenting infected stones releases endotoxin). Stone culture predicts postoperative sepsis better than bladder urine culture.
- An obstructed infected kidney is (behaves like) an **abscess with a collecting system**. Antibiotics barely reach it, and bladder urine may be sterile.

2. Host response `Endotoxin (LPS, lipopolysaccharide) / (or) Gram-positive wall products → TLR4 (Toll-like receptor 4) on macrophages → NF-κB (nuclear factor κB) → cytokines TNF-α (tumour necrosis factor α), interleukins IL-1, IL-6, IL-8 → complement + coagulation activation + endothelial injury (activation)`

- **Vasodilatation:** inducible nitric oxide causes (lowers) low systemic vascular resistance ("warm" shock).
- **Capillary leak:** (endothelial) glycocalyx damage leads to hypovolaemia and tissue oedema.
- **Coagulopathy:** tissue factor release causes microthrombi, platelet consumption and DIC (disseminated intravascular coagulation).
- **Myocardial depression** and **mitochondrial dysfunction (failure):** cells cannot use oxygen, so lactate rises.
- **Immunoparalysis:** a later anti-inflammatory phase predisposes to (invites) secondary infection.

3. Organ dysfunction: AKI (acute kidney injury), ARDS (acute respiratory distress syndrome), encephalopathy, liver dysfunction (failure), DIC, multi-organ failure.

Clinical features

- Fever or hypothermia, rigors, flank pain, tachycardia, tachypnoea, confusion, oliguria, hypotension. Often starts (begins) within hours of an endourological procedure (endourology).

Investigations

- **Lactate**, blood cultures ×2 (two blood cultures), urine culture (and pelvic urine at drainage), blood count, creatinine, bilirubin, platelets, coagulation, arterial (blood) gas, glucose. Procalcitonin helps decide when to stop antibiotics.
- **Imaging:** bedside ultrasound (hydronephrosis, debris); **CT (computed tomography)** for stone, gas or abscess.

Management (Surviving Sepsis Campaign: hour-1 bundle 2018, guidelines 2021)

A. Resuscitation and antibiotics (SSC, Surviving Sepsis Campaign 2021) Recognise (SSC: do not use qSOFA alone for screening) → Lactate → blood cultures ×2 + urine culture → IV (intravenous) broad-spectrum antibiotics within 1 hour (shock or probable sepsis; within 3 h if sepsis only possible) → 30 mL/kg (balanced) crystalloid if hypotensive or lactate ≥4 mmol/L (within 3 h if hypotensive or hypoperfused) → noradrenaline first line for MAP ≥65 (target MAP 65); add vasopressin → SOURCE CONTROL: stent or nephrostomy (equally effective), drain abscess, change catheter; stone surgery later → refractory shock (persistent vasopressor need): hydrocortisone 200 mg/day; ICU (intensive care unit) → re-measure (repeat) lactate, capillary refill → de-escalate by culture at 48–72 h

- **Empirical antibiotics:** by local resistance: third-generation cephalosporin or piperacillin–tazobactam ± aminoglycoside; carbapenem if ESBL risk. Adjust for renal function.

B. Source control: the key urological step

- **Drain the obstructed infected kidney urgently:** retrograde **double-J stent** or **percutaneous nephrostomy (PCN)**. Randomised trials show equal effect on sepsis (equally effective in a randomised trial (Pearle, 1998)).
- Stent: suits coagulopathy; avoid high-pressure contrast injection.
- PCN: better for (suits) large or impacted stones, pyonephrosis; allows (gives) pelvic urine (for) culture.
- **No definitive stone surgery during sepsis.** Treat the stone after (once) the infection clears.
- Other sources: drain renal or perinephric abscess; emphysematous pyelonephritis (percutaneous drainage, nephrectomy if diffuse gas without fluid); prostatic abscess drainage; change or remove (the) catheter; debride Fournier gangrene.

C. Supportive care (Support)

- Oxygen and (or) ventilation, glucose control (insulin if glucose ≥180 mg/dL), transfusion (transfuse) below haemoglobin 7 g/dL, thromboprophylaxis, stress-ulcer prophylaxis, renal replacement if needed. Watch for AKI (acute kidney injury), ARDS (acute respiratory distress syndrome) and DIC.

D. De-escalation and duration

- Narrow antibiotics by (to) culture at 48–72 h. Usual course (usually) 7–10 days (another source: 7–14 days); longer with an undrained focus or *S. aureus* bacteraemia; shorter courses suffice after good source control. Step down (switch) to oral therapy when (once afebrile and) stable.

Special situations

- **Emphysematous pyelonephritis** (diabetics): fluids, glucose control, antibiotics, drain; nephrectomy for diffuse gas without fluid.
- **Pregnancy:** stent or nephrostomy; avoid fluoroquinolones.
- **Fungal urosepsis:** remove catheter, antifungal, drain fungal balls.
- **Prognosis:** urosepsis does better than sepsis from other sites if the source is drained early; delay in drainage (delayed drainage) is the main avoidable cause of death.

Prevention

- Sterile urine before elective endourology; antibiotic prophylaxis; low irrigation pressure and access sheath; short operating time with infected stones; stage the procedure if pus is found.

Diagram to draw: pathway from *E. coli* adherence to bacteraemia and cytokine-driven shock; obstructed kidney leading to backflow, cytokine cascade and shock; obstructed kidney → pyelovenous backflow → cytokine cascade → shock and organ failure.

> **Key points to remember:**
> - *E. coli* ~85% community UTI; P pili cause pyelonephritis; *Proteus* causes struvite; *Enterococcus* resists cephalosporins.
> - Sepsis = SOFA rise ≥2; septic shock = vasopressor for MAP ≥65 plus lactate >2 mmol/L.
> - Obstruction plus infection is the emergency; bladder urine may be sterile.
> - Hour-1: lactate, cultures, antibiotics, 30 mL/kg, noradrenaline (SSC 2021).
> - Infected obstructed kidney: drain urgently; no definitive stone surgery in sepsis.
> - Stent and nephrostomy equally effective (stent = nephrostomy for drainage); delay definitive stone surgery.

SAQ Principles of antimicrobial therapy in UTI

Paper answer · P1

Type: SAQ (10 marks)
Source: 2015-Summer Paper 1, Q3f

Definitions

- **Uncomplicated** urinary tract infection (UTI): healthy non-pregnant women, normal tract. **Complicated:** men, pregnancy, abnormal tract, catheter, immunosuppression.
- **Recurrent:** ≥2 in 6 months or ≥3 in 12 months.

Principles

- **Site decides the level:** urine in cystitis, tissue elsewhere; nitrofurantoin and fosfomycin lack tissue levels: **cystitis only**.

Killing pattern	Drugs	Dosing goal
Time-dependent (time above MIC [minimum inhibitory concentration])	β-lactams, carbapenems	Frequent doses or extended infusion
Concentration-dependent (peak:MIC)	Aminoglycosides, fluoroquinolones (FQs)	High once-daily dose

- **Resistance thresholds:** trimethoprim–sulfamethoxazole (TMP-SMX) if local resistance ≤20%; oral FQ for pyelonephritis if ≤10%.
- **Collateral damage:** FQs and cephalosporins select resistance and *C. difficile*; avoid them in cystitis.
- Culture all but simple cystitis; **drain** obstructed infected kidneys; shortest course.

Empirical choices

Scenario	Drug	Duration
Cystitis (women)	Nitrofurantoin 100 mg twice daily; fosfomycin 3 g; pivmecillinam 400 mg; TMP-SMX 160/800 mg twice daily	5 d; single dose; 3–5 d; 3 d
Cystitis (men)	TMP-SMX or per culture	7 d (= 14 d, Drekonja 2021)
Pyelonephritis (oral)	Ciprofloxacin 500 mg twice daily; levofloxacin 750 mg daily	7 d; 5 d
Pyelonephritis (intravenous)	Ceftriaxone 1 g daily; gentamicin 5–7 mg/kg once daily	10–14 d; oral when afebrile
Complicated / catheter UTI	Aminoglycoside or 3rd-generation cephalosporin per culture; change catheter	7–14 d
Urosepsis	Broad-spectrum **within 1 h**, source control	–

Asymptomatic bacteriuria

- Treat **only in pregnancy and before mucosa-breaching urological procedures**. Not in elderly, diabetic, catheterised or spinal cord injury patients.
- **Pregnancy:** avoid FQs and TMP-SMX; cephalexin is suitable.

Prophylaxis

- Recurrent: vaginal oestrogen, **methenamine hippurate** (non-inferior to antibiotics, ALTAR 2022), cranberry, post-coital antibiotics.
- Surgical: single dose within 1 hour; ≤24 h for percutaneous nephrolithotomy and prostheses.
- **Also know:** ESBL (extended-spectrum β-lactamase) producers; nitrofurantoin avoided if creatinine clearance <30 mL/min; gepotidacin (US 2025).

Diagram to draw: Concentration–time curve showing peak:MIC and time above MIC.

> **Key points to remember:**
> - Recurrent = ≥2 in 6 months or ≥3 in 12 months.
> - Cystitis: nitrofurantoin 5 d, fosfomycin single, TMP-SMX 3 d (resistance ≤20%).
> - Pyelonephritis: ciprofloxacin 7 d or levofloxacin 5 d (FQ resistance ≤10%).

SAQ Ciprofloxacin

Paper answer · P1

Type: SAQ (10 marks)
Source: 2017-Summer Paper 1, Q3d

Class and mechanism

- Second-generation **fluoroquinolone**, derived from nalidixic acid.
- Inhibits **DNA gyrase** (topoisomerase II, main target in Gram-negatives) and **topoisomerase IV** (Gram-positives). This blocks DNA supercoiling and replication: **bactericidal**.

Spectrum

- Enterobacteriaceae, ***Pseudomonas aeruginosa***, *Haemophilus*, *S. saprophyticus*, *S. aureus* (methicillin-sensitive); *Legionella*, some mycobacteria.
- Weak against streptococci and enterococci; **no anaerobic cover**, so bowel and vaginal flora are partly spared.

Pharmacokinetics

- Oral bioavailability about 70%; IV (intravenous) and oral forms; half-life about 4 h.
- **Concentration-dependent killing:** AUC/MIC (area under curve to minimum inhibitory concentration) predicts effect; post-antibiotic effect.
- High urine and tissue levels; **good prostatic penetration**.
- Renal excretion: reduce dose if creatinine clearance <30 mL/min.

Doses

- Cystitis 250 mg twice daily × 3 days; pyelonephritis 500 mg twice daily × 7 days (or 1 g extended-release daily); IV 400 mg 12-hourly.
- Chronic bacterial prostatitis: 4–6 weeks.

Indications and restrictions

- Pyelonephritis and complicated UTI (urinary tract infection) **only where local resistance ≤10%**; prostatitis; prophylaxis for transrectal biopsy (falling, because of resistance).
- **Not first line for uncomplicated cystitis** (FDA (US Food and Drug Administration) 2016; European restrictions 2019): use only when no alternative.

Resistance

- *gyrA*/*parC* mutations, efflux pumps, reduced porins, plasmid *qnr* genes. Linked to ESBL (extended-spectrum β-lactamase) *E. coli* (ST131). Main cause of post-biopsy sepsis.

Adverse effects

- GI (gastrointestinal) upset (commonest), photosensitivity, *C. difficile*.
- **FDA boxed warnings:** tendinopathy and **Achilles rupture** (2008; risk higher with age, steroids, renal failure); peripheral neuropathy (2013); disabling effects (2016); psychiatric effects and dysglycaemia (2018); aortic aneurysm and dissection (2018).
- CNS (central nervous system): seizures, confusion; QT prolongation.

Interactions

- Antacids, iron, zinc, calcium block absorption (separate by ≥2 h). Raises theophylline and tizanidine levels; potentiates warfarin.

Pregnancy and children

- Avoid: **cartilage damage** in immature animals. Also avoid in breastfeeding.

> **Key points to remember:**
> - DNA gyrase and topoisomerase IV; concentration-dependent.
> - Use for pyelonephritis only if local resistance ≤10%.
> - Tendon rupture, neuropathy, aortic aneurysm; avoid in pregnancy and children.

SAQ Complicated urinary tract infections.

Paper answer · P1

Type: SAQ (10 marks)
Source: 2018-Summer Paper 1, Q3f

Definition

- **Complicated UTI (urinary tract infection):** infection in a **structurally or functionally abnormal tract** or a **compromised host**, or with a resistant organism; more likely to fail or recur.
- **FDA (US Food and Drug Administration) 2018:** pyuria + pathogen in urine or blood + fever, chills, flank pain, with an abnormality or catheter.
- **Uncomplicated:** healthy non-pregnant woman, normal tract.

Complicating factors

Category	Examples
Obstruction	BOO (bladder outlet obstruction), PUJ (pelviureteric junction) obstruction, stricture, stone, tumour
Voiding dysfunction	Neurogenic bladder, high PVR (post-void residual), VUR (vesicoureteral reflux)
Foreign body / iatrogenic	Catheter, stent, nephrostomy, recent instrumentation, hospital-acquired
Anatomical	Diverticulum, fistula, cystocele
Host	**Male sex**, pregnancy, diabetes, immunosuppression, renal failure, transplant, elderly
Other	Symptoms >7 days; recent antibiotics

- Each causes stasis, biofilm or weak defences.

Organisms

- ***E. coli*** under half; more *Klebsiella*, *Proteus*, *Pseudomonas*, *Enterococcus*, *Candida*. Often **MDR (multidrug-resistant)** or ESBL (extended-spectrum β-lactamase) producers.

Evaluation

- **Urine culture mandatory before antibiotics**; blood cultures if febrile.
- Renal function, glucose, lactate.
- **Imaging:** ultrasound first; **CT (computed tomography)** if no response in 72 hours, stones, abscess or emphysematous infection suspected.

Treatment

- **Empirical IV (intravenous) therapy** by local resistance (EAU (European Association of Urology)): amoxicillin + aminoglycoside, 2nd-generation cephalosporin + aminoglycoside, or IV 3rd-generation cephalosporin. Avoid fluoroquinolones if used in the last 6 months or resistance >10%. Carbapenem or newer agents for ESBL (ceftolozane–tazobactam, ceftazidime–avibactam).
- **Source control is essential:** drain obstruction (**stent or nephrostomy, equally effective**), change catheter, drain abscess, treat stones later.
- Switch to oral when afebrile and stable; de-escalate by culture.
- **Duration:** usually 7–14 days (14 days in men if prostatitis not excluded); 7 days may suffice once afebrile ≥48 hours.
- **Sepsis:** antibiotics within 1 hour, fluids, vasopressors, ICU (intensive care unit).
- Correct the underlying cause, or infection recurs.

Special forms

- **Also know:** emphysematous pyelonephritis and cystitis (diabetics); pyonephrosis; renal and perinephric abscess; XGP (xanthogranulomatous pyelonephritis); CAUTI (catheter-associated UTI); funguria; unresolved bacteriuria (resistance, staghorn "critical mass").

Diagram to draw: Flowchart: suspect cUTI → culture → imaging → empirical IV → drain obstruction → oral step-down.

> **Key points to remember:**
> - Complicated = abnormal tract or compromised host; male sex alone qualifies.
> - Culture mandatory; often MDR.
> - Drain obstruction: stent = nephrostomy.
> - 7–14 days; oral step-down when afebrile.

SAQ Drug Fosfomycin

Paper answer · P1

Type: SAQ (10 marks)
Source: 2022-Summer Paper 1, Q3c

Introduction

- Bactericidal **phosphonic acid derivative** antibiotic, chemically unrelated to other classes. Oral form: **fosfomycin trometamol**; intravenous form: fosfomycin disodium.

Mechanism of action

- Enters bacteria through glycerol-3-phosphate and glucose-6-phosphate transporters.
- Irreversibly inhibits **MurA** (enolpyruvyl transferase), blocking the **first step of peptidoglycan (cell wall) synthesis**. β-lactams and vancomycin act at later steps, so **cross-resistance is limited**.

Pharmacokinetics

- Oral bioavailability is modest, but the drug is excreted **unchanged in urine**. One 3 g dose keeps high urinary concentrations for about 1–2 days, which permits single-dose therapy. Low tissue levels.

- Concentrations fall in renal failure; avoid at very low creatinine clearance.

Spectrum

- **Escherichia coli**, including **ESBL (extended-spectrum β-lactamase)** producers; most Enterobacteriaceae (Klebsiella variable); *Enterococcus* including **VRE (vancomycin-resistant enterococcus)**; staphylococci.
- **Not reliable:** *Pseudomonas aeruginosa*, *Acinetobacter*.

Uses in urology

Indication	Dose
Uncomplicated cystitis in women (first-line, EAU (European Association of Urology) and IDSA (Infectious Diseases Society of America))	3 g oral sachet, single dose
Cystitis in pregnancy, asymptomatic bacteriuria (safe, category B)	3 g single dose
MDR (multidrug-resistant) lower UTI (urinary tract infection)	Oral option avoiding IV (intravenous) therapy; repeated doses used off-label
Prophylaxis for transrectal prostate biopsy	As effective as fluoroquinolone
Recurrent UTI prophylaxis	3 g every 10 days (EAU)

- **Not for pyelonephritis** or urosepsis orally (poor tissue levels). Emerging data suggest IV fosfomycin may help in bacteraemic MDR E. coli UTI.
- Not needed as prophylaxis for flexible cystoscopy.

Adverse effects

- Diarrhoea, GI (gastrointestinal) upset, headache, vaginitis; rare hypersensitivity. IV form: **sodium load**, hypokalaemia.

Resistance

- Transporter mutations, MurA changes, plasmid **fosA** enzymes. Resistance develops quickly in vitro, so systemic infections use combinations.

Evidence

- Huttner (2018) RCT (randomised controlled trial): nitrofurantoin 5 days gave higher clinical resolution than single-dose fosfomycin (70% vs 58%).

Diagram to draw: peptidoglycan synthesis showing fosfomycin (MurA), β-lactams and vancomycin sites.

> **Key points to remember:**
> - Inhibits MurA: first step of cell wall synthesis; no cross-resistance.
> - 3 g single dose for cystitis; active against ESBL E. coli and VRE.
> - Not for pyelonephritis; poor Pseudomonas cover.

SAQ Papillary necrosis.

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 7 Paper 1, Q3a

Definition

- **RPN (renal papillary necrosis):** ischaemic necrosis of the renal papillae (inner medulla), which may slough into the collecting system.

Pathophysiology

- The papilla is vulnerable: it depends on the **vasa recta**, has **low oxygen tension** and a **hypertonic** environment. Almost every cause reduces medullary blood flow or oxygenation.
- **Sickle cell:** hypoxia and hypertonicity make cells sickle and block the vasa recta, causing microinfarcts.
- **Analgesics:** NSAIDs (non-steroidal anti-inflammatory drugs) block vasodilator prostaglandins; metabolites concentrate in the medulla.
- **Diabetes:** microangiopathy plus infection.

Causes ("POSTCARDS")

Letter	Cause
P	Pyelonephritis
O	Obstruction
S	Sickle cell disease or trait
T	Tuberculosis
C	Cirrhosis
A	Analgesic abuse
R	Renal transplant rejection; renal vein thrombosis
D	**Diabetes mellitus**
S	Systemic vasculitis
- **Also know:** infant dehydration and hypoxia; shock; renal candidiasis.	

Clinical features

- **Acute:** fever, flank pain, haematuria, **ureteric colic from a sloughed papilla**, sepsis; bilateral cases may cause AKI (acute kidney injury).
- **Chronic or incidental:** polyuria (poor concentration), sterile pyuria, recurrent UTI (urinary tract infection), CKD (chronic kidney disease). Retained papillae may **calcify** and harbour infection, including fungi.

Investigations

- Urine microscopy and culture; **strain urine** and send tissue for histology; creatinine, glucose, haemoglobin electrophoresis.
- **CT urography or IVU (intravenous urography):** early contrast tracking into the papilla ("**ball-on-tee**", "**lobster claw**"); "**ring sign**" (contrast around a detached papilla); clubbed calyces; filling defects; triangular ring-shaped calcification.
- Retrograde pyelography when renal function is poor.
- **Differential diagnosis:** tuberculosis, medullary sponge kidney, calyceal diverticulum, upper tract urothelial carcinoma.

Management

- **Treat the cause:** stop analgesics, control diabetes, hydrate, manage sickle crisis.
- **Culture-guided antibiotics.**
- **Obstruction by a sloughed papilla:** urgent drainage (stent or nephrostomy), then ureteroscopic extraction.
- Remove calcified retained papillae that cause persistent infection.
- **Follow-up:** renal function; analgesic abusers risk urothelial cancer.

Diagram to draw: calyx showing ring sign and lobster-claw appearance.

> **Key points to remember:**
> - Vasa recta, low oxygen and hypertonicity make the papilla vulnerable.
> - POSTCARDS; diabetes, analgesics, sickle cell, obstruction with infection.
> - Ring sign; sloughed papilla causes colic and obstruction: drain urgently.

SAQ Cephalsporins

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 8 Paper 1, Q3c

Assumed meaning: cephalosporins (generations, spectrum, urological use).

Introduction and mechanism

- **β-lactam** antibiotics. They bind **PBPs (penicillin-binding proteins)** and block peptidoglycan cross-linking, so the cell wall fails: **bactericidal**, time-dependent killing.
- Mostly renal excretion (adjust dose in renal failure); **ceftriaxone** also has biliary excretion (no adjustment).
- Resistance: β-lactamases (**ESBL [extended-spectrum β-lactamase]**, AmpC), altered PBPs, reduced porin entry.

Generations

Generation	Examples	Spectrum	Urological use
First	Cefazolin (intravenous), cephalexin, cefadroxil (oral)	Gram-positive (not MRSA [methicillin-resistant *Staphylococcus aureus*]); *E. coli*, *Klebsiella*, *P. mirabilis*	**Surgical prophylaxis (cefazolin)**; cystitis in pregnancy; paediatric prophylaxis
Second	Cefuroxime, cefaclor; cefoxitin, cefotetan	Wider Gram-negative; cefoxitin and cefotetan add **anaerobes**	Prophylaxis for vaginal and bowel surgery; SWL (shock-wave lithotripsy)
Third	Ceftriaxone, cefotaxime, cefixime, cefpodoxime; **ceftazidime**	Most Enterobacterales; **only ceftazidime covers *Pseudomonas***	Pyelonephritis, urosepsis; **gonorrhoea (ceftriaxone 500 mg IM [intramuscular])**; prostate biopsy prophylaxis
Fourth	Cefepime	Gram-negatives incl. *Pseudomonas*, stable to AmpC; Gram-positive	Complicated, nosocomial UTI (urinary tract infection)
Fifth	Ceftaroline	MRSA + Gram-negatives	Limited urological role

- **Newer combinations** for complicated UTI with multidrug resistance: ceftazidime–avibactam, ceftolozane–tazobactam, cefiderocol.

Gaps in spectrum

- **Enterococci** (intrinsically resistant; never treat with cephalosporins), *Listeria*, MRSA (except ceftaroline), atypical organisms.
- ESBL and AmpC producers hydrolyse third-generation agents.

Adverse effects

- Hypersensitivity, less than penicillins. Cross-reactivity with third-generation agents is <1%; avoid after immediate (IgE [immunoglobulin E]-mediated) penicillin reaction.
- ***C. difficile* colitis**; "collateral damage" (selection of ESBL organisms).
- Positive Coombs test; bleeding with cefotetan and cefamandole.
- Ceftriaxone: biliary sludge; avoid in jaundiced neonates and with intravenous calcium.

Paediatric use

- Febrile UTI in neonates: **ampicillin + third-generation cephalosporin** (covers enterococcus). Oral cefixime 8 mg/kg/day. A first-generation agent is a good narrow-spectrum choice for many outpatient UTIs.

Diagram to draw: β-lactam ring binding a PBP and blocking peptidoglycan cross-links.

> **Key points to remember:**
> - Cefazolin = standard surgical prophylaxis.
> - Ceftriaxone misses *Pseudomonas*; ceftazidime and cefepime cover it.
> - No cephalosporin covers enterococci; ESBL hydrolyses third-generation agents.

SAQ Biofilm

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 18 SAQ Sheet, Q3c (also answered in part in 2015-Summer Paper 3, Q3d)

Definition

- A **structured microbial community in a self-made polymeric matrix**, adherent to a living or inert surface. The matrix contains exopolysaccharide, extracellular DNA and proteins.

Stages of formation

1. **Conditioning film:** host proteins coat the device within minutes.
2. **Attachment:** reversible, then irreversible (pili, adhesins, flagella).
3. **Matrix production and microcolonies.**
4. **Maturation:** 3-D (three-dimensional) structure with water channels; cell-to-cell signalling (quorum sensing).
5. **Dispersion:** free bacteria seed new infection.

Why antibiotics and immunity fail

- Poor drug penetration through the matrix.
- Slow-growing, dormant cells resist drugs that need active growth.

- Altered genes, envelope and targets.
- Survive **1000–1500 times** the usual killing concentration. Neutrophils are excluded.

Urological examples

Site	Point
Urethral catheter	Forms within minutes; **the catheter, not the bladder, is infected**. Proteus urease makes crystalline biofilm → encrustation and blockage
Ureteric stent, nephrostomy	Stent colonised more often than urine shows; nephrostomy changed every 3 months
Penile prosthesis, AUS (artificial urinary sphincter)	Revision of an "uninfected" device: about 10% infection
Infection stones	Bacteria inside struvite; antibiotics cannot sterilise
Chronic bacterial prostatitis	Biofilm commoner in prostatitis *E. coli*; needs longer, penetrating therapy
Recurrent cystitis	**Intracellular bacterial communities**: biofilm-like pods inside umbrella cells

- Common organisms: Pseudomonas, Proteus, *E. coli*, Enterococcus, coagulase-negative staphylococci, Candida.

Prevention and management

- Avoid devices; shortest dwell; aseptic insertion; closed dependent drainage.
- **Change the catheter** when treating symptomatic CAUTI (catheter-associated UTI (urinary tract infection)); remove infected stent or implant.
- Silver-alloy catheters reduce bacteriuria, not clinical UTI.
- Implants: antibiotic-coated devices, no-touch technique, **antiseptic washout** at revision (infection about 10% → 3%).
- Stones: complete clearance. Chemodissolution for catheter encrustation. Macrolides penetrate biofilm (prostatitis).
- **Also know (research):** quorum-sensing inhibitors, bacteriophages, new coatings.

Diagram to draw: five-stage biofilm cycle on a catheter surface.

> **Key points to remember:**
> - Biofilm: microbes in self-made matrix on a surface.
> - Survives 1000–1500 times normal antibiotic levels.
> - Treat CAUTI by changing the catheter.
> - Revision washout cuts prosthesis infection from about 10% to 3%.

SAQ Renal mucormycosis

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 19 SAQ Sheet, Q3c

Introduction

- Mucormycosis: an **angioinvasive** fungal infection by Mucorales (*Rhizopus* commonest; *Mucor*, *Lichtheimia*).

- Kidney involvement occurs in **disseminated** disease, or as **isolated renal mucormycosis**, reported particularly from India, sometimes in apparently immunocompetent young adults. Cases rose during the 2021 COVID-19 (coronavirus disease 2019) wave.

Risk factors

- Uncontrolled diabetes, especially **ketoacidosis**; corticosteroids; neutropenia and haematological cancer; transplantation; iron overload or deferoxamine; intravenous drug use; AIDS (acquired immunodeficiency syndrome).

Pathogenesis

- Spores inhaled or ingested → **haematogenous** seeding of the kidney.
- Hyphae invade arteries and veins → **thrombosis → infarction and haemorrhagic necrosis**.
- Acidosis and free iron favour growth (hence the ketoacidosis risk).

Clinical features

- Fever, flank pain, tender flank mass, haematuria, pyuria with sterile bacterial cultures.
- **Bilateral** disease → AKI (acute kidney injury) or anuria.
- Suspect when "pyelonephritis" fails antibiotics in a high-risk patient.

Investigations

- **CECT (contrast-enhanced computed tomography):** enlarged kidney(s), wedge-shaped or diffuse **non-enhancing** areas (infarcts), perinephric stranding or collection, renal vessel thrombosis.
- **Tissue diagnosis** (biopsy or nephrectomy specimen): **broad, ribbon-like, aseptate or pauciseptate hyphae branching at right angles**, invading vessels.
- KOH (potassium hydroxide) mount, calcofluor stain, fungal culture (often negative), PCR (polymerase chain reaction).
- Differential: *Aspergillus* has narrow septate hyphae with acute-angle branching.
- Image sinuses, lungs, brain.

Management (three pillars)

1. **Reverse predisposition:** control glucose, correct ketoacidosis, taper steroids and immunosuppression, stop deferoxamine.
2. **Early surgical debridement:** nephrectomy for a unilateral necrotic kidney; drain collections.
3. **Antifungal: liposomal amphotericin B 5 mg/kg/day** first line (ECMM/MSG ERC (European Confederation of Medical Mycology/Mycoses Study Group Education and Research Consortium) guideline, 2019). Step-down or salvage: **posaconazole or isavuconazole**. - **Fluconazole, voriconazole and echinocandins are inactive.** - Bilateral: antifungal plus dialysis.

Prognosis

- Mortality is high, especially with bilateral or disseminated disease and delayed diagnosis.

Diagram to draw: right-angle branching aseptate hypha invading a renal vessel.

Key points to remember:

- Angioinvasion → infarction; diabetic ketoacidosis is the key risk.
- CECT: non-enhancing enlarged kidney; histology: broad aseptate right-angle hyphae.
- Liposomal amphotericin B 5 mg/kg/day plus nephrectomy and glucose control.
- Voriconazole and echinocandins do not work.

SAQ Nitrofurantoin.

Paper answer · P4

Type: SAQ (10 marks)
Source: 2020-Summer Paper 4, Q3e

Introduction

- A synthetic **nitrofuran**; **first-line for acute uncomplicated cystitis** (EAU (European Association of Urology)) because it concentrates in urine and spares bowel and vaginal flora.

Mechanism

- Reduced by bacterial flavoproteins (nitroreductases) to **reactive intermediates** that damage ribosomal proteins, DNA and several enzyme systems.
- **Bacteriostatic**, bactericidal at high urinary concentration.
- Multiple targets keep **acquired resistance low**.

Pharmacokinetics

- Well absorbed (better with food); **rapidly excreted in urine**. Serum and tissue levels are sub-therapeutic.
- Hence **useless for pyelonephritis, prostatitis, febrile or bloodstream infection**.
- Macrocrystals: slower absorption, less nausea.

Spectrum

Susceptible	Resistant
E. coli (most), ***Enterococcus faecalis***, *Staphylococcus saprophyticus*, *S. aureus* (not MRSA (methicillin-resistant *S. aureus*)), many *Klebsiella*, *Citrobacter*	***Proteus*** (intrinsic), *Pseudomonas*, *Serratia*, *Providencia*, *Morganella*, *Acinetobacter*

- Active against many ESBL (extended-spectrum β-lactamase)-producing *E. coli*.

Uses and doses

Indication	Regimen
Uncomplicated cystitis (women)	**100 mg twice daily × 5 days** (modified release)
Recurrent UTI (urinary tract infection) prophylaxis, adults	50–100 mg at night, or post-coital
Children: treatment / prophylaxis	3–5 mg/kg/day in 2 doses / 1–2 mg/kg at night (VUR (vesicoureteral reflux), after 3 months of age)
Pregnancy	Second and third trimesters; first trimester only if no alternative; **avoid near term**

Adverse effects

- **GI (gastrointestinal) upset** (commonest).
- **Pulmonary**: acute hypersensitivity (fever, cough, dyspnoea); **chronic interstitial fibrosis** with long-term use.
- **Peripheral neuropathy** (worse with renal impairment, diabetes, anaemia, B-vitamin deficiency).
- **Haemolysis in G6PD (glucose-6-phosphate dehydrogenase) deficiency**, including the neonate (hence avoid at term and <3 months of age).
- Hepatotoxicity (cholestatic or chronic hepatitis), rare.

Precautions

- **Renal impairment**: inadequate urine levels and more toxicity. Historically avoided if CrCl (creatinine clearance) <60; Beers Criteria (2015) allow short courses down to **CrCl 30 mL/min**; avoid long-term use in the elderly.
- Interactions: probenecid (blocks excretion), magnesium antacids, quinolones (antagonism).

Diagram to draw: spectrum table (susceptible versus resistant).

> **Key points to remember:**
>
> - Urine-only drug: cystitis yes, pyelonephritis or prostatitis no.
> - 100 mg twice daily × 5 days; prophylaxis 50–100 mg at night (children 1–2 mg/kg).
> - Proteus intrinsically resistant; resistance in E. coli low.
> - Toxicity: pulmonary fibrosis, neuropathy, G6PD haemolysis; CrCl ≥30 (Beers 2015).

Rapid Revision Sheet · Section 3

Antimicrobial resistance (SAQ)

- Plasmid-mediated resistance is commonest.
- ESBL: plasmid, co-resistant; KPC/NDM: carbapenem resistance.
- TMP-SMX if resistance ≤20%; fluoroquinolone for pyelonephritis if ≤10%.
- Stewardship: culture, narrow, short, no ASB treatment.

ESBL producing enterobacterial UTI (SAQ)

- Plasmid CTX-M; resists 3rd-gen (third-generation) cephalosporins; inhibited by clavulanate (clavulanate inhibits).
- Oral: nitrofurantoin, fosfomycin, pivmecillinam.
- Serious infection: carbapenem; MERINO: pip-taz (piperacillin–tazobactam) inferior.
- Stewardship prevents spread.

Emphysematous pyelonephritis: prognostic factors and conservative management (SAQ)

- Huang–Tseng: 1 pelvis, 2 parenchyma, 3A perinephric, 3B pararenal, 4 bilateral or solitary.
- Four risk factors: thrombocytopenia, AKI, altered sensorium, shock.
- Wan type I (no fluid) 50–60% mortality; type II <20%.
- Drainage first; nephrectomy for extensive disease with ≥2 risk factors or drainage failure.

Renal and perinephric abscess (SAQ)

- Now mostly ascending Gram-negative abscess in a damaged kidney.
- Perinephric: within Gerota fascia; diabetes in a third.
- Fever beyond 4–5 days of antibiotics: get a CT.
- CT is the investigation of choice; follow until resolution.
- ≤3 cm antibiotics; 3–5 cm either; >5 cm percutaneous drainage.
- Percutaneous drainage first line; nephrectomy if non-functioning.

Xanthogranulomatous pyelonephritis. (SAQ)

- Stone, obstruction, Proteus infection → foamy macrophages.
- CT: non-functioning enlarged kidney, contracted pelvis, "bear paw" sign.
- Mimics RCC; treat by complete nephrectomy, not drainage.

Malakoplakia (SAQ)

- Defective macrophage digestion of *E. coli*.
- Michaelis–Gutmann bodies are pathognomonic.
- Fluoroquinolones + bethanechol/vitamin C; nephrectomy for unilateral renal disease. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

Catheter Associated Urinary Tract Infections (CAUTI) (SAQ)

- Bacteriuria 5–10% per catheter-day; duration is the key risk.
- Diagnosis: symptoms + ≥10^3 CFU/mL from a fresh catheter.
- Remove catheter early; closed drainage; treat only symptomatic CAUTI, 7 or 10–14 days.

Fungal UTI (SAQ)

- Candida commonest; *C. glabrata* often fluconazole-resistant.
- Asymptomatic candiduria: remove catheter, no antifungal (except neutropenia, neonate, procedure).
- Fluconazole 2 weeks; echinocandins fail in urine.
- Fungus ball: drain, antifungal, remove.

Chronic (non-bacterial) prostatitis: diagnosis, classification and management (SAQ)

- NIH I–IV; III = pain ≥3 of 6 months, no infection; IIIA inflammatory, IIIB not.
- CBP: same organism, fluoroquinolone 4–6 weeks.
- CP/CPPS: NIH-CPSI to score, UPOINT to phenotype and treat.
- Multimodal therapy; α-blocker plus antibiotic only early.

Interstitial cystitis / bladder pain syndrome (LAQ)

- AUA definition: bladder pain with LUTS >6 weeks, no other cause; ESSIC >6 months.
- AUA 2022: no tiers; phenotype-based; cystoscopy if over 50 or HL suspected; anaesthetic challenge, not potassium test (obsolete).
- HL: fulguration or triamcinolone; ciclosporin if refractory.
- Amitriptyline, PPS, DMSO; PPS causes maculopathy; antibiotics and BCG not offered.

Lymphogranuloma venereum (SAQ)

- Serovars L1–L3; invasive and lymphotropic.
- Proctitis is now the commonest presentation.
- Groove sign; esthiomene; rectal stricture.
- Doxycycline 100 mg twice daily for 21 days.

Describe clinical implication of AIDS on the Genito-Urinary Tract? (LAQ)

- AIDS = CD4 <200; UTI risk rises as CD4 falls.
- HIV-TB: disseminated, less fibrosis, fewer strictures.
- Indinavir stones: radiolucent, may be CT-invisible; stop drug and hydrate.
- PEP within 72 hours for 28 days; circumcision cuts acquisition 50–60%.

Lichen sclerosus (BXO, balanitis xerotica obliterans) (SAQ)

- Genital skin disease; meatus is the sentinel site.
- Spares proximal urethra; perineal urethrostomy easy.
- Clobetasol first; oral mucosa, never skin, for repair.
- SCC in 2.3–9%: lifelong follow-up.

Premalignant conditions in urology (including premalignant lesions of penis) (SAQ)

- PeIN: intact basement membrane; WHO 2022 all high grade.
- Queyrat 10–33% vs Bowen (about) 5% progression; bowenoid papulosis does not progress.
- Lichen sclerosus: SCC 2.3–9%.
- Untreated bladder CIS: 40–83% invade.
- GCNIS: 50% invasive in 5 years; radiotherapy ≥20 Gy.

Genitourinary tuberculosis: pathogenesis, diagnosis, management and thimble bladder (LAQ)

- Kidney 80%; periglomerular seeding; bladder disease descends from the kidney; heals by fibrosis, so obstruction worsens on treatment.
- Sterile pyuria; 3–5 early-morning urine AFB cultures; culture is the gold standard and the only test that monitors treatment.
- GeneXpert (Xpert) detects rifampicin resistance in 2 hours; cannot monitor treatment.
- LF-LAM for HIV with CD4 under 200.
- TST and IGRA show infection, not active disease.
- 2HRZE/4HR with pyridoxine; surgery after 4–6 weeks of ATT.
- Thimble bladder <20 mL: augment if capacity <100 mL; substitute if <20 mL; needs creatinine <2 mg/dL and a good urethra.
- Relapse up to 22%; follow up 10 years.

Discuss the Aetiopathogenesis, Clinical feature and management of Chyluria. (LAQ)

- Chyluria = pyelolymphatic fistula, usually from *W. bancrofti* obstruction.
- Ether test and urine triglycerides confirm; cystoscopy after a fatty meal lateralises.
- Diet with MCT and DEC → silver nitrate sclerotherapy → laparoscopic hilar lymphatic disconnection.
- Chyluria precedes elephantiasis and is often intermittent.

Hydatid disease of kidney (SAQ)

- Kidney: third site (about 2–3%).
- Hydatiduria (grape-skin membranes) is pathognomonic of a communicating cyst.
- Albendazole 400 mg twice daily around surgery; no scolicide if communicating. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

Genitourinary infections, common organisms, bacteraemia and urosepsis: pathophysiology and management (LAQ)

- *E. coli* ~85% community UTI; P pili cause pyelonephritis; *Proteus* causes struvite; *Enterococcus* resists cephalosporins.
- Sepsis = SOFA rise ≥2; septic shock = vasopressor for MAP ≥65 plus lactate >2 mmol/L.
- Obstruction plus infection is the emergency; bladder urine may be sterile.
- Hour-1: lactate, cultures, antibiotics, 30 mL/kg, noradrenaline (SSC 2021).
- Infected obstructed kidney: drain urgently; no definitive stone surgery in sepsis.
- Stent and nephrostomy equally effective (stent = nephrostomy for drainage); delay definitive stone surgery.

Principles of antimicrobial therapy in UTI (SAQ)

- Recurrent = ≥2 in 6 months or ≥3 in 12 months.
- Cystitis: nitrofurantoin 5 d, fosfomycin single, TMP-SMX 3 d (resistance ≤20%).
- Pyelonephritis: ciprofloxacin 7 d or levofloxacin 5 d (FQ resistance ≤10%).

Ciprofloxacin (SAQ)

- DNA gyrase and topoisomerase IV; concentration-dependent.
- Use for pyelonephritis only if local resistance ≤10%.
- Tendon rupture, neuropathy, aortic aneurysm; avoid in pregnancy and children.

Complicated urinary tract infections. (SAQ)

- Complicated = abnormal tract or compromised host; male sex alone qualifies.
- Culture mandatory; often MDR.
- Drain obstruction: stent = nephrostomy.
- 7–14 days; oral step-down when afebrile.

Drug Fosfomycin (SAQ)

- Inhibits MurA: first step of cell wall synthesis; no cross-resistance.
- 3 g single dose for cystitis; active against ESBL E. coli and VRE.
- Not for pyelonephritis; poor Pseudomonas cover.

Papillary necrosis. (SAQ)

- Vasa recta, low oxygen and hypertonicity make the papilla vulnerable.
- POSTCARDS; diabetes, analgesics, sickle cell, obstruction with infection.
- Ring sign; sloughed papilla causes colic and obstruction: drain urgently.

Cephalsporins (SAQ)

- Cefazolin = standard surgical prophylaxis.
- Ceftriaxone misses *Pseudomonas*; ceftazidime and cefepime cover it.
- No cephalosporin covers enterococci; ESBL hydrolyses third-generation agents.

Biofilm (SAQ)

- Biofilm: microbes in self-made matrix on a surface.
- Survives 1000–1500 times normal antibiotic levels.
- Treat CAUTI by changing the catheter.
- Revision washout cuts prosthesis infection from about 10% to 3%.

Renal mucormycosis (SAQ)

- Angioinvasion → infarction; diabetic ketoacidosis is the key risk.
- CECT: non-enhancing enlarged kidney; histology: broad aseptate right-angle hyphae.
- Liposomal amphotericin B 5 mg/kg/day plus nephrectomy and glucose control.
- Voriconazole and echinocandins do not work.

Nitrofurantoin. (SAQ)

- Urine-only drug: cystitis yes, pyelonephritis or prostatitis no.
- 100 mg twice daily × 5 days; prophylaxis 50–100 mg at night (children 1–2 mg/kg).
- Proteus intrinsically resistant; resistance in E. coli low.
- Toxicity: pulmonary fibrosis, neuropathy, G6PD haemolysis; CrCl ≥30 (Beers 2015). *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*